previously blocked side back to the origin and then descend again, establishing a circus movement. An example of this in a ring of tissue is shown in **Figure 29–12**. If the reentry is in the AV node, the reentrant activity depolarizes the atrium, and the resulting atrial beat is called an echo beat. In addition, the reentrant activity in the node propagates back down to the ventricle, producing paroxysmal nodal tachycardia. Circus movements can also become established in the atrial or ventricular muscle fibers. In individuals with an abnormal extra bundle of conducting tissue connecting the atria to the ventricles (bundle of Kent), the circus activity can pass in one direction through the AV node and in the other direction through the bundle, thus involving both the atria and the ventricles.

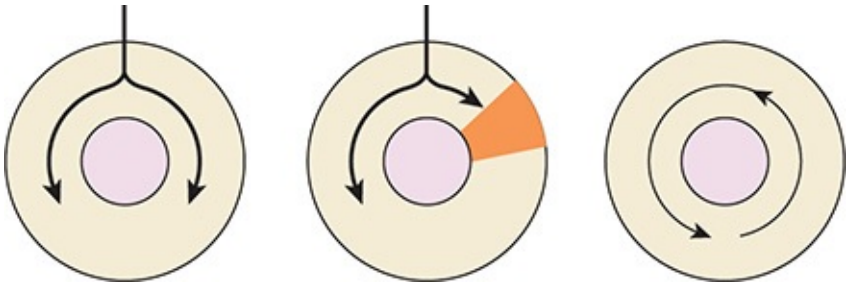


FIGURE 29–12 Depolarization of a ring of cardiac tissue. Normally, the impulse spreads in both directions in the ring **(left)** and the tissue immediately behind each branch of the impulse is refractory. When a transient block occurs on one side **(center),** the impulse on the other side goes around the ring, and if the transient block has now worn off **(right),** the impulse passes this area and continues to circle indefinitely (circus movement).

ATRIAL ARRHYTHMIAS

Excitation spreading from an independently discharging focus in the atria stimulates the AV node prematurely and is conducted to the ventricles. The P waves of atrial extrasystoles are abnormal, but the QRST configurations are usually normal **(Figure 29–13)**. The excitation may depolarize the SA node, which must repolarize and then depolarize to the firing level before it can initiate the next normal beat. Consequently, a pause occurs between the extrasystole and the next normal beat that is usually equal in length to the interval between the normal beats preceding the extrasystole, and the rhythm is “reset” (see below).

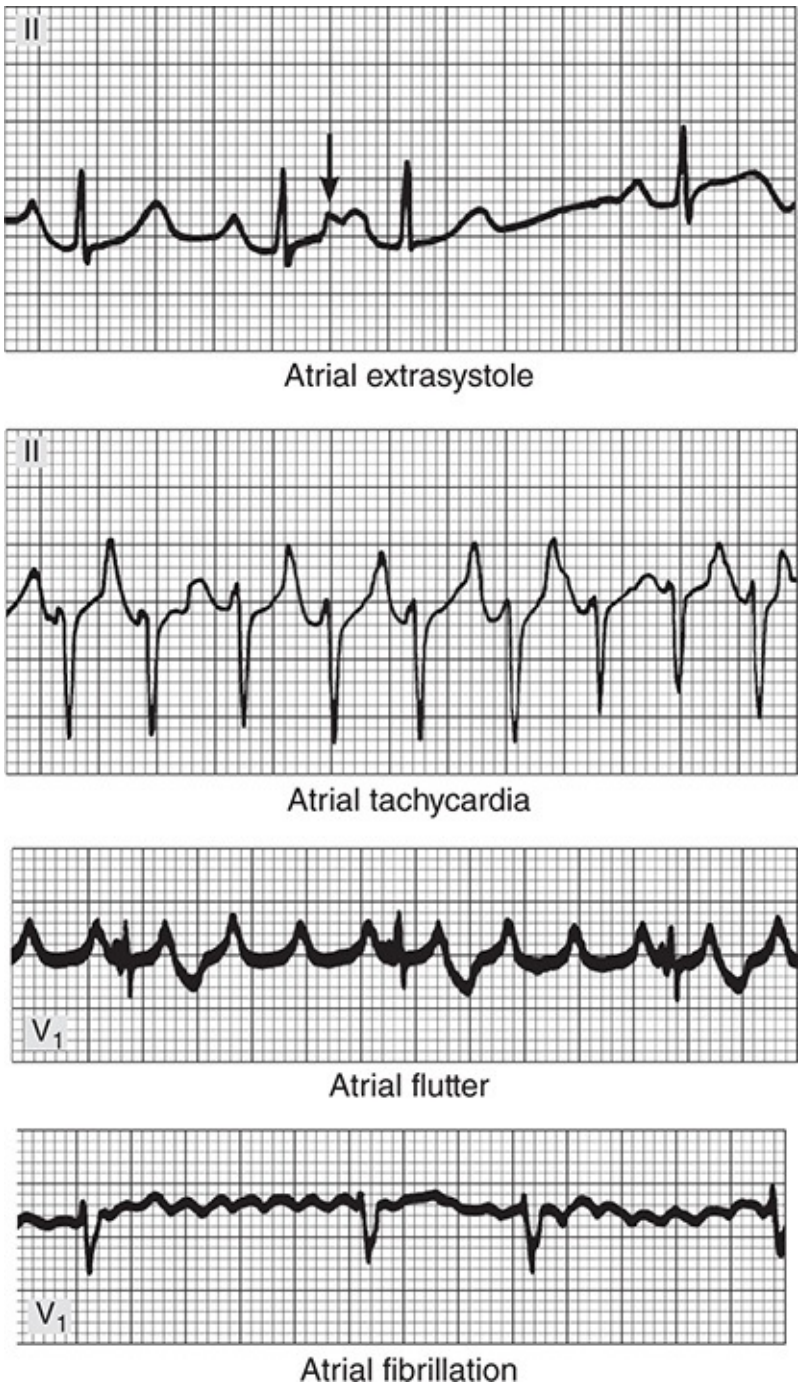


FIGURE 29–13 Atrial arrhythmias. The illustration shows an atrial premature beat with its P wave superimposed on the T wave of the preceding beat (*arrow*); atrial tachycardia; atrial flutter with 4:1 AV block; and atrial fibrillation with a totally irregular ventricular rate. Leads used to capture electrical activity are marked in each trace. AV, atrioventricular. (Tracings reproduced with permission from Goldschlager N, Goldman MJ: *Principles of Clinical Electrocardiography*, 13th ed. Originally published by Appleton & Lange. Copyright © 1989 by

McGraw-Hill.)

Atrial tachycardia occurs when an atrial focus discharges regularly or there is reentrant activity producing atrial rates up to 220/min. Sometimes, especially in digitalized patients, some degree of atrioventricular block is associated with the tachycardia **(paroxysmal atrial tachycardia with block).**

In atrial flutter, the atrial rate is 200–350/min (Figure 29–13). In the most common form of this arrhythmia, there is large counterclockwise circus movement in the right atrium. This produces a characteristic sawtooth pattern of flutter waves due to atrial contractions. It is almost always associated with 2:1 or greater AV block, because in adults the AV node cannot conduct more than about 230 impulses per minute.

In **atrial fibrillation,** the atria beat very rapidly (300–500/min) in a completely irregular and disorganized fashion. Because the AV node discharges at irregular intervals, the ventricles also beat at a completely irregular rate, usually 80–160/min (Figure 29–13). The condition can be paroxysmal or chronic, and in some cases there appears to be a genetic predisposition. The cause of atrial fibrillation is still a matter of debate, but in most cases it appears to be due to multiple concurrently circulating reentrant excitation waves in both atria. However, some cases of paroxysmal atrial fibrillation seem to be produced by discharge of one or more ectopic foci. Many of these foci appear to be located in the pulmonary veins as much as 4 cm from the heart. Atrial muscle fibers extend along the pulmonary veins and are the origin of these discharges.

CONSEQUENCES OF ATRIAL ARRHYTHMIAS

Occasional atrial extrasystoles occur from time to time in most normal humans and have no pathological significance. In paroxysmal atrial tachycardia and flutter, the ventricular rate may be so high that diastole is too short for adequate filling of the ventricles with blood between contractions. Consequently, cardiac output is reduced and symptoms of heart failure appear. Heart failure may also complicate atrial fibrillation when the ventricular rate is high. Acetylcholine liberated at vagal endings depresses conduction in the atrial musculature and AV node. This is why stimulating reflex vagal discharge by pressing on the eyeball **(oculocardiac reflex)** or massaging the carotid sinus often converts tachycardia and sometimes converts atrial flutter to normal sinus rhythm. Alternatively, vagal stimulation increases the degree of AV block, abruptly lowering the ventricular rate. Digitalis also depresses AV conduction and is used to lower a

rapid ventricular rate in atrial fibrillation.

VENTRICULAR ARRHYTHMIAS

Premature beats that originate in an ectopic ventricular focus usually have bizarrely shaped prolonged QRS complexes **(Figure 29–14)** because of the slow spread of the impulse from the focus through the ventricular muscle to the rest of the ventricle. They are usually incapable of exciting the bundle of His, and retrograde conduction to the atria therefore does not occur. In the meantime, the next succeeding normal SA nodal impulse depolarizes the atria. The P wave is usually buried in the QRS of the extrasystole. If the normal impulse reaches the ventricles, they are still in the refractory period following depolarization from the ectopic focus.

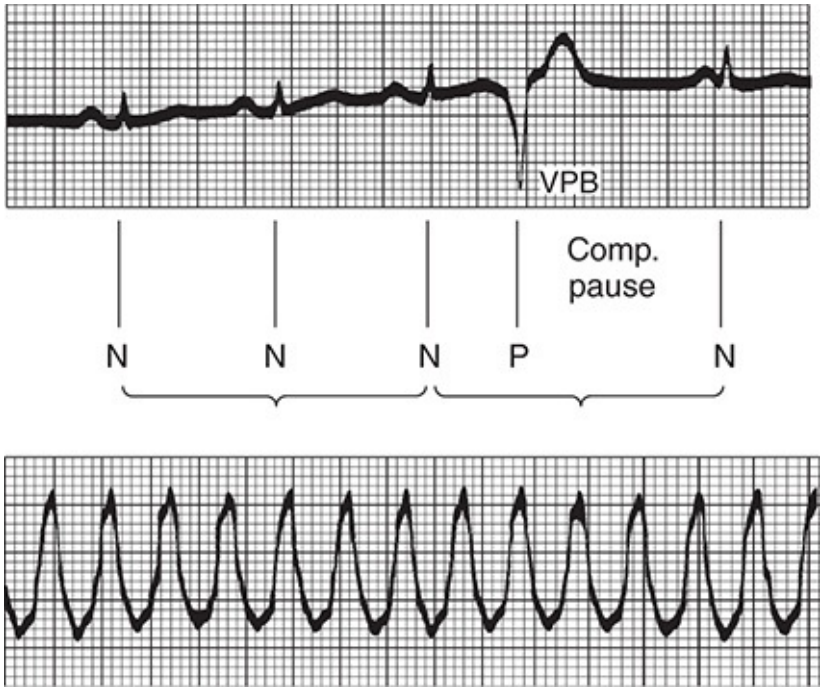


FIGURE 29–14 Top: Ventricular premature beats (VPB). The lines under the tracing illustrate the compensatory pause and show that the duration of the premature beat plus the preceding normal beat is equal to the duration of two normal beats. **Bottom:** Ventricular tachycardia.

However, the second succeeding impulse from the SA node produces a normal beat. Thus, ventricular premature beats are followed by a **compensatory pause** that is often longer than the pause after an atrial extrasystole.

Furthermore, ventricular premature beats do not interrupt the regular discharge of the SA node, whereas atrial premature beats often interrupt and “reset” the normal rhythm.

Atrial and ventricular premature beats are not strong enough to produce a pulse at the wrist if they occur early in diastole, when the ventricles have not had time to fill with blood and the ventricular musculature is still in its relatively refractory period. They may not even open the aortic and pulmonary valves, in which case there is, in addition, no second heart sound.

Paroxysmal ventricular tachycardia (Figure 29–14) is in effect a series of rapid, regular ventricular depolarizations usually due to a circus movement involving the ventricles. **Torsades de pointes** is a form of ventricular tachycardia in which the QRS morphology varies **(Figure 29–15)**. Tachycardias originating above the ventricles (supraventricular tachycardias such as paroxysmal nodal tachycardia) can be distinguished from paroxysmal ventricular tachycardia by use of the HBE; in supraventricular tachycardias, a His bundle H deflection is present, whereas in ventricular tachycardias, there is none. Ventricular premature beats are not uncommon and, in the absence of ischemic heart disease, usually benign. Ventricular tachycardia is more serious because cardiac output is decreased, and ventricular fibrillation is an occasional complication of ventricular tachycardia.

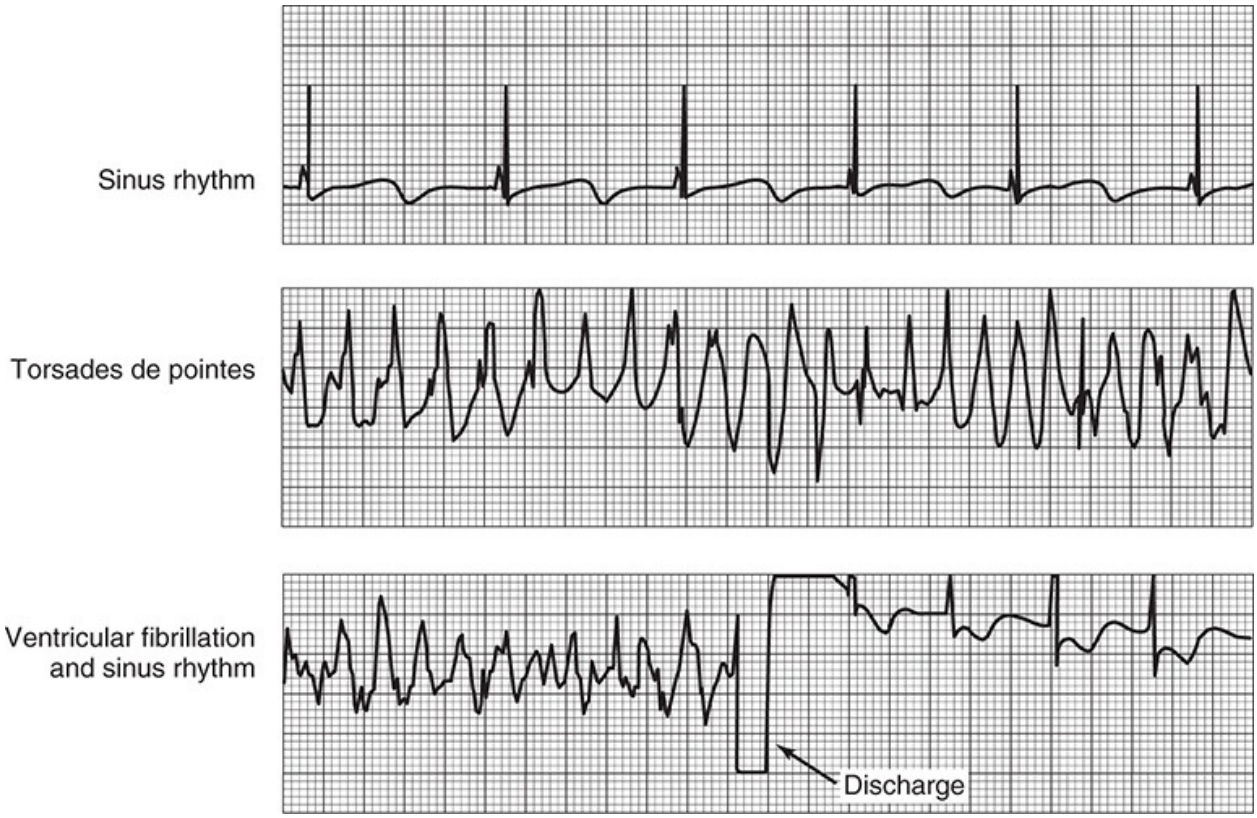


FIGURE 29–15 Record obtained from an implanted cardioverter–defibrillator in a 12-year-old boy with congenital long QT syndrome who collapsed while answering a question in school. Top: Normal sinus rhythm with long QT interval. **Middle:** Torsades de pointes. **Bottom:** Ventricular fibrillation with discharge of defibrillator, as programmed 7.5 s after the start of ventricular tachycardia, converting the heart to normal sinus rhythm. The boy recovered consciousness in 2 min and had no neurologic sequelae. (Reproduced with permission from Moss AJ, Daubert JP: Images in clinical medicine. Internal ventricular fibrillation. N Engl J Med 2000;342:398.)

In **ventricular fibrillation** (Figure 29–15), the ventricular muscle fibers contract in a totally irregular and ineffective way because of the very rapid discharge of multiple ventricular ectopic foci or a circus movement. The fibrillating ventricles, like the fibrillating atria, look like a quivering “bag of worms.” Ventricular fibrillation can be produced by an electric shock or an extrasystole during a critical interval, the **vulnerable period.** The vulnerable period coincides in time with the midportion of the T wave; that is, it occurs at a time when some of the ventricular myocardium is depolarized, some is incompletely repolarized, and some is completely repolarized. These are excellent conditions in which to establish reentry and a circus movement. The

fibrillating ventricles cannot pump blood effectively, and circulation of the blood stops. Therefore, in the absence of emergency treatment, ventricular fibrillation that lasts more than a few minutes is fatal. The most frequent cause of sudden death in patients with myocardial infarcts is ventricular fibrillation.

LONG QT SYNDROME

An indication of vulnerability of the heart during repolarization is the fact that in patients in whom the QT interval is prolonged, cardiac repolarization is irregular and the incidence of ventricular arrhythmias and sudden death increases. The syndrome can be caused by a number of different drugs, by electrolyte abnormalities, and by myocardial ischemia. It can also be congenital. Mutations of eight different genes have been reported to cause the syndrome. Six cause reduced function of various K^+ channels by alterations in their structure; one inhibits a K^+ channel by reducing the amount of the ankyrin isoform that links it to the cytoskeleton; and one increases the function of the cardiac Na^+ channel. Long QT syndrome is discussed in Clinical Box 5–5.

ACCELERATED AV CONDUCTION

An interesting condition seen in some otherwise normal individuals who are prone to attacks of paroxysmal atrial arrhythmias is **accelerated AV conduction (Wolff-Parkinson-White syndrome).** Normally, the only conducting pathway between the atria and the ventricles is the AV node. Individuals with Wolff-Parkinson-White syndrome have an additional aberrant muscular or nodal tissue connection **(bundle of Kent)** between the atria and ventricles. This conducts more rapidly than the slowly conducting AV node, and one ventricle is excited early. The manifestations of its activation merge with the normal QRS pattern, producing a short PR interval and a prolonged QRS deflection slurred on the upstroke **(Figure 29–16)**, with a normal interval between the start of the P wave and the end of the QRS complex (“PJ interval”). The paroxysmal atrial tachycardias seen in this syndrome often follow an atrial premature beat. This beat conducts normally down the AV node but spreads to the ventricular end of the aberrant bundle, and the impulse is transmitted retrograde to the atrium. A circus movement is thus established. Less commonly, an atrial premature beat finds the AV node refractory but reaches the ventricles via the bundle of Kent, setting up a circus movement in which the impulse passes from the ventricles to

the atria via the AV node.

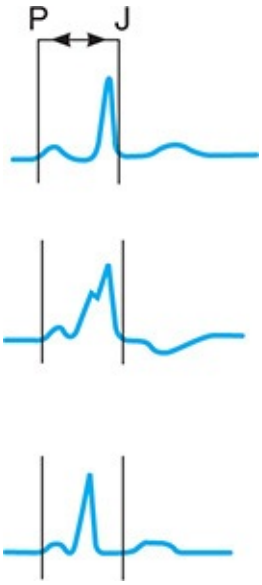


FIGURE 29–16 Accelerated AV conduction. Top: Normal sinus beat. **Middle:** Short PR interval; wide, slurred QRS complex; normal PJ interval (Wolff-Parkinson-White syndrome). **Bottom:** Short PR interval, normal QRS complex (Lown-Ganong-Levine syndrome). (Reproduced with permission from Goldschlager N, Goldman MJ: *Principles of Clinical Electrocardiography*, 13th ed. Originally published by Appleton & Lange. Copyright © 1989 by McGraw-Hill.)

In some instances, the Wolff-Parkinson-White syndrome is familial. In two such families, there is a mutation in a gene that codes for an AMP-activated protein kinase. Presumably, this kinase is normally involved in suppressing abnormal atrioventricular pathways during fetal development.

Attacks of paroxysmal supraventricular tachycardia, usually nodal tachycardia, are seen in individuals with short PR intervals and normal QRS complexes **(Lown-Ganong-Levine syndrome).** In this condition, depolarization presumably passes from the atria to the ventricles via an aberrant bundle that bypasses the AV node but enters the intraventricular conducting system distal to the node.

TREATMENTS FOR ARRHYTHMIAS

Many different drugs used in the treatment of arrhythmias slow conduction in

the conduction system and the myocardium. This depresses ectopic activity and reduces the discrepancy between normal and reentrant paths so that reentry does not occur. Drugs that target Na^+ channels (eg, quinidine) can slow I_{Na} and prolong refractoriness (eg, quinidine, disopy-ramide), decrease I_{Na} with minimal prolongation of refractoriness (eg, flecainide, propafenone) or shorten refractoriness in depolarized cells (eg, lidocaine, mexiletine). Drugs that target K^+ channels can prolong refractoriness (eg, amiodarone, sotalol, dofetilide). Drugs that block L-type voltage-gated Ca^{2+} channels can slow SA pacemaker and AV conduction (eg, nifedipine, verapamil, diltiazem). Finally, drugs that block β-adrenergic receptors thus reduce the activation of I_{CaL} (eg, propanolol, metoprolol). Interestingly, it has become clear that in some patients any of these drugs can be **proarrhythmic** rather than antiarrhythmic—that is, they can also cause various arrhythmias. Therefore, careful monitoring and alternative procedures are extremely important when using antiarrhythmic drugs.

An alternative treatment is radiofrequency catheter ablation of reentrant pathways. Catheters with electrodes at the tip can be inserted into the chambers of the heart and its environs and used to map the exact location of an ectopic focus or accessory bundle responsible for the production of reentry and supraventricular tachycardia. The pathway can then be ablated by passing radiofrequency current with the catheter tip placed close to the bundle or focus. In skilled hands, this form of treatment can be very effective and is associated with few complications. It is particularly useful in conditions that cause supraventricular tachycardias, including Wolff-Parkinson-White syndrome and atrial flutter. It has also been used with success to ablate foci in the pulmonary veins causing paroxysmal atrial fibrillation.

ELECTROCARDIOGRAPHIC FINDINGS IN OTHER CARDIAC & SYSTEMIC DISEASES

MYOCARDIAL INFARCTION

When the blood supply to part of the myocardium is interrupted, profound changes take place in the myocardium that lead to irreversible changes and death of muscle cells. The ECG is very useful for diagnosing ischemia and locating areas of infarction. The underlying electrical events and the resulting electrocardiographic changes are complex, and only a brief review can be presented here.

The three major abnormalities that cause electrocardiographic changes in acute myocardial infarction are summarized in **Table 29–3**. The first change—abnormally rapid repolarization after discharge of the infarcted muscle fibers as a result of accelerated opening of K^+ channels—develops seconds after occlusion of a coronary artery in experimental animals. It lasts only a few minutes, but before it is over the resting membrane potential of the infarcted fibers declines because of the loss of intracellular K^+. Starting about 30 min later, the infarcted fibers also begin to depolarize more slowly than the surrounding normal fibers.

TABLE 29–3 Summary of the three major abnormalities of membrane polarization associated with acute myocardial infarction.

Defect in Infarcted Cells	Current Flow	Resultant ECG Change in Leads over Infarct
Rapid repolarization	Out of infarct	ST segment elevation
Decreased resting membrane potential	Into infarct	TQ segment depression (manifested as ST segment elevation)
Delayed depolarization	Out of infarct	ST segment elevation

All three of these changes cause current flow that produces elevation of the ST segment in electrocardiographic leads recorded with electrodes over the infarcted area **(Figure 29–17)**. Because of the rapid repolarization in the infarct, the membrane potential of the area is greater than it is in the normal area during the latter part of repolarization, making the normal region negative relative to the infarct. Extracellularly, current therefore flows out of the infarct into the normal area (since, by convention, current flow is from positive to negative). This current flows toward electrodes over the injured area, causing increased positivity between the S and T waves of the ECG. Similarly, the delayed depolarization of the infarcted cells causes the infarcted area to be positive relative to the healthy tissue (Table 29–3) during the early part of repolarization, and the result is also ST segment elevation. The remaining change—the decline in resting membrane potential during diastole—causes a current flow into the infarct during ventricular diastole. The result of this current flow is a depression

of the TQ segment of the ECG. However, the electronic arrangement in electrocardiographic recorders is such that a TQ segment depression is recorded as an ST segment elevation. Thus, the hallmark of acute myocardial infarction is elevation of the ST segments in the leads overlying the area of infarction (Figure 29–17). Leads on the opposite side of the heart show ST segment depression.

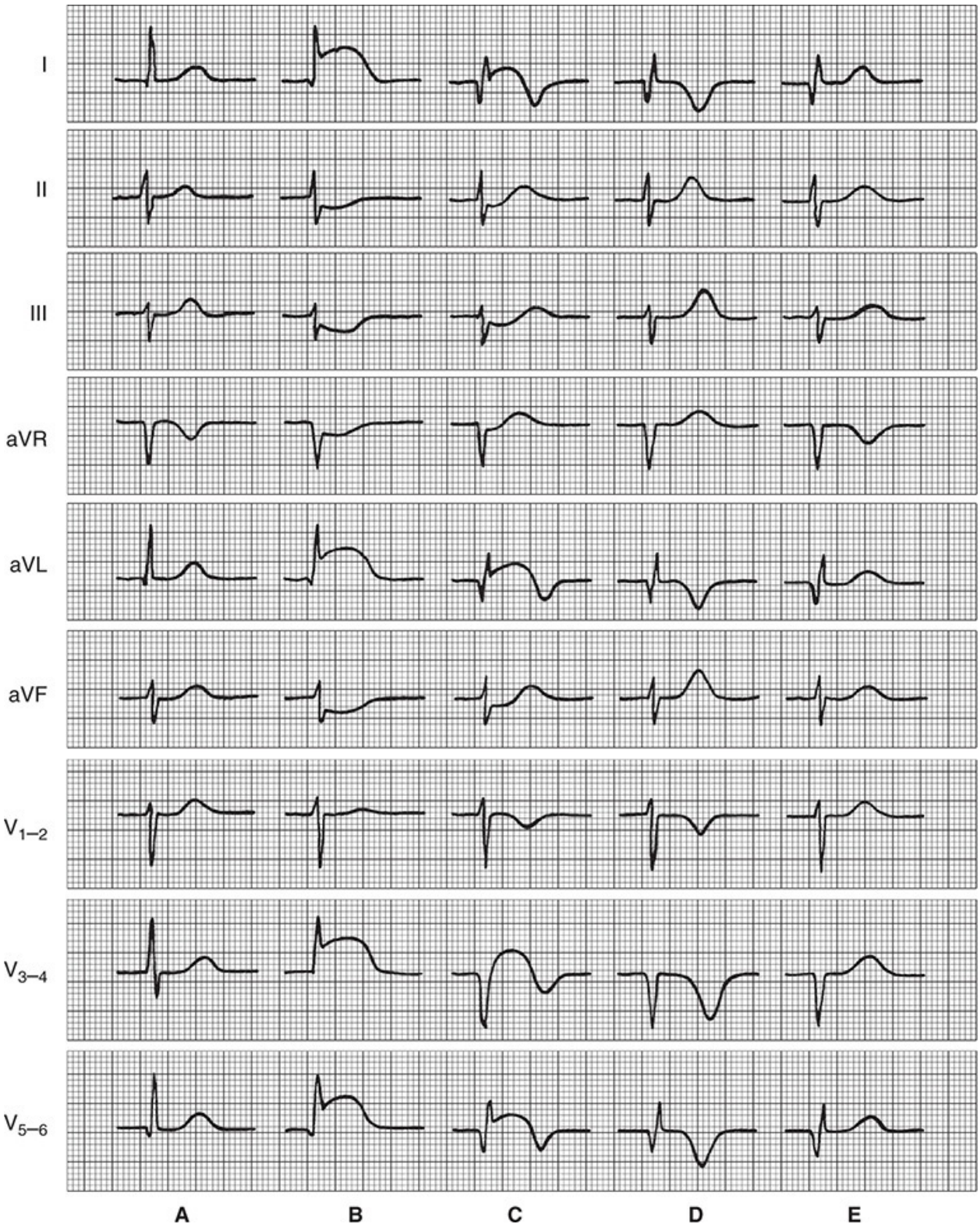


FIGURE 29–17 Diagrammatic illustration of serial electrocardiographic patterns in anterior infarction. A) Normal tracing. **B)** Very early pattern (hours

after infarction): ST segment elevation in I, aVL, and $V_{3–6}$; reciprocal ST depression in II, III, and aVF. **C)** Later pattern (many hours to a few days): Q waves have appeared in I, aVL, and $V_{5–6}$. QS complexes are present in $V_{3–4}$. This indicates that the major transmural infarction is underlying the area recorded by $V_{3–4}$; ST segment changes persist but are of lesser degree, and the T waves are beginning to invert in the leads in which the ST segments are elevated. **D)** Late established pattern (many days to weeks): The Q waves and QS complexes persist, the ST segments are isoelectric, and the T waves are symmetric and deeply inverted in leads that had ST elevation and tall in leads that had ST depression. This pattern may persist for the remainder of the patient's life. **E)** Very late pattern: This may occur many months to years after the infarction. The abnormal Q waves and QS complexes persist. The T waves have gradually returned to normal. (Reproduced with permission from Goldschlager N, Goldman MJ: *Principles of Clinical Electrocardiography,* 13th ed. Originally published by Appleton & Lange. Copyright © 1989 by McGraw-Hill.)

After some days or weeks, the ST segment abnormalities subside. The dead muscle and scar tissue become electrically silent. The infarcted area is therefore negative relative to the normal myocardium during systole, and it fails to contribute its share of positivity to the electrocardiographic complexes. The manifestations of this negativity are multiple and subtle. Common changes include the appearance of a Q wave in some of the leads in which it was not previously present and an increase in the size of the normal Q wave in some of the other leads, although so-called non–Q-wave infarcts are also seen. These latter infarcts tend to be less severe, but there is a high incidence of subsequent reinfarction. Another finding in infarction of the anterior left ventricle is "failure of progression of the R wave;" that is, the R wave fails to become successively larger in the precordial leads as the electrode is moved from right to left over the left ventricle. If the septum is infarcted, the conduction system may be damaged, causing bundle branch block or other forms of heart block.

Myocardial infarctions are often complicated by serious ventricular arrhythmias, with the threat of ventricular fibrillation and death. In experimental animals, and presumably in humans, ventricular arrhythmias occur during three periods. During the first 30 min of an infarction, arrhythmias due to reentry are common. There follows a period relatively free from arrhythmias but, starting 12 h after infarction, arrhythmias occur as a result of increased automaticity. Arrhythmias occurring 3 days to several weeks after infarction are once again

usually due to reentry. It is worth noting in this regard that infarcts that damage the epicardial portions of the myocardium interrupt sympathetic nerve fibers, producing denervation super-sensitivity to catecholamines in the area beyond the infarct. Alternatively, endocardial lesions can selectively interrupt vagal fibers, leaving the actions of sympathetic fibers unopposed.

EFFECTS OF CHANGES IN THE IONIC COMPOSITION OF THE BLOOD

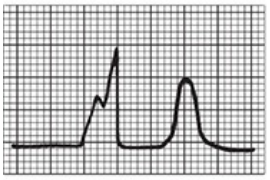
Changes in the Na^+ and K^+ concentrations of the extracellular fluids would be expected to affect the potentials of the myocardial fibers because the electrical activity of the heart depends on the distribution of these ions across the muscle cell membranes. Clinically, a fall in the plasma level of Na^+ may be associated with low-voltage electrocardiographic complexes, but changes in the plasma K^+ level produce severe cardiac abnormalities. Hyperkalemia is a very dangerous and potentially lethal condition because of its effects on the heart. As the plasma K^+ level rises, the first change in the ECG is the appearance of tall peaked T waves, a manifestation of altered repolarization **(Figure 29–18)**. At higher K^+ levels, paralysis of the atria and prolongation of the QRS complexes occur. Ventricular arrhythmias may develop. The resting membrane potential of the muscle fibers decreases as the extracellular K^+ concentration increases. The fibers eventually become unexcitable, and the heart stops in diastole. Conversely, a decrease in the plasma K^+ level causes prolongation of the PR interval, prominent U waves, and, occasionally, late T wave inversion in the precardial leads. If the T and U waves merge, the apparent QT interval is often prolonged; if the T and U waves are separated, the true QT interval is seen to be of normal duration. Hypokalemia is a serious condition, but it is not as rapidly fatal as hyperkalemia.



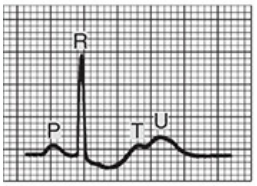
Normal tracing (plasma K^+ 4–5.5 mEq/L). PR interval = 0.16 s; QRS interval = 0.06 s; QT interval = 0.4 s (normal for an assumed heart rate of 60).



Hyperkalemia (plasma K^+ ±7.0 mEq/L). The PR and QRS intervals are within normal limits. Very tall, slender peaked T waves are now present.



Hyperkalemia (plasma K^+ ±8.5 mEq/L). There is no evidence of atrial activity; the QRS complex is broad and slurred and the QRS interval has widened to 0.2 s. The T waves remain tall and slender. Further elevation of the plasma K^+ level may result in ventricular tachycardia and ventricular fibrillation.



Hypokalemia (plasma K^+ ±3.5 mEq/L). PR interval = 0.2 s; QRS interval = 0.06 s; ST segment depression. A prominent U wave is now present immediately following the T. The actual QT interval remains 0.4 s. If the U wave is erroneously considered a part of the T, a falsely prolonged QT interval of 0.6 s will be measured.



Hypokalemia (plasma K^+ ±2.5 mEq/L). The PR interval is lengthened to 0.32 s; the ST segment is depressed; the T wave is inverted; a prominent U wave is seen. The true QT interval remains normal.

FIGURE 29–18 Correlation of plasma K^+ level and the ECG, assuming that the plasma Ca^{2+} level is normal. The diagrammed complexes are left ventricular epicardial leads. ECG, electrocardiogram. (Reproduced with permission from Goldman MJ: *Principles of Clinical Electrocardiography*, 12th ed. Originally published by Appleton & Lange. Copyright © 1986 by McGraw-Hill.)

Increases in extracellular Ca^{2+} concentration enhance myocardial contractility. When large amounts of Ca^{2+} are infused into experimental animals, the heart relaxes less during diastole and eventually stops in systole **(calcium rigor).** However, in clinical conditions associated with hypercalcemia, the plasma calcium level is rarely if ever high enough to affect the heart. Hypocalcemia causes prolongation of the ST segment and consequently of the QT interval, a change that is also produced by phenothiazines and tricyclic antidepressant drugs and by various diseases of the central nervous system.

CHAPTER SUMMARY

- Contractions in the heart are controlled via a well-regulated electrical signaling cascade that originates in pacemaker cells in the sinoatrial (SA) node and is passed via internodal atrial pathways to the atrioventricular (AV) node, the bundle of His, the Purkinje system, and to all parts of the ventricle.
- Most cardiac cells have an action potential that includes a rapid depolarization, an initial rapid repolarization, a plateau, and a slow repolarization process to return to resting potential. These changes are defined by sequential activation, inactivation and deactivation of Na^+, Ca^{2+}, and K^+ channels.
- Compared to typical myocytes, pacemaker cells have a slightly different sequence of events. After repolarization to the resting potential, there is a slow depolarization that occurs due to a channel that can pass both Na^+ and K^+. As this "funny" current continues to depolarize the cell the activation threshold of the voltage-gated Ca^{2+} channels, Ca^{2+} channels are activated to rapidly depolarize the cell. The hyperpolarization phase is again dominated by K^+ current.
- Spread of the electrical signal from cell to cell is via gap junctions. The rate of spread is dependent on anatomic features, but also can be altered (to a certain extent) via neural input.
- The electrocardiogram (ECG or EKG) is an algebraic sum of the electrical activity in the heart. The normal ECG includes well-defined waves and segments, including the P wave (atrial depolarization), the QRS complex (ventricular depolarization), and the T wave (ventricular repolarization). Various arrhythmias can be detected in irregular ECG recordings.
- Because of the contribution of ionic movement to cardiac muscle contraction, heart tissue is sensitive to ionic composition of the blood. Most serious are increases in $[K^+]$ that can produce severe cardiac abnormalities, including paralysis of the atria and ventricular arrhythmias.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which part of the ECG (eg, Figure 29–5) corresponds to ventricular

repolarization?

A. The P wave

B. The QRS duration

C. The T wave

D. The U wave

E. The PR interval

2. Which of the following normally has a slowly depolarizing “prepotential”?

A. Sinoatrial node

B. Atrial muscle cells

C. Bundle of His

D. Purkinje fibers

E. Ventricular muscle cells

3. In second-degree heart block

A. the ventricular rate is lower than the atrial rate.

B. the ventricular ECG complexes are distorted.

C. there is a high incidence of ventricular tachycardia.

D. stroke volume is decreased.

E. cardiac output is increased.

4. Currents caused by opening of which of the following channels contribute to the repolarization phase of the action potential of ventricular muscle fibers?

A. Na^+ channels

B. Cl^- channels

C. Ca^{2+} channels

D. K^+ channels

E. HCO^{3-} channels

5. In complete heart block

A. fainting may occur because the atria are unable to pump blood into the ventricles.

B. ventricular fibrillation is common.

C. the atrial rate is lower than the ventricular rate.

D. fainting may occur because of prolonged periods during which the ventricles fail to contract.

CHAPTER 30

The Heart as a Pump

OBJECTIVES

After studying this chapter, you should be able to:

- Describe how the sequential pattern of contraction and relaxation in the heart results in a normal pattern of blood flow.
- Understand the pressure, volume, and flow changes that occur during the cardiac cycle.
- Explain the basis of the arterial pulse, heart sounds, and murmurs.
- Understand methods that can be used to measure cardiac output.
- Describe how the pumping action of the heart can be compromised in the setting of specific disease states such as heart failure and shock.
- Delineate the ways by which cardiac output can be upregulated in the setting of specific physiologic demands for increased oxygen supply to the tissues, such as exercise, and how the heart's own demand for oxygen is adjusted.

INTRODUCTION

Of course, the electrical activity of the heart discussed in the previous chapter is

designed to subserve the heart's primary physiologic role—to pump blood through the lungs, where gas exchange can occur, and thence to the remainder of the body. This is accomplished when the orderly depolarization process described in the previous chapter triggers a wave of contraction that spreads through the myocardium. In single muscle fibers, contraction starts just after depolarization and lasts until about 50 ms after repolarization is completed (see Figure 5–15). Atrial systole starts after the P wave of the electrocardiogram (ECG); ventricular systole starts near the end of the R wave and ends just after the T wave. This chapter will cover the ways in which these changes in contraction produce sequential changes in pressures and flows in the heart chambers and blood vessels, thereby propelling blood appropriately as needed by whole body demands for oxygen and nutrients. As an aside, it should be noted that the term **systolic pressure** in the vascular system refers to the peak pressure reached during systole, not the mean pressure; similarly, the **diastolic pressure** refers to the lowest pressure during diastole.

MECHANICAL EVENTS OF THE CARDIAC CYCLE

EVENTS IN LATE DIASTOLE

Late in diastole, the mitral (bicuspid) and tricuspid valves between the atria and ventricles (atrioventricular [AV] valves) are open and the aortic and pulmonary valves are closed. Blood flows into the heart throughout diastole, filling the atria and ventricles. The rate of filling declines as the ventricles become distended and, especially when the heart rate is low, the cusps of the AV valves drift toward the closed position **(Figure 30–1)**. The pressure in the ventricles remains low. About 70% of the ventricular filling occurs passively during diastole.

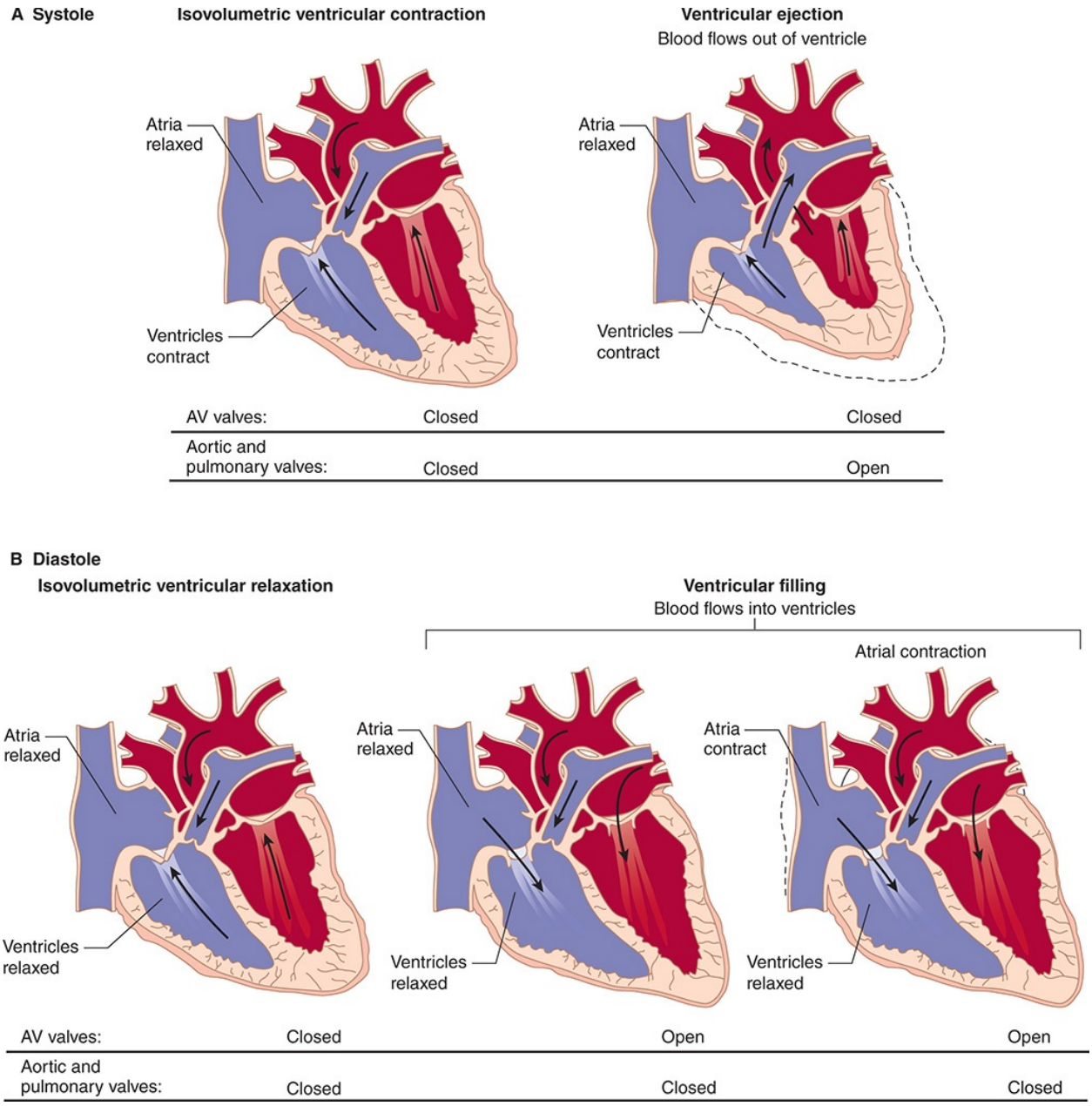


FIGURE 30–1 Divisions of the cardiac cycle: A) systole and B) diastole. The phases of the cycle are identical in both halves of the heart. The direction in which the pressure difference favors flow is denoted by an arrow; note, however, that flow will not actually occur if a valve prevents it. AV, atrioventricular. Note that the rightmost panel of those shown in B corresponds to atrial systole, even though the cardiac cycle overall remains in diastole.

ATRIAL SYSTOLE

Contraction of the atria propels some additional blood into the ventricles. Contraction of the atrial muscle narrows the orifices of the superior and inferior vena cava and pulmonary veins, and the inertia of the blood moving toward the heart tends to keep blood in it. However, despite these inhibitory influences, there is some regurgitation of blood into the veins.

VENTRICULAR SYSTOLE

At the start of ventricular systole, the AV valves close. Ventricular muscle initially shortens relatively little, but intraventricular pressure rises sharply as the myocardium presses on the blood in the ventricle **(Figure 30–2)**. This period of **isovolumetric (isovolumic, isometric) ventricular contraction** lasts about 0.05 s, until the pressures in the left and right ventricles exceed the pressures in the aorta (80 mm Hg; 10.6 kPa) and pulmonary artery (10 mm Hg) and the aortic and pulmonary valves open. During isovolumetric contraction, the AV valves bulge into the atria, causing a small but sharp rise in atrial pressure **(Figure 30–3)**.

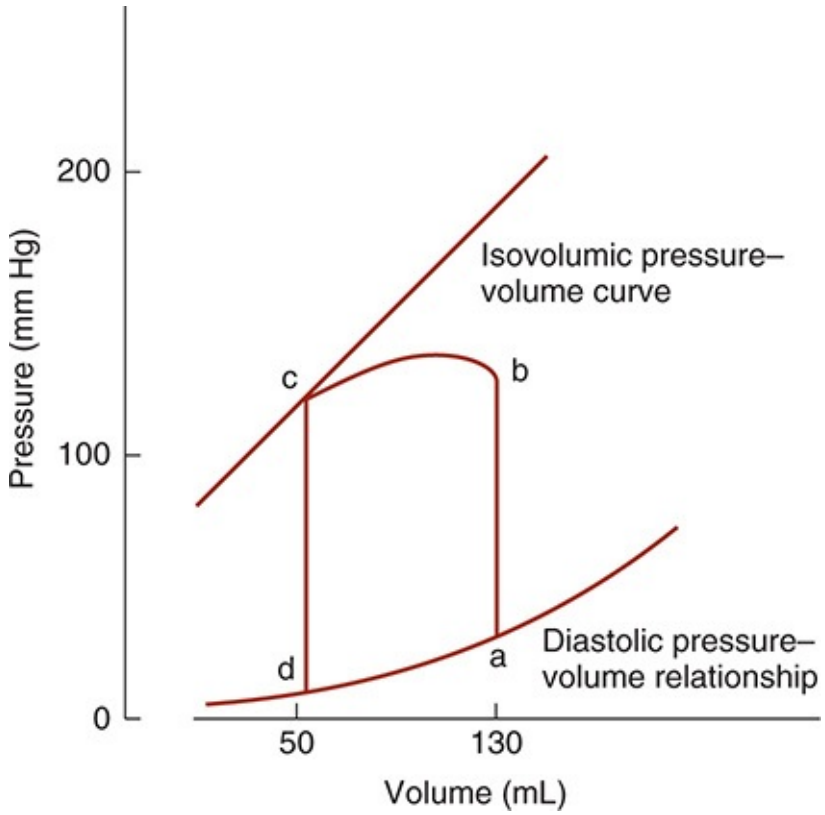


FIGURE 30–2 Normal pressure–volume loop of the left ventricle. During

diastole, the ventricle fills and pressure increases from d to a. Pressure then rises sharply from a to b during isovolumetric contraction and from b to c during ventricular ejection. At c, the aortic valves close and pressure falls during isovolumetric relaxation from c back to d. (Reproduced with permission from McPhee SJ, Lingappa VR, Ganong WF [editors]: *Pathophysiology of Disease*, 6th ed. New York, NY: McGraw-Hill; 2010.)

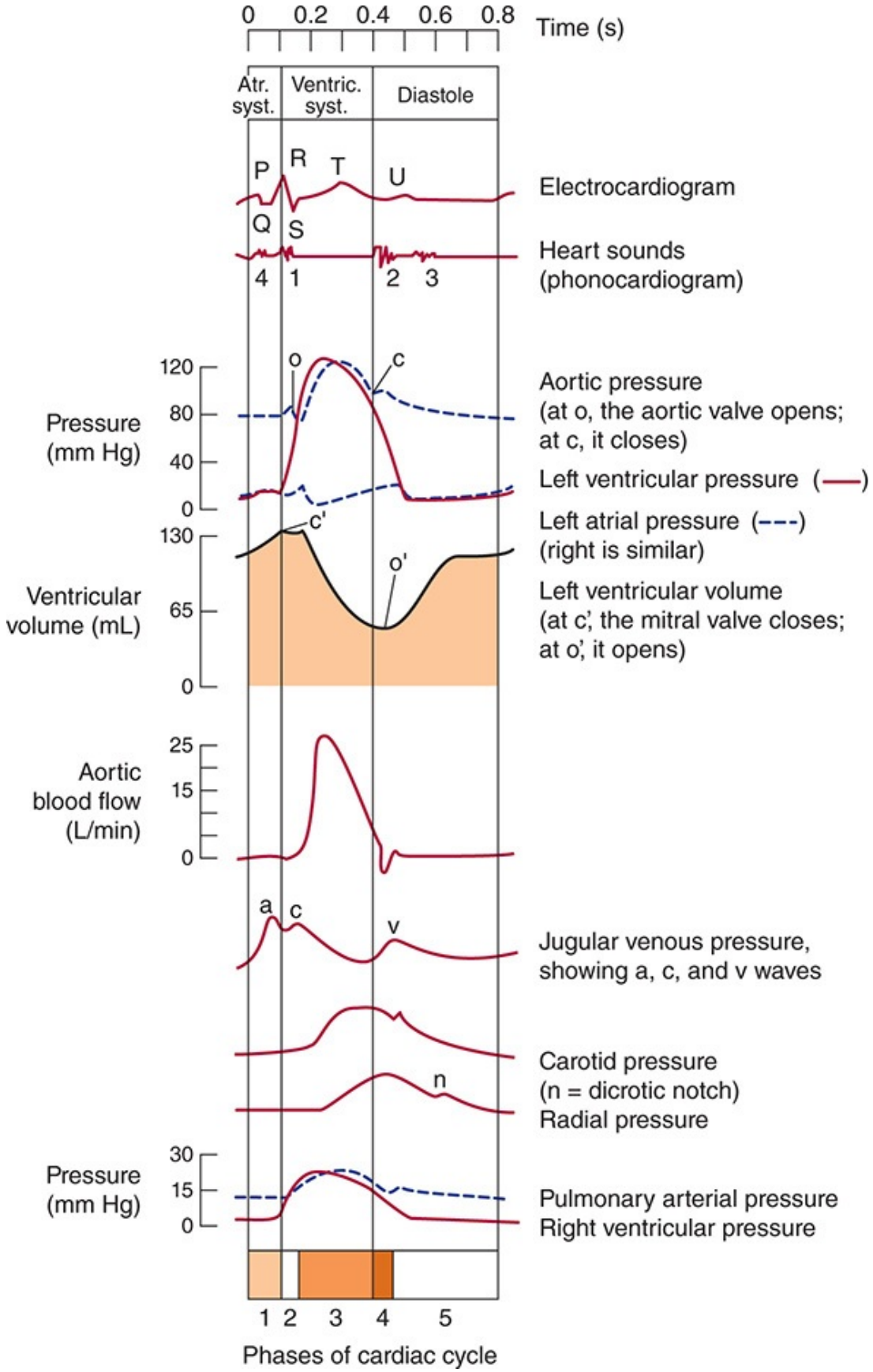

0
0.2
0.4
0.6
0.8
Time (s)
Atr. syst.
Ventric. syst.
Diastole
P
R
T
U
Q
S
Electrocardiogram
4
1
2
3
Heart sounds (phonocardiogram)
Pressure (mm Hg)
120
80
40
0
o
c
Aortic pressure (at o, the aortic valve opens; at c, it closes)
Left ventricular pressure (——)
Left atrial pressure (- - -) (right is similar)
Ventricular volume (mL)
130
65
0
c'
o'
Left ventricular volume (at c', the mitral valve closes; at o', it opens)
Aortic blood flow (L/min)
25
15
0
a
c
v
Jugular venous pressure, showing a, c, and v waves
n
Carotid pressure (n = dicrotic notch)
Radial pressure
Pressure (mm Hg)
30
15
0
Pulmonary arterial pressure
Right ventricular pressure
1
2
3
4
5
Phases of cardiac cycle

FIGURE 30–3 Events of the cardiac cycle at a heart rate of 75 beats/min. The phases of the cardiac cycle identified by the numbers at the bottom are as follows: 1, atrial systole; 2, isovolumetric ventricular contraction; 3, ventricular ejection; 4, isovolumetric ventricular relaxation; 5, ventricular filling. Note that late in systole, aortic pressure actually exceeds left ventricular pressure. However, the momentum of the blood keeps it flowing out of the ventricle for a short period. The pressure relationships in the right ventricle and pulmonary artery are similar.

When the aortic and pulmonary valves open, the phase of **ventricular ejection** begins. Ejection is rapid at first, slowing down as systole progresses. The intraventricular pressure rises to a maximum and then declines somewhat before ventricular systole ends. Peak pressures in the left and right ventricles are about 120 and 25 mm Hg, respectively. Late in systole, pressure in the aorta actually exceeds that in the left ventricle, but for a short period momentum keeps the blood moving forward. The AV valves are pulled down by the contractions of the ventricular muscle, and atrial pressure drops. The amount of blood ejected by each ventricle per stroke at rest is 70–90 mL. The **end-diastolic ventricular volume** is about 130 mL. Thus, about 50 mL of blood remains in each ventricle at the end of systole **(end-systolic ventricular volume),** and the **ejection fraction,** the percentage of the end-diastolic ventricular volume that is ejected with each stroke, is about 65%. The ejection fraction is a valuable index of ventricular function. It can be measured by injecting radionuclide-labeled red blood cells and imaging the cardiac blood pool at the end of diastole and the end of systole (equilibrium radionuclide angiocardiography), or by computed tomography.

EARLY DIASTOLE

Once the ventricular muscle is fully contracted, the already falling ventricular pressures drop more rapidly. This is the period of **protodiastole,** which lasts about 0.04 s. It ends when the momentum of the ejected blood is overcome and the aortic and pulmonary valves close, setting up transient vibrations in the blood and blood vessel walls. After the valves are closed, pressure continues to drop rapidly during the period of **isovolumetric ventricular relaxation.** Isovolumetric relaxation ends when the ventricular pressure falls below the atrial pressure and the AV valves open, permitting the ventricles to fill. Filling is rapid

at first, then slows as the next cardiac contraction approaches. Atrial pressure continues to rise after the end of ventricular systole until the AV valves open, then drops and slowly rises again until the next atrial systole.

TIMING

Although events on the two sides of the heart are similar, they are somewhat asynchronous. Right atrial systole precedes left atrial systole, whereas contraction of the right ventricle starts after that of the left (see Chapter 29). However, since pulmonary arterial pressure is lower than aortic pressure, right ventricular ejection begins before that of the left. During expiration, the pulmonary and aortic valves close at the same time; but during inspiration, the aortic valve closes slightly before the pulmonary valve. The slower closure of the pulmonary valve is due to lower impedance of the pulmonary vascular tree. When measured over a period of minutes, the outputs of the two ventricles are, of course, equal, but transient differences in output during the respiratory cycle occur in normal individuals.

LENGTH OF SYSTOLE & DIASTOLE

Cardiac muscle has the unique property of contracting and repolarizing faster when the heart rate is high (see Chapter 5), and the duration of systole decreases from 0.27 s at a heart rate of 65 beats/min to 0.16 s at a rate of 200 beats/min **(Table 30–1)**. The reduced time interval is mainly due to a decrease in the duration of systolic ejection. However, the duration of systole is much more fixed than that of diastole, so when the heart rate is increased, diastole is shortened to a much greater degree. For example, at a heart rate of 65 beats/min, the duration of diastole is 0.62 s, whereas at a heart rate of 200 beats/min, it is only 0.14 s. This fact has important physiologic and clinical implications. It is during diastole that the heart muscle rests, and coronary blood flow to the subendocardial portions of the left ventricle also can only occur during diastole (see Chapter 33). Furthermore, most of the ventricular filling occurs in diastole. At heart rates up to about 180 beats/min, filling is adequate as long as there is ample venous return, and cardiac output per minute is increased by an increase in rate. However, at very high heart rates, filling may be compromised to such a degree that cardiac output per minute falls.

TABLE 30–1 Variation in length of action potential and associated phenomena with cardiac rate.[a]

	Heart Rate 75/min	Heart Rate 200/min	Skeletal Muscle
Duration, each cardiac cycle	0.80	0.30	…
Duration of systole	0.27	0.16	…
Duration of action potential	0.25	0.15	0.007
Duration of absolute refractory period	0.20	0.13	0.004
Duration of relative refractory period	0.05	0.02	0.003
Duration of diastole	0.53	0.14	…

[a]All values are in seconds.

Used with permission of AC Barger and GS Richardson.

Because it has a prolonged action potential, cardiac muscle cannot contract in response to a second stimulus until near the end of the initial contraction (see Figure 5–16). Therefore, cardiac muscle cannot be tetanized like skeletal muscle. The highest rate at which the ventricles can contract is theoretically about 400/min, but in adults the AV node will not conduct more than about 230 impulses/min because of its long refractory period. A ventricular contraction rate of more than 230/min is seen only in paroxysmal ventricular tachycardia (see Chapter 29).

Exact measurement of the duration of isovolumetric ventricular contraction is difficult in clinical situations, but it is relatively easy to measure the duration of **total electromechanical systole (QS2),** the **preejection period (PEP),** and the **left ventricular ejection time (LVET)** by recording the ECG, phonocardiogram, and carotid pulse simultaneously. QS_2 is the period from the onset of the QRS complex to the closure of the aortic valves, as determined by the onset of the second heart sound. LVET is the period from the beginning of the carotid pressure rise to the dicrotic notch (discussed later). PEP is the difference between QS_2 and LVET and represents the time for the electrical as well as the mechanical events that precede systolic ejection. The ratio PEP/LVET is normally about 0.35, and it increases without a change in QS_2

when left ventricular performance is compromised in a variety of cardiac diseases.

ARTERIAL PULSE

The blood forced into the aorta during systole not only moves the blood in the vessels forward but also sets up a pressure wave that travels along the arteries. The pressure wave expands the arterial walls as it travels, and the expansion is palpable as the **pulse.** The rate at which the wave travels, which is independent of and much higher than the velocity of blood flow, is about 4 m/s in the aorta, 8 m/s in the large arteries, and 16 m/s in the small arteries of young adults. Consequently, the pulse is felt in the radial artery at the wrist about 0.1 s after the peak of systolic ejection into the aorta (Figure 30–3). With advancing age, the arteries become more rigid, and the pulse wave moves faster.

The strength of the pulse is determined by the pulse pressure and bears little relation to the mean pressure. The pulse is weak ("thready") in shock. It is strong when stroke volume is large; for example, during exercise or after the administration of histamine. When the pulse pressure is high, the pulse waves may be large enough to be felt or even heard by the individual (palpitation, "pounding heart"). When the aortic valve is incompetent (aortic regurgitation), the pulse is particularly strong, and the force of systolic ejection may be sufficient to make the head nod with each heartbeat. The pulse in aortic regurgitation is called a **Corrigan** or **water-hammer pulse.**

The **dicrotic notch,** a small oscillation in the falling phase of the pulse wave caused by vibrations set up when the aortic valve snaps shut (Figure 30–3), is visible if the pressure wave is recorded but is not palpable at the wrist. The pulmonary artery pressure curve also has a dicrotic notch produced by the closure of the pulmonary valves.

ATRIAL PRESSURE CHANGES & THE JUGULAR PULSE

Atrial pressure rises during atrial systole and continues to rise during isovolumetric ventricular contraction when the AV valves bulge into the atria. When the AV valves are pulled down by the contracting ventricular muscle, pressure falls rapidly and then rises as blood flows into the atria until the AV valves open early in diastole. The return of the AV valves to their relaxed

position also contributes to this pressure rise by reducing atrial capacity. The atrial pressure changes are transmitted to the great veins, producing three characteristic waves in the record of jugular pressure (Figure 30–3). The **a wave** is due to atrial systole. As noted above, some blood regurgitates into the great veins when the atria contract. In addition, venous inflow stops, and the resultant rise in venous pressure contributes to the **a wave**. The **c wave** is the transmitted manifestation of the rise in atrial pressure produced by the bulging of the tricuspid valve into the atria during isovolumetric ventricular contraction. The **v wave** mirrors the rise in atrial pressure before the tricuspid valve opens during diastole. The jugular pulse waves are superimposed on the respiratory fluctuations in venous pressure. Venous pressure falls during inspiration as a result of the increased negative intrathoracic pressure and rises again during expiration.

HEART SOUNDS

Two sounds are normally heard through a stethoscope during each cardiac cycle. The first is a low, slightly prolonged “lub” **(first sound),** caused by vibrations set up by the sudden closure of the AV valves at the start of ventricular systole (Figure 30–3). The second is a shorter, high-pitched “dup” **(second sound),** caused by vibrations associated with closure of the aortic and pulmonary valves just after the end of ventricular systole. A soft, low-pitched **third sound** is heard about one-third of the way through diastole in many normal young individuals. It coincides with the period of rapid ventricular filling and is probably due to vibrations set up by the inrush of blood. A **fourth sound** can sometimes be heard immediately before the first sound when atrial pressure is high or the ventricle is stiff in conditions such as ventricular hypertrophy. It is due to ventricular filling and is rarely heard in normal adults.

The first sound has a duration of about 0.15 s and a frequency of 25–45 Hz. It is soft when the heart rate is low, because the ventricles are well filled with blood and the leaflets of the AV valves float together before systole. The second sound lasts about 0.12 s, with a frequency of 50 Hz. It is loud and sharp when the diastolic pressure in the aorta or pulmonary artery is elevated, causing the respective valves to shut briskly at the end of systole. The interval between aortic and pulmonary valve closure during inspiration is frequently long enough for the second sound to be reduplicated (physiologic splitting of the second sound). Splitting also occurs in various diseases. The third sound, when present, has a duration of 0.1 s.

MURMURS

Murmurs, or **bruits,** are abnormal sounds heard in various parts of the vascular system. The two terms are used interchangeably, though "murmur" is more commonly used to denote noise heard over the heart than over blood vessels. As discussed in detail in Chapter 31, blood flow is laminar, nonturbulent, and silent up to a critical velocity; above this velocity (such as beyond an obstruction), blood flow is turbulent and creates sounds. Blood flow speeds up when an artery or a heart valve is narrowed.

Examples of vascular sounds outside the heart are the bruit heard over a large, highly vascular goiter, the bruit heard over a carotid artery when its lumen is narrowed and distorted by atherosclerosis, and the murmurs heard over an aneurysmal dilation of one of the large arteries, an arteriovenous (A-V) fistula, or a patent ductus arteriosus.

The major—but certainly not the only—cause of cardiac murmurs is disease of the heart valves. When the orifice of a valve is narrowed **(stenosis),** blood flow through it is accelerated and turbulent. When a valve is incompetent, blood flows through it backward **(regurgitation),** again through a narrow orifice that accelerates flow. The timing (systolic or diastolic) of a murmur due to any particular valve **(Table 30–2)** can be predicted from a knowledge of the mechanical events of the cardiac cycle. Murmurs due to disease of a particular valve can generally be heard best when the stethoscope is directly over the valve. There are also other aspects of the duration, character, accentuation, and transmission of the sound that help locate its origin in one valve or another. Most murmurs can be heard only with the aid of the stethoscope, but one of the loudest murmurs is that produced when blood flows backward in diastole through a hole in a cusp of the aortic valve. This high-pitched musical diastolic murmur is sometimes audible to the unaided ear several feet from the patient.

TABLE 30–2 Heart murmurs.

Valve	Abnormality	Timing of Murmur
Aortic or pulmonary	Stenosis Regurgitation	Systolic Diastolic
Mitral or tricuspid	Stenosis Regurgitation	Diastolic Systolic

In patients with congenital interventricular septal defects, flow from the left to the right ventricle causes a systolic murmur. Soft murmurs may also be heard in patients with interatrial septal defects, although they are not a constant finding.

Soft systolic murmurs are also common in individuals, especially children, who have no cardiac disease. Systolic murmurs are also heard in anemic patients as a result of the low viscosity of the blood and associated rapid flow (see Chapter 31).

ECHOCARDIOGRAPHY

Wall movement and other aspects of cardiac function can be evaluated by the noninvasive technique of **echocardiography.** Pulses of ultrasonic waves are emitted from a transducer that also functions as a receiver to detect waves reflected back from various parts of the heart. Reflections occur wherever acoustic impedance changes, and a recording of the echoes displayed against time on an oscilloscope provides a record of the movements of the ventricular wall, septum, and valves during the cardiac cycle. When combined with Doppler techniques, echocardiography can be used to measure velocity and volume of flow through valves. It has considerable clinical usefulness, particularly in evaluating and planning therapy in patients with valvular lesions.

CARDIAC OUTPUT

METHODS OF MEASUREMENT

In experimental animals, cardiac output can be measured with an electromagnetic flow meter placed on the ascending aorta. Two methods of measuring output that are applicable to humans, in addition to Doppler combined with echocardiography, are the **direct Fick's method** and the **indicator dilution method.**

The **Fick's principle** states that the amount of a substance taken up by an organ (or by the whole body) per unit of time is equal to the arterial level of the substance minus the venous level **(A-V difference)** times the blood flow. This principle can be applied, of course, only in situations in which the arterial blood is the sole source of the substance taken up. The principle can be used to determine cardiac output by measuring the amount of O_2 consumed by the body

in a given period and dividing this value by the A-V difference across the lungs. Because systemic arterial blood has effectively the same O_2 content in all parts of the body, the arterial O_2 content can be measured in a sample obtained from any convenient artery. A sample of venous blood in the pulmonary artery is obtained by means of a cardiac catheter. It has now become commonplace to insert a long catheter through a forearm vein and to guide its tip into the heart with the aid of a fluoroscope. Catheters can thus be inserted through the right atrium and ventricle into the small branches of the pulmonary artery. An example of the calculation of cardiac output using a typical set of values is as follows:

$$\text{Output of left ventricle}$$

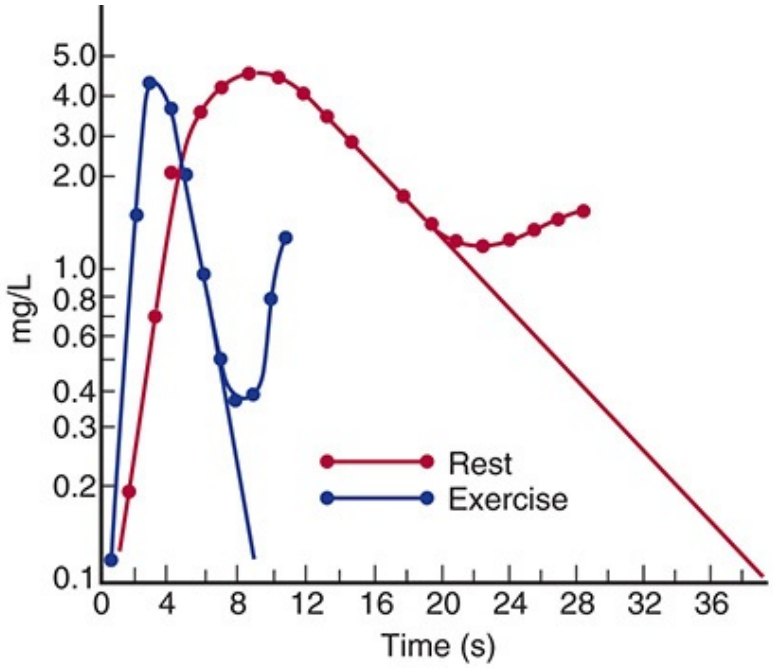
$$= \frac{O_2 \text{ consumption (mL/min)}}{[A_{O_2}] - [V_{O_2}]}$$

$$= \frac{250 \text{ mL/min}}{190 \text{ mL/L arterial blood} - 140 \text{ mL/L venous blood in pulmonary artery}}$$

$$= \frac{250 \text{ mL/min}}{50 \text{ mL/L}}$$

$$= 5 \text{ L/min}$$

In the indicator dilution technique, a known amount of a substance such as a dye or, more commonly, a radioactive isotope is injected into an arm vein and the concentration of the indicator in serial samples of arterial blood is determined. The output of the heart is equal to the amount of indicator injected divided by its average concentration in arterial blood after a single circulation through the heart **(Figure 30–4)**. The indicator must, of course, be a substance that stays in the bloodstream during the test and has no harmful or hemodynamic effects. In practice, the log of the indicator concentration in the serial arterial samples is plotted against time as the concentration rises, falls, and then rises again as the indicator recirculates. The initial decline in concentration, linear on a semilog plot, is extrapolated to the abscissa, giving the time for first passage of the indicator through the circulation. The cardiac output for that period is calculated (Figure 30–4) and then converted to output per minute.



$$F = \frac{E}{\int_0^{\alpha} C\,dt}$$

F = flow
E = amount of indicator injected
C = instantaneous concentration of indicator in arterial blood

In the **rest** example above,

$$\text{Flow in 39 s (time of first passage)} = \frac{\text{5 mg injection}}{\text{1.6 mg/L (avg concentration)}}$$

Flow = 3.1 L in 39 s

$$\text{Flow (cardiac output)/min} = 3.1 \times \frac{60}{39} = 4.7 \text{ L}$$

For the **exercise** example,

$$\text{Flow in 9 s} = \frac{5 \text{ mg}}{1.51 \text{ mg/L}} = 3.3 \text{ L}$$

$$\text{Flow/min} = 3.3 \times \frac{60}{9} = 22.0 \text{ L}$$

FIGURE 30–4 Determination of cardiac output by indicator (dye) dilution. Two examples are shown—at rest and during exercise.

A popular indicator dilution technique is **thermodilution,** in which the indicator used is cold saline. The saline is injected into the right atrium through one channel of a double-lumen catheter, and the temperature change in the blood is recorded in the pulmonary artery, using a thermistor in the other, longer side of the catheter. The temperature change is inversely proportional to the amount of blood flowing through the pulmonary artery; that is, to the extent that the cold saline is diluted by blood. This technique has two important advantages: (1) the saline is completely innocuous and (2) the cold is dissipated in the tissues so recirculation is not a problem, and it is easy to make repeated determinations.

CARDIAC OUTPUT IN VARIOUS CONDITIONS

The amount of blood pumped out of the heart per beat, the **stroke volume,** is about 70 mL from each ventricle in a resting man of average size in the supine position. The output of the heart per unit of time is the **cardiac output.** In a resting, supine man, it averages about 5.0 L/min (70 mL × 72 beats/min). There is a correlation between resting cardiac output and body surface area. The output per minute per square meter of body surface (the **cardiac index**) averages 3.2 L. The effects of various conditions on cardiac output are summarized in **Table 30–3**.

TABLE 30–3 Effect of various conditions on cardiac output.

	Condition or Factor[a]
No change	Sleep
	Moderate changes in environmental temperature
Increase	Anxiety and excitement (50–100%)
	Eating (30%)
	Exercise (up to 700%)
	High environmental temperature
	Pregnancy
	Epinephrine
Decrease	Sitting or standing from lying position (20–30%)
	Rapid arrhythmias
	Heart disease

[a]Approximate percentage changes are shown in parentheses.

FACTORS CONTROLLING CARDIAC OUTPUT

Predictably, changes in cardiac output that are called for by physiologic conditions can be produced by changes in cardiac rate, or stroke volume, or both **(Figure 30–5)**. The cardiac rate is controlled primarily by the autonomic nerves, with sympathetic stimulation increasing the rate and parasympathetic stimulation decreasing it (see Chapter 29). Stroke volume is also determined in part by neural input, with sympathetic stimuli making the myocardial muscle fibers contract with greater strength at any given length and parasympathetic stimuli having the opposite effect. When the strength of contraction increases without an increase in fiber length, more of the blood that normally remains in the ventricles is expelled; that is, the ejection fraction increases. The cardiac accelerator action of the catecholamines liberated by sympathetic stimulation is referred to as their **chronotropic action,** whereas their effect on the strength of cardiac contraction is called their **inotropic action.**

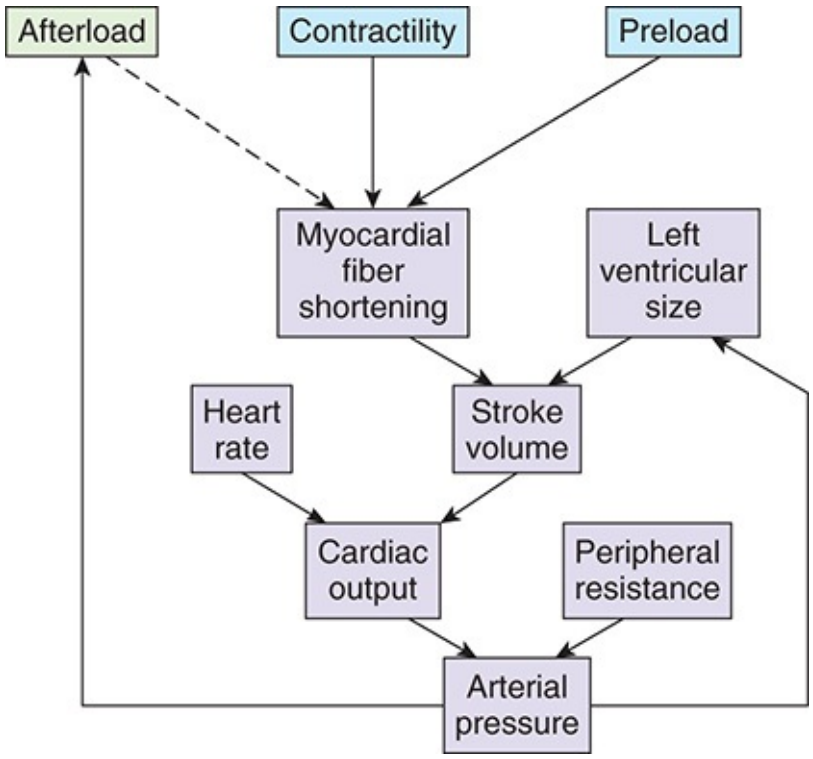


FIGURE 30–5 Interactions between the components that regulate cardiac output and arterial pressure. Solid arrows indicate increases, and the dashed arrow indicates a decrease.

The force of contraction of cardiac muscle depends on its preloading and its afterloading. These factors are illustrated in **Figure 30–6**, in which a muscle strip is stretched by a load (the **preload**) that rests on a platform. The initial phase of the contraction is isometric; the elastic component in series with the contractile element is stretched, and tension increases until it is sufficient to lift the load. The tension at which the load is lifted is the **afterload.** The muscle then contracts isotonically without developing further tension. In vivo, the preload is the degree to which the myocardium is stretched before it contracts and the afterload is the resistance against which blood is expelled.

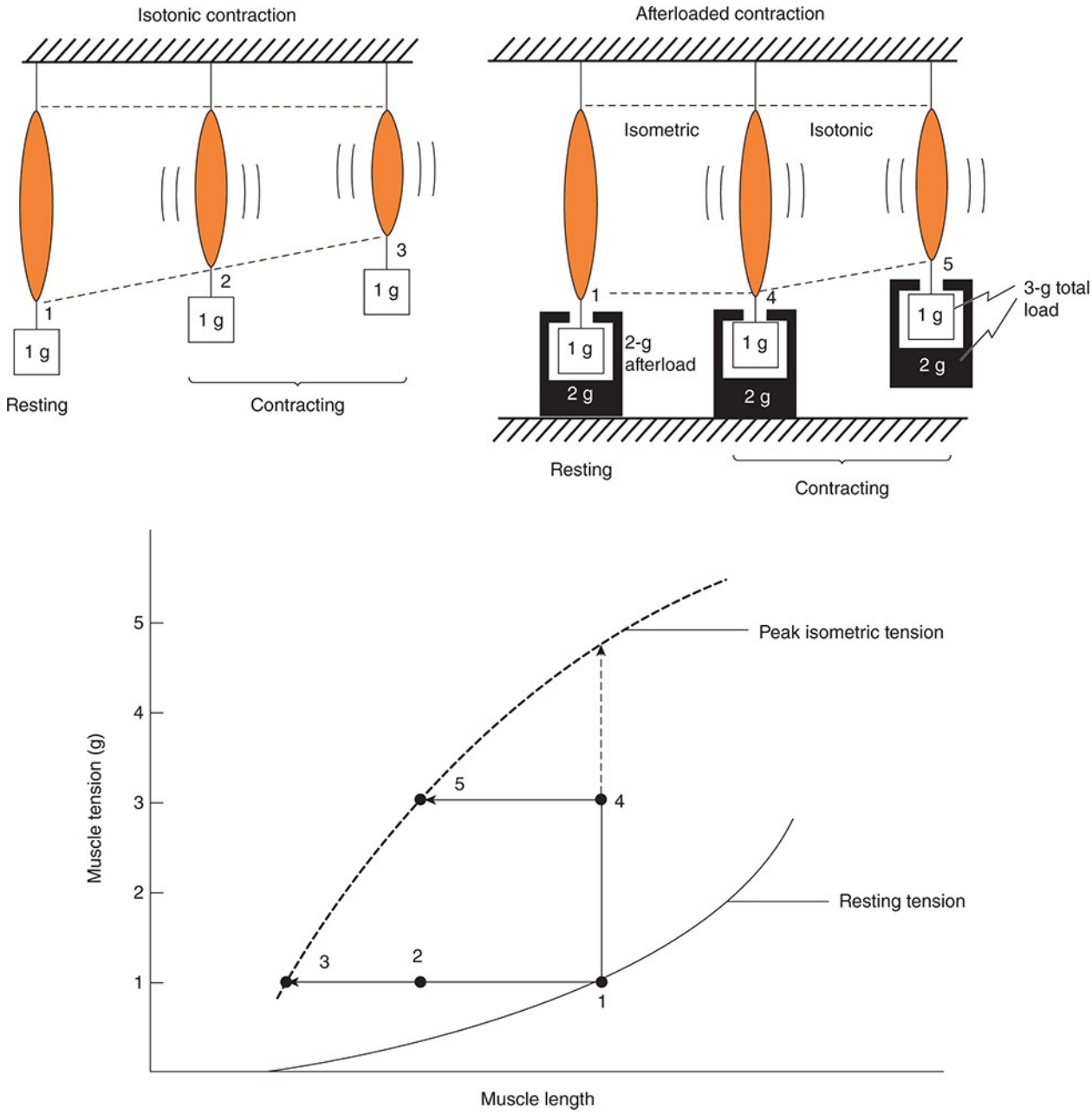


FIGURE 30–6 Model for contraction of afterloaded muscles. The figure depicts isotonic and afterloaded contractions within the constraints of the cardiac muscle length–tension diagram. The numbered points on the diagram correspond to the various conditions shown in the upper part. (Adapted with permission from Mohrman DE, Heller LJ: *Cardiovascular Physiology*, 8th ed. New York, NY: McGraw-Hill; 2014.)

RELATION OF TENSION TO LENGTH IN

CARDIAC MUSCLE

The length–tension relationship in cardiac muscle (see Figure 5–17) is similar to that in skeletal muscle (see Figure 5–10); when the muscle is stretched, the developed tension increases to a maximum and then declines as stretch becomes more extreme. Starling pointed this out when he stated that the "energy of contraction is proportional to the initial length of the cardiac muscle fiber" **(Starling's law of the heart** or the **Frank-Starling law).** For the heart, the length of the muscle fibers (ie, the extent of the preload) is proportional to the end-diastolic volume. The relation between ventricular stroke volume and end-diastolic volume is called the Frank-Starling curve.

When cardiac output is regulated by changes in cardiac muscle fiber length, this is referred to as **heterometric regulation.** Conversely, regulation due to changes in contractility independent of length is sometimes called **homometric regulation.**

FACTORS AFFECTING END-DIASTOLIC VOLUME

Alterations in systolic and diastolic function have different effects on the heart **(Clinical Box 30–1)**. When systolic contractions are reduced, there is a primary reduction in stroke volume. Diastolic function also affects stroke volume, but in a different way.

The myocardium is covered by a fibrous layer known as the epicardium. This, in turn, is surrounded by the pericardium, which separates the heart from the rest of the thoracic viscera. The space between the epicardium and pericardium (the pericardial sac) normally contains 5–30 mL of clear fluid, which lubricates the heart and permits it to contract with minimal friction.

An increase in intrapericardial pressure (eg, as a result of infection or pressure from a tumor) limits the extent to which the ventricle can fill, as does a decrease in ventricular compliance (ie, an increase in ventricular stiffness produced by myocardial infarction, infiltrative disease, and other abnormalities). Atrial contractions aid ventricular filling. Factors affecting the amount of blood returning to the heart likewise influence the degree of cardiac filling during diastole. An increase in total blood volume increases venous return **(Clinical Box 30–2)**. Constriction of the veins reduces the size of the venous reservoirs, decreasing venous pooling, and thus increasing venous return. An increase in the

normal negative intrathoracic pressure increases the pressure gradient along which blood flows to the heart, whereas a decrease impedes venous return. Standing decreases venous return, and muscular activity increases it as a result of the pumping action of skeletal muscle.

The effects of systolic and diastolic dysfunction on the pressure–volume loop of the left ventricle are summarized in **Figure 30–7**.

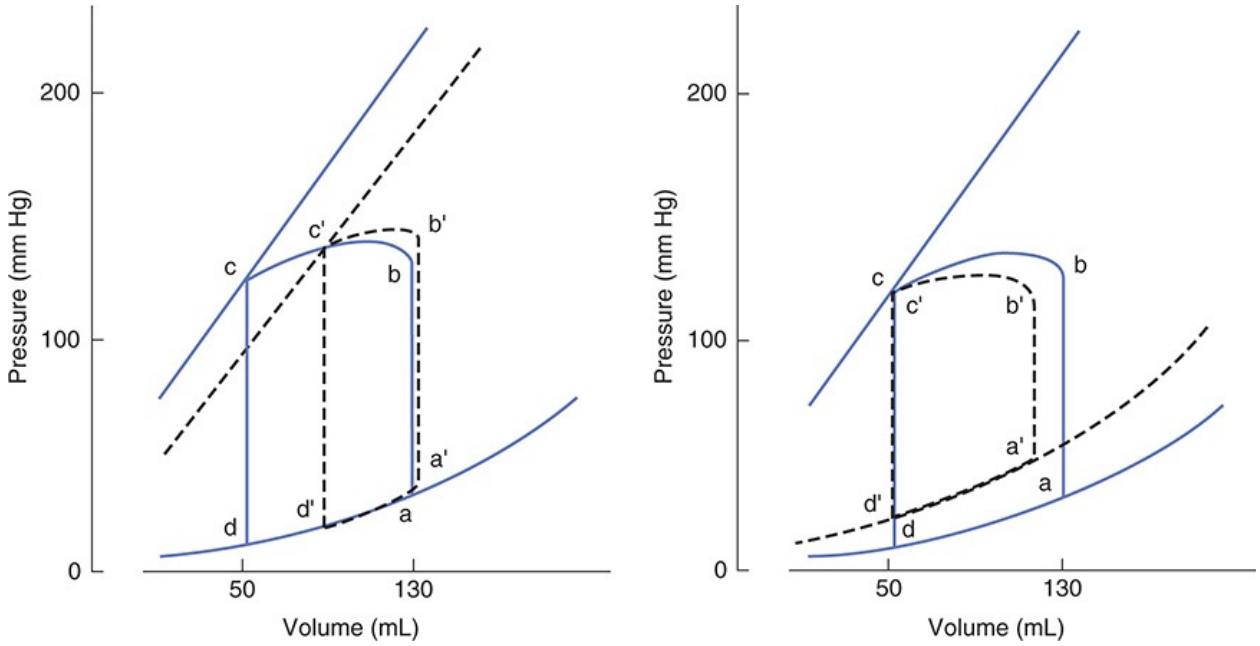


FIGURE 30–7 Effect of systolic and diastolic dysfunction on the pressure–volume loop of the left ventricle. In both panels, the solid lines represent the normal pressure–volume loop (equivalent to that shown in Figure 30–2) and the dashed lines show how the loop is shifted by the disease process represented. **Left:** Systolic dysfunction shifts the isovolumic pressure–volume curve to the right, decreasing the stroke volume from b–c to b'–c'. **Right:** Diastolic dysfunction increases end-diastolic volume and shifts the diastolic pressure–volume relationship upward and to the left. This reduces the stroke volume from b–c to b'–c'. (Reproduced with permission from McPhee SJ, Lingappa VR, Ganong WF [editors]: *Pathophysiology of Disease*, 6th ed. New York, NY: McGraw-Hill; 2010.)

MYOCARDIAL CONTRACTILITY

The contractility of the myocardium exerts a major influence on stroke volume. When the sympathetic nerves to the heart are stimulated, the whole length–

tension curve shifts upward and to the left **(Figure 30–8)**. The positive inotropic effect of norepinephrine liberated at the nerve endings is augmented by circulating norepinephrine, and epinephrine has a similar effect. Conversely, there is a negative inotropic effect of vagal stimulation on both atrial and (to a lesser extent) ventricular muscle.

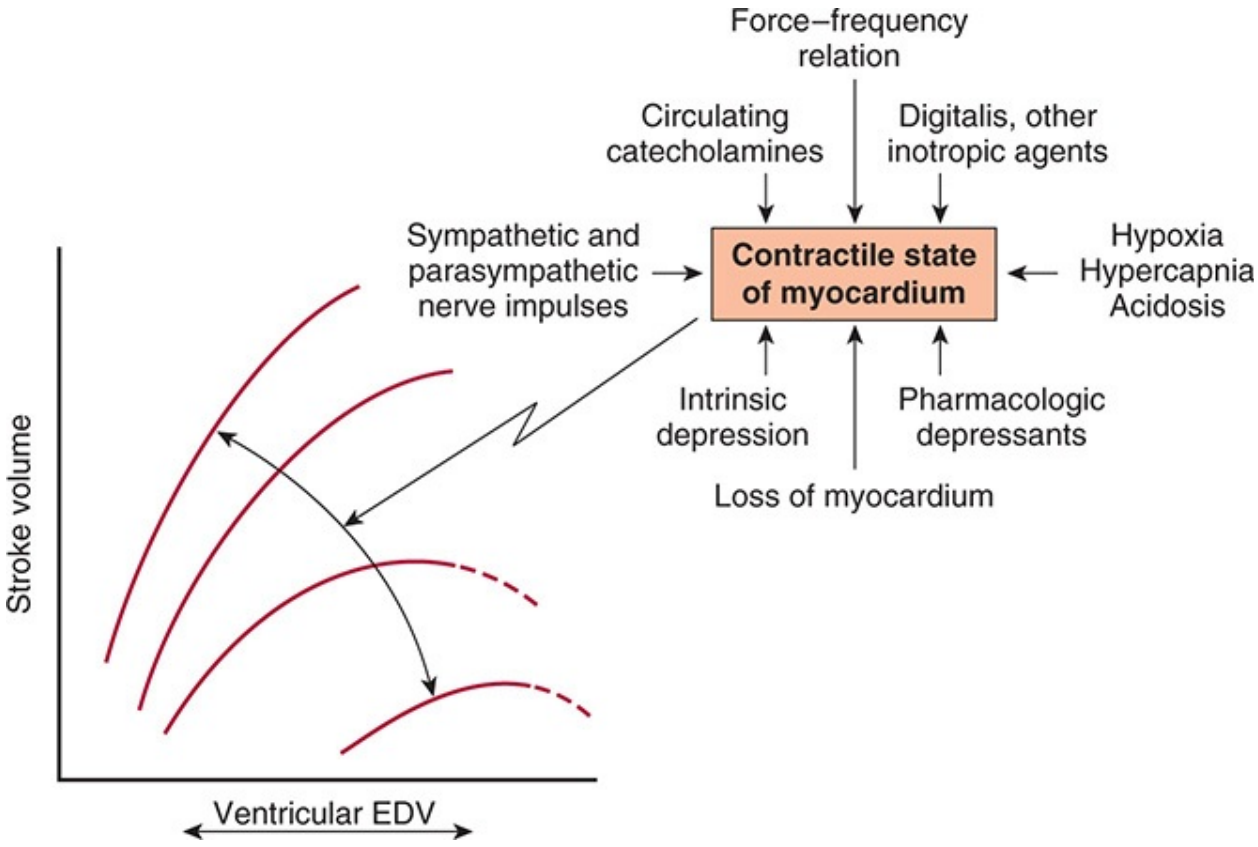


FIGURE 30–8 Effect of changes in myocardial contractility on the Frank-Starling curve. The curve shifts downward and to the right as contractility is decreased. The major factors influencing contractility are summarized on the right. The dashed lines indicate portions of the ventricular function curves where maximum contractility has been exceeded; that is, they identify points on the "descending limb" of the Frank-Starling curve. EDV, end-diastolic volume. (Reproduced with permission from Braunwald E, Ross J, Sonnenblick EH: Mechanisms of contraction of the normal and failing heart. N Engl J Med 1967; October 12; 277(15):794–800.)

Changes in cardiac rate and rhythm also affect myocardial contractility (known as the force–frequency relation, Figure 30–8). Ventricular extrasystoles condition the myocardium in such a way that the next succeeding contraction is

stronger than the preceding normal contraction. This **postextrasystolic potentiation** is independent of ventricular filling, since it occurs in isolated cardiac muscle and is due to increased availability of intracellular Ca^{2+}. A sustained increment in contractility can be produced therapeutically by delivering paired electrical stimuli to the heart in such a way that the second stimulus is delivered shortly after the refractory period of the first. It has also been shown that myocardial contractility increases as the heart rate increases, although this effect is relatively small.

Catecholamines exert their inotropic effect via an action on cardiac β_1-adrenergic receptors and Gs, with resultant activation of adenylyl cyclase and increased intracellular cyclic adenosine 3′,5′-monophosphate (cAMP). Xanthines such as caffeine and theophylline that inhibit the breakdown of cAMP are predictably positively inotropic. The positively inotropic effect of digitalis and related drugs (Figure 30–8), on the other hand, is due to their inhibitory effect on the Na^+, K^+ ATPase in the myocardium, and a subsequent decrease in calcium removal from the cytosol by Na^+/Ca^{2+} exchange (see Chapter 5). Hypercapnia; hypoxia; acidosis; and drugs, such as quinidine, procainamide, and barbiturates, depress myocardial contractility. The contractility of the myocardium is also reduced in heart failure (intrinsic depression). The causes of this depression are not fully understood but may reflect down-regulation of β-adrenergic receptors and associated signaling pathways and impaired calcium liberation from the sarcoplasmic reticulum. In acute heart failure, such as that associated with sepsis, this response could be considered an appropriate adaptation (so-called “myocardial hibernation”) to a situation where energy supply to the heart is limited, thereby reducing energy expenditure and avoiding cell death.

CLINICAL BOX 30–1

Heart Failure

Heart failure occurs when the heart is unable to put out an amount of blood that is adequate for the needs of the tissues. It can be acute and associated with sudden death, or chronic. The failure may involve primarily the right ventricle (cor pulmonale), but much more commonly it involves the larger, thicker left ventricle or both ventricles. Heart failure may also be systolic or diastolic. In **systolic failure,** stroke volume is reduced because ventricular contraction is weak. This causes an increase in the end-systolic ventricular volume, so that the **ejection fraction** falls from 65% to as low as 20%. The

initial response to failure is activation of the genes that cause cardiac myocytes to hypertrophy, and thickening of the ventricular wall **(cardiac remodeling).** The incomplete filling of the arterial system leads to increased discharge of the sympathetic nervous system and increased secretion of renin and aldosterone, so Na^+ and water are retained. These responses are initially compensatory, but eventually the failure worsens and the ventricles dilate.

In **diastolic failure,** the ejection fraction is initially maintained, but the elasticity of the myocardium is reduced so filling during diastole is reduced. This leads to inadequate stroke volume and the same cardiac remodeling and Na^+ and water retention that occur in systolic failure. It should be noted that the inadequate cardiac output in heart failure may be relative rather than absolute. When a large arteriovenous fistula is present, in thyrotoxicosis or in thiamine deficiency, cardiac output may be elevated in absolute terms but still be inadequate to meet the needs of the tissues **(high-output failure)**.

THERAPEUTIC HIGHLIGHTS

Treatment of heart failure is aimed at improving cardiac contractility, treating the symptoms, and decreasing the load on the heart. Currently, the most effective treatment in general use is inhibition of the production of angiotensin II with angiotensin-converting enzyme (ACE) inhibitors. Blockade of the effects of angiotensin II on AT_1 receptors with nonpeptide antagonists is also of value. Blocking the production of angiotensin II or its effects also reduces the circulating aldosterone level and decreases blood pressure, reducing the afterload against which the heart pumps. The effects of aldosterone can be further reduced by administering aldosterone receptor blockers. Reducing venous tone with nitrates or hydralazine increases venous capacity so that the amount of blood returned to the heart is reduced, lowering the preload. Diuretics reduce the fluid overload. Drugs that block β-adrenergic receptors have been shown to decrease mortality and morbidity. Digitalis derivatives such as digoxin have classically been used to treat heart failure because of their ability to increase stores of intracellular Ca^{2+} and hence exert a positive inotropic effect, but they are now used in a secondary role to treat systolic dysfunction and slow the ventricular rate in patients with atrial fibrillation.

INTEGRATED CONTROL OF CARDIAC OUTPUT

The mechanisms listed above operate in an integrated way to maintain cardiac output. For example, during muscular exercise, there is increased sympathetic discharge, so that myocardial contractility is increased and the heart rate rises. The increase in heart rate is particularly prominent in normal individuals, and there is only a modest increase in stroke volume **(see Table 30–4 and Clinical Box 30–3)**. However, patients with transplanted hearts are able to increase their cardiac output during exercise in the absence of cardiac innervation through the operation of the Frank-Starling mechanism **(Figure 30–9)**. Circulating catecholamines also contribute. If venous return increases and there is no change in sympathetic tone, venous pressure rises, diastolic inflow is greater, ventricular end-diastolic pressure increases, and the heart muscle contracts more forcefully. During muscular exercise, venous return is increased by the pumping action of the muscles and the increase in respiration (see Chapter 32). In addition, because of vasodilation in the contracting muscles, peripheral resistance, and, consequently, afterload are decreased. The end result in both normal and transplanted hearts is thus a prompt and marked increase in cardiac output.

TABLE 30–4 Changes in cardiac function with exercise. Note that stroke volume levels off, then falls somewhat (as a result of the shortening of diastole) when the heart rate rises to high values.

Work (kg-m/min)	O_2 Usage (mL/min)	Pulse Rate (per min)	Cardiac Output (L/min)	Stroke Volume (mL)	A-V O_2 Difference (mL/dL)
Rest	267	64	6.4	100	4.3
288	910	104	13.1	126	7.0
540	1430	122	15.2	125	9.4
900	2143	161	17.8	110	12.3
1260	3007	173	20.9	120	14.5

Reproduced with permission from Asmussen E, Nielsen M: The cardiac output in rest and work determined by the acetylene and the dye injection methods. Acta Physiol Scand 1952;27(2–3):217–230.

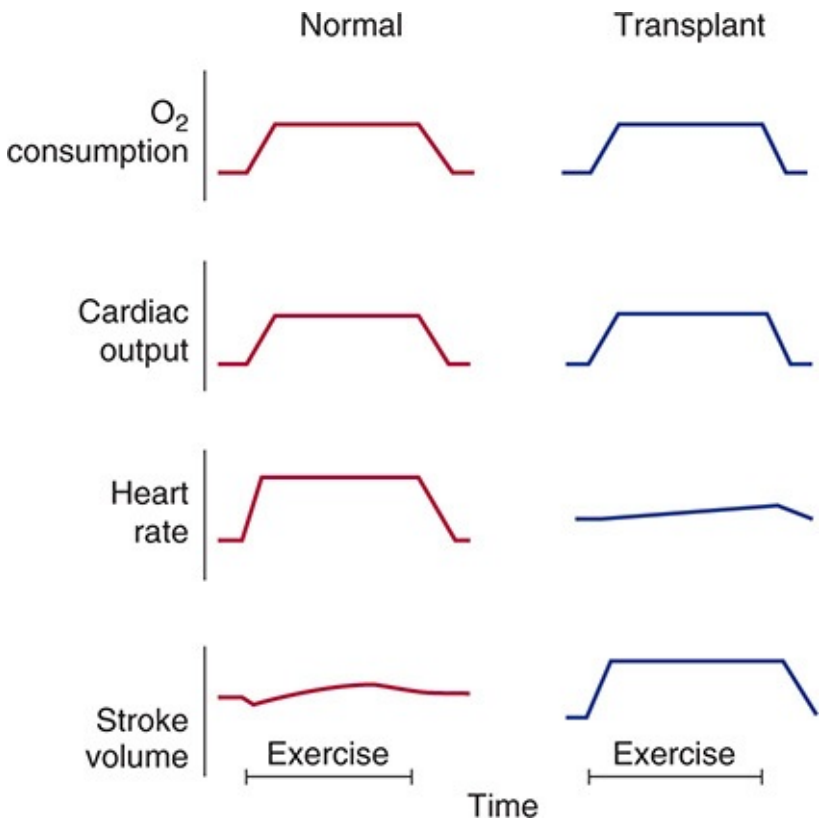


FIGURE 30–9 Cardiac responses to moderate supine exercise in normal humans and patients with transplanted and hence denervated hearts. Note that the transplanted heart, without the benefit of neural input, relies primarily on an increase in stroke volume rather than heart rate to raise cardiac output in the setting of exercise. (Reproduced with permission from Kent KM, Cooper T: The denervated heart. N Engl J Med 1974; Nov7; 291(19):1017–1021.)

One of the differences between untrained individuals and trained athletes is that the athletes have lower heart rates, greater end-systolic ventricular volumes, and greater stroke volumes at rest. Therefore, they can potentially achieve a given increase in cardiac output by further increases in stroke volume without increasing their heart rate to as great a degree as an untrained individual.

CLINICAL BOX 30–2

Shock

Circulatory shock comprises a collection of different entities that share certain common features; however, the feature that is common to all the entities is

inadequate tissue perfusion with a relatively or absolutely inadequate cardiac output. The cardiac output may be inadequate because the amount of fluid in the vascular system is inadequate to fill it **(hypovolemic shock).** Alternatively, it may be inadequate in the relative sense because the size of the vascular system is increased by vasodilation even though the blood volume is normal **(distributive, vasogenic,** or **low-resistance shock).** Shock may also be caused by inadequate pumping action of the heart as a result of myocardial abnormalities **(cardiogenic shock),** and by inadequate cardiac output as a result of obstruction of blood flow in the lungs or heart **(obstructive shock).**

Hypovolemic shock is also called “cold shock.” It is characterized by hypotension; a rapid, thready pulse; cold, pale, clammy skin; intense thirst; rapid respiration; and restlessness or, alternatively, torpor. None of these findings, however, are invariably present. Hypovolemic shock is commonly subdivided into categories on the basis of cause. Of these, it is useful to consider the effects of hemorrhage in some detail because of the multiple compensatory reactions that come into play to defend extracellular fluid (ECF) volume. Thus, the decline in blood volume produced by bleeding decreases venous return, and cardiac output falls. The heart rate is increased, and with severe hemorrhage, a fall in blood pressure always occurs. With moderate hemorrhage (5–15 mL/kg body weight), pulse pressure is reduced but mean arterial pressure may be normal. The blood pressure changes vary from individual to individual, even when exactly the same amount of blood is lost. The skin is cool and pale and may have a grayish tinge because of stasis in the capillaries and a small amount of cyanosis. Inadequate perfusion of the tissues leads to increased anaerobic glycolysis, with the production of large amounts of lactic acid. In severe cases, the blood lactate level rises from the normal value of about 1 mmol/L to 9 mmol/L or more. The resulting **lactic acidosis** depresses the myocardium, decreases peripheral vascular responsiveness to catecholamines, and may be severe enough to cause coma. When blood volume is reduced and venous return is decreased, moreover, stimulation of arterial baroreceptors is reduced, increasing sympathetic output. Even if there is no drop in mean arterial pressure, the decrease in pulse pressure decreases the rate of discharge in the arterial baroreceptors, and reflex tachycardia and vasoconstriction result.

With more severe blood loss, tachycardia is replaced by bradycardia; this occurs while shock is still reversible. The bradycardia is presumably due to unmasking a vagally mediated depressor reflex, and the response may have

evolved as a mechanism for stopping further blood loss. With even greater hemorrhage, the heart rate rises again. Vasoconstriction is generalized, sparing only the vessels of the brain and heart. A widespread reflex venoconstriction also helps maintain the filling pressure of the heart. In the kidneys, both afferent and efferent arterioles are constricted, but the efferent vessels are constricted to a greater degree. The glomerular filtration rate is depressed, but renal plasma flow is decreased to a greater extent, so that the filtration fraction increases. Na^+ retention is marked, and the nitrogenous products of metabolism are retained in the blood **(azotemia** or **uremia).** If the hypotension is prolonged, renal tubular damage may be severe **(acute kidney injury).** After a moderate hemorrhage, the circulating plasma volume is restored in 12–72 h. Preformed albumin also enters rapidly from extravascular stores, but most of the tissue fluids that are mobilized are protein-free. After the initial influx of preformed albumin, the rest of the plasma protein losses are replaced, presumably by hepatic synthesis, over a period of 3–4 days. Erythropoietin appears in the circulation, and the reticulocyte count increases, reaching a peak in 10 days. The red cell mass is restored to normal in 4–8 weeks.

THERAPEUTIC HIGHLIGHTS

The treatment of shock is aimed at correcting the cause and helping the physiologic compensatory mechanisms to restore an adequate level of tissue perfusion. If the primary cause of the shock is blood loss, the treatment should include early and rapid transfusion of adequate amounts of compatible whole blood. In shock due to burns and other conditions in which there is hemoconcentration, plasma is the treatment of choice to restore the fundamental defect, the loss of plasma. Concentrated human serum albumin and other hypertonic solutions expand the blood volume by drawing fluid out of the interstitial spaces. They are valuable in emergency treatment but have the disadvantage of further dehydrating the tissues of an already dehydrated patient.

OXYGEN CONSUMPTION BY THE HEART

Basal O_2 consumption by the myocardium is about 2 mL/100 g/min. This value is considerably higher than that of resting skeletal muscle. O_2 consumption by the beating heart is about 9 mL/100 g/min at rest. Increases occur during

exercise and in a number of different states. Cardiac venous O_2 tension is low, and little additional O_2 can be extracted from the blood in the coronaries, so increases in O_2 consumption require increases in coronary blood flow. The regulation of coronary flow is discussed in Chapter 33.

CLINICAL BOX 30–3

Circulatory Changes during Exercise

The blood flow of resting skeletal muscle is low (2–4 mL/100 g/min). When a muscle contracts, it compresses the vessels in it if it develops more than 10% of its maximal tension; when it develops more than 70% of its maximal tension, blood flow is completely stopped. Between contractions, however, flow is so greatly increased that blood flow per unit of time in a rhythmically contracting muscle is increased as much as 30-fold. Local mechanisms maintaining a high blood flow in exercising muscle include a fall in tissue P_{O_2}, a rise in tissue P_{CO_2}, and accumulation of K^+ and other vasodilator metabolites. The temperature rises in active muscle, and this further dilates the vessels. Dilation of the arterioles and precapillary sphincters causes a 10- to 100-fold increase in the number of open capillaries. The average distance between the blood and the active muscle cells—and the distance O_2 and metabolic products must diffuse—is thus greatly decreased. The dilation increases the cross-sectional area of the vascular bed, and the velocity of flow therefore decreases.

The systemic cardiovascular response to exercise that provides for the additional blood flow to contracting muscle depends on whether the muscle contractions are primarily isometric or primarily isotonic with the performance of external work. With the start of an isometric muscle contraction, the heart rate rises, probably as a result of psychic stimuli acting on the medulla oblongata. The increase is largely due to decreased vagal tone, although increased discharge of the cardiac sympathetic nerves plays some role. Within a few seconds of the onset of an isometric muscle contraction, systolic and diastolic blood pressures rise sharply. Stroke volume changes relatively little, and blood flow to the steadily contracting muscles is reduced as a result of compression of their blood vessels. The response to exercise involving isotonic muscle contraction is similar in that there is a prompt increase in heart rate, but different in that a marked increase in stroke volume

occurs. In addition, there is a net fall in total peripheral resistance due to vasodilation in exercising muscles. Consequently, systolic blood pressure rises only moderately, whereas diastolic pressure usually remains unchanged or falls.

The difference in response to isometric and isotonic exercise is explained in part by the fact that the active muscles are tonically contracted during isometric exercise and consequently contribute to increased total peripheral resistance. Cardiac output is increased during isotonic exercise to values that may exceed 35 L/min, the amount being proportional to the increase in O_2 consumption. The maximal heart rate achieved during exercise decreases with age. In children, it rises to 200 or more beats/min; in adults it rarely exceeds 195 beats/min, and in elderly individuals the rise is even smaller. Both at rest and at any given level of exercise, trained athletes have a larger stroke volume and lower heart rate than untrained individuals and they tend to have larger hearts. Training increases the maximal oxygen consumption ($V{\cdot}O_{2max}$) that can be produced by exercise in an individual. $V{\cdot}O_{2max}$ averages about 38 mL/kg/min in active healthy men and about 29 mL/kg/min in active healthy women. It is lower in sedentary individuals. $V{\cdot}O_{2max}$ is the product of maximal cardiac output and maximal O_2 extraction by the tissues, and both increase with training.

A great increase in venous return also takes place with exercise, although the increase in venous return is not the primary cause of the increase in cardiac output. Venous return is increased by the activity of the muscle and thoracic pumps; by mobilization of blood from the viscera; by increased pressure transmitted through the dilated arterioles to the veins; and by noradrenergically mediated venoconstriction, which decreases the volume of blood in the veins. Blood mobilized from the splanchnic area and other reservoirs may increase the amount of blood in the arterial portion of the circulation by as much as 30% during strenuous exercise. After exercise, the blood pressure may transiently drop to subnormal levels, presumably because accumulated metabolites keep the muscle vessels dilated for a short period. However, the blood pressure soon returns to the preexercise level. The heart rate returns to normal more slowly.

O_2 consumption by the heart is determined primarily by the intramyocardial tension, the contractile state of the myocardium, and the heart rate. Ventricular

work per beat correlates with O_2 consumption. The work is the product of stroke volume and mean arterial pressure in the pulmonary artery or the aorta (for the right and left ventricle, respectively). Because aortic pressure is seven times greater than pulmonary artery pressure, the stroke work of the left ventricle is approximately seven times the stroke work of the right. In theory, a 25% increase in stroke volume without a change in arterial pressure should produce the same increase in O_2 consumption as a 25% increase in arterial pressure without a change in stroke volume. However, for reasons that are incompletely understood, pressure work produces a greater increase in O_2 consumption than volume work. In other words, an increase in afterload causes a greater increase in cardiac O_2 consumption than does an increase in preload. This is why angina pectoris due to deficient delivery of O_2 to the myocardium is more common in aortic stenosis than in aortic regurgitation. In aortic stenosis, intraventricular pressure must be increased to force blood through the stenotic valve, whereas in aortic regurgitation, an increase in stroke volume occurs with little change in aortic impedance.

It is worth noting that the increase in O_2 consumption produced by increased stroke volume when the myocardial fibers are stretched is an example of the operation of the law of Laplace. This law, which is discussed in detail in Chapter 31, states that the tension developed in the wall of a hollow viscus is proportional to the radius of the viscus. When the heart is dilated, its radius is increased. O_2 consumption per unit time increases when the heart rate is increased by sympathetic stimulation because of the increased number of beats and the increased velocity and strength of each contraction. However, this is somewhat offset by the decrease in end-systolic volume and hence in the radius of the heart.

CHAPTER SUMMARY

- Blood flows into the atria and then the ventricles of the heart during diastole and atrial systole, and is ejected during systole when the ventricles contract and pressure exceeds the pressures in the pulmonary artery and aorta.
- Careful timing of the opening and closing of the atrioventricular (AV), pulmonary, and aortic valves allows blood to move in an appropriate direction through the heart with minimal regurgitation.
- The proportion of blood leaving the ventricles in each cardiac cycle is called

the ejection fraction and is a sensitive indicator of cardiac health.

- The arterial pulse represents a pressure wave set up when blood is forced into the aorta; it travels much faster than the blood itself.
- Heart sounds reflect the normal vibrations set up by abrupt valve closures; heart murmurs can arise from abnormal flow often (although not exclusively) caused by diseased valves.
- Changes in cardiac output reflect variations in heart rate, stroke volume, or both; these are controlled, in turn, by neural and hormonal input to cardiac myocytes.
- Cardiac output can be measured by assessing the amount of oxygen consumed by the body during a given period divided by the difference in oxygen concentration between arterial blood and the pulmonary artery (Fick's principle). Alternatively, dilution of an inert dye or thermodilution of cold saline can be used.
- In heart failure, the ejection fraction of the heart is reduced due to impaired contractility in systole or reduced filling during diastole; this results in inadequate blood supplies to meet the body's needs. Initially, this is manifested only during exercise, but eventually the heart will not be able to supply sufficient blood flow even at rest.
- Cardiac output is strikingly increased during exercise.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her peak left ventricular pressure occur?

   A. Rapid filling

   B. Isovolumetric contraction

   C. Ventricular ejection

   D. Atrial systole

   E. Isovolumetric relaxation

2. The work performed by the left ventricle is substantially greater than that performed by the right ventricle, because in the left ventricle

A. the contraction is slower.
B. the wall is thicker.
C. the stroke volume is greater.
D. the preload is greater.
E. the afterload is greater.

3. The dicrotic notch on the aortic pressure curve is caused by
A. closure of the mitral valve.
B. closure of the tricuspid valve.
C. closure of the aortic valve.
D. closure of the pulmonary valve.
E. rapid filling of the left ventricle.

4. The second heart sound is caused by
A. closure of the aortic and pulmonary valves.
B. vibrations in the ventricular wall during systole.
C. ventricular filling.
D. closure of the mitral and tricuspid valves.
E. retrograde flow in the vena cava.

5. The fourth heart sound is caused by
A. closure of the aortic and pulmonary valves.
B. vibrations in the ventricular wall during systole.
C. ventricular filling.
D. closure of the mitral and tricuspid valves.
E. retrograde flow in the vena cava.

6. During exercise, a man consumes 1.8 L of oxygen per minute. His arterial O_2 content is 190 mL/L, and the O_2 content of his mixed venous blood is 134 mL/L. His cardiac output is approximately
A. 3.2 L/min.
B. 16 L/min.
C. 32 L/min.
D. 54 L/min.
E. 160 mL/min.

7. Starling’s law of the heart
A. does not operate in the failing heart.

B. does not operate during exercise.

C. explains the increase in heart rate produced by exercise.

D. explains the increase in cardiac output that occurs when venous return is increased.

E. explains the increase in cardiac output when the sympathetic nerves supplying the heart are stimulated.

8. A 65-year-old woman comes to her physician complaining of chest pain on exertion and is diagnosed with angina pectoris. An examination reveals an aortic abnormality. This is more likely to be aortic stenosis as opposed to aortic regurgitation because:

A. An increase in preload causes a greater increase in cardiac O_2 consumption than an increase in afterload.

B. Cardiac O_2 consumption is unrelated to ventricular work per beat.

C. Increased pressure work causes a greater increase in cardiac O_2 consumption than an increase in volume work.

D. Cardiac venous O_2 tension is low.

E. Aortic stenosis decreases pressure in the aortic valve.

CHAPTER 31

Blood as a Circulatory Fluid & the Dynamics of Blood & Lymph Flow

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the cellular components of blood, and their origins, how the fragility of red blood cells is controlled, the role of hemoglobin in transporting oxygen in red blood cells, and the basis of hemoglobinopathies.
- Define the composition of plasma and lymph and identify the sources of their constituents.
- Understand the molecular basis of blood groups and the reasons for transfusion reactions.
- Delineate the process of hemostasis that restricts blood loss when vessels are damaged, the adverse consequences of intravascular thrombosis, and the basis of clotting disorders.
- Identify the types of blood and lymphatic vessels that make up the circulatory system and the regulation and function of their primary constituent cell types.
- Describe how physical principles dictate the flow of blood and lymph around the body.

- Understand the basis of methods used to measure blood flow and blood pressure in various vascular segments.
- Understand the basis and consequences of a sustained elevation of systemic arterial pressure (hypertension).

INTRODUCTION

The **circulatory system** supplies inspired O_2 as well as substances absorbed from the gastrointestinal tract to the tissues, returns CO_2 to the lungs and other products of metabolism to the kidneys, functions in the regulation of body temperature, and distributes hormones and other agents that regulate cell function. The blood, the carrier of these substances, is pumped through a closed system of blood vessels by the heart. From the left ventricle, blood is pumped through the arteries and arterioles to the capillaries, where it equilibrates with the interstitial fluid. The capillaries drain through venules into the veins and back to the right atrium. Some tissue fluids enter another system of closed vessels, the lymphatics, which drain lymph via the thoracic duct and the right lymphatic duct into the venous system. The circulation is controlled by multiple regulatory systems that function in general to maintain adequate capillary blood flow when possible in all organs, but particularly in the heart and brain.

Blood flows through the circulation primarily because of the forward motion imparted to it by the pumping of the heart, although in the case of the systemic circulation, diastolic recoil of the walls of the arteries, compression of the veins by skeletal muscles during exercise, and the negative pressure in the thorax during inspiration also move the blood forward. The resistance to flow depends to a minor degree on the viscosity of the blood but mostly on the diameter of the vessels, and principally that of the arterioles. The blood flow to each tissue is regulated by local chemical and general neural and humoral mechanisms that dilate or constrict its vessels. All of the blood flows through the lungs, but the systemic circulation is made up of numerous different circuits in parallel **(Figure 31–1)**. The arrangement permits wide variations in regional blood flow without changing total systemic flow.

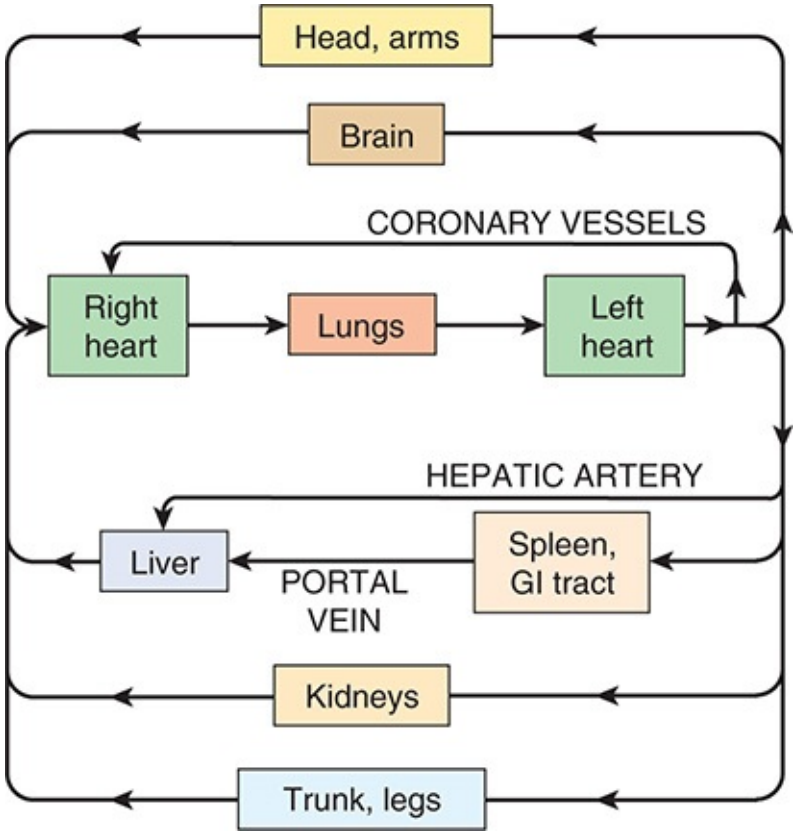


FIGURE 31–1 Diagram of the circulation in the adult.

This chapter is concerned with blood and lymph and with the multiple functions of the cells they contain. It will also address general principles that apply to all parts of the circulation and pressure and flow in the systemic circulation. The homeostatic mechanisms operating to adjust flow are the subject of Chapter 32. The special characteristics of the pulmonary and renal circulations are discussed in Chapters 34 and 37. Likewise, the role of blood as the carrier of many immune effector cells will not be discussed here, but rather was covered in Chapter 3.

BLOOD AS A CIRCULATORY FLUID

Blood consists of a protein-rich fluid known as plasma, in which are suspended cellular elements: white blood cells, red blood cells, and platelets. The normal total circulating blood volume is about 8% of the body weight (5600 mL in a 70-kg man). About 55% of this volume is plasma.

BONE MARROW

In the adult, red blood cells, many white blood cells, and platelets are formed in the bone marrow. In the fetus, blood cells are also formed in the liver and spleen, and in adults such **extramedullary hematopoiesis** may occur in diseases in which the bone marrow becomes destroyed or fibrosed. In children, blood cells are actively produced in the marrow cavities of all the bones. By age 20, the marrow in the cavities of the long bones, except for the upper humerus and femur, has become inactive **(Figure 31–2)**.

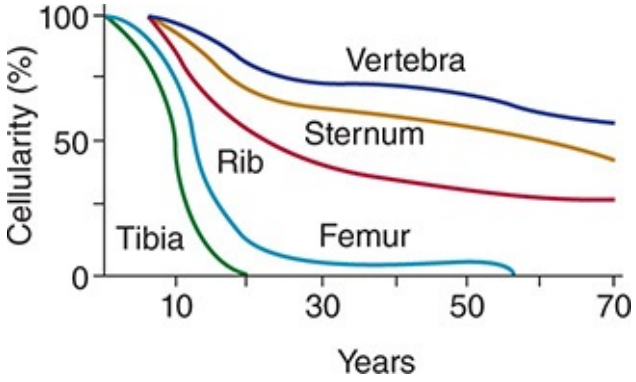


FIGURE 31–2 Changes in red bone marrow cellularity in various bones with age. Hundred percent equals the degree of cellularity at birth. (Reproduced with permission from Whitby LEH, Britton CJC: *Disorders of the Blood*, 10th ed. Churchill Livingstone; 1969.)

The bone marrow is actually one of the largest organs in the body, approaching the size and weight of the liver. It is also one of the most active. Normally, 75% of the cells in the marrow belong to the white blood cell–producing myeloid series and only 25% are maturing red cells, even though there are over 500 times as many red cells in the circulation as there are white cells. This difference in the marrow reflects the fact that the average life span of white cells is short, whereas that of red cells is long.

Hematopoietic stem cells (HSCs) are bone marrow cells that are capable of producing all types of blood cells. They differentiate into one or another type of committed stem cells **(progenitor cells).** These in turn form the various differentiated types of blood cells. There are separate pools of progenitor cells for megakaryocytes, lymphocytes, erythrocytes, eosinophils, and basophils; neutrophils and monocytes arise from a common precursor. The bone marrow stem cells are also the source of osteoclasts (see Chapter 21), Kupffer cells (see Chapter 28), mast cells, dendritic cells, and Langerhans cells. The HSCs are few

in number but are capable of completely replacing the bone marrow when injected into a patient whose own bone marrow has been entirely destroyed.

The HSCs are derived from uncommitted, totipotent stem cells that can be stimulated to form any cell in the body. Adults have a few of these, but they are more readily obtained from the blastocysts of embryos. There is not surprisingly immense interest in stem cell research due to its potential to regenerate diseased tissues, but ethical issues are involved, and debate on these issues will undoubtedly continue.

WHITE BLOOD CELLS

Normally, human blood contains 4000–11,000 white blood cells per microliter **(Table 31–1)**. Of these, the **granulocytes (polymorphonuclear leukocytes, PMNs)** are the most numerous. Young granulocytes have horseshoe-shaped nuclei that become multilobed as the cells grow older **(Figure 31–3)**. Most of them contain neutrophilic granules **(neutrophils),** but a few contain granules that stain with acidic dyes **(eosinophils),** and some have basophilic granules **(basophils).** The other two cell types found normally in peripheral blood are **lymphocytes,** which have large round nuclei and scanty cytoplasm, and **monocytes,** which have abundant agranular cytoplasm and kidney-shaped nuclei (Figure 31–3). Acting together, these cells provide the body with the powerful defenses against tumors and viral, bacterial, and parasitic infections that were discussed in Chapter 3.

TABLE 31–1 Normal values for the cellular elements in human blood.

Cell	Cells/μL (average)	Approximate Normal Range	Percentage of Total White Cells
Total white blood cells	9000	4000–11,000	…
Granulocytes			
Neutrophils	5400	3000–6000	50–70
Eosinophils	275	150–300	1–4
Basophils	35	0–100	0.4
Lymphocytes	2750	1500–4000	20–40
Monocytes	540	300–600	2–8
Erythrocytes			
Females	4.8×10^6	…	…
Males	5.4×10^6	…	…
Platelets	300,000	200,000–500,000	…

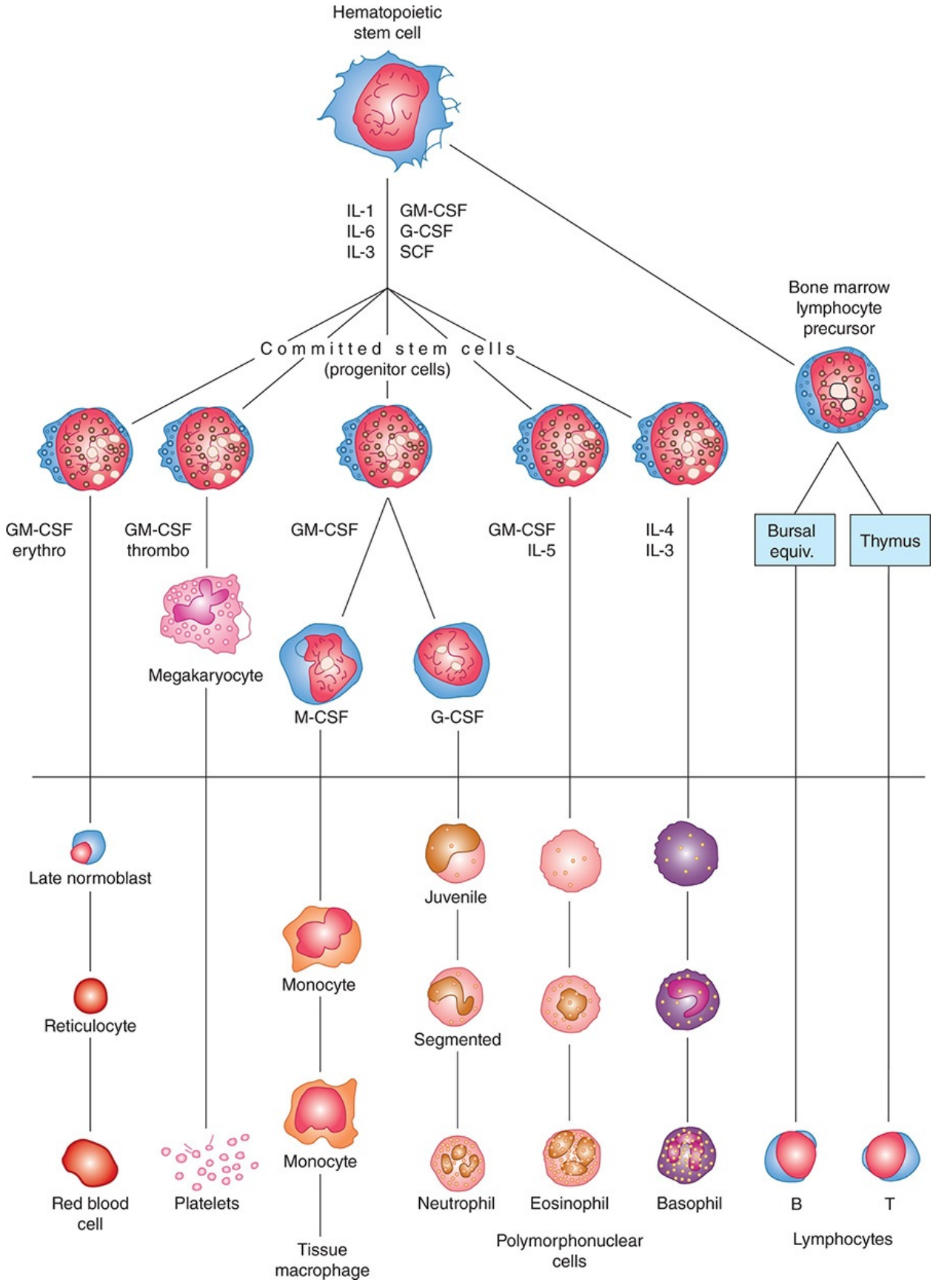

Hematopoietic stem cell
IL-1
IL-6
IL-3
GM-CSF
G-CSF
SCF
Committed stem cells (progenitor cells)
Bone marrow lymphocyte precursor
GM-CSF erythro
GM-CSF thrombo
GM-CSF
GM-CSF IL-5
IL-4 IL-3
Bursal equiv.
Thymus
Megakaryocyte
M-CSF
G-CSF
Late normoblast
Reticulocyte
Red blood cell
Platelets
Monocyte
Monocyte
Tissue macrophage
Juvenile
Segmented
Neutrophil
Eosinophil
Basophil
Polymorphonuclear cells
B
T
Lymphocytes

FIGURE 31–3 Development of various formed elements of the blood from bone marrow cells. Cells below the horizontal line are found in normal peripheral blood. The principal sites of action of erythropoietin (erythro) and the various colony-stimulating factors (CSFs) that stimulate the differentiation of the components are indicated. G, granulocyte; M, macrophage; IL, interleukin; thrombo, thrombopoietin; erythro, erythropoietin; SCF, stem cell factor.

PLATELETS

Platelets are small, granulated bodies that aggregate at sites of vascular injury. They lack nuclei and are 2–4 µm in diameter (Figure 31–3). There are about 300,000/µL of circulating blood, and they normally have a half-life of about 4 days. The **megakaryocytes,** giant cells in the bone marrow, form platelets by pinching off bits of cytoplasm and extruding them into the circulation. Between 60% and 75% of the platelets that have been extruded from the bone marrow are in the circulating blood, and the remainder are mostly in the spleen. Splenectomy causes an increase in the platelet count **(thrombocytosis).**

RED BLOOD CELLS

The red blood cells **(erythrocytes)** carry hemoglobin in the circulation. They are biconcave disks **(Figure 31–4)** that are manufactured in the bone marrow. In mammals, they lose their nuclei before entering the circulation. In humans, they survive in the circulation for an average of 120 days. The average normal red blood cell count is 5.4 million/µL in men and 4.8 million/µL in women. The number of red cells is also conveniently expressed as the **hematocrit,** or the percentage of the blood, by volume, that is occupied by erythrocytes. Each human red blood cell is about 7.5 µm in diameter and 2 µm thick, and each contains approximately 29 pg of hemoglobin **(Table 31–2)**. There are thus about 3×10^{13} red blood cells and about 900 g of hemoglobin in the circulating blood of an adult man **(Figure 31–5)**.

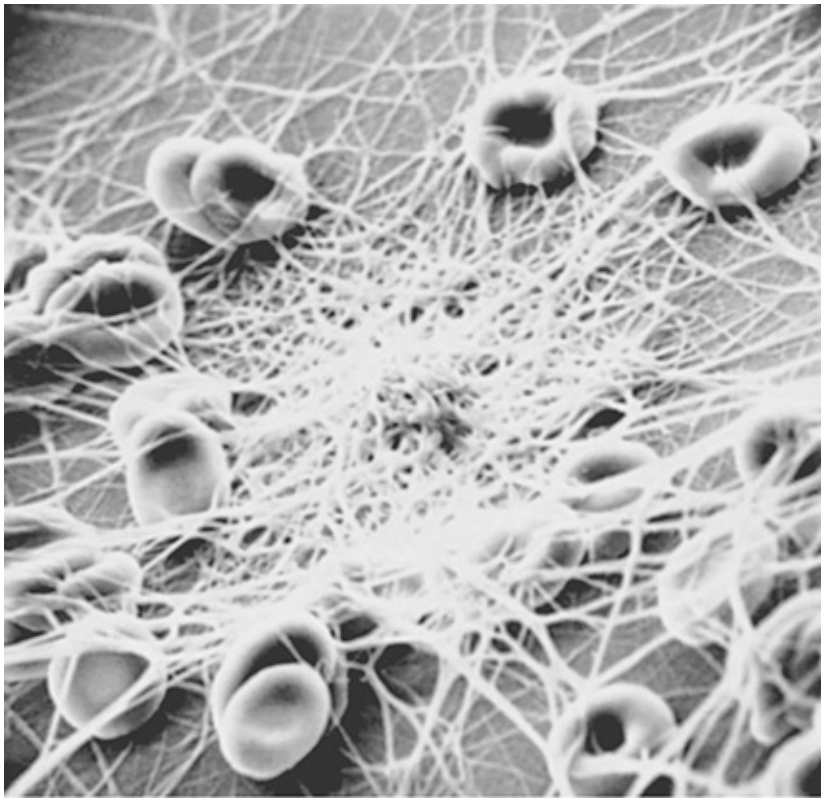


FIGURE 31–4 Human red blood cells and fibrin fibrils. Blood was placed on a polyvinyl chloride surface, fixed, and photographed with a scanning electron microscope. Reduced from ×2590. (Used with permission of NF Rodman.)

TABLE 31–2 Characteristics of human red cells.[a]

		Male	Female
Hematocrit (Hct) (%)		47	42
Red blood cells (RBC) (10^6/μL)		5.4	4.8
Hemoglobin (Hb) (g/dL)		16	14
Mean corpuscular volume (MCV) (fL)	$= \frac{\text{Hct} \times 10}{\text{RBC } (10^6/\mu\text{L})}$	87	87
Mean corpuscular hemoglobin (MCH) (pg)	$= \frac{\text{Hb} \times 10}{\text{RBC } (10^6/\mu\text{L})}$	29	29
Mean corpuscular hemoglobin concentration (MCHC) (g/dL)	$= \frac{\text{Hb} \times 100}{\text{Hct}}$	34	34
Mean cell diameter (MCD) (μm)	= Mean diameter of 500 cells in smear	7.5	7.5

[a]Cells with MCVs > 95 fL are called macrocytes; cells with MCVs < 80 fL are called microcytes; cells with MCHCs < 25 g/dL are called hypochromic.

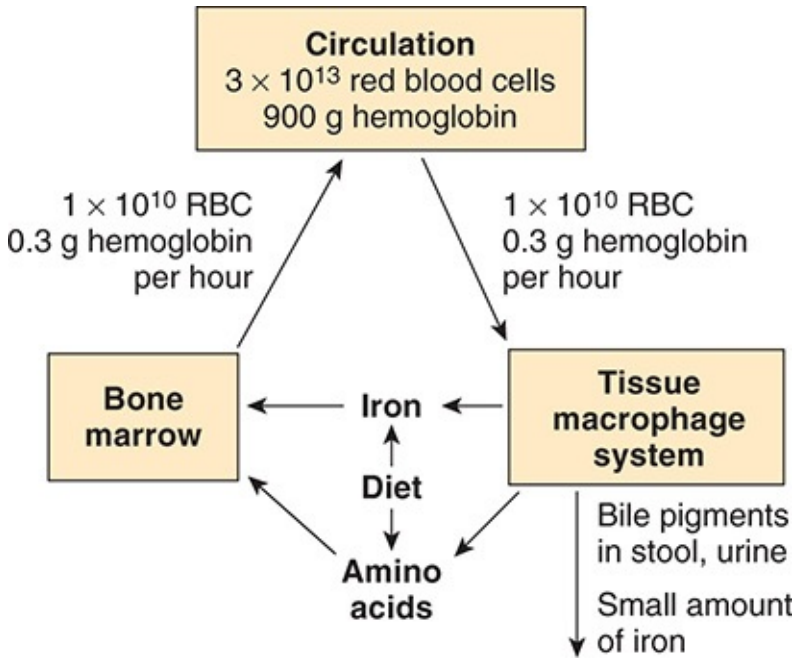


FIGURE 31–5 Red cell formation and destruction. RBC, red blood cells.

The feedback control of erythropoiesis by erythropoietin is discussed in Chapter 38, and the role of IL-1, IL-3, IL-6 (interleukin), and GM-CSF (granulocyte-macrophage colony-stimulating factor) in development of the relevant erythroid stem cells is shown in Figure 31–3.

ROLE OF THE SPLEEN

The spleen is an important blood filter that removes aged or abnormal red cells. It also contains many platelets and plays a significant role in the immune system. Abnormal red cells are removed if they are not as flexible as normal red cells and consequently are unable to squeeze through the slits between the endothelial cells that line the splenic sinuses **(Clinical Box 31–1)**.

HEMOGLOBIN

The red, oxygen-carrying pigment in the red blood cells of vertebrates is the protein **hemoglobin**. Hemoglobin is a globular molecule made up of four

subunits **(Figure 31–6)**. Each subunit contains a **heme** moiety conjugated to a polypeptide. Heme is an iron-containing porphyrin derivative **(Figure 31–7)**. The polypeptides are referred to collectively as the **globin** portion of the hemoglobin molecule. There are two pairs of polypeptides in each hemoglobin molecule. In normal adult human hemoglobin **(hemoglobin A),** the two polypeptides are called α chains and β chains. Thus, hemoglobin A is designated $\alpha_2\beta_2$. Not all the hemoglobin in the blood of normal adults is hemoglobin A. About 2.5% of the hemoglobin is hemoglobin A_2, in which β chains are replaced by δ chains ($\alpha_2\delta_2$).

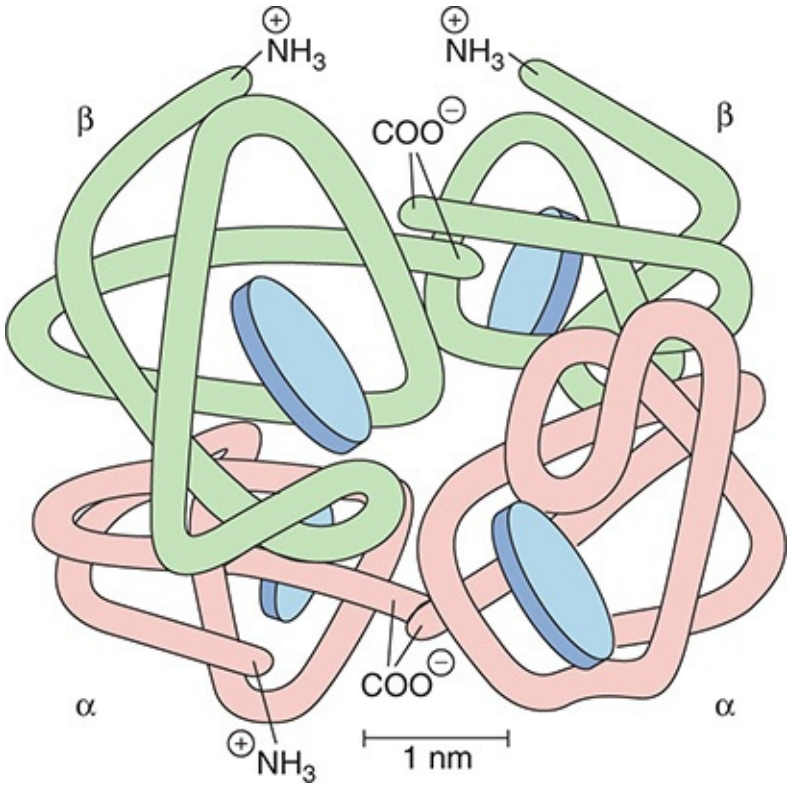


FIGURE 31–6 Diagrammatic representation of a molecule of hemoglobin A, showing the four subunits. There are two α and two β polypeptide chains, each containing a heme moiety. These moieties are represented by the blue disks. (Reproduced with permission from Harper HA, et al: *Physiologische Chemie*. Springer; 1975.)

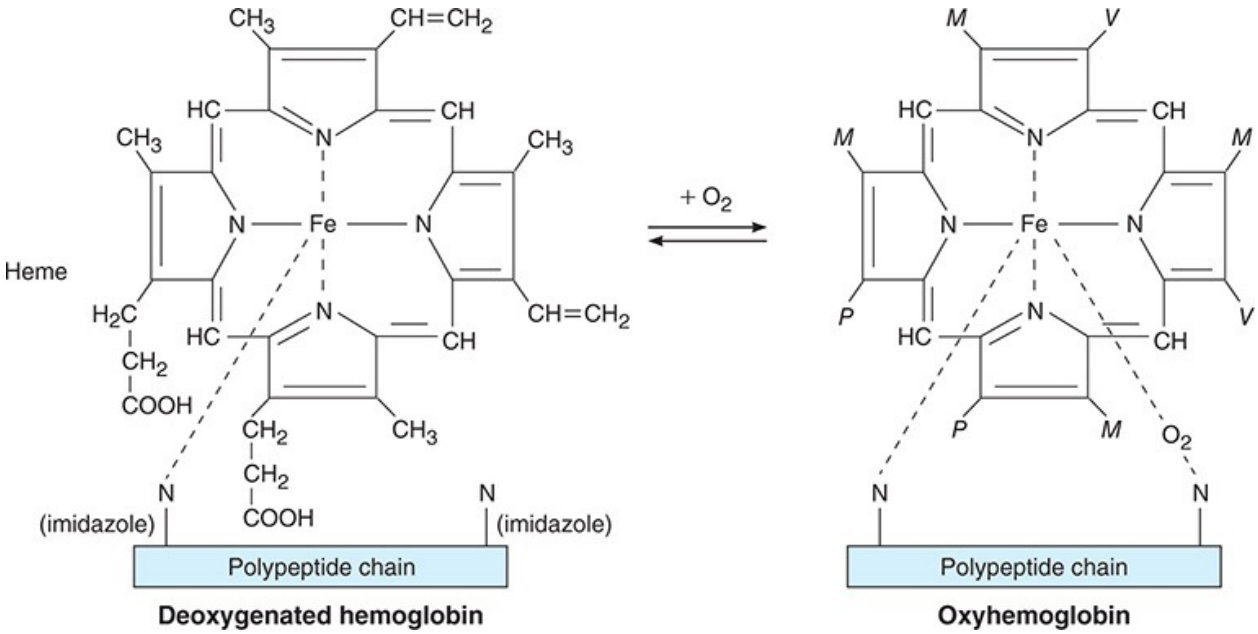


FIGURE 31–7 Reaction of heme with O_2. The abbreviations *M*, *V*, and *P* stand for the groups shown on the molecule on the left.

There are small amounts of hemoglobin A derivatives closely associated with hemoglobin A that represent glycated hemoglobins. One of these, hemoglobin A_{1c} (HbA_{1c}), has a glucose attached to the terminal valine in each β chain and is of special interest because it increases in the blood of patients with poorly controlled diabetes mellitus (see Chapter 24), and is measured clinically as a marker of the progression of that disease and/or the effectiveness of treatment.

REACTIONS OF HEMOGLOBIN

O_2 binds to the Fe^{2+} in the heme moiety of hemoglobin to form **oxyhemoglobin.** The affinity of hemoglobin for O_2 is affected by pH, temperature, and the concentration in the red cells of 2,3-bisphosphoglycerate (2,3-BPG). 2,3-BPG and H^+ compete with O_2 for binding to deoxygenated hemoglobin, decreasing the affinity of hemoglobin for O_2 by shifting the positions of the four peptide chains (quaternary structure). The details of the oxygenation and deoxygenation of hemoglobin and the physiologic role of these reactions in O_2 transport are discussed in Chapter 35.

CLINICAL BOX 31–1

Red Cell Fragility

Red blood cells, like other cells, shrink in solutions with an osmotic pressure greater than that of normal plasma. In solutions with a lower osmotic pressure they swell, become spherical rather than disk-shaped, and eventually lose their hemoglobin **(hemolysis).** The hemoglobin of hemolyzed red cells dissolves in the plasma, coloring it red. A 0.9% sodium chloride solution is isotonic with plasma. When **osmotic fragility** is normal, red cells begin to hemolyze when suspended in 0.5% saline; 50% lysis occurs in 0.40–0.42% saline, and lysis is complete in 0.35% saline. In **hereditary spherocytosis** (congenital hemolytic icterus), the cells are spherocytic in normal plasma and hemolyze more readily than normal cells in hypotonic sodium chloride solutions. Abnormal spherocytes are also trapped and destroyed in the spleen, meaning that hereditary spherocytosis is one of the most common causes of **hereditary hemolytic anemia.** The spherocytosis is caused by mutations in proteins that make up the membrane skeleton of the erythrocyte, which normally maintain the shape and flexibility of the red cell membrane, including **spectrin,** the transmembrane protein **band 3,** and the linker protein, **ankyrin.** Red cells can also be lysed by drugs (especially penicillin and sulfa drugs) and infections. The susceptibility of red cells to hemolysis by these agents is increased by deficiency of the enzyme glucose-6-phosphate dehydrogenase (G6PD), which catalyzes the initial step in the oxidation of glucose via the hexose monophosphate pathway (see Chapter 1). This pathway generates dihydronicotinamide adenine dinucleotide phosphate (NADPH), which is needed for the maintenance of normal red cell fragility. Severe G6PD deficiency also inhibits the killing of bacteria by granulocytes and predisposes to severe infections.

THERAPEUTIC HIGHLIGHTS

Severe cases of hereditary spherocytosis can be treated by splenectomy, but this is not without other risks, such as sepsis. Milder cases can be treated with dietary folate supplementation and/or blood transfusions. Treatment of other forms of hemolytic anemia depends on the underlying cause. Some forms are autoimmune in nature, and have been shown to benefit from treatment with corticosteroids.

When blood is exposed to various drugs and other oxidizing agents in vitro or in vivo, the ferrous iron (Fe^{2+}) that is normally present in hemoglobin is converted to ferric iron (Fe^{3+}), forming **methemoglobin.** Methemoglobin is dark-colored, and when it is present in large quantities in the circulation, it causes a dusky discoloration of the skin resembling cyanosis (see Chapter 35). Some oxidation of hemoglobin to methemoglobin occurs normally, but an enzyme system in the red cells, the dihydronicotinamide adenine dinucleotide (NADH)-methemoglobin reductase system, converts methemoglobin back to hemoglobin. Congenital absence of this system is one cause of hereditary methemoglobinemia.

Carbon monoxide reacts with hemoglobin to form **carbon monoxyhemoglobin (carboxyhemoglobin).** The affinity of hemoglobin for O_2 is much lower than its affinity for carbon monoxide, which consequently displaces O_2 on hemoglobin, reducing the oxygen-carrying capacity of blood (see Chapter 35).

HEMOGLOBIN IN THE FETUS

The blood of the human fetus normally contains **fetal hemoglobin (hemoglobin F).** Its structure is similar to that of hemoglobin A except that the β chains are replaced by γ chains; that is, hemoglobin F is $\alpha_2\gamma_2$. Fetal hemoglobin is normally replaced by adult hemoglobin soon after birth **(Figure 31–8)**. In certain individuals, it fails to disappear and persists throughout life. In the body, its O_2 content at a given PO_2 is greater than that of adult hemoglobin because it binds 2,3-BPG less avidly. Hemoglobin F is therefore critical to facilitate movement of O_2 from the maternal to the fetal circulation, particularly at later stages of gestation where oxygen demand increases (see Chapter 33). In young embryos there are, in addition, ζ and ε chains, forming Gower 1 hemoglobin ($\zeta_2\varepsilon_2$) and Gower 2 hemoglobin ($\alpha_2\varepsilon_2$). Switching from one form of hemoglobin to another during development seems to be regulated largely by oxygen availability, with relative hypoxia favoring the production of hemoglobin F both via direct effects on globin gene expression, as well as upregulated production of erythropoietin.

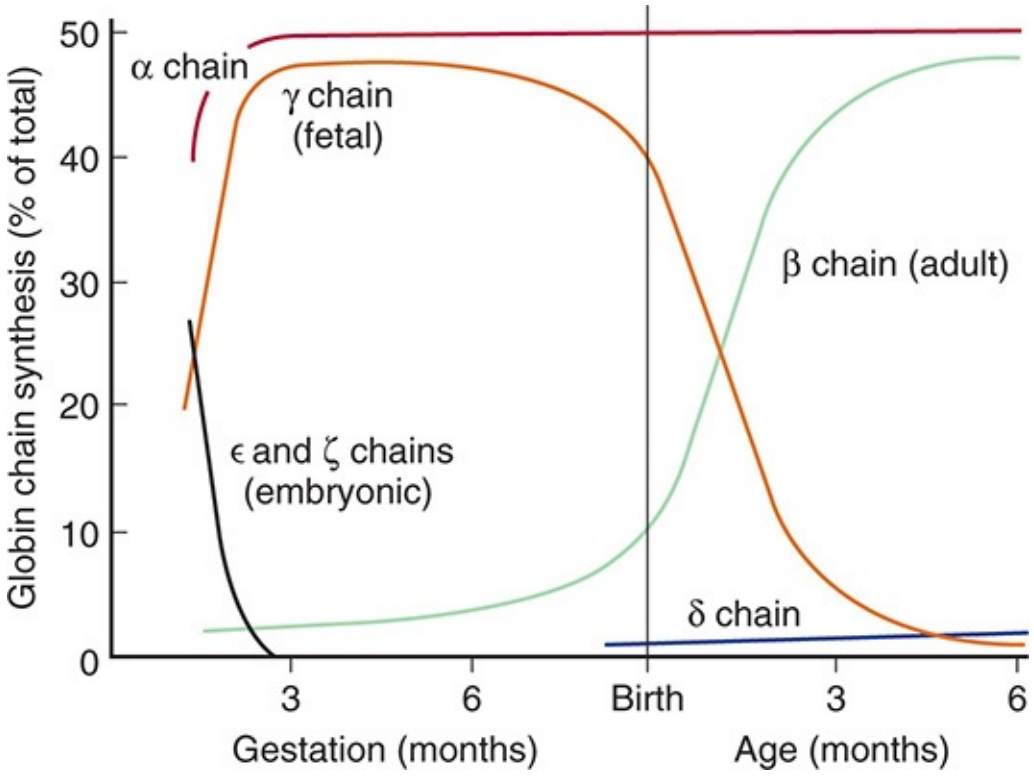


FIGURE 31–8 Development of human hemoglobin chains. The graph shows the normal rate of synthesis of the various hemoglobin chains in utero, and how these change after birth.

SYNTHESIS OF HEMOGLOBIN

The average normal hemoglobin content of blood is 16 g/dL in men and 14 g/dL in women, all of it in red cells. In the body of a 70-kg man, there are about 900 g of hemoglobin, and 0.3 g of hemoglobin is destroyed and 0.3 g synthesized every hour (Figure 31–5). The heme portion of the hemoglobin molecule is synthesized from glycine and succinyl-CoA **(Clinical Box 31–2)**.

CATABOLISM OF HEMOGLOBIN

When old red blood cells are destroyed by tissue macrophages, the globin portion of the hemoglobin molecule is split off, and the heme is converted to **biliverdin.** The enzyme involved is a subtype of heme oxygenase (see Figure 28–4), and CO is formed in the process. CO is an intercellular messenger, like NO (see Chapters 2 and 3). In humans, most of the biliverdin is converted to

bilirubin and excreted in the bile (see Chapter 28). The iron from the heme is reused for hemoglobin synthesis.

Exposure of the skin to white light converts bilirubin to lumirubin, which has a shorter half-life than bilirubin. **Phototherapy** (exposure to light) is of value in treating infants with jaundice due to hemolysis. Iron is essential for hemoglobin synthesis; if blood is lost from the body and the iron deficiency is not corrected, **iron deficiency anemia** results.

CLINICAL BOX 31–2

Abnormalities of Hemoglobin Production

There are two major types of inherited disorders of hemoglobin in humans: the **hemoglobinopathies,** in which abnormal globin polypeptide chains are produced, and the **thalassemias** and related disorders, in which the chains are normal in structure but produced in decreased amounts or absent because of defects in the regulatory portion of the globin genes. Mutant genes that cause the production of abnormal hemoglobins are widespread, and over 1000 abnormal hemoglobins have been described in humans. In one of the most common examples, hemoglobin S, the α chains are normal but the β chains have a single substitution of a valine residue for one glutamic acid, leading to **sickle cell anemia** (Table 31–3). When an abnormal gene inherited from one parent dictates formation of an abnormal hemoglobin (ie, when the individual is heterozygous), half the circulating hemoglobin is abnormal and half is normal. When identical abnormal genes are inherited from both parents, the individual is homozygous and all the hemoglobin is abnormal. It is theoretically possible to inherit two different abnormal hemoglobins, one from the father and one from the mother. Studies of the inheritance and geographic distribution of abnormal hemoglobins have made it possible in some cases to decide where the mutant gene originated and approximately how long ago the mutation occurred. In general, harmful mutations tend to die out, but mutant genes that confer traits with survival value persist and spread in the population. Many of the abnormal hemoglobins are harmless; however, some have abnormal O_2 equilibriums, while others cause anemia. For example, hemoglobin S polymerizes at low O_2 tensions, and this causes the red cells to become sickle-shaped, hemolyze, and form aggregates that block blood vessels. The sickle cell gene is an example of a gene that has persisted and spread in the population due to its beneficial effect when present

in heterozygous form. It originated in Africa, and confers resistance to one type of malaria. In some parts of Africa, 40% of the population is heterozygous for hemoglobin S. There is a corresponding prevalence of 10% among African Americans in the United States.

THERAPEUTIC HIGHLIGHTS

Hemoglobin F decreases the polymerization of deoxygenated hemoglobin S, and hydroxyurea stimulates production of hemoglobin F in children and adults. Hydroxyurea has therefore proven to be a very valuable agent for the treatment of sickle cell disease. In patients with severe sickle cell disease, hematopoietic stem cell transplantation has also been shown to have some benefit, and prophylactic treatment with antibiotics has also been shown to be helpful. The clinically important thalassemias result in severe anemia, often requiring repeated blood transfusions. However, these run the risk of iron overload, and often must be accompanied by treatment with drugs that chelate iron. Hematopoietic stem cell transplantation is also being explored for treatment of the thalassemias.

TABLE 31–3 Partial amino acid composition of normal human β chain, and some hemoglobins with abnormal β chains.[a]

	Positions on Polypeptide Chain of Hemoglobin						
Hemoglobin	**123**	**67**	**26**	**63**	**67**	**121**	**146**
A (normal)	Val-His-Leu	Glu-Glu	Glu	His	Val	Glu	His
S (sickle cell)		Val					
C		Lys					
$G_{San\ Jose}$		Gly					
E			Lys				
$M_{Saskatoon}$				Tyr			
$M_{Milwaukee}$					Glu		
O_{Arabia}						Lys	

[a]Other hemoglobins have abnormal α chains. Abnormal hemoglobins that are very similar electrophoretically but differ slightly in composition are indicated by the same letter and a subscript indicating the geographic location where they were first discovered; hence, $M_{Saskatoon}$ and $M_{Milwaukee}$.

PLASMA

The fluid portion of the blood, the **plasma,** is a remarkable solution containing an immense number of ions, inorganic molecules, and organic molecules that are in transit to various parts of the body or aid in the transport of other substances. Normal plasma volume is about 5% of body weight, or roughly 3500 mL in a 70-kg man. Plasma clots on standing, remaining fluid only if an anticoagulant is added. If whole blood is allowed to clot and the clot is removed, the remaining fluid is called **serum.** Serum has essentially the same composition as plasma, except that its fibrinogen and clotting factors II, V, and VIII **(Table 31–4)** have been removed and it has a higher serotonin content because of the breakdown of platelets during clotting.

TABLE 31–4 System for naming blood-clotting factors.

Factor[a]	Names
I	Fibrinogen
II	Prothrombin
III	Thromboplastin
IV	Calcium
V	Proaccelerin, labile factor, accelerator globulin
VII	Proconvertin, SPCA, stable factor
VIII	Antihemophilic factor (AHF), antihemophilic factor A, antihemophilic globulin (AHG)
IX	Plasma thromboplastic component (PTC), Christmas factor, antihemophilic factor B
X	Stuart-Prower factor
XI	Plasma thromboplastin antecedent (PTA), antihemophilic factor C
XII	Hageman factor, glass factor
XIII	Fibrin-stabilizing factor, Laki-Lorand factor
HMW-K	High-molecular-weight kininogen, Fitzgerald factor
Pre-Ka	Prekallikrein, Fletcher factor
Ka	Kallikrein
PL	Platelet phospholipid

[a]Factor VI is not a separate entity and has been dropped.

PLASMA PROTEINS

The plasma proteins consist of **albumin, globulin,** and **fibrinogen** fractions. Most capillary walls are relatively impermeable to the proteins in plasma, and the proteins therefore exert an osmotic force of about 25 mm Hg across the capillary wall (**oncotic pressure;** see Chapter 1) that pulls water into the blood. The plasma proteins are also responsible for 15% of the buffering capacity of proteins in the blood (including hemoglobin; see Chapter 39) because of the

weak ionization of their substituent COOH and NH_2 groups. At the normal plasma pH of 7.40, the proteins are mostly in the anionic form (see Chapter 1). Plasma proteins may have specific functions (eg, antibodies and the proteins concerned with blood clotting), whereas others function as nonspecific carriers for various hormones, other solutes, and drugs.

ORIGIN OF PLASMA PROTEINS

Circulating antibodies are manufactured by lymphocytes. Most of the other plasma proteins are synthesized in the liver. These proteins and their principal functions are listed in **Table 31–5**.

TABLE 31–5 Some of the proteins synthesized by the liver: Physiologic functions and properties.

Name	Principal Function	Binding Characteristics	Serum or Plasma Concentration
Albumin	Binding and carrier protein; osmotic regulator	Hormones, amino acids, steroids, vitamins, fatty acids	4500–5000 mg/dL
Orosomucoid	Uncertain; may have a role in inflammation		Trace; rises in inflammation
α_1-Antiprotease	Trypsin and general protease inhibitor	Proteases in serum and tissue secretions	1.3–1.4 mg/dL
α-Fetoprotein	Osmotic regulation; binding and carrier protein[a]	Hormones, amino acids	Found normally in fetal blood
α_2-Macroglobulin	Inhibitor of serum endoproteases	Proteases	150–420 mg/dL
Antithrombin-III	Protease inhibitor of intrinsic coagulation system	1:1 binding to proteases	17–30 mg/dL
Ceruloplasmin	Transport of copper	Six atoms copper/molecule	15–60 mg/dL
C-reactive protein	Uncertain; has role in tissue inflammation	Complement C1q	<1 mg/dL; rises in inflammation
Fibrinogen	Precursor to fibrin in hemostasis		200–450 mg/dL
Haptoglobin	Binding, transport of cell-free hemoglobin	Hemoglobin 1:1 binding	40–180 mg/dL
Hemopexin	Binds to porphyrins, particularly heme for heme recycling	1:1 with heme	50–100 mg/dL
Transferrin	Transport of iron	Two atoms iron/molecule	3.0–6.5 mg/dL
Apolipoprotein B	Assembly of lipoprotein particles	Lipid carrier	
Angiotensinogen	Precursor to pressor peptide angiotensin II		
Proteins, coagulation factors II, VII, IX, X	Blood clotting		20 mg/dL
Antithrombin C, protein C	Inhibition of blood clotting		
Insulin-like growth factor I	Mediator of anabolic effects of growth hormone	IGF-I receptor	
Steroid hormone-binding globulin	Carrier protein for steroids in bloodstream	Steroid hormones	3.3 mg/dL
Thyroxine-binding globulin	Carrier protein for thyroid hormone in bloodstream	Thyroid hormones	1.5 mg/dL
Transthyretin (thyroid-binding prealbumin)	Carrier protein for thyroid hormone in bloodstream	Thyroid hormones	25 mg/dL

[a]The function of α-fetoprotein is uncertain, but because of its structural homology to albumin it is often assigned these functions.

Data on the turnover of albumin show that synthesis plays an important role in the maintenance of normal levels. In normal adult humans, the plasma albumin level is 3.5–5.0 g/dL, and the total exchangeable albumin pool is 4.0–5.0 g/kg body weight; 38–45% of this albumin is intravascular, and much of the rest of it is in the skin. Between 6% and 10% of the exchangeable pool is degraded per day, and the degraded albumin is replaced by hepatic synthesis of 200–400 mg/kg/day. The albumin is probably transported to the extravascular areas by vesicular transport across the walls of the capillaries (see Chapter 2). Albumin synthesis is carefully regulated. It is decreased during fasting and increased in conditions such as nephrosis in which there is excessive albumin

loss.

HYPOPROTEINEMIA

Plasma protein levels are maintained during starvation until body protein stores are markedly depleted. However, in prolonged starvation and in malabsorption syndromes due to intestinal diseases, plasma protein levels are low **(hypoproteinemia).** They are also low in liver disease, because hepatic protein synthesis is depressed, and in nephrosis, because large amounts of albumin are lost in the urine. Because of the decrease in the plasma oncotic pressure, edema tends to develop. Rarely, there is congenital absence of one or another plasma protein. An example of congenital protein deficiency is the congenital form of **afibrinogenemia,** characterized by defective blood clotting.

LYMPH

Lymph is tissue fluid that enters the lymphatic vessels. It drains into the venous blood via the thoracic and right lymphatic ducts. It contains clotting factors and clots on standing in vitro. In most locations, it also contains proteins that have traversed capillary walls and can then return to the blood via the lymph. Nevertheless, its protein content is generally lower than that of plasma, which contains about 7 g/dL, but lymph protein content varies with the region from which the lymph drains **(Table 31–6)**. Water-insoluble fats are absorbed from the intestine into the lymphatics, and the lymph in the thoracic duct after a meal is milky because of its high fat content (see Chapter 26). Lymphocytes also enter the circulation principally through the lymphatics, and there are appreciable numbers of lymphocytes in thoracic duct lymph.

TABLE 31–6 Approximate protein content of lymph in humans.

Source of Lymph	Protein Content (g/dL)
Choroid plexus	0
Ciliary body	0
Skeletal muscle	2
Skin	2
Lung	4
Gastrointestinal tract	4.1
Heart	4.4
Liver	6.2

Data from JN Diana.

BLOOD TYPES

The membranes of human red cells contain a variety of **blood group antigens,** which are also called **agglutinogens.** The most important and best known of these are the A and B antigens, but there are many more.

THE ABO SYSTEM

The A and B antigens are inherited as mendelian dominants, and individuals are divided into four major **blood types** on this basis. Type A individuals have the A antigen, type B have the B, type AB have both, and type O have neither. The A and B antigens are complex oligosaccharides that differ in their terminal sugar. An *H* gene codes for a fucose transferase that adds a terminal fucose, forming the H antigen that is usually present in individuals of all blood types **(Figure 31–9)**. Individuals who are type A also express a second transferase that catalyzes placement of a terminal N-acetylgalactosamine on the H antigen, whereas individuals who are type B express a transferase that places a terminal galactose. Individuals who are type AB have both transferases. Individuals who are type O have neither, so the H antigen persists.

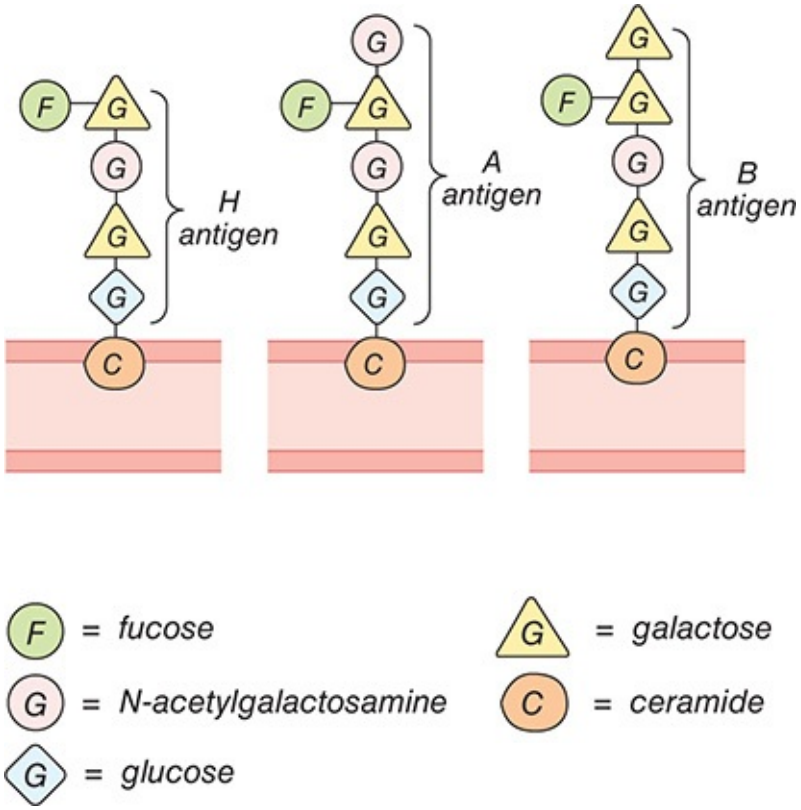


FIGURE 31–9 Antigens of the ABO system on the surface of red blood cells.

Antibodies against red cell agglutinogens are called **agglutinins.** Antigens very similar to A and B are common in intestinal bacteria and possibly in foods to which newborn individuals are exposed. Therefore, infants rapidly develop antibodies against the antigens not present in their own cells. Thus, type A individuals develop anti-B antibodies, type B individuals develop anti-A antibodies, type O individuals develop both, and type AB individuals develop neither **(Table 31–7)**. When the plasma of a type A individual is mixed with type B red cells, the anti-B antibodies cause the type B red cells to clump (agglutinate), as shown in **Figure 31–10**. The other agglutination reactions produced by mismatched plasma and red cells are summarized in Table 31–7. ABO **blood typing** is performed by mixing an individual's red blood cells with antisera containing the various agglutinins on a slide and seeing whether agglutination occurs.

TABLE 31–7 Summary of ABO system.

Blood Type	Agglutinins in Plasma	Frequency in United States (%)	Plasma Agglutinates Red Cells of Type:
O	Anti-A, anti-B	45	A, B, AB
A	Anti-B	41	B, AB
B	Anti-A	10	A, AB
AB	None	4	None

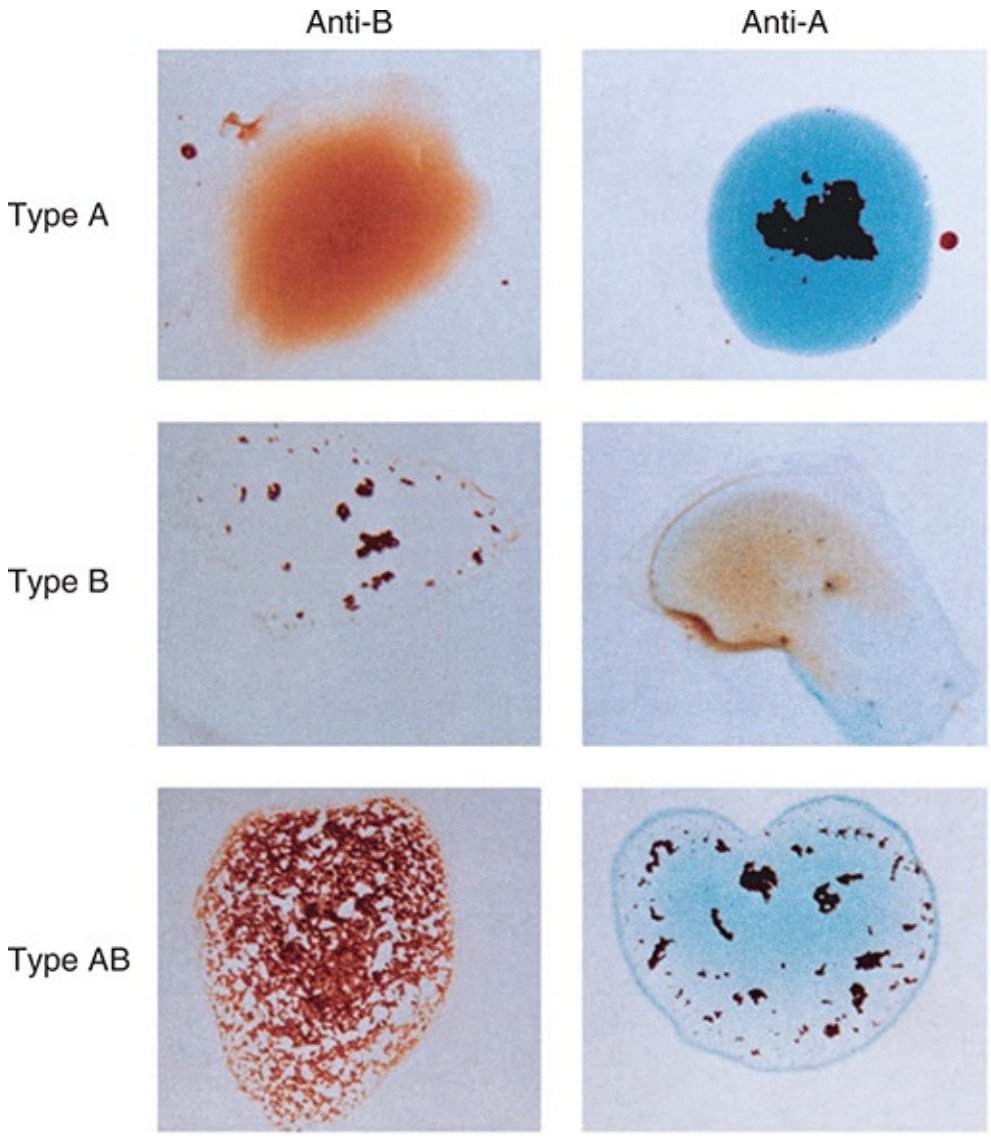


FIGURE 31–10 Red cell agglutination in incompatible plasma.

TRANSFUSION REACTIONS

Dangerous **hemolytic transfusion reactions** occur when blood is transfused into an individual with an incompatible blood type; that is, an individual who has agglutinins against the red cells in the transfusion. The plasma in the transfusion is usually so diluted in the recipient that it rarely causes agglutination even when the titer of agglutinins against the recipient's cells is high. However, when the recipient's plasma has agglutinins against the donor's red cells, the cells agglutinate and hemolyze. Free hemoglobin is liberated into the plasma. The severity of the resulting transfusion reaction may vary from an asymptomatic minor rise in the plasma bilirubin level to severe jaundice and renal tubular damage leading to anuria and death.

Incompatibilities in the ABO blood group system are summarized in Table 31–7. Persons with type AB blood are "universal recipients" because they have no circulating agglutinins and can be given blood of any type without developing a transfusion reaction due to ABO incompatibility. Type O individuals are "universal donors" because they lack A and B antigens, and type O blood can be given to anyone without producing a transfusion reaction due to ABO incompatibility. This does not mean, however, that blood should ever be transfused without being cross-matched except in the most extreme emergencies, since the possibility of reactions or sensitization due to incompatibilities in systems other than ABO systems always exists. In cross-matching, donor red cells are mixed with recipient plasma on a slide and checked for agglutination.

A procedure that has recently become popular is to withdraw the patient's own blood in advance of elective surgery and then infuse this blood back **(autologous transfusion)** if a transfusion is needed during the surgery. With iron treatment, 1000–1500 mL can be withdrawn over a 3-week period. The popularity of banking one's own blood is primarily due to fear of transmission of infectious diseases by heterologous transfusions, but of course another advantage is elimination of the risk of transfusion reactions.

OTHER AGGLUTINOGENS

In addition to the ABO system of antigens in human red cells, there are systems such as the Rh, MNSs, Lutheran, Kell, Kidd, and many others. There are over 500 billion possible known blood group phenotypes, and because undiscovered antigens undoubtedly exist, it has been calculated that the number of phenotypes is actually in the trillions.

The number of blood groups in animals is as large as it is in humans. An interesting question is why this degree of polymorphism developed and has persisted through evolution. Certain diseases are more common in individuals with one blood type or another, but the differences are not great. The significance of a recognition code of this complexity is therefore unknown.

THE RH GROUP

Aside from the antigens of the ABO system, those of the Rh system are of the greatest clinical importance. The Rh factor, named for the rhesus monkey because it was first studied using the blood of this animal, is a system composed primarily of the C, D, and E antigens, although it actually contains many more. Unlike the ABO antigens, the system has not been detected in tissues other than red cells. D is by far the most antigenic component, and the term Rh-positive as it is generally used means that the individual has agglutinogen D. The Rh-negative individual has no D antigen and forms the anti-D agglutinin when injected with D-positive cells. The Rh typing serum used in routine blood typing is anti-D serum. Eighty-five percent of whites are D-positive and 15% are D-negative; over 99% of Asians are D-positive. Unlike the antibodies of the ABO system, anti-D antibodies do not develop without exposure of a D-negative individual to D-positive red cells by transfusion or entrance of fetal blood into the maternal circulation. However, D-negative individuals who have received a transfusion of D-positive blood (even years previously) can have appreciable anti-D titers and thus may develop transfusion reactions when transfused again with D-positive blood.

HEMOLYTIC DISEASE OF THE NEWBORN

Another complication due to Rh incompatibility arises when an Rh-negative mother carries an Rh-positive fetus. Small amounts of fetal blood leak into the maternal circulation at the time of delivery, and some mothers develop significant titers of anti-Rh agglutinins during the postpartum period. During the next pregnancy, the mother’s agglutinins cross the placenta to the fetus. When anti-Rh agglutinins cross the placenta to an Rh-positive fetus, they can cause hemolysis and various forms of **hemolytic disease of the newborn (erythroblastosis fetalis).** If hemolysis in the fetus is severe, the infant may die in utero or may develop anemia, severe jaundice, and edema **(hydrops fetalis). Kernicterus,** a neurologic syndrome in which unconjugated bilirubin is

deposited in the basal ganglia, may also develop, especially if birth is complicated by a period of hypoxia. Bilirubin rarely penetrates the brain in adults, but it does in infants with erythroblastosis, possibly in part because the blood-brain barrier is more permeable in infancy. However, the main reasons that the concentration of unconjugated bilirubin is very high in this condition are that production is increased and the bilirubin-conjugating system is not yet mature.

About 50% of Rh-negative individuals are sensitized (develop an anti-Rh titer) by transfusion of Rh-positive blood. Because sensitization of Rh-negative mothers by carrying an Rh-positive fetus generally occurs at birth, the first child is usually normal. However, hemolytic disease occurs in about 17% of the Rh-positive fetuses born to Rh-negative mothers who have previously been pregnant one or more times with Rh-positive fetuses. Fortunately, it is usually possible to prevent sensitization from occurring the first time by administering a single dose of anti-Rh antibodies in the form of Rh immune globulin during the postpartum period. Such passive immunization does not harm the mother and has been demonstrated to prevent active antibody formation by the mother. In obstetric clinics, the institution of such treatment on a routine basis to unsensitized Rh-negative women who have delivered an Rh-positive baby has reduced the overall incidence of hemolytic disease by more than 90%. In addition, fetal Rh typing with material obtained by amniocentesis or chorionic villus sampling is now possible, and treatment with a small dose of Rh immune serum will prevent sensitization during pregnancy.

HEMOSTASIS

Hemostasis is the process of forming clots in the walls of damaged blood vessels and preventing blood loss while maintaining blood in a fluid state within the vascular system. A collection of complex interrelated systemic mechanisms operates to maintain a balance between coagulation and anticoagulation.

RESPONSE TO INJURY

When a small blood vessel is transected or damaged, the injury initiates a series of events **(Figure 31–11)** that lead to the formation of a clot. This seals off the damaged region and prevents further blood loss. A temporary **hemostatic plug** of platelets is triggered when platelets bind to collagen and aggregate. There is also constriction of the vessel, with constriction of an injured arteriole or small

artery occasionally so marked that its lumen is obliterated, at least temporarily. The vasoconstriction is due to serotonin (5-HT) and other vasoconstrictors liberated from the platelets adhered to the walls of the damaged vessel. This is then followed by conversion of the plug into the definitive clot.

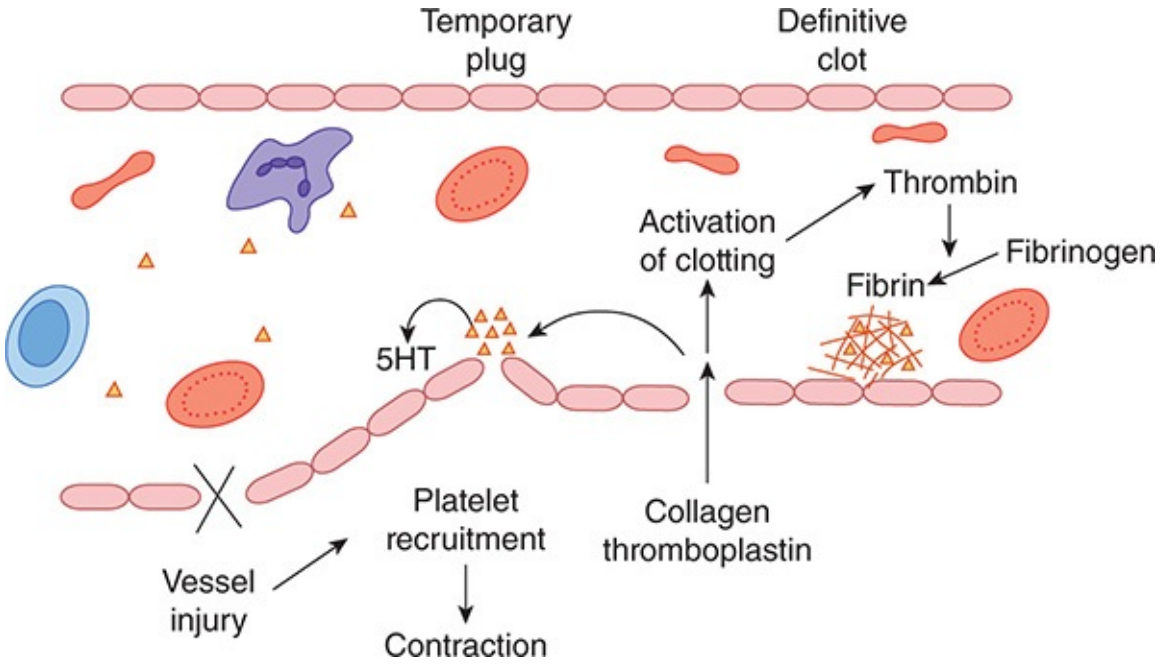


FIGURE 31–11 Summary of reactions involved in hemostasis. Injury to a blood vessel exposes collagen and thromboplastin, recruiting platelets to the site of injury to form a temporary plug. Platelets release 5-hydroxytryptamine, among other factors, resulting in smooth muscle contraction and vasoconstriction. Activation of the clotting cascade in response to collagen and thromboplastin activates thrombin, which converts circulating fibrinogen to fibrin monomers. Fibrin monomers polymerize and are cross-linked and accumulate with platelets at the site of injury to form the definitive clot.

THE CLOTTING MECHANISM

The loose aggregation of platelets in the temporary plug is bound together and converted into the definitive clot by **fibrin.** Fibrin formation involves a cascade of enzymatic reactions and a series of numbered clotting factors (Table 31–4). The fundamental reaction is conversion of the soluble plasma protein fibrinogen to insoluble fibrin **(Figure 31–12)**. The process involves the release of two pairs of polypeptides from each fibrinogen molecule. The remaining portion, **fibrin monomer,** then polymerizes with other monomer molecules to form **fibrin.** The

fibrin is initially a loose mesh of interlacing strands. It is converted by the formation of covalent cross-linkages to a dense, tight aggregate (stabilization). This latter reaction is catalyzed by activated factor XIII and requires Ca^{2+}.

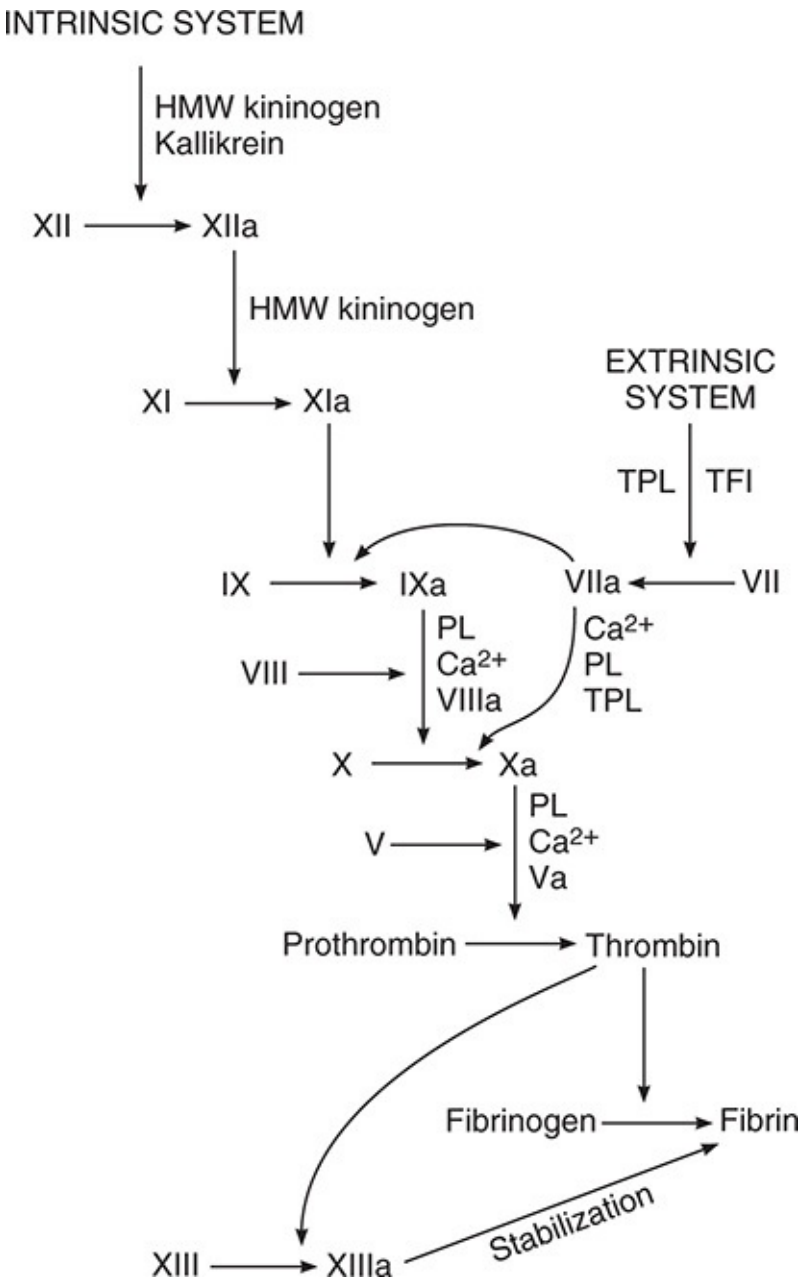


FIGURE 31–12 The clotting mechanism. a, active form of clotting factor. TFI, tissue factor pathway inhibitor; TPL, tissue thromboplastin. For other abbreviations, see Table 31–4.

The conversion of fibrinogen to fibrin is catalyzed by thrombin. Thrombin is

a serine protease that is formed from its circulating precursor, prothrombin, by the action of activated factor X. It has additional actions, including activation of platelets, endothelial cells, and leukocytes via so-called proteinase activated receptors, which are G-protein-coupled.

Factor X can be activated by either of two systems, known as intrinsic and extrinsic (Figure 31–12). The initial reaction in the **intrinsic system** is conversion of inactive factor XII to active factor XII (XIIa). This activation, which is catalyzed by high-molecular-weight kininogen and kallikrein (see Chapter 32), can be brought about in vitro by exposing the blood to glass, or in vivo by collagen fibers underlying the endothelium. Active factor XII then activates factor XI, and active factor XI activates factor IX. Activated factor IX forms a complex with active factor VIII, which is activated when it is separated from von Willebrand factor. The complex of IXa and VIIIa activate factor X. Phospholipids from aggregated platelets (PL) and Ca^{2+} are necessary for full activation of factor X. The **extrinsic system** is triggered by the release of tissue thromboplastin (TPL), a protein–phospholipid mixture that activates factor VII. Tissue thromboplastin and factor VII activate factors IX and X. In the presence of PL, Ca^{2+}, and factor V, activated factor X catalyzes the conversion of prothrombin to thrombin. The extrinsic pathway is inhibited by a **tissue factor pathway inhibitor** (TFI) that forms a quaternary structure with TPL, factor VIIa, and factor Xa.

ANTICLOTTING MECHANISMS

The tendency of blood to clot is balanced in vivo by reactions that prevent clotting inside the blood vessels, break down any clots that do form, or both. These reactions include the interaction between the platelet-aggregating effect of thromboxane A_2 and the antiaggregating effect of prostacyclin, which causes clots to form at the site when a blood vessel is injured but keeps the vessel lumen free of clot (see Chapter 32 and **Clinical Box 31–3**).

Antithrombin III is a circulating protease inhibitor that binds to serine proteases in the coagulation system, blocking their activity as clotting factors. This binding is facilitated by **heparin,** a naturally occurring anticoagulant that is a mixture of sulfated polysaccharides. The clotting factors that are inhibited are the active forms of factors IX, X, XI, and XII.

CLINICAL BOX 31–3

Abnormalities of Hemostasis

In addition to clotting abnormalities due to platelet disorders, hemorrhagic diseases can be produced by selective deficiencies of most of the clotting factors **(Table 31–8)**. Hemophilia A, which is caused by factor VIII deficiency, is relatively common. von Willebrand factor deficiency likewise causes a bleeding disorder (von Willebrand disease) by reducing platelet adhesion and by lowering plasma factor VIII. The condition can be congenital or acquired. The large von Willebrand molecule is subject to cleavage and resulting inactivation by the plasma metalloprotease ADAM-13 in vascular areas where fluid shear stress is elevated. Finally, when absorption of vitamin K is depressed along with absorption of other fat-soluble vitamins (see Chapter 26), the resulting clotting factor deficiencies may cause the development of a significant bleeding tendency.

Formation of clots inside blood vessels is called **thrombosis** to distinguish it from the normal extravascular clotting of blood. Thromboses are a major medical problem. They are particularly prone to occur where blood flow is sluggish because the slow flow permits activated clotting factors to accumulate instead of being washed away. They also occur in vessels when the intima is damaged by atherosclerotic plaques, and over areas of damage to the endocardium. They frequently occlude the arterial supply to the organs in which they form, and bits of thrombus **(emboli)** sometimes break off and travel in the bloodstream to distant sites, damaging other organs. An example is obstruction of the pulmonary artery or its branches by thrombi from the leg veins **(pulmonary embolism).** Congenital absence of protein C leads to uncontrolled intravascular coagulation and, in general, death in infancy. If this condition is diagnosed and treatment is instituted, the coagulation defect disappears. Resistance to activated protein C is another cause of thrombosis, and this condition is common. It is due to a point mutation in the gene for factor V, which prevents activated protein C from inactivating the factor.

Disseminated intravascular coagulation is another serious complication of septicemia, extensive tissue injury, and other diseases in which fibrin is deposited in the vascular system and many small- and medium-sized vessels are thrombosed. The increased consumption of platelets and coagulation factors causes bleeding to occur at the same time. The cause of the condition appears to be increased generation of thrombin due to increased TPL activity without adequate TFI activity.

THERAPEUTIC HIGHLIGHTS

Hemophilia has been treated with factor VIII–rich preparations made from plasma, or, more recently, factor VIII produced by recombinant DNA techniques. Some patients with von Willebrand disease are treated with desmopressin, which stimulates production of factor VIII, particularly prior to dental procedures or surgery. Thrombotic disorders, on the other hand, are treated with anticoagulants such as heparin.

TABLE 31–8 Examples of diseases due to deficiency of clotting factors.

Deficiency of Factor	Clinical Syndrome	Cause
I	Afibrinogenemia	Depletion during pregnancy with premature separation of placenta; also congenital (rare)
II	Hypoprothrombinemia (hemorrhagic tendency in liver disease)	Decreased hepatic synthesis, usually secondary to vitamin K deficiency
V	Parahemophilia	Congenital
VII	Hypoconvertinemia	Congenital
VIII	Hemophilia A (classic hemophilia)	Congenital defect due to various abnormalities of the gene on X chromosome that codes for factor VIII; disease is therefore inherited as a sex-linked characteristic
IX	Hemophilia B (Christmas disease)	Congenital
X	Stuart-Prower factor deficiency	Congenital
XI	PTA deficiency	Congenital
XII	Hageman trait	Congenital

The endothelium of the blood vessels also plays an active role in preventing the extension of clots. All endothelial cells except those in the cerebral microcirculation produce **thrombomodulin,** a thrombin-binding protein, on their surfaces. In circulating blood, thrombin is a procoagulant that activates factors V and VIII, but when it binds to thrombomodulin, it becomes an anticoagulant in that the thrombomodulin-thrombin complex activates protein C **(Figure 33–13)**. Activated protein C (APC), along with its cofactor protein S, inactivates factors V and VIII and inactivates an inhibitor of tissue plasminogen activator,

increasing the formation of plasmin.

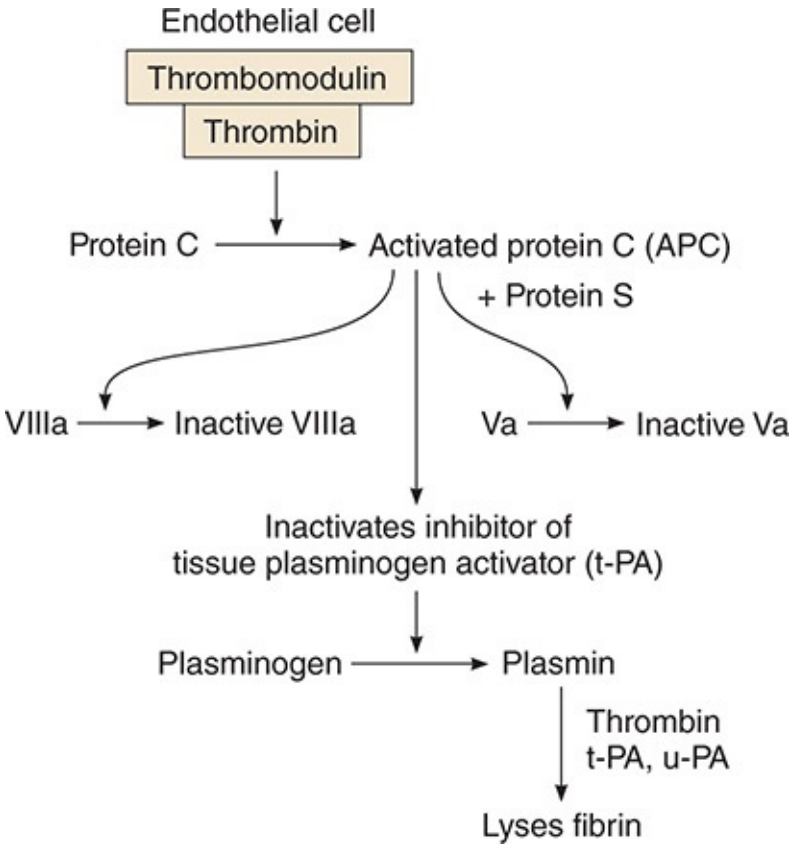


FIGURE 31–13 The fibrinolytic system and its regulation by protein C. u-PA, urokinase-type plasminogen activator.

Plasmin (fibrinolysin) is the active component of the **plasminogen (fibrinolytic) system** (Figure 31–13). This enzyme lyses fibrin and fibrinogen, with the production of fibrinogen degradation products (FDPs) that inhibit thrombin. Plasmin is formed from its inactive precursor, plasminogen, by the action of thrombin and **tissue-type plasminogen activator (t-PA).** Plasminogen is converted to active plasmin when t-PA hydrolyzes the bond between Arg 560 and Val 561. It is also activated by **urokinase-type plasminogen activator (u-PA).** If the t-PA gene or the u-PA gene is knocked out in mice, some fibrin deposition occurs and clot lysis is slowed. However, when both are knocked out, spontaneous fibrin deposition is extensive.

Plasminogen receptors are located on the surfaces of many different types of cells and are plentiful on endothelial cells. When plasminogen binds to its receptors, it becomes activated, so intact blood vessel walls are provided with a mechanism that discourages clot formation.

Human t-PA is produced by recombinant DNA techniques for clinical use in myocardial infarction and stroke. Streptokinase, a bacterial enzyme, is also fibrinolytic and is also used in the treatment of early myocardial infarction (see Chapter 33).

ANTICOAGULANTS

As noted above, heparin is a naturally occurring anticoagulant that facilitates the action of antithrombin III. Low-molecular-weight fragments have been produced from unfractionated heparin, and these are seeing increased clinical use because they have a longer half-life and produce a more predictable anticoagulant response than unfractionated heparin. The highly basic protein protamine forms an irreversible complex with heparin and is used clinically to neutralize heparin.

In vivo, a plasma Ca^{2+} level low enough to interfere with blood clotting is incompatible with life, but clotting can be prevented in vitro if Ca^{2+} is removed from the blood by the addition of substances such as oxalates, which form insoluble salts with Ca^{2+}, or **chelating agents,** which bind Ca^{2+}. Coumarin derivatives such as **dicumarol** and **warfarin** are also effective anticoagulants. They inhibit the action of vitamin K, which is a necessary cofactor for the enzyme that catalyzes the conversion of glutamic acid residues to γ-carboxyglutamic acid residues. Factors II (prothrombin), VII, IX, and X, protein C, and protein S (see above) require conversion of a number of glutamic acid residues to γ-carboxyglutamic acid residues before being released into the circulation, and hence all six are vitamin K–dependent.

STRUCTURAL FEATURES OF THE CIRCULATION

Here, we will discuss the two major cell types that make up the blood vessels and how they are arranged into the various vessel types that subserve the needs of the circulation.

ENDOTHELIUM

Located between the circulating blood and the media and adventitia of the blood vessels, the endothelial cells constitute a large and important organ. They

respond to flow changes, stretch, a variety of circulating substances, and inflammatory mediators. They secrete growth regulators and vasoactive substances (see below and Chapter 32).

VASCULAR SMOOTH MUSCLE

The smooth muscle in blood vessel walls has been one of the most-studied forms of visceral smooth muscle because of its importance in the regulation of blood pressure and hypertension. The membranes of the muscle cells contain various types of K^+, Ca^{2+}, and Cl^- channels. Contraction is produced primarily by the myosin light chain mechanism described in Chapter 5. However, vascular smooth muscle also undergoes prolonged contractions that determine vascular tone. These may be due in part to the latch-bridge mechanism (see Chapter 5), but other factors also play a role. Some of the molecular mechanisms that appear to be involved in contraction and relaxation are shown in **Figure 31–14**.

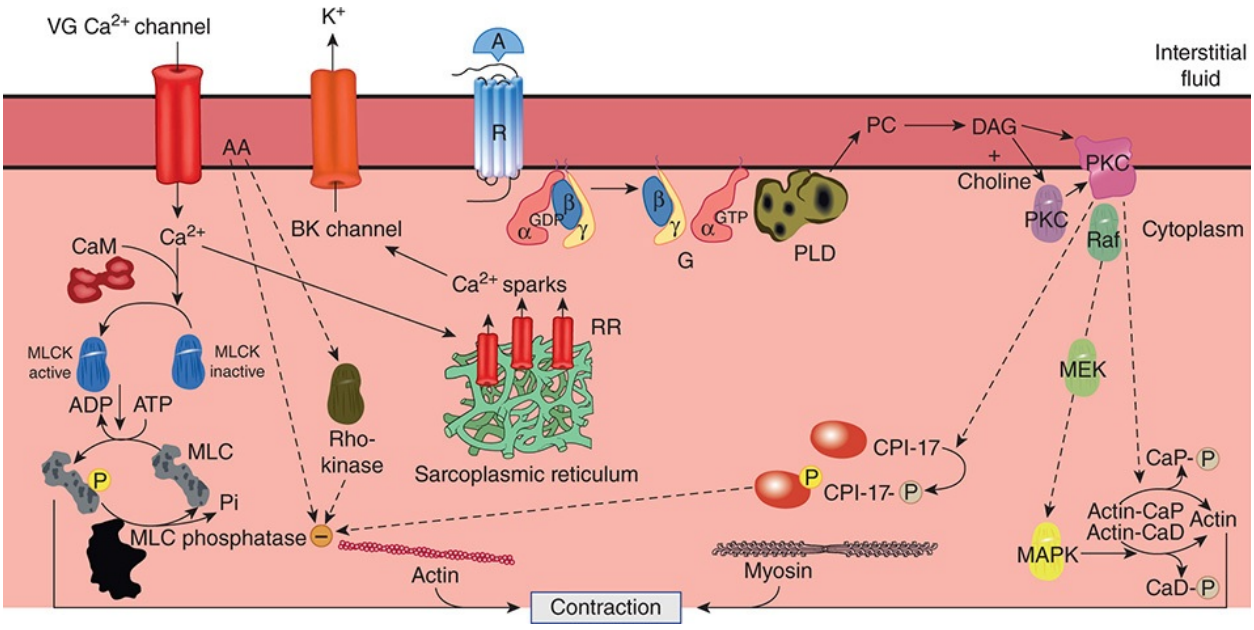


FIGURE 31–14 Some of the established and postulated mechanisms involved in the contraction and relaxation of vascular smooth muscle. A, agonist; AA, arachidonic acid; BK, Ca^+-activated K^+ channel; G, heterotrimeric G-protein; MLC, myosin light chain; MLCK, myosin light chain kinase; PLD, phospholipase D; R, receptor; RR, ryanodine receptors; SR, sarcoplasmic reticulum; VGCC, voltage-gated Ca^{2+} channel. For other abbreviations, see Chapter 2.

Vascular smooth muscle cells provide an interesting example of the way high and low cytosolic Ca^{2+} can have different and even opposite effects (see Chapter 2). In these cells, influx of Ca^{2+} via voltage-gated Ca^{2+} channels produces a diffuse increase in cytosolic Ca^{2+} that initiates contraction. However, the Ca^{2+} influx also initiates Ca^{2+} release from the sarcoplasmic reticulum via ryanodine receptors (see Chapter 5), and the high local Ca^{2+} concentration produced by these Ca^{2+} sparks increases the activity of **Ca^{2+}-activated K^+ channels** in the cell membrane. These are also known as big K or **BK channels** because K^+ flows through them at a high rate. The resulting K^+ efflux increases the membrane potential, shutting off voltage-gated Ca^{2+} channels and producing relaxation. The site of action of the Ca^{2+} sparks is the β_1-subunit of the BK channel, and mice in which this subunit is knocked out develop increased vascular tone and blood pressure.

ARTERIES & ARTERIOLES

The characteristics of the various types of blood vessels are listed in **Table 31–9**. The walls of all arteries are made up of an outer layer of connective tissue, the adventitia; a middle layer of smooth muscle, the media; and an inner layer, the intima, made up of the endothelium and underlying connective tissue **(Figure 31–15)**. The walls of the aorta and other large diameter arteries contain a relatively large amount of elastic tissue, primarily located in the inner and external elastic laminas. They are stretched during systole and recoil on the blood during diastole. The walls of the arterioles contain less elastic tissue but much more smooth muscle. The muscle is innervated by noradrenergic nerve fibers, which function as constrictors, and in some instances by cholinergic fibers, which dilate the vessels. The arterioles are the major site of the resistance to blood flow, and small changes in their caliber cause large changes in the total peripheral resistance.

TABLE 31–9 Characteristics of various types of blood vessels in humans.

Vessel	Lumen Diameter	Wall Thickness	All Vessels of Each Type	
			Approximate Total Cross-Sectional Area (cm^2)	Percentage of Blood Volume Contained[a]
Aorta	2.5 cm	2 mm	4.5	2
Artery	0.4 cm	1 mm	20	8
Arteriole	30 µm	20 µm	400	1
Capillary	5 µm	1 µm	4500	5
Venule	20 µm	2 µm	4000	
Vein	0.5 cm	0.5 mm	40	54
Vena cava	3 cm	1.5 mm	18	

[a]In systemic vessels; there is an additional 12% in the heart and 18% in the pulmonary circulation.

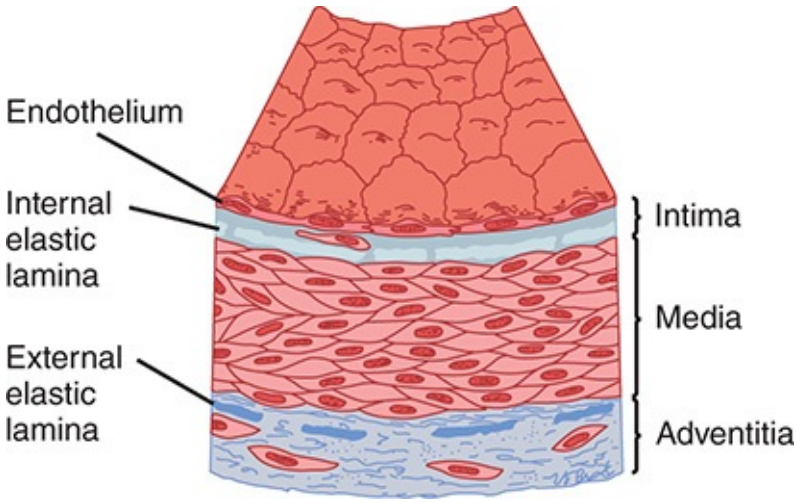


FIGURE 31–15 Structure of a normal muscle artery. (Reproduced with permission from Ross R, Glomset JA: The pathogenesis of atherosclerosis. N Engl J Med 1976; August 12; 295(7):369–377.)

CAPILLARIES

The arterioles divide into smaller muscle-walled vessels called **metarterioles,** and these in turn feed into capillaries **(Figure 31–16)**. The openings of the capillaries are surrounded on the upstream side by minute smooth muscle **precapillary sphincters.** It is unsettled whether the metarterioles are innervated, and it appears that the precapillary sphincters are not. However, they can of course respond to local or circulating vasoconstrictor substances. The capillaries are about 5 µm in diameter at the arterial end and 9 µm in diameter at the venous end. When the sphincters are dilated, the diameter of the capillaries is just sufficient to permit red blood cells to squeeze through in “single file.” As they

pass through the capillaries, the red cells become thimble- or parachute-shaped, with the flow pushing the center ahead of the edges. This configuration appears to be simply due to the pressure in the center of the vessel, whether or not the edges of the red blood cell are in contact with the capillary walls.

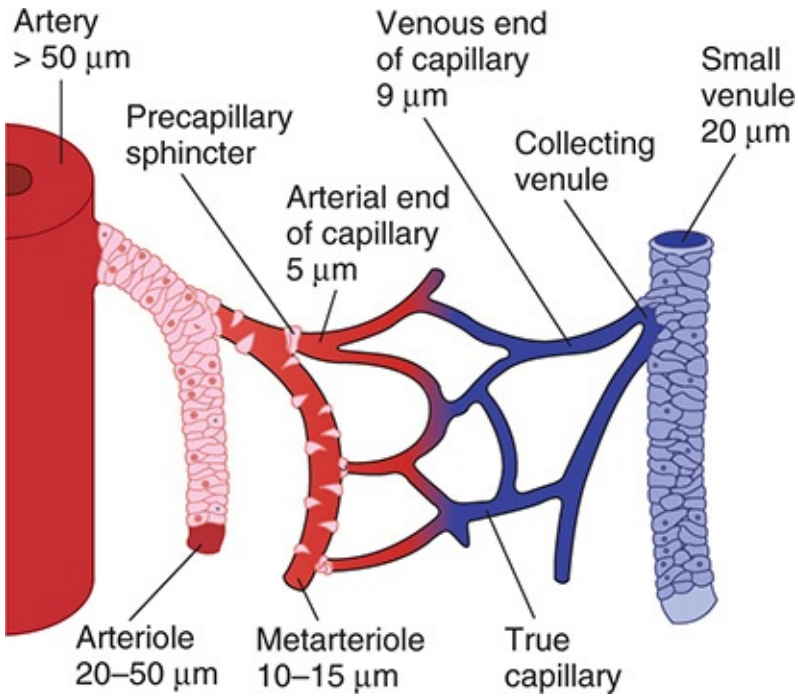


FIGURE 31–16 The microcirculation. Arterioles give rise to metarterioles, which give rise to capillaries. The capillaries drain via short collecting venules to the venules. The walls of the arteries, arterioles, and small venules contain relatively large amounts of smooth muscle. There are scattered smooth muscle cells in the walls of the metarterioles, and the openings of the capillaries are guarded by muscular precapillary sphincters. The diameters of the various vessels are also shown. (Used with permission of JN Diana.)

The total area of all the capillary walls in the body exceeds 6300 m^2 in the adult. The walls, which are about 1 µm thick, are made up of a single layer of endothelial cells. The structure of the walls varies from organ to organ. In many beds, including those in skeletal, cardiac, and smooth muscle, the junctions between the endothelial cells **(Figure 31–17)** permit the passage of molecules up to 10 nm in diameter. It also appears that plasma and its dissolved proteins can be taken up by endocytosis, transported across the endothelial cells, and discharged by exocytosis (**vesicular transport;** see Chapter 2). However, this process can account for only a small portion of the transport across the

endothelium. In the brain, the capillaries resemble the capillaries in muscle, but the junctions between endothelial cells are tighter, and transport across them is largely limited to small molecules (although pathological conditions may open these junctions). In most endocrine glands, the intestinal villi, and parts of the kidneys, on the other hand, the cytoplasm of the endothelial cells is attenuated to form gaps called **fenestrations.** These fenestrations are 20–100 nm in diameter and may be opened to permit the passage of larger molecules, although their permeability is likely significantly reduced under normal circumstances by a thick layer of endothelial glycocalyx. An exception to this, however, is found in the liver, where the sinusoidal capillaries are extremely porous, the endothelium is discontinuous, and gaps occur between endothelial cells that are not closed by membranes (see Figure 28–2). Some of the gaps are 600 nm in diameter, and others may be as large as 3000 nm. They therefore permit the passage of large molecules, including plasma proteins, which is important for hepatic function (see Chapter 28). The permeability of capillaries in various parts of the body, expressed in terms of their hydraulic conductivity, are summarized in **Table 31–10**.

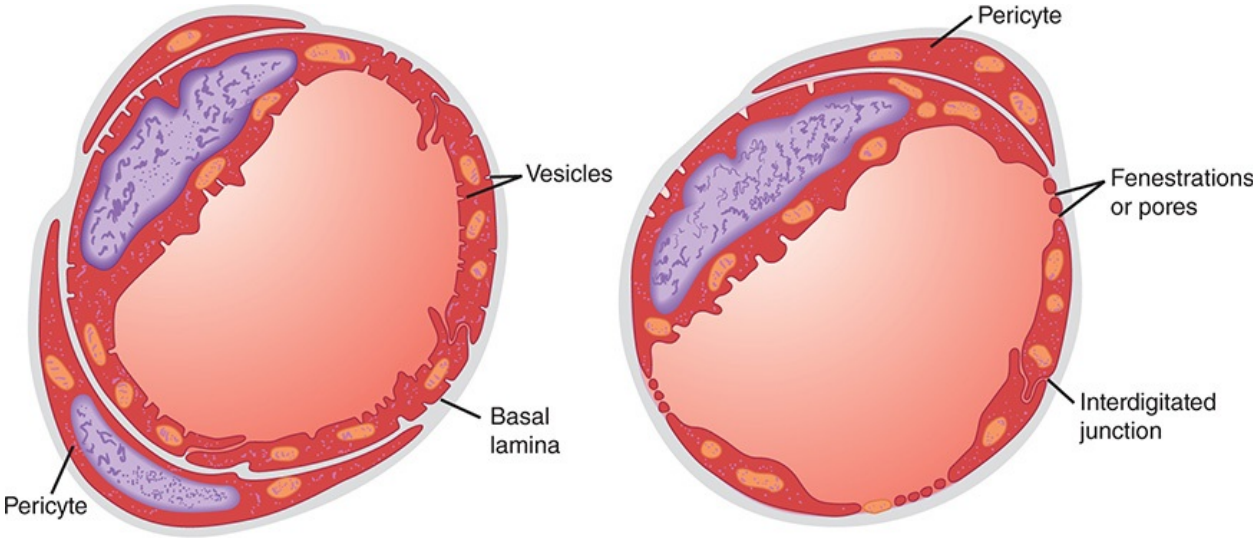


FIGURE 31–17 Cross-sections of capillaries. Left: Type of capillary found in muscle. **Right:** Fenestrated type of capillary. (Reproduced with permission from Orbison JL and D Smith, editors: *Peripheral Blood Vessels*. Baltimore: Williams & Wilkins, 1962.)

TABLE 31–10 Hydraulic conductivity of capillaries in various parts of the body.

Organ	Conductivity[a]	Type of Endothelium
Brain (excluding circumventricular organs)	3	Continuous
Skin	100	
Skeletal muscle	250	
Lung	340	
Heart	860	
Gastrointestinal tract (intestinal mucosa)	13,000	Fenestrated
Glomerulus in kidney	15,000	

[a]Units of conductivity are 10^{-13} cm^3 s^{-1} $dyne^{-1}$.

Data from JN Diana.

Capillaries and postcapillary venules have **pericytes** around their endothelial cells (Figure 31–17). These cells have long processes that wrap around the vessels. They are contractile and release a wide variety of vasoactive agents. They also synthesize and release constituents of the basement membrane and extracellular matrix. One of their physiologic functions appears to be regulation of flow through the junctions between endothelial cells, particularly in the presence of inflammation. They are closely related to the mesangial cells in the renal glomeruli (see Chapter 37).

LYMPHATICS

The lymphatics serve to collect plasma and its constituents that have exuded from the capillaries into the interstitial space (ie, the lymph). They drain from the body tissues via a system of vessels that coalesce and eventually enter the right and left subclavian veins at their junctions with the respective internal jugular veins. The lymph vessels contain valves and regularly traverse lymph nodes along their course. The ultrastructure of the small lymph vessels differs from that of the capillaries in several details: No fenestrations are visible in the lymphatic endothelium; very little if any basal lamina is present under the endothelium; and the junctions between endothelial cells are open, with no tight intercellular connections.

ARTERIOVENOUS ANASTOMOSES

In the fingers, palms, and ear lobes, short channels connect arterioles to venules, bypassing the capillaries. These **arteriovenous (A-V) anastomoses,** or **shunts,** have thick, muscular walls and are abundantly innervated, presumably by vasoconstrictor nerve fibers.

VENULES & VEINS

The walls of the venules are only slightly thicker than those of the capillaries. The walls of the veins are also thin and easily distended. They contain relatively little smooth muscle, but considerable venoconstriction is produced by activity in the noradrenergic nerves to the veins and by circulating vasoconstrictors such as endothelins. Variations in venous tone are important in circulatory adjustments.

The intima of the limb veins is folded at intervals to form **venous valves** that prevent retrograde flow. However, no valves are present in very small veins, the great veins (superior and inferior vena cava plus the pulmonary veins), or the veins from the brain and viscera.

ANGIOGENESIS

When tissues grow, blood vessels must proliferate if the tissue is to maintain a normal blood supply. Therefore, angiogenesis, the formation of new blood vessels, is important during fetal life and growth to adulthood. It is also important in adulthood for processes such as wound healing, formation of the corpus luteum after ovulation, and formation of new endometrium after menstruation. Abnormally, it is important in tumor growth since tumors that do not develop a blood supply cannot grow.

During embryonic development, a network of leaky capillaries is formed in tissues from angioblasts: this process is sometimes called **vasculogenesis.** Branches from nearby vessels then hook up with the capillaries and provide them with smooth muscle, which brings about their maturation. Angiogenesis in adults is presumably similar, but consists of new vessel formation by branching from preexisting vessels rather than from angioblasts.

Many factors are involved in angiogenesis. A key compound is **vascular endothelial growth factor (VEGF).** VEGF is primarily responsible for

vasculogenesis, whereas the budding of vessels that connect to the immature capillary network is regulated by other as yet unidentified factors. Some VEGF isoforms and receptors may play a more prominent role in the formation of lymphatic vessels **(lymphangiogenesis)** than that of blood vessels.

The actions of VEGF and related factors have received considerable attention in recent years because of the requirement for angiogenesis in the development of tumors. VEGF antagonists and other angiogenesis inhibitors have entered clinical practice as adjunctive therapies for malignancies and are being tested as first-line therapies as well.

BIOPHYSICAL CONSIDERATIONS FOR CIRCULATORY PHYSIOLOGY

FLOW, PRESSURE, & RESISTANCE

Blood always flows, of course, from areas of high pressure to areas of low pressure, except in certain situations when momentum transiently sustains flow (see Figure 30–3). The relationship between mean flow, mean pressure, and resistance in the blood vessels is analogous in a general way to the relationship between the current, electromotive force (voltage), and resistance in an electrical circuit as expressed in Ohm law:

$$\text{Current (I)} = \frac{\text{Electromotive force (E)}}{\text{Resistance (R)}}$$

$$\text{Flow (F)} = \frac{\text{Pressure (P)}}{\text{Resistance (R)}}$$

Flow in any portion of the vascular system is therefore equal to the **effective perfusion pressure** in that portion divided by the **resistance.** The effective perfusion pressure is the mean intraluminal pressure at the arterial end minus the mean pressure at the venous end. The units of resistance (pressure divided by flow) are dyne·s/cm^5. To avoid dealing with such complex units, resistance in the cardiovascular system is sometimes expressed in **R units,** which are obtained by dividing pressure in mm Hg by flow in mL/s (see also Table 33–1). Thus, for example, when the mean aortic pressure is 90 mm Hg and the left ventricular output is 90 mL/s, the total peripheral resistance is

$$\frac{90 \text{ mm Hg}}{90 \text{ mL/s}} = 1 \text{ R unit}$$

METHODS FOR MEASURING BLOOD FLOW

Various noninvasive devices have been developed to measure blood flow. Most commonly, blood velocity can be measured with **Doppler flow meters.** Ultrasonic waves are sent into a vessel diagonally, and the waves reflected from the red and white blood cells are picked up by a downstream sensor. The frequency of the reflected waves is higher by an amount that is proportional to the rate of flow toward the sensor because of the Doppler effect.

Indirect methods for measuring the blood flow in various organs in humans include adaptations of the Fick and indicator dilution techniques described in Chapter 30. One example is the use of the Kety N_2O method for measuring cerebral blood flow (see Chapter 33). Another is determination of the renal blood flow by measuring the clearance of para-aminohippuric acid (see Chapter 37). A considerable amount of data on blood flow in the extremities have been obtained by **plethysmography.** The forearm, for example, is sealed in a watertight chamber **(plethysmograph).** Changes in the volume of the forearm, reflecting changes in the amount of blood and interstitial fluid it contains, displace the water, and this displacement is measured with a volume recorder. When the venous drainage of the forearm is occluded, the rate of increase in the volume of the forearm is a function of the arterial blood flow **(venous occlusion plethysmography).**

APPLICABILITY OF PHYSICAL PRINCIPLES TO FLOW IN BLOOD VESSELS

Physical principles and equations that describe the behavior of perfect fluids in rigid tubes have often been used indiscriminately to explain the behavior of blood in blood vessels, whereas blood vessels are not rigid tubes, and the blood is not a perfect fluid but a two-phase system of liquid and cells. Therefore, the behavior of the circulation deviates, sometimes markedly, from that predicted by physical principles. However, these principles are of value when used as an aid to understanding what goes on in the body.

LAMINAR FLOW

The flow of blood in straight blood vessels, like the flow of liquids in narrow rigid tubes, is normally **laminar.** Within the blood vessels, an infinitely thin layer of blood in contact with the wall of the vessel does not move. The next layer within the vessel has a low velocity, the next a higher velocity, and so forth, velocity being greatest in the center of the stream **(Figure 31–18)**. Laminar flow occurs at velocities up to a certain **critical velocity.** At or above this velocity, flow is turbulent. The probability of turbulence is also related to the diameter of the vessel and the viscosity of the blood. This probability can be expressed by the ratio of inertial to viscous forces as follows:

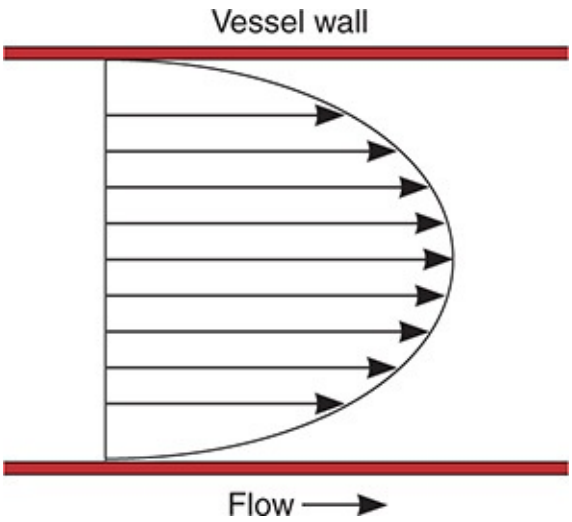


FIGURE 31–18 Diagram of the velocities of concentric laminas of a viscous fluid flowing in a tube, illustrating the parabolic distribution of velocities.

$$Re = \frac{\rho D \dot{V}}{\eta}$$

where Re is the Reynolds number; ρ is the density of the fluid; D is the diameter of the tube; V. is the velocity of the flow; and η is the viscosity of the fluid. The higher the value of Re, the greater the probability of turbulence. When D is in cm, V. is in cm/s^{-1}, and η is in poise; flow is usually not turbulent if Re is less than 2000. When Re is more than 3000, turbulence is almost always present. Laminar flow can be disturbed at the branching points of arteries, and the resulting turbulence may increase the likelihood that atherosclerotic plaques will be deposited. Constriction of an artery likewise increases the velocity of blood flow through the constriction, producing turbulence and sound beyond the

constriction **(Figure 31–19)**. Examples are bruits heard over arteries constricted by atherosclerotic plaques and the sounds of Korotkoff heard when measuring blood pressure (see below). In healthy humans, the critical velocity is sometimes exceeded in the ascending aorta at the peak of systolic ejection, but it is usually exceeded only when an artery is constricted.

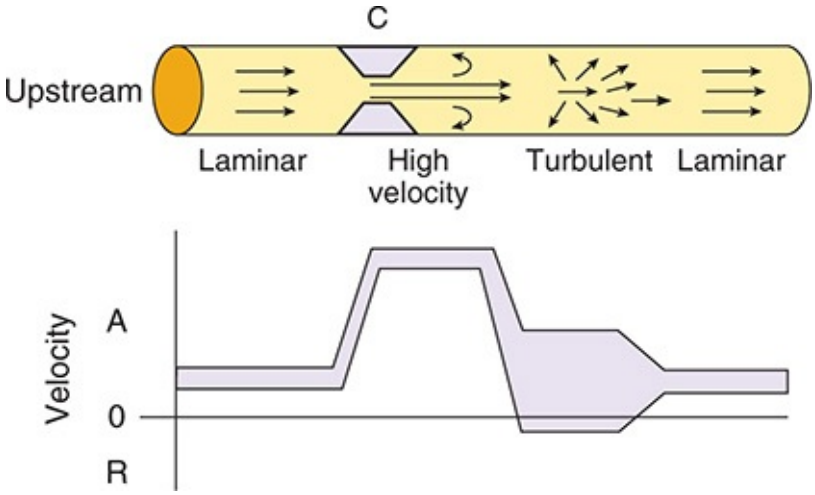


FIGURE 31–19 **Top:** Effect of constriction (C) on the profile of velocities in a blood vessel. The arrows indicate direction of velocity components, and their length is proportional to their magnitude. **Bottom:** Range of velocities at each point along the vessel. In the area of turbulence, there are many different anterograde (A) and some retrograde (R) velocities. (Modified with permission from Richards KE: Doppler echocardiographic diagnosis and quantification of vascular heart disease. Curr Probl Cardiol 1985 February; 10(2):1–49.)

SHEAR STRESS & GENE ACTIVATION

Flowing blood creates a force on the endothelium that is parallel to the long axis of the vessel. This **shear stress** (γ) is proportional to viscosity (η) times the shear rate (dy/dr), which is the rate at which the axial velocity increases from the vessel wall toward the lumen.

$$\gamma = \eta \, (dy/dr)$$

Change in shear stress and other physical variables, such as cyclic strain and stretch, produce marked changes in the expression of genes by endothelial cells. The genes that are activated include those that produce growth factors, integrins, and related molecules **(Figure 31–20)**. Recent evidence has implicated

endothelial surface structures known as **primary cilia** as the sensors of shear stress, which in turn activate signaling pathways to alter cell function. However, other components may also be involved, such as ion channels, adhesion molecules, and the cytoskeleton.

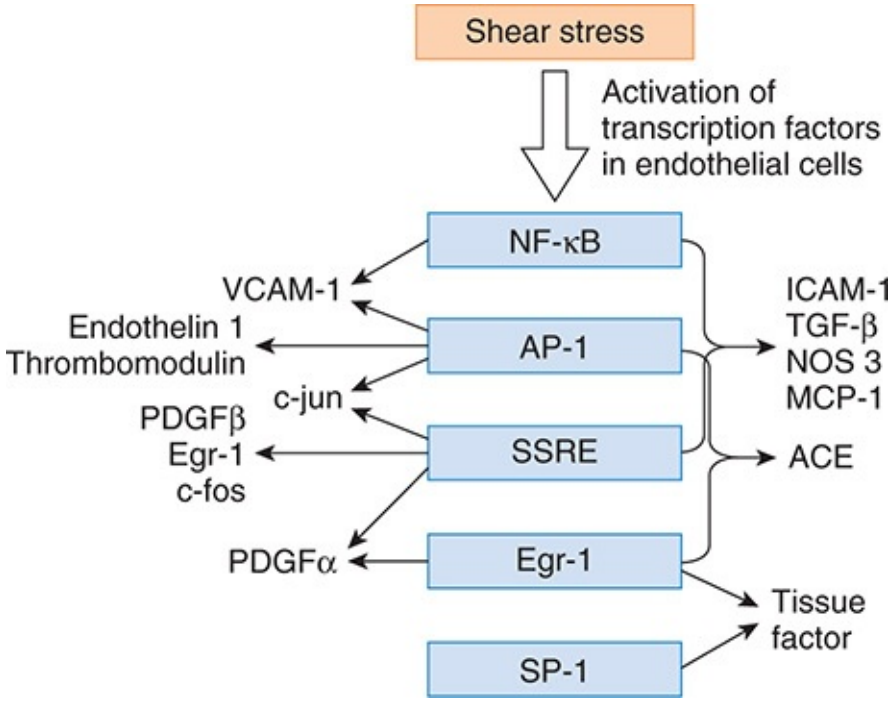


FIGURE 31–20 Endothelial transcription factors and target genes activated by shear stress. ACE, angiotensin-converting enzyme; AP-1, activator protein-1; Egr-1, early growth response protein-1; ICAM-1; intercellular adhesion molecule-1; MCP-1, monocyte chemoattractant protein-1; NF-κB, nuclear factor-κB; NOS 3, nitric oxide synthase 3; PDGF, platelet-derived growth factor; SP1, specificity protein 1; SSRE, shear stress responsive element; TGF-β, transforming growth factor-β; VCAM-1, vascular cell adhesion molecule-1. (Data from Braddock M, et al: Fluid shear stress modulation of gene expression in endothelial cells. News Physiol Sci 1998;13:241.)

AVERAGE VELOCITY

When considering flow in a system of tubes, it is important to distinguish between velocity, which is displacement per unit time (eg, cm/s), and flow, which is volume per unit time (eg, cm^3/s). Velocity ($\dot{V}$) is proportional to flow (Q) divided by the area of the conduit (A):

$$\dot{V} = \frac{Q}{A}$$

Therefore, $Q = A \times \dot{V}$, and if flow stays constant, velocity increases in direct proportion to any decrease in A (Figure 31–19).

The average velocity of fluid movement at any point in a system of tubes in parallel is inversely proportional to the *total* cross-sectional area at that point. Therefore, the average velocity of the blood is high in the aorta, declines steadily in the smaller vessels, and is lowest in the capillaries, which have 1000 times the *total* cross-sectional area of the aorta (Table 31–9). The average velocity of blood flow increases again as the blood enters the veins and is relatively high in the vena cava, although not so high as in the aorta. Clinically, the velocity of the circulation can be measured by injecting a bile salt preparation into an arm vein and timing the first appearance of the bitter taste it produces **(Figure 31–21)**. The average normal arm-to-tongue **circulation time** is 15 s.

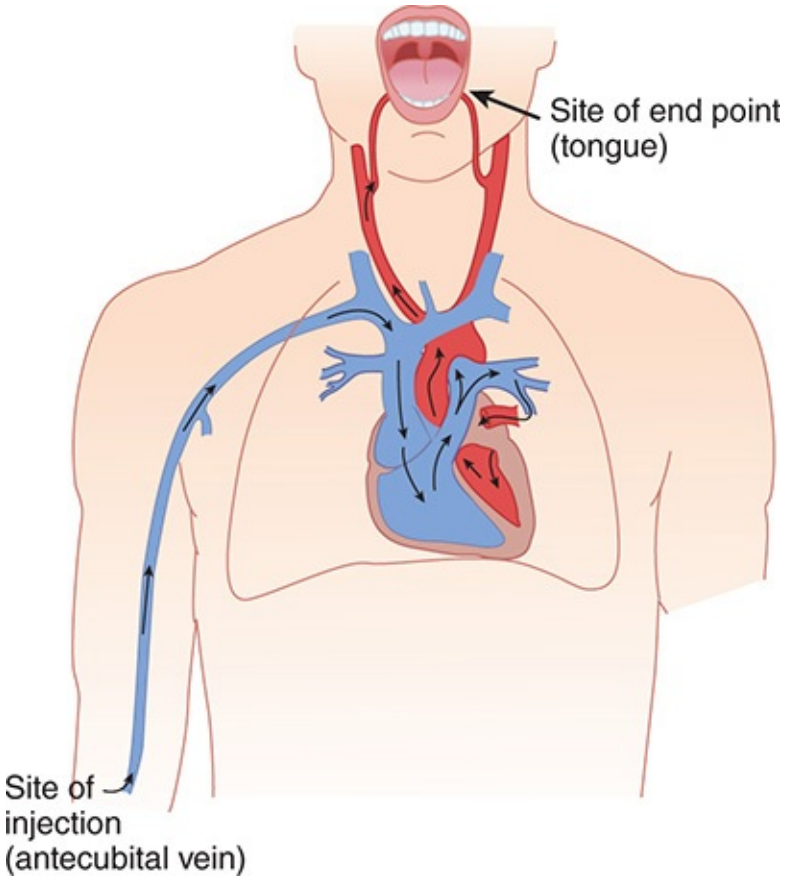


FIGURE 31–21 Pathway traversed by the injected material when the arm-to-tongue circulation time is measured.

POISEUILLE-HAGEN FORMULA

The relationship between the flow in a long narrow tube, the viscosity of the fluid, and the radius of the tube is expressed mathematically in the **Poiseuille-Hagen formula:**

$$F = (P_A - P_B) \times \left(\frac{\pi}{8}\right) \times \left(\frac{1}{\eta}\right) \times \left(\frac{r^4}{L}\right)$$

where

F = flow

$P_A - P_B$ = pressure difference between two ends of the tube

η = viscosity

r = radius of tube

L = length of tube

Because flow is equal to pressure difference divided by resistance (R),

$$R = \frac{8\eta L}{\pi r^4}$$

Because flow varies directly and resistance inversely with the fourth power of the radius, blood flow and resistance in vivo are markedly affected by small changes in the caliber of the vessels. Thus, for example, flow through a vessel is doubled by an increase of only 19% in its radius; and when the radius is doubled, resistance is reduced to 6% of its previous value. This is why organ blood flow is so effectively regulated by small changes in the caliber of the arterioles and why variations in arteriolar diameter have such a pronounced effect on systemic arterial pressure.

VISCOSITY & RESISTANCE

The resistance to blood flow is determined not only by the radius of the blood vessels but also by the viscosity of the blood. Plasma is about 1.8 times as viscous as water, whereas whole blood is 3–4 times as viscous as water. Thus, viscosity depends for the most part on the **hematocrit.** The effect of viscosity in vivo deviates from that predicted by the Poiseuille-Hagen formula. In large vessels, increases in hematocrit cause appreciable increases in viscosity. However, in vessels smaller than 100 μm in diameter—that is, in arterioles,

capillaries, and venules—the viscosity change per unit change in hematocrit is much less than it is in large-bore vessels. This is due to a difference in the nature of flow through the small vessels, known as the Fahraeus-Lindqvist effect. In small vessels, erythrocytes move to the center of the vessel, leaving cell-free plasma at the vessel wall. Therefore, the net change in viscosity per unit change in hematocrit is considerably smaller in the body than if it is measured in vitro **(Figure 31–22)**. This is why hematocrit changes have relatively little effect on the peripheral resistance except when the changes are large. In severe polycythemia (excessive production of red blood cells), the increase in resistance does increase the work of the heart. Conversely, in marked anemia, peripheral resistance is decreased, in part because of the decline in viscosity. Of course, the decrease in hemoglobin decreases the O_2-carrying ability of the blood, but the improved blood flow due to the decrease in viscosity partially compensates for this.

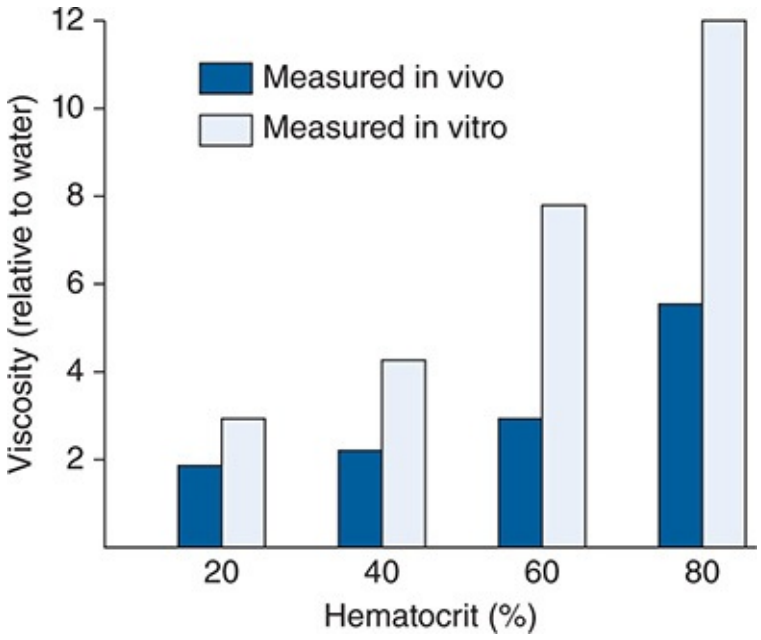


FIGURE 31–22 Effect of changes in hematocrit on the relative viscosity of blood measured in vivo and in vitro. (Data simplified and replotted from Whittaker SRF, Winton FR: The apparent viscosity of blood flowing in the isolated hind limb of the dog, and its variation with corpuscular concentration. J Physiol [Lond] 1933; 78:338.)

Viscosity is also affected by the composition of the plasma and the resistance of the cells to deformation. Clinically significant increases in viscosity are seen in diseases in which plasma proteins such as immunoglobulins are markedly

elevated as well as when red blood cells are abnormally rigid (hereditary spherocytosis).

CRITICAL CLOSING PRESSURE

In rigid tubes, the relationship between pressure and flow of homogeneous fluids is linear, but in thin-walled blood vessels in vivo it is not. When the pressure in a small blood vessel is reduced, a point is reached at which no blood flows, even though the pressure is not zero **(Figure 31–23)**. This is because the vessels are surrounded by tissues that exert a small but definite pressure on them, and when the intraluminal pressure falls below the tissue pressure, they collapse. In inactive tissues, for example, the pressure in many capillaries is low because the precapillary sphincters and metarterioles are constricted, and many of these capillaries are collapsed. The pressure at which flow ceases is called the **critical closing pressure.**

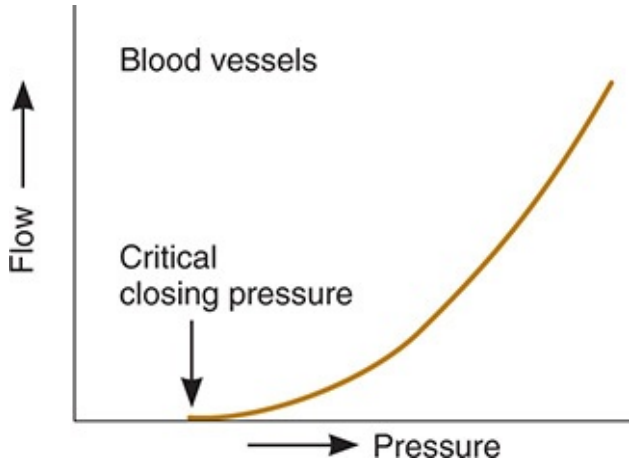


FIGURE 31–23 Relation of pressure to flow in thin-walled blood vessels.

LAW OF LAPLACE

It is perhaps surprising that structures as thin-walled and delicate as the capillaries are not more prone to rupture. The principal reason for their relative invulnerability is their small diameter. The protective effect of small size in this case is an example of the operation of the **law of Laplace,** an important physical principle with several other applications in physiology. This law states that tension in the wall of a cylinder (T) is equal to the product of the transmural pressure (P) and the radius (r) divided by the wall thickness (w):

$$T = Pr/w$$

The **transmural pressure** is the pressure inside the cylinder minus the pressure outside the cylinder, but because tissue pressure in the body is low, it can generally be ignored and P equated to the pressure inside the blood vessel. In a thin-walled vessel, w is very small and it too can be ignored, but it becomes a significant factor in vessels such as arteries. Therefore, in a thin-walled vessel, P = T divided by the two principal radii of curvature of the structure:

$$P = T\left(\frac{1}{r_1} + \frac{1}{r_2}\right)$$

In a cylinder such as a blood vessel, one radius is infinite, so

$$P = \frac{T}{r}$$

Consequently, the smaller the radius of a blood vessel, the lower the tension in the wall necessary to balance the distending pressure. In the human aorta, for example, the tension at normal pressures is about 170,000 dynes/cm, and in the vena cava it is about 21,000 dynes/cm; but in the capillaries, it is approximately 16 dynes/cm.

The law of Laplace also makes clear a disadvantage faced by dilated hearts. When the radius of a cardiac chamber is increased, a greater tension must be developed in the myocardium to produce any given pressure; consequently, a dilated heart must do more work than a nondilated heart.

RESISTANCE & CAPACITANCE VESSELS

In vivo, the veins are an important blood reservoir. Normally, they are partially collapsed and oval in cross-section. A large amount of blood can be added to the venous system before the veins become distended to the point where further increments in volume produce a large rise in venous pressure. The veins are therefore called **capacitance vessels.** The small arteries and arterioles are referred to as **resistance vessels** because they are the principal site of the peripheral resistance (see below).

At rest, at least 50% of the circulating blood volume is in the systemic veins, 12% is in the heart cavities, and 18% is in the low-pressure pulmonary circulation. Only 2% is in the aorta, 8% in the arteries, 1% in the arterioles, and

5% in the capillaries (Table 31–9). When extra blood is administered by transfusion, less than 1% of it is distributed in the arterial system (the **"high-pressure system"**), and all the rest is found in the systemic veins, pulmonary circulation, and heart chambers other than the left ventricle (the **"low-pressure system").**

ARTERIAL & ARTERIOLAR CIRCULATION

The pressure and velocities of the blood in the various parts of the systemic circulation are summarized in **Figure 31–24**. The general relationships in the pulmonary circulation are similar, but the pressure in the pulmonary artery is 25/10 mm Hg or less.

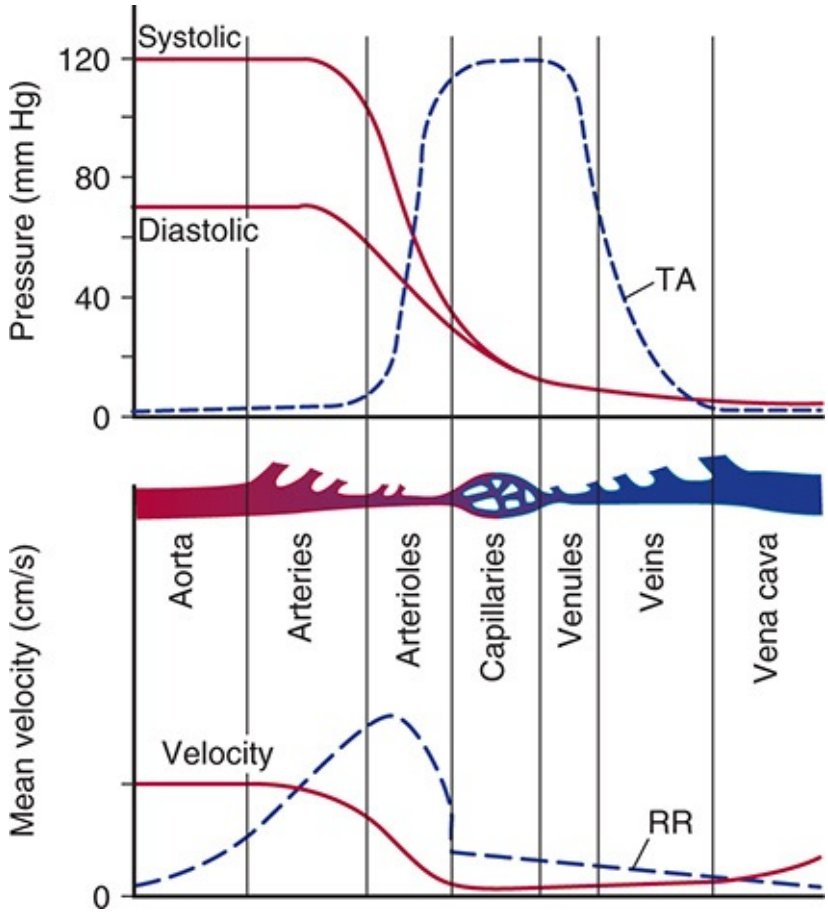


FIGURE 31–24 Diagram of the changes in pressure and velocity as blood flows through the systemic circulation. TA, total cross-sectional area of the vessels, which increases from 4.5 cm^2 in the aorta to 4500 cm^2 in the capillaries (Table 31–9). RR, relative resistance, which is highest in the arterioles.

PULSATILE FLOW OF BLOOD

Although the mean velocity of the blood in the proximal portion of the aorta is 40 cm/s, the flow is phasic, and velocity ranges from 120 cm/s during systole to a negative value at the time of the transient backflow before the aortic valve closes in diastole. In the distal portions of the aorta and in the large arteries, velocity is also much greater in systole than it is in diastole. However, the vessels are elastic, and forward flow is continuous because of the recoil during diastole of the vessel walls that have been stretched during systole **(Figure 31–25)**. Pulsatile flow appears to maintain optimal function of the tissues, apparently via distinct effects on gene transcription.

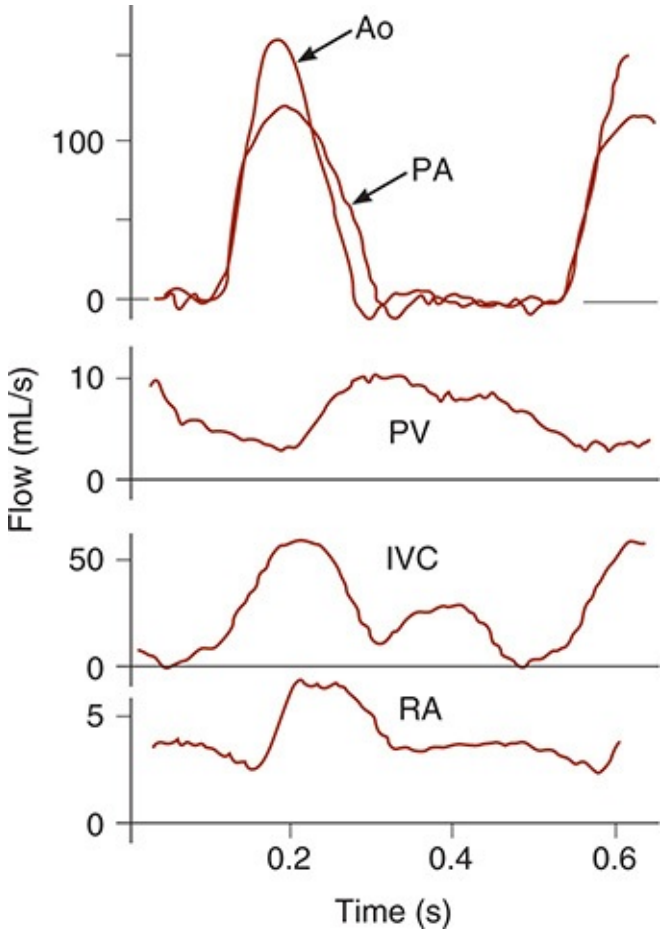


FIGURE 31–25 Changes in blood flow during the cardiac cycle in the dog. Diastole is followed by systole starting at 0.1 and again at 0.5 s. Flow patterns in humans are similar. Ao, aorta; IVC, inferior vena cava; PA, pulmonary artery; PV, pulmonary vein; RA, renal artery. (Reproduced with permission from Milnor WR: Pulsatile blood flow. N Engl J Med 1972; July 6; 287(1):27–34.)

ARTERIAL PRESSURE

The pressure in the aorta and in the brachial and other large arteries in a young adult human rises to a peak value **(systolic pressure)** of about 120 mm Hg during each heart cycle and falls to a minimum **(diastolic pressure)** of about 70 mm Hg. The arterial pressure is conventionally written as systolic pressure over diastolic pressure, for example, 120/70 mm Hg. The **pulse pressure,** the difference between the systolic and diastolic pressures, is normally about 50 mm Hg. The **mean pressure** is the average pressure throughout the cardiac cycle. Because systole is shorter than diastole, the mean pressure is slightly less than the value halfway between systolic and diastolic pressure. It can be determined only by integrating the area of the pressure curve **(Figure 31–26)**; however, as an approximation, mean pressure equals the diastolic pressure plus one-third of the pulse pressure.

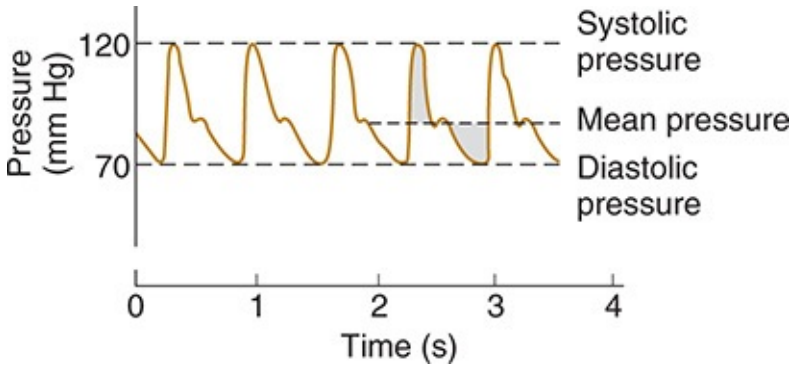


FIGURE 31–26 Brachial artery pressure curve of a normal young human, showing the relation of systolic and diastolic pressure to mean pressure. The shaded area above the mean pressure line is equal to the shaded area below it.

The pressure falls very slightly in the large- and medium-sized arteries because their resistance to flow is small, but it falls rapidly in the small arteries and arterioles, which are the main sites of the peripheral resistance against which the heart pumps. The mean pressure at the end of the arterioles is 30–38 mm Hg. Pulse pressure also declines rapidly to about 5 mm Hg at the ends of the arterioles. The magnitude of the pressure drop along the arterioles varies considerably depending on whether they are constricted or dilated.

EFFECT OF GRAVITY

The pressures in Figure 31–24 are those in blood vessels at heart level. The pressure in any vessel below heart level is increased and that in any vessel above heart level is decreased by the effect of gravity. The magnitude of the gravitational effect is 0.77 mm Hg/cm of vertical distance above or below the heart. Thus, in an adult human in the upright position, when the mean arterial pressure at heart level is 100 mm Hg, the mean pressure in a large artery in the head (50 cm above the heart) is 62 mm Hg (100 − [0.77 × 50]) and the pressure in a large artery in the foot (105 cm below the heart) is 180 mm Hg (100 + [0.77 × 105]). The effect of gravity on venous pressure is similar **(Figure 31–27)**.

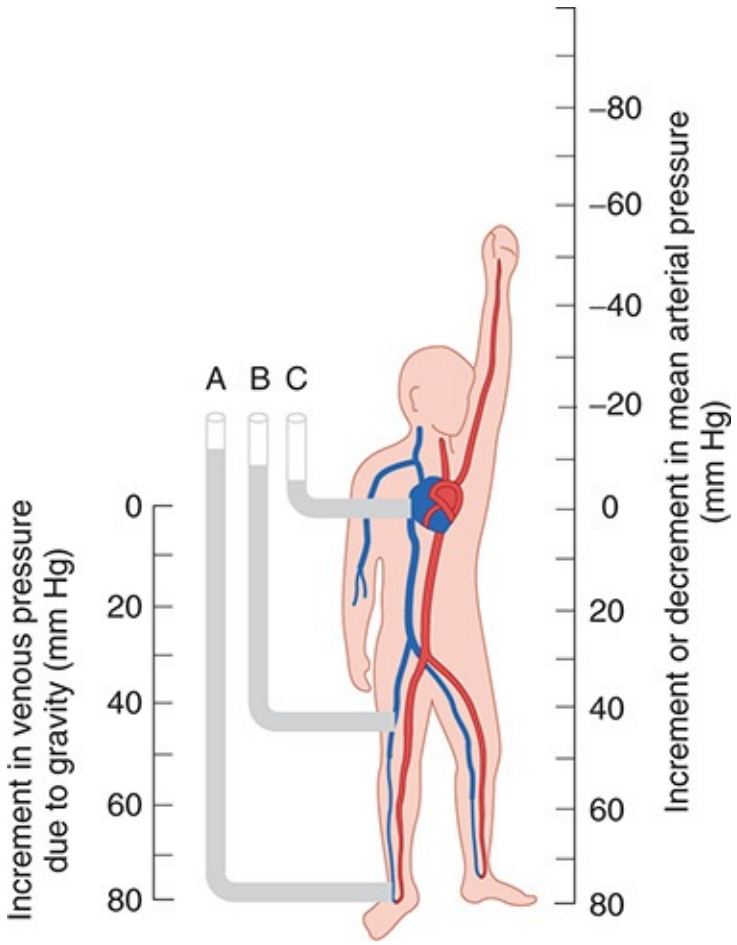


FIGURE 31–27 Effects of gravity on arterial and venous pressure. The scale on the right indicates the increment (or decrement) in mean pressure in a large artery at each level. The mean pressure in all large arteries is approximately 100 mm Hg when they are at the level of the left ventricle. The scale on the left indicates the increment in venous pressure at each level due to gravity. The manometers on the left of the figure indicate the height to which a column of blood in a tube would rise if connected to an ankle vein (A), the femoral vein

(B), or the right atrium (C), with the subject in the standing position. The approximate pressures in these locations in the recumbent position; that is, when the ankle, thigh, and right atrium are at the same level, are A, 10 mm Hg; B, 7.5 mm Hg; and C, 4.6 mm Hg.

METHODS OF MEASURING BLOOD PRESSURE

If a cannula is inserted into an artery, the arterial pressure can be measured directly with a mercury manometer or a suitably calibrated strain gauge. When an artery is tied off beyond the point at which the cannula is inserted, an **end pressure** is recorded, flow in the artery is interrupted, and all the kinetic energy of flow is converted into pressure energy. If, alternatively, a T tube is inserted into a vessel and the pressure is measured in the side arm of the tube, the recorded **side pressure,** under conditions where pressure drop due to resistance is negligible, is lower than the end pressure by the kinetic energy of flow. This is because in a tube or a blood vessel the total energy—the sum of the kinetic energy of flow and the potential energy—is constant **(Bernoulli principle).**

It is worth noting that the pressure drop in any segment of the arterial system is due both to resistance and to conversion of potential into kinetic energy. The pressure drop due to energy lost in overcoming resistance is irreversible, since the energy is dissipated as heat; but the pressure drop due to conversion of potential to kinetic energy as a vessel narrows is reversed when the vessel widens out again **(Figure 31–28)**.

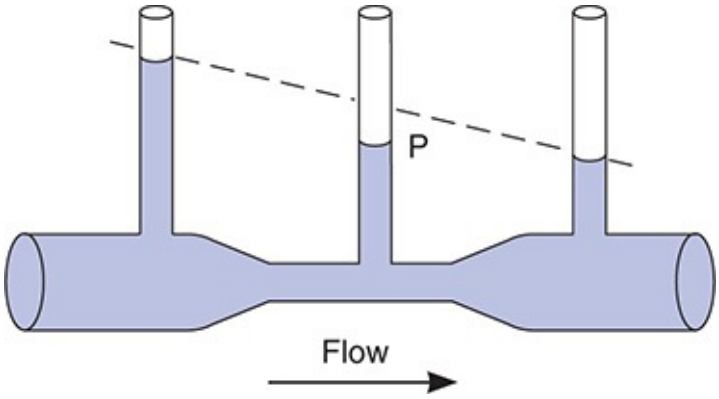


FIGURE 31–28 Bernoulli principle. When fluid flows through the narrow portion of the tube, the kinetic energy of flow is increased as the velocity increases, and the potential energy is reduced. Consequently, the measured pressure (P) is lower than it would have been at that point if the tube had not

been narrowed. The dashed line indicates what the pressure drop due to frictional forces would have been if the tube had been of uniform diameter.

The Bernoulli principle also has a significant application in pathophysiology. According to the principle, the greater the velocity of flow in a vessel, the lower the lateral pressure distending its walls. When a vessel is narrowed, the velocity of flow in the narrowed portion increases and the distending pressure decreases. Therefore, when a vessel is narrowed by a pathologic process such as an atherosclerotic plaque, the lateral pressure at the constriction is decreased and the narrowing tends to maintain itself.

AUSCULTATORY METHOD

The arterial blood pressure in humans is routinely measured by the **auscultatory method.** An inflatable cuff attached to a mercury manometer **(sphygmomanometer)** is wrapped around the arm and a stethoscope is placed over the brachial artery at the elbow. The cuff is rapidly inflated until the pressure is well above the expected systolic pressure in the brachial artery. The artery is occluded by the cuff, and no sound is heard with the stethoscope. The pressure in the cuff is then lowered slowly. At the point at which systolic pressure in the artery just exceeds the cuff pressure, a spurt of blood passes through with each heartbeat and, synchronously with each beat, a tapping sound is heard below the cuff. The cuff pressure at which the sounds are first heard is the systolic pressure. As the cuff pressure is lowered further, the sounds become louder, then dull and muffled. These are the **sounds of Korotkoff.** Finally, in most individuals, they disappear. When direct and indirect blood pressure measurements are made simultaneously, the diastolic pressure in resting adults correlates best with the pressure at which the sound disappears. However, in adults after exercise and in children, the diastolic pressure correlates best with the pressure at which the sounds become muffled. This is also true in diseases such as hyperthyroidism and aortic regurgitation.

The sounds of Korotkoff are produced by turbulent flow in the brachial artery. When the artery is narrowed by the cuff, the velocity of flow through the constriction exceeds the **critical velocity** and turbulent flow results (Figure 31–19). At cuff pressures just below the systolic pressure, flow through the artery occurs only at the peak of systole, and the intermittent turbulence produces a tapping sound. As long as the pressure in the cuff is above the diastolic pressure in the artery, flow is interrupted at least during part of diastole, and the

intermittent sounds have a staccato quality. When the cuff pressure is near the arterial diastolic pressure, the vessel is still constricted, but the turbulent flow is continuous. Continuous sounds have a muffled rather than a staccato quality.

CHANGES IN ARTERIAL BLOOD PRESSURE

Because the arterial pressure is the product of the cardiac output and the peripheral resistance, it is affected by conditions that affect either or both of these factors. Emotion increases the cardiac output and peripheral resistance, and about 20% of hypertensive patients have blood pressures that are higher in the doctor's office than at home ("white coat hypertension"). Blood pressure normally falls up to 20 mm Hg during sleep. This fall is reduced or absent in hypertension.

There is general agreement that blood pressure rises with advancing age, but the magnitude of this rise is uncertain because hypertension is a common disease and its incidence also increases with advancing age **(Clinical Box 31–4)**. Individuals who have systolic blood pressures < 120 mm Hg at age 50–60 and never develop clinical hypertension still have systolic pressures that rise throughout life **(Figure 31–29)**. This rise may be the closest approximation to the rise in normal individuals. Diastolic pressure also rises, but then starts to fall in middle age as the stiffness of arteries increases. Consequently, pulse pressure rises with advancing age. It is interesting that systolic and diastolic blood pressures are lower in women than in men until age 55–65, after which they become comparable. Because there is a positive correlation between blood pressure and the incidence of heart attacks and strokes (see below), the lower blood pressure before menopause in women may be one reason that, on average, they live longer than men.

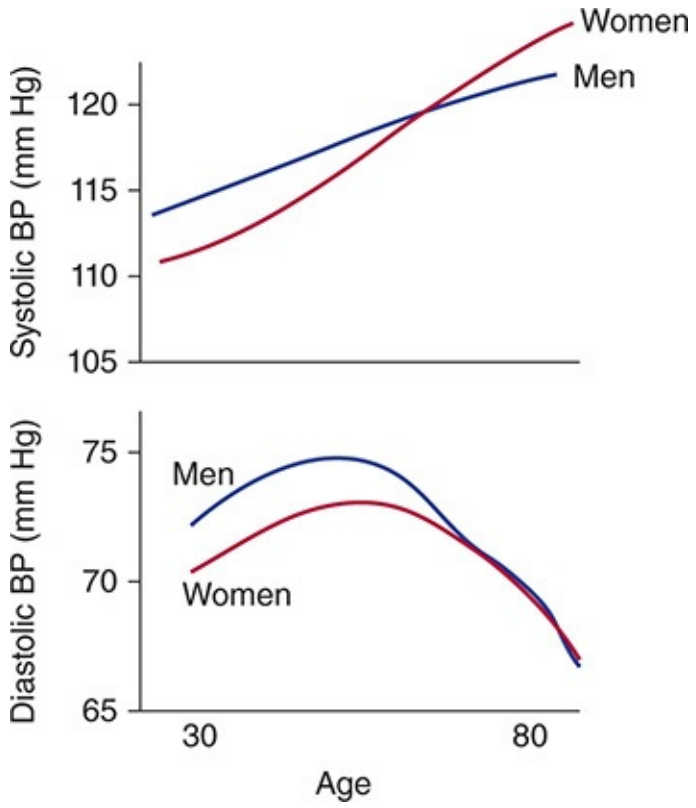


FIGURE 31–29 Effects of age and sex on arterial pressure components in humans. Data are from a large group of individuals who were studied every 2 years throughout their adult lives and who were found not be hypertensive (ie, they had systolic blood pressures < 120 mm Hg) at age 50–60. (Data replotted from Franklin SS, et al: Hemodynamic patterns of age-related changes in blood pressure: The Framingham Heart Study. Circulation 1997; 96:308 as well as the NHANES III study.)

CAPILLARY CIRCULATION

At any one time, only 5% of the circulating blood is in the capillaries, but this 5% is in a sense the most important part of the blood volume because it is the only pool from which O_2 and nutrients can enter the interstitial fluid and into which CO_2 and waste products can enter the bloodstream. Exchange across the capillary walls is essential to the survival of the tissues.

METHODS OF STUDY

It is difficult to obtain accurate measurements of capillary pressures and flows.

Experimentally, capillary pressure has been estimated by determining the amount of external pressure necessary to occlude the capillaries or the amount of pressure necessary to make saline start to flow through a micropipette inserted so that its tip faces the arteriolar end of the capillary.

CAPILLARY PRESSURE & FLOW

Capillary pressures vary considerably, but typical values in human nail bed capillaries are 32 mm Hg at the arteriolar end and 15 mm Hg at the venous end. The pulse pressure is approximately 5 mm Hg at the arteriolar end and zero at the venous end. The capillaries are short, but blood moves slowly (about 0.07 cm/s) because the total cross-sectional area of the capillary bed is large. Transit time from the arteriolar to the venular end of an average-sized capillary is 1–2 s.

CLINICAL BOX 31–4

Hypertension

Hypertension is a sustained elevation of the systemic arterial pressure. It is most commonly due to increased peripheral resistance and is a very common abnormality in humans. It can be produced by many diseases **(Table 31–11)** and causes a number of serious disorders. When the resistance against which the left ventricle must pump (afterload) is elevated for a long period, the cardiac muscle hypertrophies. The initial response is activation of immediate-early genes in the ventricular muscle, followed by activation of a series of genes involved in growth during fetal life. Left ventricular hypertrophy is associated with a poor prognosis. The total O_2 consumption of the heart, already increased by the work of expelling blood against a raised pressure (see Chapter 30), is increased further because there is more muscle. Therefore, any decrease in coronary blood flow has more serious consequences in hypertensive patients than it does in normal individuals, and degrees of coronary vessel narrowing that do not produce symptoms when the size of the heart is normal may produce myocardial infarction when the heart is enlarged.

The incidence of atherosclerosis increases in hypertension, and myocardial infarcts are common even when the heart is not enlarged. Eventually, the ability to compensate for the high peripheral resistance is exceeded, and the

heart fails. Hypertensive individuals are also predisposed to thromboses of cerebral vessels and cerebral hemorrhage. An additional complication is chronic kidney disease. However, the incidence of heart failure, strokes, and chronic kidney disease can be markedly reduced by active treatment of hypertension, even when the hypertension is relatively mild. In most patients with elevated blood pressure, the cause of the hypertension is unknown, and they are said to have **essential hypertension** (Table 31–11). At present, essential hypertension is treatable but not curable. In other, less common forms of hypertension, the cause is known. A review of these is helpful because it emphasizes ways disordered physiology can lead to disease. Pathology that compromises the renal blood supply leads to renal hypertension, as does narrowing (coarctation) of the thoracic aorta, which both increases renin secretion and increases peripheral resistance. Pheochromocytomas, adrenal medullary tumors that secrete norepinephrine and epinephrine, can cause sporadic or sustained hypertension (see Chapter 20). Estrogens increase angiotensinogen secretion, and contraceptive pills containing large amounts of estrogen cause hypertension (pill hypertension) on this basis (see Chapter 22). Increased secretion of aldosterone or other mineralocorticoids causes renal Na^+ retention, which leads to hypertension. A primary increase in plasma mineralocorticoids inhibits renin secretion. For unknown reasons, plasma renin is also low in 10–15% of patients with essential hypertension and normal circulating mineralocortical levels (low renin hypertension). Mutations in a number of single genes are also known to cause hypertension. These cases of monogenic hypertension are rare but informative. One of these is glucocorticoid-remediable aldosteronism (GRA), in which a hybrid gene encodes an adrenocorticotropic hormone (ACTH)-sensitive aldosterone synthase, with resulting hyperaldosteronism (see Chapter 20). 11-β Hydroxylase deficiency also causes hypertension by increasing the secretion of deoxycorticosterone (see Chapter 20). Normal blood pressure is restored when ACTH secretion is inhibited by administering a glucocorticoid. Mutations that decrease 11-β hydroxysteroid dehydrogenase cause loss of specificity of the mineralocorticoid receptors (see Chapter 20) with stimulation of them by cortisol and, in pregnancy, by the elevated circulating levels of progesterone. Finally, mutations of the genes for ENaCs that reduce degradation of the β or γ subunits increase ENaC activity and lead to excess renal Na^+ retention and hypertension (Liddle syndrome; see Chapter 37).

THERAPEUTIC HIGHLIGHTS

Effective lowering of the blood pressure can be produced by drugs that block α-adrenergic receptors, either in the periphery or in the central nervous system; drugs that block β-adrenergic receptors; drugs that inhibit the activity of angiotensin-converting enzyme; and calcium channel blockers that relax vascular smooth muscle.

TABLE 31–11 Estimated frequency of various forms of hypertension in the general hypertensive population.

	Percentage of Population
Essential hypertension	88
Renal hypertension	
Renovascular	2
Parenchymal	3
Endocrine hypertension	
Primary aldosteronism	5
Cushing syndrome	0.1
Pheochromocytoma	0.1
Other adrenal forms	0.2
Estrogen treatment ("pill hypertension")	1
Miscellaneous (Liddle syndrome, coarctation of the aorta, etc)	0.6

Data from Braunwald E, et al (eds): *Harrison's Principles of Internal Medicine*, 15th ed. New York, NY: McGraw-Hill; 2001.

EQUILIBRATION WITH INTERSTITIAL FLUID

As noted above, the capillary wall is a thin membrane made up of endothelial cells. Substances pass through the junctions between endothelial cells and through fenestrations when they are present. Some also pass through the cells by vesicular transport.

The factors other than vesicular transport that are responsible for transport

across the capillary wall are diffusion and filtration (see Chapter 1). Diffusion is quantitatively much more important. O_2 and glucose are in higher concentration in the bloodstream than in the interstitial fluid and diffuse into the interstitial fluid, whereas CO_2 diffuses in the opposite direction.

The rate of filtration at any point along a capillary depends on a balance of forces sometimes called the **Starling forces**. One of these is the **hydrostatic pressure gradient** (the hydrostatic pressure in the capillary minus the hydrostatic pressure of the interstitial fluid). The interstitial fluid pressure varies from one organ to another. It is positive in the liver and kidneys and as high as 6 mm Hg in the brain. The other force is the **osmotic pressure gradient** across the capillary wall (colloid osmotic pressure of plasma minus colloid osmotic pressure of interstitial fluid). This component is directed inward.

Thus:

$$\text{Fluid movement} = k\,[(P_c - P_i) - (\pi_c - \pi_i)]$$

where

k = capillary filtration coefficient

P_c = capillary hydrostatic pressure

P_i = interstitial hydrostatic pressure

π_c = capillary colloid osmotic pressure

π_i = interstitial colloid osmotic pressure

π_i is usually negligible, so the osmotic pressure gradient ($\pi_c - \pi_i$) usually equals the oncotic pressure. The capillary filtration coefficient takes into account, and is proportional to, the permeability of the capillary wall and the area available for filtration. The magnitude of the Starling forces along a typical muscle capillary is shown in **Figure 31–30**. Fluid moves into the interstitial space at the arteriolar end of the capillary and into the capillary at the venular end. In other capillaries, the balance of Starling forces may be different. For example, fluid moves out of almost the entire length of the capillaries in the renal glomeruli. On the other hand, fluid moves into the capillaries through almost their entire length in the intestines. About 24 L of fluid is filtered through the capillaries per day. This is about 0.3% of the cardiac output. About 85% of the filtered fluid is reabsorbed into the capillaries, and the remainder returns to the circulation via the lymphatics.

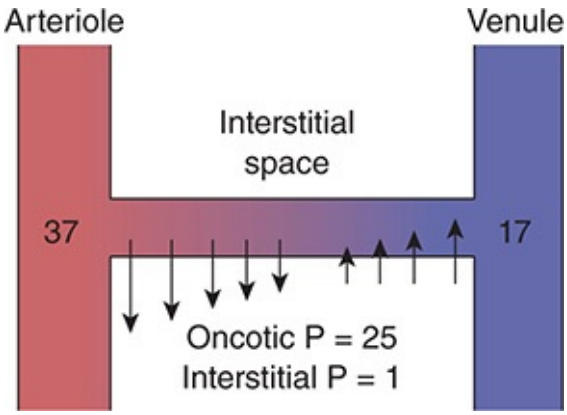


FIGURE 31–30 Schematic representation of pressure gradients across the wall of a muscle capillary. The numbers at the arteriolar and venular ends of the capillary are the hydrostatic pressures in mm Hg at these locations. The arrows indicate the approximate magnitude and direction of fluid movement. In this example, the pressure differential at the arteriolar end of the capillary is 11 mm Hg ([37 − 1] −25) outward; at the opposite end, it is 9 mm Hg (25 − [17 − 1]) inward.

It is worth noting that small molecules often equilibrate with the tissues near the arteriolar end of each capillary. In this situation, total diffusion can be increased by increasing blood flow; that is, exchange is **flow-limited** (Figure 31–31). Conversely, transfer of substances that do not reach equilibrium with the tissues during their passage through the capillaries is said to be **diffusion-limited.**

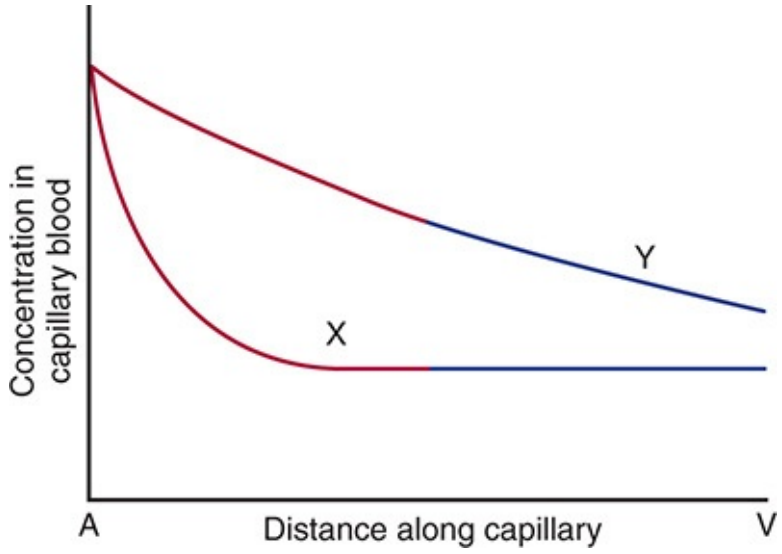


FIGURE 31–31 Flow-limited and diffusion-limited exchange across

capillary walls. A and V indicate the arteriolar and venular ends of the capillary. Substance X equilibrates with the tissues (movement into the tissues equals movement out) well before the blood leaves the capillary, whereas substance Y does not equilibrate. If other factors stay constant, the amount of X entering the tissues can be increased only by increasing blood flow; that is, it is flow-limited. The movement of Y is diffusion-limited.

ACTIVE & INACTIVE CAPILLARIES

In resting tissues, most of the capillaries are collapsed. In active tissues, the metarterioles and the precapillary sphincters dilate. The intracapillary pressure rises, overcoming the critical closing pressure of the vessels, and blood flows through all of the capillaries. Relaxation of the smooth muscle of the metarterioles and precapillary sphincters is due to the action of vasodilator metabolites formed locally in active tissue (see Chapter 32).

After noxious stimulation, substance P released by the axon reflex (see Chapter 33) increases capillary permeability. Bradykinin and histamine also increase capillary permeability. When capillaries are stimulated mechanically, they empty (white reaction; see Chapter 33), probably due to contraction of the precapillary sphincters.

VENOUS CIRCULATION

Blood flows through the blood vessels, including the veins, primarily because of the pumping action of the heart. However, venous flow is aided by the heartbeat, the increase in the negative intrathoracic pressure during each inspiration, and contractions of skeletal muscles that compress the veins **(muscle pump).**

VENOUS PRESSURE & FLOW

The pressure in the venules is 12–18 mm Hg. It falls steadily in the larger veins to about 5.5 mm Hg in the great veins outside the thorax. The pressure in the great veins at their entrance into the right atrium **(central venous pressure)** averages 4.6 mm Hg, but fluctuates with respiration and heart action.

Peripheral venous pressure, like arterial pressure, is affected by gravity. It is increased by 0.77 mm Hg for each centimeter below the right atrium and decreased by a like amount for each centimeter above the right atrium the

pressure is measured (Figure 31–27). Thus, on a proportional basis, gravity has a greater effect on venous than on arterial pressures.

When blood flows from the venules to the large veins, its average velocity increases as the total cross-sectional area of the vessels decreases. In the great veins, the velocity of blood is about one-fourth that in the aorta, averaging about 10 cm/s.

THORACIC PUMP

During inspiration, the intrapleural pressure falls from −2.5 to −6 mm Hg. This negative pressure is transmitted to the great veins and, to a lesser extent, the atria, so that central venous pressure fluctuates from about 6 mm Hg during expiration to approximately 2 mm Hg during quiet inspiration. The drop in venous pressure during inspiration aids venous return. When the diaphragm descends during inspiration, intra-abdominal pressure rises, and this also squeezes blood toward the heart because backflow into the leg veins is prevented by the venous valves.

EFFECTS OF HEARTBEAT

The variations in atrial pressure are transmitted to the great veins, producing the **a, c,** and **v waves** of the venous pressure-pulse curve (see Chapter 30). Atrial pressure drops sharply during the ejection phase of ventricular systole because the atrioventricular valves are pulled downward, increasing the capacity of the atria. This action sucks blood into the atria from the great veins. The sucking of the blood into the atria during systole contributes appreciably to venous return, especially at rapid heart rates.

Close to the heart, venous flow becomes pulsatile. When the heart rate is slow, two periods of peak flow are detectable, one during ventricular systole, due to pulling down of the atrioventricular valves, and one in early diastole, during the period of rapid ventricular filling (Figure 31–25).

MUSCLE PUMP

In the limbs, the veins are surrounded by skeletal muscles, and contraction of these muscles during activity compresses the veins. Pulsations of nearby arteries may also compress veins. Because the venous valves prevent reverse flow, the

blood moves toward the heart. During quiet standing, when the full effect of gravity is manifest, venous pressure at the ankle is 85–90 mm Hg (Figure 31–27). Pooling of blood in the leg veins reduces venous return, with the result that cardiac output is reduced, sometimes to the point where fainting occurs. Rhythmic contractions of the leg muscles while the person is standing serve to lower the venous pressure in the legs to less than 30 mm Hg by propelling blood toward the heart. This heartward movement of the blood is decreased in patients with **varicose veins** because their valves are incompetent. These patients may develop stasis and ankle edema. However, even when the valves are incompetent, muscle contractions continue to produce a basic heartward movement of the blood because the resistance of the larger veins in the direction of the heart is less than the resistance of the small vessels away from the heart.

VENOUS PRESSURE IN THE HEAD

In the upright position, the venous pressure in the parts of the body above the heart is decreased by the force of gravity. The neck veins collapse above the point where the venous pressure is close to zero. However, the dural sinuses have rigid walls and cannot collapse. The pressure in them in the standing or sitting position is therefore subatmospheric. The magnitude of the negative pressure is proportional to the vertical distance above the top of the collapsed neck veins, and in the superior sagittal sinus may be as much as −10 mm Hg. This fact must be kept in mind by neurosurgeons. Neurosurgical procedures are sometimes performed with the patient seated. If one of the sinuses is opened during such a procedure it sucks air, causing **air embolism.**

AIR EMBOLISM

Because air, unlike fluid, is compressible, its presence in the circulation has serious consequences. The forward movement of the blood depends on the fact that blood is incompressible. Large amounts of air fill the heart and effectively stop the circulation, causing sudden death because most of the air is compressed by the contracting ventricles rather than propelled into the arteries. Small amounts of air are swept through the heart with the blood, but the bubbles lodge in the small blood vessels. The surface capillarity of the bubbles markedly increases the resistance to blood flow, and flow is reduced or abolished. Blockage of small vessels in the brain leads to serious and even fatal neurologic abnormalities. Treatment with hyperbaric oxygen (see Chapter 35) is of value

because the pressure reduces the size of the gas emboli. In experimental animals, the amount of air that produces fatal air embolism varies considerably, depending in part on the rate at which it enters the veins. Sometimes as much as 100 mL can be injected without ill effects, whereas at other times as little as 5 mL is lethal.

MEASURING VENOUS PRESSURE

Central venous pressure can be measured directly by inserting a catheter into the thoracic great veins. **Peripheral venous pressure** correlates well with central venous pressure in most conditions. To measure peripheral venous pressure, a needle attached to a manometer containing sterile saline is inserted into an arm vein. The peripheral vein should be at the level of the right atrium (a point half the chest diameter from the back in the supine position). The values obtained in millimeters of saline can be converted into millimeters of mercury (mm Hg) by dividing by 13.6 (the density of mercury). The amount by which peripheral venous pressure exceeds central venous pressure increases with the distance from the heart along the veins. The mean pressure in the antecubital vein is normally 7.1 mm Hg, compared with a mean pressure of 4.6 mm Hg in the central veins.

Central venous pressure is decreased during negative pressure breathing and shock. It is increased by positive pressure breathing, straining, expansion of the blood volume, and heart failure. In advanced heart failure or obstruction of the superior vena cava, the pressure in the antecubital vein may reach values of 20 mm Hg or more.

LYMPHATIC CIRCULATION & INTERSTITIAL FLUID VOLUME

LYMPHATIC CIRCULATION

Fluid efflux normally exceeds influx across the capillary walls, but the extra fluid enters the lymphatics and drains through them back into the blood. This keeps the interstitial fluid pressure from rising and promotes the turnover of tissue fluid. The normal 24-h lymph flow is 2–4 L.

Lymphatic vessels can be divided into two types: initial lymphatics and collecting lymphatics **(Figure 31–32)**. The former lack valves and smooth

muscle in their walls, and they are found in regions such as the intestine or skeletal muscle. Tissue fluid enters them through loose junctions between the endothelial cells that form their walls. The fluid in them apparently is massaged by muscle contractions of the organs and contraction of arterioles and venules, with which they are often associated. They drain into the collecting lymphatics, which have valves and smooth muscle in their walls and contract in a peristaltic manner, propelling the lymph along the vessels. Flow in the collecting lymphatics is further aided by movements of skeletal muscle, the negative intrathoracic pressure during inspiration, and the suction effect of high velocity flow of blood in the veins in which the lymphatics terminate. However, the contractions are the principal factor propelling the lymph.

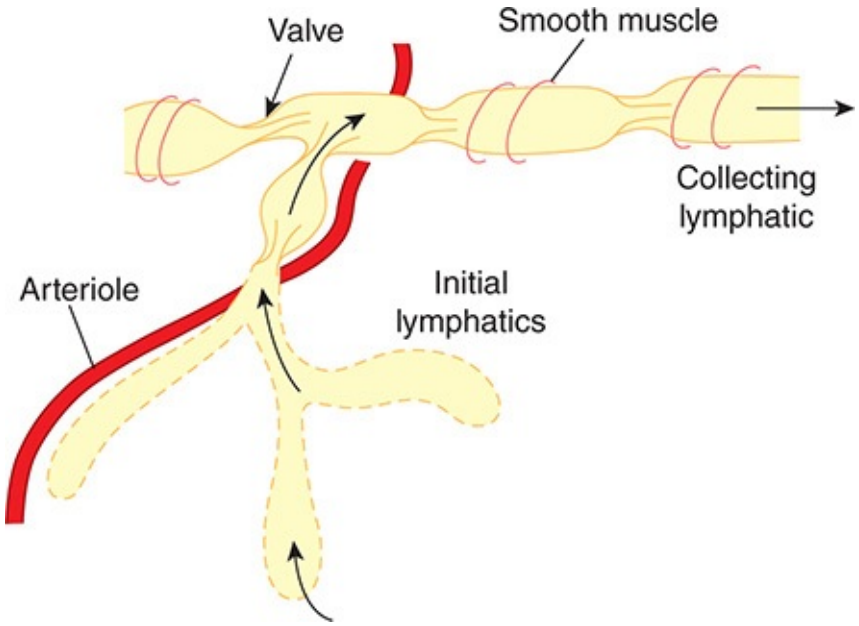


FIGURE 31–32 Schematic of the lymphatic system. Initial lymphatics are highly permeable structures without smooth muscle or valves. Unidirectional flow is accomplished in the collecting lymphatics, which appear like a string of beads due to the regular valves. Lymphatics are often close to blood vessels, whose contractions also encourage flow in the lymphatic system.

OTHER FUNCTIONS OF THE LYMPHATIC SYSTEM

Appreciable quantities of protein enter the interstitial fluid in the liver and intestine, and smaller quantities enter from the blood in other tissues. The

macromolecules enter the lymphatics, presumably at the junctions between the endothelial cells, and the proteins are returned to the bloodstream via the lymphatics. The amount of protein returned in this manner in 1 day is equal to 25–50% of the total circulating plasma protein. The transport of lipids absorbed from the intestine via the lymphatics has been discussed in Chapter 26.

INTERSTITIAL FLUID VOLUME

The amount of fluid in the interstitial spaces depends on the capillary pressure, the interstitial fluid pressure, the oncotic pressure, the capillary filtration coefficient, the number of active capillaries, the lymph flow, and the total extracellular fluid (ECF) volume. The ratio of precapillary to postcapillary venular resistance is also important. Precapillary constriction lowers filtration pressure, whereas postcapillary constriction raises it. Changes in any of these variables lead to changes in the volume of interstitial fluid. Factors promoting an increase in this volume are summarized in **Table 31–12**. **Edema** is the accumulation of interstitial fluid in abnormally large amounts.

TABLE 31–12 Causes of increased interstitial fluid volume and edema.

Increased filtration pressure
Venular constriction
Increased venous pressure (heart failure, incompetent valves, venous obstruction, increased total ECF volume, effect of gravity, etc)
Decreased osmotic pressure gradient across capillary
Decreased plasma protein level
Accumulation of osmotically active substances in interstitial space
Increased capillary permeability
Substance P
Histamine and related substances
Kinins, etc
Inadequate lymph flow

In active tissues, capillary pressure rises, often to the point where it exceeds the oncotic pressure throughout the length of the capillary. In addition, osmotically active metabolites may temporarily accumulate in the interstitial

fluid because they cannot be washed away as rapidly as they are formed. To the extent that they accumulate, they exert an osmotic effect that decreases the magnitude of the osmotic gradient due to the oncotic pressure. The amount of fluid leaving the capillaries is therefore markedly increased and the amount entering them reduced. Lymph flow is increased, decreasing the degree to which the fluid would otherwise accumulate, but exercising muscle, for example, still increases in volume by as much as 25%.

Interstitial fluid tends to accumulate in dependent parts because of the effect of gravity. In the upright position, the capillaries in the legs are protected from the high arterial pressure by the arterioles, but the high venous pressure is transmitted to them through the venules. Skeletal muscle contractions keep the venous pressure low by pumping blood toward the heart (see above) when the individual moves about; however, if one stands still for long periods, fluid accumulates and edema eventually develops. The ankles also swell during long trips when travelers sit for prolonged periods with their feet in a dependent position. Whenever there is abnormal retention of salt in the body, water is also retained. The salt and water are distributed throughout the ECF, and since the interstitial fluid volume is therefore increased, there is a predisposition to edema. Salt and water retention is a factor in the edema seen in heart failure, nephrosis, and cirrhosis, but there are also variations in the mechanisms that govern fluid movement across the capillary walls in these diseases. In heart failure, for example, venous pressure is usually elevated, with a consequent elevation in capillary pressure. In cirrhosis of the liver, oncotic pressure is low because hepatic synthesis of plasma proteins is depressed; and in nephrosis, oncotic pressure is low because large amounts of protein are lost in the urine.

Another cause of edema is inadequate lymphatic drainage. Edema caused by lymphatic obstruction is called **lymphedema,** and the edema fluid has a high protein content. If it persists, it causes a chronic inflammatory condition that can lead to fibrosis of the interstitial tissue. One cause of lymphedema is radical mastectomy, during which removal of the axillary lymph nodes leads to reduced lymph drainage.

CHAPTER SUMMARY

- Blood consists of a suspension of red blood cells (erythrocytes), white blood cells, and platelets. The cells arise in the bone marrow and are subject to regular renewal. Mutations in proteins that make up the cytoskeleton of the red cells make them more fragile and susceptible to lysis.

- Hemoglobin, stored in red blood cells, transports oxygen to peripheral tissues. Fetal hemoglobin is specialized to facilitate diffusion of oxygen from mother to fetus during development. Mutated forms of hemoglobin lead to red cell abnormalities and anemia.
- The blood cells are suspended in a protein-rich fluid known as plasma. The majority of plasma proteins are synthesized by the liver. Lymph is an exudate of plasma with lower protein concentration that serves to recover fluids from the tissues.
- Complex oligosaccharide structures on the membrane of red blood cells, specific to groups of individuals, form the basis of the ABO blood group system. AB blood group oligosaccharides, as well as other blood group molecules, can trigger the production of antibodies in naïve individuals following inappropriate transfusions, with potentially serious consequences due to erythrocyte agglutination.
- When a blood vessel is injured, a complex cascade of molecular interactions among clotting factors ensures that the vessel is sealed and blood loss is minimized (hemostasis). Loss of specific clotting factors may compromise this process and lead to excessive bleeding and bruising. Abnormal clotting that occurs inside an intact vessel is called thrombosis and may occlude blood supply to vital regions.
- Blood flows from the heart to arteries and arterioles, thence to capillaries, and eventually to venules and veins and back to the heart. Each segment of the vasculature has specific contractile properties and regulatory mechanisms that subserve its physiologic functions. Transfer of oxygen and nutrients from the blood to tissues, as well as collection of metabolic wastes, occurs exclusively in the capillary beds.
- Fluid also leaves the circulation across the walls of capillaries. Some is reabsorbed; the remainder enters the lymphatic system, which eventually drains into the subclavian veins to return fluid to the bloodstream.
- Physical principles of pressure, wall tension, and vessel caliber govern the flow of blood through each segment of the circulation. These same physical principles can be applied to measure blood flow and blood pressure.
- Hypertension is an increase in mean blood pressure that is usually chronic and is common in humans. Hypertension can result in serious health consequences if left untreated. The majority of hypertension is of unknown cause, but several gene mutations underlie rare forms of the disease and are informative about mechanisms that control the dynamics of the circulatory

system.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. A 2-year-old-African-American boy is brought to his pediatrician because of acute fever, apparent bone pain, and painful swelling of his hands and feet. A physical examination reveals splenomegaly and a blood test reveals the presence of hemoglobin S. After the acute symptoms subside with rest, hydroxyurea is prescribed. This drug would be expected to reduce the risk of further acute symptoms by which mechanism?

   A. Increased synthesis of hemoglobin S

   B. Increasing the oxygen-carrying capacity of hemoglobin

   C. Increasing expression of hypoxia-inducible factor 1 alpha

   D. Decreasing the oxygen-carrying capacity of hemoglobin

   E. Increasing the synthesis of hemoglobin F

2. A patient is brought to the emergency room suffering from severe blood loss following a motor vehicle accident. Because of a natural disaster, the lab is closed and it is not possible to determine the patient's blood type before initiating a blood transfusion to save his life. However, banked blood is available. Blood of which type would be least likely to provoke a dangerous hemolytic transfusion reaction?

   A. A

   B. B

   C. AB

   D. O

   E. A mixture of A and B

3. Which of the following is not required for hemostasis following injury to a blood vessel?

   A. Platelets

   B. 5-Hydroxytryptamine

   C. Nitric oxide

   D. Collagen

   E. Fibrin monomers

4. A pharmacologist discovers a drug that stimulates the production of VEGF

receptors. He is excited because the drug might be of value in the treatment of

A. coronary artery disease.

B. cancer.

C. emphysema.

D. diabetes insipidus.

E. dysmenorrhea.

5. Which of the following has the highest *total* cross-sectional area in the body?

A. Arteries

B. Arterioles

C. Capillaries

D. Venules

E. Veins

6. The velocity of blood flow

A. is higher in the capillaries than the arterioles.

B. is higher in the veins than in the venules.

C. is higher in the veins than the arteries.

D. falls to zero in the descending aorta during diastole.

E. is reduced in a constricted area of a blood vessel.

7. When the radius of the resistance vessels is increased, which of the following is increased?

A. Systolic blood pressure

B. Diastolic blood pressure

C. Viscosity of the blood

D. Hematocrit

E. Capillary blood flow

8. The pressure in a capillary in skeletal muscle is 35 mm Hg at the arteriolar end and 14 mm Hg at the venular end. The interstitial pressure is 0 mm Hg. The colloid osmotic pressure is 25 mm Hg in the capillary and 1 mm Hg in the interstitium. The net force producing fluid movement across the capillary wall at its arteriolar end is

A. 3 mm Hg out of the capillary.

B. 3 mm Hg into the capillary.

C. 10 mm Hg out of the capillary.

D. 11 mm Hg out of the capillary.

E. 11 mm Hg into the capillary.

9. A 40-year-old man with long-standing diabetes comes to his primary care physician complaining of swelling in his feet and ankles that is so bad that he cannot don his shoes. He states that his urine has developed a foamy appearance and tests reveal that his blood urea nitrogen is decreased. The swelling in this patient is most likely primarily caused by a reduction in which of the following:

A. Lymph flow

B. Barrier function of the capillaries

C. Venous hydrostatic pressure

D. Capillary oncotic pressure

E. Interstitial oncotic pressure

10. A 30-year-old patient comes to her primary care clinician complaining of headaches and vertigo. A blood test reveals a hematocrit of 65%, and a diagnosis of polycythemia is made. Which of the following would also be increased?

A. Mean blood pressure

B. Radius of the resistance vessels

C. Radius of the capacitance vessels

D. Central venous pressure

E. Capillary blood flow

11. Lymph flow from the foot is

A. increased when an individual rises from the supine to the standing position.

B. increased by massaging the foot.

C. increased when capillary permeability is decreased.

D. decreased when the valves of the leg veins are incompetent.

E. decreased by exercise.

12. A 60-year-old woman is evaluated for a new finding of chest pain on mild exertion and is incidentally found to be suffering from primary hypertension. Evaluation of her coronary arteries reveals only minimal obstruction. This patient most likely suffers from angina on mild exertion despite only modest coronary artery disease because, compared with a normotensive individual, which of the following would be increased?

A. Oxygen supply to the coronary muscle

B. The size of heart
C. The risk of stroke
D. The propensity to thrombosis
E. Nitric oxide production by the vascular endothelium

CHAPTER 32

Cardiovascular Regulatory Mechanisms

OBJECTIVES

After studying this chapter, you should be able to:

- Contrast the sympathetic and parasympathetic control of the heart and vasculature.
- Identify the components and neurotransmitters in the central neural pathways that control arterial blood pressure and heart rate and list factors that determine the activity within the pathways.
- Identify the locations of the receptors and afferent and efferent pathways, the effector responses, and the functions of the baroreceptor reflex pathway.
- Describe the role of atrial and cardiopulmonary receptors in the control of the cardiovascular system in health and disease.
- Contrast the locations, properties, and functions of the peripheral and central chemoreceptor reflexes.
- Define how the process of autoregulation contributes to control of vascular caliber.
- Identify the paracrine factors, particularly those secreted by the endothelium, that regulate vascular tone and their mechanisms of action.
- List circulating vasodilators and vasoconstrictors.

INTRODUCTION

Multiple cardiovascular regulatory mechanisms exist that help increase the blood supply to active tissues and increase or decrease heat loss from the body by redistributing the blood. In the face of challenges such as hemorrhage, they maintain blood flow to the heart and brain. When the challenge faced is severe, flow to these vital organs is maintained at the expense of the circulation to the rest of the body.

Circulatory adjustments are effected by altering the output of the pump (the heart), changing the diameter of the resistance vessels (primarily the arterioles), or altering the amount of blood pooled in the capacitance vessels (the veins). Regulation of cardiac output is discussed in Chapter 30. The caliber of the arterioles is adjusted in part by autoregulation **(Table 32–1)**. It is also increased in active tissues by locally produced vasodilator metabolites and substances secreted by the endothelium. The diameter of the arterioles is also regulated by circulating vasoactive substances and the sympathetic nerves that innervate them. Circulating vasoactive substances and sympathetic nerves also determine the caliber of the capacitance vessels. The systemic regulatory mechanisms synergize with the local mechanisms and adjust vascular responses throughout the body.

TABLE 32–1 Summary of factors affecting the caliber of the arterioles.

Vasoconstriction	Vasodilation
Local factors	
Decreased local temperature	Increased CO_2 and decreased O_2
Autoregulation	Increased K^+, adenosine, lactate
	Decreased local pH
	Increased local temperature
Endothelial products	
Endothelin-1	Nitric oxide
Locally released platelet serotonin	Kinins
Thromboxane A_2	Prostacyclin
Circulating neurohumoral agents	
Epinephrine (except in skeletal muscle and liver)	Epinephrine in skeletal muscle and liver
Norepinephrine	Calcitonin G-related protein
Arginine vasopressin	Substance P
Angiotensin II	Histamine
Endogenous digitalis-like substance	Atrial natriuretic peptide
Neuropeptide Y	Vasoactive intestinal polypeptide
Neural factors	
Increased discharge of sympathetic nerves	Decreased discharge of sympathetic nerves
	Activation of β_2-adrenoceptors on the vasculature of skeletal muscles of the limbs

The terms **vasoconstriction** and **vasodilation** are generally used to refer to constriction and dilation of the resistance vessels. Changes in the caliber of the veins are referred to as **venoconstriction** or **venodilation.**

NEURAL CONTROL OF THE CARDIOVASCULAR SYSTEM

INNERVATION OF THE BLOOD VESSELS

Most of the vasculature is an example of an autonomic effector organ that receives innervation from the sympathetic but not the parasympathetic division of the autonomic nervous system (see Chapter 13). Sympathetic postganglionic fibers terminate on vascular smooth muscle in all parts of the body and release norepinephrine to act on α_1-adrenoceptors to mediate vasoconstriction. In the vasculature of exercising muscles, activation of the sympathetic nervous system and release of epinephrine from the adrenal medulla can also mediate vasodilation by the activation of β_2-adrenoceptors on skeletal muscle blood vessels.

Although the arterioles and the other resistance vessels are most densely innervated, all blood vessels except capillaries and venules contain smooth muscle and receive motor nerve fibers from the sympathetic division of the autonomic nervous system. The fibers to the resistance vessels regulate tissue blood flow and arterial pressure. The fibers to the venous capacitance vessels vary the volume of blood "stored" in the veins. The innervation of most veins is sparse, but the splanchnic veins are well innervated. Venoconstriction is produced by stimuli that also activate the vasoconstrictor nerves to the arterioles. The resultant decrease in venous capacity increases venous return, shifting blood to the arterial side of the circulation.

When the sympathetic nerves are damaged **(sympathectomy** or **neuropathy**), the blood vessels dilate. A change in the level of activity (increase or decrease) in sympathetic nerves is just one of the many factors that mediate vasoconstriction or vasodilation (Table 32–1).

INNERVATION OF THE HEART

The heart is one example of an effector organ that receives opposing influences from the sympathetic and parasympathetic divisions of the autonomic nervous system (see Chapter 13). Release of norepinephrine from postganglionic sympathetic nerves activates β_1-adrenoceptors in the heart, notably on the sinoatrial (SA) node, atrioventricular (AV) node, His-Purkinje conductive tissue, and atrial and ventricular contractile tissue. Stimulation of sympathetic nerves leads to an increase in heart rate **(chronotropy),** rate of transmission in the cardiac conductive tissue **(dromotropy),** and the force of ventricular contraction **(inotropy).** On the other hand, release of acetylcholine from postganglionic

parasympathetic (vagus) nerves activates muscarinic receptors in the heart, notably on the SA and AV nodes and atrial muscle. Stimulation of the vagus nerve reduces the heart rate, the rate of transmission through the AV node, and atrial contractility.

The above description presents an oversimplified explanation of autonomic control of cardiac function. There are adrenergic and cholinergic receptors on autonomic nerve terminals that modulate transmitter release from nerve endings. For example, release of acetylcholine from vagal nerve terminals inhibits the release of norepinephrine from sympathetic nerve terminals, so this can enhance the effects of vagal nerve activation on the heart.

There is a moderate amount of tonic discharge in the cardiac sympathetic nerves at rest, but there is considerable tonic vagal discharge **(vagal tone)** in humans and other large animals. After the administration of muscarinic receptor antagonists such as atropine, the heart rate in humans increases from 70 beats/min, its normal resting value, to 150–180 beats/min because the sympathetic tone is unopposed. In humans in whom both noradrenergic and cholinergic systems are blocked, the heart rate is approximately 100 beats/min.

CENTRAL NEURAL CONTROL OF THE CARDIOVASCULAR SYSTEM

The cardiovascular system is under neural influences coming from several parts of the brainstem, forebrain, and insular cortex. The brainstem receives feedback from sensory receptors in the vasculature (eg, baroreceptors and chemoreceptors). A simplified model of the feedback control circuit is shown in **Figure 32–1**. An increase in neural output from the brainstem to sympathetic nerves leads to a decrease in blood vessel diameter (arteriolar vasoconstriction) and increases in stroke volume and heart rate, which contribute to a rise in blood pressure. This in turn causes an increase in baroreceptor activity, which signals the brainstem to reduce the neural output to sympathetic nerves. The baroreceptor reflex is described in detail below.

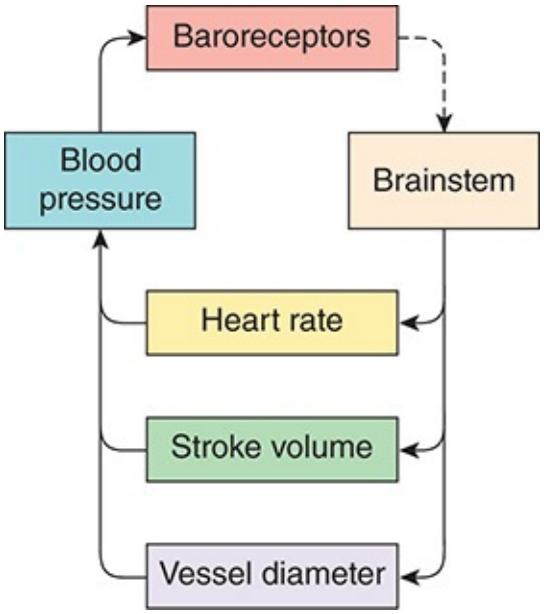


FIGURE 32–1 Feedback control of blood pressure. Brainstem excitatory input to sympathetic nerves to the heart and vasculature increases heart rate and stroke volume and reduces vessel diameter. Together these increase blood pressure, which activates the baroreceptor reflex to reduce the activity in the brainstem.

Venoconstriction and a decrease in the stores of blood in the venous reservoirs usually accompany increases in arteriolar constriction, although changes in the capacitance vessels do not always parallel changes in the resistance vessels. In the presence of an increase in sympathetic nerve activity to the heart and vasculature, there is usually an associated decrease in the activity of vagal fibers to the heart. Conversely, a decrease in sympathetic activity causes vasodilation, a fall in blood pressure, and an increase in the storage of blood in the venous reservoirs. There is usually a concomitant decrease in heart rate, but this is mostly due to stimulation of the vagal innervation of the heart.

One of the major sources of excitatory input to sympathetic nerves controlling the vasculature is a group of neurons located near the pial surface of the medulla in the **rostral ventrolateral medulla (RVLM; Figure 32–2)**. The axons of RVLM neurons course dorsally and medially and then descend in the lateral column of the spinal cord to the thoracolumbar intermediolateral cell column (IML). They contain phenylethanolamine-N-methyltransferase (PNMT; see Chapter 7), but glutamate is the excitatory transmitter they secrete to activate preganglionic sympathetic neurons. Neurovascular compression of the RVLM has been linked to some cases of **essential hypertension** in humans **(Clinical**

Box 32–1).

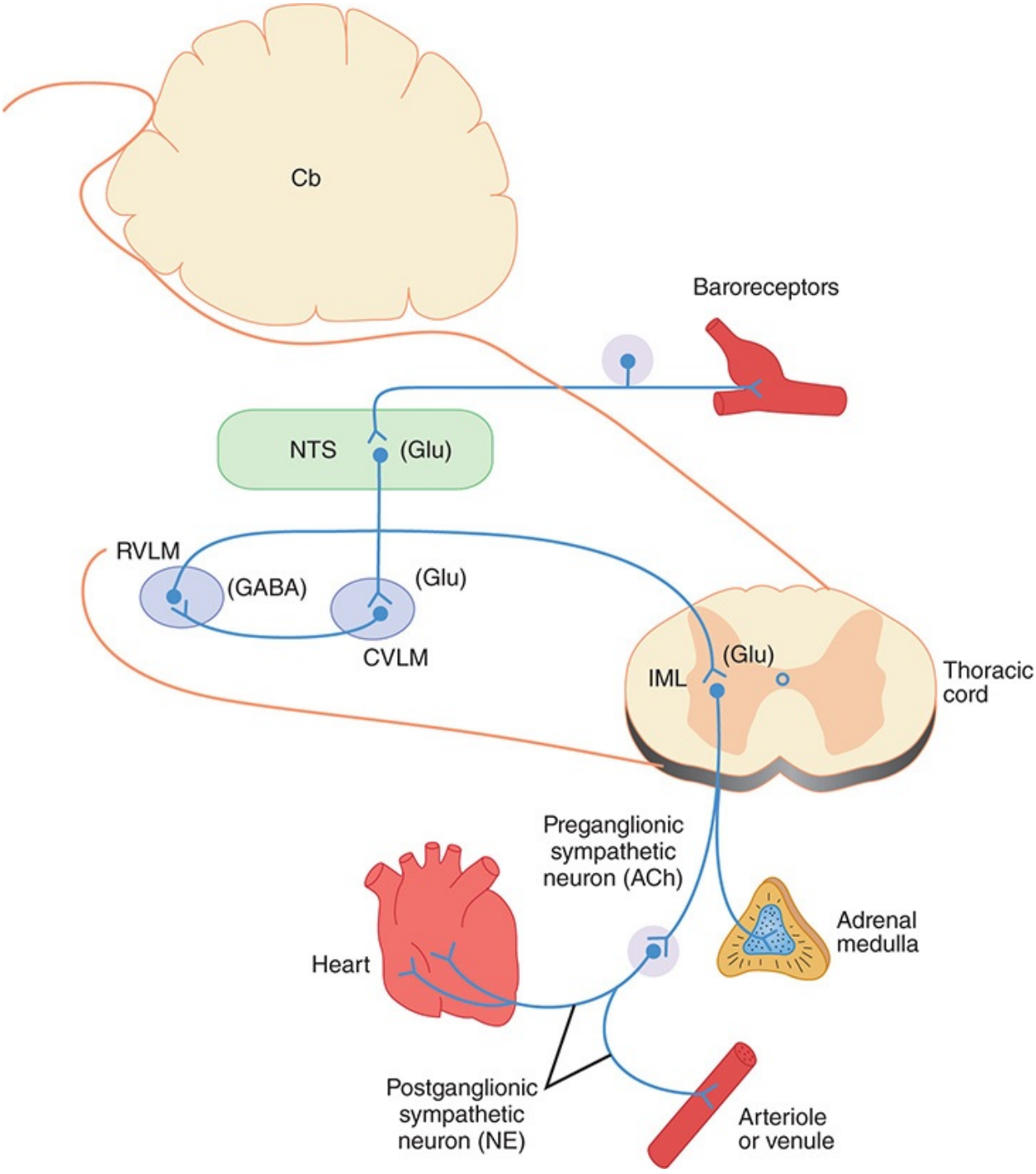


FIGURE 32–2 Basic neural pathways involved in the control of blood pressure. The drawing shows a parasagittal section of the brainstem and its connections to the thoracic spinal cord. The rostral ventrolateral medulla (RVLM) is one of the major sources of excitatory input to sympathetic nerves

controlling the vasculature. These neurons receive inhibitory input from the baroreceptors via an inhibitory (GABAergic) neuron in the caudal ventrolateral medulla (CVLM). The nucleus of the tractus solitarius (NTS) is the site of termination of baroreceptor afferent fibers that release glutamate. ACh, acetylcholine; GABA, γ-aminobutyric acid; Glu, glutamate; IML, intermediolateral cell column; NE, norepinephrine.

The activity of RVLM neurons is determined by many factors **(see Table 32–2)**. They include input from a pathway that originates in the cerebral cortex (particularly the limbic cortex) and has a relay in the hypothalamus. This pathway is responsible for the blood pressure rise and tachycardia produced by emotions such as stress, sexual excitement, and anger. The RVLM also receives input from other brain regions and from arterial baroreceptors and the carotid and aortic chemoreceptors. In addition, some stimuli act directly on RVLM neurons to increase their firing rate.

TABLE 32–2 Factors affecting the activity of the RVLM.

Direct stimulation
CO_2
Hypoxia
Excitatory inputs
Cortex via hypothalamus
Mesencephalic periaqueductal gray
Brainstem reticular formation
Pain pathways
Somatic afferents (somatosympathetic reflex)
Carotid and aortic chemoreceptors
Inhibitory inputs
Cortex via hypothalamus
Caudal ventrolateral medulla
Caudal medullary raphe nuclei
Lung inflation afferents

Carotid, aortic, and cardiopulmonary baroreceptors

Inflation of the lungs causes vasodilation and a decrease in blood pressure. This response is mediated via vagal afferents from the lungs that inhibit RVLM and sympathetic nerve activities. Pain usually causes a rise in blood pressure via afferent impulses converging in the RVLM. However, prolonged severe pain may cause vasodilation and fainting. The activity in afferents from exercising muscles probably exerts a similar pressor effect via a pathway to the RVLM. The pressor response to stimulation of somatic afferent nerves is called the **somatosympathetic reflex.**

The medulla is also a major site of origin of excitatory input to cardiac vagal motor neurons in the nucleus ambiguus **(Figure 32–3)**. **Table 32–3** is a summary of factors that affect heart rate. In general, stimuli that increase heart rate also increase blood pressure, whereas those that decrease heart rate lower blood pressure. However, there are exceptions, such as combined hypotension and tachycardia with stimulation of atrial stretch receptors and hypertension and bradycardia during increased intracranial pressure.

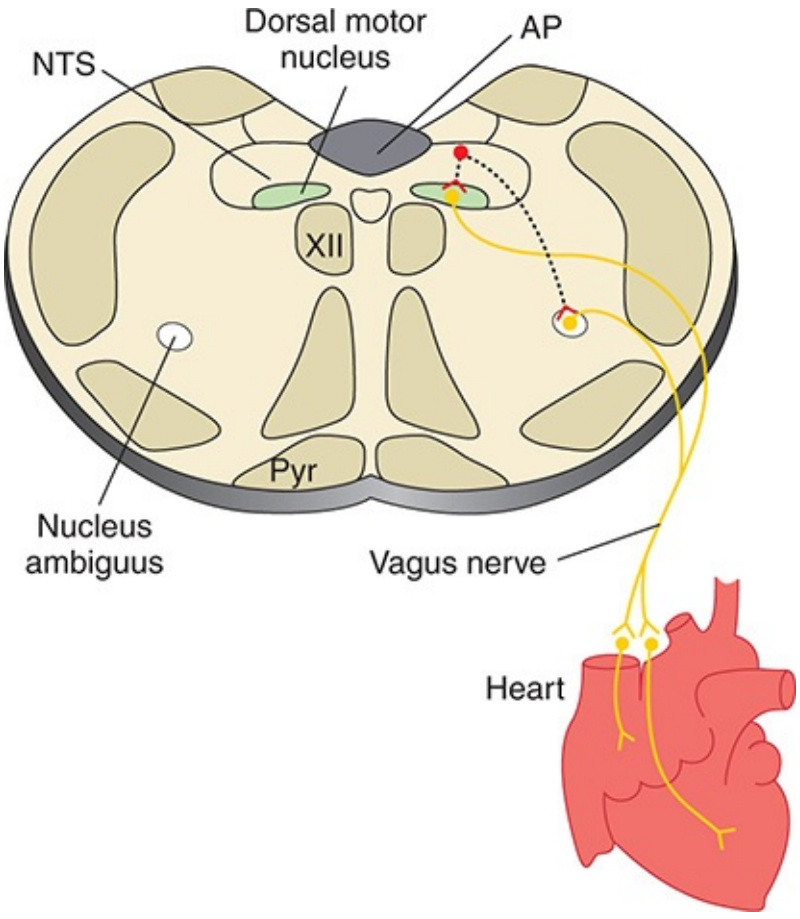


FIGURE 32–3 Basic pathways involved in the control of heart rate by the vagus nerves. Neurons in the nucleus of the tractus solitarius (NTS) project to and excite cardiac preganglionic parasympathetic neurons primarily in the nucleus ambiguus. Some are also located in the dorsal motor nucleus of the vagus; however, this nucleus primarily contains vagal motor neurons that project to the gastrointestinal tract. AP, area postrema; Pyr, pyramid; XII, hypoglossal nucleus.

TABLE 32–3 Factors affecting heart rate.

Heart rate accelerated by:
Decreased activity of arterial baroreceptors
Increased activity of atrial stretch receptors
Inspiration
Excitement
Anger
Most painful stimuli
Hypoxia
Exercise
Thyroid hormones
Fever
Heart rate slowed by:
Increased activity of arterial baroreceptors
Expiration
Fear
Grief
Stimulation of pain fibers in trigeminal nerve
Increased intracranial pressure

BARORECEPTORS

The **baroreceptors** are stretch receptors in the walls of the heart and blood vessels. The **carotid sinus** and **aortic arch** receptors monitor the arterial

circulation. There are also stretch receptors in the low-pressure part of the circulation **(cardiopulmonary receptors).** These include receptors in the walls of the right and left atria at the entrance of the superior and inferior vena cava, in the pulmonary veins, and in the pulmonary circulation.

The carotid sinus is a small dilation of the internal carotid artery just above the bifurcation of the common carotid into external and internal carotid branches **(Figure 32–4)**. Baroreceptors are in the adventitia of the vessel within this dilation as well as in the wall of the arch of the aorta. The afferent nerve fibers from the carotid sinus form a distinct branch of the glossopharyngeal nerve, the **carotid sinus nerve.** The fibers from the aortic arch form a branch of the vagus nerve, the **aortic depressor nerve.**

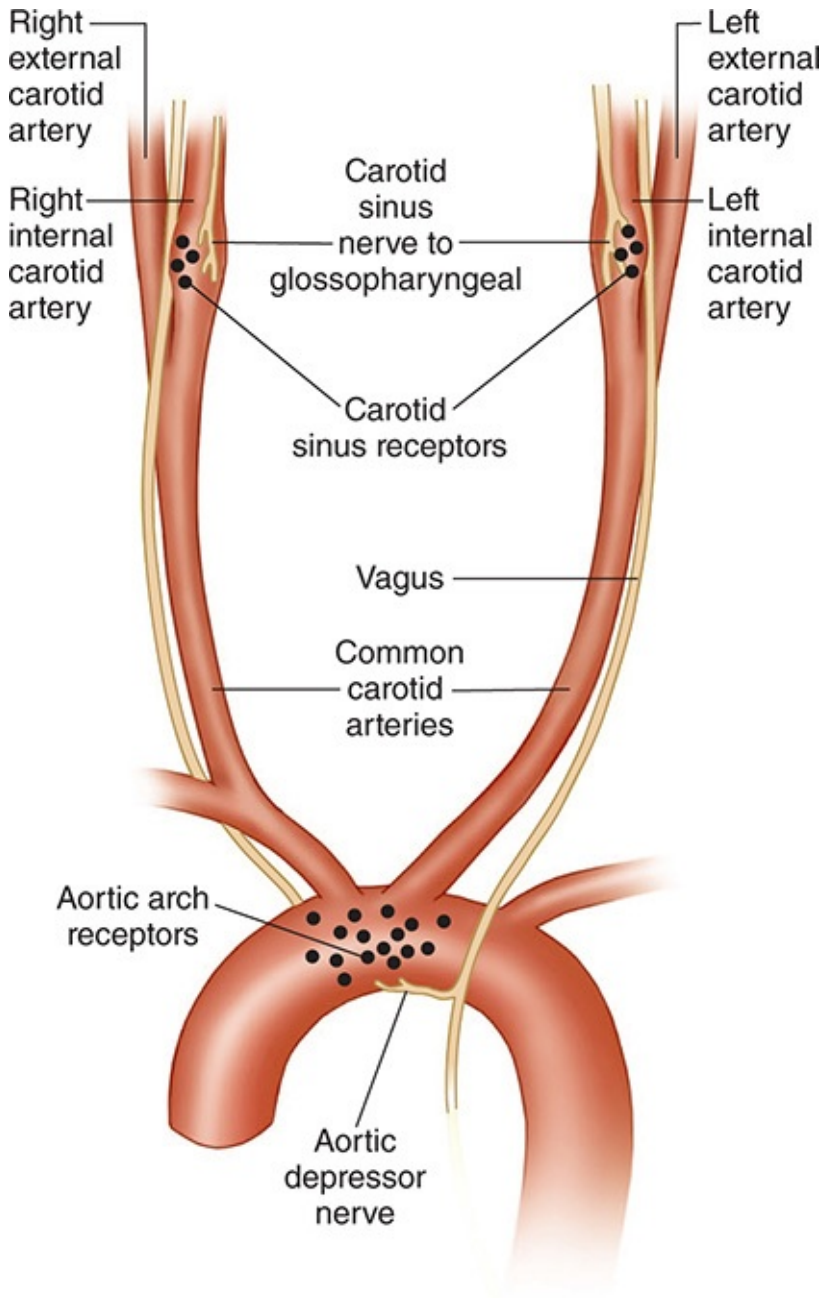


FIGURE 32–4 Baroreceptor areas in the carotid sinus and aortic arch. One set of baroreceptors (stretch receptors) is located in the carotid sinus, a small dilation of the internal carotid artery just above the bifurcation of the common carotid into external and internal carotid branches. These receptors are innervated by a branch of the glossopharyngeal nerve, the carotid sinus nerve. A second set of baroreceptors is located in the wall of the arch of the aorta. These receptors are innervated by a branch of the vagus nerve, the aortic depressor nerve.

The baroreceptors are stimulated by distension of the structures in which they are located; they discharge at an increased rate when the pressure in the carotid sinus and aortic arch rises. Their afferent fibers pass via the glossopharyngeal and vagus nerves to the medulla. These fibers release glutamate to excite neurons in the **nucleus of the tractus solitarius** (NTS) as the first synapse in the baroreceptor reflex (Figure 32–2). These NTS neurons then excite neurons in the **caudal ventrolateral medulla** (CVLM). These CVLM neurons release the inhibitory neurotransmitter γ-aminobutyric acid (GABA) into the RVLM to reduce the firing rate of these. Excitatory projections also extend from the NTS to the vagal motor neurons in the nucleus ambiguus and dorsal motor nucleus (Figure 32–3). Thus, increased baroreceptor discharge *inhibits* the tonic discharge of sympathetic nerves and *excites* the vagal innervation of the heart. These neural changes produce vasodilation, venodilation, hypotension, bradycardia, and a decrease in cardiac output.

BARORECEPTOR NERVE ACTIVITY

Baroreceptors are more sensitive to pulsatile pressure than to constant pressure. A decline in pulse pressure without any change in mean pressure decreases the rate of baroreceptor discharge and provokes a rise in systemic blood pressure and tachycardia. At normal blood pressure levels (about 100 mm Hg mean pressure), a burst of action potentials appears in a single baroreceptor fiber during systole, but there are few action potentials in early diastole **(Figure 32–5)**. At lower mean pressures, this phasic change in firing is even more dramatic with activity only occurring during systole. At these lower pressures, the overall firing rate is considerably reduced. The threshold for eliciting activity in the carotid sinus nerve is approximately 50 mm Hg; maximal activity occurs at approximately 200 mm Hg, with activity throughout the cardiac cycle.

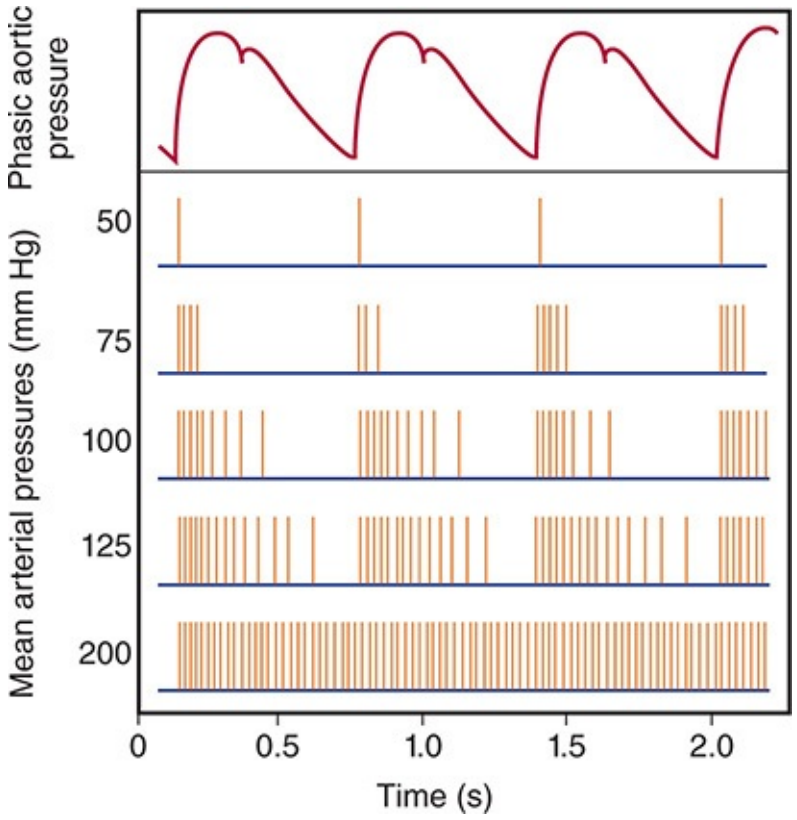


FIGURE 32–5 Discharges (vertical lines) in a single afferent nerve fiber from the carotid sinus at various levels of mean arterial pressures, plotted against changes in aortic pressure with time. Baroreceptors are very sensitive to changes in pulse pressure as shown by the record of phasic aortic pressure. (Reproduced with permission from Levy MN, Pappano AJ: *Cardiovascular Physiology*, 9th ed. Mosby; 2007.)

From the foregoing discussion, it is apparent that the baroreceptors on the arterial side of the circulation, their afferent connections to the medullary cardiovascular areas, and the efferent pathways from these areas constitute a reflex feedback mechanism that operates to stabilize blood pressure and heart rate. Any drop in systemic arterial pressure decreases the discharge in the buffer nerves, and there is a compensatory rise in blood pressure and cardiac output. Any rise in pressure produces dilation of the arterioles and decreases cardiac output until the blood pressure returns to its previous baseline level.

Clinical Box 32–1

Essential Hypertension & Neurovascular Compression of the RVLM

In about 88% of patients with elevated blood pressure, the cause of the hypertension is unknown, and they are said to have **essential hypertension** (see Chapter 31). **Neurovascular compression** of the RVLM is associated with essential hypertension in some persons. For example, patients with a schwannoma (acoustic neuroma) or meningioma lying close to the RVLM also have hypertension. Magnetic resonance angiography (MRA) has been used to compare the incidence of neurovascular compression in hypertensive and normotensive individuals and to correlate indices of increased sympathetic nerve activity with the presence or absence of compression. Some of these studies showed a higher incidence of coexistence of neurovascular compression with essential hypertension than in other forms of hypertension or normotension, but others showed the presence of a compression in normotensive subjects. On the other hand, there was a strong positive relationship between the presence of neurovascular compression and increased sympathetic nerve activity.

THERAPEUTIC HIGHLIGHTS

In the 1970s, Dr. Peter Jannetta, a neurosurgeon in Pittsburgh, PA, developed a technique for “microvascular decompression” of the medulla to treat trigeminal neuralgia and hemifacial spasm, which he attributed to pulsatile compression of the vertebral and posterior inferior cerebellar arteries impinging on the fifth and seventh cranial nerves. Moving the arteries away from the nerves led to reversal of the neurologic symptoms in many cases. Some of these patients were also hypertensive, and they showed reductions in blood pressure postoperatively. Later, a few human studies showed that surgical decompression of the RVLM could sometimes relieve hypertension. There are also reports that hypertension is relieved after surgical decompression in patients with a schwannoma or meningioma in the vicinity of the RVLM.

ROLE OF BARORECEPTORS IN SHORT-TERM CONTROL OF BLOOD PRESSURE

The changes in heart rate and blood pressure that occur on standing up or lying down are due for the most part to baroreceptor reflexes. Baroreceptors are very

important in short-term control of arterial pressure. Activation of the reflex allows for rapid adjustments in blood pressure in response to abrupt changes in posture, blood volume, cardiac output, or peripheral resistance during exercise. Removal of the baroreceptor reflex prevents an individual from adjusting their blood pressure in response to stimuli that cause abrupt changes in blood volume, cardiac output, or peripheral resistance, including exercise and postural changes.

In chronic hypertension, the baroreceptor reflex mechanism is “reset” to maintain an elevated rather than a normal blood pressure. In perfusion studies on hypertensive experimental animals, raising the pressure in the isolated carotid sinus lowers the elevated systemic pressure; and decreasing the perfusion pressure raises the elevated pressure. A long-term change in blood pressure resulting from loss of baroreceptor reflex control is called **neurogenic hypertension.**

ATRIAL STRETCH & CARDIOPULMONARY RECEPTORS

The stretch receptors in the atria are of two types: those that discharge primarily during atrial systole (type A), and those that discharge primarily in late diastole, at the time of peak atrial filling (type B). The discharge of type B baroreceptors is increased when venous return is increased and decreased by positive-pressure breathing, indicating that these baroreceptors respond primarily to distension of the atrial walls. The reflex circulatory adjustments initiated by activation of these receptors include vasodilation and a fall in blood pressure. However, the heart rate is increased rather than decreased.

Receptors in the endocardial surfaces of the ventricles are activated during ventricular distension. The response is a vagal bradycardia and hypotension, comparable to a baroreceptor reflex. Left ventricular stretch receptors may play a role in the maintenance of vagal tone that keeps the heart rate low at rest. Various chemicals are known to elicit reflexes due to activation of cardiopulmonary chemoreceptors and may play a role in various cardiovascular disorders **(Clinical Box 32–2)**.

VALSALVA MANEUVER

The function of the receptors can also be tested by monitoring the changes in pulse pressure and blood pressure that occur in response to brief periods of

straining (forced expiration against a closed glottis: the **Valsalva maneuver**). Valsalva maneuvers occur regularly during coughing, defecation, and heavy lifting. The blood pressure rises at the onset of straining **(Figure 32–6)** because the increase in intrathoracic pressure is added to the pressure of the blood in the aorta. It then falls because the high intrathoracic pressure compresses the veins, decreasing venous return and cardiac output. The decreases in arterial pressure and pulse pressure inhibit the baroreceptors, causing tachycardia and a rise in peripheral resistance. When the glottis is opened and the intrathoracic pressure returns to normal, cardiac output is restored but the peripheral vessels are constricted. The blood pressure therefore rises above normal, and this stimulates the baroreceptors, causing bradycardia and a drop in pressure to normal levels.

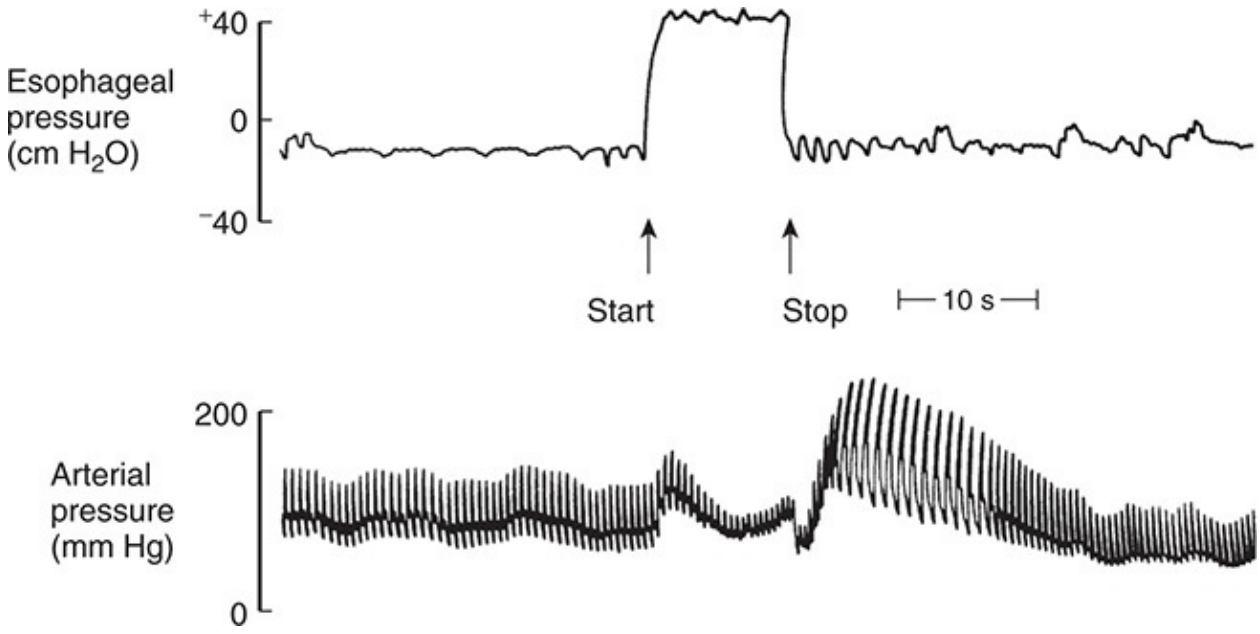


FIGURE 32–6 Diagram of the response to straining (the Valsalva maneuver) in a healthy man, recorded with a needle in the brachial artery. Blood pressure rises at the onset of straining because increased intrathoracic pressure is added to the pressure of the blood in the aorta. It then falls because the high intrathoracic pressure compresses veins, decreasing venous return and cardiac output. (Used with permission of M Mcllroy.)

In patients whose sympathetic nervous system is not functional, heart rate changes still occur because the baroreceptors and the vagi are intact. However, in patients with autonomic insufficiency, a syndrome in which autonomic function is widely disrupted, the heart rate changes are absent. For unknown reasons, patients with primary hyperaldosteronism also fail to show the heart rate changes

and the blood pressure rise when the intrathoracic pressure returns to normal. Their response to the Valsalva maneuver returns to normal after removal of the aldosterone-secreting tumor.

PERIPHERAL CHEMORECEPTOR REFLEX

Peripheral arterial chemoreceptors in the **carotid** and **aortic bodies** are exposed to very high rates of blood flow. These receptors are primarily activated by a reduction in arterial partial pressure of oxygen (PaO_2), but they also respond to an increase in the arterial partial pressure of carbon dioxide ($PaCO_2$) and pH. Chemoreceptors exert their main effects on respiration; however, their activation also leads to vasoconstriction. Heart rate changes are variable and depend on various factors, including changes in respiration. A direct effect of chemoreceptor activation is to increase vagal nerve activity. However, hypoxia also produces hyperpnea and increased catecholamine release from the adrenal medulla, both of which produce tachycardia and an increase in cardiac output. Hemorrhage that produces hypotension leads to chemoreceptor stimulation due to decreased blood flow to the chemoreceptors and consequent stagnant anoxia of these organs. Chemoreceptor activity may also contribute to the production of **Mayer waves,** which are slow, regular oscillations in arterial pressure that occur at the rate of about one every 20–40 s during hypotension. Under these conditions, hypoxia stimulates the chemoreceptors to increase blood pressure; this improves the blood flow in the receptor organs and eliminates the stimulus to the chemoreceptors; blood pressure then falls and a new cycle is initiated. Mayer waves should not be confused with **Traube–Hering waves,** which are fluctuations in blood pressure synchronized with respiration.

Clinical Box 32–2

Cardiopulmonary Chemosensitive Receptors

Activation of chemosensitive vagal C fibers in the cardiopulmonary region (eg, juxtacapillary region of alveoli, ventricles, atria, great veins, and pulmonary artery) causes profound bradycardia, hypotension, and a brief period of apnea followed by rapid shallow breathing. This response pattern is called the **Bezold–Jarisch reflex** and can be elicited by a variety of substances including capsaicin, serotonin, phenylbiguanide, and veratridine.

The Bezold–Jarisch reflex is activated during certain pathophysiologic conditions. For example, this reflex may be activated during myocardial ischemia and reperfusion as a result of increased production of oxygen radicals and by agents used as radiocontrast for coronary angiography. This can contribute to the hypotension that is frequently a stubborn complication of heart disease. Activation of cardiopulmonary chemosensitive receptors may also be part of a defense mechanism protecting individuals from toxic chemical hazards. Activation of cardiopulmonary reflexes may help reduce the amount of inspired pollutants that gets absorbed into the blood, protecting vital organs from potential toxicity of these pollutants, and facilitating their elimination. Finally, the syndrome of cardiac slowing with hypotension **(vasovagal syncope)** has also been attributed to activation of the Bezold–Jarisch reflex. Vasovagal syncope can occur after prolonged upright posture that results in pooling of blood in the lower extremities and diminished intracardiac blood volume (also called **postural syncope**). This phenomenon is exaggerated if combined with dehydration. The resultant arterial hypotension is sensed in the carotid sinus baroreceptors, and afferent fibers from these receptors trigger autonomic signals that increase cardiac rate and contractility. However, pressure receptors in the wall of the left ventricle respond by sending signals that trigger paradoxical bradycardia and decreased contractility, resulting in sudden marked hypotension. The individual also feels lightheaded and may experience a brief episode of loss of consciousness.

THERAPEUTIC HIGHLIGHTS

The most critical intervention for individuals who experience episodes of neurogenic syncope is to avoid dehydration and to avoid situations that trigger the adverse event. Episodes of syncope may be reduced or prevented by an increased dietary salt intake or administration of mineralocorticoids. Vasovagal syncope has been treated with the use of **β-adrenoceptor antagonists** and **disopyramide,** an antiarrhythmic agent that blocks Na^+ channels. Cardiac pacemakers have also been used to stabilize the heart rate during episodes that normally trigger bradycardia.

CENTRAL CHEMORECEPTORS

When intracranial pressure is increased, the blood supply to the brain is compromised, and the local hypercapnia activates central chemoreceptors located along the ventrolateral medullary surface. Their activation induces a rise in systemic arterial pressure **(Cushing reflex)** that tends to restore the blood flow to the brain. Over a considerable range, the blood pressure rise is proportional to the increase in intracranial pressure. The rise in blood pressure causes a reflex decrease in heart rate via the arterial baroreceptors. This is why bradycardia rather than tachycardia is characteristically seen in patients with increased intracranial pressure.

Hypercapnia directly stimulates the RVLM (Table 32–3), but the direct peripheral effect of hypercapnia is vasodilation. Therefore, the peripheral and central actions tend to cancel each other out. Moderate hyperventilation that significantly lowers the CO_2 tension of the blood causes cutaneous and cerebral vasoconstriction, but there is little change in blood pressure. Exposure to high concentrations of CO_2 is associated with marked cutaneous and cerebral vasodilation, but vasoconstriction occurs elsewhere and usually there is a slow rise in blood pressure.

LOCAL REGULATION

AUTOREGULATION

The capacity of tissues to regulate their own blood flow is referred to as **autoregulation.** Most vascular beds have an intrinsic capacity to compensate for moderate changes in perfusion pressure by changes in vascular resistance, so that blood flow remains relatively constant. This capacity is well developed in the kidneys (see Chapter 37), but it has also been observed in the mesentery, skeletal muscle, brain, liver, and myocardium. It is due in part to the intrinsic contractile response of smooth muscle to stretch **(myogenic theory of autoregulation).** As the pressure rises, the blood vessels are distended and the vascular smooth muscle fibers that surround the vessels contract. If it is postulated that the muscle responds to the tension in the vessel wall, this theory explains the greater degree of contraction at higher pressures; the wall tension is proportional to the distending pressure times the radius of the vessel (law of Laplace; see Chapter 31), and the maintenance of a stable wall tension as the pressure rises would require a decrease in radius. Vasodilator substances tend to accumulate in active tissues, and these “metabolites” also contribute to autoregulation **(metabolic theory of autoregulation).** When blood flow decreases, they accumulate and the

vessels dilate; when blood flow increases, they tend to be washed away.

VASODILATOR METABOLITES

The metabolic changes that produce vasodilation include, in most tissues, decreases in O_2 tension and pH. These changes cause relaxation of the arterioles and precapillary sphincters. A local fall in O_2 tension, in particular, can initiate a program of vasodilatory gene expression secondary to production of hypoxia-inducible factor-1α (HIF-1α), a transcription factor with multiple targets. Increases in CO_2 tension and osmolality also dilate the vessels. The direct dilator action of CO_2 is most pronounced in the skin and brain. The neutrally mediated vasoconstrictor effects of systemic, as opposed to local, hypoxia and hypercapnia have been discussed above. A rise in temperature exerts a direct vasodilator effect, and the temperature rise in active tissues (due to the heat of metabolism) may contribute to vasodilation. K^+ is another substance that accumulates locally, and has demonstrated dilator activity secondary to the hyperpolarization of vascular smooth muscle cells. Lactate may also contribute to the dilation. In injured tissues, histamine released from damaged cells increases capillary permeability. Thus, it is probably responsible for some of the swelling in areas of inflammation. Adenosine plays a vasodilator role in cardiac muscle but not in skeletal muscle. It also inhibits the release of norepinephrine.

LOCALIZED VASOCONSTRICTION

Injured arteries and arterioles constrict strongly. The constriction appears to be due in part to the local liberation of serotonin from platelets that stick to the vessel wall in the injured area. Injured veins also constrict.

A drop in tissue temperature causes vasoconstriction, and this local response to cold plays a part in temperature regulation (see Chapter 17).

SUBSTANCES SECRETED BY THE ENDOTHELIUM

ENDOTHELIAL CELLS

As noted in Chapter 31, the endothelial cells constitute a large and important

tissue. They secrete many growth factors and vasoactive substances. The vasoactive substances include prostaglandins and thromboxanes, nitric oxide (NO), and endothelins.

PROSTACYCLIN & THROMBOXANE A_2

Prostacyclin is produced by endothelial cells and thromboxane A_2 by platelets from their common precursor, arachidonic acid, via the cyclooxygenase pathway. Thromboxane A_2 promotes platelet aggregation and vasoconstriction, whereas prostacyclin inhibits platelet aggregation and promotes vasodilation. The balance between thromboxane A_2 and prostacyclin fosters localized platelet aggregation and consequent clot formation (see Chapter 31) while preventing excessive extension of the clot and maintaining blood flow around it.

The thromboxane A_2–prostacyclin balance can be shifted toward prostacyclin by administration of low doses of aspirin. Aspirin produces irreversible inhibition of cyclooxygenase by acetylating a serine residue in its active site. Obviously, this reduces production of both thromboxane A_2 and prostacyclin. However, endothelial cells produce new cyclooxygenase in a matter of hours, whereas platelets cannot manufacture the enzyme, and the level rises only as new platelets enter the circulation. This is a slow process because platelets have a half-life of about 4 days. Therefore, administration of small amounts of aspirin for prolonged periods reduces clot formation and has been shown to be of value in preventing myocardial infarctions, unstable angina, transient ischemic attacks, and stroke.

NITRIC OXIDE

Many different stimuli act on endothelial cells to produce **endothelium-derived relaxing factor (EDRF),** a substance that is now known to be **nitric oxide.** NO is synthesized from arginine **(Figure 32–7)** in a reaction catalyzed by nitric oxide synthase (NO synthase, NOS). Three isoforms of NOS have been identified: NOS 1, found in the nervous system; NOS 2, found in macrophages and other immune cells; and NOS 3, found in endothelial cells. NOS 1 and NOS 3 are activated by agents that increase intracellular Ca^{2+} concentrations, including the vasodilators acetylcholine and bradykinin. The NOS in immune cells is not activated by Ca^{2+} but is induced by cytokines. NO formed in the

endothelium diffuses to smooth muscle cells, where it activates soluble guanylyl cyclase, producing cyclic 3,5-guanosine monophosphate (cGMP; see Figure 32–7), which in turn mediates the relaxation of vascular smooth muscle. NO is short-lived and is inactivated by hemoglobin.

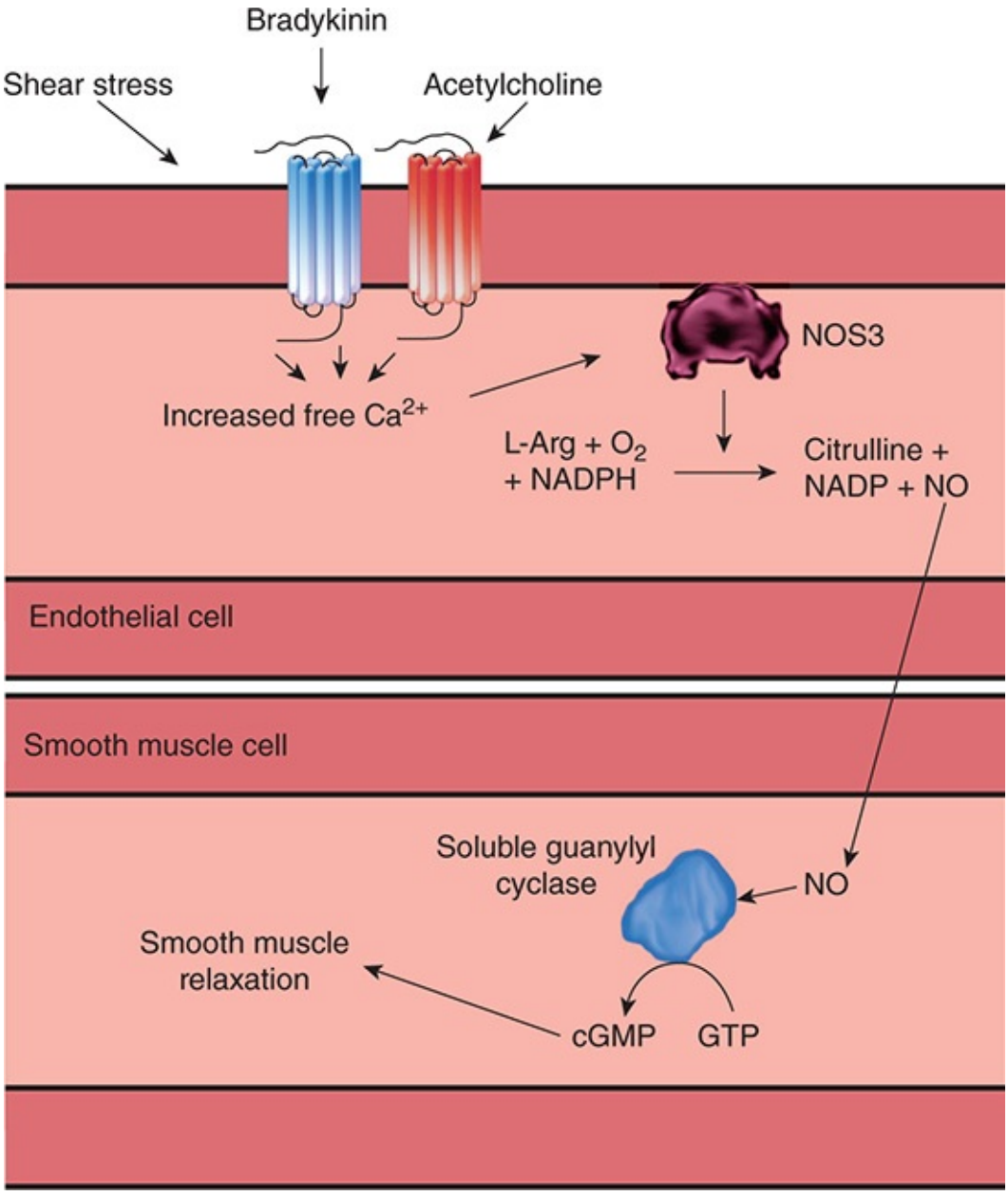


FIGURE 32–7 Endothelial-derived NO leads to smooth muscle relaxation. The endothelial form of nitric oxide synthase (NOS3) is activated by increased free Ca^{2+} and can be induced by a variety of extracellular signals. NOS3 acts on L-arginine, NADPH, and O_2 to yield citrulline, NADP, and NO. A variety of

cofactors are required for this reaction (not shown). NO can diffuse to adjacent smooth muscle cells where it activates soluble guanylyl cyclase. This leads to the production of cGMP and subsequent smooth muscle relaxation.

Adenosine, atrial natriuretic peptide (ANP), and histamine via H_2 receptors produce relaxation of vascular smooth muscle that is independent of the endothelium. However, acetylcholine, histamine via H_1 receptors, bradykinin, vasoactive intestinal peptide (VIP), substance P, and some other polypeptides act via the endothelium, and various vasoconstrictors that act directly on vascular smooth muscle would produce much greater constriction if their effects were not limited by their ability simultaneously to cause release of NO. When flow to a tissue is suddenly increased by arteriolar dilation, the large arteries to the tissue also dilate. This flow-induced dilation is due to local release of NO. Products of platelet aggregation also cause release of NO, and the resulting vasodilation helps keep blood vessels with an intact endothelium patent. This is in contrast to injured blood vessels, where the endothelium is damaged at the site of injury and platelets therefore aggregate and produce vasoconstriction (see Chapter 31).

Further evidence for a physiologic role of NO is the observation that mice lacking NOS 3 are hypertensive. This suggests that tonic release of NO is necessary to maintain normal blood pressure.

NO is also involved in vascular remodeling and angiogenesis, and NO may be involved in the pathogenesis of atherosclerosis. It is interesting in this regard that in some patients with heart transplants, an accelerated form of atherosclerosis develops in the vessels of the transplant, and there is reason to believe that this is triggered by endothelial damage. Nitroglycerin and other nitrovasodilators that are of great value in the treatment of angina act by stimulating guanylyl cyclase in the same manner as NO.

Penile erection is also produced by release of NO, with consequent vasodilation and engorgement of the corpora cavernosa (see Chapter 23). This accounts for the efficacy of drugs such as Viagra that slow the breakdown of cGMP.

OTHER FUNCTIONS OF NO

NO is present in the brain and, acting via cGMP, it is important in brain function (see Chapter 7). NO is also necessary for the antimicrobial and cytotoxic activity of various inflammatory cells, although the net effect of NO in inflammation and tissue injury depends on the amount and kinetics of release, which in turn may

depend on the specific NOS isoform involved. In the gastrointestinal tract, NO is important in the relaxation of smooth muscle. Other functions of NO are mentioned in other parts of this book.

CARBON MONOXIDE

The production of carbon monoxide (CO) from heme is shown in Figure 28–4. Heme oxygenase-2 (HO-2), the enzyme that catalyzes the reaction, is also present in cardiovascular tissues, and there is growing evidence that CO as well as NO produces local dilation in blood vessels. Interestingly, hydrogen sulfide is likewise emerging as a third gas transmitter that regulates vascular tone, although the relative roles of NO, CO, and H_2S have yet to be established.

ENDOTHELINS

Endothelial cells also produce **endothelin-1,** one of the most potent vasoconstrictor agents yet isolated. Endothelin-1 (ET-1), endothelin-2 (ET-2), and endothelin-3 (ET-3) are the members of a family of three similar 21-amino-acid polypeptides. Each is encoded by a different gene.

ENDOTHELIN-1

In endothelial cells, the product of the endothelin-1 gene is processed to a 39-amino-acid prohormone, **big endothelin-1,** which has about 1% of the activity of endothelin-1. The prohormone is cleaved by **endothelin-converting enzyme** at a tryptophan-valine (Trp-Val) bond to form endothelin-1. Small amounts of big endothelin-1 and endothelin-1 are secreted into the blood, but for the most part, they are secreted locally and act in a paracrine fashion.

Two different endothelin receptors have been cloned, both of which are coupled via G-proteins to phospholipase C (see Chapter 2). The ET_A receptor, which is specific for endothelin-1, is found in many tissues and mediates the vasoconstriction produced by endothelin-1. The ET_B receptor responds to all three endothelins and is coupled to G_i. It may mediate vasodilation, and it appears to mediate the developmental effects of the endothelins (see below).

REGULATION OF SECRETION

Endothelin-1 is not stored in secretory granules, and most regulatory factors alter the transcription of its gene, with changes in secretion occurring promptly thereafter. Factors activating and inhibiting the gene are summarized in **Table 32–4**.

TABLE 32–4 Regulation of endothelin-1 secretion via transcription of its gene.

Stimulators
Angiotensin II
Catecholamines
Growth factors
Hypoxia
Insulin
Oxidized LDL
HDL
Shear stress
Thrombin
Inhibitors
NO
ANP
PGE_2
Prostacyclin

ANP, atrial natriuretic peptide; HDL, high-density lipoprotein; LDL, low-density lipoprotein; NO, nitric oxide; PGE_2, prostaglandin E_2; VIP, vasoactive intestinal polypeptide.

CARDIOVASCULAR FUNCTIONS

As noted above, endothelin-1 appears primarily to be a paracrine regulator of vascular tone. However, endothelin-1 is not increased in hypertension, and in mice in which one allele of the endothelin-1 gene is knocked out, blood pressure

is actually elevated rather than reduced. The concentration of circulating endothelin-1 is, however, elevated in heart failure and after myocardial infarction, so it may play a role in the pathophysiology of these diseases.

OTHER FUNCTIONS OF ENDOTHELINS

Endothelin-1 is found in the brain and kidneys as well as the endothelial cells. Endothelin-2 is produced primarily in the kidneys and intestine. Endothelin-3 is present in the blood and is found in high concentrations in the brain. It is also found in the kidneys and gastrointestinal tract. In the brain, endothelins are abundant and, in early life, are produced by both astrocytes and neurons. They are found in the dorsal root ganglia, ventral horn cells, the cortex, the hypothalamus, and cerebellar Purkinje cells. They also play a role in regulating transport across the blood-brain barrier. There are endothelin receptors on mesangial cells (see Chapter 37), and the polypeptide participates in tubuloglomerular feedback.

SYSTEMIC REGULATION BY NEUROHUMORAL AGENTS

Many circulating substances affect the vascular system. The vasodilator regulators include kinins, VIP, and ANP. Circulating vasoconstrictor hormones include vasopressin, norepinephrine, epinephrine, and angiotensin II.

KININS

Two related vasodilator peptides called **kinins** are found in the body. One is the nonapeptide **bradykinin,** and the other is the decapeptide **kallidin,** also known as **lysylbradykinin** **(Figure 32–8)**. Kallidin can be converted to bradykinin by aminopeptidase. Both peptides are metabolized to forms that are active at the type 1 bradykinin receptor by **kininase I,** a carboxypeptidase that removes the carboxyl terminal arginine (Arg). In addition, the dipeptidylcarboxypeptidase **kininase II** inactivates bradykinin and kallidin by removing phenylalanine-arginine (Phe-Arg) from the carboxyl terminal as well as additional cleavages. Kininase II is the same enzyme as **angiotensin-converting enzyme (ACE),** which removes histidine-leucine (His-Leu) from the carboxyl terminal end of angiotensin I.

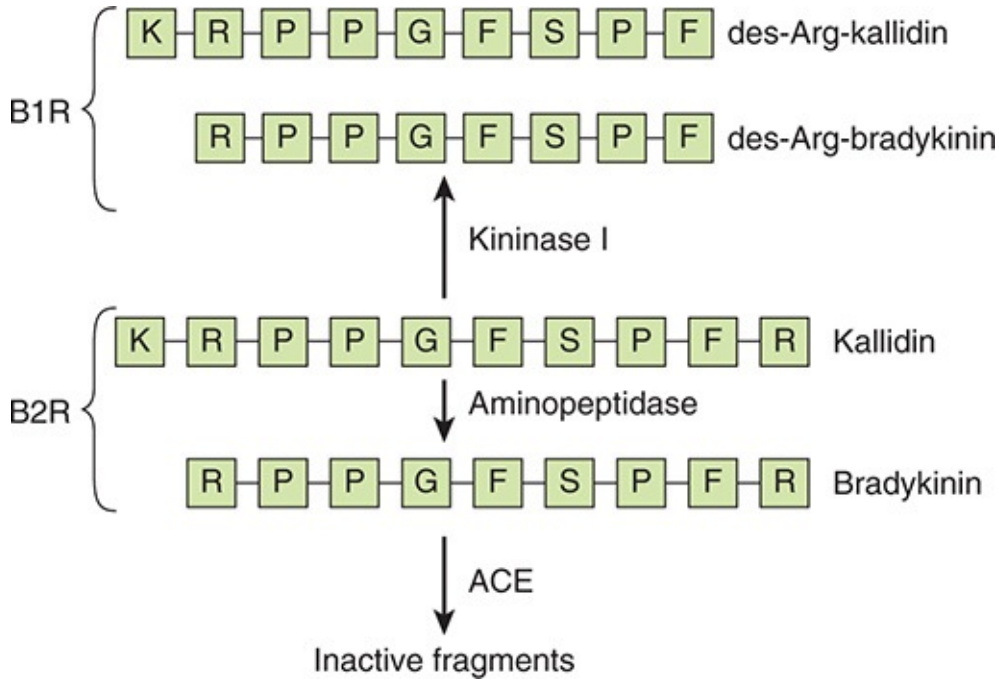


FIGURE 32–8 Kinins. Kallidin (lysylbradykinin) can be converted to bradykinin by aminopeptidase. Both kallidin and bradykinin act at type 2 bradykinin receptors (B2R). The peptides are converted by kininase I to their des-Arg derivatives that are ligands for type 1 bradykinin receptors (B1R). They can also be cleaved by angiotensin-converting enzyme (ACE, also known as kininase II) to shorter fragments that lack biologic activity.

Bradykinin and kallidin are formed from two precursor proteins: **high-molecular-weight kininogen** and **low-molecular-weight kininogen** **(Figure 32–9)**. They are formed by alternative splicing of a single gene located on chromosome 3. Proteases called **kallikreins** release the peptides from their precursors. There are two types of kallikreins: **plasma kallikrein,** which circulates in an inactive form, and **tissue kallikrein,** which appears to be located primarily on the apical membranes of cells concerned with transcellular electrolyte transport. Tissue kallikrein is found in many tissues, including sweat and salivary glands, the pancreas, the prostate, the intestine, and the kidneys. Tissue kallikrein acts on high-molecular-weight kininogen to form bradykinin and low-molecular-weight kininogen to form kallidin. When activated, plasma kallikrein acts on high-molecular-weight kininogen to form bradykinin.

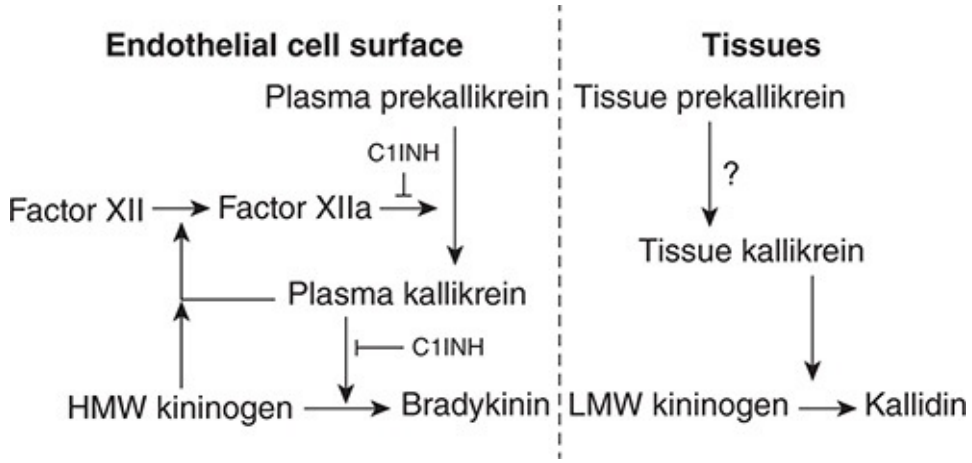


FIGURE 32–9 Formation of kinins from high-molecular-weight (HMW) and low-molecular-weight (LMW) kininogens. Note that the formation of bradykinin is subject to a positive feedback loop that can be interrupted in two places by C1-esterase inhibitor (C1INH).

Inactive plasma kallikrein **(prekallikrein)** is converted to the active form, kallikrein, by active factor XII, the factor that initiates the intrinsic blood clotting cascade. Kallikrein also activates factor XII in a positive feedback loop, and high-molecular-weight kininogen has a factor XII–activating action (see Figure 31–12). This system is normally kept in check by the C1-esterase inhibitor (C1INH), which is a member of the serpin protein family that inhibits several enzymes in the clotting cascade by binding irreversibly to their active sites.

The actions of both kinins resemble those of histamine. They are primarily paracrines, although small amounts are also found in the circulating blood. They cause contraction of visceral smooth muscle, but they relax vascular smooth muscle via NO, lowering blood pressure. They also significantly increase vascular permeability resulting in edema, attract leukocytes, and cause pain upon injection under the skin. They are formed during active secretion in sweat glands, salivary glands, and the exocrine portion of the pancreas, and they are probably responsible for the increase in blood flow when these tissues are actively secreting their products. In keeping with the scheme for bradykinin production outlined above, mutations in C1INH lead to the rare bradykinin-mediated swelling disorder known as hereditary angioedema.

Two bradykinin receptors, B1R and B2R, have been identified. The B1R may mediate the pain-producing effects of the kinins, and antagonists have entered clinical trials for the treatment of pain and inflammation. The B2 receptor has strong homology to the histamine H_2-receptor and is found in many different

tissues.

NATRIURETIC HORMONES

A family of natriuretic peptides is also involved in vascular regulation, including ANP secreted by the heart, B-type natriuretic peptide (BNP), and C-type natriuretic peptide (CNP). They are released in response to hypervolemia. ANP and BNP circulate, whereas CNP acts predominantly in a paracrine fashion. In general, these peptides antagonize the action of various vasoconstrictor agents and lower blood pressure. ANP and BNP also serve to coordinate the control of vascular tone with fluid and electrolyte homeostasis via actions on the kidney.

CIRCULATING VASOCONSTRICTORS

Vasopressin is a potent vasoconstrictor, but when it is injected in healthy individuals, there is a compensating decrease in cardiac output, so that there is little change in blood pressure. Its role in blood pressure regulation is discussed in Chapter 17.

Norepinephrine has a generalized vasoconstrictor action, whereas epinephrine dilates the vessels in skeletal muscle and the liver. The relative unimportance of circulating norepinephrine, as opposed to norepinephrine released from vasomotor nerves, is pointed out in Chapter 19, where the cardiovascular actions of catecholamines are discussed in detail.

Angiotensin II has a generalized vasoconstrictor action. It is formed by the action of ACE on angiotensin I, which itself is liberated by the action of renin from the kidney on circulating angiotensinogen (see Chapter 38). Renin secretion, in turn, is increased when the blood pressure falls or extracellular fluid (ECF) volume is reduced, and angiotensin II therefore helps maintain blood pressure. Angiotensin II also increases water intake and stimulates aldosterone secretion, and increased formation of angiotensin II is part of a homeostatic mechanism that operates to maintain ECF volume (see Chapter 19). In addition, there are renin–angiotensin systems in many different organs, including the walls of blood vessels. Angiotensin II produced in blood vessel walls could be important in some forms of clinical hypertension. The role of angiotensin II in cardiovascular regulation is also amply demonstrated in the widespread use of ACE inhibitors as antihypertensive medications.

Urotensin-II, a polypeptide first isolated from the spinal cord of fish, is

present in human cardiac and vascular tissue. It is one of the most potent mammalian vasoconstrictors known, and is being explored for its role in a large range of different human disease states. For example, levels of both urotensin-II and its receptor have been shown to be elevated in hypertension and heart failure, and may be markers of disease in these and other conditions.

CHAPTER SUMMARY

- The vasculature is innervated by sympathetic nerves that mediate vasoconstriction; the heart is innervated both by sympathetic nerves that mediate an increase in heart rate, AV conduction, and ventricular contractility and by parasympathetic (vagal) nerves that produce a reduction in heart rate and conduction through the AV node.
- RVLM neurons project to the thoracolumbar IML and release glutamate on preganglionic sympathetic neurons that innervate the heart and vasculature. The NTS is the major excitatory input to cardiac vagal motor neurons in the nucleus ambiguus. RVLM and NTS neurons receive input from other brain regions, baroreceptors, chemoreceptors, and other sources.
- Carotid sinus and aortic depressor baroreceptors are innervated by branches of the ninth and tenth cranial nerves, respectively (glossopharyngeal and aortic depressor nerves). These receptors are most sensitive to changes in pulse pressure but also respond to changes in mean arterial pressure. Baroreceptor nerves terminate in the NTS and release glutamate. NTS neurons project to the CVLM and nucleus ambiguus and release glutamate. CVLM neurons project to RVLM and release GABA. This leads to a reduction in sympathetic activity and an increase in vagal activity (ie, the baroreceptor reflex).
- The reflex circulatory adjustments initiated by activation of atrial stretch receptors include a reduction in blood pressure and an increase in heart rate. Activation of ventricular stretch receptors induces hypotension and bradycardia. Activation of cardiopulmonary chemoreceptors induces the Bezold–Jarisch reflex that plays a role in various cardiovascular disorders (eg, myocardial ischemia and reperfusion and vasovagal syncope).
- Activation of peripheral chemoreceptors in the carotid and aortic bodies, primarily by a reduction in PaO_2, leads to an increase in vasoconstriction. Heart rate changes are variable and depend on various factors including changes in respiration. Central chemoreceptors are activated primarily by

hypercapnia; activation leads to an increase in blood pressure and a baroreceptor reflex-induced bradycardia.

- Most vascular beds have a capacity to respond to changes in blood pressure within a certain range by altering vascular resistance to maintain stable blood flow. This property is known as autoregulation and is related to intrinsic properties of the vascular smooth muscle as well as the actions of local vasoactive metabolites.
- Local factors such as oxygen tension, pH, temperature, and metabolic products contribute to vascular regulation; many produce vasodilation to restore blood flow.
- The endothelium is an important source of vasoactive mediators that act to either contract or relax vascular smooth muscle. In particular, three gaseous mediators—NO, CO, and H_2S—are important regulators of vasodilation.
- Endothelins and angiotensin II induce vasoconstriction and may be involved in the pathogenesis of some forms of hypertension.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. When a pheochromocytoma (tumor of the adrenal medulla) suddenly discharges a large amount of epinephrine into the circulation, the patient's heart rate would be expected to

   A. increase because the increase in blood pressure stimulates the carotid and aortic baroreceptors.

   B. increase because epinephrine has a direct chronotropic effect on the heart.

   C. increase because of increased tonic parasympathetic discharge to the heart.

   D. decrease because the increase in blood pressure stimulates the carotid and aortic chemoreceptors.

   E. decrease because of increased tonic parasympathetic discharge to the heart.

2. A 45-year-old woman had a blood pressure of 155/95 mm Hg when she was at her physician's office for a physical. She was worried as this was her first physical in over 10 years. The clinician suggested that she begin monitoring her blood pressure at home. Stress-induced activation of which neurons might contribute to the higher BP while in her physician's office?

   A. Limbic cortex and NTS

B. Hypothalamus and CVLM

C. NTS and CVLM

D. Hypothalamus and RVLM

E. Limbic cortex and nucleus ambiguus

3. Which group of neurons within central autonomic pathways release glutamate on their postsynaptic targets?

A. CVLM neurons projecting to the IML and NTS neurons projecting to the nucleus ambiguus.

B. RVLM neurons projecting to the IML and NTS neurons projecting to the CVLM.

C. NTS neurons projecting to the nucleus ambiguus and CVLM neurons projecting to the RVLM.

D. NTS neurons projecting to the RVLM and CVLM neurons projecting to the RVLM.

E. CVLM neurons projecting to the NTS and RVLM neurons projecting to the IML.

4. List the location of the arterial baroreceptors, their afferent innervation, and the central site of termination of the afferent fibers.

A. Atria and pulmonary veins, vagus nerve, and nucleus ambiguus

B. Carotid body and aortic body, glossopharyngeal nerve and vagus nerve, and NTS

C. Carotid sinus and aortic arch, carotid sinus nerve and aortic depressor nerve, and NTS

D. Carotid sinus and atria, glossopharyngeal and vagus nerves, and nucleus ambiguus

E. Carotid body and ventricle, dorsal root fibers, and RVLM

5. A 30-year-old woman experienced profound bradycardia, hypotension, and a brief period of apnea followed by rapid shallow breathing. This description is consistent with the activation of which of the following reflexes?

A. Cushing reflex

B. Central chemoreceptor reflex

C. Peripheral chemoreceptor reflex

D. Atrial stretch receptor reflex

E. Bezold–Jarisch reflex

6. A 53-year-old woman with chronic lung disease was experiencing difficulty

breathing. Her PaO_2 and $PaCO_2$ were 50 mm Hg and 60 mm Hg, respectively. Peripheral chemoreceptors are very sensitive to ____ and central chemoreceptors are very sensitive to ____. Fill in the missing information

A. Small increases in $PaCO_2$ and small decreases in local partial pressure of O_2.

B. Small decreases in $PaCO_2$ and small increases in local partial pressure of O_2.

C. Small decreases in PaO_2 and small increases in local partial pressure of CO_2.

D. Small increases in PaO_2 and small decreases in local partial pressure of CO_2.

E. Small decreases in arterial pH and small increases in local pH.

7. Due to a partial vascular occlusion, a portion of the small bowel becomes hypoxic. Factors that would likely contribute to an adaptive increase in blood flow in the affected segment would include all of the following except

A. HIF-1α

B. Increased local tension of CO_2

C. Relaxation of precapillary sphincters

D. Increased local concentration of K^+

E. Changes in gene expression

8. A 55-year-old man comes to his primary care clinician complaining of erectile dysfunction. He is given a prescription for Viagra, and on follow-up, reports that his ability to sustain an erection has been improved markedly by this treatment. The action of which of the following vasoactive mediators would primarily be increased in this patient?

A. Histamine

B. Endothelin-1

C. Prostacyclin

D. Nitric oxide

E. Atrial natriuretic peptide

9. Why is the dilator response to injected acetylcholine changed to a constrictor response when the endothelium is damaged?

A. More Na^+ is generated.

B. More bradykinin is generated.
C. The damage lowers the pH of the remaining layers of the artery.
D. The damage augments the production of endothelin by the endothelium.
E. The damage interferes with the production of NO by the endothelium.

CHAPTER 33

Circulation Through Special Regions

OBJECTIVES

After studying this chapter, you should be able to:

- Define the special features of the circulation in the brain. Describe how cerebrospinal fluid (CSF) is formed and reabsorbed, and its role in protecting the brain from injury. Understand how the blood-brain barrier impedes the entry of specific substances into the brain. Explain how cerebral blood flow can be measured, how it is regulated by intracranial pressure, and how it protects the supply of oxygen to the brain.
- Understand the anatomy of the coronary circulation of the heart and how the cardiac cycle regulates flow in the coronary blood vessels. Delineate how the oxygen needs of the contracting myocardium are met by the coronary arteries, and the consequences of their occlusion.
- Describe how the cutaneous circulation participates in heat loss from the body and how this function is regulated. List the vascular reactions of the skin and the reflexes that mediate them.
- Describe the uterine circulation and how it changes during the course of pregnancy and parturition. Understand how the fetus is supplied with oxygen and nutrients in utero, and the circulatory events required for a transition to independent life after birth.

INTRODUCTION

The distribution of the cardiac output to various parts of the body at rest in a normal man is shown in **Table 33–1**. The general principles described in preceding chapters apply to the circulation of all these regions, but the vascular supplies of many organs have additional special features that are important to their physiology. This chapter is concerned with the special circulations of the brain, the heart, and the skin, as well as the placenta and fetus. In addition, the portal circulation of the anterior pituitary is discussed in Chapter 18; the pulmonary circulation in Chapter 34; the renal circulation in Chapter 37; and the circulation of the splanchnic area, particularly the intestines and liver, in Chapters 25 and 28.

TABLE 33–1 Resting blood flow and O_2 consumption of various organs in a 63-kg adult man with a mean arterial blood pressure of 90 mm Hg and an O_2 consumption of 250 mL/min.

Region	Mass (kg)	Blood Flow		Arteriovenous Oxygen Difference (mL/L)	Oxygen Consumption		Resistance (R units)[a]		Percentage of Total	
		mL/min	mL/100 g/min		mL/min	mL/100 g/min	Absolute	per kg	Cardiac Output	Oxygen Consumption
Liver	2.6	1500	57.7	34	51	2.0	3.6	9.4	27.8	20.4
Kidneys	0.3	1260	420.0	14	18	6.0	4.3	1.3	23.3	7.2
Brain	1.4	750	54.0	62	46	3.3	7.2	10.1	13.9	18.4
Skin	3.6	462	12.8	25	12	0.3	11.7	42.1	8.6	4.8
Skeletal muscle	31.0	840	2.7	60	50	0.2	6.4	198.4	15.6	20.0
Heart muscle	0.3	250	84.0	114	29	9.7	21.4	6.4	4.7	11.6
Rest of body	23.8	336	1.4	129	44	0.2	16.1	383.2	6.2	17.6
Whole body	63.0	5400	8.6	46	250	0.4	1.0	63.0	100.0	100.0

[a]R units are pressure (mm Hg) divided by blood flow (mL/s).

Reproduced with permission from Bard P (editor): *Medical Physiology,* 11th ed. Mosby, 1961.

CEREBRAL CIRCULATION: ANATOMIC CONSIDERATIONS

VESSELS

The principal arterial inflow to the brain in humans is via four arteries: two internal carotid arteries and two vertebral arteries. In humans, the carotid arteries are quantitatively the most significant. The vertebral arteries unite to form the basilar artery, and the basilar artery and the carotids form the **circle of Willis** below the hypothalamus. The circle of Willis is the origin of the six large vessels

supplying the cerebral cortex. Substances injected into one carotid artery are distributed almost exclusively to the cerebral hemisphere on that side. Normally no crossing over occurs, probably because the pressure is equal on both sides. Even when it is not, the anastomotic channels in the circle do not permit a very large flow. Occlusion of one carotid artery, particularly in older patients, often causes serious symptoms of cerebral ischemia. There are precapillary anastomoses between the cerebral vessels, but flow through these channels is generally insufficient to maintain the circulation and prevent infarction when a cerebral artery is occluded.

Venous drainage from the brain by way of the deep veins and dural sinuses empties principally into the internal jugular veins in humans, although a small amount of venous blood drains through the ophthalmic and pterygoid venous plexuses, through emissary veins to the scalp, and down the system of paravertebral veins in the spinal canal.

The cerebral vessels have a number of unique anatomic features. In the choroid plexuses, there are gaps between the endothelial cells of the capillary wall, but the choroid epithelial cells that separate them from the cerebrospinal fluid (CSF) are connected to one another by tight junctions. The capillaries in the brain substance resemble the nonfenestrated capillaries found in muscle (see Chapter 31), and there are tight junctions between the endothelial cells that limit the passage of substances via the paracellular route. In addition, there are relatively few vesicles in the endothelial cytoplasm, and presumably little vesicular transport. However, multiple transport systems are present in the capillary cells. The brain capillaries are surrounded by the endfeet of astrocytes **(Figure 33–1)**. These endfeet are closely applied to the basal lamina of the capillaries, but they do not cover the entire capillary wall, and gaps of about 20 nm occur between endfeet **(Figure 33–2)**. However, the endfeet induce formation of tight junctions in the capillaries (see Chapter 31).

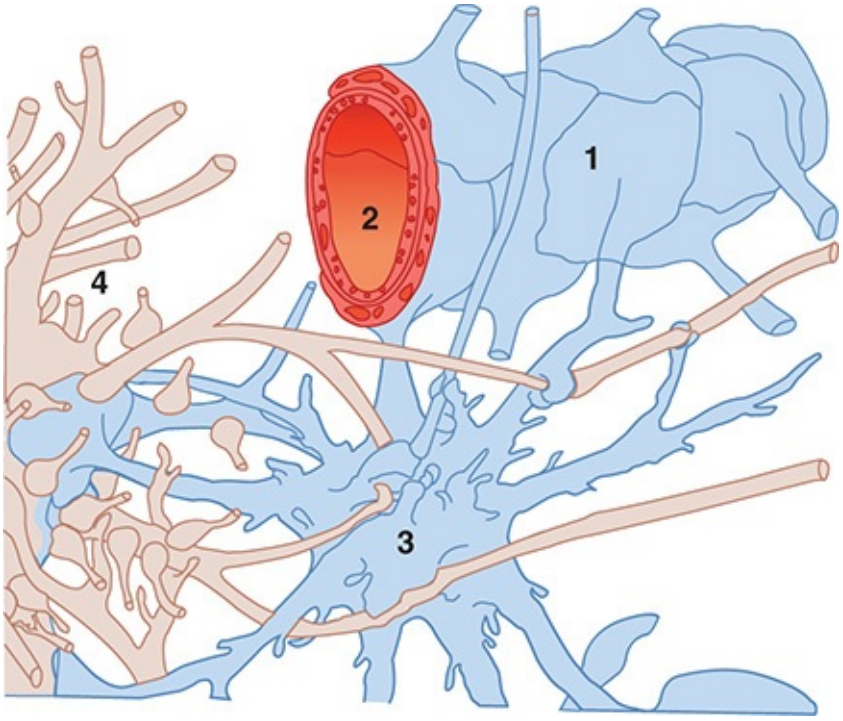


FIGURE 33–1 Relation of fibrous astrocyte (3) to a capillary (2) and neuron (4) in the brain. The endfeet of the astrocyte processes form a discontinuous membrane around the capillary (1). Astrocyte processes also envelop the neuron. (Adapted with permission from Krstic RV: *Die Gewebe des Menschen und der Säugetiere*. Springer; 1978.)

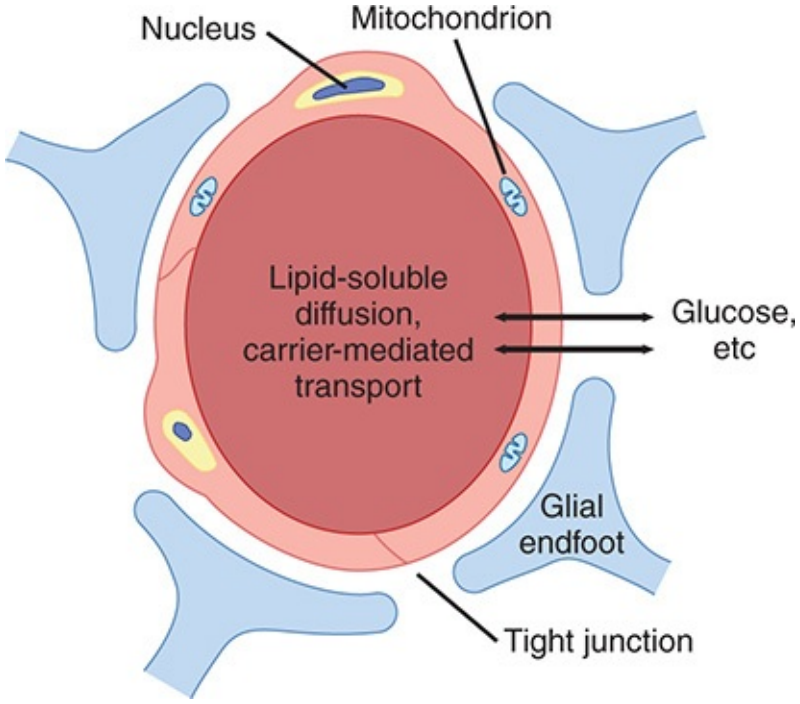


FIGURE 33–2 Transport across cerebral capillaries. Only free lipid-soluble substances can move passively across the endothelial cells. Water-soluble solutes, such as glucose, require active transport mechanisms. Proteins and protein-bound lipids are excluded.

INNERVATION

Three systems of nerves innervate the cerebral blood vessels. Postganglionic sympathetic neurons have their cell bodies in the superior cervical ganglia, and their endings contain norepinephrine. Many also contain neuropeptide Y. Cholinergic neurons that probably originate in the sphenopalatine ganglia also innervate the cerebral vessels, and the postganglionic cholinergic neurons acting on the blood vessels contain acetylcholine. Many also contain vasoactive intestinal peptide (VIP) (see Chapter 7) and peptide histidyl methionine (PHM-27). These nerves end primarily on large arteries. Sensory nerves are found on more distal arteries. They have their cell bodies in the trigeminal ganglia and contain substance P, neurokinin A, and calcitonin gene-related peptide (CGRP). Substance P, CGRP, VIP, and PHM-27 cause vasodilation, whereas neuropeptide Y is a vasoconstrictor. Touching or pulling on the cerebral vessels causes pain.

CEREBROSPINAL FLUID

FORMATION & ABSORPTION

CSF fills the ventricles and subarachnoid space. In humans, the volume of CSF is about 150 mL, and the rate of CSF production is about 550 mL/d. Thus, the CSF turns over about 3.7 times a day. It has been estimated that 50–70% of the CSF is formed in the choroid plexuses and the remainder is formed around blood vessels and along the ventricular walls. The CSF in the ventricles flows through the foramens of Magendie and Luschka to the subarachnoid space and is absorbed through the **arachnoid villi** into veins, primarily the cerebral venous sinuses. The villi consist of projections of the fused arachnoid membrane and endothelium of the sinuses into low pressure venous sinuses. Similar, smaller villi project into veins around spinal nerve routes. These projections may contribute to the outflow of CSF into venous blood by a process known as **bulk flow,** which is unidirectional. An additional and perhaps more important route for CSF reabsorption into the bloodstream in health is via the cribriform plate above the nose and thence into the cervical lymphatics. However, reabsorption

via one-way valves (of uncertain structural basis) in the arachnoid villi may assume a greater role if CSF pressure is elevated. Likewise, when CSF builds up abnormally, aquaporin water channels may be expressed in the choroid plexus and brain microvessels as a compensatory adaptation.

CSF is formed continuously by the choroid plexus in two stages. First, plasma is passively filtered across the choroidal capillary endothelium. Next, secretion of water and ions across the choroidal epithelium provides for active control of CSF composition and quantity. Bicarbonate, chloride, and potassium ions enter the CSF via channels in the epithelial cell apical membranes. Aquaporins provide for water movement to balance osmotic gradients. The composition of CSF **(Table 33–2)** is essentially the same as that of brain extracellular fluid (ECF), which in living humans makes up 15% of the brain volume. In adults, free communication appears to take place between the brain interstitial fluid and CSF, although the diffusion distances from some parts of the brain to the CSF are appreciable. Consequently, equilibration may take some time to occur, and local areas of the brain may have extracellular microenvironments that are transiently different from CSF.

TABLE 33–2 Concentration of various substances in human cerebrospinal fluid (CSF) and plasma.

	Units	CSF	Plasma	Ratio CSF/Plasma
Na^+	(mEq/kg H_2O)	147.0	150.0	0.98
K^+	(mEq/kg H_2O)	2.9	4.6	0.62
Mg^{2+}	(mEq/kg H_2O)	2.2	1.6	1.39
Ca^{2+}	(mEq/kg H_2O)	2.3	4.7	0.49
Cl^-	(mEq/kg H_2O)	113.0	99.0	1.14
HCO_3^-	(mEq/L)	25.1	24.8	1.01
PCO_2	(mm Hg)	50.2	39.5	1.28
pH		7.33	7.40	…
Osmolality	(mOsm/kg H_2O)	289.0	289.0	1.00
Protein	(mg/dL)	20.0	6000.0	0.003
Glucose	(mg/dL)	64.0	100.0	0.64
Inorganic phosphate	(mg/dL)	3.4	4.7	0.73
Urea	(mg/dL)	12.0	15.0	0.80
Creatinine	(mg/dL)	1.5	1.2	1.25
Uric acid	(mg/dL)	1.5	5.0	0.30
Cholesterol	(mg/dL)	0.2	175.0	0.001

Lumbar CSF pressure is normally 70–180 mm H_2O. Up to pressures well above this range, the rate of CSF formation is independent of intraventricular pressure. However, absorption is proportional to the pressure **(Figure 33–3)**. At

a pressure of 112 mm H_2O, which is the average normal CSF pressure, formation and absorption are equal. Below a pressure of approximately 68 mm H_2O, absorption stops. Large amounts of fluid accumulate when the capacity for CSF reabsorption is decreased **(external hydrocephalus, communicating hydrocephalus).** Fluid also accumulates proximal to the block and distends the ventricles when the foramens of Luschka and Magendie are blocked or there is obstruction within the ventricular system **(internal hydrocephalus, noncommunicating hydrocephalus).**

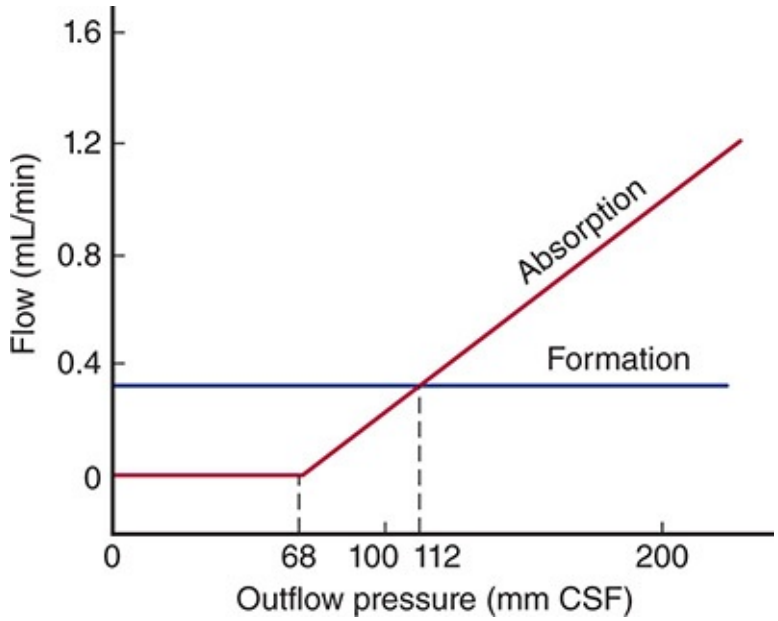


FIGURE 33–3 CSF formation and absorption in humans at various CSF pressures. Note that at 112 mm CSF, formation and absorption are equal, and at 68 mm CSF, absorption is zero. CSF, cerebrospinal fluid. (Modified with permission from Cutler RWP, et al: Formation and absorption of cerebrospinal fluid in man. Brain 1968;91(4):707-720.)

PROTECTIVE FUNCTION

The most critical role for CSF (and the meninges) is to protect the brain. The dura is attached firmly to bone. Normally, there is no "subdural space," with the arachnoid being held to the dura by the surface tension of the thin layer of fluid between the two membranes. As shown in **Figure 33–4**, the brain itself is supported within the arachnoid by the blood vessels and nerve roots and by the multiple fine fibrous **arachnoid trabeculae.** The brain weighs about 1400 g in

air, but in its “water bath” of CSF it has a net weight of only 50 g. The buoyancy of the brain in the CSF permits its relatively flimsy attachments to suspend it very effectively. When the head receives a blow, the arachnoid slides on the dura and the brain moves, but its motion is gently checked by the CSF cushion and by the arachnoid trabeculae.

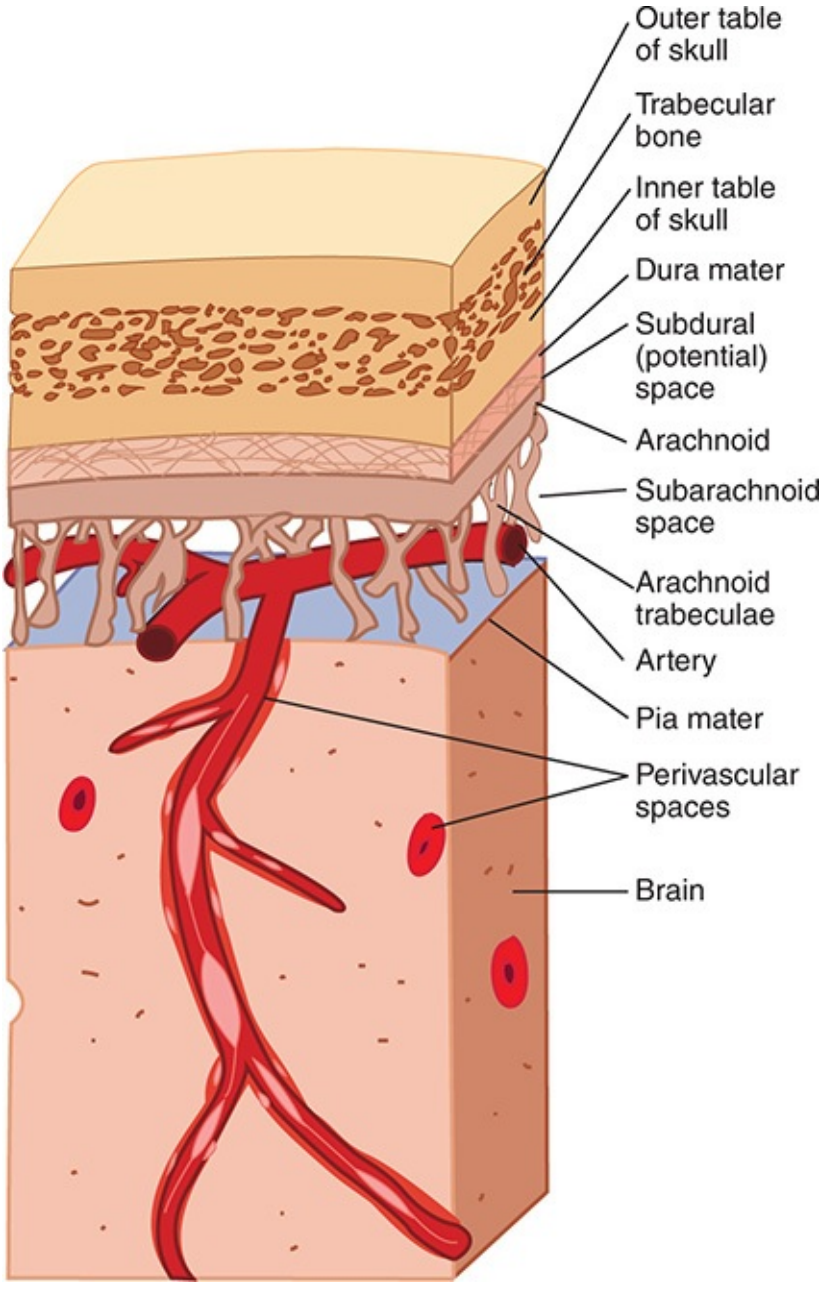


FIGURE 33–4 Investing membranes of the brain, showing their relation to the skull and to brain tissue. (Reproduced with permission from Young B, Heath JW: *Wheater’s Functional Histology*, 4th ed. Churchill Livingstone;

2000.)

The pain produced by spinal fluid deficiency illustrates the importance of CSF in supporting the brain. Removal of CSF during lumbar puncture can cause a severe headache after the fluid is removed, because the brain hangs on the vessels and nerve roots, and traction on them stimulates pain fibers. The pain can be relieved by intrathecal injection of sterile isotonic saline.

HEAD INJURIES

Without the protection of the spinal fluid and the meninges, the brain would probably be unable to withstand even the minor traumas of everyday living; but with the protection afforded, it takes a fairly severe blow to produce cerebral damage. The brain is damaged most commonly when the skull is fractured and bone is driven into neural tissue (depressed skull fracture), when the brain moves far enough to tear the delicate bridging veins from the cortex to the bone, or when the brain is accelerated by a blow on the head and is driven against the skull or the tentorium at a point opposite where the blow was struck **(contrecoup injury).**

THE BLOOD-BRAIN BARRIER

The tight junctions between capillary endothelial cells in the brain and between the epithelial cells in the choroid plexus effectively prevent proteins from entering the brain in adults and slow the penetration of some smaller molecules as well. An example is the slow penetration of urea **(Figure 33–5)**. This uniquely limited exchange of substances into the brain is referred to as the **blood-brain barrier,** a term most commonly used to encompass this barrier overall and more specifically the barrier in the choroid epithelium between blood and CSF.

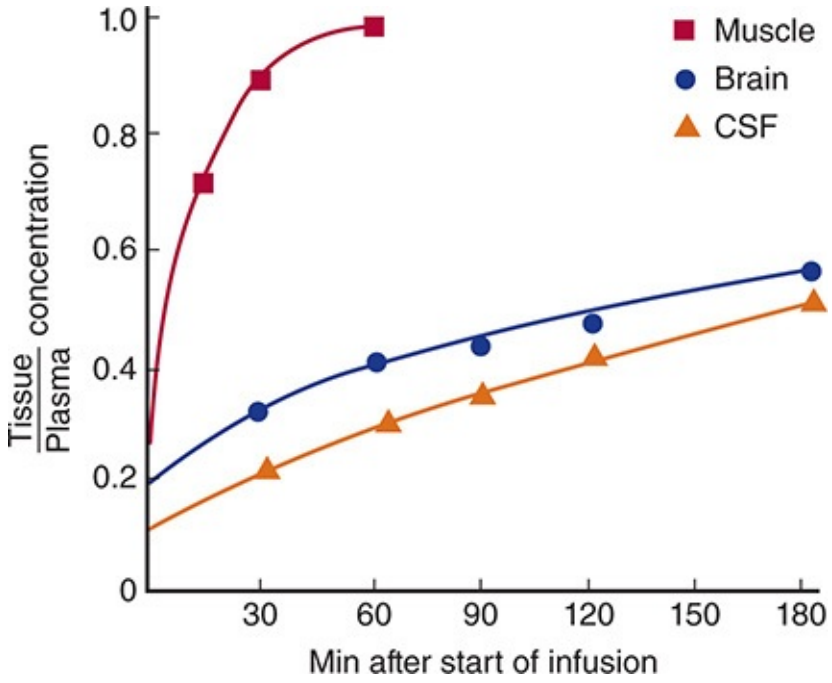


FIGURE 33–5 Penetration of urea into muscle, brain, and cerebrospinal fluid (CSF). Urea was administered by constant infusion.

Passive diffusion across the tight cerebral capillaries is very limited, and little vesicular transport takes place. However, there are numerous carrier-mediated and active transport systems in the cerebral capillaries.

PENETRATION OF SUBSTANCES INTO THE BRAIN

Water, CO_2, and O_2 penetrate the brain with ease, as do the lipid-soluble free forms of steroid hormones, whereas their protein-bound forms and, in general, all proteins and polypeptides do not. The rapid passive penetration of CO_2 contrasts with the regulated transcellular penetration of H^+ and HCO_3^- and has physiologic significance in the regulation of respiration (see Chapter 35).

Glucose is the major ultimate source of energy for nerve cells. Its diffusion across the blood-brain barrier would be very slow, but the rate of transport into the CSF is markedly enhanced by the presence of specific transporters, including the glucose transporter 1 (GLUT1). The brain contains two forms of GLUT1: GLUT1 55K and GLUT1 45K. Both are encoded by the same gene, but they differ in the extent to which they are glycosylated. GLUT1 55K is present in

high concentration in brain capillaries **(Figure 33–6)**. Infants with congenital GLUT1 deficiency have low CSF glucose concentrations in the presence of normal plasma glucose, and they have seizures and delayed development. In addition, transporters for thyroid hormones; several organic acids; choline; nucleic acid precursors; and neutral, basic, and acidic amino acids are present at the blood-brain barrier.

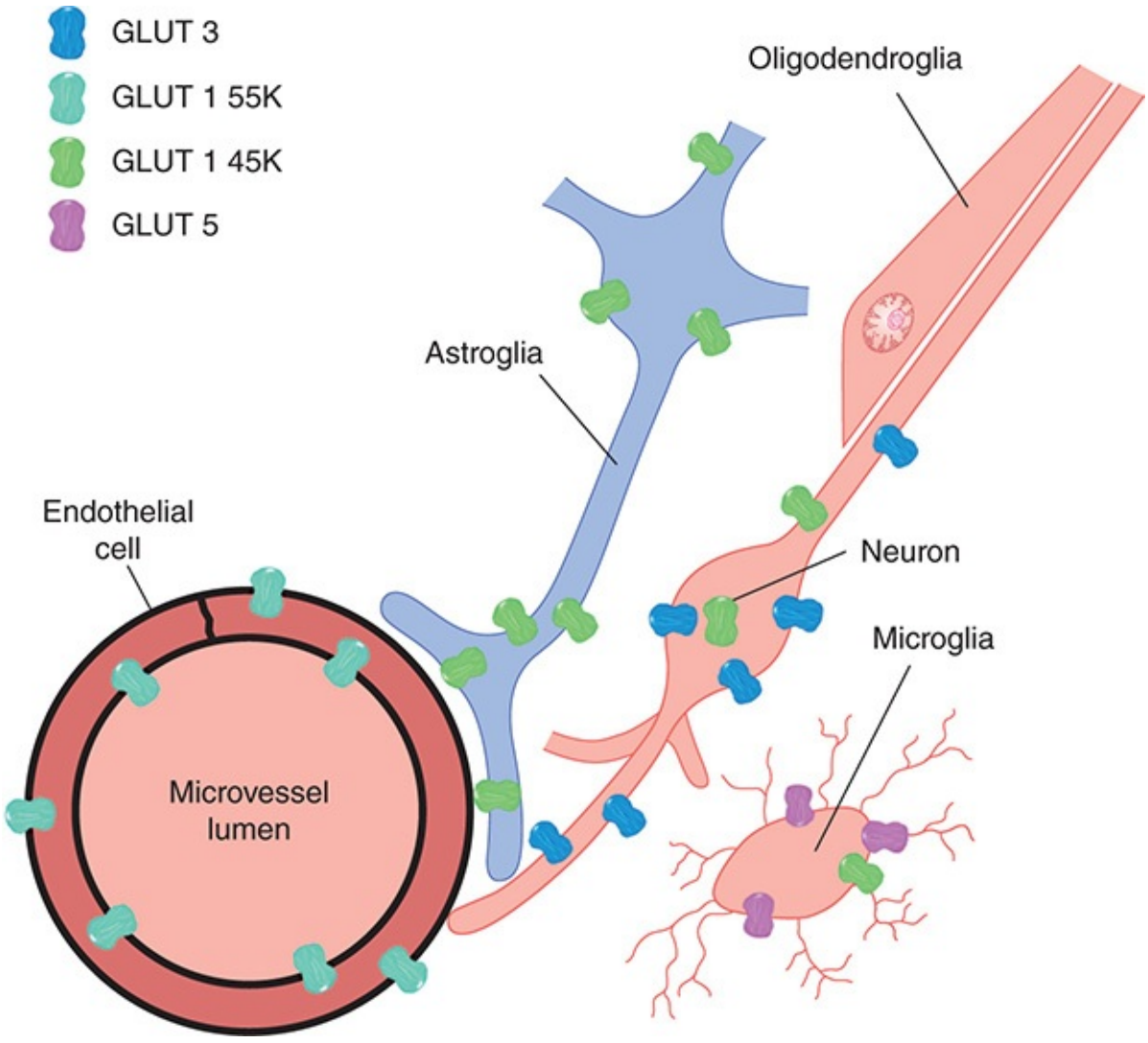


FIGURE 33–6 Localization of the various GLUT transporters in the brain. (Adapted with permission from Maher F, Vannucci SJ, Simpson IA: Glucose transporter proteins in brain. FASEB J 1994 Oct; 8(13):1003-1011.)

A variety of drugs and peptides actually permeate into the endothelial cells of the cerebral capillaries but are promptly transported back into the blood by a nonspecific multidrug transporter in the apical membrane. This **P-glycoprotein**

is a member of the family of adenosine triphosphate (ATP) binding cassette proteins that transport various proteins and lipids across cell membranes (see Chapter 2). In the absence of this transporter in mice, much larger proportions of systemically administered doses of various chemotherapeutic drugs, analgesics, and opioid peptides are found in the brain than in controls. If pharmacologic agents that inhibit this transporter can be developed, they could be of value in the treatment of brain tumors and other central nervous system (CNS) diseases in which it is difficult to introduce adequate amounts of therapeutic agents into the brain.

CIRCUMVENTRICULAR ORGANS

When dyes that bind to proteins in the plasma are injected, they stain many tissues but spare most of the brain. However, four small areas in or near the brainstem do take up the stain. These areas are: (1) the **posterior pituitary** (neurohypophysis) and the adjacent ventral part of the **median eminence** of the hypothalamus, (2) the **area postrema,** (3) the **organum vasculosum of the lamina terminalis (OVLT,** supraoptic crest), and (4) the **subfornical organ (SFO).**

These areas are referred to collectively as the **circumventricular organs (Figure 33–7)**. All have fenestrated capillaries, and because of their permeability they are said to be "outside the blood-brain barrier." Some of them function as **neurohemal organs;** that is, areas in which polypeptides secreted by neurons enter the circulation. Others contain receptors for many different peptides and other substances, and function as chemoreceptor zones in which substances in the circulating blood can act to trigger changes in brain function without penetrating the blood-brain barrier. For example, the area postrema is a chemoreceptor trigger zone that initiates vomiting in response to chemical changes in the plasma (see Chapter 27). It is also concerned with cardiovascular control, where circulating angiotensin II acts to produce a neurally mediated increase in blood pressure. Angiotensin II also acts on the SFO and possibly on the OVLT to increase water intake. In addition, it appears that the OVLT is the site of the osmoreceptor controlling vasopressin secretion (see Chapter 38), and evidence suggests that circulating interleukin-1 (IL-1) produces fever by acting here too.

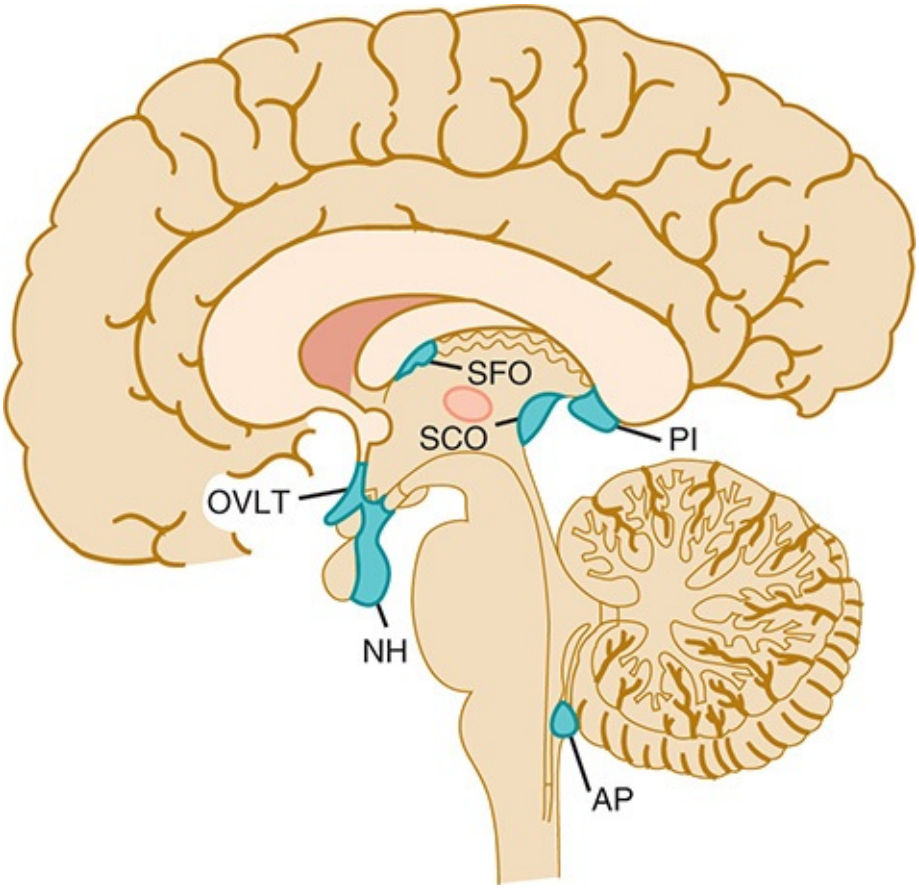


FIGURE 33–7 Circumventricular organs. The neurohypophysis (NH), organum vasculosum of the lamina terminalis (OVLT, organum vasculosum of the lamina terminalis), subfornical organ (SFO), and area postrema (AP) are shown projected on a sagittal section of the human brain. PI, pineal; SCO, subcommissural organ.

The subcommissural organ (Figure 33–7) is closely associated with the pineal gland and histologically resembles the circumventricular organs. However, it does not have fenestrated capillaries, and is not highly permeable. It may, instead, be involved in guiding axons during brain development. Conversely, the pineal and the anterior pituitary do have fenestrated capillaries and are outside the blood-brain barrier, but both are endocrine glands and are not part of the brain.

FUNCTION OF THE BLOOD-BRAIN BARRIER

The blood-brain barrier strives to maintain the constancy of the environment of

the neurons in the CNS **(Clinical Box 33–1)**. Even minor variations in the concentrations of K^+, Ca^{2+}, Mg^{2+}, H^+, and other ions can have far-reaching consequences. The constancy of the composition of the ECF in all parts of the body is maintained by multiple homeostatic mechanisms (see Chapters 1 and 38), but because of the sensitivity of the cortical neurons to ionic change, it is not surprising that an additional defense has evolved to protect them. Other functions of the blood-brain barrier include protection of the brain from endogenous and exogenous toxins in the blood and prevention of the escape of neurotransmitters into the general circulation.

DEVELOPMENT OF THE BLOOD-BRAIN BARRIER

In experimental animals, many small molecules penetrate the brain more readily during the fetal and neonatal period than they do in the adult. On this basis, it is often stated that the blood-brain barrier is immature at birth. Humans are more mature at birth, and not surprisingly, detailed data on passive permeability of the normal human neonatal blood-brain barrier are not available. However, in severely jaundiced infants with high plasma levels of free bilirubin and an immature hepatic bilirubin-conjugating system, free bilirubin enters the brain and, in the presence of asphyxia, damages the basal ganglia **(kernicterus).** The counterpart of this situation in later life is the Crigler-Najjar syndrome in which there is a congenital deficiency of glucuronyl transferase. These individuals can have very high free bilirubin levels in the blood and develop encephalopathy. In other conditions, however, free bilirubin levels are generally not high enough to produce brain damage.

CLINICAL BOX 33–1

Clinical Implications of the Blood-Brain Barrier

Clinicians must know the degree to which drugs penetrate the brain in order to treat diseases of the nervous system intelligently. For example, it is clinically relevant that the amines dopamine and serotonin penetrate brain tissue to a very limited degree but their corresponding acid precursors, L-dopa and 5-hydroxytryptophan, respectively, enter with relative ease (see Chapters 7 and 12). Another important clinical consideration is the fact that

the blood-brain barrier tends to break down in areas of infection or injury. Tumors develop new blood vessels, and the capillaries that are formed lack contact with normal astrocytes. Therefore, there are no tight junctions, and the vessels may even be fenestrated. The lack of a barrier helps in identifying the location of tumors; substances such as radioactive iodine-labeled albumin penetrate normal brain tissue very slowly, but they enter tumor tissue, making the tumor stand out as an island of radioactivity in the surrounding normal brain. The blood-brain barrier can also temporarily be disrupted by sudden marked increases in blood pressure or by intravenous injection of hypertonic fluids.

CEREBRAL BLOOD FLOW & ITS REGULATION

KETY METHOD

According to the **Fick's principle** (see Chapter 30), the blood flow of any organ can be measured by determining the amount of a given substance (Q_x) removed from the bloodstream by the organ per unit of time and dividing that value by the difference between the concentration of the substance in arterial blood and the concentration in the venous blood from the organ ($[A_x] - [V_x]$). Thus:

$$\text{Cerebral blood flow (CBF)} = \frac{Q_x}{[A_x] - [V_x]}$$

This can be applied clinically using inhaled nitrous oxide (N_2O) **(Kety method).** The average cerebral blood flow in young adults is 54 mL/100 g/min. The average adult brain weighs about 1400 g, so the flow for the whole brain is about 756 mL/min. Note that the Kety method provides an average value for perfused areas of brain and gives no information about regional differences in blood flow. It also can only measure flow to perfused parts of the brain. If the blood flow to a portion of the brain is occluded, the measured flow does not change because the nonperfused area does not take up any N_2O.

In spite of the marked local fluctuations in brain blood flow with neural activity, the cerebral circulation is regulated in such a way that total blood flow remains relatively constant. The factors involved in regulating the flow are summarized in **Figure 33–8**.

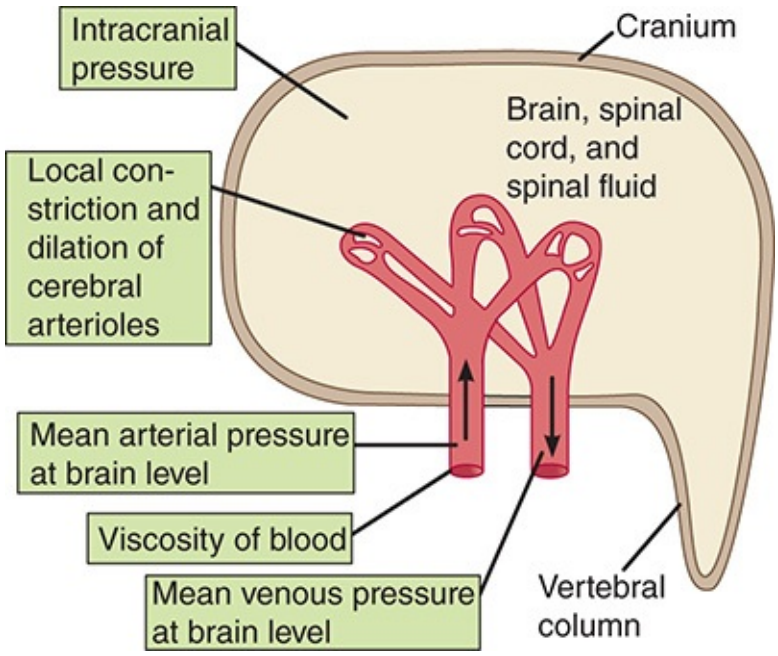


FIGURE 33–8 Diagrammatic summary of the factors affecting overall cerebral blood flow.

ROLE OF INTRACRANIAL PRESSURE

In adults, the brain, spinal cord, and spinal fluid are encased, along with the cerebral vessels, in a rigid bony enclosure. The cranial cavity normally contains a brain weighing approximately 1400 g, 75 mL of blood, and 75 mL of spinal fluid. Because brain tissue and spinal fluid are essentially incompressible, the volume of blood, spinal fluid, and brain in the cranium at any time must be relatively constant **(Monro-Kellie doctrine).** More importantly, the cerebral vessels are compressed whenever the intracranial pressure rises. Any change in venous pressure promptly causes a similar change in intracranial pressure. Thus, a rise in venous pressure decreases cerebral blood flow both by decreasing the effective perfusion pressure and by compressing the cerebral vessels. This relationship helps compensate for changes in arterial blood pressure at the level of the head. For example, if the body is accelerated upward (positive *g*), blood moves toward the feet and arterial pressure at the level of the head decreases. However, venous pressure also falls and intracranial pressure falls, so that the pressure on the vessels decreases and blood flow is much less severely compromised than it would otherwise be. Conversely, during acceleration downward, force acting toward the head (negative *g*) increases arterial pressure at head level, but intracranial pressure also rises, so that the vessels are

supported and do not rupture. The cerebral vessels are protected during the straining associated with defecation or delivery in the same way.

AUTOREGULATION

As seen in other vascular beds, autoregulation is prominent in the brain **(Figure 33–9)**. This process, by which the flow to many tissues is maintained at relatively constant levels despite variations in perfusion pressure, is discussed in Chapter 32. In the brain, autoregulation maintains a constant cerebral blood flow at arterial pressures of 65–140 mm Hg.

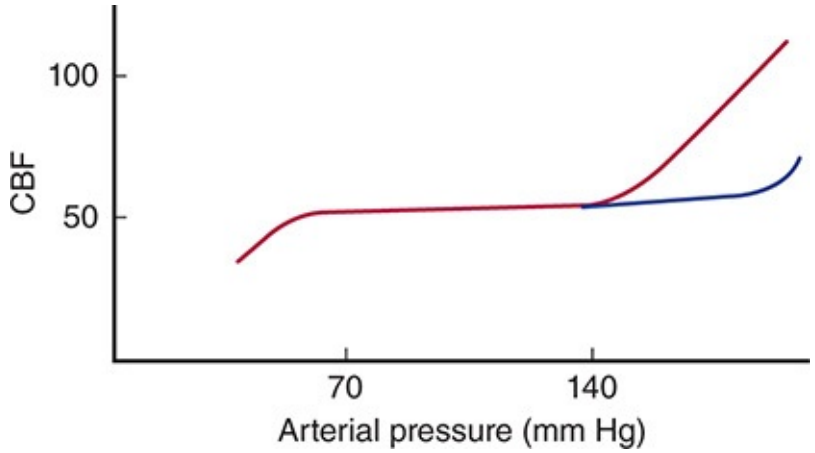


FIGURE 33–9 Autoregulation of cerebral blood flow (CBF) during steady-state conditions. The blue line shows the alteration produced by sympathetic stimulation during autoregulation.

ROLE OF VASOMOTOR & SENSORY NERVES

The innervation of large cerebral blood vessels by postganglionic sympathetic and parasympathetic nerves and the additional distal innervation by sensory nerves have been described above. The nerves may also modulate tone indirectly, via the release of paracrine substances from astrocytes. The precise role of these nerves, however, remains a matter of debate. It has been argued that noradrenergic discharge occurs when the blood pressure is markedly elevated. This reduces the resultant passive increase in blood flow and helps protect the blood-brain barrier from the disruption that could otherwise occur (see above). Thus, vasomotor discharges affect autoregulation. With sympathetic stimulation, the constant-flow, or plateau, part of the pressure-flow curve is extended to the

right (Figure 33–9); that is, greater increases in pressure can occur without an increase in flow. On the other hand, the vasodilator hydralazine and the angiotensin-converting enzyme (ACE) inhibitor captopril reduce the length of the plateau. Finally, neurovascular coupling may adjust local perfusion in response to changes in brain activity (see below).

BLOOD FLOW IN VARIOUS PARTS OF THE BRAIN

A major advance in recent decades has been the development of techniques for monitoring regional blood flow in living, conscious humans. Among the most valuable methods are **positron emission tomography (PET)** and related techniques in which a short-lived radioisotope is used to label a compound and the compound is injected. The arrival and clearance of the tracer are monitored by scintillation detectors placed over the head. Because blood flow is tightly coupled to brain metabolism, local uptake of 2-deoxyglucose is also a good index of blood flow (see below and Chapter 1). If the 2-deoxyglucose is labeled with a short half-life positron emitter such as ^{18}F, ^{11}O, or ^{15}O, its concentration in any part of the brain can be monitored.

Another valuable technique involves magnetic resonance imaging (MRI). MRI is based on detecting resonant signals from different tissues in a magnetic field. **Functional magnetic resonance imaging (fMRI)** measures the amount of blood in a tissue area. When neurons become active, their increased discharge alters the local field potential. A still unsettled mechanism triggers an increase in local blood flow and oxygen. The increase in oxygenated blood is detected by fMRI. PET scanning can be used to measure not only blood flow but also the concentration of molecules, such as dopamine, in various regions of the living brain. On the other hand, fMRI does not involve the use of radioactivity. Consequently, it can be used at frequent intervals to measure changes in regional blood flow in a single individual.

In resting humans, the average blood flow in gray matter is 69 mL/100 g/min compared with 28 mL/100 g/min in white matter. A striking feature of cerebral function is the marked variation in local blood flow with changes in brain activity. In persons who are awake but at rest, blood flow is greatest in the premotor and frontal regions. This is the part of the brain that is believed to be concerned with decoding and analyzing afferent input and with intellectual activity. During voluntary clenching of the right hand, flow is increased in the

hand area of the left motor cortex and the corresponding sensory areas in the postcentral gyrus. Especially when the movements being performed are sequential, the flow is also increased in the supplementary motor area. When persons talk, there is a bilateral increase in blood flow in the face, tongue, and mouth-sensory and motor areas and the upper premotor cortex in the categorical (usually the left) hemisphere. When the speech is stereotyped, Broca and Wernicke areas do not show increased flow, but when the speech is creative—that is, when it involves ideas—flow increases in both these areas. Reading produces widespread increases in blood flow. Problem solving, reasoning, and motor ideation without movement produce increases in selected areas of the premotor and frontal cortex. In anticipation of a cognitive task, many of the brain areas that will be activated during the task are activated beforehand, as if the brain produces an internal model of the expected task. In right-handed individuals, blood flow to the left hemisphere is greater when a verbal task is being performed and blood flow to the right hemisphere is greater when a spatial task is being performed **(Clinical Box 33–2)**. An example of how blood flow changes are regionalized during a mental task is shown in **Figure 33–10**.

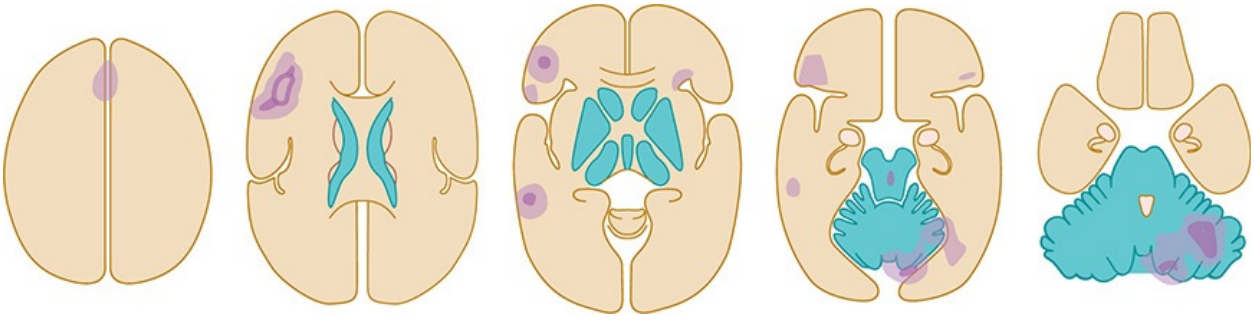


FIGURE 33–10 Activity in the human brain at five different horizontal levels while a subject generates a verb that is appropriate for each noun presented by an examiner. This mental task activates the frontal cortex (slices 1–4), anterior cingulate gyrus (slice 1), and posterior temporal lobe (slice 3) on the left side and the cerebellum (slices 4 and 5) on the right side. Light purple, moderate activation; dark purple, marked activation. (Based on PET scans in Posner MI, Raichle ME: *Images of Mind*. Scientific American Library, 1994.)

CLINICAL BOX 33–2

Changes in Cerebral Blood Flow in Disease

Several disease states are known to be associated with localized or general

changes in cerebral blood flow, as revealed by PET scanning and fMRI. For example, epileptic foci are hyperemic during seizures, whereas flow is reduced in other parts of the brain. Between seizures, flow is sometimes reduced in the foci that generate the seizures. Parietooccipital flow is decreased in patients with symptoms of agnosia (see Chapter 15). In Alzheimer disease, the earliest change is decreased metabolism and blood flow in the superior parietal cortex, with later spread to the temporal and finally the frontal cortex. The precentral and postcentral gyri, basal ganglia, thalamus, brainstem, and cerebellum are relatively spared. In Huntington disease, blood flow is reduced bilaterally in the caudate nucleus, and this alteration in flow occurs early in the disease. In manic depressives (but interestingly, not in patients with unipolar depression), there is a general decrease in cortical blood flow when the patients are depressed. In schizophrenia, some evidence suggests decreased blood flow in the frontal lobes, temporal lobes, and basal ganglia. Finally, during the aura in patients with migraine, a bilateral decrease in blood flow starts in the occipital cortex and spreads anteriorly to the temporal and parietal lobes.

BRAIN METABOLISM & OXYGEN REQUIREMENTS

UPTAKE & RELEASE OF SUBSTANCES BY THE BRAIN

If the cerebral blood flow is known, it is possible to calculate the consumption or production by the brain of O_2, CO_2, glucose, or any other substance present in the bloodstream by multiplying the cerebral blood flow by the difference between the concentration of the substance in arterial blood and cerebral venous blood **(Table 33–3)**. When calculated in this manner, a negative value indicates that the brain is producing the substance.

TABLE 33–3 Utilization and production of substances by the adult human brain in vivo.

Substance	Uptake (+) or Output (–) per 100 g of Brain/min	Total/min
Substances utilized		
Oxygen	+3.5 mL	+49 mL
Glucose	+5.5 mg	+77 mg
Glutamate	+0.4 mg	+5.6 mg
Substances produced		
Carbon dioxide	–3.5 mL	–49 mL
Glutamine	–0.6 mL	–8.4 mg

Substances not used or produced in the fed state: lactate, pyruvate, total ketones, and α-ketoglutarate.

OXYGEN CONSUMPTION

O_2 consumption by the human brain (**cerebral** metabolic rate for O_2, $CMRO_2$) averages approximately 20% of the total body resting O_2 consumption (Table 33–1). The brain is extremely sensitive to hypoxia, and occlusion of its blood supply produces unconsciousness in a period as short as 10 s. The vegetative structures in the brainstem are more resistant to hypoxia than the cerebral cortex, and patients may recover from accidents such as cardiac arrest and other conditions causing fairly prolonged hypoxia with normal vegetative functions but severe, permanent intellectual deficiencies. The basal ganglia use O_2 at a very high rate, and symptoms of Parkinson disease as well as intellectual deficits can be produced by chronic hypoxia. The thalamus and the inferior colliculus are also very susceptible to hypoxic damage **(Clinical Box 33–3)**.

ENERGY SOURCES

Glucose is the major ultimate source of energy for the brain; under normal conditions, 90% of the energy needed to maintain ion gradients across cell membranes and transmit electrical impulses comes from this source. Glucose enters the brain via GLUT1 in cerebral capillaries (see above). Other transporters

then distribute it to neurons and glial cells.

Glucose is taken up from the blood in large amounts, and the RQ (respiratory quotient; see Chapter 26) of cerebral tissue is 0.95–0.99 in normal individuals. Importantly, insulin is not required for most cerebral cells to utilize glucose. In general, glucose utilization at rest parallels blood flow and O_2 consumption. This does not mean that the total source of energy is always glucose. During prolonged starvation, appreciable utilization of other substances occurs. Indeed, evidence indicates that as much as 30% of the glucose taken up under normal conditions is converted to amino acids, lipids, and proteins, and those substances other than glucose are metabolized for energy during convulsions. Some utilization of amino acids from the circulation may also take place even though the amino acid arteriovenous difference across the brain is normally minute.

The consequences of hypoglycemia in terms of neural function are discussed in Chapter 24.

GLUTAMATE & AMMONIA REMOVAL

The brain's uptake of glutamate is approximately balanced by its output of glutamine. Glutamate entering the brain associates with ammonia and leaves as glutamine. The glutamate-glutamine conversion in the brain—the opposite of the reaction in the kidney that produces some of the ammonia entering the tubules—serves as a detoxifying mechanism to keep the brain free of ammonia. Ammonia is very toxic to nerve cells, and ammonia intoxication is believed to be at least among the causes of neurologic symptoms that may emerge in advanced liver disease (**hepatic encephalopathy**, see Chapter 28).

CORONARY CIRCULATION

ANATOMIC CONSIDERATIONS

The two coronary arteries that supply the myocardium arise from sinuses behind two of the cusps of the aortic valve at the root of the aorta **(Figure 33–11)**. Most of the venous blood returns to the heart through the coronary sinus and anterior cardiac veins **(Figure 33–12)**, which drain into the right atrium. In addition, there are other vessels that empty directly into the heart chambers. These include **arteriosinusoidal vessels,** sinusoidal capillary-like vessels that connect arterioles to the chambers; **Thebesian veins** that connect capillaries to the

chambers; and a few **arterioluminal vessels** that are small arteries draining directly into the chambers. A few anastomoses occur between the coronary arterioles and extracardiac arterioles, especially around the mouths of the great veins.

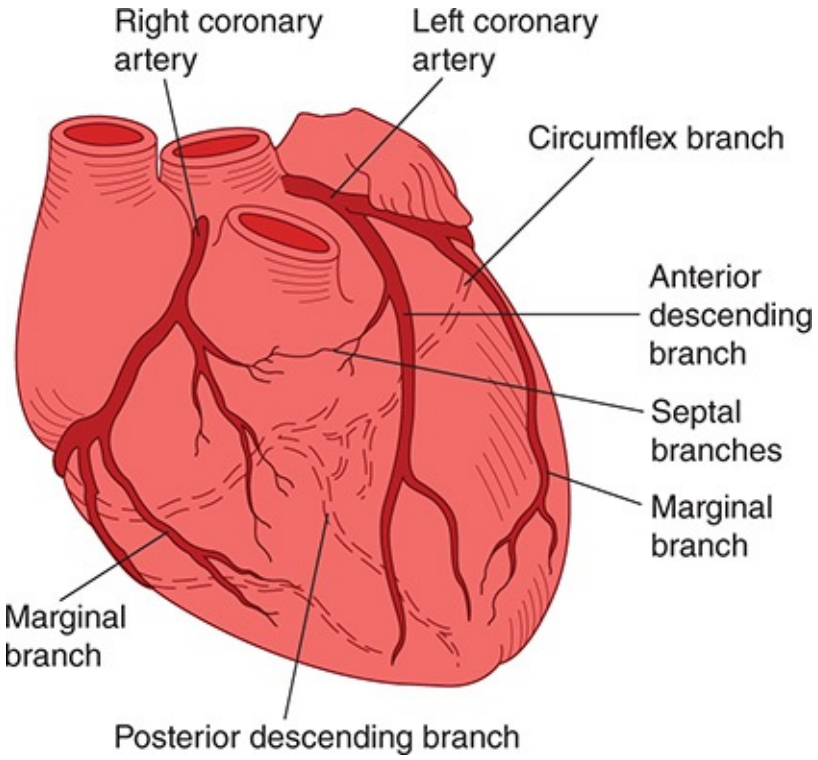


FIGURE 33–11 Coronary arteries and their principal branches in humans. (Reproduced with permission from Ross G: *Essentials of Human Physiology*. Year Book Medical Publishers; 1978.)

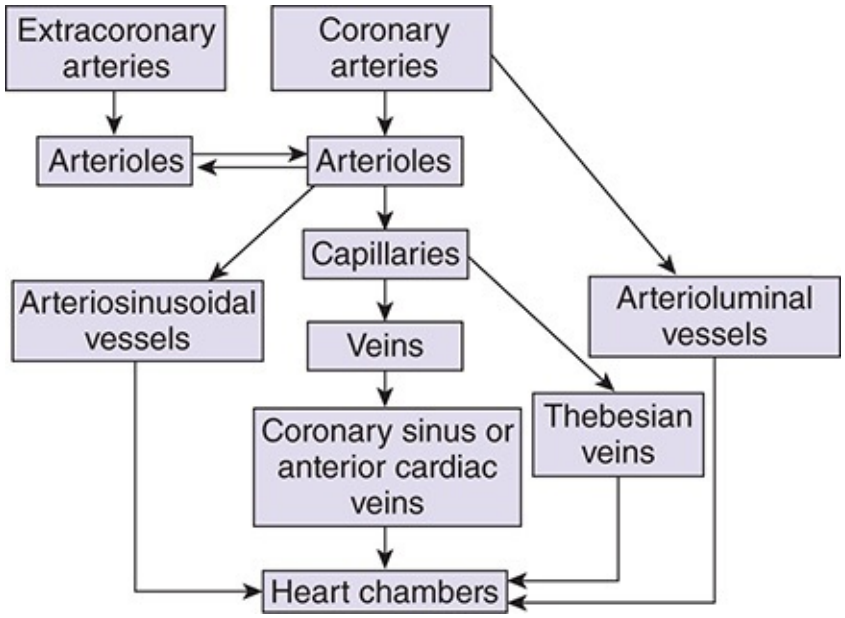


FIGURE 33–12 Diagram of the coronary circulation.

CLINICAL BOX 33–3

Stroke

When the blood supply to a part of the brain is interrupted, ischemia damages or kills the cells in the area, producing the signs and symptoms of a stroke. There are two general types of strokes: hemorrhagic and ischemic. Hemorrhagic stroke occurs when a cerebral artery or arteriole ruptures, sometimes but not always at the site of a small aneurysm. Ischemic stroke occurs when flow in a vessel is compromised by atherosclerotic plaques on which thrombi form. Thrombi may also be produced elsewhere (eg, in the atria in patients with atrial fibrillation) and pass to the brain as emboli where they then lodge and interrupt flow. In the past, little could be done to modify the course of a stroke and its consequences. However, it has become clear that in the penumbra (the area surrounding the most severe brain damage) ischemia reduces glutamate uptake by astrocytes, and the increase in local glutamate causes excitotoxic damage and death to neurons (see Chapter 7).

THERAPEUTIC HIGHLIGHTS

The clot-lysing drug, tissue-type plasminogen activator (t-PA) (see Chapter 31) is of great benefit in ischemic strokes. Drugs that prevent excitotoxic damage

have had equivocal effects in human clinical trials, but growing insights into the mechanism of injury are yielding more targeted agents that may be of benefit via this pathway in the future. However, t-PA (and presumably any effective antiexcitotoxic treatment) must be given early in the course of a stroke to be of maximum benefit. This is why stroke has become a condition in which rapid diagnosis and treatment are extremely important. In addition, of course, it is important to determine if a stroke is thrombotic or hemorrhagic, since clot lysis is contraindicated in the latter. Trials are ongoing of therapies that enhance hemostasis in hemorrhagic stroke.

PRESSURE GRADIENTS & FLOW IN THE CORONARY VESSELS

The heart is a muscle that, like skeletal muscle, compresses its blood vessels when it contracts. The pressure inside the left ventricle is slightly higher than in the aorta during systole **(Table 33–4)**. Consequently, in the arteries supplying the subendocardial portion of the left ventricle, flow occurs only during diastole, although the force is sufficiently dissipated in the more superficial portions of the left ventricular myocardium to permit some flow in this region throughout the cardiac cycle. Because the relative length of diastole is shorter when the heart rate is high, left ventricular coronary flow is reduced during tachycardia. On the other hand, the pressure differential between the aorta and the right ventricle, and between the aorta and the atria, is somewhat greater during systole than during diastole. Consequently, coronary flow in those parts of the heart is not appreciably reduced during systole. Flow in the right and left coronary arteries is shown in **Figure 33–13**. Because no blood flow occurs during systole in the subendocardial portion of the left ventricle, this region is prone to ischemic damage and is the most common site of myocardial infarction. Blood flow to the left ventricle is decreased in patients with stenotic aortic valves because the pressure in the left ventricle must be much higher than that in the aorta to eject the blood. Consequently, the coronary vessels are more severely compressed during systole. Patients with aortic stenosis are particularly prone to develop symptoms of myocardial ischemia, in part because of this compression and in part because the myocardium requires more O_2 to expel blood through the stenotic aortic valve. Coronary flow is also decreased when the aortic diastolic pressure is low. The rise in venous pressure in conditions such as heart failure

reduces coronary flow because it decreases effective coronary perfusion pressure **(Clinical Box 33–4)**.

TABLE 33–4 Pressure in aorta and left and right ventricles (vent) in systole and diastole.

	Pressure (mm Hg) in			Pressure Differential (mm Hg) between Aorta and	
	Aorta	Left Vent	Right Vent	Left Vent	Right Vent
Systole	120	121	25	–1	95
Diastole	80	0	0	80	80

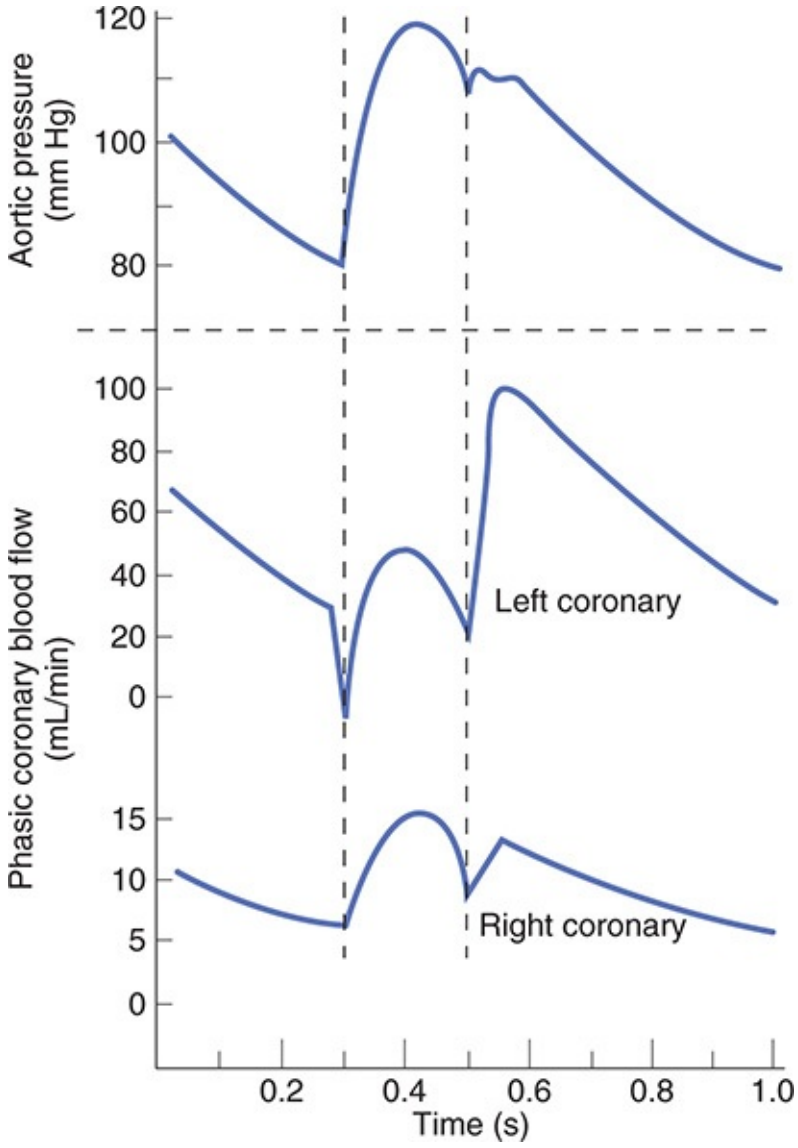


FIGURE 33–13 Blood flow in the left and right coronary arteries during various phases of the cardiac cycle. Systole occurs between the two vertical dashed lines. (Reproduced with permission from Berne RM, Levy MN: *Physiology*, 2nd ed. Mosby; 1988.)

Coronary blood flow has been measured by inserting a catheter into the coronary sinus and applying the Kety method to the heart on the assumption that the N_2O content of coronary venous blood is typical of the entire myocardial effluent. Coronary flow at rest in humans is about 250 mL/min (5% of the cardiac output). A number of techniques utilizing **radionuclides** detected with radiation detectors over the chest have been used to study regional blood flow in the heart and to detect areas of ischemia and infarct as well as to evaluate

ventricular function. Radionuclides such as thallium-201 (^{201}Tl) are pumped into cardiac muscle cells by Na^+, K^+ ATPase and equilibrate with the intracellular K^+ pool. For the first 10–15 min after intravenous injection, ^{201}Tl distribution is directly proportional to myocardial blood flow, and areas of ischemia can be detected by their low uptake. The uptake of this isotope is often determined soon after exercise and again several hours later to bring out areas in which exertion leads to compromised flow. Conversely, radiopharmaceuticals such as technetium-99m stannous pyrophosphate (^{99m}Tc-PYP) are selectively taken up by infarcted tissue and make infarcts stand out as “hot spots” on scintigrams of the chest. Coronary angiography can be combined with measurement of ^{133}Xe washout (see above) to provide detailed analysis of coronary blood flow. Radiopaque contrast medium is first injected into the coronary arteries, and radiographs are used to outline their distribution. The angiographic camera is then replaced with a scintillation camera, and ^{133}Xe washout is measured.

VARIATIONS IN CORONARY FLOW

At rest, the heart extracts 70–80% of the O_2 from each unit of blood delivered to it (Table 33–1). O_2 delivery can only be increased significantly by increasing blood flow. Therefore, it is not surprising that blood flow increases when the metabolism of the myocardium is increased. The caliber of the coronary vessels, and consequently the rate of coronary blood flow, is influenced not only by pressure changes in the aorta but also by chemical and neural factors. The coronary circulation also shows considerable autoregulation.

CHEMICAL FACTORS

The close relationship between coronary blood flow and myocardial O_2 consumption implies that one or more metabolic products causes coronary vasodilation. Factors suspected of playing this role include a lack of O_2 and increased local concentrations of CO_2, H^+, K^+, lactate, prostaglandins, adenine nucleotides, and adenosine. Likely several or all of these vasodilator metabolites act in an integrated manner, redundant manner, or both. Asphyxia, hypoxia, and intracoronary injections of cyanide all increase coronary blood flow 200–300% in denervated as well as intact hearts, and the feature common to these three stimuli is hypoxia of the myocardial fibers. A similar increase in flow is

produced in the area supplied by a coronary artery if the artery is occluded and then released. This **reactive hyperemia** is similar to that seen in the skin (see below) and appears to be due to the release of adenosine.

CLINICAL BOX 33–4

Coronary Artery Disease

When flow through a coronary artery is reduced to the point that the myocardium it supplies becomes hypoxic, **angina pectoris** develops (see Chapter 30). If the myocardial ischemia is severe and prolonged, irreversible changes occur in the muscle, and the result is **myocardial infarction.** Many individuals have angina only on exertion, and coronary blood flow is normal at rest. Others have more severe restriction of blood flow and have anginal pain at rest as well. Partially occluded coronary arteries can be constricted further by vasospasm, producing myocardial infarction. However, it is now clear that the most common cause of myocardial infarction is rupture of an **atherosclerotic plaque,** or hemorrhage into it, which triggers the formation of a coronary-occluding clot at the site of the plaque. The electrocardiographic changes in myocardial infarction are discussed in Chapter 29. When myocardial cells die, they leak enzymes into the circulation, and measuring the rise in serum enzymes and isoenzymes produced by infarcted myocardial cells also plays an important role in the diagnosis of myocardial infarction. The enzymes most commonly measured are the MB isomer of creatine kinase (CK-MB), troponin T, and troponin I. Myocardial infarction is a very common cause of death in developed countries because of the widespread occurrence of atherosclerosis. In addition, there is a relation between atherosclerosis and circulating levels of **lipoprotein(a) (Lp[a]).** Lp(a) has an outer coat consisting of apo(a). It interferes with fibrinolysis by down-regulating plasmin generation (see Chapter 31). It now appears that atherosclerosis has an important inflammatory component as well. The lesions of the disease contain inflammatory cells, and there is a positive correlation between increased levels of C-reactive protein and other **inflammatory markers** in the circulation with subsequent myocardial infarction.

THERAPEUTIC HIGHLIGHTS

Treatment of myocardial infarction aims to restore flow to the affected area as

rapidly as possible while minimizing reperfusion injury. Needless to say, it should be started as promptly as possible to avoid irreversible changes in heart function. In acute disease, antithrombotic agents are often given, but these can be problematic, leading to increased mortality due to bleeding if cardiac surgery is subsequently needed. Mechanical/surgical approaches to coronary artery disease include balloon angioplasty and/or implantation of stents to hold the vessels open, or coronary artery bypass grafting (CABG).

NEURAL FACTORS

The coronary arterioles contain α-adrenergic receptors, which mediate vasoconstriction, and β-adrenergic receptors, which mediate vasodilation. Activity in the noradrenergic nerves to the heart and injections of norepinephrine cause coronary vasodilation. Mechanistically, norepinephrine increases the heart rate and the force of cardiac contraction, and the vasodilation is due to production of vasodilator metabolites by the myocardium secondary to the increase in its activity. When the inotropic and chronotropic effects of noradrenergic discharge are blocked by a β-adrenergic blocking drug, stimulation of the noradrenergic nerves or injection of norepinephrine in unanesthetized animals elicits coronary vasoconstriction. Thus, the direct effect of noradrenergic stimulation is constriction rather than dilation of the coronary vessels. On the other hand, stimulation of vagal fibers to the heart dilates the coronaries.

When the systemic blood pressure falls, the overall effect of the reflex increase in noradrenergic discharge is increased coronary blood flow secondary to the metabolic changes in the myocardium at a time when the cutaneous, renal, and splanchnic vessels are constricted. In this way the circulation of the heart, like that of the brain, is preserved when flow to other organs is compromised.

CUTANEOUS CIRCULATION

The amount of heat lost from the body is regulated to a large extent by varying the amount of blood flowing through the skin. The fingers, toes, palms, and earlobes contain well-innervated anastomotic connections between arterioles and venules (arteriovenous anastomoses; see Chapter 31). Blood flow in response to thermoregulatory stimuli can vary from 1 to as much as 150 mL/100 g of

skin/min, and it has been postulated that these wide variations are possible because blood can be shunted through the anastomoses. The subdermal capillary and venous plexus is a blood reservoir of some importance, and the skin is one of the few places where the reactions of blood vessels can be observed visually.

WHITE REACTION

When a pointed object is drawn lightly over the skin, the stroke lines become pale **(white reaction).** The mechanical stimulus apparently initiates contraction of the precapillary sphincters, and blood drains out of the capillaries and small veins. The response appears in about 15 s.

TRIPLE RESPONSE

When the skin is stroked more firmly with a pointed instrument, instead of the white reaction there is reddening at the site that appears in about 10 s **(red reaction).** This is followed in a few minutes by local swelling and diffuse, mottled reddening around the site. The initial redness is due to capillary dilation, a direct response of the capillaries to pressure. The swelling **(wheal)** is local edema due to increased permeability of the capillaries and postcapillary venules, with consequent extravasation of fluid. The redness spreading out from the site **(flare)** is due to arteriolar dilation. This three-part response—the red reaction, wheal, and flare—is called the **triple response** and is part of the normal reaction to injury (see Chapter 3). It persists after total sympathectomy. On the other hand, the flare is absent in locally anesthetized skin and in denervated skin after the sensory nerves have degenerated, but it is present immediately after nerve block or section above the site of the injury. This, plus other evidence, indicates that it is due to an **axon reflex,** a response in which impulses initiated in sensory nerves by the injury are relayed antidromically down other branches of the sensory nerve fibers **(Figure 33–14)**. This is the one situation in the body in which there is substantial evidence for a physiologic effect due to antidromic conduction. The transmitter released at the central termination of the sensory C fiber neurons is substance P (see Chapter 7), and substance P and CGRP are present in all parts of the neurons. Both dilate arterioles and, in addition, substance P cause extravasation of fluid. Effective nonpeptide antagonists to substance P have been developed, and they reduce the extravasation. Thus, it appears that substance P produces the wheal.

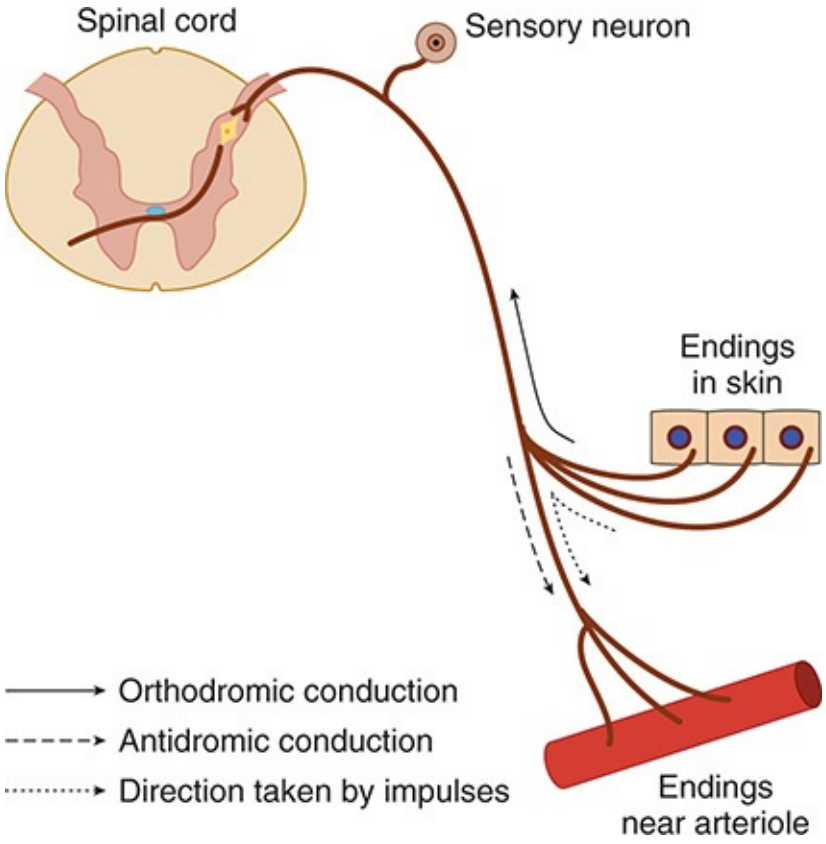


FIGURE 33–14 Axon reflex.

REACTIVE HYPEREMIA

A response of the blood vessels that occurs in many organs but is visible in the skin is **reactive hyperemia,** an increase in the amount of blood in a region when its circulation is reestablished after a period of occlusion. When the blood supply to a limb is occluded, the cutaneous arterioles below the occlusion dilate. When the circulation is reestablished, blood flowing into the dilated vessels makes the skin fiery red. O_2 in the atmosphere can diffuse a short distance through the skin, and reactive hyperemia is prevented if the circulation of the limb is occluded in an atmosphere of 100% O_2. Therefore, the arteriolar dilation is apparently due to a local effect of hypoxia.

GENERALIZED RESPONSES

Noradrenergic nerve stimulation and circulating epinephrine and norepinephrine constrict cutaneous blood vessels. No known vasodilator nerve fibers extend to

the cutaneous vessels, and thus vasodilation is brought about by a decrease in constrictor tone as well as the local production of vasodilator metabolites. Skin color and temperature also depend on the state of the capillaries and venules. A cold blue or gray skin is one in which the arterioles are constricted and the capillaries dilated; a warm red skin is the one in which both are dilated.

Because painful stimuli cause diffuse noradrenergic discharge, a painful injury causes generalized cutaneous vasoconstriction in addition to the local triple response. When the body temperature rises during exercise, the cutaneous blood vessels dilate in spite of continuing noradrenergic discharge in other parts of the body. Dilation of cutaneous vessels in response to a rise in hypothalamic temperature overcomes other reflex activity. Shock is more profound in patients with elevated temperatures because of cutaneous vasodilation, and patients in shock should not be warmed to the point that their body temperature rises. Conversely, cold causes cutaneous vasoconstriction; however, with severe cold, superficial vasodilation may supervene. This latter vasodilation is the cause of the ruddy complexion seen on a cold day.

PLACENTAL & FETAL CIRCULATION

UTERINE CIRCULATION

The blood flow of the uterus parallels the metabolic activity of the myometrium and endometrium and undergoes cyclic fluctuations that correlate with the menstrual cycle in nonpregnant women. The function of the spiral and basilar arteries of the endometrium in menstruation is discussed in Chapter 22. During pregnancy, blood flow increases rapidly as the uterus increases in size **(Figure 33–15)**. Vasodilator metabolites are undoubtedly produced in the uterus, as they are in other active tissues. In early pregnancy, the arteriovenous O_2 difference across the uterus is small, and it has been suggested that estrogens act on the blood vessels to increase uterine blood flow in excess of tissue O_2 needs. Corticotrophin-releasing hormone also appears to play an important role in up-regulating uterine blood flow, as well as in the eventual timing of birth. However, even though uterine blood flow increases 20-fold during pregnancy, the size of the conceptus increases much more, changing from a single cell to a fetus plus a placenta that can weigh 4–5 kg at term in humans. Consequently, more O_2 is extracted from the uterine blood during the latter part of pregnancy, and the O_2 saturation of uterine blood falls.

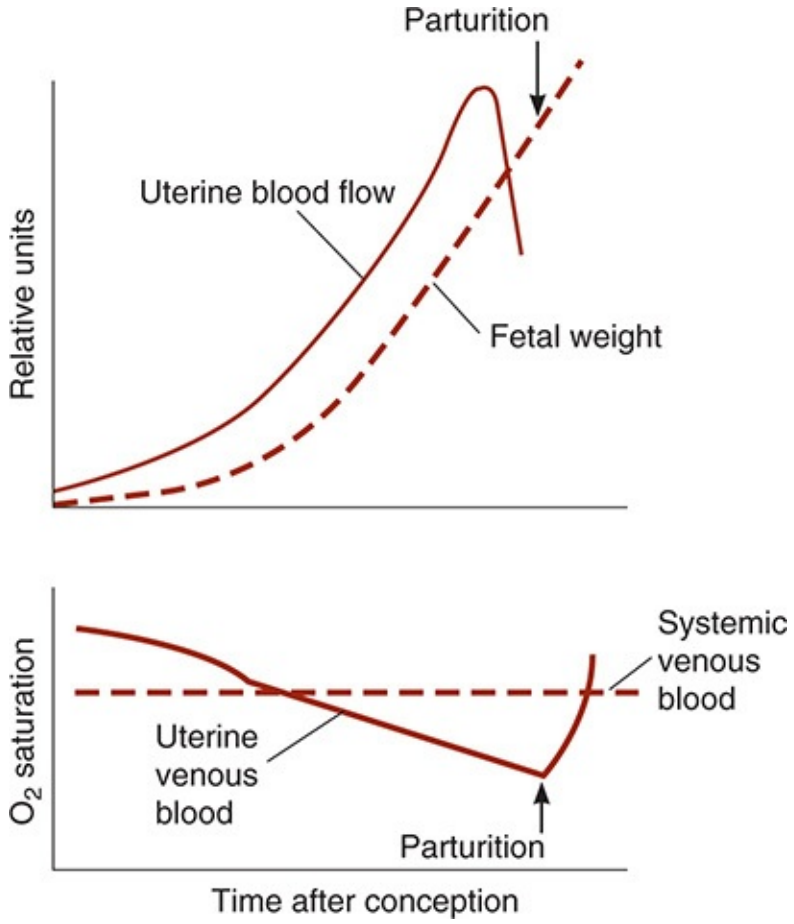


FIGURE 33–15 Changes in uterine blood flow and the amount of O_2 in uterine venous blood during pregnancy. (Modified with permission from Keele CA, Neil *E: Samson Wright's Applied Physiology*, 12th ed. Oxford University Press; 1971.)

PLACENTA

The placenta is the "fetal lung" (**Figures 33–16** and **33–17**). Its maternal portion is in effect a large blood sinus. Into this "lake" project the villi of the fetal portion containing the small branches of the fetal umbilical arteries and vein (Figure 33–16). O_2 is taken up by the fetal blood and CO_2 is discharged into the maternal circulation across the walls of the villi in a manner analogous to O_2 and CO_2 exchange in the lungs (see Chapter 35). However, the cellular layers covering the villi are thicker and less permeable than the alveolar membranes in the lungs, and exchange is much less efficient. The placenta is also the route by which all nutritive materials enter the fetus and by which fetal wastes are

discharged to the maternal blood.

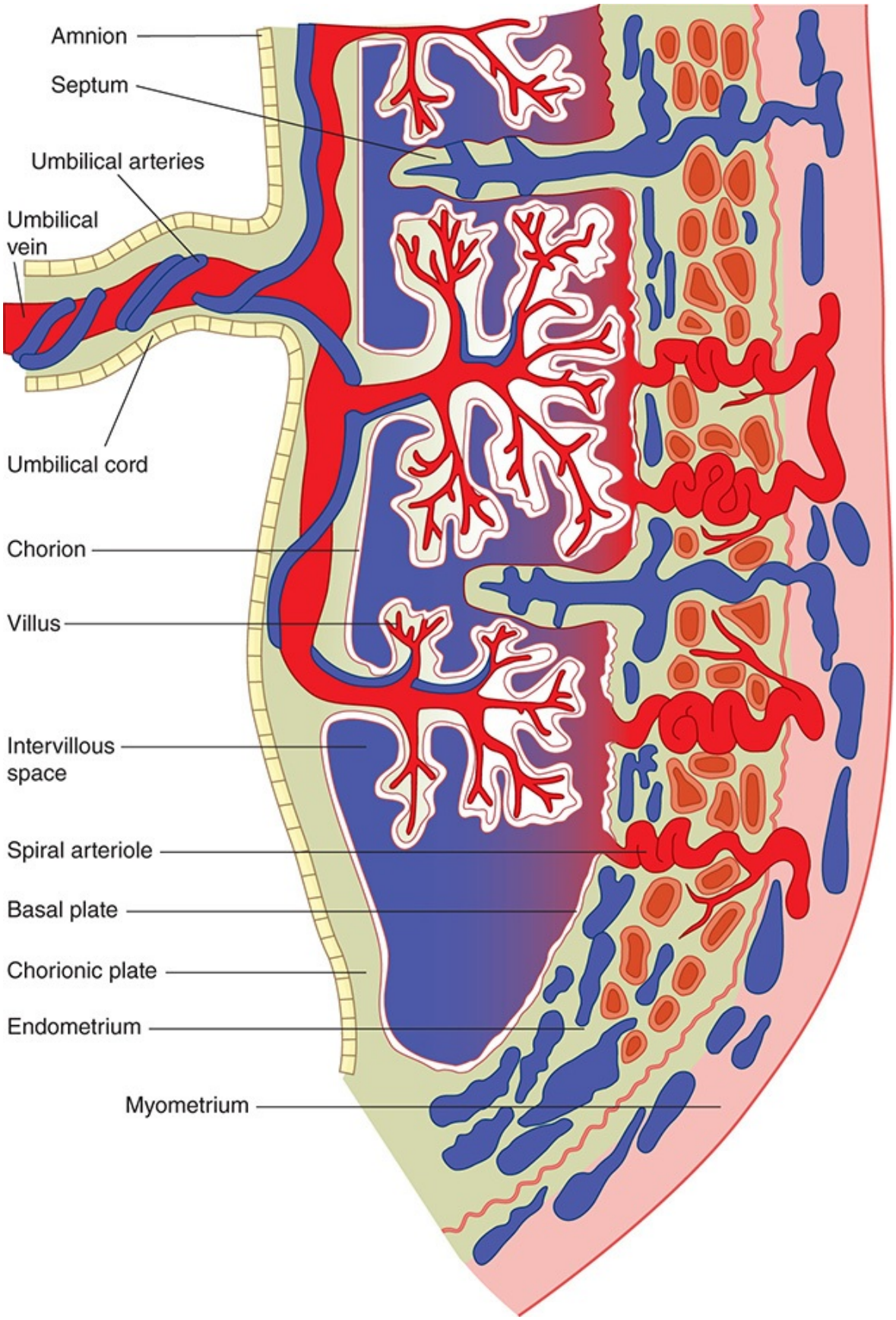

Amnion
Septum
Umbilical arteries
Umbilical vein
Umbilical cord
Chorion
Villus
Intervillous space
Spiral arteriole
Basal plate
Chorionic plate
Endometrium
Myometrium

FIGURE 33–16 Diagram of a section through the human placenta, showing the way the fetal villi project into the maternal sinuses. (©Elsevier, Inc., Netterimages.com.)

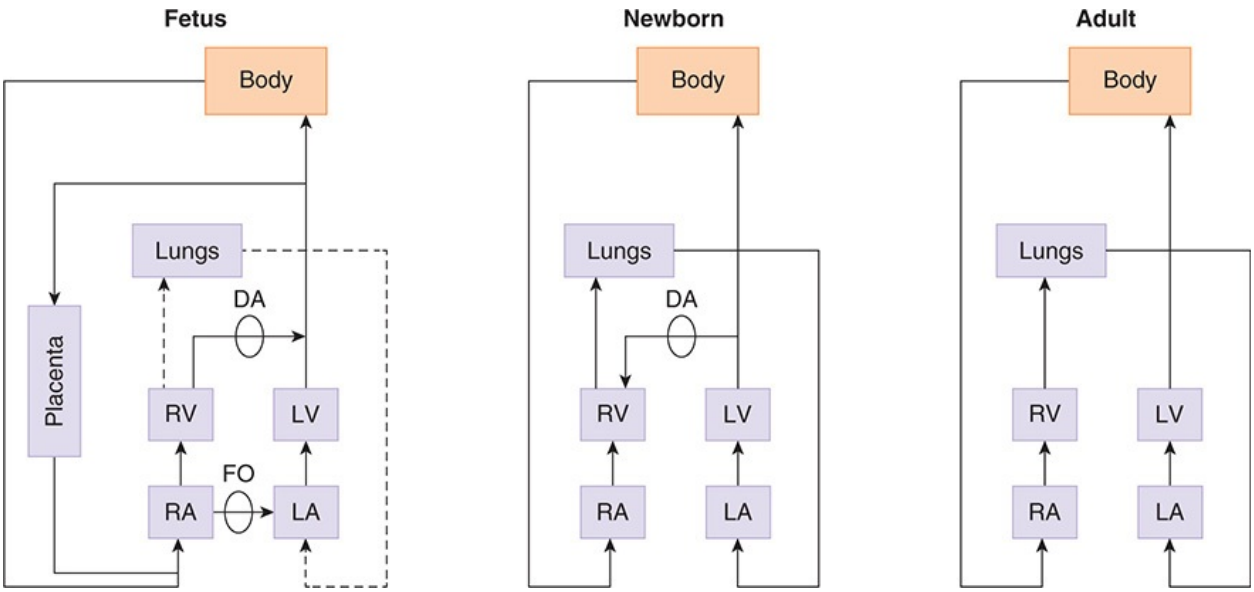


FIGURE 33–17 Diagram of the circulation in the fetus, the newborn infant, and the adult. DA, ductus arteriosus; FO, foramen ovale; LA, left atrium; LV, left ventricle; RA, right atrium; RV, right ventricle. Note that circulation to the fetal liver is not shown for simplicity (see Figure 33–18).

FETAL CIRCULATION

The arrangement of the circulation in the fetus is shown diagrammatically in Figure 33–17. The blood in the umbilical vein in humans is believed to be about 80% saturated with O_2, compared with 98% saturation in the arterial circulation of the adult. The ductus venosus **(Figure 33–18)** diverts some of this blood directly to the inferior vena cava, and the remainder mixes with the portal blood of the fetus. The portal and systemic venous blood of the fetus is only 26% saturated, and the saturation of the mixed blood in the inferior vena cava is approximately 67%. Most of the blood entering the heart through the inferior vena cava is diverted directly to the left atrium via the patent foramen ovale. Most of the blood from the superior vena cava enters the right ventricle and is expelled into the pulmonary artery. The resistance of the collapsed lungs is high, and the pressure in the pulmonary artery is several mm Hg higher than it is in the

aorta, so that most of the blood in the pulmonary artery passes through the **ductus arteriosus** to the aorta. In this manner, the relatively unsaturated blood from the right ventricle is diverted to the trunk and lower body of the fetus, while the head of the fetus receives the better-oxygenated blood from the left ventricle. From the aorta, some of the blood is pumped into the umbilical arteries and back to the placenta. The O_2 saturation of the blood in the lower aorta and umbilical arteries of the fetus is approximately 60%.

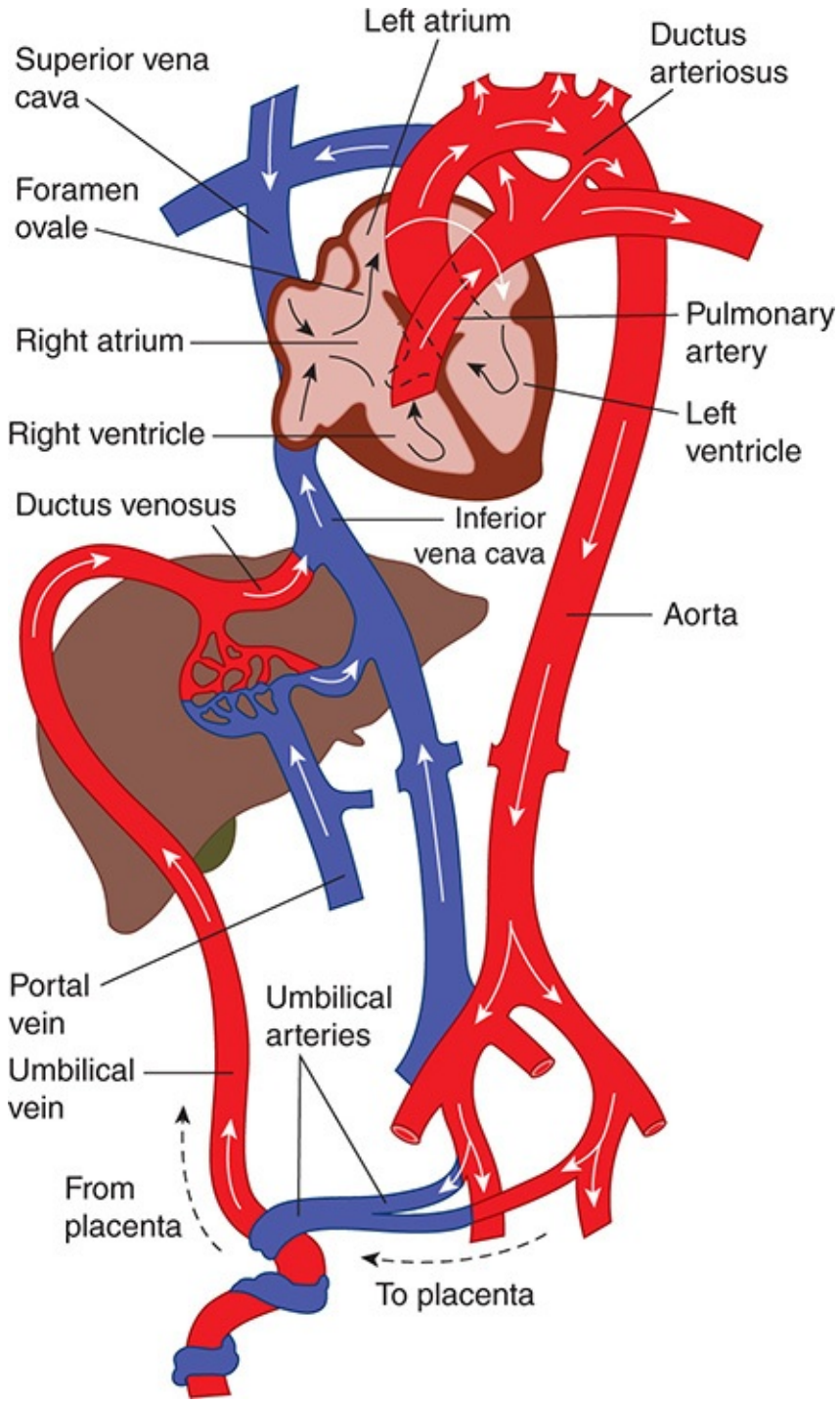


FIGURE 33–18 Circulation in the fetus. Most of the oxygenated blood reaching the heart via the umbilical vein and inferior vena cava is diverted through the foramen ovale and pumped out the aorta to the head, while the deoxygenated blood returned via the superior vena cava is mostly pumped through the pulmonary artery and ductus arteriosus to the feet and the umbilical arteries.

FETAL RESPIRATION

The tissues of fetal and newborn mammals have a remarkable but poorly understood resistance to hypoxia. However, the O_2 saturation of the maternal blood in the placenta is so low that the fetus might suffer hypoxic damage if fetal red cells did not have a greater O_2 affinity than adult red cells **(Figure 33–19)**. The fetal red cells contain fetal hemoglobin (hemoglobin F), whereas the adult cells contain adult hemoglobin (hemoglobin A). The cause of the difference in O_2 affinity between the two is that hemoglobin F binds 2, 3-DPG less effectively than hemoglobin A does. The decrease in O_2 affinity due to the binding of 2, 3-DPG is discussed in Chapter 31.

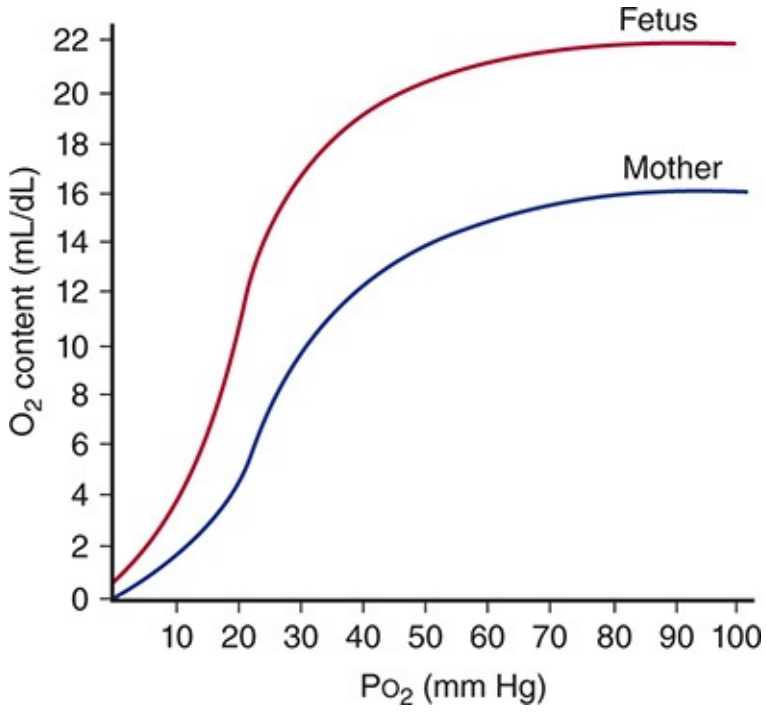


FIGURE 33–19 Dissociation curves of hemoglobin in human maternal and fetal blood.

CHANGES IN FETAL CIRCULATION & RESPIRATION AT BIRTH

Because of the patent ductus arteriosus and foramen ovale (Figure 33–18), the left heart and right heart pump in parallel in the fetus rather than in series as they

do in the adult. At birth the circulation has to change from a parallel to a serial system and from placental to pulmonary gas exchange. This has to be largely completed within minutes of birth, although the complete changes may not be achieved for many hours. The essential element initiating the changes is the fall in the pulmonary vascular resistance when the lungs are first filled with a gas. The pulmonary vascular resistance falls to less than 20% of the value in utero after the lungs are expanded by the first few breaths.

The stimulus for the baby to initiate breathing after birth is not fully understood. In a normal labor, there is no evidence that hypoxia plays a significant role and it is likely that the sudden exposure to light, sound, cold air, and other tactile stimuli are the most important triggers. Oxygen is a strong vasodilator of the pulmonary vascular tree, operating through the production of nitric oxide (NO). The high arterial oxygen tension and bradykinin, which is released from the lungs during their initial inflation, are vasoconstrictors for the umbilical arteries and ductus arteriosus.

Blood returning from the lungs raises the pressure in the left atrium, closing the foramen ovale by pushing the valve that guards it against the interatrial septum. The ductus arteriosus constricts and this, together with the fall in pulmonary vascular resistance, directs most of the output of the right ventricle through the lungs. The ductus arteriosus is functionally closed within a few hours of birth and permanent anatomic closure follows in the next 24–48 h due to extensive intimal thickening. In addition, relatively high concentrations of vasodilators are present in the ductus in utero—especially prostaglandin $F_{2\alpha}$—and synthesis of this prostaglandin is blocked by inhibition of cyclooxygenase at birth. In many premature infants the ductus fails to close spontaneously but closure can be produced by drugs that inhibit cyclooxygenase.

After pulmonary respiration is established, highly oxygenated blood containing pulmonary bradykinin reaches the umbilical arteries, initiating constriction. Further constriction is also stimulated by cold and handling of the cord and the arteries are completely constricted within 3−5 minutes after birth. Venous flow continues as a result of the increased placental pressure within the contracted uterus and the negative intrathoracic pressure of the baby’s respiratory efforts (initially −30 to −50 mm Hg). This redistribution of blood within the fetoplacental circulation after birth is called the “placental transfusion.” Within 5 minutes, however, flow ceases within the umbilical cord and there is little residual blood remaining within the placenta.

Finally, in the fetal circulation the right ventricular output is slightly greater than that of the left. This is possible because of the parallel pump system and the

shunts. After birth the serial system requires both ventricular outputs to be equal before the shunts can close.

CHAPTER SUMMARY

- Blood flow to the brain is via four arteries that ultimately unite to form the circle of Willis; from which six arteries supply the cerebral cortex. There is little cross-over of substances from one side of the arterial system to the other side of the brain. Venous drainage is via the deep veins and dural sinuses into the internal jugular veins.
- Cerebrospinal fluid is produced predominantly in the choroid plexus of the brain, driven mainly by active transport mechanisms in the choroid epithelial cells. Fluid is reabsorbed into the bloodstream to maintain appropriate pressure in the setting of continuous production.
- The permeation of circulating substances into the brain is tightly controlled. Water, CO_2, and O_2 permeate freely. Other substances (such as glucose) require specific transport mechanisms, whereas entry of macromolecules is negligible. The effectiveness of the blood-brain barrier in preventing entry of xenobiotics is bolstered by active efflux mediated by P-glycoprotein.
- The brain consumes approximately 20% of the total body resting O_2 consumption, and occlusion of its blood supply can produce unconsciousness in seconds. Glucose is also critical to supply energy for brain function. When blood flow to a part of the brain is interrupted, the local injury results in a stroke.
- The coronary circulation supplies oxygen to the contracting myocardium. Metabolic products and neural input induce vasodilation as needed for oxygen demand.
- Blockage of coronary arteries may lead to irreversible injury to heart tissue. Blood flows to the subendocardial portion of the left ventricle only during diastole, making this region most susceptible to myocardial infarction in the setting of coronary artery disease.
- Control of cutaneous blood flow is a key facet of temperature regulation, and is underpinned by varying levels of shunting through arteriovenous anastomoses. Hypoxia, axon reflexes, and sympathetic input are all important determinants of flow through the cutaneous vasculature.
- The fetal circulation cooperates with that of the placenta and uterus to

deliver oxygen and nutrients to the growing fetus, as well as carrying away waste products. Unique anatomic features of the fetal circulation as well as biochemical properties of fetal hemoglobin serve to ensure adequate O_2 supply, particularly to the head.

- At birth, the circulation must shift rapidly to a serial system where the right and left heart no longer pump in parallel. The foramen ovale and the ductus arteriosus close such that the neonatal lungs take over as the site for oxygen exchange. Following the establishment of pulmonary respiration, flow in the umbilical system rapidly ceases due to vasoconstrictive signals.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following organs has the greatest blood flow per 100 g of tissue?

   A. Brain

   B. Heart muscle

   C. Skin

   D. Liver

   E. Kidneys

2. A baby boy is brought to the hospital because of convulsions. In the course of a workup, his body temperature and plasma glucose are found to be normal, but his cerebrospinal fluid glucose is 12 mg/dL (normal, 65 mg/dL). A possible explanation of his condition is

   A. constitutive activation of GLUT3 in neurons.

   B. SGLT-1 deficiency in astrocytes.

   C. GLUT5 deficiency in cerebral capillaries.

   D. GLUT1 55K deficiency in cerebral capillaries.

   E. GLUT1 45K deficiency in microglia.

3. A scientist injects a rat intravenously with Evans blue, a dye that binds to serum albumin, then euthanizes the animal one hour later. When she examines sections of the brain, where would blue staining be expected?

   A. Medulla

   B. Area postrema

   C. Motor area

D. Parietal lobe

E. Thalamus

4. The pressure differential between the heart and the aorta is least in the

A. left ventricle during systole.

B. left ventricle during diastole.

C. right ventricle during systole.

D. right ventricle during diastole.

E. left atrium during systole.

5. Injection of tissue plasminogen activator (t-PA) would probably be most beneficial

A. after at least 1 year of uncomplicated recovery following occlusion of a coronary artery.

B. after at least 2 months of rest and recuperation following occlusion of a coronary artery.

C. during the second week after occlusion of a coronary artery.

D. during the second day after occlusion of a coronary artery.

E. during the second hour after occlusion of a coronary artery.

6. Which of the following does *not* dilate arterioles in the skin?

A. Increased body temperature

B. Acetylcholine

C. Bradykinin

D. Substance P

E. Epinephrine

7. Blood in which of the following vessels normally has the lowest P_{O_2}?

A. Maternal artery

B. Maternal uterine vein

C. Maternal femoral vein

D. Umbilical artery

E. Umbilical vein

8. A female infant is delivered at a gestational age of 32 weeks and weighing 1 kg. Forty-eight hours after birth, she is noted to have a heart murmur and a bounding pulse, and is feeding poorly. Her symptoms resolve with administration of indomethacin. What is the most likely diagnosis?

A. Necrotizing enterocolitis
B. Bronchopulmonary dysplasia
C. Apnea
D. Patent ductus arteriosus
E. Respiratory distress syndrome

SECTION VI

Respiratory Physiology

Respiration, or the uptake of O_2 and removal of CO_2 from the body as a whole, is the primary goal of the lungs. At rest, a normal human breathes 12–15 times a minute. With each breath containing ~500 mL of air, this translates to 6–8 L of air that is inspired and expired every minute. Once the air reaches the depths of the lungs in the alveoli, simple diffusion allows O_2 to enter the blood in the pulmonary capillaries and CO_2 to enter the alveoli, from where it can be expired. Using some basic math, on average, 250 mL of O_2 enters the body per minute and 200 mL of CO_2 is excreted. In addition to the O_2 that enters the respiratory system, inspired air also contains a variety of particulates that must be properly filtered and/or removed to maintain lung health. Finally, although humans have a certain amount of control over breathing, most of the minute to minute function, including the fine adjustments necessary for proper lung function, are accomplished independent of voluntary control. The goal of this section is to review basic concepts that underlie important aspects of the control and outcome of breathing as well as to highlight other important functions in respiratory physiology.

The respiratory system is connected to the outside world by the upper airway that leads down a set of conduits before reaching the gas-exchanging areas (the alveoli). The function of the lungs is supported by a variety of anatomic features that serve to inflate/deflate the lungs, thereby allowing the movement of gases to and from the rest of the body. Supporting features include the chest wall, the respiratory muscles (which increase and decrease the size of the thoracic cavity), the areas in the brain that control the muscles, and the tracts and nerves that connect the brain to the muscles. Finally, the lungs support the pulmonary circulation, which allows for movement of gases to other organs and tissues of the body. In the first chapter of this section, the unique anatomic, histological,

and cellular makeup of the respiratory system and how the intricate structure of the lungs contributes to respiratory physiology will be explored. This examination will lead into basic lung measurements that both define and allow for lung inflation/deflation as well as some of the nonrespiratory functions essential for lung health.

The discussion will continue with an overview of the primary function of the respiratory system—the capture of O_2 from the outside environment and its delivery to tissues, as well as the simultaneous removal of CO_2 from the tissues to the outside environment. During this discussion, the critical role of pH in gas exchange as well as the ability of the lungs to contribute to pH regulation of the blood is examined. A discussion of respiratory responses to altered O_2 or CO_2 concentrations, caused by environmental and/or pathophysiological changes, is used to better understand the overall control of coordinated uptake of O_2 and excretion of CO_2.

Control of breathing is quite complex and includes not only the repetitive neuronal firing that controls muscle movements that inflate/deflate the lungs but also a series of feedback loops that increase/decrease deflation depending on the gas content of the blood. The final chapter in this section begins with an overview of some of the key factors that aid in this regulation of respiration. Specific examples of common respiratory abnormalities and how they relate to altered regulation of breathing are also discussed to better understand the intricate feedback loops that help regulate breathing.

Due to the complexity of the lungs and thus the variety of working parts that can be compromised, there is a wide-ranging list of diseases that impact its function. Such diseases include common (and uncommon) respiratory infections, asthma, chronic obstructive pulmonary disease (COPD), acute respiratory distress syndrome, pulmonary hypertension, lung cancer, and many more. The health burden from such a diverse collection of disorders cannot be overstated. Using COPD as an example, current conservative estimates hold that over 12 million adults in the United States suffer from the condition. In fact, COPD is the fourth leading cause of death (and rising) in the world and a contributory factor in an equal number of non-COPD deaths. Although treatment strategies for COPD, largely based on continuing research efforts and understanding, have contributed to an improved lifestyle, the underlying causes are, as of yet, untreatable. The continued and improved understanding of respiratory physiology and lung

function (and its dysfunction) will provide opportunities to develop new strategies for treatment of COPD, as well as the myriad of other lung diseases.

CHAPTER 34

Introduction to Pulmonary Structure & Mechanics

OBJECTIVES

After studying this chapter, you should be able to:

- List the passages through which air passes from the exterior to the alveoli, and describe the cells that line each of them.
- List the major muscles involved in respiration, and state the role of each.
- Define the basic measures of lung volume and give approximate values for each in a normal adult.
- Define lung compliance and airway resistance.
- Compare the pulmonary and systemic circulations and list some major differences between them.
- Describe basic lung defense and metabolic functions.
- Define partial pressure and calculate the partial pressure of each of the important gases in the atmosphere at sea level.

INTRODUCTION

The structure of the respiratory system is uniquely suited to its primary function, the transport of gases in (O_2) and out (CO_2) of the body. In addition, the respiratory system provides a large volume of tissue that is constantly exposed to the outside environment, and thus, to potential infection and injury. Finally, the pulmonary system includes a unique circulation that must handle the blood flow; the lungs are the only organ that receives the whole cardiac output in the body. This chapter begins with the basic anatomy and cellular physiology that contribute to the respiratory system and their unique features. The chapter also includes discussion of how the anatomic features contribute to the basic mechanics of breathing, as well as highlights of nonrespiratory functions in the pulmonary system.

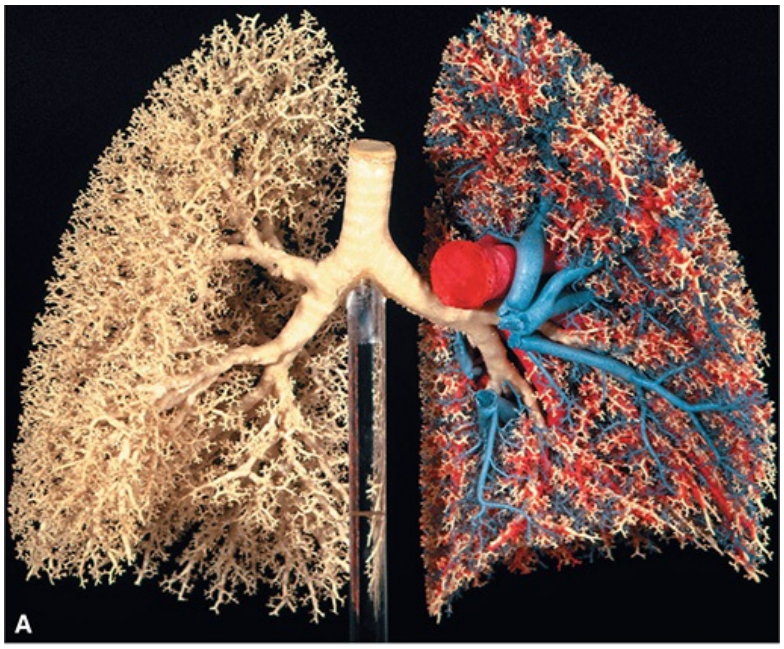
ANATOMY OF THE LUNGS

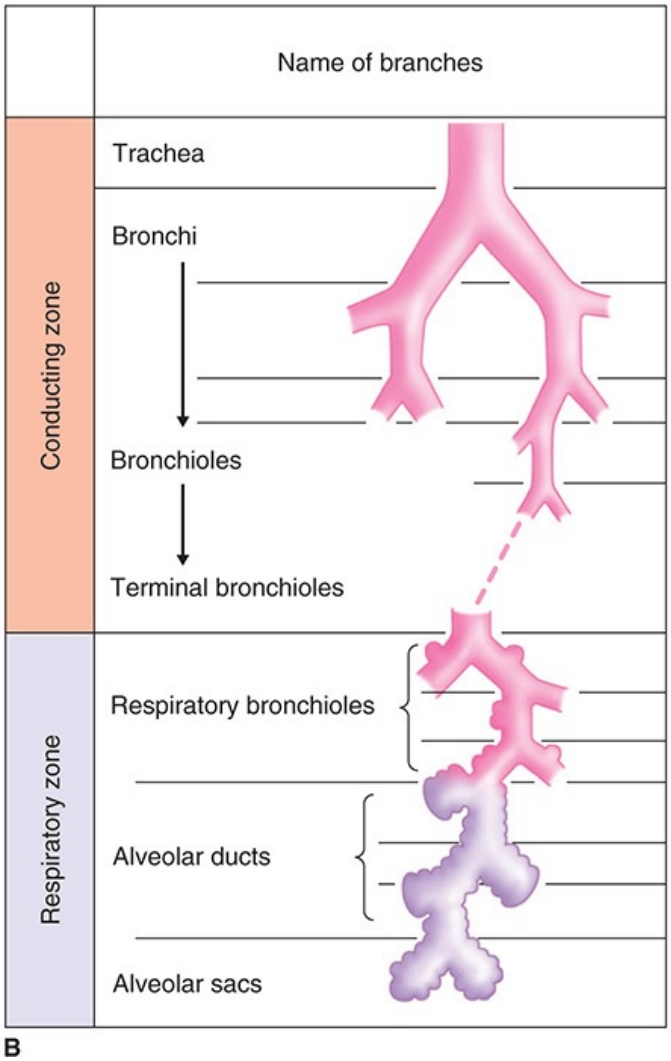
REGIONS OF THE RESPIRATORY TRACT

Airflow through the respiratory system can be broken down into three interconnected regions: the **upper airway,** the **conducting airway,** and the **alveolar airway** (also known as the **lung parenchyma** or **acinar tissue**). The upper airway consists of the entry systems, the nose/nasal cavity and mouth that lead into the pharynx. The larynx extends from the lower part of the pharynx to complete the upper airway. The nose is the primary point of entry for inhaled air; therefore, the mucosal epithelium lining the nasopharyngeal airways is exposed to the highest concentration of inhaled allergens, toxicants, and particulate matter. With this in mind, it is easy to understand that in addition to olfaction, the nose and upper airway provides two additional crucial functions in airflow: (1) filtering out large particulates to prevent them from reaching the conducting and alveolar airways and (2) serving to warm and humidify air as it enters the body. Particulates larger than 30–50 μm in size (diameter) tend to not to be inhaled through the nose whereas particulates on the order of 5–10 μm impact on the nasopharynx and do enter the conducting airway. Most of these latter particles settle on mucous membranes in the nose and pharynx. Because of their momentum, they do not follow the airstream as it curves downward into the lungs, and they impact on or near the **tonsils** and **adenoids,** large collections of immunologically active lymphoid tissue in the back of the pharynx.

CONDUCTING AIRWAY

The conducting airway begins at the trachea and branches dichotomously to greatly expand the surface area of the tissue in the lungs. The first 16 generations of passages form the conducting zone of the airways that transports gas from and to the upper airway described above **(Figure 34–1)**. These branches are made up of bronchi, bronchioles, and terminal bronchioles. The conducting airway is made up of a variety of specialized cells that provide more than simply a conduit for air to reach the lungs **(Figure 34–2)**. The mucosal epithelium is attached to a thin basement membrane, and beneath this, the lamina propria. Collectively these are referred to as the "airway mucosa." Smooth muscle cells are found beneath the epithelium and an enveloping connective tissue is likewise interspersed with cartilage that is more predominant in the portions of the conducting airway of greater caliber. The epithelium is organized as a pseudostratified epithelium and contains several cell types, including ciliated and secretory cells (eg, goblet cells and glandular acini) that provide key components for airway innate immunity, and basal cells that can serve as progenitor cells for repair during injury. As the conducting airway transitions to terminal and transitional bronchioles, the histological appearance of the conducting tubes change. Secretory glands are absent from the epithelium of the bronchioles and terminal bronchioles, smooth muscle plays a more prominent role and cartilage is largely absent from the underlying tissue. Club cells (formerly termed "Clara cells"), nonciliated cuboidal epithelial cells that secrete important defense markers and serve as progenitor cells after injury, make up a large portion of the epithelial lining in the latter portions of the conducting airway.


A


Name of branches
Conducting zone
Trachea
Bronchi
Bronchioles
Terminal bronchioles
Respiratory zone
Respiratory bronchioles
Alveolar ducts
Alveolar sacs

B

FIGURE 34–1 Conducting and respiratory zones in the airway. A) Resin cast of the human airway tree shows dichotomous branching beginning at the trachea. Note the added pulmonary arteries (red) and veins (blue) displayed in the left lung. **B)** The branching pattern of the airway is sketched with individual regions of the conducting and respiratory airways identified. (Reproduced with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008.)

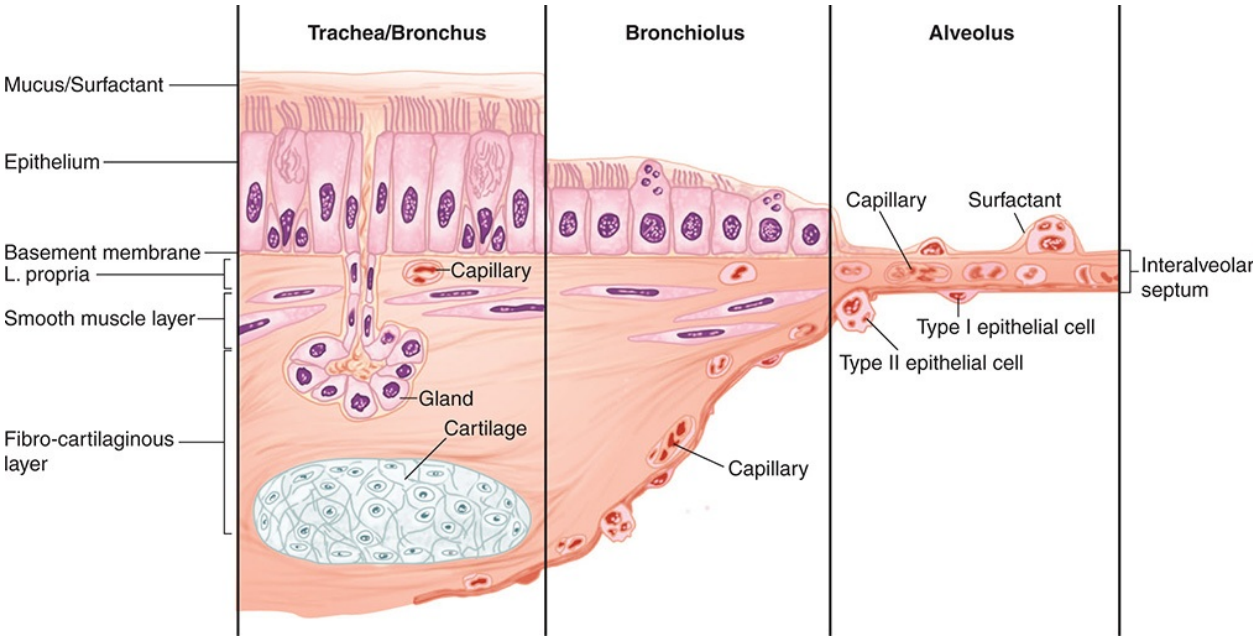


FIGURE 34–2 Cellular transition from conducting airway to the alveolus. The epithelial layer transitions from pseudostratified layer with submucosal glands to a cuboidal epithelium and then to a squamous epithelium. The underlying mesynchyme tissue and capillary structure also changes with the airway transition. (Adapted with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008.)

Epithelial cells in the conducting airway can secrete a variety of molecules that aid in lung defense. Secretory immunoglobulins (IgA), collectins (including surfactant protein (SP) A and SP-D), defensins and other peptides and proteases, reactive oxygen species (ROS), and reactive nitrogen species are all generated by airway epithelial cells. These secretions can act directly as antimicrobials to help keep the airway free of infection. Airway epithelial cells also secrete a variety of chemokines and cytokines that recruit traditional immune cells and

other immune effector cells to site of infections. The smaller particles that make it through the upper airway, ~2–5 µm in diameter, generally fall on the walls of the bronchi as the airflow slows in the smaller passages. There they can initiate reflex bronchial constriction and coughing. Alternatively, they can be moved away from the lungs by the "mucociliary escalator." The epithelium of the respiratory passages from the anterior third of the nose to the beginning of the respiratory bronchioles is ciliated (Figure 34–2). The cilia are bathed in a periciliary fluid where they typically beat at rates of 10–15 Hz. On top of the periciliary layer and the beating cilia rests a mucus layer, a complex mixture of proteins and polysaccharides secreted from specialized cells, glands, or both in the conducting airway. This combination allows for the trapping of foreign particles (in the mucus) and their transport out of the airway (powered by ciliary beat). The ciliary mechanism is capable of moving particles away from the lungs at a rate of at least 16 mm/min. When ciliary motility is defective, as can occur in smokers, or as a result of other environmental conditions or genetic deficiencies, mucus transport is virtually absent. This can lead to chronic sinusitis, recurrent lung infections, and bronchiectasis. Some of these symptoms are evident in cystic fibrosis **(Clinical Box 34–1)**.

The walls of the bronchi and bronchioles are innervated by the autonomic nervous system. Nerve cells in the airways sense mechanical stimuli or the presence of unwanted substances in the airways such as inhaled dusts, cold air, noxious gases, and cigarette smoke. These neurons can signal the respiratory centers to contract the respiratory muscles and initiate sneeze or cough reflexes. The receptors show rapid adaptation when they are continuously stimulated to limit sneeze and cough under normal conditions. The β_2-adrenergic receptors help mediate bronchodilation. They also increase bronchial secretions (eg, mucus), while α_1-adrenergic receptors inhibit secretions.

ALVEOLAR AIRWAY

Between the trachea and the alveolar sacs, the airways divide 23 times. The last seven generations form the transitional and respiratory zones where gas exchange occurs are made up of transitional and respiratory bronchioles, alveolar ducts, and alveoli (Figure 34–1). These multiple divisions greatly increase the total cross-sectional area of the airways, from 2.5 cm^2 in the trachea to 11,800 cm^2 in the alveoli. Consequently, the velocity of airflow in the small airways declines to very low values. The transition from the conducting to the

respiratory region that ends in the alveoli also includes a change in cellular arrangements (Figure 34–2 and **Figure 34–3**). There are 300 million alveoli in humans, and the total area of the alveolar walls in contact with capillaries in both lungs is about 70 m^2.

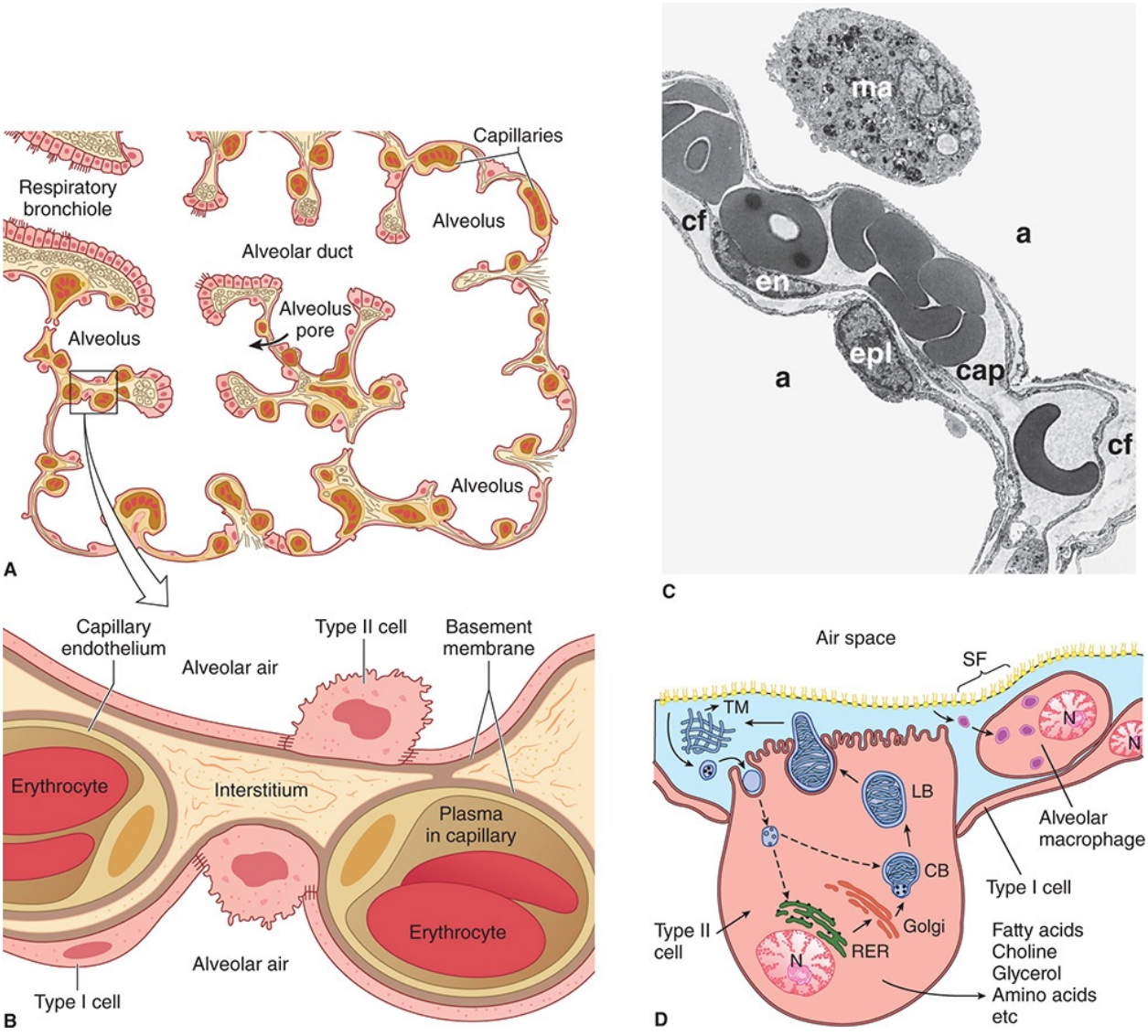


FIGURE 34–3 Prominent cells in the adult human alveolus. A) A cross-section of the respiratory zone shows the relationship between capillaries and the airway epithelium. Only 4 of the 18 alveoli are labeled. **B)** Enlargement of the boxed area from (A) displaying intimate relationship between capillaries, the interstitium, and the alveolar epithelium. **C)** Electron micrograph displaying a typical area depicted in (B). The pulmonary capillary (cap) in the septum contains plasma with red blood cells. Note the closely apposed endothelial and

pulmonary epithelial cell membranes separated at places by additional connective tissue fibers (cf); en, nucleus of endothelial cell; epl, nucleus of type I alveolar epithelial cell; a, alveolar space; ma, alveolar macrophage. **D)** Type II cell formation and metabolism of surfactant. Lamellar bodies (LBs) are formed in type II alveolar epithelial cells and secreted by exocytosis into the fluid lining the alveoli. The released lamellar body material is converted to tubular myelin (TM), and the TM is the source of the phospholipid surface film (SF). Surfactant is taken up by endocytosis into alveolar macrophages and type II epithelial cells. CB, composite body; Golgi, Golgi apparatus; N, nucleus; RER, rough endoplasmic reticulum. (For **(A)** Reproduced with permission from Greep RO, Weiss L: *Histology*, 3rd ed. New York: New York, NY: McGraw-Hill; 1973; **(B)** Adapted with permission from Gong H, Drage CW: *The Respiratory System: A Core Curriculum*. Norwalk, CT: Appleton-Century-Crofts: 1982; **(C)** Reproduced with permission from Burri PA: Development and growth of the human lung. In: *Handbook of Physiology*, Section 3, *The Respiratory System*. Fishman AP, Fisher AB [editors]. American Physiological Society, 1985; and **(D)** Reproduced with permission from Wright JR: Metabolism and turnover of lung surfactant. Am Rev Respir Dis1987 Aug; 136(2):444, 1987.)

The alveoli are lined by two types of epithelial cells. **Type I cells** are flat cells with large cytoplasmic extensions and are the primary lining cells of the alveoli, covering approximately 95% of the alveolar epithelial surface area. **Type II cells (granularpneumocytes)** are thicker and contain numerous lamellar inclusion bodies. Although these cells make up only 5% of the surface area, they represent approximately 60% of the epithelial cells in the alveoli. Type II cells are important in alveolar repair as well as other lung cellular functions. One prime function of the type II cell is the production of **surfactant** (Figure 34–3D). Typical **lamellar bodies,** membrane-bound organelles containing whorls of phospholipid, are formed in these cells and secreted into the alveolar lumen by exocytosis. Tubes of lipid called **tubular myelin** form from the extruded bodies, and the tubular myelin in turn forms a phospholipid film. Following secretion, the phospholipids of surfactant line up in the alveoli with their hydrophobic fatty acid tails facing the alveolar lumen. This surfactant layer plays an important role in maintaining alveolar structure by reducing surface tension (see below). Surface tension is inversely proportional to the surfactant concentration per unit area. The surfactant molecules move further apart as the alveoli enlarge during inspiration, and surface tension increases, whereas it decreases when they move closer together during expiration. Some of the protein–lipid complexes in surfactant are taken up by endocytosis in type II alveolar cells and recycled.

CLINICAL BOX 34–1

Cystic Fibrosis

Among whites, cystic fibrosis is one of the most common genetic disorders; greater than 3% of the United States population are carriers for this autosomal recessive disease.

The gene that is abnormal in cystic fibrosis is located on the long arm of chromosome 7 and encodes the **cystic fibrosis transmembrane conductance regulator (CFTR),** a regulated Cl^- channel located on the apical membrane of various secretory and absorptive epithelia. The number of reported mutations in the *CFTR* gene that cause cystic fibrosis is large (> 1000) and the mutations are now grouped into five classes (I–V) based on their effects on cellular function. Class I mutations do not allow for synthesis of the protein. Class II mutations have protein processing defects. Class III mutations have a block in their channel regulation. Class IV mutations display altered conductance of the ion channel. Class V mutations display reduced synthesis of the protein. The severity of the defect varies with the class and the individual mutation. The most common mutation causing cystic fibrosis is loss of the phenylalanine residue at amino acid position 508 of the protein (ΔF508), a Class II mutation that limits the amount of CFTR protein that gets to the plasma membrane.

One outcome of cystic fibrosis is repeated pulmonary infections, particularly with *Pseudomonas aeruginosa*, and progressive, eventually fatal destruction of the lungs. There is also suppressed chloride secretion across the wall of the airways. One would expect Na^+ reabsorption to be depressed as well, and indeed in sweat glands it is. However, in the lungs, it is enhanced, so that the Na^+ and water move out of airways, leaving their other secretions inspissated and sticky. This results in a reduced periciliary layer that inhibits function of the mucociliary and escalator and alters the local environment to reduce the effectiveness of antimicrobial secretions.

THERAPEUTIC HIGHLIGHTS

Traditional treatments of cystic fibrosis address the various symptoms. Chest physiotherapy and mucolytics are used to loosen thick mucus and aid lung clearance. Antibiotics are used to prevent new infections and keep chronic infections in check. Bronchodilators and anti-inflammatory medications are

used to help expand and clear air passages. Pancreatic enzymes and nutritive supplements are used to increase nutrient absorption and promote weight gain. Because of the “single gene” mutation of this disease, gene therapy has been closely examined; however, results have not been successful. More recently, drugs that target the molecular defects have been advancing in clinical trials and are showing great promise for better treatments. The newly developed genome editing tool, CRISPR-Cas9 (clustered regularly interspaced short palindromic repeats (CRISPR) associated nuclease 9), will become a highly promising in vivo gene therapy for cystic fibrosis by correcting mutation(s) of *CFTR* gene.

The alveoli are surrounded by pulmonary capillaries. In most areas, air and blood are separated only by the alveolar epithelium and the capillary endothelium, so they are about 0.5 µm apart (Figure 34–3). The alveoli also contain other specialized cells, including pulmonary alveolar macrophages (PAMs, or AMs), lymphocytes, plasma cells, neuroendocrine cells, and mast cells. PAMs are an important component of the pulmonary defense system. Like other macrophages, these cells come originally from the bone marrow. PAMs are actively phagocytic and ingest small particles that evade the mucociliary escalator and reach the alveoli. They also help process inhaled antigens for immunologic attack, and they secrete substances that attract granulocytes to the lungs as well as substances that stimulate granulocyte and monocyte formation in the bone marrow. PAM function can also be detrimental—when they ingest large amounts of the substances in cigarette smoke or other irritants, they may release lysosomal products into the extracellular space to cause inflammation.

RESPIRATORY MUSCLES

The lungs are positioned within the thoracic cavity, which is defined by the rib cage and the spinal column. The lungs are surrounded by a variety of muscles that contribute to breathing **(Figure 34–4)**. Movement of the diaphragm accounts for 75% of the change in intrathoracic volume during quiet inspiration. Attached around the bottom of the thoracic cage, this muscle arches over the liver and moves downward like a piston when it contracts. The distance it moves ranges from 1.5 cm to as much as 7 cm with deep inspiration.

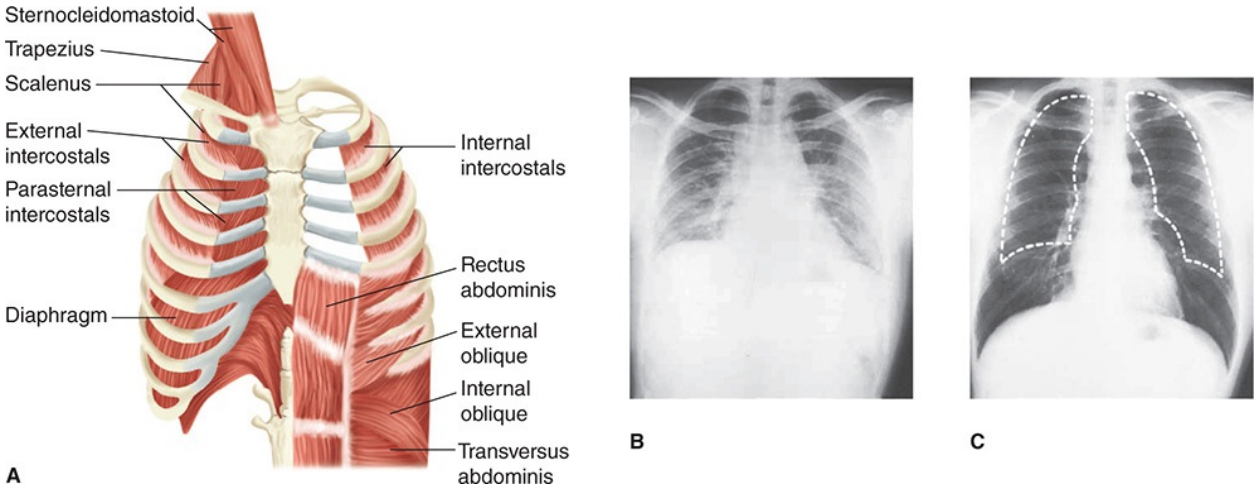


FIGURE 34–4 Muscles and movement in respiration. A) An idealized diagram of respiratory muscles surrounding the rib cage. The diaphragm and intercostals play prominent roles in respiration. **B)** and **C)** Radiograph of chest in full expiration (B) and full inspiration (C). The dashed white line in C is an outline of the lungs in full expiration. Note the difference in intrathoracic volume. (For **(A)** Reproduced with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008; **(B, C)** Reproduced with permission from Comroe JH Jr: *Physiology of Respiration*, 2nd ed., Year Book; 1975.)

The diaphragm has three parts: the portion, made up of muscle fibers that are attached to the ribs around the bottom of the thoracic cage; the crural portion, made up of fibers that are attached to the ligaments along the vertebrae; and the central tendon, into which the costal and the crural fibers insert. The central tendon is also the inferior part of the pericardium. The crural fibers pass on either side of the esophagus and can compress it when they contract. The costal and crural portions are innervated by different parts of the phrenic nerve and can contract separately. For example, during vomiting and eructation, intra-abdominal pressure is increased by contraction of the costal fibers but the crural fibers remain relaxed, allowing material to pass from the stomach into the esophagus.

The other important **inspiratory muscles** are the **external intercostal muscles,** which run obliquely downward and forward from rib to rib. The ribs pivot as if hinged at the back, so that when the external intercostals contract they elevate the lower ribs. This pushes the sternum outward and increases the anteroposterior diameter of the chest. The transverse diameter also increases but to a lesser degree. Either the diaphragm or the external intercostal muscles alone

can maintain adequate ventilation at rest. Transection of the spinal cord above the third cervical segment is fatal without artificial respiration, but transection below the fifth cervical segment is not because it leaves the phrenic nerves that innervate the diaphragm intact; the phrenic nerves arise from cervical segments 3–5. Conversely, in patients with bilateral phrenic nerve palsy but intact innervation of their intercostal muscles, respiration is somewhat labored but adequate to maintain life. The scalene and sternocleidomastoid muscles in the neck are accessory inspiratory muscles that help elevate the thoracic cage during deep labored respiration.

A decrease in intrathoracic volume and forced expiration result when the **expiratory muscles** contract. The internal intercostals have this action because they pass obliquely downward and posteriorly from rib to rib and therefore pull the rib cage downward when they contract. Contractions of the muscles of the anterior abdominal wall also aid expiration by pulling the rib cage downward and inward and by increasing the intra-abdominal pressure, which pushes the diaphragm upward.

In order for air to get into the conducting airway it must pass through the **glottis,** defined as the area including and between the vocal folds within the larynx. The abductor muscles in the larynx contract early in inspiration, pulling the vocal cords apart and opening the glottis. During swallowing or gagging, a reflex contraction of the adductor muscles closes the glottis and prevents aspiration of food, fluid, or vomitus into the lungs. In unconscious or anesthetized patients, glottic closure may be incomplete and vomitus may enter the trachea, causing an inflammatory reaction in the lungs **(aspiration pneumonia).**

LUNG PLEURA

The **pleural cavity** and its infoldings serve as a lubricating fluid/area that allows for lung movement within the thoracic cavity **(Figure 34–5A)**. There are two layers that contribute to the pleural cavity: the **parietal pleura** and the **visceral pleura.** The parietal pleura is a membrane that lines the chest cavity containing the lungs. The **visceral pleura** is a membrane that lines the lung surface. The pleural fluid (∼15–20 mL) forms a thin layer between the pleural membranes and prevents friction between surfaces during inspiration and expiration.

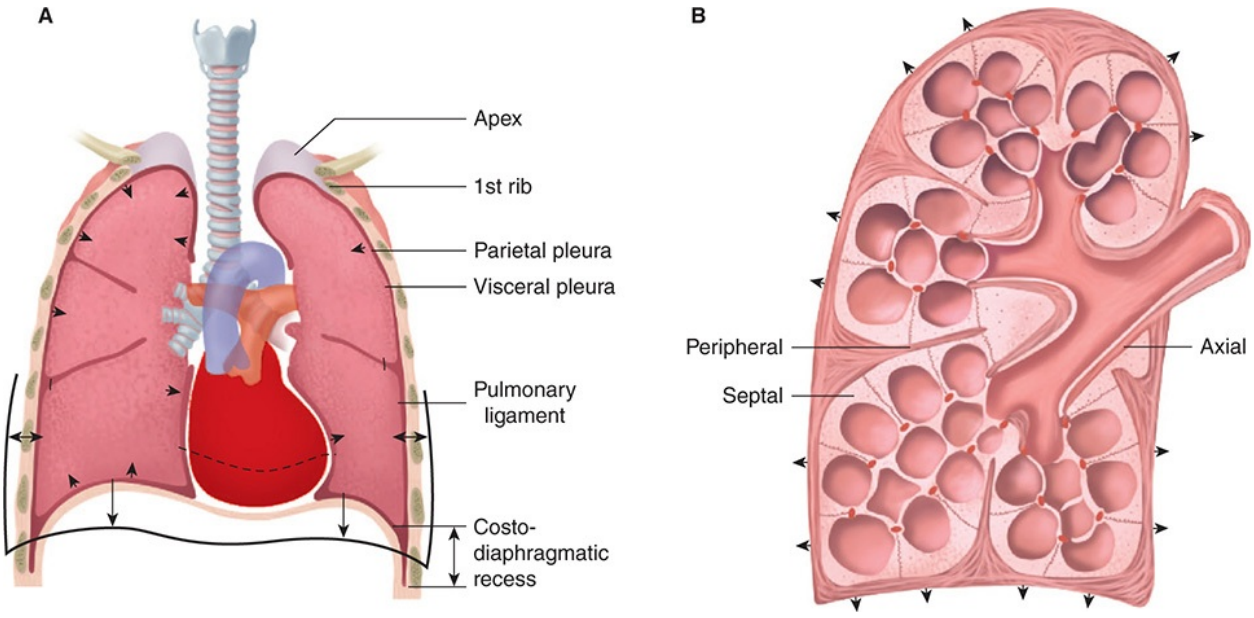


FIGURE 34–5 Pleural space and connective fibers. A) Front sectional drawing of lungs within the rib cage. Note the parietal and visceral pleura and the infoldings around the lung lobes that include pleural space. **B)** Connective fiber tracts of the lung are highlighted. Note the axial fibers along the airways, the peripheral fibers in the pleura, and the septal fibers. (Adapted with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008.)

The lungs themselves contain a vast amount of free space—it is ~80% air. Although this maximizes surface area for gas exchange, it also requires an extensive support network to maintain lung shape and function. The **connective tissue** within the visceral pleura contains three layers that help support the lungs. Elastic fibers that follow the mesothelium effectively wrap the three lobes of the right lung and the two lobes of the left lung **(Figure 34–5B)**. A deep sheet of fine fibers that follow the outline of the alveoli provide support to individual air sacks. Between these two separate sheets lies connective tissue that is interspersed with individual cells for support and lung maintenance/function.

Blood & Lymph in the Lungs

Both the **pulmonary circulation** and the **bronchial circulation** contribute to blood flow in the lungs. In the pulmonary circulation **(Figure 34–6)**, almost all the blood in the body passes via the pulmonary artery to the pulmonary capillary bed, where it is oxygenated and returned to the left atrium via the pulmonary

veins. The pulmonary arteries strictly follow the branching of the bronchi down to the respiratory bronchioles. The pulmonary veins, however, are spaced between the bronchi on their return to the heart. The separate and much smaller bronchial circulation includes the bronchial arteries that come from systemic arteries. They form capillaries, which drain into bronchial veins or anastomose with pulmonary capillaries or veins. The bronchial veins drain into the azygos vein. The bronchial circulation nourishes the trachea down to the terminal bronchioles and also supplies the pleura and hilar lymph nodes. It should be noted that lymphatic channels are more abundant in the lungs than in any other organ. Lymph nodes are arranged along the bronchial tree and extend down until the bronchi are ~5 mm in diameter. Lymph node sizes can range from 1 mm at the bronchial periphery to 10 mm along the trachea. The nodes are connected by lymph vessels and allow for unidirectional flow of lymph to the subclavian veins.

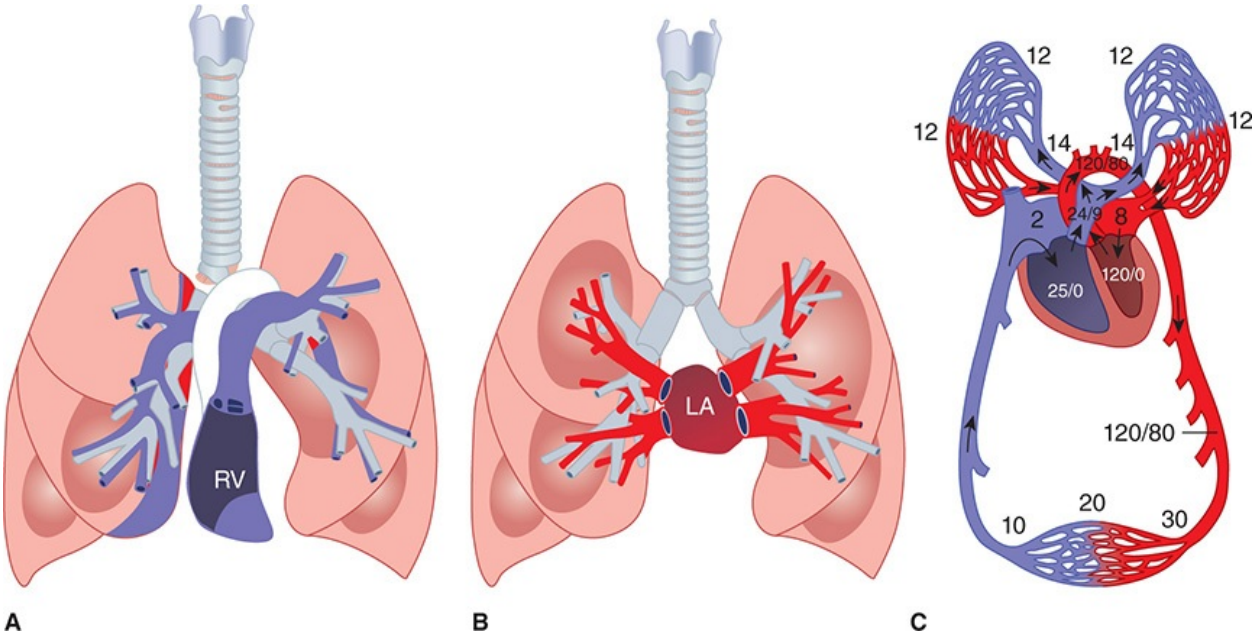


FIGURE 34–6 Pulmonary circulation. A) Schematic diagram of the relation of the main branches of the pulmonary arteries to the bronchial tree. **B)** Schematic of the pulmonary veins in relation to the bronchial tree. **C)** Representative areas of blood flow are labeled with corresponding blood pressure (mm Hg). In all diagrams, blue represents deoxygenated blood and red represents oxygenated blood. The deoxygenated/oxygenated shift in **C** occurs as blood traverses capillaries that wrap individual alveoli and is not as abrupt as shown. LA, left atrium; RV, right ventricle. (For **(A, B)** Reproduced with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed.

New York: McGraw-Hill; 2008; **(C)** Modified with permission from Comroe JH Jr: *Physiology of Respiration*, 2nd ed. Year Book; 1974.)

MECHANICS OF RESPIRATION

INSPIRATION & EXPIRATION

The lungs and the chest wall are elastic structures. Normally, no more than a thin layer of fluid is present between the lungs and the chest wall (intrapleural space). The lungs slide easily on the chest wall, but resist being pulled away from it in the same way that two moist pieces of glass slide on each other but resist separation. The pressure in the “space” between the lungs and chest wall (intrapleural pressure) is subatmospheric **(Figure 34–7)**. The lungs are stretched when they expand at birth, and at the end of quiet expiration their tendency to recoil from the chest wall is just balanced by the tendency of the chest wall to recoil in the opposite direction. If the chest wall is opened, the lungs collapse; and if the lungs lose their elasticity, the chest expands and becomes barrel-shaped.

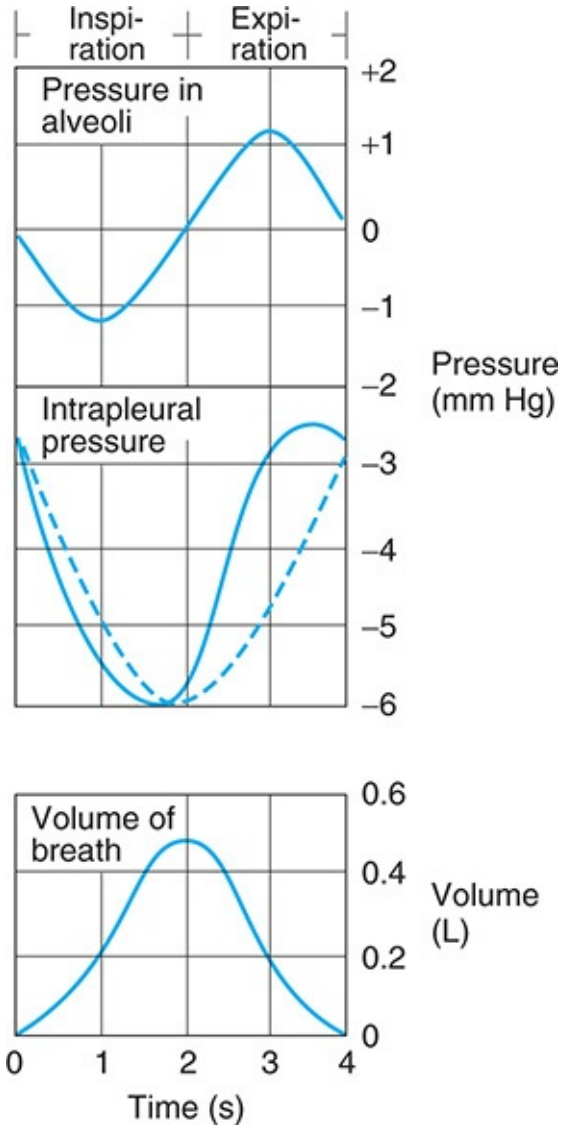


FIGURE 34–7 Pressure in the alveoli and the pleural space relative to atmospheric pressure during inspiration and expiration. The dashed line indicates what the intrapleural pressure would be in the absence of airway and tissue resistance; the actual curve (solid line) is skewed to the left by the resistance. Volume of breath during inspiration/expiration is graphed for comparison.

Inspiration is an active process. The contraction of the inspiratory muscles increases intrathoracic volume. The intrapleural pressure at the base of the lungs, which is normally about –2.5 mm Hg (relative to atmospheric) at the start of inspiration, decreases to about –6 mm Hg. The lungs are pulled into a more expanded position. The pressure in the airway becomes slightly negative, and air flows into the lungs. At the end of inspiration, the lung recoil begins to pull the

chest back to the expiratory position, where the recoil pressures of the lungs and chest wall balance (see below). The pressure in the airway becomes slightly positive, and air flows out of the lungs. Expiration during quiet breathing is passive in the sense that no muscles that decrease intrathoracic volume contract. However, some contraction of the inspiratory muscles occurs in the early part of expiration. This contraction exerts a braking action on the recoil forces and slows expiration. Strong inspiratory efforts reduce intrapleural pressure to values as low as –30 mm Hg, producing correspondingly greater degrees of lung inflation. When ventilation is increased, the extent of lung deflation is also increased by active contraction of expiratory muscles that decrease intrathoracic volume.

QUANTITATING RESPIRATORY PHENOMENA

Modern spirometers permit direct measurement of gas intake and output. Since gas volumes vary with temperature and pressure and since the amount of water vapor in them varies, these devices have the ability to correct respiratory measurements involving volume to a stated set of standard conditions. It should be noted that correct measurements are highly dependent on the ability for the practitioner to properly encourage the patient to fully utilize the device. Modern techniques for gas analysis make possible rapid, reliable measurements of the composition of gas mixtures and the gas content of body fluids. For example, O_2 and CO_2 electrodes, small probes sensitive to O_2 or CO_2, can be inserted into the airway or into blood vessels or tissues and the P_{O_2} and P_{CO_2} recorded continuously. Chronic assessment of oxygenation is carried out noninvasively with a **pulse oximeter,** which can be easily placed on a fingertip.

Lung Volumes & Capacities

Important quantitation of lung function can be gleaned from the displacement of air volume during inspiration and/or expiration. Lung capacities refer to subdivisions that contain two or more volumes. Volumes and capacities recorded on a spirometer from a healthy individual are shown in **Figure 34–8**. Diagnostic spirometry is used to assess a patient's lung function for purposes of comparison with a normal population, or with previous measures from the same patient. The amount of air that moves into the lungs with each inspiration (or the amount that moves out with each expiration) during quiet breathing is called the **tidal**

volume (TV). Typical values for TV are on the order of 500–750 mL. The air inspired with a maximal inspiratory effort in excess of the TV is the **inspiratory reserve volume** (IRV; ~2 L). The volume expelled by an active expiratory effort after passive expiration is the **expiratory reserve volume** (ERV; ~1 L), and the air left in the lungs after a maximal expiratory effort is the **residual volume** (RV; ~1.3 L). When all four of the above components are taken together, they make up the **total lung capacity** (TLC, ~5 L). The TLC can be broken down into alternative capacities that help define functioning lungs. The **vital capacity** (VC, ~3.5 L) refers to the maximum amount of air expired from the fully inflated lungs, or maximum inspiratory level (this represents TV + IRV + ERV). The **inspiratory capacity** (IC, ~2.5 L) is the maximum amount of air inspired from the end-expiratory level (IRV + TV). The **functional residual capacity** (FRC; ~2.5 L) represents the volume of the air remaining in the lungs after expiration of a normal breath (RV + ERV).

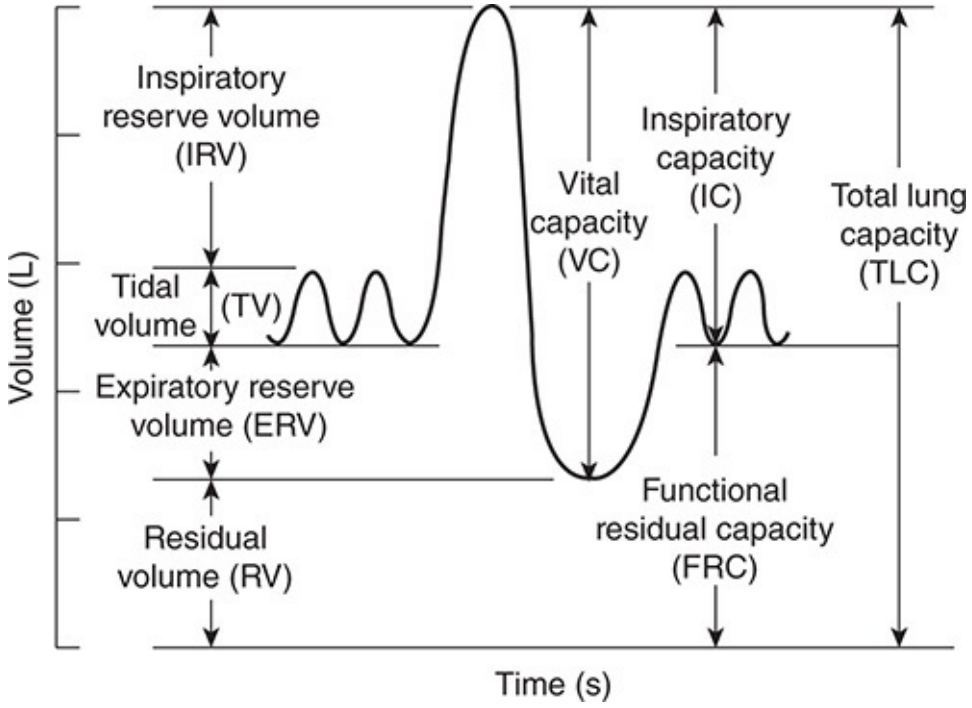


FIGURE 34–8 Lung volumes and capacity measurements. Lung volumes recorded by a spirometer. Lung capacities are determined from volume recordings. See text for definitions. (Reproduced with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008.)

Dynamic measurements of lung volumes and capacities have been used to

determine lung dysfunction. The **forced vital capacity (FVC),** the largest amount of air that can be expired after a maximal inspiratory effort, is frequently measured clinically as an index of pulmonary function. It gives useful information about the strength of the respiratory muscles and other aspects of pulmonary function. The fraction of the VC expired during the first second of a forced expiration is referred to as **FEV_1** (forced expiratory volume in the first second; **Figure 34–9**). The FEV_1 to FVC ratio (FEV_1/FVC) is a useful tool in the recognizing classes of airway disease **(Clinical Box 34–2)**. Other dynamic measurements include the **respiratory minute volume (RMV)** and the **maximal voluntary ventilation (MVV).** RMV is normally ~6 L (500 mL/breath × 12 breaths/min). The **MVV** is the largest volume of gas that can be moved into and out of the lungs in 1 min by voluntary effort. Typically this is measured over a 15-s period and prorated to a minute; normal values range from 140 L/min to 180 L/min for healthy adult men. Changes in RMV and MVV in a patient can be indicative of lung dysfunction.

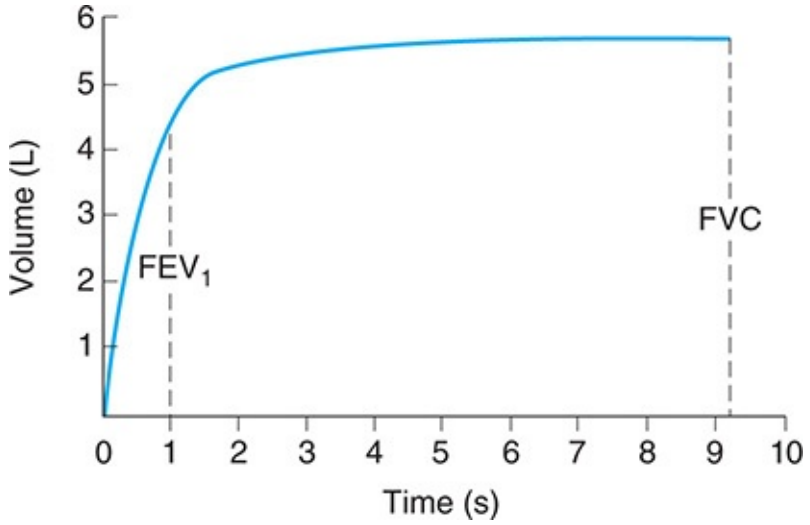


FIGURE 34–9 Volume of gas expired by a normal adult man during a forced expiration, demonstrating the FEV_1 and the FVC. From the graph, the forced expiratory volume in the first second (FEV_1) to the forced vital capacity (FVC) ration (FEV_1/FVC) can be calculated (4L/5L = 80%). (Reproduced with permission from Crapo RO: Pulmonary-function testing, N Engl J Med 1994; July 7; 331(1):25–30.)

COMPLIANCE OF THE LUNGS & CHEST WALL

Compliance is developed due to the tendency for tissue to resume its original position after an applied force has been removed. After an expiration during quiet breathing (eg, at the FRC), the lungs have a tendency to collapse and the chest wall has a tendency to expand. The interaction between the recoil of the lungs and recoil of the chest can be measured through a spirometer that has a valve just beyond the mouthpiece. The mouthpiece contains a pressure-measuring device. After the person inhales a given amount, the valve is shut, closing off the airway. The respiratory muscles are then relaxed while the pressure in the airway is recorded. The procedure is repeated after inhaling or actively exhaling various volumes. The curve of airway pressure obtained in this way, plotted against volume, is the **pressure–volume curve** of the total respiratory system (P_{TR} in **Figure 34–10**). The pressure is zero at a lung volume that corresponds to the volume of gas in the lungs at **FRC (**ie, **relaxation volume).** As can be noted from Figure 34–10, this relaxation pressure is the sum of slightly negative pressure component from the chest wall (P_w) and a slightly positive pressure from the lungs (P_L). P_{TR} is positive at greater volumes and negative at smaller volumes. **Compliance** of the lungs and chest wall is measured as the slope of the P_{TR} curve, or, as a change in lung volume per unit change in airway pressure ($\Delta V/\Delta P$). It is normally measured in the pressure range where the relaxation pressure curve is steepest, and normal values are ~0.2 L/cm H_2O in a healthy adult man. However, compliance depends on lung volume and thus can vary. In an extreme example, an individual with only one lung has approximately half the ΔV for a given ΔP. Compliance is also slightly greater when measured during deflation than when measured during inflation. Consequently, it is more informative to examine the whole pressure–volume curve. The curve is shifted downward and to the right (compliance is decreased) by pulmonary edema and interstitial pulmonary fibrosis **(Figure 34–11)**. Pulmonary fibrosis is a progressive restrictive airway disease in which there is stiffening and scarring of the lungs. The curve is shifted upward and to the left (compliance is increased) in emphysema.

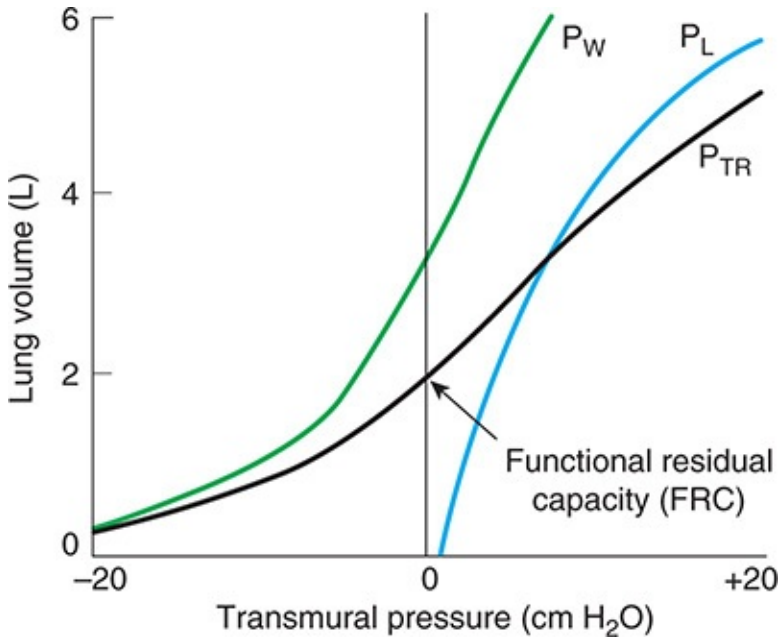


FIGURE 34–10 Pressure–volume curves in the lungs. The pressure–volume curves of the total respiratory system (P_{TR}, black line), the lungs (P_L, blue line), and the chest (P_W, green line) are plotted together with standard volumes for functional residual capacity and tidal volume. The transmural pressure is intrapulmonary pressure minus intrapleural pressure in the case of the lungs, intrapleural pressure minus outside (barometric) pressure in the case of the chest wall, and intrapulmonary pressure minus barometric pressure in the case of the total respiratory system. From these curves, the total and actual elastic work associated with breathing can be derived. (Modified with permission from Mines AH: *Respiratory Physiology*, 3rd ed. New York, NY: Raven Press; 1993.)

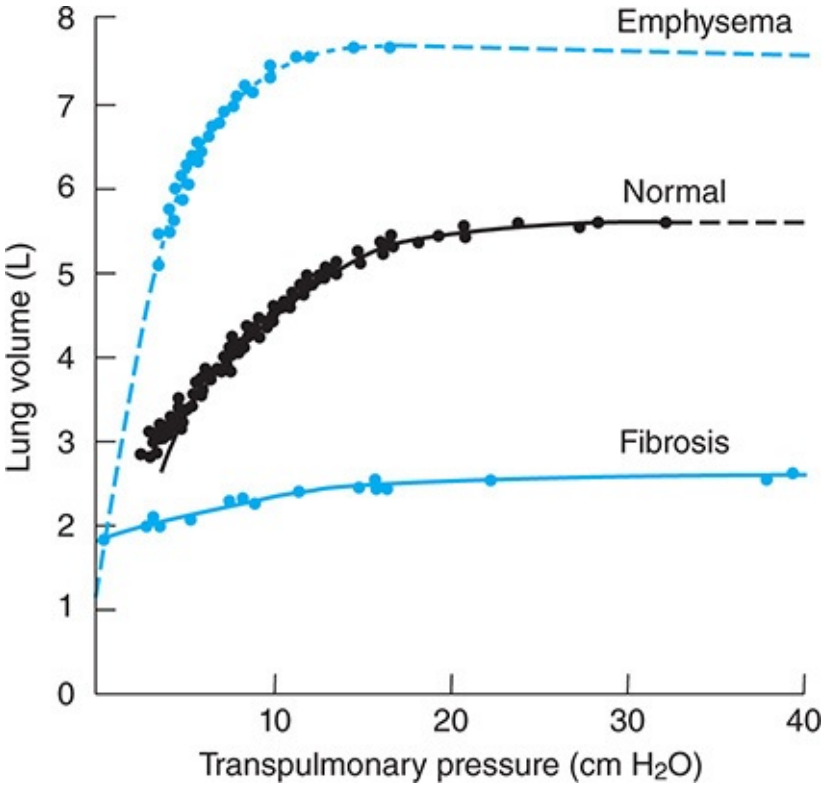
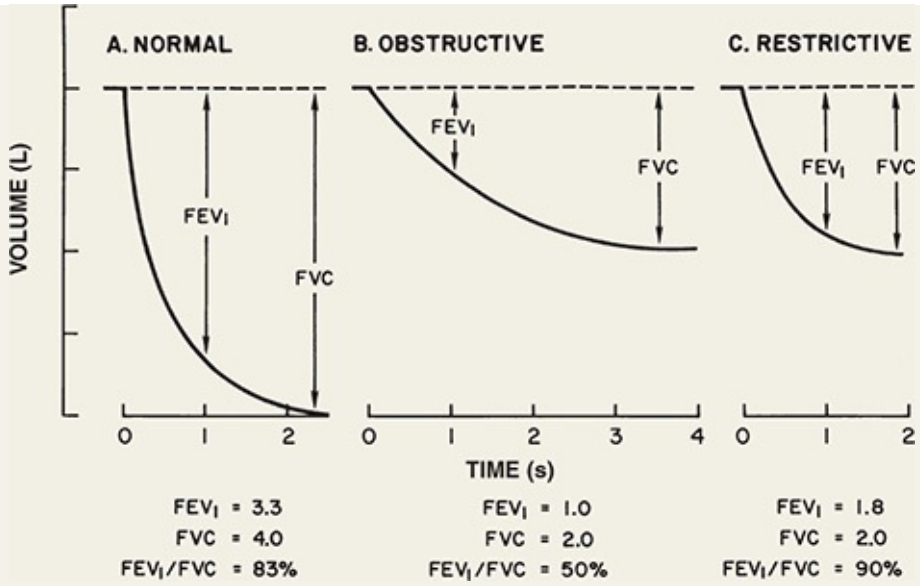


FIGURE 34–11 Static expiratory pressure–volume curves of lungs in healthy persons and persons with severe emphysema and pulmonary fibrosis. (Modified with permission from Pride NB, Macklem PT: Lung mechanics in disease. In: *Handbook of Physiology*. Section 3, *The Respiratory System*. Vol III, part 2. Fishman AP [editor]. American Physiological Society, 1986.)

CLINICAL BOX 34–2

Altered Airflow in Disease:



Representative lung volume expired over time in normal and diseased lungs during respiratory testing or pulmonary function test (PFT). A) Healthy normal subject. **B)** Patient with obstructive lung disease. **C)** Patient with restrictive lung disease. Note the differences in FEV_1, FVC and FEV_1/FVC at the bottom of the figure. FEV_1, forced expiratory volume in the first second; FVC, forced vital capacity. (Reproduced with permission from Fishman AP et al (eds): *Fishman's Pulmonary Diseases and Disorders*. 4th ed. New York: McGraw-Hill; 2008.)

Airflow Measurements of Obstructive & Restrictive Disease

In the example above, a healthy or normal FVC is ~4.0 L and a healthy FEV_1 is ~3.3 L. The calculated FEV_1/FVC is ~80%. Patients with obstructive or restrictive lung diseases can display reduced FVC, on the order of 2.0 L in the example above. Measurement of FEV_1, however, tends to vary significantly between the two diseases. In obstructive lung disorders, patients tend to show a slow, steady slope to the FVC, resulting in a small FEV_1, on the order of 1.0 L in the example. However, in the restrictive lung disorder, patient's airflow tends to be fast at first, and then quickly level out to approach FVC. The resultant FEV_1 is much greater, on the order of 1.8 L in the example, even though FVC is equivalent (compare B, C above). A quick calculation of FEV_1/FVC for patients with obstructive (50%) versus restrictive (90%) patterns defines the hallmark measurements in evaluating these two diseases. Obstructive disorders result in a marked decrease in both FVC and

FEV_1/FVC, whereas restrictive disorders result in a loss of FVC without loss in FEV_1/FVC. It should be noted that these examples are idealized and several disorders can show mixed readings.

Obstructive Disease—Asthma

Asthma is characterized by episodic or chronic wheezing, cough, and a feeling of tightness in the chest as a result of bronchoconstriction. Although the disease is not fully understood, three airway abnormalities are present: **airway obstruction** that is at least partially reversible, airway inflammation, and airway hyperresponsiveness to a variety of stimuli. A link to allergy has long been recognized, and plasma IgE levels are often elevated. Proteins released from eosinophils in the inflammatory reaction may damage the airway epithelium and contribute to the hyperresponsiveness. Leukotrienes are released from eosinophils and mast cells, and can enhance bronchoconstriction. Numerous other amines, neuropeptides, chemokines, and interleukins have effects on bronchial smooth muscle or produce inflammation, and they may be involved in asthma.

Restrictive Disease—Idiopathic Pulmonary Fibrosis

Idiopathic pulmonary fibrosis (IPF), a type of pulmonary fibrosis, is a rare and irreversible interstitial lung disease that usually affects middle-aged and older adults (> 50 years old). The most common cause of death related to IPF is respiratory failure. Other causes of death include pulmonary hypertension (PH) and PH-associated right heart failure, pulmonary embolism and lung cancer. Up to 132,000 people in the United States are currently affected by IPF, and there are 30,000–40,000 new cases each year. Alveolar epithelial injury due to epithelial cell apoptosis and endoplasmic reticulum (ER) stress and epithelial-to-mesenchymal transition (EMT) have been implicated as cellular mechanisms involved in the development of IPF.

THERAPEUTIC HIGHLIGHTS

Asthma

Because β_2-adrenergic receptors mediate bronchodilation, β_2-adrenergic agonists have long been the mainstay of “rescue” treatment for mild to moderate asthma attacks. Inhaled steroids are used even in mild to moderate cases to reduce inflammation; they are very effective, but their side effects can be a problem. Agents that block synthesis of leukotrienes or their $CysLT_1$

receptor inhibit acetylcholine-mediated activation of muscarinic (M) receptors and blockade Cl^- channels and Ca^{2+} channels in mast cells have also proved useful in certain cases.

IPF

Clinical management and treatment have been focused on suppressing the inflammatory and fibrotic process of IPF. Agents with anti-fibrotic, anti-inflammation, and anti-oxidant effects (eg, nintedanib, pirfenidone, thalidomide) are often used for treatment of IPF. Lung transplantation (replacement of one or both lungs) is a surgical therapeutic intervention of proven benefits in IPF, which is often reserved for patients at the advanced stage of IPF; the 5-year survival rate is approximately 50% for lung transplant patients.

Airway Resistance

Airway resistance is defined as the change of pressure (ΔP) from the alveoli to the mouth divided by the change in flow rate ($\dot{V}$). Because of the structure of the bronchial tree, and thus the pathway for air that contributes to its resistance, it is difficult to apply mathematical estimates of the movement through the bronchial tree. However, measurements where alveolar and intrapleural pressure can be compared to actual pressure (eg, Figure 34–7 middle panel) show the contribution of airway resistance. Airway resistance is significantly increased as lung volume is reduced. Also, bronchi and bronchioles significantly contribute to airway resistance. Thus, contraction of the smooth muscle that lines the bronchial airways will increase airway resistance and make breathing more difficult.

Role of Surfactant in Alveolar Surface Tension

An important factor affecting the compliance of the lungs is the surface tension of the film of fluid that lines the alveoli. The magnitude of this component at various lung volumes can be measured by removing the lungs from the body of an experimental animal and distending them alternately with saline and with air while measuring the intrapulmonary pressure. Because saline reduces the surface tension to nearly zero, the pressure–volume curve obtained with saline measures

only the tissue elasticity **(Figure 34–12)**, whereas the curve obtained with air measures both tissue elasticity and surface tension. The difference between the saline and air curves is much smaller when lung volumes are small. Differences are also obvious in the curves generated during inflation and deflation. This difference is termed **hysteresis,** and notably is not present in the saline generated curves (Figure 34–12). The alveolar environment, and specifically the secreted factors that help reduce surface tension and keep alveoli from collapsing, contribute to hysteresis.

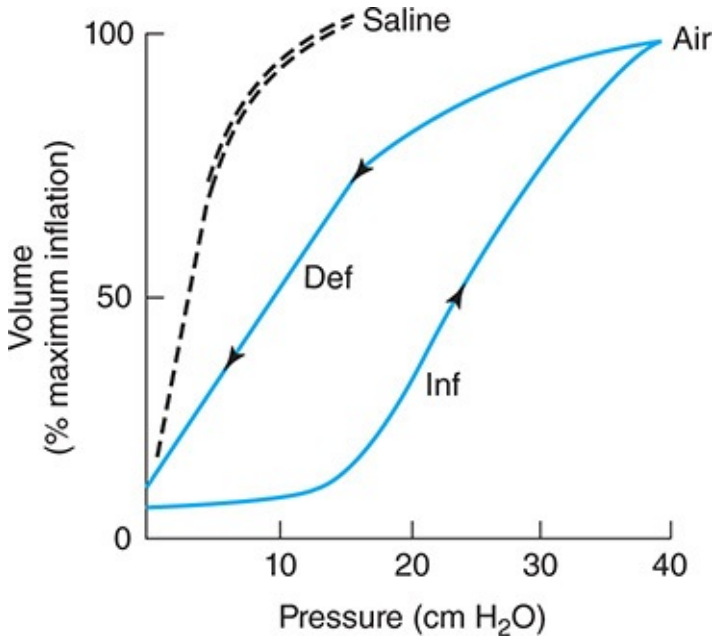


FIGURE 34–12 Pressure–volume curves in the lungs of a cat after removal from the body. Saline: lungs inflated and deflated with saline to reduce surface tension, resulting in a measurement of tissue elasticity. **Air:** lungs inflated (Inf) and deflated (Def) with air results in a measure of both tissue elasticity and surface tension. (Reproduced with permission from Morgan TE: Pulmonary surfactant. N Engl J Med 1971; May 27; 284(21):1185–1193.)

The low surface tension when the alveoli are small is due to the presence of **surfactant** in the fluid lining the alveoli. Surfactant is a mixture of dipalmitoylphosphatidylcholine (DPPC), other lipids, and proteins. If the surface tension is not kept low when the alveoli become smaller during expiration, they collapse in accordance with the law of Laplace. In spherical structures like an alveolus, the distending pressure (P) equals two times the tension (T) divided by the radius (r): P = 2T/r, if T is not reduced as r is reduced, the tension overcomes the distending pressure. Surfactant also helps prevent pulmonary edema. It has

been calculated that if it were not present, the unopposed surface tension in the alveoli would produce a 20 mm Hg force; such a force would greatly favor transudation of fluid from the blood into the alveoli.

Formation of the phospholipid film is greatly facilitated by the proteins in surfactant. This material contains four unique proteins: surfactant protein (SP)-A, SP-B, SP-C, and SP-D. SP-A is a large glycoprotein and has a collagen-like domain within its structure. It has multiple functions, including regulation of the feedback uptake of surfactant by the type II alveolar epithelial cells that secrete it. SP-B and SP-C are smaller proteins, which are the key protein members of the monomolecular film of surfactant. Like SP-A, SP-D is a glycoprotein. Its full function is uncertain; however, it plays an important role in the organization of SP-B and SP-C into the surfactant layer. Both SP-A and SP-D are members of the collectin family of proteins that are involved in innate immunity in the conducting airway as well as in the alveoli. Some clinical aspects of surfactant are discussed in **Clinical Box 34–3**.

WORK OF BREATHING

Work is performed by the respiratory muscles in stretching the elastic tissues of the chest wall and lungs (elastic work; approximately 65% of the total work), moving inelastic tissues (viscous resistance; 7% of total), and moving air through the respiratory passages (airway resistance; 28% of total). Because the dimensions of pressure × volume ($g/cm^2 \times cm^3 = g \times cm$) has the same dimensions as work (force × distance; g × cm), the work of breathing can be calculated from the previously presented pressure-volume curve (Figure 34–10). Note that the relaxation pressure curve of the total respiratory system (P_{TR}) differs from that of the lungs alone (P_L). The amount of elastic work required to inflate the whole respiratory system is less than the amount required to inflate the lungs alone because part of the work comes from elastic energy stored in the thorax.

CLINICAL BOX 34–3

Surfactant

Surfactant is important at birth. The fetus makes respiratory movements in utero, but the lungs remain collapsed until birth. After birth, the infant makes

several strong inspiratory movements and the lungs expand. Surfactant keeps them from collapsing again. Surfactant deficiency is an important cause of **infant respiratory distress syndrome** (IRDS, also known as **hyaline membrane disease**), the serious pulmonary disease that develops in infants born before their surfactant system is functional. Surface tension in the lungs of these infants is high, and the alveoli are collapsed in many areas **(atelectasis).** An additional factor in IRDS is retention of fluid in the lungs. During fetal life, Cl^- is secreted with fluid by the pulmonary epithelial cells. At birth, there is a shift to Na^+ absorption by these cells via the epithelial Na^+ channels (ENaCs), and fluid is absorbed with the Na^+. Prolonged immaturity of the ENaCs contributes to the pulmonary abnormalities in IRDS.

Overproduction/dysregulation of surfactant proteins can also lead to respiratory distress and is the cause of pulmonary alveolar proteinosis.

THERAPEUTIC HIGHLIGHTS

Treatment of IRDS is commonly done with surfactant replacement therapy. Interestingly, such surfactant replacement therapy has not been as successful in clinical trials for adults experiencing respiratory distress due to surfactant dysfunction.

Estimates of the total work of quiet breathing range from 0.3 to 0.8 kg-m/min. The value rises markedly during exercise, but the energy cost of breathing in normal individuals represents less than 3% of the total energy expenditure during exercise. The work of breathing is greatly increased in diseases such as emphysema, asthma, and heart failure with dyspnea and orthopnea. The respiratory muscles have length-tension relations like those of other skeletal and cardiac muscles, and when they are severely stretched, they contract with less strength. They can also become fatigued and fail (pump failure), leading to inadequate ventilation.

DIFFERENCES IN VENTILATION & BLOOD FLOW IN DIFFERENT PARTS OF THE LUNG

In the upright position, ventilation ($\dot{V}$) per unit lung volume is greater at the base of the lung than at the apex. The reason for this is that at the start of inspiration,

intrapleural pressure is less negative at the base than at the apex, and since the intrapulmonary intrapleural pressure difference is less than at the apex, the lung is less expanded. Conversely, at the apex, the lung is more expanded; that is, the percentage of maximum lung volume is greater. Because of the stiffness of the lung, the increase in lung volume per unit increase in pressure is smaller when the lung is initially more expanded, and ventilation is consequently greater at the base. Blood flow is also greater at the base than the apex. The relative change in blood flow from the apex to the base is greater than the relative change in ventilation, so the ventilation/perfusion ($\dot{V}/\dot{Q}$) ratio is low at the base and high at the apex.

The ventilation and perfusion differences from the apex to the base of the lungs have usually been attributed to gravity: they tend to disappear in the supine position, and the weight of the lungs would be expected to create pressure at the base in the upright position. However, the inequalities of ventilation and blood flow in human lungs were found to persist to a remarkable degree in the weightlessness of space. Therefore, other factors also play a role in producing the inequalities.

DEAD SPACE & UNEVEN VENTILATION

Because gaseous exchange in the respiratory system occurs only in the terminal portions of the airways, the gas that occupies the rest of the respiratory system is not available for gas exchange with pulmonary capillary blood. Normally, the volume (in mL) of this **anatomic dead space** is approximately equal to the body weight in pounds. As an example, in a man who weighs 150 lb (68 kg), only the first 350 mL of the 500 mL inspired with each breath at rest mixes with the air in the alveoli. Conversely, with each expiration, the first 150 mL expired is gas that occupied the dead space, and only the last 350 mL is gas from the alveoli. Consequently, the **alveolar ventilation,** that is, the amount of air reaching the alveoli per minute, is less than the RMV. Note that because of the dead space, rapid shallow breathing produces much less alveolar ventilation than slow deep breathing at the same RMV **(Table 34–1)**.

TABLE 34–1 Effect of variations in respiratory rate and depth on alveolar ventilation.

Respiratory rate	30/min	10/min
Tidal volume	200 mL	600 mL
Minute volume	6 L	6 L
Alveolar ventilation	(200 – 150) × 30 = 1500 mL	(600 – 150) × 10 = 4500 mL

It is important to distinguish between the **anatomic dead space** (respiratory system volume exclusive of alveoli) and the **total (physiological) dead space** (volume of gas not equilibrating with blood; ie, wasted ventilation). In healthy individuals, the two dead spaces are identical and can be estimated by body weight. However, in disease states, no exchange may take place between the gas in some of the alveoli and the blood, and some of the alveoli may be overventilated. The volume of gas in nonperfused alveoli and any volume of air in the alveoli in excess of that necessary to oxygenate the blood in the alveolar capillaries is part of the dead space (nonequilibrating) gas volume. The anatomic dead space can be measured by analysis of the single-breath N_2 curve **(Figure 34–13)**. From mid-inspiration, the patient takes as deep a breath as possible of pure O_2, then exhales steadily while the N_2 content of the expired gas is continuously measured. The initial gas exhaled (phase I) is the gas that filled the dead space and that consequently contains no N_2. This is followed by a mixture of dead space and alveolar gas (phase II) and then by alveolar gas (phase III). The volume of the dead space is the volume of the gas expired from peak inspiration to the midportion of phase II.

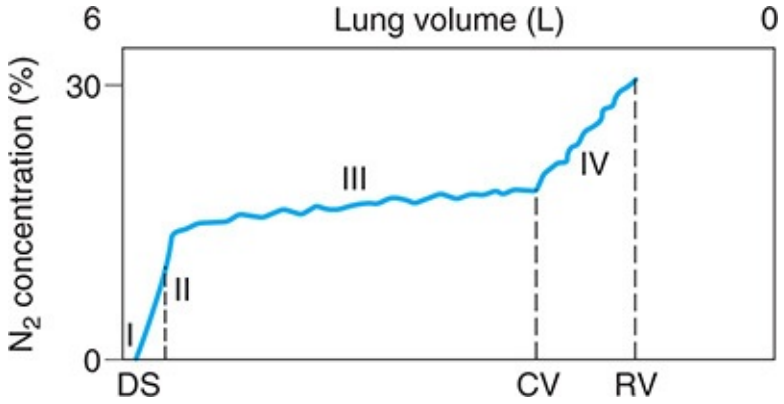


FIGURE 34–13 Single-breath N_2 curve. From mid-inspiration, the patient takes a deep breath of pure O_2 then exhales steadily. The changes in the N_2

concentration of expired gas during expiration are shown, with the various phases of the curve indicated by roman numerals. Notably, region I is representative of the dead space (DS); from I–II is a mixture of DS and alveolar gas; the transition from III–IV is the closing volume (CV), and the end of IV is the residual volume (RV).

Phase III of the single-breath N_2 curve terminates at the **closing volume (CV)** and is followed by phase IV, during which the N_2 content of the expired gas is increased. The CV is the lung volume above RV at which airways in the lower, dependent parts of the lungs begin to close off because of the lesser transmural pressure in these areas. The gas in the upper portions of the lungs is richer in N_2 than the gas in the lower, dependent portions because the alveoli in the upper portions are more distended at the start of the inspiration of O_2 and, consequently, the N_2 in them is less diluted with O_2. It is also worth noting that in most normal individuals, phase III has a slight positive slope even before phase IV is reached. This indicates that even during phase III there is a gradual increase in the proportion of the expired gas coming from the relatively N_2-rich upper portions of the lungs.

The total dead space can be calculated from the P_{CO_2} of expired air, the P_{CO_2} of arterial blood, and the TV. The tidal volume (V_T) times the P_{CO_2} of the expired gas (P_{ECO_2}) equals the arterial P_{CO_2} (Pa_{CO_2}) times the difference between the TV and the dead space (V_D) plus the P_{CO_2} of inspired air (P_{ICO_2}) times V_D **(Bohr equation):**

$$P_{ECO_2} \times V_T = Pa_{CO_2} \times (V_T - V_D) + P_{ICO_2} \times V_D$$

The term $P_{ICO_2} \times V_D$ is so small that it can be ignored and the equation solved for V_D, where $V_D = V_T - (P_{ECO_2} \times V_T)/(Pa_{CO_2})$

If, for example: P_{ECO_2} = 28 mm Hg; Pa_{CO_2} = 40 mm Hg; and V_T = 500 mL, then V_D = 150 mL.

The equation can also be used to measure the anatomic dead space if one replaces Pa_{CO_2} with alveolar P_{CO_2} (P_{ACO_2}), which is the P_{CO_2} of the last 10 mL of expired gas. P_{CO_2} is an average of gas from different alveoli in proportion to their ventilation regardless of whether they are perfused. This is in contrast to Pa_{CO_2}, which is gas equilibrated only with perfused alveoli, and consequently, in individuals with underperfused alveoli, is greater than P_{ACO_2}.

GAS EXCHANGE IN THE LUNGS

PARTIAL PRESSURES

Unlike liquids, gases expand to fill the volume available to them, and the volume occupied by a given number of gas molecules at a given temperature and pressure is (ideally) the same regardless of the composition of the gas. **Partial pressures** are frequently used to describe gases in respiration. The pressure of a gas is proportional to its temperature and number of moles occupying a certain volume **(Table 34–2)**. The pressure exerted by any one gas in a mixture of gases (its partial pressure) is equal to the total pressure times the fraction of the total amount of gas it represents.

TABLE 34–2 Properties of gases.

$P = \frac{nRT}{V}$ (from equation of state of ideal gas)

The pressure (P) of a gas is proportional to its temperature (T) and the number of moles (n) per volume (V); R, Gas constant.

The composition of dry air, or the fraction (F) of gas molecules in the dry air, is 20.98% O_2 (FO_2), 0.04% CO_2 (FCO_2), 78.06% N_2 (FN_2), and 0.92% other inert constituents such as argon and helium. The barometric pressure (P_B) at sea level is 760 mm Hg (1 atmosphere). The partial pressure (indicated by the symbol P) of O_2 (ie, P_{O_2}) in dry air is therefore 0.21 × 760, or 160 mm Hg at sea level. The P_{N_2} and the other inert gases is 0.79 × 760, or 600 mm Hg; and the P_{CO_2} is 0.0004 × 760, or 0.3 mm Hg. The water vapor in the air in most climates reduces these percentages, and therefore the partial pressures, to a slight degree. Air equilibrated with water is saturated with water vapor, and inspired air is saturated by the time it reaches the lungs. The P_{H_2O} at body temperature (37°C) is 47 mm Hg. Therefore, the partial pressures at sea level of the other gases in the air reaching the lungs are P_{O_2}, 150 mm Hg; P_{CO_2}, 0.3 mm Hg; and P_{N_2} (including the other inert gases), 563 mm Hg.

Gas diffuses from areas of high partial pressure to areas of low partial pressure, with the rate of diffusion depending on the concentration gradient and the nature of the barrier between the two areas. When a mixture of gases is in

contact with, and permitted to, equilibrate with a liquid, each gas in the mixture dissolves in the liquid to an extent determined by its partial pressure and its solubility in the fluid. The partial pressure of a gas in a liquid is the pressure that, in the gaseous phase in equilibrium with the liquid, would produce the concentration of gas molecules found in the liquid.

SAMPLING ALVEOLAR AIR

Theoretically, all but the first 150 mL expired from a healthy 150 lb man (ie, the dead space) with each expiration is the gas that was in the alveoli **(alveolar air),** but some mixing always occurs at the interface between the dead-space gas and the alveolar air (Figure 34–13). A later portion of expired air is therefore the portion taken for analysis. Using modern apparatus with a suitable automatic valve, it is possible to collect the last 10 mL expired gas during quiet breathing. The composition of alveolar gas is compared with that of inspired and expired air in **Figure 34–14**.

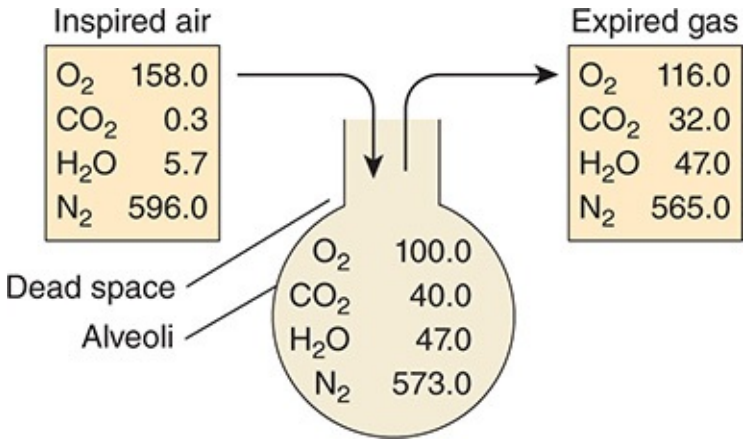


FIGURE 34–14 Partial pressures of gases (mm Hg) in various parts of the respiratory system. Typical partial pressures for inspired air, alveolar air, and expired air are given. See text for additional details.

P_{AO_2} can also be calculated from the **alveolar gas equation:**

$$P_{AO_2} = P_{IO_2} - P_{ACO_2}\left(F_{IO_2} + \frac{1 - F_{IO_2}}{R}\right)$$

where F_{IO_2} is the fraction of O_2 molecules in the inspired (I) dry gas, P_{IO_2} is the

inspired P_{O_2}, and R is the respiratory exchange ratio; that is, the flow of CO_2 molecules across the alveolar membrane per minute divided by the flow of O_2 molecules across the membrane per minute.

COMPOSITION OF ALVEOLAR AIR

Oxygen continuously diffuses out of the gas in the alveoli into the bloodstream, and CO_2 continuously diffuses into the alveoli from the blood. In the steady state, inspired air mixes with the alveolar gas, replacing the O_2 that has entered the blood and diluting the CO_2 that has entered the alveoli. Part of this mixture is expired. The O_2 content of the alveolar gas then falls and its CO_2 content rises until the next inspiration. Because the volume of gas in the alveoli is about 2 L at the end of expiration (FRC), each 350 mL increment of inspired and expired air has relatively little effect on P_{O_2} and P_{CO_2}. Indeed, the composition of alveolar gas remains remarkably constant, not only at rest but also under a variety of other conditions.

DIFFUSION ACROSS THE ALVEOLOCAPILLARY MEMBRANE

Gases diffuse from the alveoli to the blood in the pulmonary capillaries or vice versa across the thin alveolocapillary membrane made up of the pulmonary epithelium, the capillary endothelium, and their fused basement membranes (Figure 34–3). Whether or not substances passing from the alveoli to the capillary blood reach equilibrium in the 0.75 s that blood takes to traverse the pulmonary capillaries at rest depends on their reaction with substances in the blood. Thus, for example, the anesthetic gas nitrous oxide (N_2O) does not react and reaches equilibrium in about 0.1 s **(Figure 34–15)**. In this situation, the amount of N_2O taken up is not limited by diffusion but by the amount of blood flowing through the pulmonary capillaries; that is, it is **flow-limited.** On the other hand, carbon monoxide (CO) is taken up by hemoglobin in the red blood cells at such a high rate that the partial pressure of CO in the capillaries stays very low and equilibrium is not reached in the 0.75 s the blood is in the pulmonary capillaries. Therefore, the transfer of CO is not limited by perfusion at rest and instead is **diffusion-limited.** O_2 is intermediate between N_2O and CO;

it is taken up by hemoglobin, but much less avidly than CO, and it reaches equilibrium with capillary blood in about 0.3 s. Thus, its uptake is **perfusion-limited.**

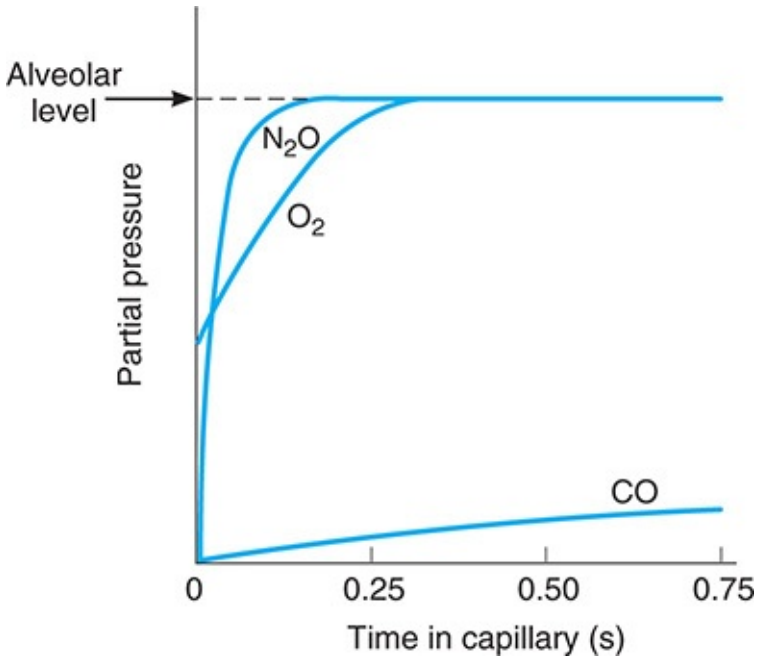


FIGURE 34–15 Uptake of various substances during the 0.75 s they are in transit through a pulmonary capillary. N_2O is not bound in blood, so its partial pressure in blood rises rapidly to its partial pressure in the alveoli. Conversely, CO is avidly taken up by red blood cells, so its partial pressure reaches only a fraction of its partial pressure in the alveoli. O_2 is intermediate between the two.

The **diffusing capacity** of the lungs for a given gas is directly proportional to the surface area of the alveolo-capillary membrane and inversely proportional to its thickness. The diffusing capacity for CO (D_{LCO}) is measured as an index of diffusing capacity because its uptake is diffusion-limited. D_{LCO} is proportional to the amount of CO entering the blood ($\dot{V}_{CO}$) divided by the partial pressure of CO in the alveoli minus the partial pressure of CO in the blood entering the pulmonary capillaries. Except in habitual cigarette smokers, this latter term is close to zero, so it can be ignored and the equation becomes:

$$D_{LCO} = \frac{\dot{V}_{CO}}{P_{ACO}}$$

The normal value of D_{LCO} at rest is about 25 mL/min/mm Hg. It increases up to threefold during exercise because of capillary dilation and an increase in the number of active capillaries. The P_{O_2} of alveolar air is normally 100 mm Hg, and the P_{O_2} of the blood entering the pulmonary capillaries is 40 mm Hg. The diffusing capacity for O_2, like that for CO at rest, is about 25 mL/min/mm Hg, and the P_{O_2} of blood is raised to 97 mm Hg, a value just under the alveolar P_{O_2}.

The P_{CO_2} of venous blood is 46 mm Hg, whereas that of alveolar air is 40 mm Hg, and CO_2 diffuses from the blood into the alveoli along this gradient. The P_{CO_2} of blood leaving the lungs is 40 mm Hg. CO_2 passes through all biological membranes with ease, and the diffusing capacity of the lungs for CO_2 is much greater than the capacity for O_2. It is for this reason that CO_2 retention is rarely a problem in patients with alveolar fibrosis even when the reduction in diffusing capacity for O_2 is severe.

PULMONARY CIRCULATION

PULMONARY BLOOD VESSELS

The pulmonary vascular bed resembles the systemic one, except that the walls of the pulmonary artery and its large branches are about 30% as thick as the wall of the aorta, and the small arterial vessels, unlike the systemic arterioles, contain relatively little smooth muscle in their walls. The walls of the pulmonary venous vessels also contain some smooth muscle. The pulmonary capillaries are large, and there are multiple anastomoses, so that each alveolus sits in a capillary basket.

PRESSURE, VOLUME, & FLOW

With two quantitatively minor exceptions, the blood put out by the left ventricle returns to the right atrium and is ejected by the right ventricle, making the pulmonary vasculature unique in that it accommodates a blood flow that is almost equal to that of all the other organs in the body.

One of the exceptions is part of the bronchial blood flow. There are anastomoses between the bronchial capillaries and the pulmonary capillaries and veins, and although some of the bronchial blood enters the bronchial veins, some enters the pulmonary capillaries and veins, bypassing the right ventricle. The

other exception is blood that flows from the coronary arteries into the chambers of the left side of the heart. Because of the small **physiological shunt** created by those two exceptions, the blood in systemic arteries has a P_{O_2} about 2 mm Hg lower than that of blood that has equilibrated with alveolar air, and the saturation of hemoglobin is 0.5% less.

The pressure in the various parts of the pulmonary portion of the pulmonary circulation is shown in Figure 34–6C. The pressure gradient in the pulmonary circulation system is about 7 mm Hg, compared with a gradient of about 90 mm Hg in the systemic circulation. Pulmonary capillary pressure is about 10 mm Hg, whereas the oncotic pressure is 25 mm Hg, so that an inward-directed pressure gradient of about 15 mm Hg keeps the alveoli free of all but a thin film of fluid. When the pulmonary capillary pressure is more than 25 mm Hg, pulmonary congestion and edema result.

The volume of blood in the pulmonary vessels at any one time is about 1 L, of which less than 100 mL is in the capillaries. The mean velocity of the blood in the root of the pulmonary artery is the same as that in the aorta (about 40 cm/s). It falls off rapidly, then rises slightly again in the larger pulmonary veins. It takes a red cell about 0.75 s to traverse the pulmonary capillaries at rest and 0.3 s or less during exercise.

EFFECT OF GRAVITY

Gravity has a relatively marked effect on the pulmonary circulation. In the upright position, the upper portions of the lungs are well above the level of the heart, and the bases are at or below it. Consequently, in the upper part of the lungs, the blood flow is less, the alveoli are larger, and ventilation is less than at the base **(Figure 34–16)**. The pressure in the capillaries at the top of the lungs is close to the atmospheric pressure in the alveoli. Pulmonary arterial pressure is normally just sufficient to maintain perfusion, but if it is reduced or if alveolar pressure is increased, some of the capillaries collapse. Under these circumstances, no gas exchange takes place in the affected alveoli and they become part of the physiological dead space.

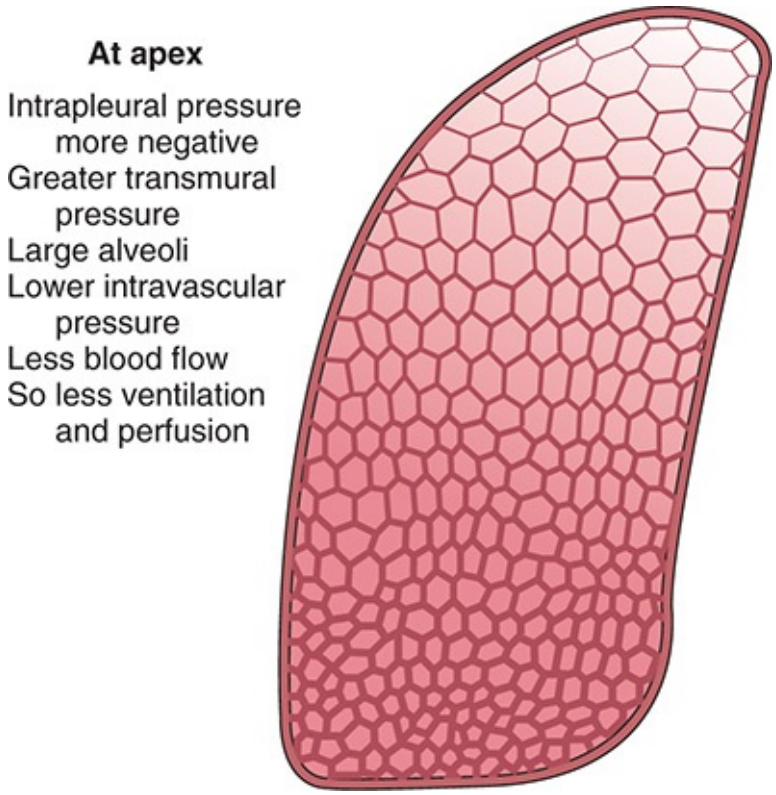


FIGURE 34–16 Diagram of normal differences in ventilation and perfusion of the lung in the upright position. Outlined areas are representative of changes in alveolar size (not actual size). Note the gradual change in alveolar size from top (apex) to bottom. Characteristic differences of alveoli at the apex of the lung are stated. (Modified with permission from Levitzky MG: *Pulmonary Physiology*, 6th ed. New York, NY: McGraw-Hill; 2003.)

In the middle portions of the lungs, the pulmonary arterial pressure and capillary pressure exceed alveolar pressure, but the pressure in the pulmonary venules may be lower than alveolar pressure during normal expiration, so they are collapsed. Under these circumstances, blood flow is determined by the difference between pulmonary arterial pressure and alveolar pressure rather than the difference between pulmonary arterial pressure and pulmonary venous pressure. Beyond the constriction, blood "falls" into the pulmonary veins, which are compliant and take whatever amount of blood the constriction lets flow into them. This has been called the **waterfall effect.** Obviously, the compression of vessels produced by alveolar pressure decreases and pulmonary blood flow increases as the arterial pressure increases toward the base of the lungs. In the lower portions of the lungs, alveolar pressure is lower than the pressure in all parts of the pulmonary circulation and blood flow is determined by the arterial-

venous pressure difference. Examples of diseases affecting the pulmonary circulation are given in **Clinical Box 34–4**.

VENTILATION/PERFUSION RATIOS

The ratio of pulmonary ventilation ($\dot{V}$) to pulmonary blood flow or perfusion ($\dot{Q}$) for the whole lungs at rest is about 0.8 (4.2 L/min ventilation divided by 5.5 L/min blood flow). However, relatively marked differences occur in this **ventilation/perfusion Q($\dot{V}/\dot{Q}$) ratio** in various parts of the normal lungs as a result of the effect of gravity, and local changes in the ventilation/perfusion ratio are common in disease. If the ventilation to an alveolus is reduced relative to its perfusion, the P_{O_2} in the alveolus falls because less O_2 is delivered to it and the P_{CO_2} rises because less CO_2 is expired. Conversely, if perfusion is reduced relative to ventilation, the P_{CO_2} falls because less CO_2 is delivered and the P_{O_2} rises because less O_2 enters the blood.

As noted above, ventilation, as well as perfusion in the upright position, declines in a linear fashion from the bases to the apices of the lungs. However, the ventilation/perfusion ratios are high in the upper portions of the lungs. When widespread, nonuniformity of ventilation and perfusion in the lungs can cause CO_2 retention and lowers systemic arterial P_{O_2}.

CLINICAL BOX 34–4

Diseases Affecting the Pulmonary Circulation

Pulmonary Hypertension

Pulmonary hypertension (PH) is classified into five groups: pulmonary arterial hypertension (PAH; including persistent PH of the newborn), PH associated with left heart disease, PH due to lung diseases and/or hypoxia, chronic thromboembolic pulmonary hypertension (CTEPH), and PH with unclear multifactorial mechanisms. PH, as a syndrome with multiple causes, can occur at any age, but idiopathic PAH (a fatal and progressive form of PAH) predominantly affects young women. The causes for PH or PAH are different from those causing systemic arterial hypertension. For example, acute hypoxia causes pulmonary vasoconstriction and chronic hypoxia causes pulmonary hypertension, whereas hypoxia causes systemic vasodilation.

Other unique causes for PH include inhalation of cocaine, treatment with dexfenfluramine and related appetite-suppressing drugs, HIV/HPV infection, systemic lupus erythematosus and scleroderma, and mutations of the genes encoding the bone morphogenetic protein receptor II (BMPRII), the pH-sensitive two-pore-domain K^+ channel (KCNK3), the serotonin transporter (SERT), the activin receptor kinase-like 1 (ALK1), the transient receptor potential canonical channel 6 (TRPC6), the glycoprotein endoglin (ENG), and the scaffolding protein caveolin 1 (CAV1). Sustained pulmonary vasoconstriction, concentric pulmonary vascular remodeling, in situ pulmonary thrombosis, and increased pulmonary vascular wall stiffness all contribute to the increased pulmonary vascular resistance and pulmonary arterial pressure in patients with PH or PAH. If appropriate therapy is not initiated, the increased right ventricular afterload can lead eventually to right heart failure and death.

THERAPEUTIC HIGHLIGHTS

Treatment with vasodilators and antiproliferative agents, such as continuous intravenous infusion of prostacyclin (Flolan) and oral intake or inhalation of prostacyclin analogs (eg, beroprost, ilorost, treprostinil), is effective for patients with severe PH (eg, idiopathic, heritable and associated PAH). Conventional treatment includes diuretics, oxygen therapy, anticoagulation, digoxin, and exercise. Inhaled nitric oxide (NO) is effective for many pediatric patients. Advanced treatment includes voltage-gated Ca^{2+} channel blockers (eg, nifedipine, diltiazem, verapamil, amlodipine), prostanoids (eg, epoprostenol/PGI_2, treprostinil, iloprost), endothelin receptor antagonists (eg, bosentan, ambrisentan), phosphodiesterase (PDE) inhibitors (eg, sildenafil/Viagra, tadalafil/Cialis), and combination therapy. Surgical treatment includes atrial septostomy, whole lung transplantation, living-donor lobar lung transplantation, pulmonary endarterectomy, and pulmonary arterioplasty.

A unique mechanism that ensures the optimal ventilation/perfusion ratio in the lungs is hypoxic pulmonary vasoconstriction (HPV). HPV is a homeostatic mechanism that, in the adult men, diverts desaturated, mixed venous blood in the pulmonary artery from poorly ventilated (hypoxic) areas of the lungs to other better ventilated (normoxic) areas of the lungs. The alveolar hypoxia-induced vasoconstriction helps maintain an optimal ventilation/perfusion ratio and

maximize the oxygenation of the mixed venous blood in the lungs.

REGULATION OF PULMONARY BLOOD FLOW

It is unsettled whether pulmonary veins and pulmonary arteries are regulated separately, although constriction of the veins increases pulmonary capillary pressure and constriction of pulmonary arteries increases the load on the right side of the heart.

Pulmonary blood flow is affected by both active and passive factors. There is an extensive autonomic innervation of the pulmonary vessels, and stimulation of the cervical sympathetic ganglia reduces pulmonary blood flow by as much as 30%. The vessels also respond to circulating humoral agents. A diversity of the receptors involved in the regulation of pulmonary vasomotor tone and their effect on pulmonary smooth muscle are summarized in **Table 34–3**. Many of the dilator responses are endothelium-dependent and presumably operate via release of nitric oxide (NO).

TABLE 34–3 Receptors affecting smooth muscle in pulmonary arteries and veins.

Receptor	Subtype	Response	Endothelium Dependency
Autonomic			
Adrenergic	α_1	Contraction	No
	α_2	Relaxation	Yes
	β_2	Relaxation	Yes
Muscarinic	M_3	Relaxation	Yes
Purinergic	P2X	Contraction	No
	P2Y	Relaxation	Yes
Tachykinin	NK_1	Relaxation	Yes
	NK_2	Contraction	No
VIP	?	Relaxation	?
CGRP	?	Relaxation	No
Humoral			
Adenosine	A_1	Contraction	No
	A_2	Relaxation	No
Angiotensin II	AT_1	Contraction	No
ANP	ANP_A	Relaxation	No
	ANP_B	Relaxation	No
Bradykinin	B_1?	Relaxation	Yes
	B_2	Relaxation	Yes
Endothelin	ET_A	Contraction	No
	ET_B	Relaxation	Yes
Histamine	H_1	Relaxation	Yes
	H_2	Relaxation	No
5-HT	5-HT_1	Contraction	No
	5-HT_1C	Relaxation	Yes
Prostacyclin (PGI_2)	IP	Relaxation	No
Thromboxane	TP	Contraction	No
Vasopressin	V_1	Relaxation	Yes

Modified and reproduced with permission from Barnes PJ, Lin SF: Regulation of pulmonary vascular tone. Pharmacol Rev 1995;47:88.

Passive factors such as cardiac output and gravitational forces also have significant effects on pulmonary blood flow. Local adjustments of perfusion to

ventilation occur with local changes in O_2. With exercise, cardiac output increases and pulmonary arterial pressure rises. More red cells move through the lungs without any reduction in the O_2 saturation of the hemoglobin in them, and consequently, the total amount of O_2 delivered to the systemic circulation is increased. Capillaries dilate (also referred as capillary extension), and previously underperfused capillaries are “recruited” (also referred to as capillary recruitment) to carry blood. The net effect is a marked increase in pulmonary blood flow with few, if any, alterations in autonomic outflow to the pulmonary vessels.

When a bronchus or a bronchiole is obstructed, hypoxia develops in the underventilated alveoli beyond the obstruction. The O_2 reduction apparently acts directly on vascular smooth muscle in the area to produce constriction (ie, HPV), shunting blood away from the hypoxic area. Accumulation of CO_2 leads to a drop in pH in the area, and a decline in pH also produces vasoconstriction in the lungs, as opposed to the vasodilation it produces in other tissues. Conversely, reduction of the blood flow to a portion of the lung lowers the alveolar P_{CO_2} in that area, and this leads to constriction of the bronchi supplying it, shifting ventilation away from the poorly perfused area. Systemic hypoxia also causes the pulmonary arterioles to constrict, with a resultant increase in pulmonary arterial pressure.

METABOLIC & ENDOCRINE FUNCTIONS OF THE LUNGS

In addition to their functions in gas exchange, the lungs have a number of metabolic functions. They manufacture surfactant for local use, as noted above. They also contain a fibrinolytic system that lyses clots in the pulmonary vessels. They release a variety of substances that enter the systemic arterial blood **(Table 34–4)**, and they remove other substances from the systemic venous blood that reach them via the pulmonary artery. Prostaglandins are removed from the circulation, but they are also synthesized in the lungs and released into the blood when lung tissue is stretched.

TABLE 34–4 Biologically active substances metabolized by the lungs.

Synthesized and used in the lungs
Surfactant

Synthesized or stored and released into the blood
Prostaglandins Histamine Kallikrein
Partially removed from the blood
Prostaglandins Bradykinin Adenine nucleotides Serotonin Norepinephrine Acetylcholine
Activated in the lungs
Angiotensin II (from angiotensin I by the angiotensin-converting enzyme, ACE)

The lungs play an important role in activating angiotensin. The physiologically inactive decapeptide angiotensin I is converted to the pressor, aldosterone-stimulating octapeptide angiotensin II in the pulmonary circulation. The reaction occurs in other tissues as well, but it is particularly prominent in the lungs. Large amounts of the angiotensin-converting enzyme (ACE) responsible for this activation are located on the surface of the endothelial cells of the pulmonary capillaries. The converting enzyme also inactivates bradykinin, an endothelium-dependent vasodilator. Circulation time through the pulmonary capillaries is less than 1 s, yet 70% of the angiotensin I reaching the lungs is converted to angiotensin II in a single trip through the capillaries. Four other peptidases have been identified on the surface of the pulmonary endothelial cells, but their full physiological role is unsettled.

Removal of serotonin and norepinephrine reduces the amounts of these vasoactive substances reaching the systemic circulation. However, many other vasoactive hormones pass through the lungs without being metabolized. These include epinephrine, dopamine, oxytocin, vasopressin, and angiotensin II. In addition, various amines and polypeptides are secreted by neuroendocrine cells

in the lungs.

CHAPTER SUMMARY

- Air enters the respiratory system in the upper airway, proceeds to the conducting airway and then on to the respiratory airway that ends in the alveoli. The cross-sectional area of the airway gradually increases through the conducting zone, and then rapidly increases during the transition from conducting to respiratory zones.
- The mucociliary escalator in the conducting airway helps keep particulates out of the respiratory zone.
- There are several important measures of lung volume, including tidal volume (TV), inspiratory reserve volume (IRV), expiratory reserve volume (ERV), forced vital capacity (FVC), the forced expiratory volume in first second (FEV_1), respiratory minute volume (RMV), and maximal voluntary ventilation.
- Net "driving pressure" for air movement into the lungs includes the force of muscle contraction, lung compliance ($\Delta P/\Delta V$), and airway resistance ($\Delta P/\Delta$ .V).
- Surfactant decreases surface tension in the alveoli and helps keep them from deflating.
- Not all air that enters the airway is available for gas exchange. The regions where gas is not exchanged in the airway are termed "dead space." The conducting airway represents anatomic dead space. Increased dead space can occur in response to disease that affects air exchange in the respiratory zone (physiological dead space).
- The pressure gradient in the pulmonary circulation system is much less than that in the systemic circulation.
- Hypoxic pulmonary vasoconstriction is a unique mechanism that directs the mixed venous blood from poorly ventilated (hypoxic) areas of the lungs to well (normoxic) ventilated areas to assure maximal oxygenation of the venous blood.
- There are a variety of biologically activated substances that are metabolized in the lungs. These include substances that are made and function in the lungs (eg, surfactant), substances that are released or removed from the blood (eg, prostaglandins), and substances that are activated as they pass

through the lungs (eg, angiotensin II).

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. On the summit of Mt. Everest, where the barometric pressure is about 250 mm Hg, the partial pressure of O_2 in mm Hg is about

   A. 0.1.
   B. 0.5.
   C. 5.
   D. 50.
   E. 100.

2. The forced vital capacity is

   A. the amount of air that normally moves into (or out of) the lungs with each respiration.
   B. the amount of air that enters the lungs but does not participate in gas exchange.
   C. the amount of air expired after maximal expiratory effort.
   D. the largest amount of gas that can be moved into and out of the lungs in 1 min.

3. The tidal volume is

   A. the amount of air that normally moves into (or out of) the lungs with each respiration.
   B. the amount of air that enters the lungs but does not participate in gas exchange.
   C. the amount of air expired after maximal expiratory effort.
   D. the amount of gas that can be moved into and out of the lungs in 1 min.

4. Which of the following is responsible for the movement of O_2 from the alveoli into the blood in the pulmonary capillaries?

   A. Active transport
   B. Filtration
   C. Secondary active transport
   D. Facilitated diffusion
   E. Passive diffusion

5. Airway resistance
   A. is increased if the lungs are removed and inflated with saline.
   B. does not affect the work of breathing.
   C. is increased in paraplegic patients.
   D. is increased following bronchial smooth muscle contraction.
   E. makes up 80% of the work of breathing.
6. Surfactant lining the alveoli
   A. helps prevent alveolar collapse.
   B. is produced in alveolar type I cells and secreted into the alveolus.
   C. is increased in the lungs of heavy smokers.
   D. is a glycolipid complex.

CHAPTER 35

Gas Transport & pH

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the manner in which O_2 flows "downhill" from the lungs to the tissues and CO_2 flows "downhill" from the tissues to the lungs.
- List the important factors affecting the affinity of hemoglobin for O_2 and the physiological significance of each.
- List the reactions that increase the amount of CO_2 in the blood and draw the CO_2 dissociation curve for arterial and venous blood.
- Define alkalosis and acidosis and list typical causes and compensatory responses to each.
- Define hypoxia and describe differences in subtypes of hypoxia.
- Describe the effects of hypercapnia and hypocapnia and give examples of conditions that can cause them.

INTRODUCTION

The concentrations for O_2 and CO_2 (measured as partial pressures, or P_{O_2} and

P_{CO_2}) change within each region of the lung, allowing for these gases to flow "downhill" or from higher partial pressures to lower partial pressures. For example, P_{O_2} is highest in the alveoli upon inspiration and lowest in the deoxygenated blood, whereas, P_{CO_2} is exactly opposite. This allows for O_2 to cross the alveoli and reoxygenate the blood in the pulmonary vasculature, while CO_2 leaves the bloodstream and enters the alveoli where it is expired. However, the amount of both of these gases transported to and from the tissues would be grossly inadequate if it were not for the fact that about 99% of the O_2 that dissolves in the blood combines with the O_2-carrying protein hemoglobin and that about 94.5% of the CO_2 that dissolves enters into a series of reversible chemical reactions that convert it into other compounds. Thus, the presence of hemoglobin increases the O_2-carrying capacity of the blood 70-fold, and the reactions of CO_2 increase the blood CO_2 content 17-fold. In this chapter, physiological details that underlie O_2 and CO_2 movement under various conditions are discussed.

OXYGEN TRANSPORT

OXYGEN DELIVERY TO THE TISSUES

Oxygen delivery, or by definition, the volume of oxygen delivered to the systemic vascular bed per minute, is the product of the cardiac output and the arterial oxygen concentration. The ability to deliver O_2 in the body depends on both the respiratory and the cardiovascular systems. O_2 delivery to a particular tissue depends on the amount of O_2 entering the lungs, the adequacy of pulmonary gas exchange, the blood flow to the tissue, and the capacity of the blood to carry O_2. Blood flow to an individual tissue depends on cardiac output and the degree of constriction of the vascular bed in the tissue. The amount of O_2 in the blood is determined by the amount of dissolved O_2, the amount of hemoglobin in the blood, and the affinity of the hemoglobin for O_2.

REACTION OF HEMOGLOBIN & OXYGEN

The dynamics of the reaction of hemoglobin with O_2 make it a particularly

suitable O_2 carrier. Hemoglobin is a protein made up of four subunits, each of which contains a **heme** moiety attached to a polypeptide chain. In normal adults, most of the hemoglobin molecules contain two α and two β chains. Heme (see Figure 31–7) is a porphyrin ring complex that includes one atom of ferrous iron. Each of the four iron atoms in hemoglobin can reversibly bind one O_2 molecule. The iron stays in the ferrous state, so that the reaction is **oxygenation** (not oxidation). It has been customary to write the reaction of hemoglobin with O_2 as $Hb + O_2 \rightleftarrows HbO_2$. Because it contains four deoxyhemoglobin (Hb) units, the hemoglobin molecule can also be represented as Hb_4, and it actually reacts with four molecules of O_2 to form Hb_4O_8.

$$Hb_4 + O_2 \rightleftarrows Hb_4O_2$$
$$Hb_4O_2 + O_2 \rightleftarrows Hb_4O_4$$
$$Hb_4O_4 + O_2 \rightleftarrows Hb_4O_6$$
$$Hb_4O_6 + O_2 \rightleftarrows Hb_4O_8$$

The reaction is rapid, requiring less than 0.01 s. The deoxygenation of Hb_4O_8 is also very rapid.

The quaternary structure of hemoglobin determines its affinity for O_2. In deoxyhemoglobin, the globin units are tightly bound in a **tense (T) configuration,** which reduces the affinity of the molecule for O_2. When O_2 is first bound, the bonds holding the globin units are released, producing a **relaxed (R) configuration,** which exposes more O_2-binding sites. The net result is a 500-fold increase in O_2 affinity. In tissues, these reactions are reversed, resulting in O_2 release. The transition from one state to another has been calculated to occur about 10^8 times in the life of a red blood cell.

The **oxygen–hemoglobin dissociation curve** relates percentage saturation of the O_2-carrying power of hemoglobin (abbreviated as SaO_2) to the PO_2 **(Figure 35–1)**. This curve has a characteristic sigmoid shape due to the T–R configuration interconversion. Combination of the first heme in the Hb molecule with O_2 increases the affinity of the second heme for O_2, and oxygenation of the second increases the affinity of the third, and so on, so that the affinity of Hb for the fourth O_2 molecule is many times that for the first. Especially note that small changes at low PO_2 lead to large changes in SaO_2.

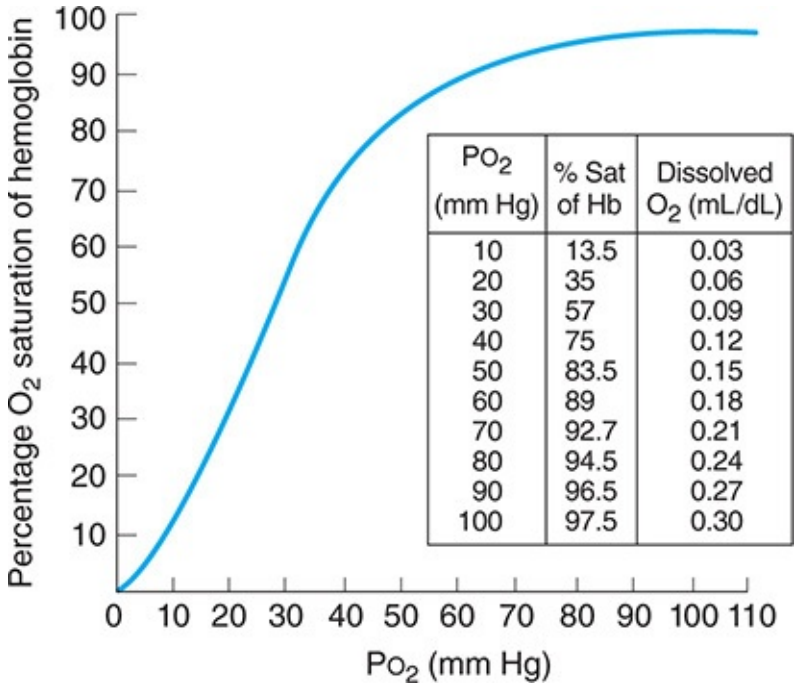


FIGURE 35–1 Oxygen–hemoglobin dissociation curve. pH 7.40, temperature 38°C. Inset table relates the percentage of saturated hemoglobin (SaO_2) to Po_2 and dissolved O_2. (Modified with permission from Comroe JH Jr, et al: *The Lung: Clinical Physiology and Pulmonary Function Tests*, 2nd ed. Year Book; 1962.)

When blood is equilibrated with 100% O_2, the normal hemoglobin becomes 100% saturated. When fully saturated, each gram (g) of normal hemoglobin contains 1.39 mL of O_2. However, blood normally contains small quantities of inactive hemoglobin derivatives, and the measured value in vivo is thus slightly lower. Using the traditional estimate of saturated hemoglobin in vivo, 1.34 mL of O_2, the hemoglobin concentration in normal blood is about 15 g/dL (14 g/dL in women and 16 g/dL in men). Therefore, 1 dL of blood contains 20.1 mL (1.34 mL × 15) of O_2 bound to hemoglobin when the hemoglobin is 100% saturated. The amount of dissolved O_2 is a linear function of the P_{O_2} (0.003 mL/dL blood/mm Hg P_{O_2}).

In vivo, the hemoglobin in the blood at the ends of the pulmonary capillaries is about 97.5% saturated with O_2 (P_{O_2} = 100 mm Hg). Because of a slight admixture with venous blood that bypasses the pulmonary capillaries (ie, physiological shunt), the hemoglobin in systemic arterial blood is only 97% saturated. The arterial blood therefore contains a total of about 19.8 mL of O_2

per dL: 0.29 mL in solution and 19.5 mL bound to hemoglobin. In venous blood at rest, the hemoglobin is 75% saturated and the total O_2 content is about 15.2 mL/dL: 0.12 mL in solution and 15.1 mL bound to hemoglobin. Thus, at rest the tissues remove about 4.6 mL of O_2 from each deciliter (dL) of blood passing through them **(Table 35–1)**; 0.17 mL of this total represents O_2 that was in solution in the blood, and the remainder represents O_2 that was liberated from hemoglobin. In this way, 250 mL of O_2 per minute is transported from the blood to the tissues at rest.

TABLE 35–1 Gas content of blood.

	mL/dL of Blood Containing 15 g of Hemoglobin			
	Arterial Blood (Po_2 95 mm Hg; Pco_2 40 mm Hg; Hb 97% Saturated)		Venous Blood (Po_2 40 mm Hg; Pco_2 46 mm Hg; Hb 75% Saturated)	
Gas	Dissolved	Combined	Dissolved	Combined
O_2	0.29	19.5	0.12	15.1
CO_2	2.62	46.4	2.98	49.7
N_2	0.98	0	0.98	0

FACTORS AFFECTING THE AFFINITY OF HEMOGLOBIN FOR OXYGEN

Three important conditions affect the oxygen–hemoglobin dissociation curve: the **pH,** the **temperature,** and the concentration of **2,3-diphosphoglycerate (DPG; 2,3-DPG).** A rise in temperature or a fall in pH shifts the curve to the right **(Figure 35–2)**. When the curve is shifted in this direction, a higher P_{O_2} is required for hemoglobin to bind a given amount of O_2. Conversely, a fall in temperature or a rise in pH shifts the curve to the left, and a lower P_{O_2} is required to bind a given amount of O_2. A convenient index for comparison of such shifts is the P_{50}, the P_{O_2} at which hemoglobin is half saturated with O_2. The higher the P_{50}, the lower the affinity of hemoglobin for O_2.

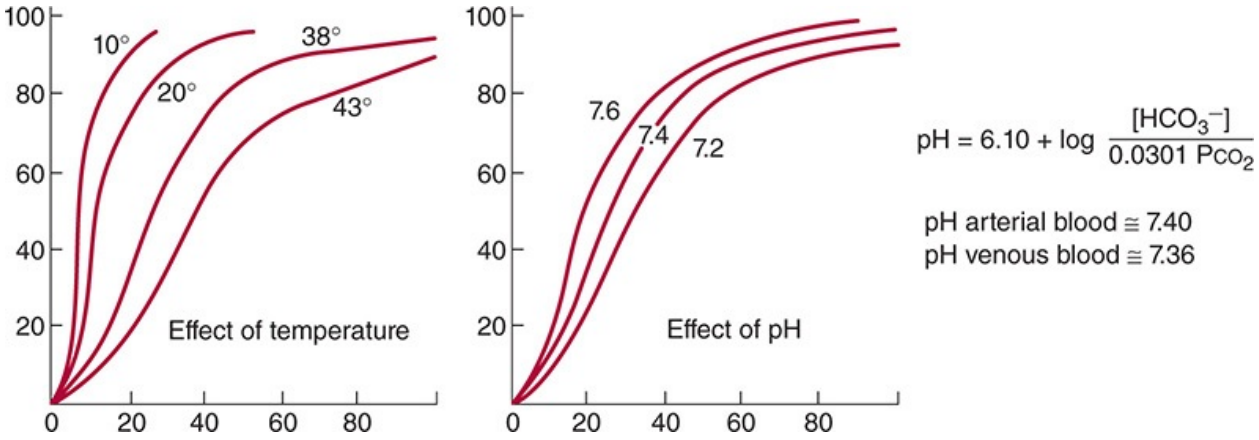


FIGURE 35–2 Effects of temperature and pH on the oxygen–hemoglobin dissociation curve. Both changes in temperature **(left)** and pH **(right)** can alter the affinity of hemoglobin for O_2. Plasma pH can be estimated using the modified Henderson–Hasselbalch equation, as shown. (Modified with permission from Comroe JH Jr, et al: *The Lung: Clinical Physiology and Pulmonary Function Tests*, 2nd ed. Year Book; 1962.)

The decrease in O_2 affinity of hemoglobin when the pH of blood falls is called the **Bohr effect** and is closely related to the fact that deoxygenated hemoglobin (deoxyhemoglobin) binds H^+ more actively than does oxygenated hemoglobin (oxyhemoglobin). The pH of blood falls as its CO_2 content increases, so that when the P_{CO_2} rises, the curve shifts to the right and the P_{50} rises. Most of the unsaturation of hemoglobin that occurs in the tissues is secondary to the decline in the P_{O_2}, but an extra 1–2% unsaturation is due to the rise in P_{CO_2} and consequent shift of the dissociation curve to the right.

2,3-DPG is very plentiful in red cells. It is formed from 3-phosphoglyceraldehyde, which is a product of glycolysis via the Embden–Meyerhof pathway. It is a highly charged anion that binds to the β chains of deoxyhemoglobin. One mole of deoxyhemoglobin binds 1 mol of 2,3-DPG. In effect,

$$HbO_2 + \text{2,3-DPG} \rightleftarrows Hb - \text{2,3-DPG} + O_2$$

In this equilibrium, an increase in the concentration of 2,3-DPG shifts the reaction to the right, causing more O_2 to be liberated.

Because acidosis inhibits red cell glycolysis, the 2,3-DPG concentration falls when the pH is low. Conversely, thyroid hormones, growth hormones, and

androgens can all increase the concentration of 2,3-DPG and the P_{50}.

Exercise has been reported to produce an increase in 2,3-DPG within 60 min (although the rise may not occur in trained athletes). The P_{50} is also increased during exercise, because the temperature rises in active tissues and CO_2 and metabolites accumulate, lowering the pH. In addition, much more O_2 is removed from each unit of blood flowing through active tissues because the tissue P_{O_2} declines. Finally, at low P_{O_2} values, the oxygen–hemoglobin dissociation curve is steep, and large amounts of O_2 are liberated per unit drop in P_{O_2}. Some clinical features of hemoglobin are discussed in **Clinical Box 35–1**.

An interesting contrast to hemoglobin is **myoglobin,** an iron-containing pigment found in skeletal muscle. Myoglobin resembles hemoglobin but binds 1 rather than 4 mol of O_2 per mole protein. The lack of cooperative binding is reflected in the myoglobin dissociation curve, a rectangular hyperbola rather than the sigmoid curve observed for hemoglobin **(Figure 35–3)**. Additionally, the leftward shift of the myoglobin O_2 binding curve when compared with hemoglobin demonstrates a higher affinity for O_2, and thus promotes a favorable transfer of O_2 from hemoglobin in the blood. The steepness of the myoglobin curve also shows that O_2 is released only at low P_{O_2} values (eg, during exercise). The myoglobin content is greatest in muscles specialized for sustained contraction. The muscle blood supply is compressed during such contractions, and myoglobin can continue to provide O_2 under reduced blood flow and/or reduced P_{O_2} in the blood.

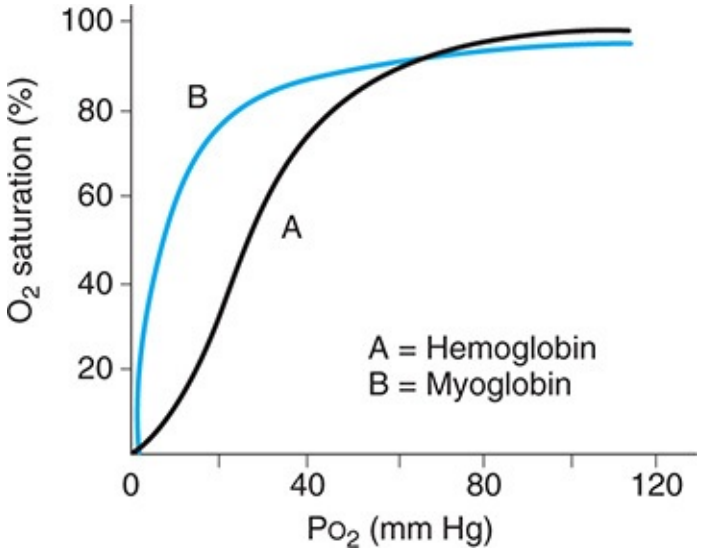


FIGURE 35–3 Comparison of dissociation curves for hemoglobin and myoglobin. The myoglobin binding curve (B) lacks the sigmoidal shape of the hemoglobin binding curve (A) because of the single O_2-binding site in each molecule. Myoglobin also has greater affinity for O_2 than hemoglobin (curve shifted left) and thus can release O_2 in muscle when P_{O_2} in blood is low (eg, during exercise).

CARBON DIOXIDE TRANSPORT

MOLECULAR FATE OF CARBON DIOXIDE IN BLOOD

The solubility of CO_2 in blood is about 20 times that of O_2; therefore, considerably more CO_2 than O_2 is present in simple solution at equal partial pressures. The CO_2 that diffuses into red blood cells is rapidly hydrated to H_2CO_3 because of the presence of carbonic anhydrase **(Figure 35–4)**. The H_2CO_3 dissociates to H^+ and HCO_3^-, and the H^+ is buffered, primarily by hemoglobin, while the HCO_3^- enters the plasma. Some of the CO_2 in the red cells reacts with the amino groups of hemoglobin and other proteins (R), forming **carbamino compounds.**

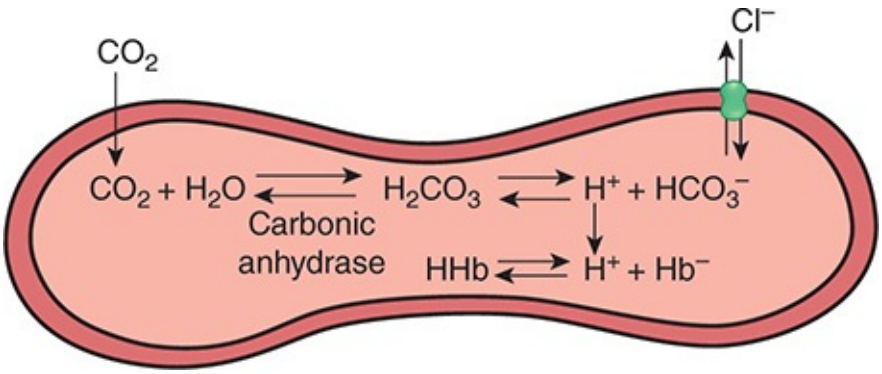
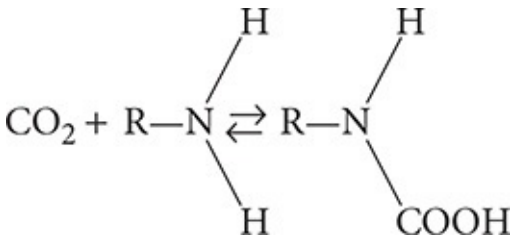


FIGURE 35–4 Fate of CO_2 in the red blood cell. Upon entering the red blood cell, CO_2 is rapidly hydrated to H_2CO_3 by carbonic anhydrase. H_2CO_3 is in equilibrium with H^+ and its conjugate base, HCO_3^-. H^+ can interact with deoxyhemoglobin, whereas HCO_3^- can be transported outside of the cell via anion exchanger 1 (AE1 or Band 3). In effect, for each CO_2 molecule that enters

the red cell, there is an additional HCO_3^- or Cl^- in the cell.



CLINICAL BOX 35–1

Hemoglobin & O_2 Binding In Vivo

Cyanosis

Reduced hemoglobin has a dark color, and a dusky bluish discoloration of the tissues, called **cyanosis,** appears when the reduced hemoglobin concentration of the blood in the capillaries is more than 5 g/dL. Its occurrence depends on the total amount of hemoglobin in the blood, the degree of hemoglobin unsaturation, and the state of the capillary circulation. Cyanosis is most easily seen in the nail beds and mucous membranes and in the earlobes, lips, and fingers, where the skin is thin and there are plenty capillaries. Although visible observation is indicative of cyanosis, it is not fully reliable. Further tests of arterial oxygen tension and saturation, blood and hemoglobin counts can provide more reliable diagnoses.

Effects of 2,3-DPG on Fetal & Stored Blood

The affinity of fetal hemoglobin (hemoglobin F) for O_2, which is greater than that for adult hemoglobin (hemoglobin A), facilitates the movement of O_2 from the mother to the fetus. The cause of this greater affinity is the poor binding of 2,3-DPG by the γ polypeptide chains that replace β chains in fetal hemoglobin. Some abnormal hemoglobins in adults have low P_{50} values, and the resulting high O_2 affinity of the hemoglobin causes enough tissue hypoxia to stimulate increased red cell formation, with resulting polycythemia. It is interesting to speculate that these hemoglobins may not bind 2,3-DPG.

Red cell 2,3-DPG concentration is increased in anemia and in a variety of diseases in which there is chronic hypoxia. This facilitates the delivery of O_2 to the tissues by raising the P_{O_2} at which O_2 is released in peripheral capillaries. In banked blood that is stored, the 2,3-DPG level falls and the

ability of this blood to release O_2 to the tissues is reduced. This decrease, which obviously limits the benefit of the blood if it is transfused into a hypoxic patient, is less if the blood is stored in citrate-phosphate-dextrose solution rather than the usual acid-citrate-dextrose solution.

THERAPEUTIC HIGHLIGHTS

Cyanosis is an indication of poorly oxygenated hemoglobin rather than a disease, and thus can have many causes, from cold exposure to drug overdose to chronic lung disease. As such, proper treatment depends on the underlying cause. For cyanosis caused by exposure to cold, maintaining a warm environment can be effective, whereas supplemental oxygen administration may be required under conditions of chronic and hypoxic disease.

Because deoxyhemoglobin binds more H^+ than oxyhemoglobin and forms carbamino compounds more readily, binding of O_2 to hemoglobin reduces its affinity for CO_2. The **Haldane effect** refers to the increased capacity of deoxygenated hemoglobin to bind and carry CO_2. Consequently, venous blood carries more CO_2 than arterial blood, CO_2 uptake is facilitated in the tissues, and CO_2 release is facilitated in the lungs. About 11% of the CO_2 added to the blood in the systemic capillaries is carried to the lungs as carbamino-CO_2.

CHLORIDE SHIFT

Because the rise in the HCO_3^- content of red cells is much greater than that in plasma as the blood passes through the capillaries, about 70% of the HCO_3^- formed in the red cells enters the plasma. The excess HCO_3^- leaves the red cells in exchange for Cl^- (Figure 35–4). This process is mediated by **anion exchanger 1 (AE1;** also called Band 3), a major membrane protein in the red blood cell. Because of this **chloride shift,** the Cl^- content of the red cells in venous blood is significantly greater than that in arterial blood. The chloride shift occurs rapidly and is essentially complete within 1 s.

Note that for each CO_2 molecule added to a red cell, there is an increase of one osmotically active particle in the cell—either an HCO_3^- or a Cl^- (Figure 35–

4). Consequently, the red cells take up water and increase in size. For this reason, plus the fact that a small amount of fluid in the arterial blood returns via the lymphatics rather than the veins, the hematocrit of venous blood is normally 3% greater than that of the arterial blood. In the lungs, the Cl^- moves back out of the cells and they shrink.

SPATIAL DISTRIBUTION OF CARBON DIOXIDE IN BLOOD

For convenience, the various fates of CO_2 in the plasma and red cells are summarized in **Table 35–2**. The extent to which they increase the capacity of the blood to carry CO_2 is indicated by the difference between the lines indicating the dissolved CO_2 and the total CO_2 in the dissociation curves for CO_2 shown in **Figure 35–5**.

TABLE 35–2 Fate of CO_2 in blood.

In plasma
1. Dissolved
2. Formation of carbamino compounds with plasma protein
3. Hydration, H^+ buffered, HCO_3^- in plasma
In red blood cells
1. Dissolved
2. Formation of carbamino-Hb
3. Hydration, H^+ buffered, 70% of HCO_3^- enters the plasma
4. Cl^- shifts into cells; mOsm in cells increases

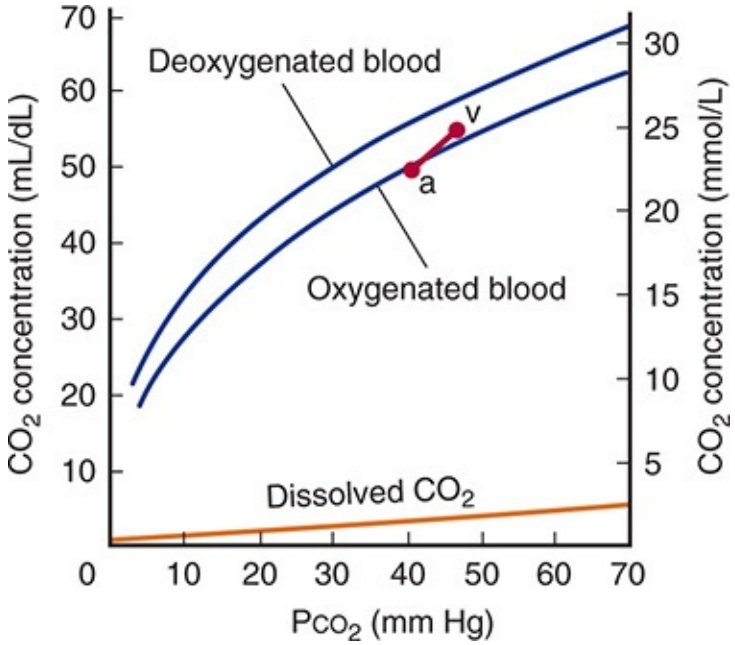


FIGURE 35–5 CO_2 dissociation curves. The arterial point (a) and the venous point (v) indicate the total CO_2 content found in arterial blood and venous blood of normal resting humans. Note the low amount of CO_2 that is dissolved (orange trace) compared to that which can be carried by other means (Table 35–2). (Modified with permission from Schmidt RF, Thews G [editors]: *Human Physiology*. Springer; 1983.)

Of the approximately 49 mL of CO_2 in each deciliter of arterial blood (Table 35–1), 2.6 mL is dissolved, 2.6 mL is in carbamino compounds, and 43.8 mL is in HCO_3^-. In the tissues, 3.7 mL of CO_2 per deciliter of blood is added, 0.4 mL stays in solution, 0.8 mL forms carbamino compounds, and 2.5 mL forms HCO_3^-. The pH of the blood drops from 7.40 to 7.36. In the lungs, the processes are reversed, and the 3.7 mL of CO_2 is discharged into the alveoli. In this manner, 200 mL of CO_2 per minute at rest and much larger amounts during exercise are transported from the tissues to the lungs and excreted. It is worth noting that this amount of CO_2 is equivalent in 24 h to over 12,500 mEq of H^+.

ACID–BASE BALANCE & GAS TRANSPORT

The major source of acids in the blood under normal conditions is through cellular metabolism. The CO_2 formed by metabolism in the tissues is in large

part hydrated to H_2CO_3, resulting in the large total H^+ load noted above (> 12,500 mEq/day). However, most of the CO_2 is excreted in the lungs, and the small quantities of the remaining H^+ are excreted by the kidneys.

BUFFERING IN THE BLOOD

Acid and base shifts in the blood are largely controlled by three main buffers in blood: (1) proteins, (2) hemoglobin, and (3) the carbonic acid-bicarbonate system. Plasma **proteins** are effective buffers because both their free carboxyl and their free amino groups dissociate:

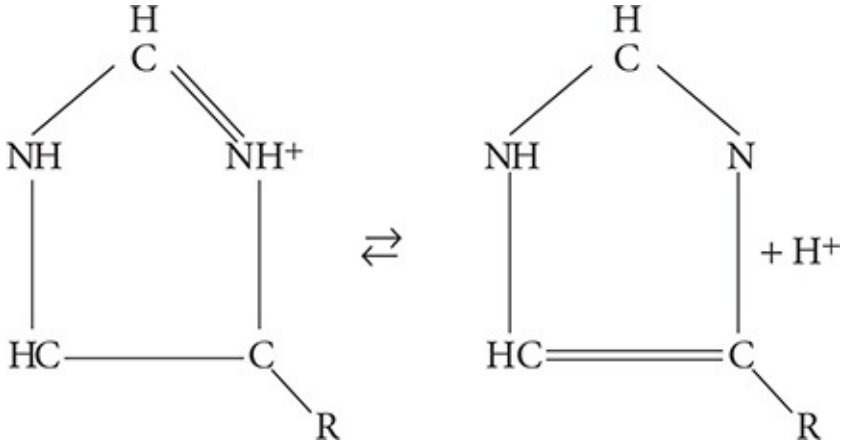
$$RCOOH \rightleftarrows RCOO^- + H^+$$

$$pH = pK'_{RCOOH} + \log \frac{[RCOO^-]}{[RCOOH]}$$

$$RNH_3^+ \rightleftarrows RNH_2 + H^+$$

$$pH = pK'_{RNH_3} + \log \frac{[RNH_2]}{[RNH_3^+]}$$

The second buffer system is provided by the dissociation of the imidazole groups of the histidine residues in **hemoglobin.**



In the pH 7.0–7.7 range, the free carboxyl and amino groups of hemoglobin contribute relatively little to its buffering capacity. However, the hemoglobin molecule contains 38 histidine residues, and on this basis—plus the fact that hemoglobin is present in large amounts—the hemoglobin in blood has six times

the buffering capacity of the plasma proteins. In addition, the action of hemoglobin is unique because the imidazole groups of deoxyhemoglobin (Hb) dissociate less than those of oxyhemoglobin (HbO_2), making Hb a weaker acid and therefore a better buffer than HbO_2. The titration curves for Hb and HbO_2 **(Figure 35–6)** illustrate the differences in H^+ buffering capacity.

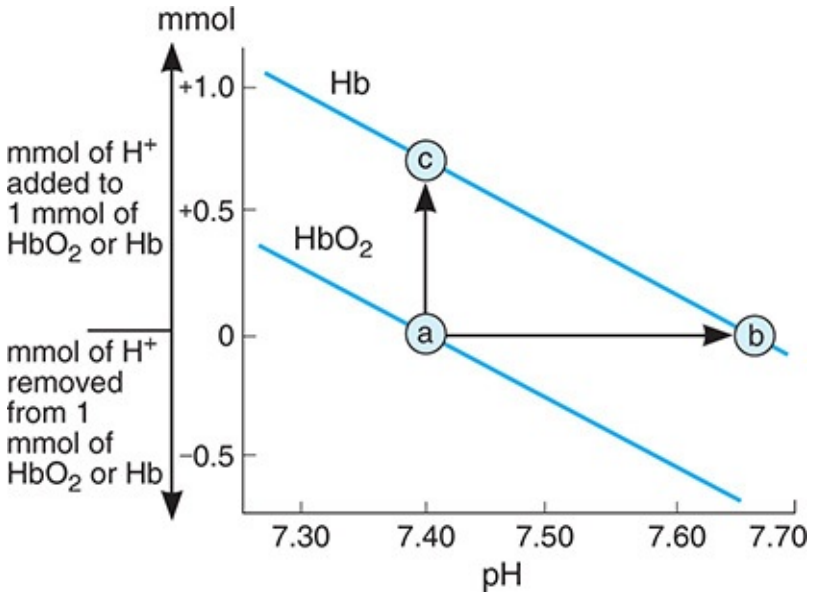


FIGURE 35–6 Comparative titration curves for oxygenated hemoglobin (HbO_2) and deoxyhemoglobin (Hb). The arrow from a to c indicates the number of additional millimoles of H^+ that Hb can buffer compared with a similar concentration of HbO_2 (ie, no shift in pH). The arrow from a to b indicates the pH shift that would occur on deoxygenation of HbO_2 without additional H^+.

The third and major buffer system in blood is the **carbonic acid-bicarbonate system:**

$$H_2CO_3 \rightleftarrows H^+ + HCO_3^-$$

The Henderson–Hasselbalch equation for this system is

$$pH = pK + \log \frac{[HCO_3^-]}{[H_2CO_3]}$$

The pK for this system in an ideal solution is low (about 3), and the amount

of H_2CO_3 is small and hard to measure accurately. However, in the body, H_2CO_3 is in equilibrium with CO_2:

$$H_2CO_3 \rightleftarrows CO_2 + H_2O$$

If the pK is changed to pK′ (apparent ionization constant; distinguished from the true pK due to less than ideal conditions for the solution) and $[CO_2]$ is substituted for $[H_2CO_3]$, the pK′ is 6.1:

$$pH = 6.10 + \log \frac{[HCO_3^-]}{[CO_2]}$$

The clinically relevant form of this equation is:

$$pH = 6.10 + \log \frac{[HCO_3^-]}{0.0301\ P_{CO_2}}$$

since the amount of dissolved CO_2 is proportional to the partial pressure of CO_2 and the solubility coefficient of CO_2 in mmol/L/mm Hg is 0.0301. $[HCO_3^-]$ cannot be measured directly, but pH and P_{CO_2} can be measured with suitable accuracy with pH and P_{CO_2} glass electrodes, and $[HCO_3^-]$ can then be calculated.

The pK' of this system is still low relative to the pH of the blood, but the system is one of the most effective buffer systems in the body because the amount of dissolved CO_2 is controlled by respiration (ie, it is an "open" system). Additional control of the plasma concentration of HCO_3^- is provided by the kidneys. When H^+ is added to the blood, HCO_3^- declines as more H_2CO_3 is formed. If the extra H_2CO_3 were not converted to CO_2 and H_2O and the CO_2 excreted in the lungs, the H_2CO_3 concentration would rise. Without CO_2 removal to reduce H_2CO_3, sufficient H^+ addition that would halve the plasma HCO_3^- would alter the pH 7.4 to 6.0. However, such a H^+ concentration increase is tolerated because: (1) extra H_2CO_3 that is formed is removed and (2) the H^+ rise stimulates respiration and therefore produces a drop in P_{CO_2}, so that some additional H_2CO_3 is removed. The net pH after such an increase in H^+ concentration is actually 7.2 or 7.3.

There are two additional factors that make the carbonic acid-bicarbonate system such a good biological buffer. First, the reaction $CO_2 + H_2O \rightleftarrows H_2CO_3$ proceeds slowly in either direction unless the enzyme **carbonic anhydrase** is present. There is no carbonic anhydrase in plasma, but there is an abundant supply in red blood cells, spatially confining and controlling the reaction. Second, the presence of hemoglobin in the blood increases the buffering of the system by binding free H^+ produced by the hydration of CO_2 and allowing for movement of the HCO_3^- into the plasma.

ACIDOSIS & ALKALOSIS

The pH of the arterial plasma is normally 7.40 and that of venous plasma slightly lower. A decrease in pH below the norm **(acidosis)** is technically present whenever the arterial pH is below 7.40 and an increase in pH **(alkalosis)** is technically present whenever pH is above 7.40. In practice, variations of up to 0.05 pH unit occur without untoward effects. Acid–base disorders are split into four categories: respiratory acidosis, respiratory alkalosis, metabolic acidosis, and metabolic alkalosis. In addition, these disorders can occur in combination. Some causes of acid–base disturbances are shown in **Table 35–3**.

TABLE 35–3 Plasma pH, HCO_3^-, and PCO_2 values in various typical disturbances of acid–base balance.[a]

Condition	Arterial Plasma			Cause
	pH	HCO_3^- (mEq/L)	P_{CO_2} (mm Hg)	
Normal	7.40	24.1	40	
Metabolic acidosis	7.28	18.1	40	NH_4Cl ingestion
	6.96	5.0	23	Diabetic acidosis
Metabolic alkalosis	7.50	30.1	40	$NaHCO_3$ ingestion
	7.56	49.8	58	Prolonged vomiting
Respiratory acidosis	7.34	25.0	48	Breathing 7% CO_2
	7.34	33.5	64	Emphysema
Respiratory alkalosis	7.53	22.0	27	Voluntary hyper-ventilation
	7.48	18.7	26	Three-week residence at 4000-m altitude

[a]In the diabetic acidosis and prolonged vomiting examples, respiratory compensation for primary metabolic acidosis and alkalosis has occurred, and the P_{CO_2} has shifted from 40 mm Hg. In the emphysema and high-altitude examples, renal compensation for primary respiratory acidosis and alkalosis has occurred and has made the deviations from normal of the plasma HCO_3^- larger than they would otherwise be.

RESPIRATORY ACIDOSIS

Any short-term rise in arterial P_{CO_2} (ie, above 40 mm Hg, due to hypoventilation) results in **respiratory acidosis.** Recall that CO_2 that is retained is in equilibrium with H_2CO_3, which in turn is in equilibrium with HCO_3^-. The effective rise in plasma HCO_3^- means that a new equilibrium is reached at a lower pH. This can be indicated graphically on a plot of plasma HCO_3^- concentration versus pH **(Figure 35–7)**. The pH change observed at any increase in P_{CO_2} during respiratory acidosis depends on the buffering capacity of the

blood. The initial changes shown in Figure 35–7 are those that occur independently of any compensatory mechanism; that is, they are those of **uncompensated respiratory acidosis.**

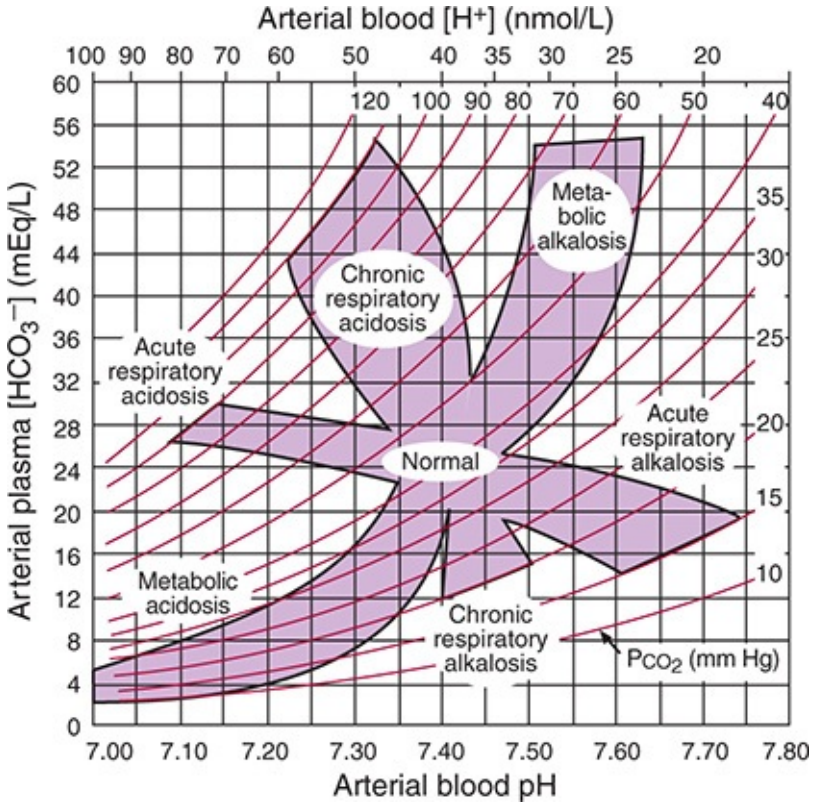


FIGURE 35–7 Acid–base nomogram. Changes in the P_{CO_2} (curved lines), plasma HCO_3^-, and pH (or $[H^+]$) of arterial blood in respiratory and metabolic acidosis are shown. Note the shifts in HCO_3^- and pH as acute respiratory acidosis and alkalosis are compensated, producing their chronic counterparts. (Reproduced with permission from Brenner BM, Rector FC Jr. [editors]: *Brenner & Rector's The Kidney*, 7th ed. St. Louis, MO: Saunders; 2004.)

RESPIRATORY ALKALOSIS

Any short-term lowering of P_{CO_2} below what is needed for proper CO_2 exchange (ie, below 35 mm Hg, as can occur during hyperventilation) results in **respiratory alkalosis.** The decreased CO_2 shifts the equilibrium of the carbonic acid-bicarbonate system to effectively lower $[H^+]$ and increase pH. As in respiratory acidosis, initial pH changes corresponding to respiratory alkalosis

(Figure 35–7) are those that occur independently of any compensatory mechanism and are thus **uncompensated respiratory alkalosis.**

METABOLIC ACIDOSIS & ALKALOSIS

Blood pH changes can also arise by nonrespiratory mechanism. **Metabolic acidosis** (or nonrespiratory acidosis) occurs when strong acids are added to blood. If, for example, a large amount of acid is ingested (eg, aspirin overdose), acids in the blood are quickly increased. The H_2CO_3 that is formed is converted to H_2O and CO_2, and the CO_2 is rapidly excreted via the lungs. This is the situation in **uncompensated metabolic acidosis** (Figure 35–7). Note that in contrast to respiratory acidosis, metabolic acidosis does not include a change in P_{CO_2}; the shift toward metabolic acidosis occurs along an isobar line **(Figure 35–8)**. When the free $[H^+]$ level falls as a result of addition of alkali, or more commonly, the removal of large amounts of acid (eg, following vomiting), **metabolic alkalosis** results. In uncompensated metabolic alkalosis the pH rises along the isobar line (Figures 35–7 and 35–8).

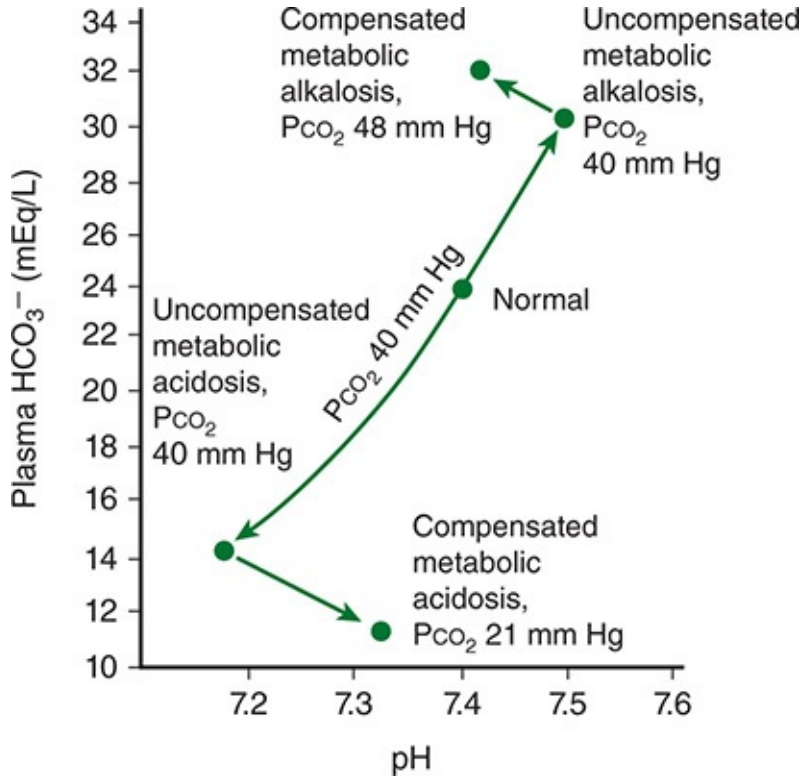


FIGURE 35–8 Acid–base paths during metabolic acidosis. Changes in true

plasma pH, HCO_3^-, and P_{CO_2} at rest, during metabolic acidosis and alkalosis, and following respiratory compensation are plotted. Metabolic acidosis or alkalosis causes changes in pH along the P_{CO_2} isobar line (middle line). Respiratory compensation moves pH toward normal by altering P_{CO_2} (top and bottom arrows). (Modified with permission from Davenport HW: *The ABC of Acid-Base Chemistry*, 6th ed. University of Chicago Press; 1974.)

RESPIRATORY & RENAL COMPENSATION

Uncompensated acidosis and alkalosis as described above are seldom seen because of compensation systems. The two main compensatory systems are **respiratory compensation** and **renal compensation.**

The respiratory system compensates for metabolic acidosis or alkalosis by altering ventilation, and consequently, the P_{CO_2}, which can directly change blood pH. Respiratory mechanisms are fast. In response to metabolic acidosis, ventilation is increased, resulting in a decrease of P_{CO_2} (eg, from 40 mm Hg to 20 mm Hg) and a subsequent increase in pH toward normal (Figure 35–8). In response to metabolic alkalosis, ventilation is decreased, P_{CO_2} is increased, and a subsequent decrease in pH occurs. Because respiratory compensation is a quick response, the graphical representation in Figure 35–8 overstates the two-step adjustment in blood pH. In actuality, as soon as metabolic acidosis begins, respiratory compensation is invoked and the large shifts in pH depicted do not occur.

For complete compensation from respiratory or metabolic acidosis/alkalosis, renal compensatory mechanisms are invoked. The kidney responds to acidosis by actively secreting fixed acids while retaining filtered HCO_3^-. In contrast, the kidney responds to alkalosis by decreasing H^+ secretion and by decreasing the retention of filtered HCO_3^-.

Renal tubule cells in the kidney have active carbonic anhydrase and thus can produce H^+ and HCO_3^- from CO_2. In response to acidosis, these cells secrete H^+ into the tubular fluid in exchange for Na^+ while the HCO_3^- is actively reabsorbed into the peritubular capillary; for each H^+ secreted, one Na^+ and one HCO_3^- are added to the blood. The result of this renal compensation for respiratory acidosis is shown graphically in the shift from acute to chronic respiratory acidosis in Figure 35–7. Conversely, in response to alkalosis, the

kidney decreases H^+ secretion and depresses HCO_3^- reabsorption. The result of this renal compensation for respiratory alkalosis is shown graphically in the shift from acute to chronic respiratory alkalosis in Figure 35–7. Clinical evaluations of acid–base status are discussed in **Clinical Box 35–2**, and the role of the kidneys in acid–base homeostasis is discussed in more detail in Chapter 38.

HYPOXIA

Hypoxia is O_2 deficiency at the tissue level. It is a more correct term than **anoxia** (lack of O_2), since there is rarely no O_2 at all left in the tissues.

Numerous classifications for hypoxia have been used, but the more traditional four-type system still has considerable utility if the definitions of the terms are kept clearly in mind. The four categories are (1) **hypoxemia** (sometimes termed **hypoxic hypoxia**), in which the P_{O_2} of the arterial blood is reduced; (2) **anemic hypoxia,** in which the arterial P_{O_2} is normal but the amount of hemoglobin available to carry O_2 is reduced; (3) **ischemic** or **stagnant hypoxia,** in which the blood flow to a tissue is so low that adequate O_2 is not delivered to it despite a normal P_{O_2} and hemoglobin concentration; and (4) **histotoxic hypoxia,** in which the amount of O_2 delivered to a tissue is adequate but, because of the action of a toxic agent, the tissue cells cannot make use of the O_2 supplied to them. Some specific effects of hypoxia on cells and tissues are discussed in **Clinical Box 35–3**.

CLINICAL BOX 35–2

Clinical Evaluation of Acid–Base Status

In evaluating disturbances of acid–base balance, it is important to know the pH and HCO_3^- content of arterial plasma. Reliable pH determinations can be made with a pH meter and a glass pH electrode. Using pH and a direct measurement of the P_{CO_2} with a CO_2 electrode, HCO_3^- concentration can be calculated. The P_{CO_2} is ~8 mm Hg higher and the pH 0.03–0.04 unit lower in venous than arterial plasma because venous blood contains the CO_2 being carried from the tissues to the lungs. Therefore, the calculated HCO_3^- concentration is about 2 mmol/L higher. However, if this is kept in mind, free-

flowing venous blood can be substituted for arterial blood in most clinical situations.

A measurement that is of some value in the differential diagnosis of metabolic acidosis is the **anion gap.** This gap, which is something of a misnomer, refers to the difference between the concentration of cations other than Na^+ and the concentration of anions other than Cl^- and HCO_3^- in the plasma. It consists for the most part of proteins in anionic form, HPO_4^{2-}, SO_4^{2-}, and organic acids; a normal value is about 12 mEq/L. It is increased when the plasma concentration of K^+, Ca^{2+}, or Mg^+ is decreased; when the concentration of (or the charge on) plasma proteins is increased; or when organic anions such as lactate or foreign anions accumulate in blood. It is decreased when cations are increased or when plasma albumin is decreased. The anion gap is increased in metabolic acidosis due to ketoacidosis, lactic acidosis, and other forms of acidosis in which organic anions are increased.

HYPOXEMIA

By definition, hypoxemia is a condition of reduced arterial P_{O_2}. Hypoxemia is a problem in normal individuals at high altitudes and is a complication of pneumonia and a variety of other diseases of the respiratory system.

CLINICAL BOX 35–3

Effects of Hypoxia on Cells and Selected Tissues

Effects on Cells

Hypoxia causes the production of transcription factors (hypoxia-inducible factors; HIFs). These are made up of α and β subunits. In normally oxygenated tissues, the α subunits are rapidly ubiquitinated and destroyed. However, in hypoxic cells, the α subunits dimerize with β subunits, and the dimers activate genes that produce several proteins including angiogenic factors and erythropoietin, among others.

Effects on the Brain

In hypoxymia and the other generalized forms of hypoxia, the brain is

affected first. A sudden drop in the inspired P_{O_2} to less than 20 mm Hg, which occurs, for example, when cabin pressure is suddenly lost in a plane flying above 16,000 m, causes loss of consciousness in 10–20 s and death in 4–5 min. Less severe hypoxia causes a variety of mental aberrations not unlike those produced by alcohol: impaired judgment, drowsiness, dulled pain sensibility, excitement, disorientation, loss of time sense, and headache. Other symptoms include anorexia, nausea, vomiting, tachycardia, and, when the hypoxia is severe, hypertension. The rate of ventilation is increased in proportion to the severity of the hypoxia of the carotid chemoreceptor cells.

Respiratory Stimulation

Dyspnea is by definition difficult or labored breathing in which the person is conscious of shortness of breath; **hyperpnea** is the general term for an increase in the rate or depth of breathing regardless of the patient's subjective sensations. **Tachypnea** is rapid, shallow breathing. In general, a normal individual is not conscious of respiration until ventilation is doubled, and breathing is not uncomfortable until ventilation is tripled or quadrupled. Whether or not a given level of ventilation is uncomfortable also appears to depend on a variety of other factors. Hypercapnia and, to a lesser extent, hypoxia cause dyspnea. An additional factor is the effort involved in moving the air in and out of the lungs (the work of breathing).

EFFECTS OF DECREASED BAROMETRIC PRESSURE

The composition of air stays the same, but the total barometric pressure falls with increasing altitude **(Figure 35–9)**, and thus, the P_{O_2} also falls. At 3000 m (~10,000 ft) above sea level, the alveolar P_{O_2} is about 60 mm Hg and there is enough hypoxic stimulation of the chemoreceptors under normal breathing to cause increased ventilation. As one ascends higher, the alveolar P_{O_2} falls less rapidly and the alveolar P_{CO_2} declines because of the hyperventilation. The resulting fall in arterial P_{CO_2} produces respiratory alkalosis. A number of compensatory mechanisms operate over a period of time to increase altitude tolerance **(acclimatization),** but in unacclimatized persons, mental symptoms such as irritability appear at about 3700 m. At 5500 m, the hypoxic symptoms are severe; and at altitudes above 6100 m (20,000 ft), consciousness is usually

lost.

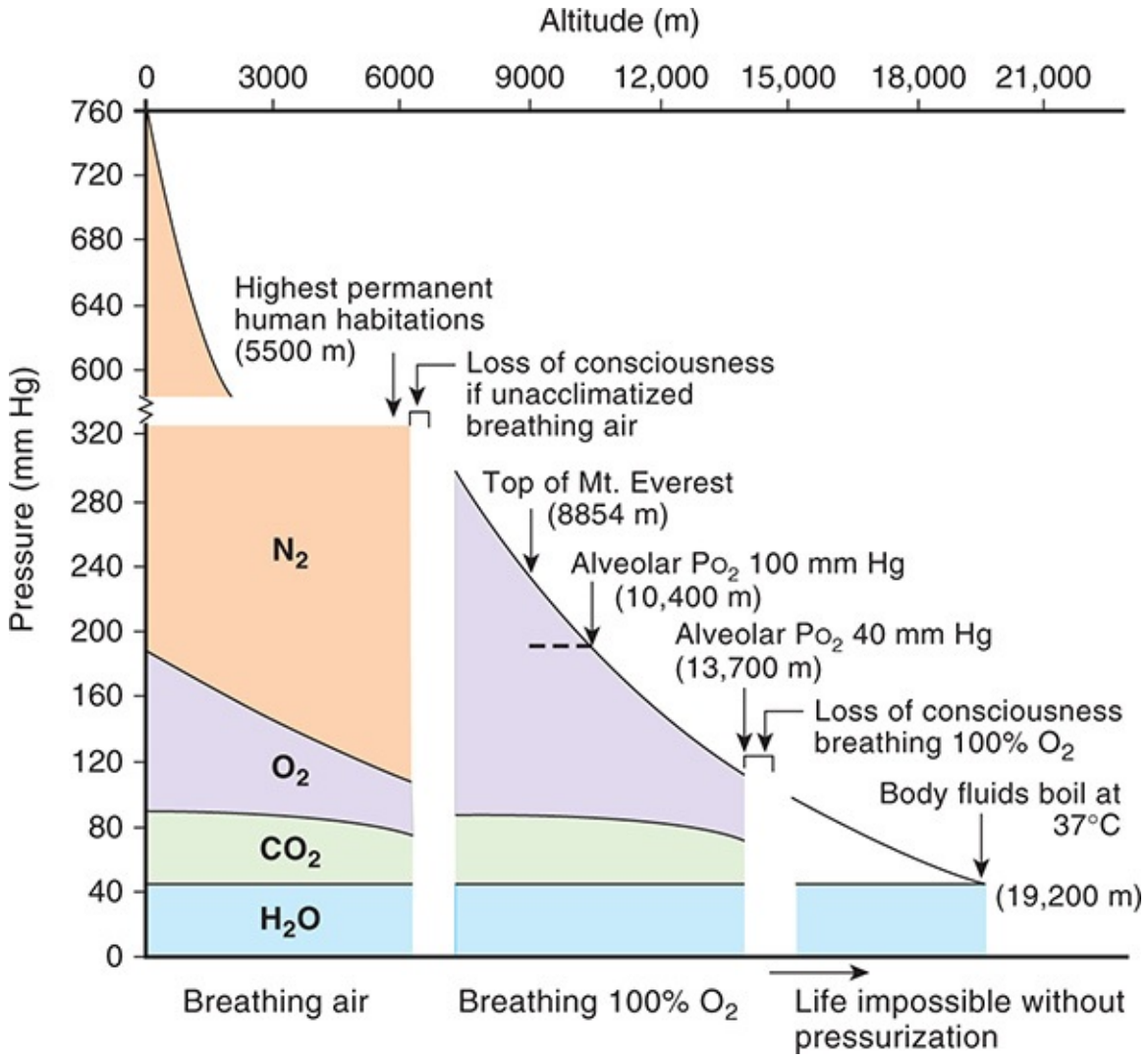


FIGURE 35–9 Composition of alveolar air in individuals breathing air (0–6100 m) and 100% O_2 (6100–13,700 m). The minimal alveolar P_{O_2} that an unacclimatized person can tolerate without loss of consciousness is about 35–40 mm Hg. Note that with increasing altitude, the alveolar P_{CO_2} drops because of the hyperventilation due to hypoxic stimulation of the carotid and aortic chemoreceptors. The fall in barometric pressure with increasing altitude is not linear because air is compressible.

HYPOXIC SYMPTOMS & BREATHING OXYGEN

Some of the effects of high altitude can be offset by breathing 100% O_2. Under

these conditions, the total atmospheric pressure becomes the limiting factor in altitude tolerance.

The partial pressure of water vapor in the alveolar air is constant at 47 mm Hg, and that of CO_2 is normally 40 mm Hg, so that the lowest barometric pressure at which a normal alveolar P_{O_2} of 100 mm Hg is possible is 187 mm Hg, the pressure at about 10,400 m (34,000 ft). At greater altitudes, the increased ventilation due to the decline in alveolar P_{O_2} lowers the alveolar P_{CO_2} somewhat, but the maximum alveolar P_{O_2} that can be attained when breathing 100% O_2 at the ambient barometric pressure of 100 mm Hg at 13,700 m is ~40 mm Hg. At ~14,000 m, consciousness is lost in spite of the administration of 100% O_2. At 19,200 m, the barometric pressure is 47 mm Hg, and at or below this pressure the body fluids boil at body temperature. The point is largely academic, however, because any individual exposed to such a low pressure would be dead of hypoxia before the bubbles of body fluids could cause death.

Of course, an artificial atmosphere can be created around an individual; in a pressurized suit or cabin supplied with O_2 and a system to remove CO_2, it is possible to ascend to any altitude and to live in the vacuum of interplanetary space. Some delayed effects of high altitude are discussed in **Clinical Box 35–4**.

ACCLIMATIZATION

Acclimatization to altitude is due to the operation of a variety of compensatory mechanisms. The respiratory alkalosis produced by the hyperventilation shifts the oxygen–hemoglobin dissociation curve to the left, but a concomitant increase in red blood cell 2,3-DPG tends to decrease the O_2 affinity of hemoglobin. The net effect is a small increase in P_{50}. The decrease in O_2 affinity makes more O_2 available to the tissues. However, the value of the increase in P_{50} is limited because when the arterial P_{O_2} is markedly reduced, the decreased O_2 affinity also interferes with O_2 uptake by hemoglobin in the lungs.

The initial ventilatory response to increased altitude is relatively small, because the alkalosis tends to counteract the stimulating effect of hypoxia. However, ventilation steadily increases over the next 4 days **(Figure 35–10)** because the active transport of H^+ into cerebrospinal fluid (CSF), or possibly a developing lactic acidosis in the brain, causes a fall in CSF pH that increases the response to hypoxia. After 4 days, the ventilatory response begins to decline slowly, but it takes years of residence at higher altitudes for it to decline to the

initial level, if it is reached at all.

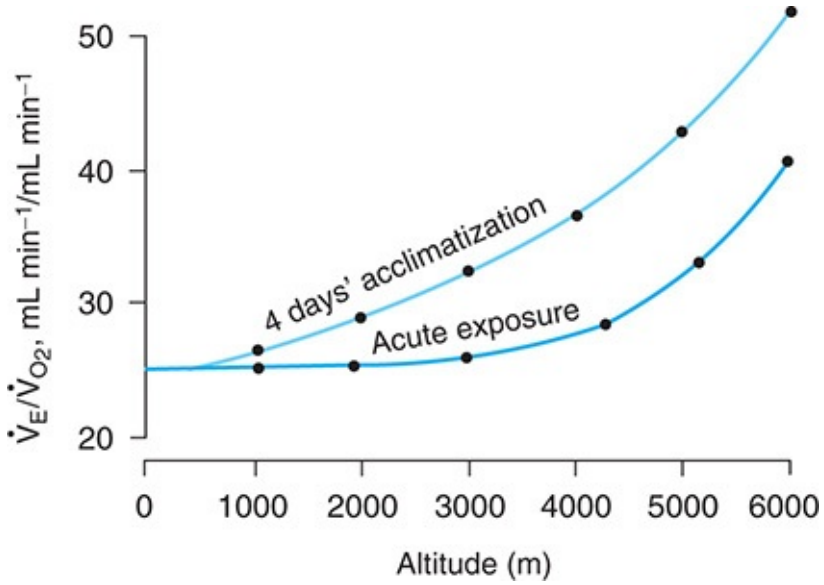


FIGURE 35–10 Effect of acclimatization on the ventilatory response at various altitudes. $\dot{V}_E/\dot{V}_{O_2}$ is the ventilatory equivalent, the ratio of expired minute volume $(\dot{V}_E)$ to the O_2 consumption $(\dot{V}_{O_2})$. (Reproduced with permission from Lenfant C, Sullivan K: Adaptation to high altitude. N Engl J Med 1971; June 10; 284(23):1298–1309.)

CLINICAL BOX 35–4

Delayed Effects of High Altitude

When they first arrive at a high altitude, many individuals develop acute "mountain sickness." This syndrome develops 8–24 h after arrival at altitude and lasts 4–8 days. It is characterized by headache, irritability, insomnia, breathlessness, and nausea and vomiting. Its cause is unsettled, but it appears to be associated with cerebral edema. The low P_{O_2} at high altitude causes systemic arteriolar dilation, and if cerebral autoregulation does not compensate, there is an increase in capillary pressure that favors increased transudation of fluid into brain tissue.

Two more serious syndromes that are associated with high-altitude illness: **high-altitude cerebral edema** and **high-altitude pulmonary edema.** In high-altitude cerebral edema, the capillary leakage in mountain sickness progresses to frank brain swelling, with ataxia, disorientation, and in some

cases coma and death due to herniation of the brain through the tentorium. High-altitude pulmonary edema is a patchy edema of the lungs that is related to the marked pulmonary hypertension that develops at high altitude. It has been argued that it occurs because not all pulmonary arteries have enough smooth muscle to constrict in response to hypoxia, and in the capillaries supplied by those arteries, the general rise in pulmonary arterial pressure causes a capillary pressure increase that disrupts their walls (stress failure).

THERAPEUTIC HIGHLIGHTS

All forms of high-altitude illness are benefited by descent to lower altitude and by treatment with the diuretic acetazolamide. This drug inhibits carbonic anhydrase and results in stimulated respiration, increased $PaCO_2$, and reduced formation of CSF. When cerebral edema is marked, large doses of glucocorticoids are often administered as well. Their mechanism of action is unsettled. In high-altitude pulmonary edema, prompt treatment with O_2 is essential—and, if available, use of a hyperbaric chamber. Portable hyperbaric chambers are now available in a number of mountain areas. Nifedipine, a Ca^{2+} channel blocker that lowers pulmonary artery pressure, can also be useful.

Erythropoietin secretion increases promptly on ascent to high altitude and then falls somewhat over the following 4 days as the ventilatory response increases and the arterial P_{O_2} rises. The increase in circulating red blood cells triggered by the erythropoietin begins in 2–3 days and is sustained as long as the individual remains at high altitude. Compensatory changes also occur in the tissues. The mitochondria, which are the site of oxidative reactions, increase in number, and myoglobin increases, which facilitates the movement of O_2 into the tissues. The tissue content of cytochrome oxidase also increases.

The effectiveness of the acclimatization process is indicated by the fact that permanent human habitations exist in the Andes and Himalayas at elevations above 5500 m (18,000 ft). The natives who live in these villages are barrel-chested and markedly polycythemic. They have low alveolar P_{O_2} values, but in most other ways they are remarkably normal.

DISEASES CAUSING HYPOXEMIA

Hypoxemia is the most common form of hypoxia seen clinically. The diseases that cause it can be roughly divided into those in which the gas exchange apparatus fails, those such as congenital heart disease in which large amounts of blood are shunted from the venous to the arterial side of the circulation, and those in which the respiratory pump fails. Lung failure occurs when conditions such as pulmonary fibrosis produce alveolar-capillary block, or there is ventilation–perfusion imbalance. Pump failure can be due to fatigue of the respiratory muscles in conditions in which the work of breathing is increased or to a variety of mechanical defects such as pneumothorax or bronchial obstruction that limit ventilation. It can also be caused by abnormalities of the neural mechanisms that control ventilation, such as depression of the respiratory neurons in the medulla by morphine and other drugs. Some specific causes of hypoxemia are discussed in the following text.

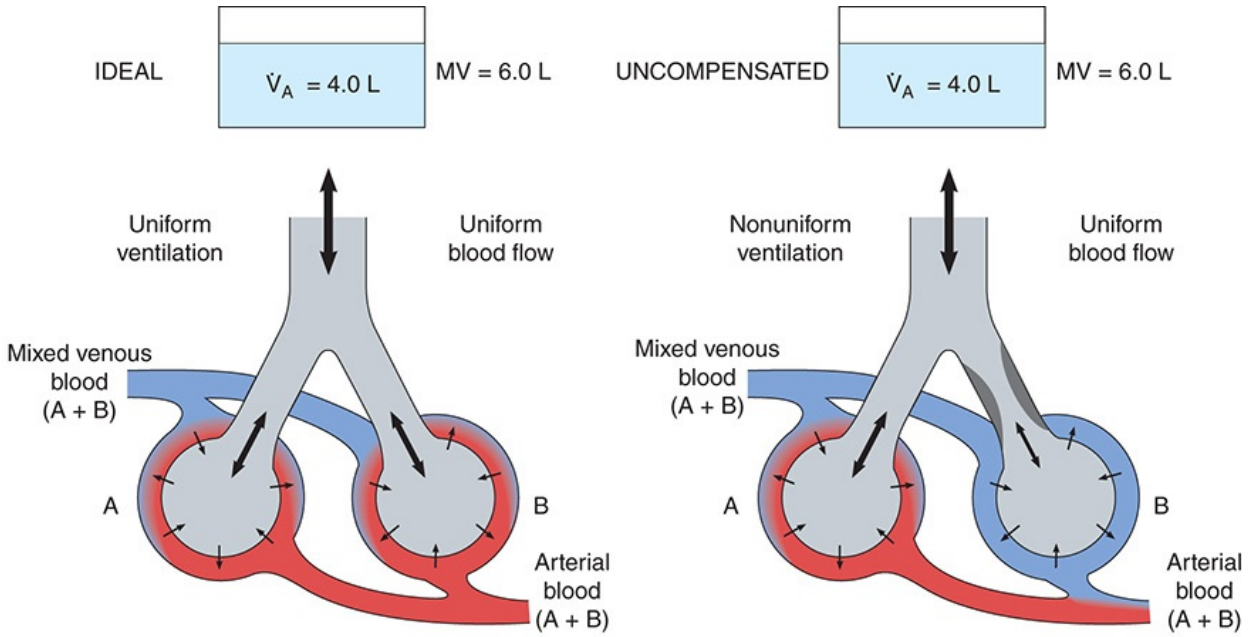
VENOUS-TO-ARTERIAL SHUNTS

When a cardiovascular abnormality such as an interatrial septal defect permits large amounts of unoxygenated venous blood to bypass the pulmonary capillaries and dilute the oxygenated blood in the systemic arteries (“right-to-left shunt”), chronic hypoxemia and cyanosis **(cyanotic congenital heart disease)** result. Administration of 100% O_2 raises the O_2 content of alveolar air but has little effect on hypoxia due to venous-to-arterial shunts. This is because the deoxygenated venous blood does not have the opportunity to get to the lung to be oxygenated.

VENTILATION–PERFUSION IMBALANCE

Patchy ventilation–perfusion imbalance is by far the most common cause of hypoxemia in clinical situations. In disease processes that prevent ventilation of some of the alveoli, the ventilation-blood flow ratios in different parts of the lung determine the extent to which systemic arterial P_{O_2} declines. If nonventilated alveoli are perfused, the nonventilated but perfused portion of the lung is in effect a right-to-left shunt, dumping unoxygenated blood into the left side of the heart. Lesser degrees of ventilation–perfusion imbalance are more common. In the example illustrated in **Figure 35–11**, the balanced ventilation–perfusion example on the left illustrates a uniform distribution throughout gas exchange. However, when ventilation is not in balance with perfusion, O_2

exchange is compromised. Note that the underventilated alveoli (B) have a low alveolar P_{O_2}, whereas the overventilated alveoli (A) have a high alveolar P_{O_2} while both have the same blood flow. The unsaturation of the hemoglobin of the blood coming from B is not completely compensated by the slightly greater saturation of the blood coming from A, because hemoglobin is normally nearly saturated in the lungs and the higher alveolar P_{O_2} adds only a little more O_2 to the hemoglobin than it normally carries. Consequently, the arterial blood is unsaturated. The CO_2 content of the arterial blood is generally normal in such situations, since extra loss of CO_2 in overventilated regions can balance diminished loss in underventilated areas.



	A	B	A + B
Alveolar ventilation (L/min)	2.0	2.0	4.0
Pulmonary blood flow (L/min)	2.5	2.5	5.0
Ventilation/blood flow ratio	0.8	0.8	0.8
Mixed venous O_2 saturation (%)	75.0	75.0	75.0
Arterial O_2 saturation (%)	97.4	97.4	97.4
Mixed venous O_2 tension (mm Hg)	40.0	40.0	40.0
Alveolar O_2 tension (mm Hg)	104.0	104.0	104.0
Arterial O_2 tension (mm Hg)	104.0	104.0	104.0

	A	B	A + B
Alveolar ventilation (L/min)	3.2	0.8	4.0
Pulmonary blood flow (L/min)	2.5	2.5	5.0
Ventilation/blood flow ratio	1.3	0.3	0.8
Mixed venous O_2 saturation (%)	75.0	75.0	75.0
Arterial O_2 saturation (%)	98.2	91.7	95.0
Mixed venous O_2 tension (mm Hg)	40.0	40.0	40.0
Alveolar O_2 tension (mm Hg)	116.0	66.0	106.0
Arterial O_2 tension (mm Hg)	116.0	66.0	84.0

FIGURE 35–11 Comparison of ventilation/blood flow relationships in health and disease. Left: "Ideal" ventilation/blood flow relationship. **Right:** Nonuniform ventilation and uniform blood flow, uncompensated. $\dot{V}_A$, alveolar ventilation; MV, respiratory minute volume. See text for details. (Reproduced with permission from Comroe JH Jr, et al: *The Lung: Clinical Physiology and*

Pulmonary Function Tests, 2nd ed. Year Book; 1962.)

OTHER FORMS OF HYPOXIA

ANEMIC HYPOXIA

Hypoxia due to anemia is not severe at rest unless the hemoglobin deficiency is marked, because 2,3-DPG increases in the red blood cells. However, anemic patients may have considerable difficulty during exercise because of a limited ability to increase O_2 delivery to the active tissues **(Figure 35–12)**.

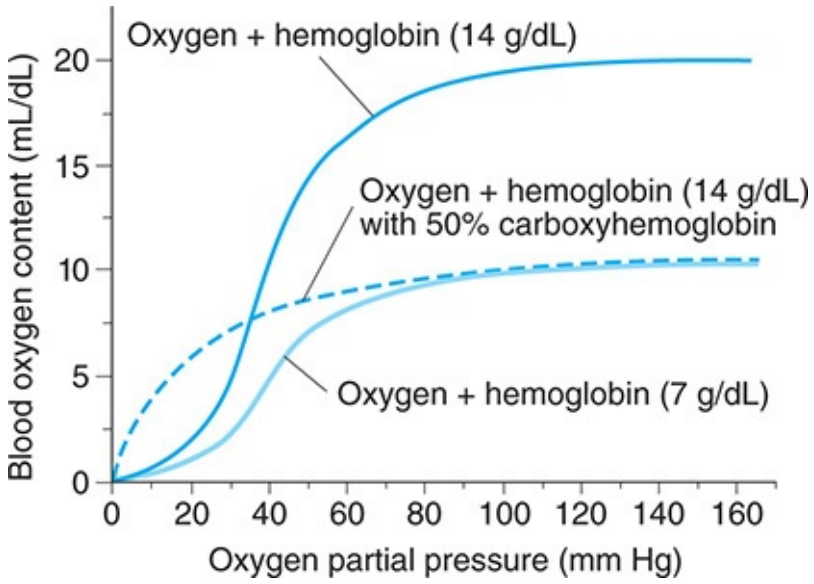


FIGURE 35–12 Effects of anemia and CO on hemoglobin binding of O_2. Normal oxyhemoglobin (14 g/dL hemoglobin) dissociation curve compared with anemia (7 g/dL hemoglobin) and with oxyhemoglobin dissociation curves in CO poisoning (50% arboxy-hemoglobin). Note that the CO-poisoning curve is shifted to the left of the anemia curve. (Reproduced with permission from Leff AR, Schumacker PT: *Respiratory Physiology: Basics and Applications*. Saunders; 1993.)

CARBON MONOXIDE POISONING

Small amounts of carbon monoxide (CO) are formed in the body, and this gas may function as a chemical messenger in the brain and elsewhere. In larger amounts, it is poisonous. Outside the body, it is formed by incomplete

combustion of carbon. It was used by the Greeks and Romans to execute criminals, and today it causes more deaths than any other gas. CO poisoning has become less common in the United States, since natural gas replaced other gases such as coal gas, which contain large amounts of CO. CO is, however, still readily available, as the exhaust of gasoline engines is 6% or more CO.

CO is toxic because it reacts with hemoglobin to form **carbon monoxyhemoglobin (carboxyhemoglobin, COHb),** and COHb does not take up O_2 (Figure 35–12). CO poisoning is often listed as a form of anemic hypoxia because the amount of hemoglobin that can carry O_2 is reduced, but the total hemoglobin content of the blood is unaffected by CO. The affinity of hemoglobin for CO is 210 times its affinity for O_2, and COHb liberates CO very slowly. An additional difficulty is that when COHb is present, the dissociation curve of the remaining HbO_2 shifts to the left, decreasing the amount of O_2 released. This is why an anemic individual who has 50% of the normal amount of HbO_2 may be able to perform moderate work, whereas an individual with HbO_2 reduced to the same level because of the formation of COHb is seriously incapacitated.

Because of the affinity of CO for hemoglobin, progressive COHb formation occurs when the alveolar P_{CO} is greater than 0.4 mm Hg. However, the amount of COHb formed depends on the duration of exposure to CO as well as the concentration of CO in the inspired air and the alveolar ventilation.

CO is also toxic to the cytochromes in the tissues, but the amount of CO required to poison the cytochromes is 1000 times the lethal dose; tissue toxicity thus plays no role in clinical CO poisoning.

The symptoms of CO poisoning are those of any type of hypoxia, especially headache and nausea, but there is little stimulation of respiration, since in the arterial blood, P_{O_2} remains normal and the carotid and aortic chemoreceptors are not stimulated. The cherry-red color of COHb is visible in the skin, nail beds, and mucous membranes. Death results when about 70–80% of the circulating hemoglobin is converted to COHb. The symptoms produced by chronic exposure to sublethal concentrations of CO are those of progressive brain damage, including mental changes and, sometimes, a parkinsonism-like state.

Treatment of CO poisoning consists of immediate termination of the exposure and adequate ventilation, by artificial respiration if necessary. Ventilation with O_2 is preferable to ventilation with fresh air, since O_2 hastens the dissociation of COHb. Hyperbaric oxygenation (see below) is useful in this condition.

ISCHEMIC HYPOXIA

Ischemic hypoxia, or stagnant hypoxia, is due to slow circulation and is a problem in organs such as the kidneys and heart during shock. The liver and possibly the brain are damaged by ischemic hypoxia in heart failure. The blood flow to the lung is normally very large, and it takes prolonged hypotension to produce significant damage. However, acute respiratory distress syndrome (ARDS) can develop when there is prolonged circulatory collapse.

HISTOTOXIC HYPOXIA

Hypoxia due to inhibition of tissue oxidative processes is most commonly the result of cyanide poisoning. Cyanide inhibits cytochrome oxidase and possibly other enzymes. Methylene blue or nitrites are used to treat cyanide poisoning. They act by forming **methemoglobin,** which then reacts with cyanide to form **cyanmethemoglobin,** a nontoxic compound. The extent of treatment with these compounds is, of course, limited by the amount of methemoglobin that can be safely formed. Hyperbaric oxygenation may also be useful.

OXYGEN TREATMENT OF HYPOXIA

Administration of oxygen-rich gas mixtures is of very limited value in hypoperfusion, anemic, and histotoxic hypoxia because all that can be accomplished in this way is an increase in the amount of dissolved O_2 in the arterial blood. This is also true in hypoxemia when it is due to shunting of unoxygenated venous blood past the lungs. In other forms of hypoxemia, O_2 is of great benefit. Treatment regimens that deliver less than 100% O_2 are of value both acutely and chronically, and administration of O_2 24 h/day for 2 years in this manner has been shown to significantly decrease the mortality of chronic obstructive pulmonary disease. O_2 toxicity and therapy are discussed in **Clinical Box 35–5**.

HYPERCAPNIA & HYPOCAPNIA

HYPERCAPNIA

Retention of CO_2 in the body **(hypercapnia)** initially stimulates respiration. Retention of larger amounts produces such symptoms as confusion, diminished sensory acuity and, eventually, coma with respiratory depression and death due to depression of the central nervous system. In patients with these symptoms, the P_{CO_2} is markedly elevated and severe respiratory acidosis is present. Large amounts of HCO_3^- are excreted, but more HCO_3^- is reabsorbed, raising the plasma HCO_3^- and partially compensating for the acidosis.

CO_2 is so much more soluble than O_2 that hypercapnia is rarely a problem in patients with pulmonary fibrosis. However, it does occur in ventilation–perfusion inequality and when for any reason alveolar ventilation is inadequate in the various forms of pump failure. It is exacerbated when CO_2 production is increased. For example, in febrile patients there is a 13% increase in CO_2 production for each 1°C rise in temperature, and a high carbohydrate intake increases CO_2 production because of the increase in the respiratory quotient. Normally, alveolar ventilation increases and the extra CO_2 is expired, but it accumulates when ventilation is compromised.

HYPOCAPNIA

Hypocapnia is the result of hyperventilation. During voluntary hyperventilation, the arterial P_{CO_2} falls from 40 mm Hg to as low as 15 mm Hg while the alveolar P_{O_2} rises to 120–140 mm Hg.

The more chronic effects of hypocapnia are seen in neurotic patients who chronically hyperventilate. Cerebral blood flow may be reduced 30% or more because of the direct constrictor effect of hypocapnia on the cerebral vessels. The cerebral ischemia causes light-headedness, dizziness, and paresthesias. Hypocapnia also increases cardiac output. It has a direct constrictor effect on many peripheral vessels, but it depresses the vasomotor center, so that the blood pressure is usually unchanged or only slightly elevated.

Other consequences of hypocapnia are due to the associated respiratory alkalosis, the blood pH being increased to 7.5 or 7.6. The plasma HCO_3^- level is low, but HCO_3^- reabsorption is decreased because of the inhibition of renal acid secretion by the low P_{CO_2}. The plasma total calcium level does not change, but the plasma Ca^{2+} level falls and hypocapnic individuals develop carpopedal spasm, a positive Chvostek sign, and other signs of tetany.

CLINICAL BOX 35–5

Administration of Oxygen & Its Potential Toxicity

It is interesting that while O_2 is necessary for life in aerobic organisms, it is also toxic. Indeed, 100% O_2 has been demonstrated to exert toxic effects not only in animals but also in bacteria, fungi, cultured animal cells, and plants. The toxicity seems to be due to the production of reactive oxygen species including superoxide anion (O_2^-) and H_2O_2. When 80–100% O_2 is administered to humans for periods of 8 h or more, the respiratory passages become irritated, causing substernal distress, nasal congestion, sore throat, and coughing.

A chronic condition characterized by lung cysts and densities **(bronchopulmonary dysplasia)** develops in some infants treated with O_2 for respiratory distress syndrome. This syndrome may be a manifestation of O_2 toxicity. Another complication in these infants is **retinopathy of prematurity (retrolental fibroplasia),** the formation of opaque vascular tissue in the eyes, which can lead to serious visual defects. The retinal receptors mature from the center to the periphery of the retina, and they use considerable O_2. This causes the retina to become vascularized in an orderly manner. Oxygen treatment before maturation is complete provides the needed O_2 to the photoreceptors, and consequently the normal vascular pattern fails to develop. Evidence indicates that this condition can be prevented or ameliorated by treatment with vitamin E, which exerts an anti-oxidant effect and, in animals, by growth hormone inhibitors.

Administration of 100% O_2 at increased pressure accelerates the onset of O_2 toxicity, with the production not only of tracheobronchial irritation but also of muscle twitching, ringing in the ears, dizziness, convulsions, and coma. The speed with which these symptoms develop is proportional to the pressure at which the O_2 is administered; for example, at 4 atm, symptoms develop in 30 min in half the patients, whereas at 6 atm, convulsions develop in a few minutes.

On the other hand, exposure to 100% O_2 at 2–3 atm can increase dissolved O_2 in arterial blood to the point that arterial O_2 tension is greater than 2000 mm Hg and tissue O_2 tension is 400 mm Hg. If exposure is limited to 5 h or

less at these pressures, O_2 toxicity is not a problem. Therefore, **hyperbaric O_2** therapy in closed tanks is used to treat diseases in which improved oxygenation of tissues cannot be achieved in other ways. It is of demonstrated value in carbon monoxide poisoning, radiation-induced tissue injury, gas gangrene, very severe blood loss anemia, diabetic leg ulcers, and other wounds that are slow to heal, and rescue of skin flaps and grafts in which the circulation is marginal. It is also the primary treatment for decompression sickness and air embolism.

In hypercapnic patients in severe pulmonary failure, the CO_2 level may be so high that it depresses rather than stimulates respiration. Some of these patients keep breathing only because the carotid and aortic chemoreceptors drive the respiratory center. If the hypoxic drive is withdrawn by administering O_2, breathing may stop. During the resultant apnea, the arterial P_{O_2} drops but breathing may not start again, as P_{CO_2} further depresses the respiratory center. Therefore, O_2 therapy in this situation must be started with care.

CHAPTER SUMMARY

- Partial pressure differences between air and blood for O_2 and CO_2 dictate a net flow of O_2 into the blood and CO_2 out of the blood in the pulmonary system.
- The amount of O_2 in the blood is determined by the amount dissolved (minor) and the amount bound (major) to hemoglobin. Each hemoglobin molecule contains four subunits that each can bind O_2. Hemoglobin O_2 binding is cooperative and also affected by pH, temperature, and the concentration of 2,3-diphosphoglycerate (2,3-DPG).
- CO_2 in blood is rapidly converted into H_2CO_3 due to the activity of carbonic anhydrase. CO_2 also readily forms carbamino compounds with blood proteins (including hemoglobin). The rapid net loss of CO_2 allows more CO_2 to dissolve in blood.
- The pH of plasma is 7.4. A decrease in plasma pH is termed **acidosis** and an increase of plasma pH is termed **alkalosis.** A short-term change in arterial P_{CO_2} due to decreased ventilation results in respiratory acidosis. A short-

term change in arterial P_{CO_2} due to increased ventilation results in respiratory alkalosis. Metabolic acidosis occurs when strong acids are added to the blood, and metabolic alkalosis occurs when strong bases are added to (or strong acids are removed from) the blood.

- Respiratory compensation to acidosis or alkalosis involves quick changes in ventilation. Such changes effectively change the P_{CO_2} in the blood plasma. Renal compensation mechanisms are much slower and involve H^+ secretion or HCO_3^- reabsorption.
- Hypoxia is a deficiency of O_2 at the tissue level. Hypoxia has powerful consequences at the cellular, tissue, and organ level: It can alter cellular transcription factors and thus protein expression; it can quickly alter brain function and produce symptoms similar to alcohol (eg, dizziness, impaired mental function, drowsiness, headache); and it can affect ventilation. Long-term hypoxia results in cell and tissue death.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Most of the CO_2 transported in the blood is

   A. dissolved in plasma.

   B. in carbamino compounds formed from plasma proteins.

   C. in carbamino compounds formed from hemoglobin.

   D. bound to Cl^-.

   E. in HCO_3^-.

2. Which of the following has the greatest effect on the ability of blood to transport oxygen?

   A. Capacity of the blood to dissolve oxygen

   B. Amount of hemoglobin in the blood

   C. pH of plasma

   D. CO_2 content of red blood cells

   E. Temperature of the blood

3. Which of the following is true of the system?

$$CO_2 + H_2O \rightleftarrows^{1} H_2CO_3 \rightleftarrows^{2} H^+ + HCO_3^-$$

A. Reaction 2 is catalyzed by carbonic anhydrase.

B. Because of reaction 2, the pH of blood declines during hyperventilation.

C. Reaction 1 occurs in the red blood cell.

D. Reaction 1 occurs primarily in plasma.

E. The reactions move to the right when there is excess H^+ in the tissues.

4. In comparing uncompensated respiratory acidosis and uncompensated metabolic acidosis which one of the following is true?

A. Plasma pH change is always greater in uncompensated respiratory acidosis compared to uncompensated metabolic acidosis.

B. There are no compensation mechanisms for respiratory acidosis, whereas there is respiratory compensation for metabolic acidosis.

C. Uncompensated respiratory acidosis involves changes in plasma [HCO_3^-], whereas plasma [HCO_3^-] is unchanged in uncompensated metabolic acidosis.

D. Uncompensated respiratory acidosis is associated with a change in P_{CO_2}, whereas in uncompensated metabolic acidosis P_{CO_2} is constant.

CHAPTER 36

Regulation of Respiration

OBJECTIVES

After studying this chapter, you should be able to:

- Locate the pre-Bötzinger complex and describe its role in producing spontaneous respiration.
- Identify the location and probable functions of the dorsal and ventral groups of respiratory neurons, the pneumotaxic center, and the apneustic center in the brainstem.
- List the specific respiratory functions of the vagus nerves and the respiratory receptors in the carotid body, the aortic body, and the ventral surface of the medulla oblongata.
- Describe and explain the ventilatory responses to increased CO_2 concentrations in the inspired air.
- Describe and explain the ventilatory responses to decreased O_2 concentrations in the inspired air.
- Describe the effects of each of the main nonchemical factors that influence respiration.
- Describe the effects of exercise on ventilation and O_2 exchange in the tissues.

- Define periodic breathing and explain its occurrence in various disease states.

INTRODUCTION

Spontaneous respiration is produced by rhythmic discharge of motor neurons that innervate the respiratory muscles. This discharge is totally dependent on nerve impulses from the brain; breathing stops if the spinal cord is transected above the origin of the phrenic nerves. The rhythmic discharges from the brain that produce spontaneous respiration are regulated by alterations in arterial P_{O_2}, P_{CO_2}, and H^+ concentration, and this chemical control of breathing is supplemented by a number of nonchemical influences. The overall respiration control system thus consists of a network of neurons in cortex and medulla/pons that exert voluntary control and automatic control, respectively. The system adjusts the rate of ventilation as environmental conditions change (eg, from normoxic to hypoxic condition) and arterial P_{O_2}, P_{CO_2}, and H^+ concentration ($[H^+]$) alter. The mechano- and chemoreceptors in the system sense the changes in force/displacement and arterial levels of gases and metabolites, and adjust the rate of ventilation to ensure optimal gas exchange in the lungs **(Figure 36–1)**. The physiological bases for these phenomena are discussed in this chapter.

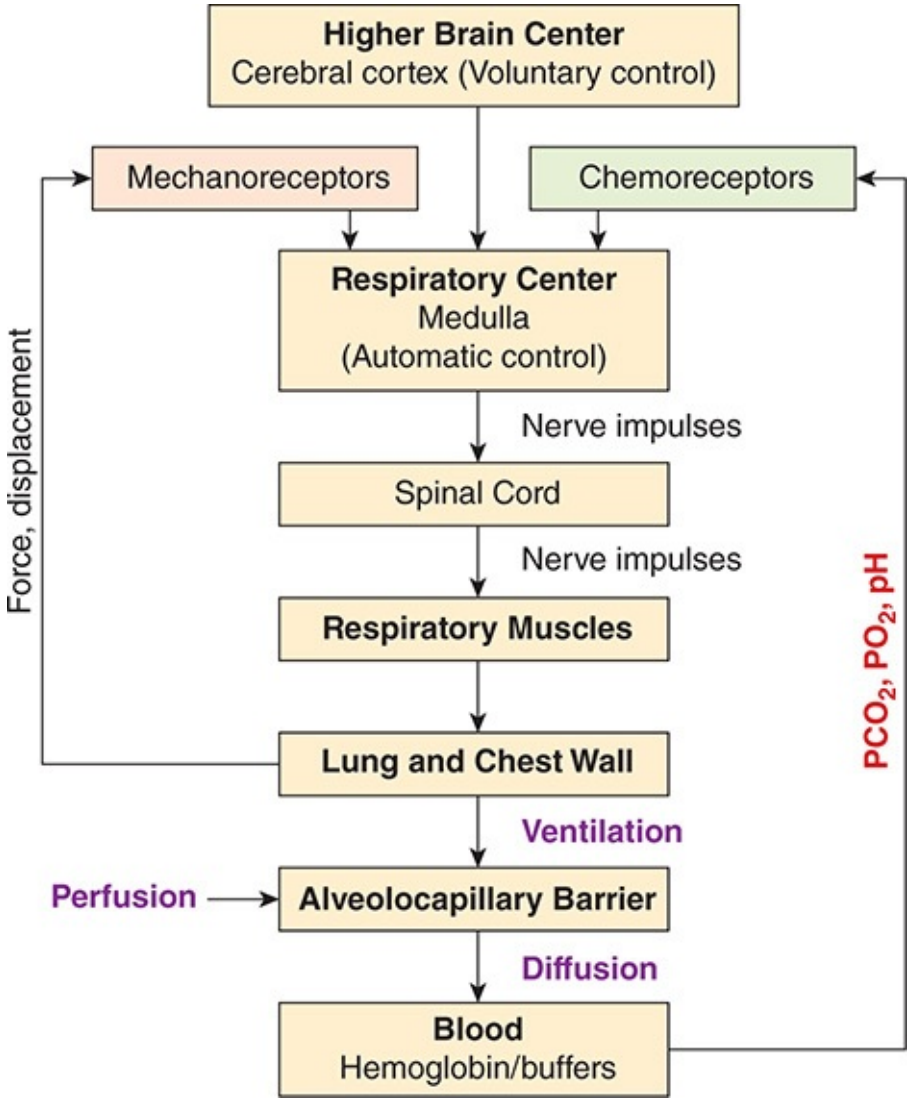


FIGURE 36–1 Diagram of the overall respiratory control system. The volunteer control center is in cerebral cortex and the automatic control centers are in medulla and pons. The nerve impulses or electrical signals from the respiratory neurons in these centers regulate the activity of respiratory muscles (including diaphragm) and thus the rate and depth of breathing (ventilation). The mechanoreceptors and chemoreceptors (central and peripheral) then sense the physical changes in the lungs and chest wall and the chemical changes in the blood (eg, arterial P_{CO_2}, arterial P_{O_2} and $[H^+]$) to further adjust the control of breathing. (Modified from Berne RM, Levy MN (eds): *Physiology*, 2nd ed. Mosby, 1988.)

NEURAL CONTROL OF BREATHING

CONTROL SYSTEMS

Two separate neural mechanisms regulate respiration. One is responsible for **voluntary control** and the other for **automatic control.** The voluntary control system is located in the cerebral cortex and sends impulses to the respiratory motor neurons via the corticospinal tracts. The automatic system is driven by a group of pacemaker cells in the medulla. Impulses from these cells activate motor neurons in the cervical and thoracic spinal cord that innervate inspiratory muscles. Those in the cervical cord activate the diaphragm via the phrenic nerves, and those in the thoracic spinal cord activate the external intercostal muscles. However, the impulses also reach the innervation of the internal intercostal muscles and other expiratory muscles.

The motor neurons to the expiratory muscles are inhibited when those supplying the inspiratory muscles are active, and vice versa. Although spinal reflexes contribute to this **reciprocal innervation,** it is due primarily to activity in descending pathways. Impulses in these descending pathways excite agonists and inhibit antagonists. The one exception to the reciprocal inhibition is a small amount of activity in phrenic axons for a short period after inspiration. The function of this postinspiratory output appears to be to brake the lung's elastic recoil and make respiration smooth.

MEDULLARY SYSTEMS

The main components of the **respiratory control pattern generator** responsible for automatic respiration are located in the medulla. Rhythmic respiration is initiated by a small group of synaptically coupled pacemaker cells in the **pre-Bötzinger complex** (pre-BÖTC) on either side of the medulla between the nucleus ambiguus and the lateral reticular nucleus **(Figure 36–2)**. These neurons discharge rhythmically, and they produce rhythmic discharges in phrenic motor neurons that are abolished by sections between the pre-BÖTC complex and these motor neurons. They also contact the hypoglossal nuclei, and the tongue is involved in the regulation of airway resistance.

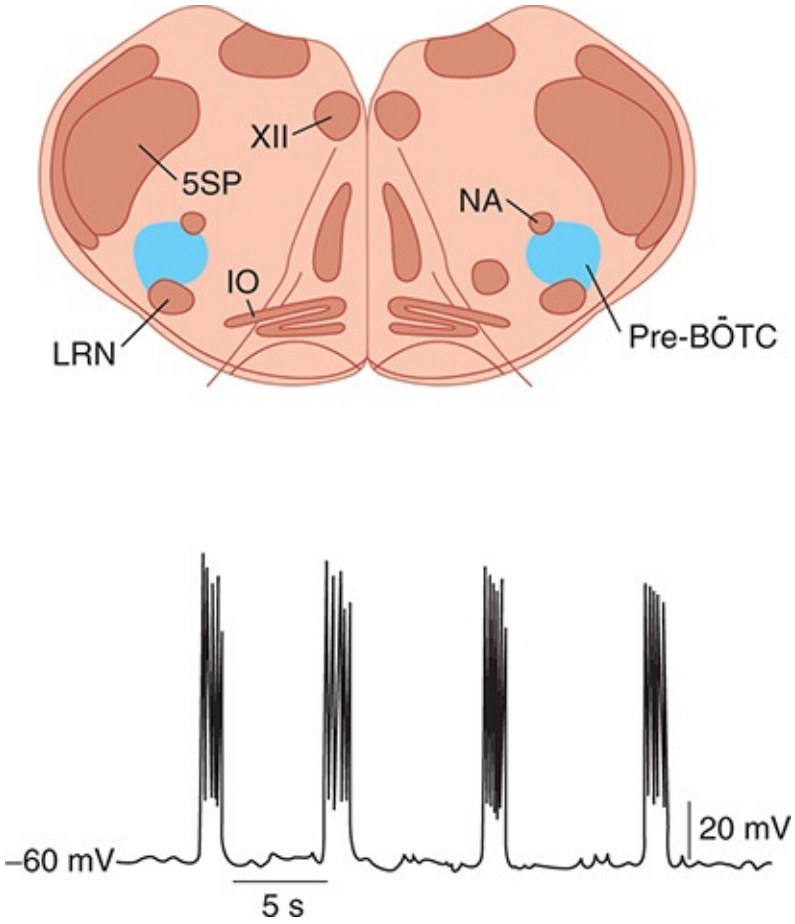


FIGURE 36–2 Pacemaker cells in the pre-Bötzinger complex (pre-BÖTC). **Top:** Anatomic diagram of the pre-BÖTC from a neonatal rat. **Bottom:** Sample rhythmic discharge tracing of neurons in the pre-BÖTC complex from a brain slice of a neonatal rat. IO, inferior olive; LRN, lateral reticular nucleus; NA, nucleus ambiguus; XII, nucleus of 12th cranial nerve; 5SP, spinal nucleus of trigeminal nerve. (Modified with permission from Feldman JC, Gray PA: Sighs and gasps in a dish. Nat Neurosci 2000;3(6):531–532.)

Neurons in the pre-BÖTC complex discharge rhythmically in brain slice preparations in vitro, and if the slices become hypoxic, discharge changes to one associated with gasping. Addition of cadmium to the slices causes occasional sigh-like discharge patterns. There are NK1 receptors and μ-opioid receptors on these neurons and, in vivo, substance P stimulates and opioids inhibit respiration. Depression of respiration is a side effect that limits the use of opioids in the treatment of pain. However, it is now known that the 5-hydroxytryptamine receptor 4 (5-HT_4 receptor) is present in the pre-BÖTC complex and treatment with 5-HT_4 agonists blocks the inhibitory effect of opioids on respiration in

experimental animals, without inhibiting their analgesic effect.

In addition, dorsal and ventral groups of respiratory neurons are present in the medulla **(Figure 36–3)**. However, lesions of these neurons do not abolish respiratory activity, and they apparently project to the pre-BÖTC pacemaker neurons.

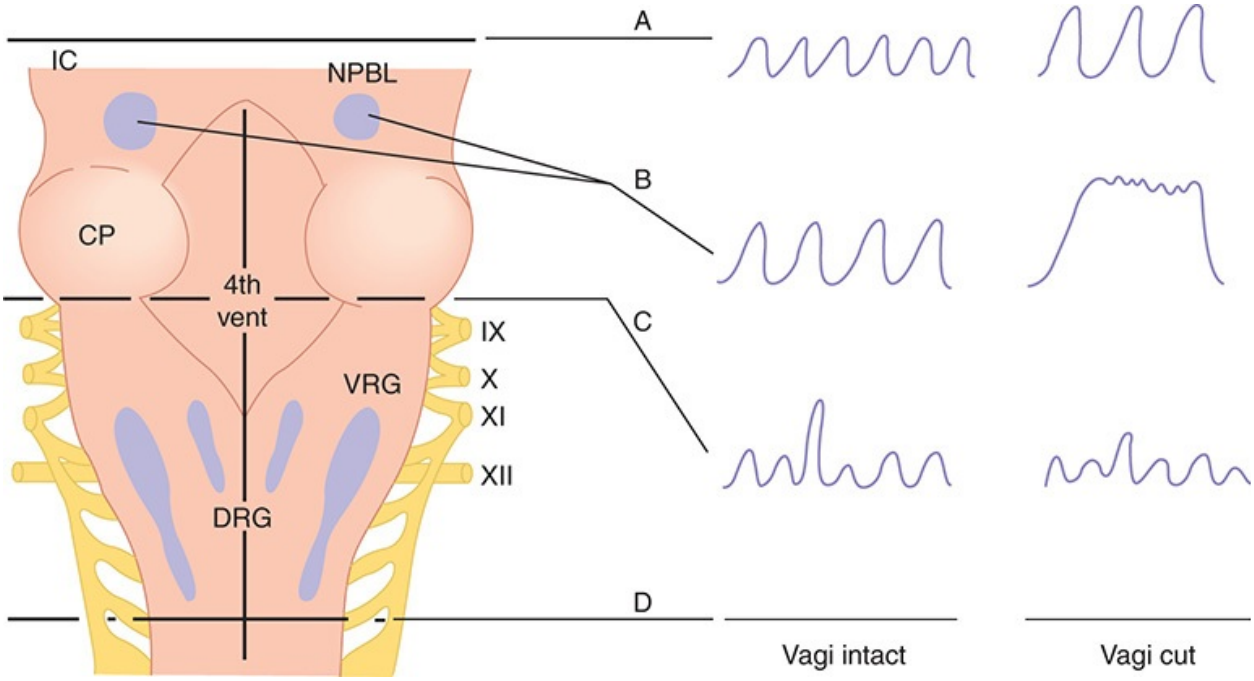


FIGURE 36–3 Respiratory neurons in the brainstem. Dorsal view of brainstem; cerebellum removed. The effects of various lesions and brainstem transections are shown; the spirometer tracings at the right indicate the depth and rate of breathing. If a lesion is introduced at D, breathing ceases. The effects of higher transections, with and without vagus nerves transection, are shown (see text for details). CP, middle cerebellar peduncle; DRG, dorsal group of respiratory neurons; IC, inferior colliculus; NPBL, nucleus parabrachialis (pneumotaxic center); VRG, ventral group of respiratory neurons; 4th vent, fourth ventricle. The roman numerals identify cranial nerves. (Modified with permission from Mitchell RA, Berger A: State of the art: Review of neural regulation of respiration. Am Rev Respir Dis 1975 Feb;111(2):206-224.)

PONTINE & VAGAL INFLUENCES

Although the rhythmic discharge of medullary neurons concerned with respiration is spontaneous, it is modified by neurons in the pons and afferents in

the vagus from receptors in the airways and lungs. An area known as the **pneumotaxic center** in the medial parabrachial and Kölliker-Fuse nuclei of the dorsolateral pons contains neurons active during inspiration and neurons active during expiration. When this area is damaged, respiration becomes slower and tidal volume greater, and when the vagi are also cut in anesthetized animals, there are prolonged inspiratory spasms that resemble breath holding (**apneusis;** section B in Figure 36–3). The normal function of the pneumotaxic center is unknown, but it may play a role in switching between inspiration and expiration.

Stretching of the lungs during inspiration initiates impulses in afferent pulmonary vagal fibers. These impulses inhibit inspiratory discharge. This is why the depth of inspiration is increased after vagotomy (Figure 36–3) and apneusis develops if the vagi are cut after damage to the pneumotaxic center. Vagal feedback activity does not alter the rate of rise of the neural activity in respiratory motor neurons **(Figure 36–4)**.

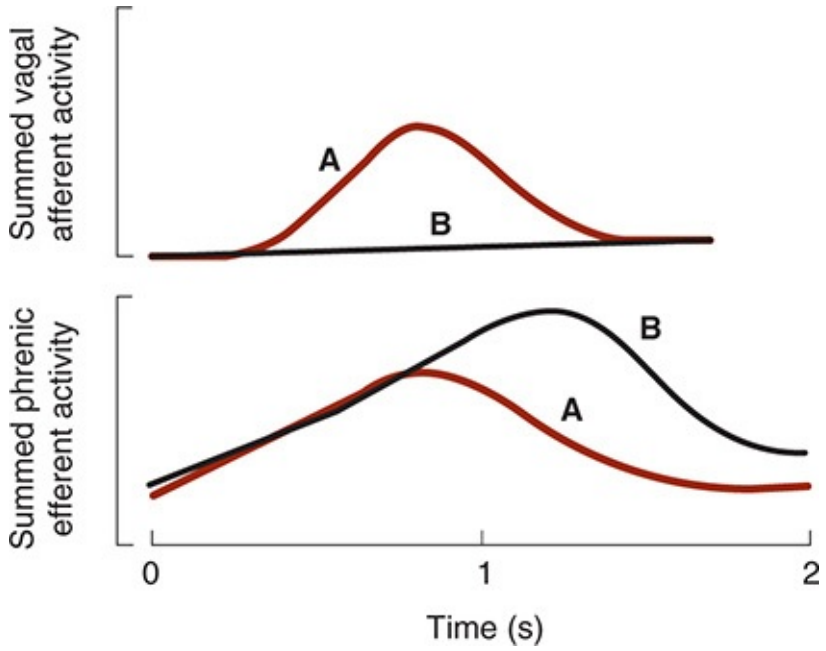


FIGURE 36–4 Afferent vagal fibers inhibit inspiratory discharge. Superimposed records of two breaths: **A)** with and **B)** without feedback vagal afferent activity from stretch receptors in the lungs. Note that the rate of rise in phrenic nerve activity to the diaphragm is unaffected but the discharge is prolonged in the absence of vagal input.

When the activity of the inspiratory neurons is increased in intact animals, the rate and the depth of breathing are increased. The depth of respiration is

increased because the lungs are stretched to a greater degree before the amount of vagal and pneumotaxic center inhibitory activity is sufficient to overcome the more intense inspiratory neuron discharge. The respiratory rate is increased because the after-discharge in the vagal and possibly the pneumotaxic afferents to the medulla is rapidly overcome.

REGULATION OF RESPIRATORY ACTIVITY

A rise in the P_{CO_2} or H^+ concentration of arterial blood or a drop in its P_{O_2} increases the level of respiratory neuron activity in the medulla, and changes in the opposite direction have a slight inhibitory effect. The effects of variations in blood chemistry on ventilation are mediated via respiratory **chemoreceptors**—the carotid and aortic bodies and collections of cells in the medulla and elsewhere that are sensitive to changes in the chemistry of the blood. They initiate impulses that stimulate the respiratory center. Superimposed on this basic **chemical control of respiration,** other afferents provide nonchemical controls that affect breathing in particular situations **(Table 36–1)**.

TABLE 36–1 Stimuli affecting the respiratory center.

Chemical control
CO_2 (via CSF and brain interstitial fluid H^+ concentration)
O_2, H^+ } (via carotid and aortic bodies)
Nonchemical control
Vagal afferents from receptors in the airways and lungs
Afferents from the pons, hypothalamus, and limbic system
Afferents from proprioceptors
Afferents from baroreceptors: arterial, atrial, ventricular, pulmonary

CHEMICAL CONTROL OF BREATHING

The chemical regulatory mechanisms adjust ventilation in such a way that the alveolar P_{CO_2} is normally held constant, the effects of excess H^+ in the blood are combated, and the P_{O_2} is raised when it falls to a potentially dangerous level. The respiratory minute volume is proportional to the metabolic rate, but the link

between metabolism and ventilation is CO_2, not O_2. The receptors in the carotid and aortic bodies are stimulated by a rise in the P_{CO_2} or H^+ concentration of arterial blood or a decline in its P_{O_2}. After denervation of the carotid chemoreceptors, the response to a drop in P_{O_2} is abolished; the predominant effect of hypoxia after denervation of the carotid bodies is a direct depression of the respiratory center. The response to changes in arterial blood H^+ concentration in the pH 7.3–7.5 range is also abolished, although larger changes exert some effect. The response to changes in arterial P_{CO_2}, on the other hand, is affected only slightly; it is reduced no more than 30–35%.

CAROTID & AORTIC BODIES

There is a carotid body near the carotid bifurcation on each side, and there are usually two or more aortic bodies near the arch of the aorta **(Figure 36–5)**. Each carotid and aortic body **(glomus)** contains islands of two types of cells, type I and type II cells, surrounded by fenestrated sinusoidal capillaries. The type I or **glomus cells** are closely associated with cuplike endings of the afferent nerves **(Figure 36–6)**. The glomus cells resemble adrenal chromaffin cells and have dense-core granules containing catecholamines that are released upon exposure to hypoxia and cyanide. The cells are excited by hypoxia (potentially by hypoxia-induced inhibition of O_2-sensitive K^+ channels), and the principal transmitter appears to be dopamine, which excites the nerve endings by way of D_2 receptors. The type II cells are glia-like, and each surrounds four to six type I cells. The function of type II cells is not fully defined.

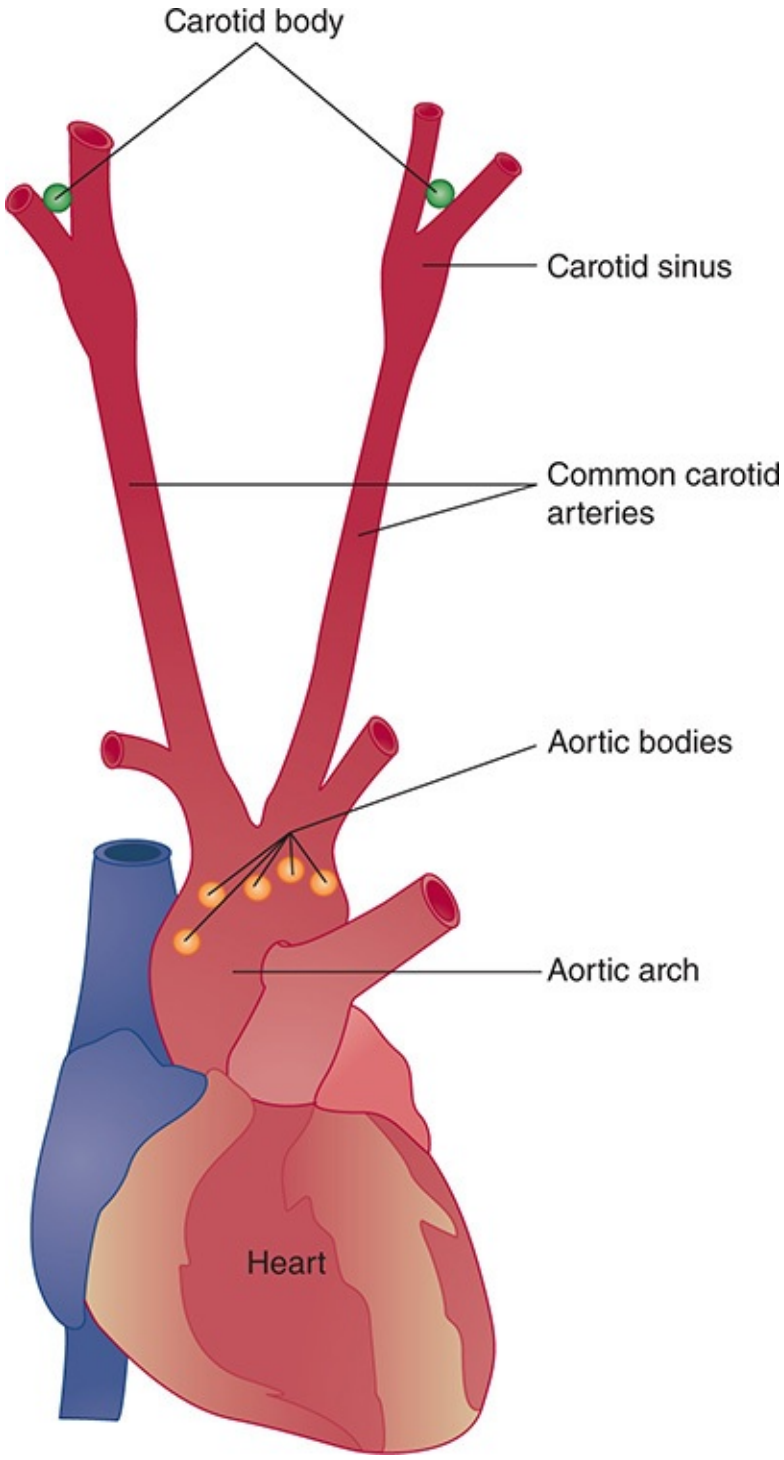


FIGURE 36–5 Location of carotid and aortic bodies. Carotid bodies are positioned near a major arterial baroreceptor, the carotid sinus. Aortic bodies are shown near the aortic arch.

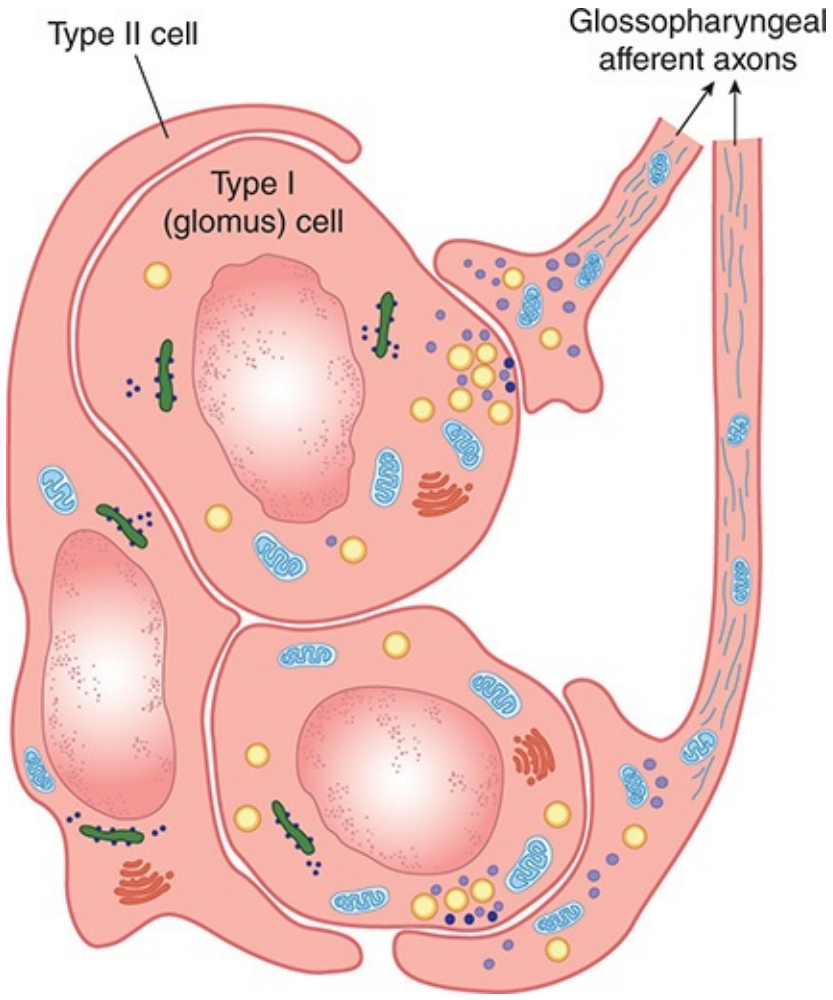


FIGURE 36–6 Organization of the carotid body. Type I (glomus) cells contain catecholamines. When exposed to hypoxia, they release their catecholamines, which stimulate the cuplike endings of the carotid sinus nerve fibers in the glossopharyngeal nerve. The glia-like type II cells surround the type I cells and probably have a sustentacular function.

Outside the capsule of each body, the nerve fibers acquire a myelin sheath; however, they are only 2–5 µm in diameter and conduct at the relatively low rate of 7–12 m/s. Afferents from the carotid bodies ascend to the medulla via the carotid sinus and glossopharyngeal nerves, and fibers from the aortic bodies ascend in the vagi. Studies in which one carotid body has been isolated and perfused while recordings are being taken from its afferent nerve fibers show that there is a graded increase in impulse traffic in these afferent fibers as the P_{O_2} of the perfusing blood is lowered **(Figure 36–7)** or the P_{CO_2} is raised.

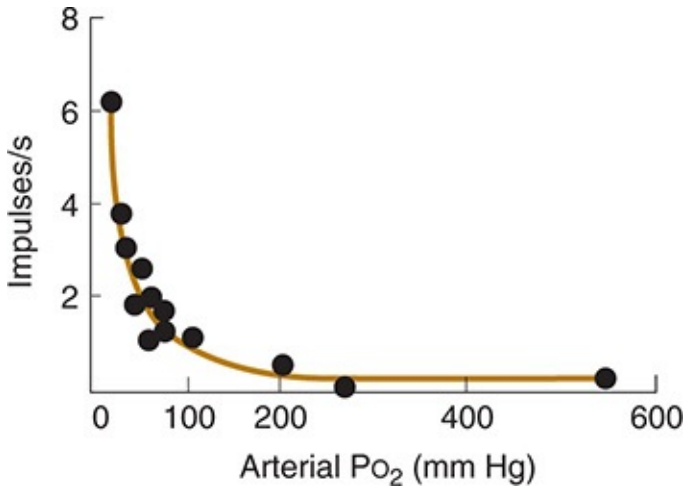


FIGURE 36–7 Effect of P_{CO_2} on afferent nerve firing. The rate of discharge of a single afferent fiber from the carotid body (circles) is plotted at several P_{O_2} values and fitted to a line. A sharp increase in firing rate is observed as P_{O_2} falls below normal resting levels (ie, near 100 mm Hg). (Used with permission of S. Sampson.)

Type I glomus cells have O_2-sensitive K^+ channels, whose conductance is reduced in proportion to the degree of hypoxia to which they are exposed. This reduces the K^+ efflux, depolarizing the cell and causing Ca^{2+} influx, primarily via L-type voltage-gated Ca^{2+} channels. The Ca^{2+} influx triggers action potentials and transmitter release, with consequent excitation of the afferent nerve endings. The smooth muscle cells of pulmonary arteries contain similar O_2-sensitive K^+ channels, which mediate the vasoconstriction caused by hypoxia. This is in contrast to systemic arteries, which contain ATP-sensitive K^+ channels that permit more K^+ efflux with hypoxia and consequently cause vasodilation instead of vasoconstriction.

The blood flow in each 2 mg carotid body is about 0.04 mL/min, or 2000 mL/100 g of tissue/min compared with a blood flow of 54 mL or 420 mL per 100 g/min in the brain and kidneys, respectively. Because the blood flow per unit of tissue is so enormous, the O_2 needs of the cells can be met largely by dissolved O_2 alone. Therefore, the receptors are not stimulated in conditions such as anemia or carbon monoxide poisoning, in which the amount of dissolved O_2 in the blood reaching the receptors is generally normal, even though the combined O_2 in the blood is markedly decreased. The receptors are stimulated when the arterial P_{O_2} is low or when, because of vascular stasis, the amount of

O_2 delivered to the receptors per unit time is decreased. Powerful stimulation is also produced by cyanide, which prevents O_2 utilization at the tissue level. In sufficient doses, nicotine and lobeline activate the chemoreceptors. It has also been reported that infusion of K^+ increases the discharge rate in chemoreceptor afferents, and because the plasma K^+ level is increased during exercise, the increase may contribute to exercise-induced hyperpnea.

Because of their anatomic location, the aortic bodies have not been studied in as detail as the carotid bodies. Their responses are probably similar but of lesser magnitude. In humans in whom both carotid bodies have been removed but the aortic bodies left intact, the responses are essentially the same as those following denervation of both carotid and aortic bodies in animals: little change in ventilation at rest, but the ventilatory response to hypoxia is lost and the ventilatory response to CO_2 is reduced by 30%.

Neuroepithelial bodies composed of innervated clusters of amine-containing cells are found in the airways. These cells have an outward K^+ current that is reduced by hypoxia, and this would be expected to produce depolarization. However, the function of these hypoxia-sensitive cells is uncertain because, as noted above, removal of the carotid bodies alone abolishes the respiratory response to hypoxia.

CHEMORECEPTORS IN THE BRAINSTEM

The chemoreceptors that mediate the hyperventilation produced by increases in arterial P_{CO_2} after the carotid and aortic bodies are denervated are located in the medulla oblongata and consequently are called **medullary chemoreceptors.** They are separate from the dorsal and ventral respiratory neurons and are located on the ventral surface of the medulla **(Figure 36–8)**. Recent evidence indicates that additional chemoreceptors are located in the vicinity of the solitary tract nuclei, the locus coeruleus, and the hypothalamus.

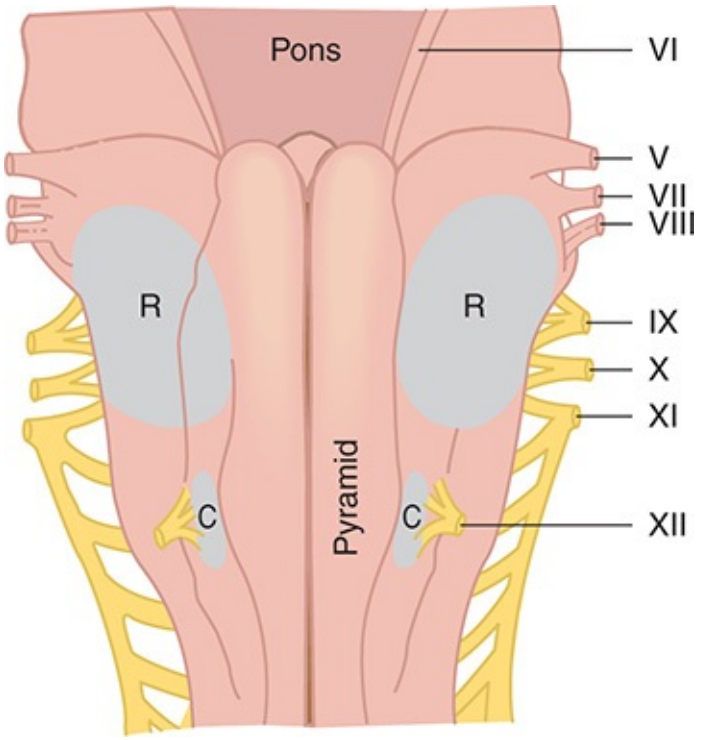


FIGURE 36–8 Rostral (R) and caudal (C) chemosensitive areas on the ventral surface of the medulla. Cranial nerves, pyramid, and pons are labeled for reference.

The chemoreceptors monitor the H^+ concentration of cerebrospinal fluid (CSF), including the brain interstitial fluid. CO_2 readily penetrates membranes, including the blood-brain barrier, whereas H^+ and HCO_3^- penetrate slowly. The CO_2 that enters the brain and CSF is promptly hydrated. The H_2CO_3 dissociates, so that the local H^+ concentration rises. The H^+ concentration in brain interstitial fluid parallels the arterial P_{CO_2}. Experimentally produced changes in the P_{CO_2} of CSF have minor, variable effects on respiration as long as the H^+ concentration is held constant, but any increase in spinal fluid H^+ concentration stimulates respiration. The magnitude of the stimulation is proportional to the rise in H^+ concentration. Thus, the effects of CO_2 on respiration are mainly due to its movement into the CSF and brain interstitial fluid, where it increases the H^+ concentration and stimulates receptors sensitive to H^+.

VENTILATORY RESPONSES TO CHANGES IN

ACID–BASE BALANCE

In metabolic acidosis due, for example, to the accumulation of acid ketone bodies in the circulation in diabetes mellitus, there is pronounced respiratory stimulation (Kussmaul breathing). The hyperventilation decreases alveolar P_{CO_2} ("blows off CO_2") and thus produces a compensatory fall in blood H^+ concentration. Conversely, in metabolic alkalosis due, for example, to protracted vomiting with loss of HCl from the body, ventilation is depressed and the arterial P_{CO_2} rises, raising the H^+ concentration toward normal. If there is an increase in ventilation that is not secondary to a rise in arterial H^+ concentration, the drop in P_{CO_2} lowers the H^+ concentration below normal **(respiratory alkalosis);** conversely, hypoventilation that is not secondary to a fall in plasma H^+ concentration causes **respiratory acidosis.**

VENTILATORY RESPONSES TO CO_2

The arterial P_{CO_2} is normally maintained at 40 mm Hg. When arterial P_{CO_2} rises as a result of increased tissue metabolism, ventilation is stimulated and the rate of pulmonary excretion of CO_2 increases until the arterial P_{CO_2} falls to normal, shutting off the stimulus. The operation of this feedback mechanism keeps CO_2 excretion and production in balance.

When a gas mixture containing CO_2 is inhaled, the alveolar P_{CO_2} rises, elevating the arterial P_{CO_2} and stimulating ventilation as soon as the blood that contains more CO_2 reaches the medulla. CO_2 elimination is increased, and the alveolar P_{CO_2} drops toward normal. This is why relatively large increments in the P_{CO_2} of inspired air (eg, 15 mm Hg) produce relatively slight increments in alveolar P_{CO_2} (eg, 3 mm Hg). However, the P_{CO_2} does not drop to normal, and a new equilibrium is reached at which the alveolar P_{CO_2} is slightly elevated and the hyperventilation persists as long as CO_2 is inhaled. The essentially linear relationship between respiratory minute volume and the alveolar P_{CO_2} is shown in **Figure 36–9**.

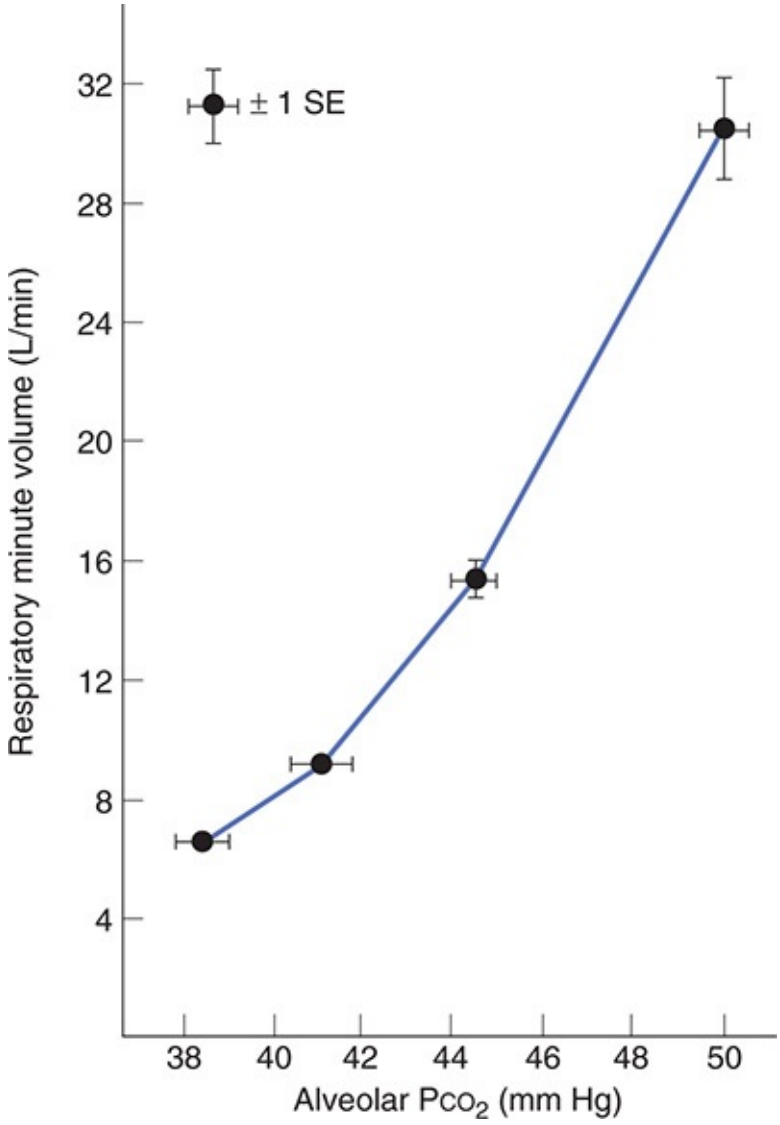


FIGURE 36–9 Responses of normal subjects to inhaling O_2 and approximately 2, 4, and 6% CO_2. The relatively linear increase in respiratory minute volume in response to increased CO_2 is due to an increase in both the depth and rate of respiration. (Reproduced with permission from Mountcastle VB (ed): *Medical Physiology*, 13th ed. Mosby; 1974.)

Of course, this linearity has an upper limit. When the P_{CO_2} of the inspired gas is close to the alveolar P_{CO_2}, elimination of CO_2 becomes difficult. When the CO_2 content of the inspired gas is more than 7%, the alveolar and arterial P_{CO_2} begin to rise abruptly in spite of hyperventilation. The resultant accumulation of CO_2 in the body **(hypercapnia)** depresses the central nervous system, including

the respiratory center, and produces headache, confusion, and eventually coma **(CO_2 narcosis).**

VENTILATORY RESPONSE TO OXYGEN DEFICIENCY

When the O_2 content of the inspired air is decreased, respiratory minute volume is increased. The stimulation is slight when the P_{O_2} of the inspired air is more than 60 mm Hg, and marked stimulation of respiration occurs only at lower P_{O_2} values **(Figure 36–10)**. However, any decline in arterial P_{O_2} below 100 mm Hg produces increased discharge in the nerves from the carotid and aortic chemoreceptors. There are two reasons why this increase in impulse traffic does not increase ventilation to any extent in normal individuals until the P_{O_2} is less than 60 mm Hg. First, because Hb is a weaker acid than HbO_2, there is a slight decrease in the H^+ concentration of arterial blood when the arterial P_{O_2} falls and hemoglobin becomes less saturated with O_2. The fall in H^+ concentration tends to inhibit respiration. In addition, any increase in ventilation that does occur lowers the alveolar P_{CO_2}, and this also tends to inhibit respiration. Therefore, the stimulatory effects of hypoxia on ventilation are not clearly manifest until they become strong enough to override the counterbalancing inhibitory effects of a decline in arterial H^+ concentration and P_{CO_2}.

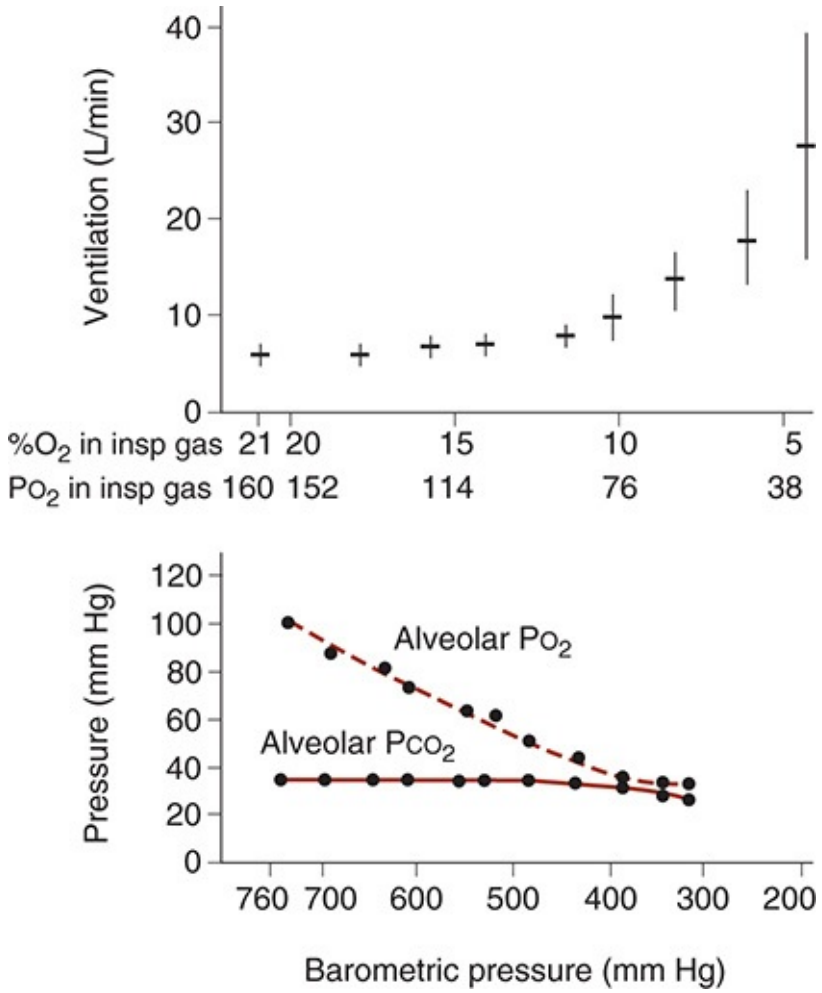


FIGURE 36–10 Effects of changes in inspired Po_2 on ventilation and alveolar gases. Top: Average respiratory minute volume during the first half hour of exposure to gases containing various amounts of O_2. Marked changes in ventilation occur at P_{O_2} values lower than 60 mm Hg. The horizontal line in each case indicates the mean; the vertical bar indicates one standard deviation. **Bottom:** Alveolar P_{O_2} and P_{CO_2} values when breathing air at various barometric pressures. The two graphs are aligned so that the P_{O_2} of the inspired gas mixtures in the upper graph correspond to the P_{O_2} at the various barometric pressures in the lower graph. (Used with permission of RH Kellogg.)

The effects on ventilation of decreasing the alveolar P_{O_2} while holding the alveolar P_{CO_2} constant are shown in **Figure 36–11**. When the alveolar P_{CO_2} is stabilized at a level 2–3 mm Hg above normal, there is an inverse relationship between ventilation and the alveolar P_{O_2} even in the 90– 110 mm Hg range; but

when the alveolar P_{CO_2} is fixed at lower than normal values, there is no stimulation of ventilation by hypoxia until the alveolar P_{O_2} falls below 60 mm Hg.

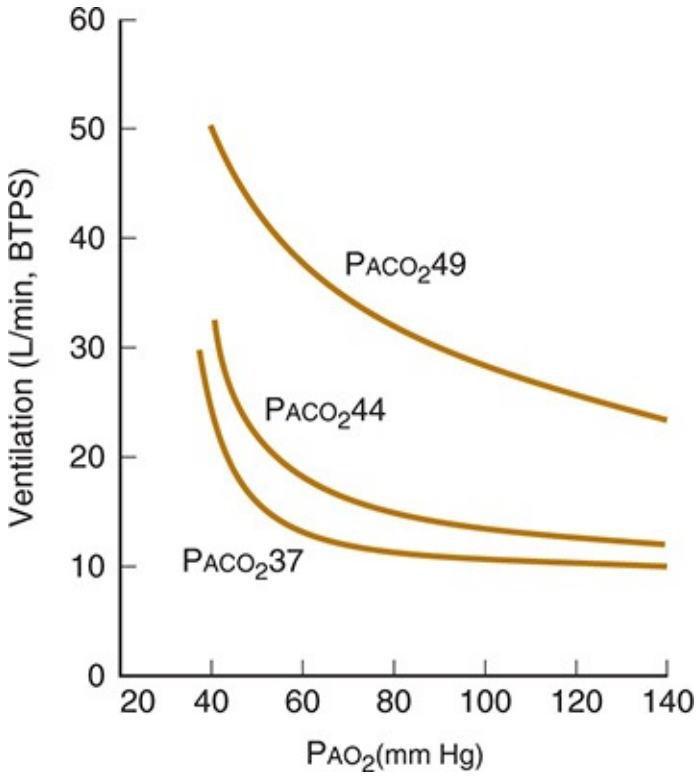


FIGURE 36–11 Ventilation at various alveolar Po_2 values when Pco_2 is held constant. Note the dramatic effect on the ventilatory response to P_{AO_2} when P_{ACO_2} is increased. (Data from Loeschke HH and Gertz KH.)

EFFECTS OF HYPOXIA ON THE CO_2 RESPONSE CURVE

When the converse experiment is performed—that is, when the alveolar P_{O_2} is held constant while the response to varying amounts of inspired CO_2 is tested—a linear response is obtained **(Figure 36–12)**. When the CO_2 response is tested at different fixed P_{O_2} values, the slope of the response curve changes, with the slope increased when alveolar P_{O_2} is decreased. In other words, hypoxia makes the individual more sensitive to an increase in arterial P_{CO_2}. However, the

alveolar P_{CO_2} level at which the curves in Figure 36–12 intersect is unaffected. In the normal individual, this threshold value is just below the normal alveolar P_{CO_2}, indicating that normally there is a very slight but definite "CO_2 drive" of the respiratory area.

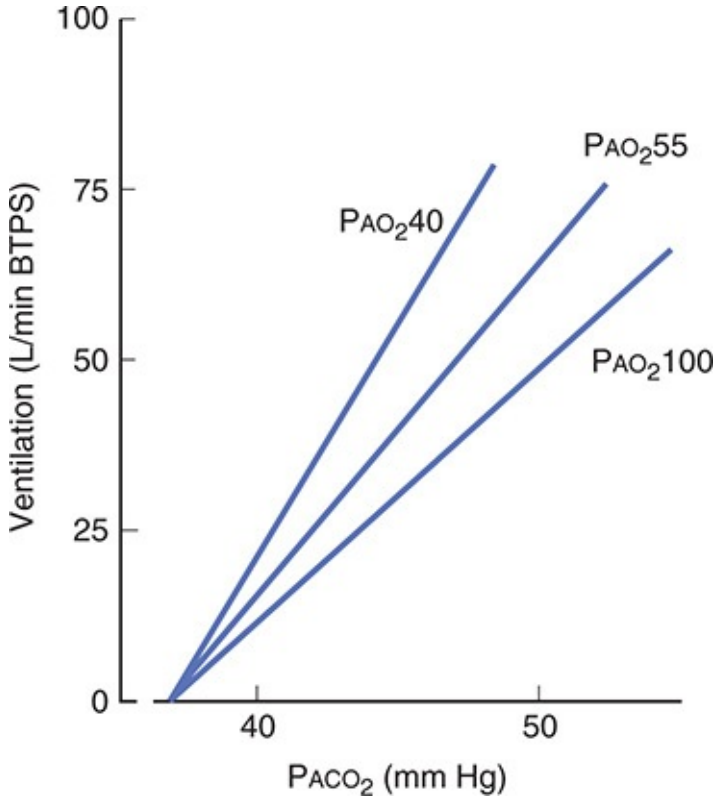


FIGURE 36–12 Fan of lines showing CO_2 response curves at various fixed values of alveolar Po_2. Decreased P_{AO_2} results in a more sensitive response to P_{ACO_2}.

EFFECT OF H^+ ON THE CO_2 RESPONSE

The stimulatory effects of H^+ and CO_2 on respiration appear to be additive and not, like those of CO_2 and O_2, complexly interrelated. In metabolic acidosis, the CO_2 response curves are similar to those in Figure 36–12, except that they are shifted to the left. In other words, the same amount of respiratory stimulation is produced by lower arterial P_{CO_2} levels. It has been calculated that the CO_2 response curve shifts 0.8 mm Hg to the left for each nanomole rise in arterial H^+.

About 40% of the ventilatory response to CO_2 is removed if the increase in arterial H^+ produced by CO_2 is prevented. As noted above, the remaining 60% is probably due to the effect of CO_2 on spinal fluid or brain interstitial fluid H^+ concentration.

BREATH HOLDING

Respiration can be voluntarily inhibited for some time, but eventually the voluntary control is overridden. The point at which breathing can no longer be voluntarily inhibited is called the **breaking point.** Breaking is due to the rise in arterial P_{CO_2} and the fall in P_{O_2}. Individuals can hold their breath longer after removal of the carotid bodies. Breathing 100% oxygen before breath holding raises alveolar P_{O_2} initially, so that the breaking point is delayed. The same is true of hyperventilating room air, because CO_2 is blown off and arterial P_{CO_2} is lower at the start. Reflex or mechanical factors appear to influence the breaking point, since persons who hold their breath as long as possible and then breathe a gas mixture low in O_2 and high in CO_2 can hold their breath for an additional 20 s or more. Psychological factors also play a role, and persons can hold their breath longer when they are told their performance is very good than when they are not.

NONCHEMICAL INFLUENCES ON RESPIRATION

RESPONSES MEDIATED BY RECEPTORS IN THE AIRWAYS & LUNGS

Receptors in the airways and lungs are innervated by myelinated and unmyelinated vagal fibers. The unmyelinated fibers are C fibers. The receptors innervated by myelinated fibers are commonly divided into **slowly adapting receptors** and **rapidly adapting receptors** on the basis of whether sustained stimulation leads to prolonged or transient discharge in their afferent nerve fibers **(Table 36–2)**. The other group of receptors presumably consists of the endings of C fibers, and they are divided into pulmonary and bronchial subgroups on the basis of their location.

TABLE 36–2 Airway and lung receptors.

Vagal Innervation	Type	Location in Interstitium	Stimulus	Response
Myelinated	Slowly adapting	Among airway smooth muscle cells (?)	Lung inflation	Inspiratory time shortening Hering-Breuer inflation and deflation reflexes Bronchodilation Tachycardia Hyperpnea
	Rapidly adapting	Among airway epithelial cells	Lung hyperinflation Exogenous and endogenous substances (eg, histamine, prostaglandins)	Cough Bronchoconstriction Mucus secretion
Unmyelinated C fibers	Pulmonary C fibers Bronchial C fibers	Close to blood vessels	Lung hyperinflation Exogenous and endogenous substances (eg, capsaicin, bradykinin, serotonin)	Apnea followed by rapid breathing Bronchoconstriction Bradycardia Hypotension Mucus secretion

Data from Berger AJ, Hornbein TF: Control of respiration. In: *Textbook of Physiology,* 21st ed, Vol 2. Patton HD, et al (editors). Saunders, 1989.

The shortening of inspiration produced by vagal afferent activity (Figure 36–4) is mediated by slowly adapting receptors, as are the **Hering-Breuer reflexes.** The Hering-Breuer inflation reflex is an increase in the duration of expiration produced by steady lung inflation, and the Hering-Breuer deflation reflex is a decrease in the duration of expiration produced by marked deflation of the lung. Because the rapidly adapting receptors are stimulated by chemicals such as histamine, they have been called **irritant receptors.** Activation of rapidly adapting receptors in the trachea causes coughing, bronchoconstriction, and mucus secretion, and activation of rapidly adapting receptors in the lung may produce hyperpnea.

Because the C fiber endings are close to pulmonary vessels, they have been called J (juxtacapillary) receptors. They are stimulated by hyperinflation of the lung, but they respond as well to intravenous or intracardiac administration of chemicals such as capsaicin. The reflex response that is produced is apnea followed by rapid breathing, bradycardia, and hypotension **(pulmonary chemoreflex)**. A similar response is produced by receptors in the heart (**Bezold-Jarisch reflex** or the **coronary chemoreflex)**. The physiological role of this reflex is uncertain, but it probably occurs in pathologic states such as pulmonary congestion or embolization, in which it is produced by endogenously released substances.

COUGHING & SNEEZING

Coughing begins with a deep inspiration followed by forced expiration against a closed glottis. This increases the intrapleural pressure to 100 mm Hg or more. The glottis is then suddenly opened, producing an explosive outflow of air at velocities up to 965 km (600 mi) per hour. Sneezing is a similar expiratory effort with a continuously open glottis. These reflexes help expel irritants and keep airways clear. Other aspects of innervation are considered in a special case **(Clinical Box 36–1)**.

AFFERENTS FROM PROPRIOCEPTORS

Carefully controlled experiments have shown that active and passive movements of joints stimulate respiration, presumably because impulses in afferent pathways from proprioceptors in muscles, tendons, and joints stimulate the inspiratory neurons. This effect probably helps increase ventilation during exercise. Other afferents are considered in **Clinical Box 36–2**.

RESPIRATORY COMPONENTS OF VISCERAL REFLEXES

Inhibition of respiration and closure of the glottis during vomiting, swallowing, and sneezing not only prevent the aspiration of food or vomitus into the trachea but, in the case of vomiting, fix the chest so that contraction of the abdominal muscles increases the intra-abdominal pressure. Similar glottic closure and inhibition of respiration occur during voluntary and involuntary straining.

Hiccup is a spasmodic contraction of the diaphragm and other inspiratory muscles that produces an inspiration during which the glottis suddenly closes. The glottic closure is responsible for the characteristic sensation and sound. Hiccups occur in the fetus in utero as well as throughout extrauterine life. Their function is unknown. Most attacks of hiccups are usually of short duration, and they often respond to breathholding or other measures that increase arterial P_{CO_2}. Intractable hiccups, which can be debilitating, sometimes respond to dopamine antagonists and perhaps to some centrally acting analgesic compounds.

Yawning is a peculiar “infectious” respiratory act whose physiological basis and significance are uncertain. Like hiccupping, it occurs in utero, and it occurs in fish and tortoises as well as mammals. The view that it is needed to increase O_2 intake has been discredited. Underventilated alveoli have a tendency to collapse, and it has been suggested that the deep inspiration and stretching them

open prevents the development of atelectasis. However, in actual experiments, no atelectasis-preventing effect of yawning could be demonstrated. Yawning increases venous return to the heart, which may benefit the circulation. It has been suggested that yawning is a nonverbal signal used for communication between monkeys in a group, and one could argue that on a different level, the same thing is true in humans.

CLINICAL BOX 36–1

Lung Innervation & Patients with Heart-Lung Transplants

Transplantation of the lung and/or heart is an established treatment for severe pulmonary disease and other conditions. In individuals with transplants, the recipient's right atrium is sutured to the donor heart, and the donor heart does not reinnervate, so the resting heart rate is elevated. The donor trachea is sutured to the recipient's just above the carina, and afferent fibers from the lungs do not regrow. Consequently, healthy patients with heart-lung transplants provide an opportunity to evaluate the role of lung innervation in normal physiology. Their cough responses to stimulation of the trachea are normal because the trachea remains innervated, but their cough responses to stimulation of the smaller airways are absent. Their bronchi tend to be dilated to a greater degree than normal. In addition, they have the normal number of yawns and sighs, indicating that these do not depend on innervation of the lungs. Finally, they lack Hering-Breuer reflexes, but their pattern of breathing at rest is normal, indicating that these reflexes do not play an important role in the regulation of resting respiration in humans.

CLINICAL BOX 36–2

Afferents from "Higher Centers"

Pain and emotional stimuli affect respiration, suggesting that afferents from the limbic system and hypothalamus signal to the respiratory neurons in the brainstem. In addition, even though breathing is not usually a conscious event, both inspiration and expiration are under voluntary control. The pathways for voluntary control pass from the neocortex to the motor neurons innervating the respiratory muscles, bypassing the medullary neurons.

Because voluntary and automatic control of respiration are separate, automatic control is sometimes disrupted without loss of voluntary control. The clinical condition that results has been called **Ondine's curse.** In German legend, Ondine was a water nymph who had an unfaithful mortal lover. The king of the water nymphs punished the lover by casting a curse on him that took away all his automatic functions. In this state, he could stay alive only by staying awake and remembering to breathe. He eventually fell asleep from sheer exhaustion, and his respiration stopped. Patients with this intriguing condition generally have bulbar poliomyelitis or disease processes that compress the medulla.

RESPIRATORY EFFECTS OF BARORECEPTOR STIMULATION

Afferent fibers from the baroreceptors in the carotid sinuses, aortic arch, atria, and ventricles relay to the respiratory neurons, as well as the vasomotor and cardioinhibitory neurons in the medulla. Impulses in them inhibit respiration, but the inhibitory effect is slight and of little physiological importance. The hyperventilation in shock is due to chemoreceptor stimulation caused by acidosis and hypoxia secondary to local stagnation of blood flow, and is not baroreceptor-mediated. The activity of inspiratory neurons affects blood pressure and heart rate, and activity in the vasomotor and cardiac areas in the medulla may have minor effects on respiration.

EFFECTS OF SLEEP

Respiration is less rigorously controlled during sleep than in the waking state, and brief periods of apnea occur in normal sleeping adults. Changes in the ventilatory response to hypoxia vary. If the P_{CO_2} falls during the waking state, various stimuli from proprioceptors and the environment maintain respiration, but during sleep, these stimuli are decreased and a decrease in P_{CO_2} can cause apnea. During rapid eye movement (REM) sleep, breathing is irregular and the CO_2 response is highly variable.

RESPIRATORY ABNORMALITIES

ASPHYXIA

In asphyxia produced by occlusion of the airway, acute hypercapnia and hypoxia develop together. Stimulation of respiration is pronounced, with violent respiratory efforts. Blood pressure and heart rate rise sharply, catecholamine secretion is increased, and blood pH drops. Eventually the respiratory efforts cease, the blood pressure falls, and the heart slows. Asphyxiated animals can still be revived at this point by artificial respiration, although they are prone to ventricular fibrillation, probably because of the combination of hypoxic myocardial damage and high circulating catecholamine levels. If artificial respiration is not started, cardiac arrest occurs in 4–5 min.

DROWNING

Drowning is asphyxia caused by immersion, usually in water. In about 10% of drownings, the first gasp of water after the losing struggle not to breathe triggers laryngospasm, and death results from asphyxia without any water in the lungs. In the remaining cases, the glottic muscles eventually relax and fluid enters the lungs. Fresh water is rapidly absorbed, diluting the plasma and causing intravascular hemolysis. Ocean water is markedly hypertonic and draws fluid from the vascular system into the lungs, decreasing plasma volume. The immediate goal in the treatment of drowning is, of course, resuscitation, but long-term treatment must also consider the circulatory effects of the water in the lungs.

PERIODIC BREATHING

The acute effects of voluntary hyperventilation demonstrate the interaction of the chemical mechanisms regulating respiration. When a normal individual hyperventilates for 2–3 min, then stops and permits respiration to continue without exerting any voluntary control over it, a period of apnea occurs. This is followed by a few shallow breaths and then by another period of apnea, followed again by a few breaths **(periodic breathing)**. The cycles may last for some time before normal breathing is resumed **(Figure 36–13)**. The apnea apparently is due to a lack of CO_2 because it does not occur following hyperventilation with gas mixtures containing 5% CO_2. During the apnea, the alveolar P_{O_2} falls and the P_{CO_2} rises. Breathing resumes because of hypoxic stimulation of the carotid and

aortic chemoreceptors before the CO_2 level has returned to normal. A few breaths eliminate the hypoxic stimulus and breathing stops until the alveolar P_{O_2} falls again. Gradually, however, the P_{CO_2} returns to normal, and normal breathing resumes. Changes in breathing patterns can be symptomatic of disease **(Clinical Box 36–3)**.

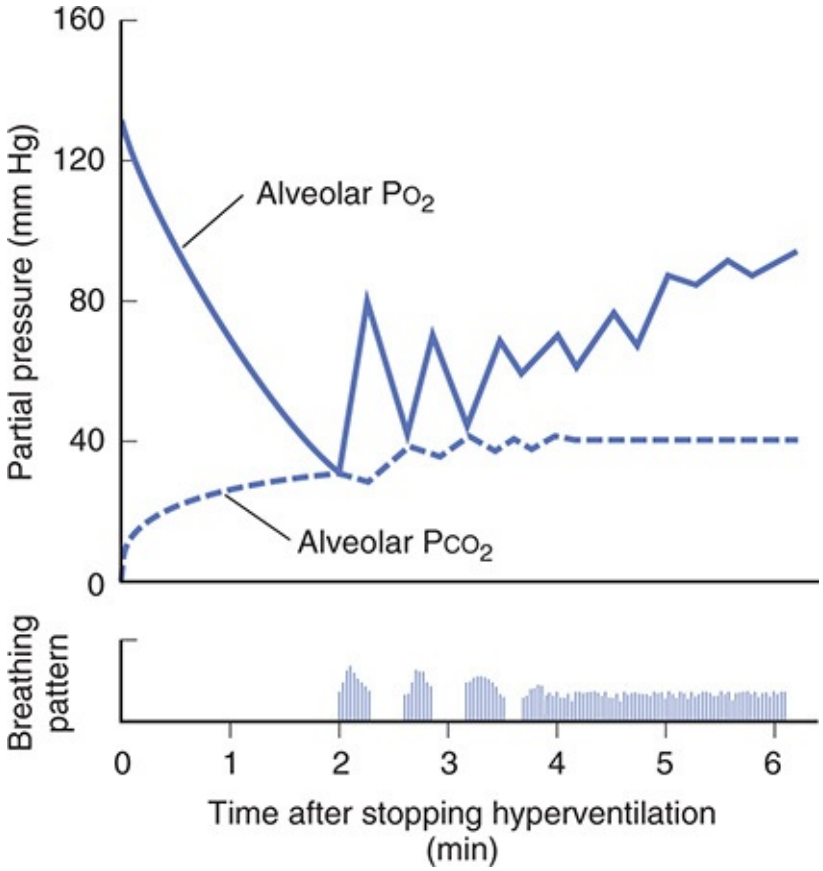


FIGURE 36–13 Changes in breathing and composition of alveolar air after forced hyperventilation for 2 min. Bars in bottom indicate breathing, whereas blank spaces are indicative of apnea.

EFFECTS OF EXERCISE

Exercise provides a physiological example to explore many of the control systems discussed above. Of course, many cardiovascular and respiratory mechanisms must operate in an integrated fashion if the O_2 needs of the active tissue are to be met and the extra CO_2 and heat removed from the body during exercise. Circulatory changes increase muscle blood flow while maintaining

adequate circulation in the rest of the body. In addition, there is an increase in the extraction of O_2 from the blood in exercising muscles and an increase in ventilation. This provides extra O_2, eliminates some of the heat, and excretes extra CO_2. A focus on regulation of ventilation and tissue O_2 is presented below, as many other aspects of regulation have been presented in previous chapters.

CLINICAL BOX 36–3

Periodic Breathing in Disease

Cheyne-Stokes Respiration

Periodic breathing occurs in various disease states and is often called Cheyne-Stokes respiration. It is seen most commonly in patients with heart failure and uremia, but it occurs also in patients with brain disease and during sleep in some normal individuals. Some of the patients with Cheyne-Stokes respiration have increased sensitivity to CO_2. The increased response is apparently due to disruption of neural pathways that normally inhibit respiration. In these individuals, CO_2 causes relative hyperventilation, lowering the arterial P_{CO_2}. During the resultant apnea, the arterial P_{CO_2} again rises to normal, but the respiratory mechanism again overresponds to CO_2. Breathing ceases, and the cycle repeats.

Another cause of periodic breathing in patients with cardiac disease is prolongation of the lung-to-brain circulation time, so that it takes longer for changes in arterial gas tensions to affect the respiratory area in the medulla. When individuals with a slower circulation hyperventilate, they lower the P_{CO_2} of the blood in their lungs, but it takes longer than normal for the blood with a low P_{CO_2} to reach the brain. During this time, the P_{CO_2} in the pulmonary capillary blood continues to be lowered, and when this blood reaches the brain, the low P_{CO_2} inhibits the respiratory area, producing apnea. In other words, the respiratory control system oscillates because the negative feedback loop from lungs to brain is abnormally long.

Sleep Apnea

Episodes of apnea during sleep can be central in origin (ie, due to failure of discharge in the nerves producing respiration) or they can be due to airway obstruction **(obstructive sleep apnea)**. Apnea can occur at any age and is

produced when the pharyngeal muscles relax during sleep. In some cases, failure of the genioglossus muscles to contract during inspiration contributes to the blockage. The genioglossus muscles pull the tongue forward, and without (or with weak) contraction the tongue can obstruct the airway. After several increasingly strong respiratory efforts, the patient wakes up, takes a few normal breaths, and falls back to sleep. Apneic episodes are most common during REM sleep, when the muscles are most hypotonic. The symptoms are loud snoring, morning headaches, fatigue, and daytime sleepiness. When severe and prolonged, the condition can lead to hypertension and its complications. Frequent apneas can lead to numerous brief awakenings during sleep and to sleepiness during waking hours. With this in mind, it is not surprising to find that the incidence of motor vehicle accidents in sleep apnea patients is seven times greater than it is in the general driving population. In addition, sleep apnea is associated with many cardiovascular diseases, such as hypertension, arrhythmia, stroke and heart failure.

THERAPEUTIC HIGHLIGHTS

Treatment of sleep apnea depends on the patient and on the cause (if known). Treatments range from mild to moderate interventions to surgery. Interventions including positional therapy, dental appliances that rearrange the architecture of the airway, avoidance of muscle relaxants (eg, alcohol) or drugs that reduce respiratory drive, or continuous positive airway pressure (C-PAP, a standard of care in the management of symptoms of moderate to severe obstructive sleep apnea). Because sleep apnea is increased in overweight or obese individuals, weight loss can also be effective.

CHANGES IN VENTILATION

During exercise, the amount of O_2 entering the blood in the lungs is increased because the amount of O_2 added to each unit of blood and the pulmonary blood flow per minute are increased. The P_{O_2} of blood flowing into the pulmonary capillaries falls from 40 to 25 mm Hg or less, so that the alveolar-capillary P_{O_2} gradient is increased and more O_2 enters the blood. Blood flow per minute is increased from 5.5 L/min to as much as 20–35 L/min. The total amount of O_2

entering the blood therefore increases from 250 mL/min at rest to values as high as 4000 mL/min. The amount of CO_2 removed from each unit of blood is increased, and CO_2 excretion increases from 200 mL/min to as much as 8000 mL/min. The increase in O_2 uptake is proportional to work load, up to a maximum. Above this maximum, O_2 consumption levels off and the blood lactate level continues to rise **(Figure 36–14)**. The lactate comes from muscles in which aerobic resynthesis of energy stores cannot keep pace with their utilization, and an **oxygen debt** is being incurred.

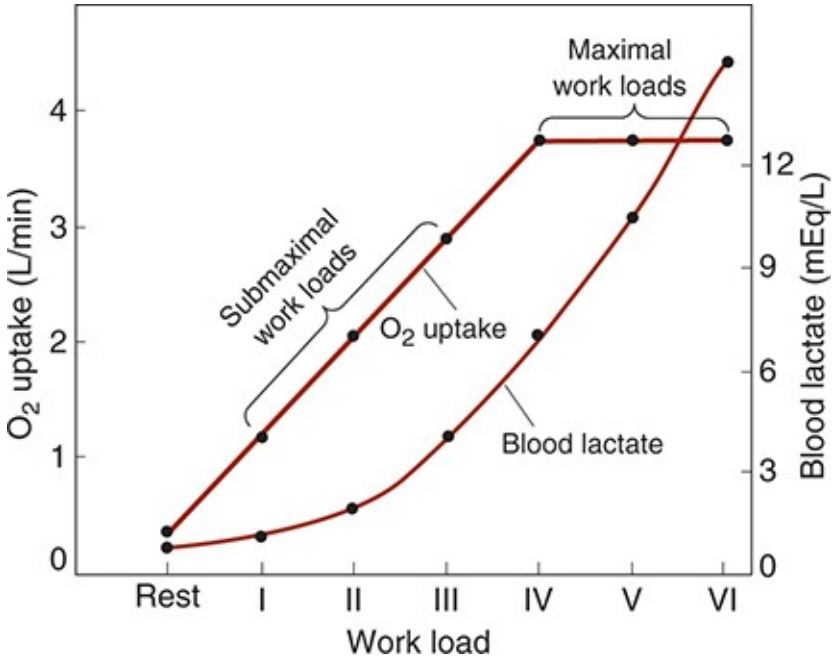


FIGURE 36–14 Relation between work load, blood lactate level, and O_2 uptake. I–VI, increasing work loads produced by increasing the speed and grade of a treadmill on which the subjects worked. (Reproduced with permission from Mitchell JH, Blomqvist G: Maximal oxygen uptake. N Engl J Med 1971 May 6;284(18):1018–1022.)

Ventilation increases abruptly with the onset of exercise, which is followed after a brief pause by a further, more gradual increase **(Figure 36–15)**. With moderate exercise, the increase is due mostly to an increase in the depth of respiration; this is accompanied by an increase in the respiratory rate when the exercise is more strenuous. Ventilation abruptly decreases when exercise ceases, which is followed after a brief pause by a more gradual decline to preexercise values. The abrupt increase at the start of exercise is presumably due to psychic

stimuli and afferent impulses from proprioceptors in muscles, tendons, and joints. The more gradual increase is presumably humoral, even though arterial pH, P_{CO_2}, and P_{O_2} remain constant during moderate exercise. The increase in ventilation is proportional to the increase in O_2 consumption, but the mechanisms responsible for the stimulation of respiration are still the subject of much debate. The increase in body temperature may play a role. Exercise increases the plasma K^+ level, and this increase may stimulate the peripheral chemoreceptors. In addition, it may be that the sensitivity of the neurons controlling the response to CO_2 is increased or that the respiratory fluctuations in arterial P_{CO_2} increase so that, even though the mean arterial P_{CO_2} does not rise, it is CO_2 that is responsible for the increase in ventilation. O_2 also seems to play some role, despite the lack of a decrease in arterial P_{O_2}, since during the performance of a given amount of work, the increase in ventilation while breathing 100% O_2 is 10–20% less than the increase while breathing air. Thus, it currently appears that a number of different factors combine to produce the increase in ventilation seen during moderate exercise.

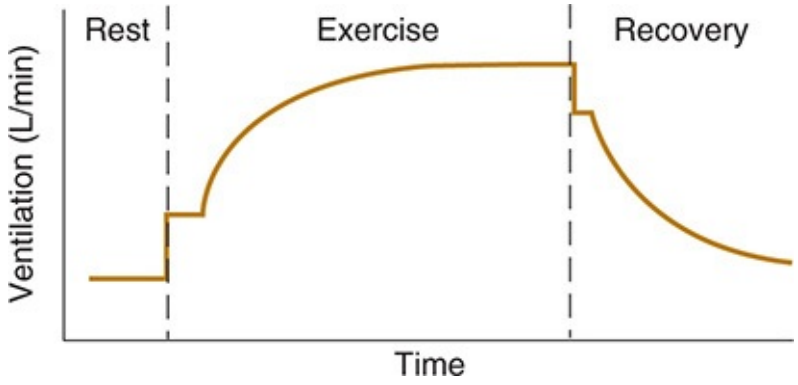


FIGURE 36–15 Diagrammatic representation of changes in ventilation during exercise. See text for details.

When exercise becomes more vigorous, buffering of the increased amounts of lactic acid that are produced liberates more CO_2, and this further increases ventilation. The response to graded exercise is shown in **Figure 36–16**. With increased production of acid, the increases in ventilation and CO_2 production remain proportional, so alveolar and arterial CO_2 change relatively little **(isocapnic buffering)**. Because of the hyperventilation, alveolar P_{O_2} increases. With further accumulation of lactic acid, the increase in ventilation outstrips CO_2

production and alveolar P_{CO_2} falls, as does arterial P_{CO_2}. The decline in arterial P_{CO_2} provides respiratory compensation for the metabolic acidosis produced by the additional lactic acid. The additional increase in ventilation produced by the acidosis depends on the carotid bodies and does not occur if they are removed.

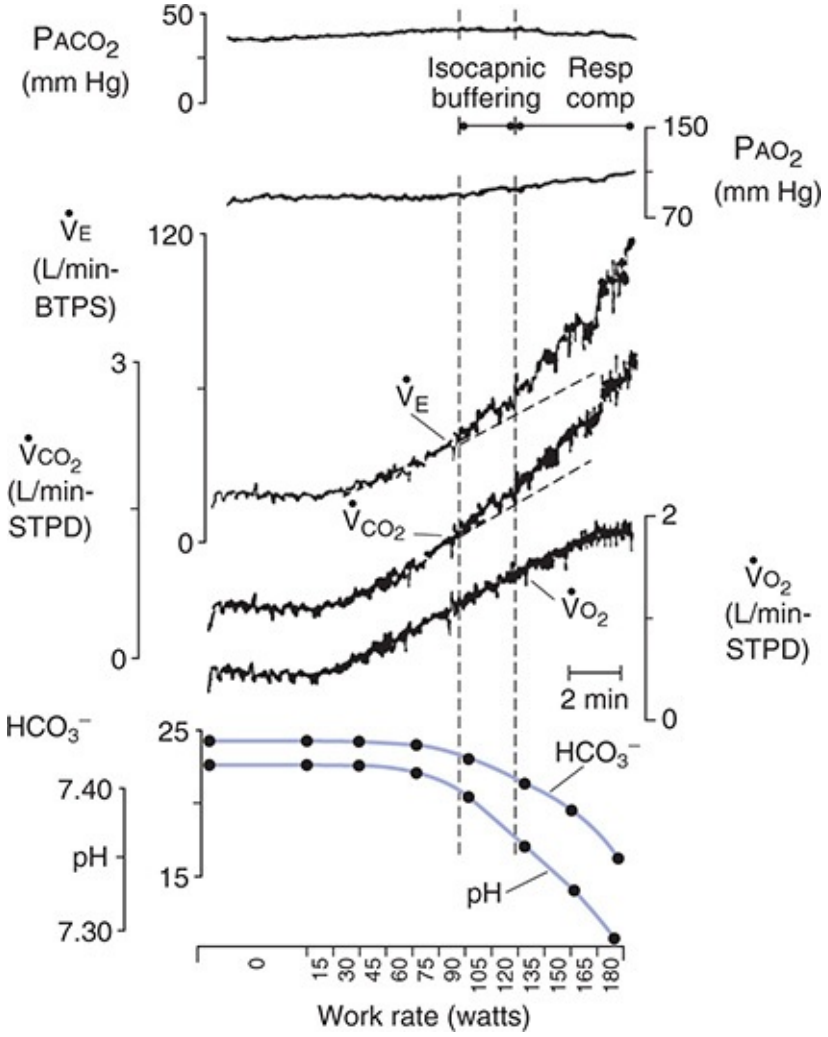


FIGURE 36–16 Physiological responses to work rate during exercise. Changes in alveolar P_{CO_2} (P_{ACO_2}), alveolar P_{O_2} (P_{AO_2}), ventilation ($\dot{V}_E$), CO_2 production ($\dot{V}_{CO_2}$), O_2 consumption ($\dot{V}_{O_2}$), arterial HCO_3^-, and arterial pH with graded increases in work by an adult male on a bicycle ergometer. Resp comp, respiratory compensation. BTPS, body temperature (BT) and pressure (760 mm Hg), saturated; STPD, standard temperature (0°C) and pressure (760 mm Hg), dry. Dashed lines emphasize deviation from linear response. See text for additional details. (Reproduced with permission from Wasserman K: Breathing during exercise. N Engl J Med 1978 Apr 6;298(14):780–785.)

The respiratory rate after exercise does not reach basal levels until the O_2 debt is repaid. This may take as long as 90 min. The stimulus to ventilation after exercise is not the arterial P_{CO_2}, which is normal or low, or the arterial P_{O_2}, which is normal or high, but the elevated arterial H^+ concentration due to the lactic acidemia. The magnitude of the O_2 debt is the amount by which O_2 consumption exceeds basal consumption from the end of exertion until the O_2 consumption has returned to preexercise basal levels. During repayment of the O_2 debt, the O_2 concentration in muscle myoglobin rises slightly. ATP and phosphorylcreatine are resynthesized, and lactic acid is removed. Eighty percent of the lactic acid is converted to glycogen and 20% is metabolized to CO_2 and H_2O.

CHANGES IN THE TISSUES

Maximum O_2 uptake during exercise is limited by the maximum rate at which O_2 is transported to the mitochondria in the exercising muscle. However, this limitation is not normally due to deficient O_2 uptake in the lungs, and hemoglobin in arterial blood is saturated even during the most severe exercise.

During exercise, the contracting muscles use more O_2, and the tissue P_{O_2} and the P_{O_2} in venous blood from exercising muscle fall nearly to zero. More O_2 diffuses from the blood, the blood P_{O_2} in the muscles drops, and more O_2 is removed from hemoglobin. Because the capillary bed of contracting muscle is dilated and many previously closed capillaries are open (capillary recruitment and extension), the mean distance from the blood to the tissue cells is greatly decreased; this facilitates the movement of O_2 from blood to cells. The oxygen–hemoglobin dissociation curve is steep in the P_{O_2} range below 60 mm Hg, and a relatively large amount of O_2 is supplied for each drop of 1 mm Hg in P_{O_2} (see Figure 35–2). Additional O_2 is supplied because, as a result of the accumulation of CO_2 and the rise in temperature in active tissues—and perhaps because of a rise in red blood cell 2,3-diphosphoglycerate (2,3-DPG)—the dissociation curve shifts to the right. The net effect is a 3-fold increase in O_2 extraction from each unit of blood (see Figure 35–3). Because this increase is accompanied by a 30-fold or greater increase in blood flow, it permits the metabolic rate of muscle to rise as much as 100-fold during exercise.

EXERCISE TOLERANCE & FATIGUE

What determines the maximum amount of exercise that can be performed by an individual? Obviously, exercise tolerance has a time as well as an intensity dimension. For example, a fit young man can produce a power output on a bicycle of about 700 W for 1 min, 300 W for 5 min, and 200 W for 40 min. It used to be argued that the limiting factors in exercise performance were the rate at which O_2 could be delivered to the tissues or the rate at which O_2 could enter the body in the lungs. These factors play a role, but it is clear that other factors also contribute, and that exercise stops when the sensation of **fatigue** progresses to the sensation of exhaustion. Fatigue is produced in part by bombardment of the brain by neural impulses from muscles, and the decline in blood pH produced by lactic acidosis also makes one feel tired, as does the rise in body temperature, dyspnea, and, perhaps, the uncomfortable sensations produced by activation of the J receptors in the lungs.

CHAPTER SUMMARY

- Breathing is under both voluntary control (located in the cerebral cortex) and automatic control (driven by pacemaker cells in the medulla). There is a reciprocal innervation to expiratory and inspiratory muscles in that motor neurons supplying expiratory muscles are inactive when motor neurons supplying inspiratory muscles are active, and vice versa.
- The pre-Bötzinger complex on either side of the medulla contains synaptically coupled pacemaker cells that allow for rhythmic generation of breathing. The spontaneous activity of these neurons can be altered by neurons in the pneumotaxic center, although the full regulatory function of these neurons on normal breathing is not understood.
- Breathing patterns are sensitive to chemicals in the blood through activation of respiratory chemoreceptors. There are chemoreceptors in the carotid and aortic bodies and in collections of cells in the medulla. These chemoreceptors respond to changes in P_{O_2} and P_{CO_2} as well as H^+ to regulate breathing.
- Receptors in the airway are additionally innervated by slowly adapting and rapidly adapting myelinated vagal fibers. Slowly adapting receptors can be activated by lung inflation. Rapidly adapting receptors, or irritant receptors, can be activated by chemicals such as histamine and result in cough or even

hyperpnea.

- Receptors in the airway are also innervated by unmyelinated vagal fibers (C fibers) that are typically found next to pulmonary vessels. They are stimulated by hyperinflation (or exogenous substances including capsaicin) and lead to the pulmonary chemoreflex. The physiological role for this response is not fully understood.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. The main respiratory control neurons
   A. send out regular bursts of impulses to expiratory muscles during quiet respiration.
   B. are unaffected by stimulation of pain receptors.
   C. are located in the pons.
   D. send out regular bursts of impulses to inspiratory muscles during quiet respiration.
   E. are unaffected by impulses from the cerebral cortex.
2. Intravenous lactic acid increases ventilation. The receptors responsible for this effect are located in the
   A. medulla oblongata.
   B. carotid bodies.
   C. lung parenchyma.
   D. aortic baroreceptors.
   E. trachea and large bronchi.
3. Spontaneous respiration ceases after
   A. transection of the brainstem above the pons.
   B. transection of the brainstem at the caudal end of the medulla.
   C. bilateral vagotomy.
   D. bilateral vagotomy combined with transection of the brainstem at the superior border of the pons.
   E. transection of the spinal cord at the level of the first thoracic segment.
4. The following physiological events that occur in vivo are listed in random order: (1) decreased CSF pH; (2) increased arterial P_{CO_2}; (3) increased CSF

P_{CO_2}; (4) stimulation of medullary chemoreceptors; and (5) increased alveolar P_{CO_2}.

What is the usual sequence in which they occur when they affect respiration?

A. 1, 2, 3, 4, 5

B. 4, 1, 3, 2, 5

C. 3, 4, 5, 1, 2

D. 5, 2, 3, 1, 4

E. 5, 3, 2, 4, 1

5. The following events that occur in the carotid bodies when they are exposed to hypoxia are listed in random order: (1) depolarization of type I glomus cells; (2) excitation of afferent nerve endings; (3) reduced conductance of hypoxia-sensitive K^+ channels in type I glomus cells; (4) Ca^{2+} entry into type I glomus cells; (5) decreased K^+ efflux.

What is the usual sequence in which they occur on exposure to hypoxia?

A. 1, 3, 4, 5, 2

B. 1, 4, 2, 5, 3

C. 3, 4, 5, 1, 2

D. 3, 1, 4, 5, 2

E. 3, 5, 1, 4, 2

6. Injection of a drug that stimulates the carotid bodies would be expected to cause

A. a decrease in the pH of arterial blood.

B. a decrease in the P_{CO_2} of arterial blood.

C. an increase in the HCO_3^- concentration of arterial blood.

D. an increase in urinary Na^+ excretion.

E. an increase in plasma Cl^-.

7. Variations in which of the following components of blood or CSF do *not* affect respiration?

A. Arterial HCO_3^- concentration

B. Arterial H^+ concentration

C. Arterial Na^+ concentration

D. CSF CO_2 concentration

E. CSF H^+ concentration

SECTION VII

Renal Physiology

The kidneys, the bladder, and the ureters make up the urinary system. Within the kidney, the functional unit is the nephron and each human kidney has approximately 1 million nephrons. The kidneys play an essential role in the regulation of water homeostasis, electrolyte composition (eg, Na, Cl, K, HCO_3), regulation of extracellular volume (thus blood pressure), and acid–base homeostasis (Chapter 39). The kidneys filter the plasma of the blood and produce urine, which allows the kidneys to excrete metabolic waste products (eg, urea; ammonium; and foreign chemicals, such as drug metabolites) from the body. The kidneys are responsible for the reabsorption of glucose and amino acids from the plasma filtrate in addition to regulated calcium and phosphate uptake (high in children). The kidneys play a role in gluconeogenesis and during fasting can synthesize and release glucose into the blood, producing almost 20% of the liver's glucose capacity. The kidneys are also endocrine organs, making kinins (see Chapter 32), 1,25-dihydroxycholecalciferol (see Chapter 21), erythropoietin (see Chapter 37), and making and secreting renin (see Chapter 38).

Specifically, in the kidneys, a fluid that resembles plasma is filtered through the glomerular capillaries into the renal tubules **(glomerular filtration).** As this glomerular filtrate passes down the tubules, its volume is reduced and its composition altered by the processes of **tubular reabsorption** (removal of water and solutes from the tubular fluid) and **tubular secretion** (secretion of solutes into the tubular fluid) to form the urine that enters the renal pelvis. From the renal pelvis, the urine passes to the bladder and is expelled to the exterior by the process of urination, or **micturition.**

Diseases of the kidney are numerous. Commonly occurring clinical conditions include acute kidney injury, chronic kidney disease (CKD), diabetic kidney

disease, nephritic and nephrotic syndromes, polycystic kidney disease, urinary tract obstruction, urinary tract infection, and renal cancer. When renal function decreases such that the kidneys are no longer functioning to maintain health, patients sometimes undergo dialysis and eventually kidney transplantation.

The prevalence of kidney disease is increasing across the world as is the cost of treating the diseases that cause kidney damage; thus kidney disease represents a large threat to healthcare resources worldwide. The two diseases that cause the largest prevalence of kidney dysfunction are diabetes and hypertension. Nearly one billion people worldwide have high blood pressure and that number is expected to increase to 1.56 billion by 2025. There are currently over 240 million people with diabetes worldwide and this figure is anticipated to rise to 380 million by 2025. Of those with diabetes, about 40% will develop CKD, which means that their risk of cardiovascular complications also increases. The cumulative global cost for dialysis and transplantation over the next decade is predicted to exceed one-trillion dollars.

CHAPTER 37

Renal Function & Micturition

OBJECTIVES

After studying this chapter, you should be able to:

- Describe the morphology of a typical nephron and its blood supply.
- Explain autoregulation and how it regulates renal function.
- Define glomerular filtration rate (GFR), describe how it can be measured, and list the major factors that alter GFR.
- Outline how each tubular segment of the nephron handles Na^+ and water.
- Explain tubular reabsorption and secretion of glucose and K^+.
- Describe how the kidney operates to produce hypertonic or hypotonic urine.
- Define the major classes of diuretics and explain where each operates to increase urine flow.
- Describe the voiding reflex and draw a cystometrogram.

FUNCTIONAL ANATOMY

THE NEPHRON

Each individual renal tubule and its glomerulus is a unit **(nephron).** The size of

the kidneys between species varies, as does the number of nephrons they contain. Each human kidney has approximately 1 million nephrons. The specific structures of the nephron are shown in diagrammatic manner in **Figure 37–1**.

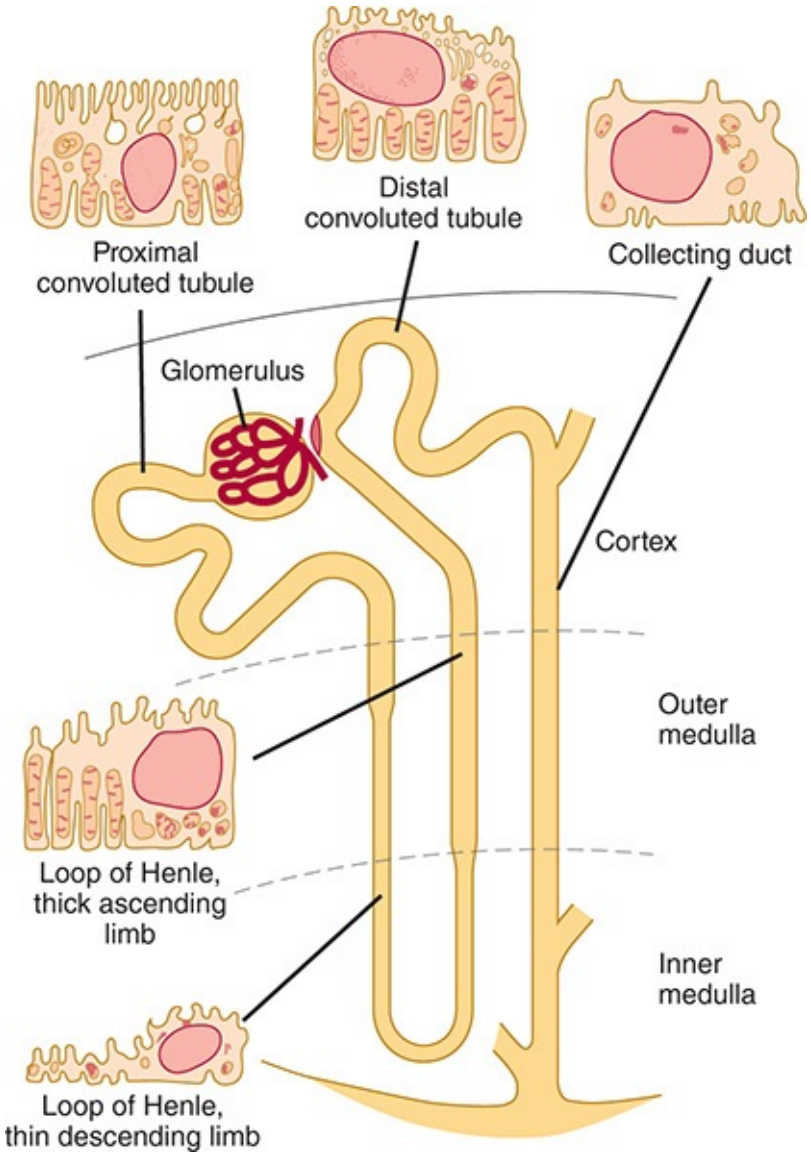


FIGURE 37–1 Diagram of a nephron. The main histologic features of the cells that make up each portion of the tubule are also shown.

The glomerulus, which is about 200 µm in diameter, is formed by the invagination of a tuft of capillaries into the dilated, blind end of the nephron **(Bowman's capsule).** The capillaries are supplied by an **afferent arteriole** and drained by the **efferent arteriole** **(Figure 37–2)**, and it is from the glomerulus that the filtrate is formed. The diameter of the afferent arteriole is larger than the

efferent arteriole. Two cellular layers separate the blood from the glomerular filtrate in Bowman's capsule: the capillary endothelium and the specialized epithelium of the capsule. The endothelium of the glomerular capillaries is fenestrated, with pores that are 70–90 nm in diameter. The endothelium of the glomerular capillaries is completely surrounded by the glomerular basement membrane along with specialized cells called podocytes. **Podocytes** have numerous pseudopodia that interdigitate (Figure 37–2) to form **filtration slits** along the capillary wall. The slits are approximately 25 nm wide, and each is closed by a thin membrane. The glomerular basement membrane, the basal lamina, does not contain visible gaps or pores. Stellate cells called **mesangial cells** are located between the basal lamina and the endothelium. They are similar to cells called **pericytes,** which are found in the walls of capillaries elsewhere in the body. Mesangial cells are especially common between two neighboring capillaries, and in these locations the basal membrane forms a sheath shared by both capillaries (Figure 37–2). The mesangial cells are contractile and play a role in the regulation of glomerular filtration. Mesangial cells secrete the extracellular matrix, take up immune complexes, and are involved in the progression of glomerular disease.

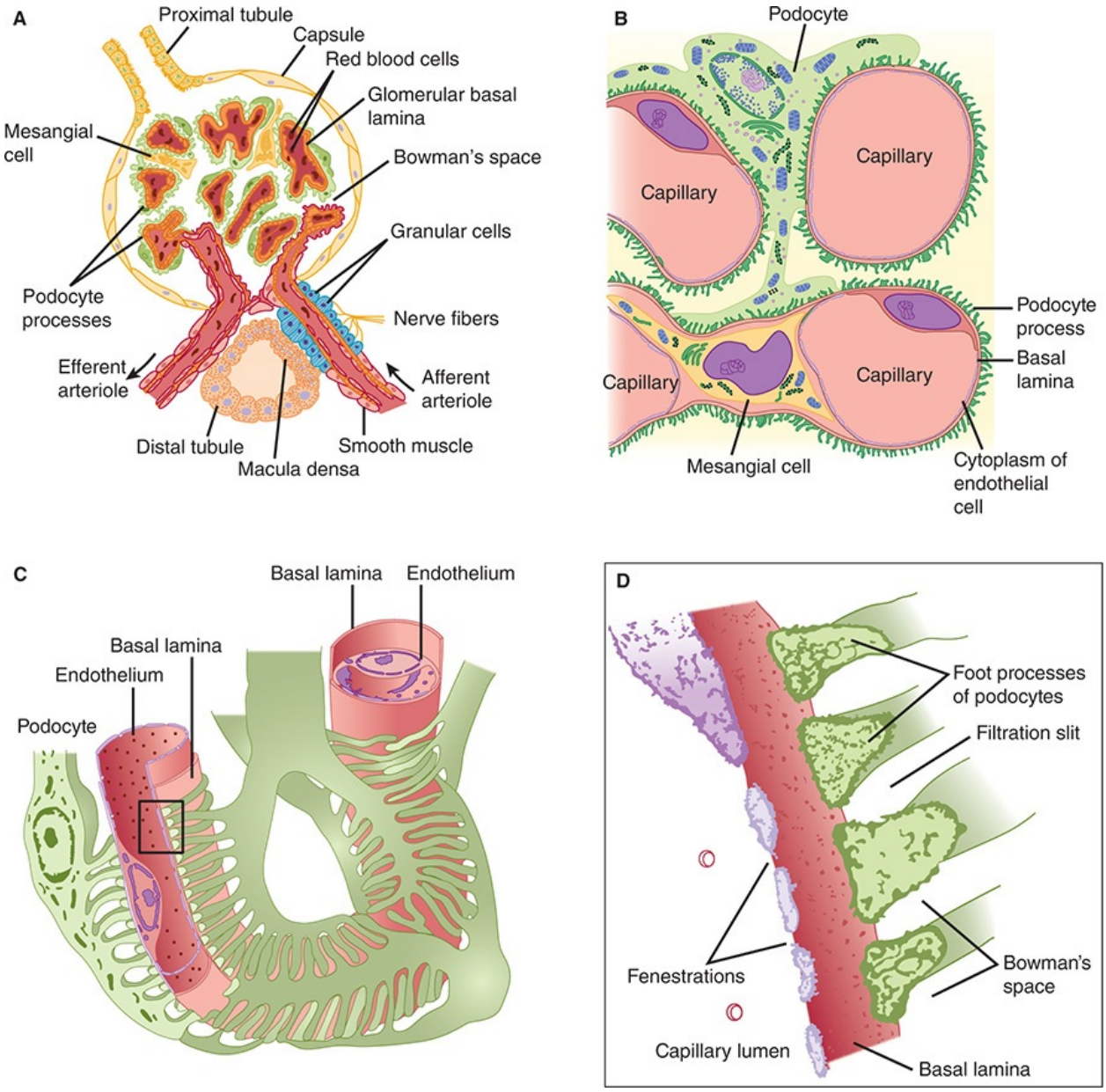


FIGURE 37–2 Structural details of glomerulus. A) Section through vascular pole, showing capillary loops. **B)** Relation of mesangial cells and podocytes to glomerular capillaries. **C)** Detail of the way podocytes form filtration slits on the basal lamina, and the relation of the lamina to the capillary endothelium. **D)** Enlargement of the rectangle in C to show the podocyte processes. The fuzzy material on their surfaces is glomerular polyanion.

Functionally, the glomerular membrane permits the free passage of neutral substances up to 4 nm in diameter and almost totally excludes those with diameters greater than 8 nm. However, the charge on molecules as well as their

diameters affects their passage into Bowman's capsule. The total area of glomerular capillary endothelium across which filtration occurs in humans is about 0.8 m^2.

The general features of the cells that make up the walls of the tubules are shown in Figure 37–1; however, there are cell subtypes in all segments, and the anatomic differences between them correlate with differences in function.

The human **proximal convoluted tubule** is about 15 mm long and 55 µm in diameter. Its wall is made up of a single layer of cells that interdigitate with one another and are united by apical tight junctions. Between the cells are extensions of the extracellular space called the **lateral intercellular spaces.** The luminal edges of the cells have a striated **brush border** due to the presence of many microvilli.

The convoluted proximal tubule straightens and the next portion of each nephron is the **loop of Henle.** The descending portion of the loop and the proximal portion of the ascending limb are made up of thin, permeable cells. On the other hand, the thick portion of the ascending limb (Figure 37–1) is made up of thick cells containing many mitochondria. The nephrons with glomeruli in the outer portions of the renal cortex have short loops of Henle **(cortical nephrons),** whereas those with glomeruli in the juxtamedullary region of the cortex **(juxtamedullary nephrons)** have long loops extending down into the medullary pyramids. In humans, only 15% of the nephrons have long loops.

The thick end of the ascending limb of the loop of Henle reaches the glomerulus of the nephron from which the tubule arose and nestles between its afferent and efferent arterioles. Specialized cells at the end form the **macula densa,** which is close to the efferent and particularly the afferent arteriole (Figure 37–2). The macula, the neighboring **lacis cells,** and the renin-secreting **granular cells** in the afferent arteriole form the **juxtaglomerular apparatus** (see Figure 38–8).

The **distal convoluted tubule,** which starts at the macula densa, is about 5 mm long. Its epithelium is lower than that of the proximal tubule, and although a few microvilli are present, there is no distinct brush border. The distal tubules coalesce to form **collecting ducts** that are about 20 mm long and pass through the renal cortex and medulla to empty into the pelvis of the kidney at the apexes of the medullary pyramids. The epithelium of the collecting ducts is made up of **principal cells (P cells)** and **intercalated cells (I cells).** The P cells, which predominate, are relatively tall and have few organelles. They are involved in Na^+ reabsorption and vasopressin-stimulated water reabsorption. The I cells, which are present in smaller numbers and are also found in the distal tubules,

have more microvilli, cytoplasmic vesicles, and mitochondria. They are concerned with acid secretion and HCO_3^- transport. The total length of the nephrons, including the collecting ducts, ranges from 45 to 65 mm.

Cells in the kidneys that appear to have a secretory function include not only the granular cells in the juxtaglomerular apparatus but also some of the cells in the interstitial tissue of the medulla. These cells are called **renal medullary interstitial cells (RMICs)** and are specialized fibroblast-like cells. They contain lipid droplets and are a major site of cyclooxygenase 2 (COX-2) and prostaglandin synthase (PGES) expression. PGE_2 is the major prostanoid synthesized in the kidney and is an important paracrine regulator of salt and water homeostasis. PGE_2 is secreted by the RMICs, by the macula densa, and by cells in the collecting ducts; prostacyclin (PGI_2) and other prostaglandins are secreted by the arterioles and glomeruli.

BLOOD VESSELS

The renal circulation is diagrammed in **Figure 37–3**. The **afferent arterioles** are short, straight branches of the interlobular arteries. Each divides into multiple capillary branches to form the tuft of vessels in the glomerulus. The capillaries coalesce to form the **efferent arteriole,** which in turn breaks up into capillaries that supply the tubules **(peritubular capillaries)** before draining into the interlobular veins. The arterial segments between glomeruli and tubules are thus technically a portal system, and the glomerular capillaries are the only capillaries in the body that drain into arterioles. However, there is relatively little smooth muscle in the efferent arterioles.

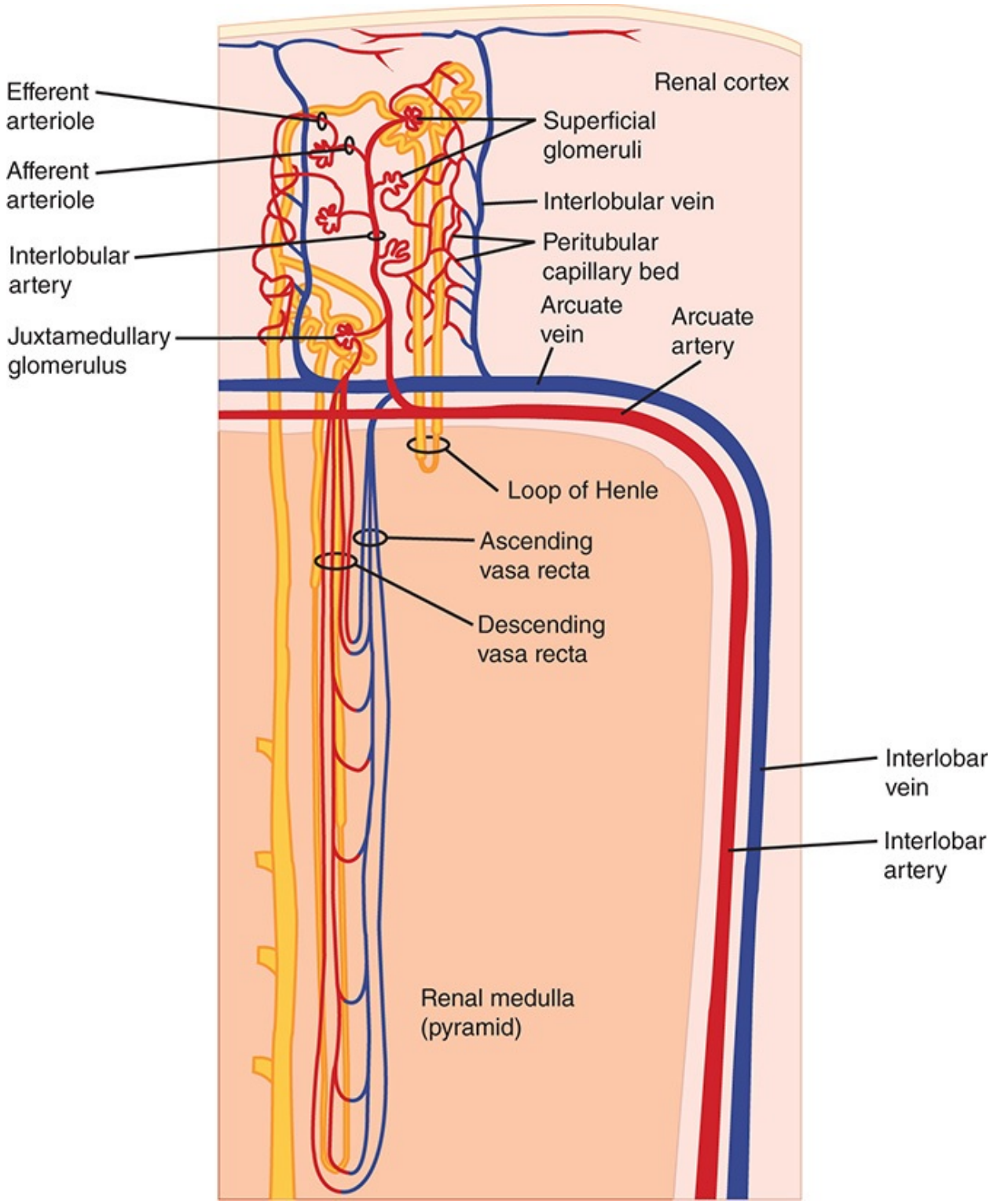


FIGURE 37–3 Renal circulation. Interlobar arteries divide into arcuate arteries, which give off interlobular arteries in the cortex. The interlobular arteries provide an afferent arteriole to each glomerulus. The efferent arteriole from each

glomerulus breaks up into capillaries that supply blood to the renal tubules. Venous blood enters interlobular veins, which in turn flow via arcuate veins to the interlobar veins. (Modified with permission from Boron WF, Boulpaep EL: *Medical Physiology*. Saunders, 2009.)

The capillaries draining the tubules of the cortical nephrons form a peritubular network, whereas the efferent arterioles from the juxtamedullary glomeruli drain not only into a peritubular network, but also into vessels that form hairpin loops (the **vasa recta**). These loops dip into the medullary pyramids alongside the loops of Henle (Figure 37–3). The descending vasa recta have a nonfenestrated endothelium that contains a facilitated transporter for urea, and the ascending vasa recta have a fenestrated endothelium, consistent with their function in conserving solutes.

The efferent arteriole from each glomerulus breaks up into capillaries that supply a number of different nephrons. Thus, the tubule of each nephron does not necessarily receive blood solely from the efferent arteriole of the same nephron. In humans, the total surface of the renal capillaries is approximately equal to the total surface area of the tubules, both being about 12 m^2. The volume of blood in the renal capillaries at any given time is 30–40 mL.

LYMPHATICS

The kidneys have an abundant lymphatic supply that drains via the thoracic duct into the venous circulation in the thorax.

CAPSULE

The renal capsule is thin but tough. If the kidney becomes edematous, the capsule limits the swelling, and the tissue pressure **(renal interstitial pressure)** rises. This decreases the glomerular filtration rate (GFR) and is claimed to enhance and prolong anuria in acute kidney injury (AKI).

INNERVATION OF THE RENAL VESSELS

The renal nerves travel along the renal blood vessels as they enter the kidney. They contain many postganglionic sympathetic efferent fibers and a few afferent fibers. There also appears to be a cholinergic innervation via the vagus nerve, but

its function is uncertain. The sympathetic preganglionic innervation comes primarily from the lower thoracic and upper lumbar segments of the spinal cord, and the cell bodies of the postganglionic neurons are in the sympathetic ganglion chain, in the superior mesenteric ganglion, and along the renal artery. The sympathetic fibers are distributed primarily to the afferent and efferent arterioles, the proximal and distal tubules, and the juxtaglomerular apparatus (see Chapter 38). In addition, there is a dense noradrenergic innervation of the thick ascending limb of the loop of Henle.

Nociceptive afferents that mediate pain in kidney disease parallel the sympathetic efferents and enter the spinal cord in the thoracic and upper lumbar dorsal roots. Other renal afferents presumably mediate a **renorenal reflex** by which an increase in ureteral pressure in one kidney leads to a decrease in efferent nerve activity to the contralateral kidney. This decrease permits an increase in its excretion of Na^+ and water.

RENAL CIRCULATION

BLOOD FLOW

In a resting adult, the kidneys receive 1.2–1.3 L of blood per minute, or just under 25% of the cardiac output. Renal blood flow can be measured with electromagnetic or other types of flow meters, or it can be determined by applying the Fick principle (see Chapter 30) to the kidney; that is, by measuring the amount of a given substance taken up per unit of time and dividing this value by the arteriovenous difference for the substance across the kidney. Because the kidney filters plasma, the **renal plasma flow** (RPF) equals the amount of a substance excreted per unit of time divided by the renal arteriovenous difference as long as the amount in the red cells is unaltered during passage through the kidney. Any excreted substance can be used if its concentration in arterial and renal venous plasma can be measured and if it is not metabolized, stored, or produced by the kidney and does not itself affect blood flow.

RPF can be measured by infusing p-aminohippuric acid (PAH) and determining its urine and plasma concentrations. PAH is filtered by the glomeruli and secreted by the tubular cells, so that its **extraction ratio** (arterial concentration minus renal venous concentration divided by arterial concentration) is high. For example, when PAH is infused at low doses, 90% of the PAH in arterial blood is removed in a single circulation through the kidney. It has therefore become commonplace to calculate the “RPF” by dividing the

amount of PAH in the urine by the plasma PAH level, ignoring the level in renal venous blood. Peripheral venous plasma can be used because its PAH concentration is essentially identical to that in the arterial plasma reaching the kidney. The value obtained should be called the **effective renal plasma flow (ERPF)** to indicate that the level in renal venous plasma was not measured. In humans, ERPF averages about 625 mL/min.

$$\text{ERPF} = \frac{U_{PAH}\dot{V}}{P_{PAH}} = \text{Clearance of PAH } (C_{PAH})$$

Example:

Concentration of PAH in urine (U_{PAH}): 14 mg/mL

Urine flow ($\dot{V}$): 0.9 mL/min

Concentration of PAH in plasma (P_{PAH}): 0.02 mg/mL

$$\text{ERPF} = \frac{14 \times 0.9}{0.02} = 630 \text{ mL/min}$$

It should be noted that the ERPF determined in this way is the **clearance** of PAH. The concept of clearance is discussed in detail below.

ERPF can be converted to actual renal plasma flow (RPF):

Average PAH extraction ratio: 0.9

$$\frac{\text{ERP}}{\text{Extraction ration}} = \frac{630}{0.9} = \text{Actual RPF} = 700 \text{ mL/min}$$

From the renal plasma flow, the renal blood flow can be calculated by dividing by 1 minus the hematocrit:

Hematocrit (Hct): 45%

$$\begin{aligned}\text{Renal blood flow} &= \text{RPF} \times \frac{1}{1 - \text{Hct}} \\ &= 700 \times \frac{1}{0.55} \\ &= 1273 \text{ mL/min}\end{aligned}$$

PRESSURE IN RENAL VESSELS

The pressure in the glomerular capillaries has been measured directly in rats and has been found to be considerably lower than predicted on the basis of indirect measurements. When the mean systemic arterial pressure is 100 mm Hg, the glomerular capillary pressure is about 45 mm Hg. The pressure drop across the glomerulus is only 1–3 mm Hg, but a further drop occurs in the efferent arteriole so that the pressure in the peritubular capillaries is about 8 mm Hg. The pressure in the renal vein is about 4 mm Hg. Pressure gradients are similar in squirrel monkeys and presumably in humans, with a glomerular capillary pressure that is about 40% of systemic arterial pressure.

REGULATION OF THE RENAL BLOOD FLOW

Norepinephrine (noradrenaline) constricts the renal vessels, with the greatest effect of injected norepinephrine being exerted on the interlobular arteries and the afferent arterioles. Dopamine is made in the kidney and causes renal vasodilation and natriuresis. Angiotensin II exerts a constrictor effect on both the afferent and efferent arterioles. Prostaglandins increase blood flow in the renal cortex and decrease blood flow in the renal medulla. Acetylcholine also produces renal vasodilation. A high-protein diet raises glomerular capillary pressure and increases renal blood flow.

FUNCTIONS OF THE RENAL NERVES

Stimulation of the renal nerves increases renin secretion by a direct action of released norepinephrine on β_1-adrenergic receptors on the juxtaglomerular cells (see Chapter 38), and it increases Na^+ reabsorption, probably by a direct action of norepinephrine on renal tubular cells. The proximal and distal tubules and the thick ascending limb of the loop of Henle are richly innervated. When the renal nerves are stimulated to increasing extents in experimental animals, the first response is an increase in the sensitivity of the granular cells in the juxtaglomerular apparatus **(Table 37–1)**, followed by increased renin secretion, then increased Na^+ reabsorption, and finally, at the highest threshold, renal vasoconstriction with decreased glomerular filtration and renal blood flow. It is still unsettled whether the effect on Na^+ reabsorption is mediated via α- or β-adrenergic receptors, and it may be mediated by both. The physiologic role of the renal nerves in Na^+ homeostasis is also unsettled, in part because most renal functions appear to be normal in patients with transplanted kidneys, and it takes

some time for transplanted kidneys to acquire a functional innervation.

TABLE 37–1 Renal responses to graded renal nerve stimulation.

Renal Nerve Stimulation Frequency (Hz)	RSR	$U_{Na}V$	GFR	RBF
0.25	No effect on basal values; augments RSR mediated by nonneural stimuli	0	0	0
0.50	Increased without changing $U_{Na}V$, GFR, or RBF	0	0	0
1.0	Increased with decreased $U_{Na}V$, without changing GFR or RBF	↓	0	0
2.50	Increased with decreased $U_{Na}V$, GFR, and RBF	↓	↓	↓

GFR, glomerular filtration rate; RBF, renal blood flow; RSR, renin secretion rate; $U_{Na}V$, urinary sodium excretion.

Reproduced from DiBona GF: Neural control of renal function: Cardiovascular implications. Hypertension 1989;13:539. By permission of the American Heart Association.

Strong stimulation of the sympathetic noradrenergic nerves to the kidneys causes a marked decrease in renal blood flow. This effect is mediated by α_1-adrenergic receptors and to a lesser extent by postsynaptic α_2-adrenergic receptors. Some tonic discharge takes place in the renal nerves at rest in animals and humans. When systemic blood pressure falls, the vasoconstrictor response produced by decreased discharge in the baroreceptor nerves includes renal vasoconstriction. Renal blood flow is decreased during exercise and, to a lesser extent, on rising from the supine position.

AUTOREGULATION OF RENAL BLOOD FLOW

When the kidney is perfused at moderate pressures (90–220 mm Hg in the dog), the renal vascular resistance varies with the pressure so that renal blood flow is

relatively constant **(Figure 37–4)**. Autoregulation of this type occurs in other organs, and several factors contribute to it (see Chapter 32). Renal autoregulation is present in denervated and in isolated, perfused kidneys but is prevented by the administration of drugs that paralyze vascular smooth muscle. It is probably produced in part by a direct contractile response to stretch of the smooth muscle of the afferent arteriole. NO may also be involved. At low perfusion pressures, angiotensin II also appears to play a role by constricting the efferent arterioles, thus maintaining the GFR. This is believed to be the explanation of the renal failure that sometimes develops in patients with poor renal perfusion who are treated with drugs that inhibit angiotensin-converting enzyme.

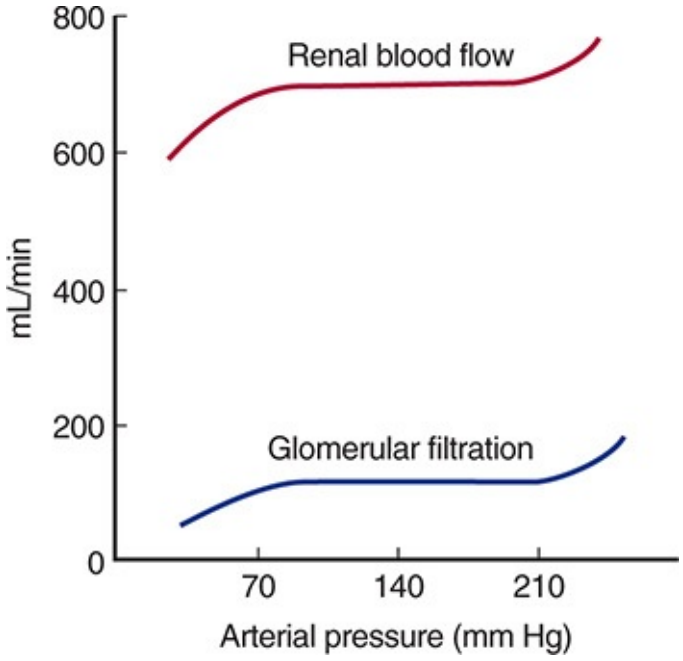


FIGURE 37–4 Autoregulation in the kidneys.

REGIONAL BLOOD FLOW & OXYGEN CONSUMPTION

The main function of the renal cortex is filtration of large volumes of blood through the glomeruli, so it is not surprising that the renal cortical blood flow is relatively great and little oxygen is extracted from the blood. Cortical blood flow is about 5 mL/g of kidney tissue/min (compared with 0.5 mL/g/min in the brain), and the arteriovenous oxygen difference for the whole kidney is only 14 mL/L of blood, compared with 62 mL/L for the brain and 114 mL/L for the heart (see

Table 33–1). The P_{O_2} of the cortex is about 50 mm Hg. On the other hand, maintenance of the osmotic gradient in the medulla requires a relatively low blood flow. It is not surprising, therefore, that the blood flow is about 2.5 mL/g/min in the outer medulla and 0.6 mL/g/min in the inner medulla. However, metabolic work is being done, particularly to reabsorb Na^+ in the thick ascending limb of Henle, so relatively large amounts of O_2 are extracted from the blood in the medulla. The P_{O_2} of the medulla is about 15 mm Hg. This makes the medulla vulnerable to hypoxia if flow is reduced further. NO, prostaglandins, and many cardiovascular peptides in this region function in a paracrine manner to maintain the balance between low blood flow and metabolic needs.

GLOMERULAR FILTRATION

MEASURING GFR

Glomerular filtration rate (GFR) is the amount of plasma ultrafiltrate formed each minute and can be measured in intact experimental animals and humans by measuring the plasma level of a substance and the amount of that substance that is excreted. A substance to be used to measure GFR must be freely filtered through the glomeruli and must be neither secreted nor reabsorbed by the tubules.

In addition to the requirement that it be freely filtered and neither reabsorbed nor secreted in the tubules, a substance suitable for measuring the GFR should be nontoxic and not metabolized by the body. Inulin, a polymer of fructose with a molecular weight of 5200, meets these criteria in humans and most animals and can be used to measure GFR.

Renal plasma clearance is the *volume of plasma* from which a substance is completely removed by the kidney in a given amount of time (usually minutes). The amount of that substance that appears in the urine per unit of time is the result of the renal filtering of a certain number of milliliters of plasma that contained this amount. GFR and clearance are measured in mL/min.

Therefore, if the substance is designated by the letter X, the GFR is equal to the concentration of X in urine (U_X) times the **urine flow** per unit of time ($\dot{V}$) divided by the **arterial plasma level** of X (P_X), or $\dot{V}/P_X$. **This value is called the clearance of X (CX).**

In practice, a loading dose of inulin is administered intravenously, followed by a sustaining infusion to keep the arterial plasma level constant. After the

inulin has equilibrated with body fluids, an accurately timed urine specimen is collected and a plasma sample obtained halfway through the collection. Plasma and urinary inulin concentrations are determined and the clearance is calculated:

$$U_{IN} = 35 \text{ mg/mL}$$
$$\dot{V} = 0.9 \text{ mL/min}$$
$$P_{IN} = 0.25 \text{ mg/mL}$$
$$C_{IN} = \frac{U_{IN}\,\dot{V}}{P_{IN}} = \frac{35 \times 0.9}{0.25}$$
$$C_{IN} = 126 \text{ mL/min}$$

Clearance of creatinine (C_{Cr}) can also be used to determine GFR; however, some creatinine is secreted by the tubules, thus the clearance of creatinine will be slightly higher than inulin. In spite of this, the clearance of endogenous creatinine is a reasonable estimate of GFR as the values agree quite well with the GFR values measured with inulin **(see Table 37–2)**. More common though is the use of **P_{Cr} values as an index of renal function (normal = 1 mg/dL).**

TABLE 37–2 Normal clearance values of different solutes.

Substance	Clearance (mL/min)
Glucose	0
Sodium	0.9
Chloride	1.3
Potassium	12
Phosphate	25
Urea	75
Inulin	125
Creatinine	140
PAH	560

NORMAL GFR

The GFR in a healthy adult of average size is approximately 125 mL/min. Its magnitude correlates fairly well with surface area, but values in women are 10% lower than those in men even after correction for surface area. A rate of 125 mL/min is 7.5 L/h, or 180 L/d, whereas the normal urine volume is about 1 L/d. Thus, 99% or more of the filtrate is normally reabsorbed. At the rate of 125 mL/min, in 1 day the kidneys filter an amount of fluid equal to four times the total body water, 15 times the ECF volume, and 60 times the plasma volume.

CONTROL OF GFR

The factors governing filtration across the glomerular capillaries are the same as those governing filtration across all other capillaries (see Chapter 31), that is, the size of the capillary bed, the permeability of the capillaries, and the hydrostatic and osmotic pressure gradients across the capillary wall. For each nephron:

$$GFR = K_f [(P_{GC} - P_T) - (\pi_{GC} - \pi_T)]$$

where K_f, the glomerular ultrafiltration coefficient, is the product of the glomerular capillary wall hydraulic conductivity (ie, its permeability) and the effective filtration surface area. P_{GC} is the mean hydrostatic pressure in the glomerular capillaries, P_T the mean hydrostatic pressure in the tubule (Bowman's space), π_{GC} the oncotic pressure of the plasma in the glomerular capillaries, and π_T the oncotic pressure of the filtrate in the tubule (Bowman's space).

PERMEABILITY

The permeability of the glomerular capillaries is about 50 times that of the capillaries in skeletal muscle. Neutral substances with effective molecular diameters of less than 4 nm are freely filtered, and the filtration of neutral substances with diameters of more than 8 nm approaches zero. Between these values, filtration is inversely proportional to diameter. However, sialoproteins in the glomerular capillary wall are negatively charged, and studies with negatively and positively charged dextrans indicate that the negative charges repel negatively charged substances in blood, with the result that filtration of anionic substances 4 nm in diameter is less than half that of neutral substances of the same size. This probably explains why albumin, with an effective molecular

diameter of approximately 7 nm, normally has a glomerular concentration only 0.2% of its plasma concentration rather than the higher concentration that would be expected on the basis of diameter alone; circulating albumin is negatively charged. Conversely, filtration of cationic substances is greater than that of neutral substances.

The amount of protein in the urine is normally less than 100 mg/day, and most of this is not filtered but comes from shed tubular cells. The presence of significant amounts of albumin in the urine is called **albuminuria.** In nephritis, the negative charges in the glomerular wall are dissipated, and albuminuria can occur for this reason without an increase in the size of the "pores" in the membrane.

SIZE OF THE CAPILLARY BED

K_f can be altered by the mesangial cells, with contraction of these cells producing a decrease in K_f that is largely due to a reduction in the area available for filtration. Contraction of points where the capillary loops bifurcate probably shifts flow away from some of the loops, and elsewhere, contracted mesangial cells distort and encroach on the capillary lumen. Agents that have been shown to affect the mesangial cells are listed in Table 37–3. Angiotensin II is an important regulator of mesangial contraction, and there are angiotensin II receptors in the glomeruli. In addition, some evidence suggests that mesangial cells make renin.

TABLE 37–3 Agents causing contraction or relaxation of mesangial cells.

Contraction	Relaxation
Endothelins	ANP
Angiotensin II	Dopamine
Vasopressin	PGE_2
Norepinephrine	cAMP
Platelet-activating factor	
Platelet-derived growth factor	
Thromboxane A_2	
PGF_2	
Leukotrienes C_4 and D_4	
Histamine	

HYDROSTATIC & OSMOTIC PRESSURE

The pressure in the glomerular capillaries is higher than that in other capillary beds because the afferent arterioles are short, straight branches of the interlobular arteries. Furthermore, the vessels “downstream” from the glomeruli, the efferent arterioles, have a relatively high resistance. The capillary hydrostatic pressure is opposed by the hydrostatic pressure in Bowman’s capsule. It is also opposed by the oncotic pressure gradient across the glomerular capillaries ($\pi_{GC} - \pi_T$). π_T is normally negligible, and the gradient is essentially equal to the oncotic pressure of the plasma proteins.

The actual pressures in one strain of rats are shown in **Figure 37–5**. The net filtration pressure (P_{UF}) is 15 mm Hg at the afferent end of the glomerular capillaries, but it falls to zero—that is, filtration equilibrium is reached—proximal to the efferent end of the glomerular capillaries. This is because fluid leaves the plasma and the oncotic pressure rises as blood passes through the glomerular capillaries. The calculated change in $\Delta\pi$ along an idealized glomerular capillary is also shown in Figure 37–5. It is apparent that portions of the glomerular capillaries do not normally contribute to the formation of the glomerular ultrafiltrate; that is, exchange across the glomerular capillaries is flow-limited rather than diffusion-limited. It is also apparent that a decrease in the rate of rise of the Δ curve produced by an increase in RPF would increase filtration because it would increase the distance along the capillary in which filtration was taking place.

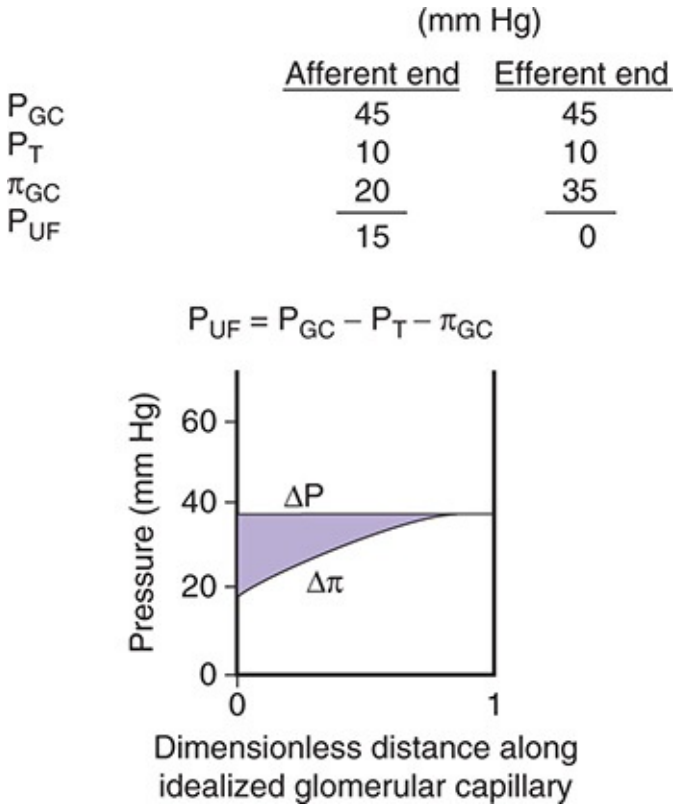


FIGURE 37–5 Hydrostatic pressure (P_{GC}) and osmotic pressure (π_{GC}) in a glomerular capillary in the rat. P_T, pressure in Bowman's capsule; P_{UF}, net filtration pressure. π_T is normally negligible, so $\Delta\pi = \pi_{GC}$. $\Delta P = P_{GC} - P_T$. (Reproduced with permission from Mercer PF, Maddox DA, Brenner BM: Current concepts of sodium chloride and water transport by the mammalian nephron. West J Med 1974; Jan; 120(1):33–45.)

There is considerable species variation in whether filtration equilibrium is reached, and some uncertainties are inherent in the measurement of K_f. It is uncertain whether filtration equilibrium is reached in humans.

CHANGES IN GFR

Variations in the factors discussed in the preceding paragraphs and listed in **Table 37–4** have predictable effects on the GFR. Changes in renal vascular resistance as a result of autoregulation tend to stabilize filtration pressure, but when the mean systemic arterial pressure drops below the autoregulatory range (Figure 37–4), GFR drops sharply. The GFR tends to be maintained when efferent arteriolar constriction is greater than afferent constriction, but either type

of constriction decreases blood flow to the tubules.

TABLE 37–4 Factors affecting the glomerular filtration rate.

Changes in renal blood flow
Changes in glomerular capillary hydrostatic pressure
Changes in systemic blood pressure
Afferent or efferent arteriolar constriction
Changes in hydrostatic pressure in Bowman capsule
Ureteral obstruction
Edema of kidney inside tight renal capsule
Changes in concentration of plasma proteins: dehydration, hypoproteinemia, etc (minor factors)
Changes in K_f
Changes in glomerular capillary permeability
Changes in effective filtration surface area

FILTRATION FRACTION

The ratio of the GFR to the RPF, the **filtration fraction,** is normally 0.16–0.20. The GFR varies less than the RPF. When there is a fall in systemic blood pressure, the GFR falls less than the RPF because of efferent arteriolar constriction, and consequently the filtration fraction rises.

TUBULAR FUNCTION

GENERAL CONSIDERATIONS

The amount of any substance (X) that is filtered is the product of the GFR and

the plasma level of the substance ($C_{ln}P_X$). The tubular cells may add more of the substance to the filtrate (tubular secretion), may remove some or all of the substance from the filtrate (tubular reabsorption), or may do both. The amount of the substance excreted per unit of ($U_X\dot{V}$) time equals the amount filtered plus the **net amount transferred** by the tubules. This latter quantity is conveniently indicated by the symbol T_X **(Figure 37–6)**. The clearance of the substance equals the GFR if there is no net tubular secretion or reabsorption, exceeds the GFR if there is net tubular secretion, and is less than the GFR if there is net tubular reabsorption.

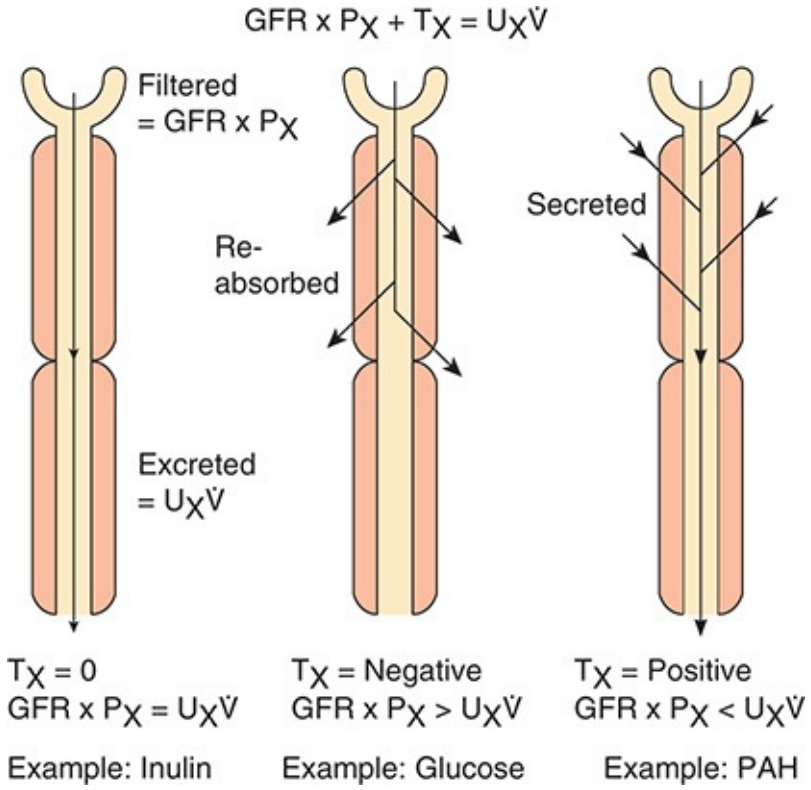


FIGURE 37–6 Tubular function. For explanation of symbols, see text.

Much of the knowledge about glomerular filtration and tubular function has been obtained by using micropuncture techniques. Micropipettes can be inserted into the tubules of the living kidney and the composition of aspirated tubular fluid determined by the use of microchemical techniques. In addition, two pipettes can be inserted in a tubule and the tubule perfused in vivo. Alternatively, isolated perfused segments of tubules can be studied in vitro, and tubular cells can be grown and studied in culture.

MECHANISMS OF TUBULAR REABSORPTION & SECRETION

Small proteins and some peptide hormones are reabsorbed in the proximal tubules by endocytosis. Other substances are secreted or reabsorbed in the tubules by passive diffusion between cells and through cells by facilitated diffusion down chemical or electrical gradients or active transport against such gradients. Movement is by way of ion channels, exchangers, cotransporters, and pumps. Many of these have now been cloned, and their regulation is being studied.

It is important to note that the pumps and other transporters in the luminal membrane are different from those in the basolateral membrane. As was discussed for the gastrointestinal epithelium, it is this polarized distribution that makes possible net movement of solutes across the epithelia.

Like transport systems elsewhere, renal active transport systems have a maximal rate, or **transport maximum (Tm),** at which they can transport a particular solute. Thus, the amount of a particular solute transported is proportional to the amount present up to the Tm for the solute, but at higher concentrations, the transport mechanism is **saturated** and there is no appreciable increment in the amount transported. However, the Tms for some systems are high, and it is difficult to saturate them.

It should also be noted that the tubular epithelium, like that of the small intestine, is a **leaky epithelium** in that the tight junctions between cells permit the passage of some water and electrolytes. The degree to which leakage by this **paracellular pathway** contributes to the net flux of fluid and solute into and out of the tubules is controversial since it is difficult to measure, but current evidence seems to suggest that it is a significant factor in the proximal tubule. One indication of this is that paracellin-1, a protein localized to tight junctions, is related to Mg^{2+} reabsorption, and a loss-of-function mutation of its gene causes severe Mg^{2+} and Ca^{2+} loss in the urine.

NA^{+} REABSORPTION

The reabsorption of Na^{+} and Cl^{-} plays a major role in body electrolyte and water homeostasis. In addition, Na^{+} transport is coupled to the movement of H^{+}, glucose, amino acids, organic acids, phosphate, and other electrolytes and substances across the tubule walls. The principal cotransporters and exchangers

in the various parts of the nephron are listed in **Table 37–5**. In the proximal tubules, the thick portion of the ascending limb of the loop of Henle, the distal tubules, and the collecting ducts, Na^+ moves by cotransport or exchange from the tubular lumen into the tubular epithelial cells down its concentration and electrical gradients, and is then actively pumped from these cells into the interstitial space. Na^+ is pumped into the interstitium by Na, K ATPase in the basolateral membrane. Thus, Na^+ is actively transported out of all parts of the renal tubule except the thin portions of the loop of Henle. The operation of the ubiquitous Na^+ pump is considered in detail in Chapter 2. It extrudes three Na^+ in exchange for two K^+ that are pumped into the cell.

TABLE 37–5 Transport proteins involved in the movement of Na^+ and Cl^- across the apical membranes of renal tubular cells.[a]

Site	Apical Transporter	Function
Proximal tubule	Na/glucose CT	Na^+ uptake, glucose uptake
	Na^+/P_i CT	Na^+ uptake, P_i uptake
	Na^+ amino acid CT	Na^+ uptake, amino acid uptake
	Na/lactate CT	Na^+ uptake, lactate uptake
	Na/H exchanger	Na^+ uptake, H^+ extrusion
	Cl/base exchanger	Cl^- uptake
Thick ascending limb	Na–K–2Cl CT	Na^+ uptake, Cl^- uptake, K^+ uptake
	Na/H exchanger	Na^+ uptake, H^+ extrusion
	K^+ channels	K^+ extrusion (recycling)
Distal convoluted tubule	NaCl CT	Na^+ uptake, Cl^- uptake
Collecting duct	Na^+ channel (ENaC)	Na^+ uptake

[a]Uptake indicates movement from tubular lumen to cell interior, extrusion is movement from cell interior to tubular lumen. CT, cotransporter; P_i, inorganic phosphate.

Data from Schnermann JB, Sayegh EI: *Kidney Physiology*. Lippincott-Raven, 1998.

The tubular cells along the nephron are connected by tight junctions at their luminal edges, but there is space between the cells along the rest of their lateral borders. Much of the Na^+ is actively transported into these **lateral intercellular spaces** (Figure 37–7).

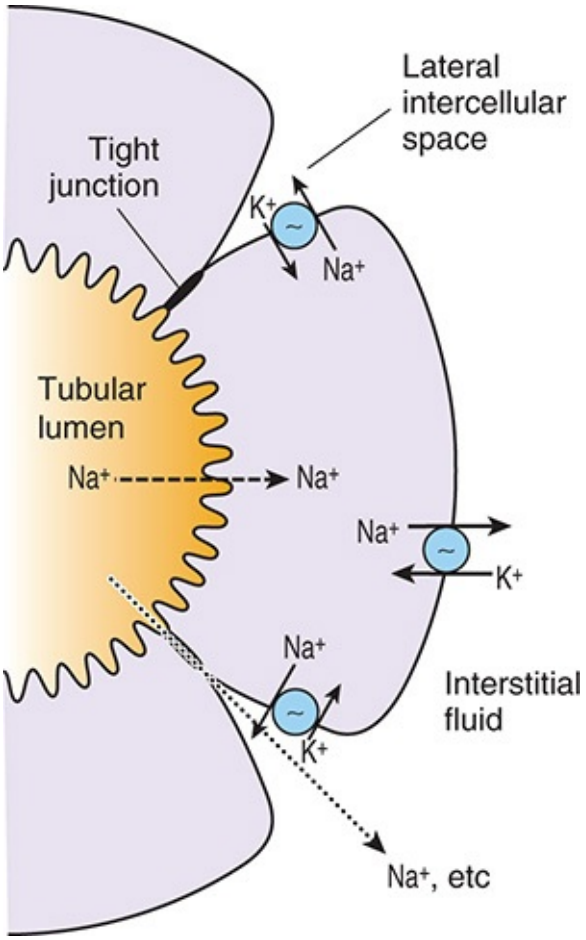


FIGURE 37–7 Mechanism of Na^+ reabsorption in the proximal tubule. Na^+ moves out of the tubular lumen by cotransport and exchange mechanism through the apical membrane of the tubule (dashed line). The Na^+ is then actively transported into the interstitial fluid by Na, K ATPase in the basolateral membrane (solid lines). K^+ enters the interstitial fluid via K^+ channels. A small amount of Na^+, other solutes, and H_2O reenter the tubular lumen by passive transport through the tight junctions (dotted lines).

Normally about 60% of the filtered Na^+ is reabsorbed in the proximal tubule, primarily by Na–H exchange. Another 30% is absorbed via the Na–2Cl–K cotransporter in the thick ascending limb of the loop of Henle. In both of these

segments of the nephron, passive paracellular movement of Na^+ also contributes to overall Na^+ reabsorption. In the distal convoluted tubule 7% of the filtered Na^+ is absorbed by the Na–Cl cotransporter. The remainder of the filtered Na^+, about 3%, is absorbed via ENaC channels in the collecting ducts, and this is the portion that is regulated by aldosterone to permit homeostatic adjustments in Na^+ balance.

GLUCOSE REABSORPTION

Glucose, amino acids, and bicarbonate are reabsorbed along with Na^+ in the early portion of the proximal tubule **(Figure 37–8)**. Glucose is typical of substances removed from the urine by secondary active transport. It is filtered at a rate of approximately 100 mg/min (80 mg/dL of plasma × 125 mL/min). Essentially all of the glucose is reabsorbed, and no more than a few milligrams appear in the urine per 24 h. The amount reabsorbed is proportional to the amount filtered and hence to the plasma glucose level (P_G) times the GFR up to the transport maximum (Tm_G). When the Tm_G is exceeded, the amount of glucose in the urine rises **(Figure 37–9)**. The Tm_G is about 375 mg/min in men and 300 mg/min in women.

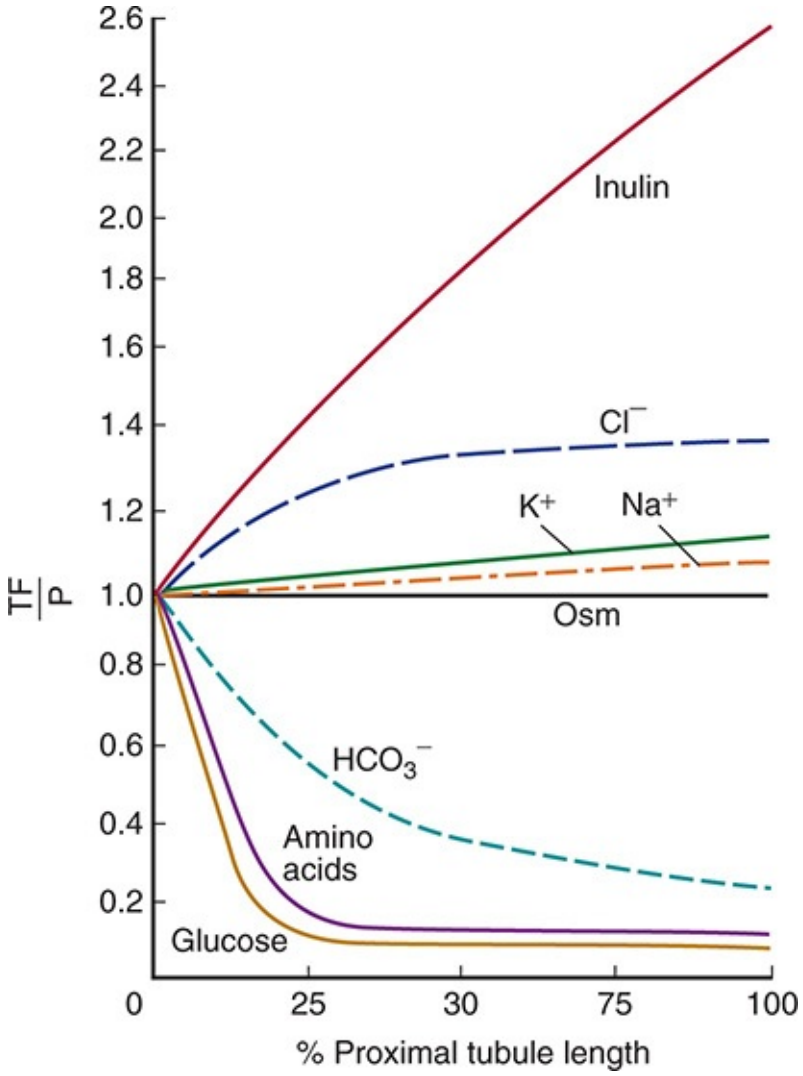


FIGURE 37–8 Reabsorption of various solutes in the proximal tubule. TF/P, tubular fluid:plasma concentration ratio. (Used with permission of FC Rector Jr.)

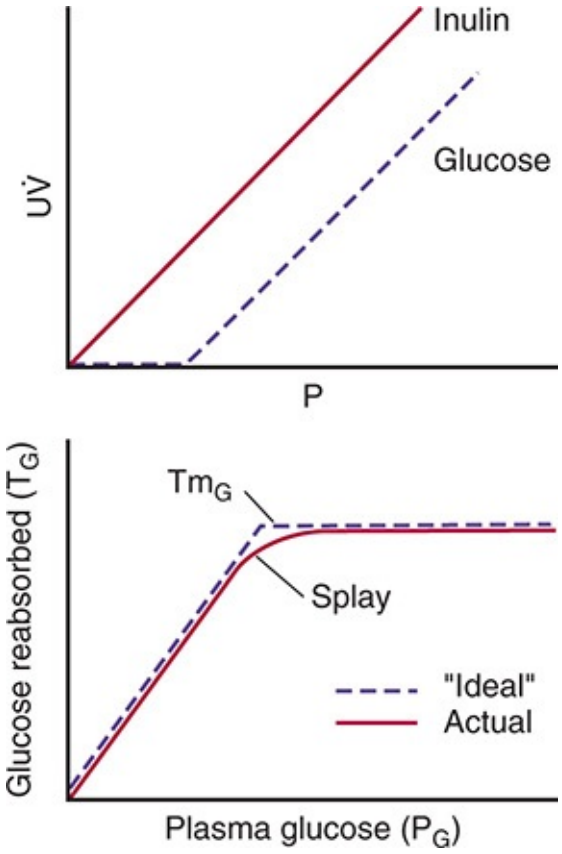


FIGURE 37–9 Renal glucose transport. Top: Relation between the plasma level (P) and excretion (UV) of glucose and inulin. **Bottom:** Relation between the plasma glucose level (P_G) and amount of glucose reabsorbed (T_G).

The **renal threshold** for glucose is the plasma level at which the glucose first appears in the urine in more than the normal minute amounts. One would predict that the renal threshold would be about 300 mg/dL, that is, 375 mg/min (Tm_G) divided by 125 mL/min (GFR). However, the actual renal threshold is about 200 mg/dL of arterial plasma, which corresponds to a venous level of about 180 mg/dL. Figure 37–9 shows why the actual renal threshold is less than the predicted threshold. The "ideal" curve shown in this diagram would be obtained if the Tm_G in all the tubules was identical and if all the glucose were removed from each tubule when the amount filtered was below the Tm_G. This is not the case, and in humans, for example, the actual curve is rounded and deviates considerably from the "ideal" curve. This deviation is called **splay.** The magnitude of the splay is inversely proportional to the avidity with which the transport mechanism binds the substance it transports.

GLUCOSE TRANSPORT MECHANISM

Glucose reabsorption in the kidneys is similar to glucose reabsorption in the intestine (see Chapter 26). Glucose and Na^+ bind to the sodium-dependent glucose transporter (SGLT)-2 in the apical membrane, and glucose is carried into the cell as Na^+ moves down its electrical and chemical gradient. The Na^+ is then pumped out of the cell into the interstitium, and the glucose exits by facilitated diffusion via glucose transporter (GLUT)-2 into the interstitial fluid. At least in the rat, there is some transport by SGLT-1 and GLUT-1 as well.

SGLT-2 specifically binds the D isomer of glucose, and the rate of transport of D-glucose is many times greater than that of L-glucose. Glucose transport in the kidneys is inhibited, as it is in the intestine, by the plant glucoside **phlorhizin,** which competes with D-glucose for binding to the carrier.

ADDITIONAL EXAMPLES OF SECONDARY ACTIVE TRANSPORT

Like glucose reabsorption, amino acid reabsorption is most marked in the early portion of the proximal convoluted tubule. Absorption in this location resembles absorption in the intestine (see Chapter 26). The main carriers in the apical membrane cotransport Na^+, whereas the carriers in the basolateral membranes are not Na^+-dependent. Na^+ is pumped out of the cells by Na, K ATPase and the amino acids leave by passive or facilitated diffusion to the interstitial fluid.

Some Cl^- is reabsorbed with Na^+ and K^+ in the thick ascending limb of the loop of Henle. In addition, two members of the family of **Cl channels** have been identified in the kidney. Mutations in the gene for one of the renal channels is associated with Ca^{2+}-containing kidney stones and hypercalciuria **(Dent disease),** but how tubular transport of Ca^{2+} and Cl^- are linked is still unsettled.

PAH TRANSPORT

The dynamics of PAH transport illustrate the operation of the active transport mechanisms that secrete substances into the tubular fluid **(Clinical Box 37–1)**. The filtered load of PAH is a linear function of the plasma level, but PAH secretion increases as P_{PAH} rises only until a maximal secretion rate (Tm_{PAH}) is reached **(Figure 37–10)**. When P_{PAH} is low, C_{PAH} is high; but as P_{PAH} rises

above Tm_{PAH}, C_{PAH} falls progressively. It eventually approaches the clearance of inulin (C_{In}) **(Figure 37–11)** because the amount of PAH secreted becomes a smaller and smaller fraction of the total amount excreted. Conversely, the clearance of glucose is essentially zero at P_G levels below the renal threshold; but above the threshold, C_G rises to approach C_{In} as P_G is raised.

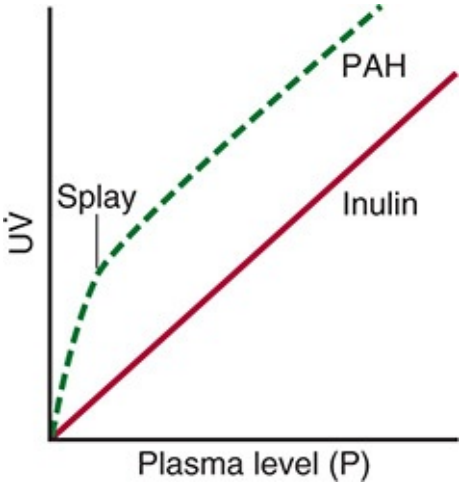


FIGURE 37–10 Relation between plasma levels and excretion ($U\dot{V}$) of PAH and inulin.

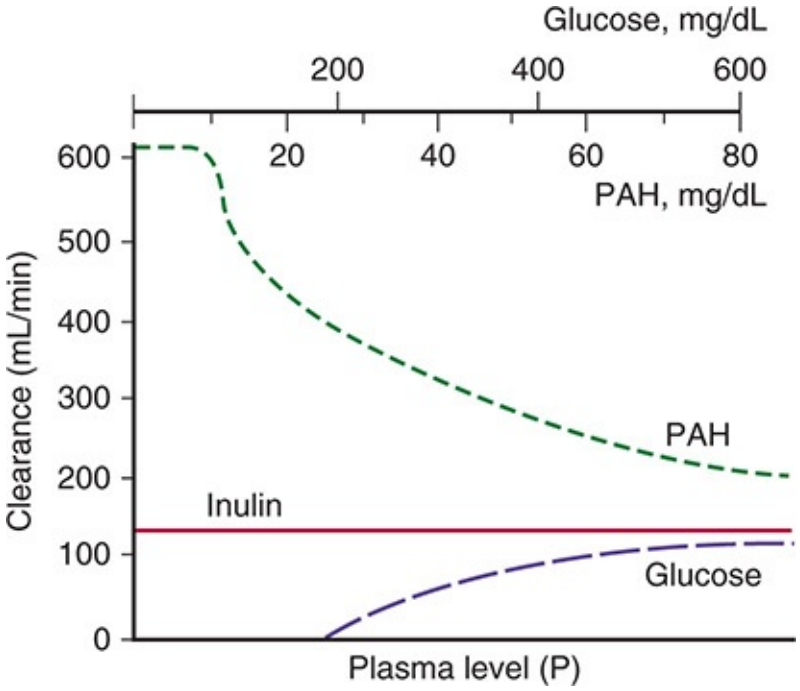


FIGURE 37–11 Clearance of inulin, glucose, and PAH at various plasma

levels of each substance in humans.

CLINICAL BOX 37–1

Substances Secreted by the Tubules

Derivatives of hippuric acid in addition to PAH, phenol red and other sulfonphthalein dyes, penicillin, and a variety of iodinated dyes are actively secreted into the tubular fluid. Substances that are normally produced in the body and secreted by the tubules include various ethereal sulfates, steroid and other glucuronides, and 5-hydroxyindoleacetic acid, the principal metabolite of serotonin.

THERAPEUTIC HIGHLIGHTS

The loop diuretic, furosemide, and the thiazide diuretics are organic anions that gain access to their tubular sites of action (thick ascending limb and distal convoluted tubule, respectively) when they are secreted into the urine by the proximal tubule.

The use of C_{PAH} to measure ERPF is discussed above.

TUBULOGLOMERULAR FEEDBACK & GLOMERULOTUBULAR BALANCE

Signals from the renal tubule in each nephron feed back to affect filtration in its glomerulus. As the rate of flow through the ascending limb of the loop of Henle and first part of the distal tubule increases, glomerular filtration in the same nephron decreases, and, conversely, a decrease in flow increases the GFR **(Figure 37–12)**. This process, which is called **tubuloglomerular feedback,** tends to maintain the constancy of the load delivered to the distal tubule.

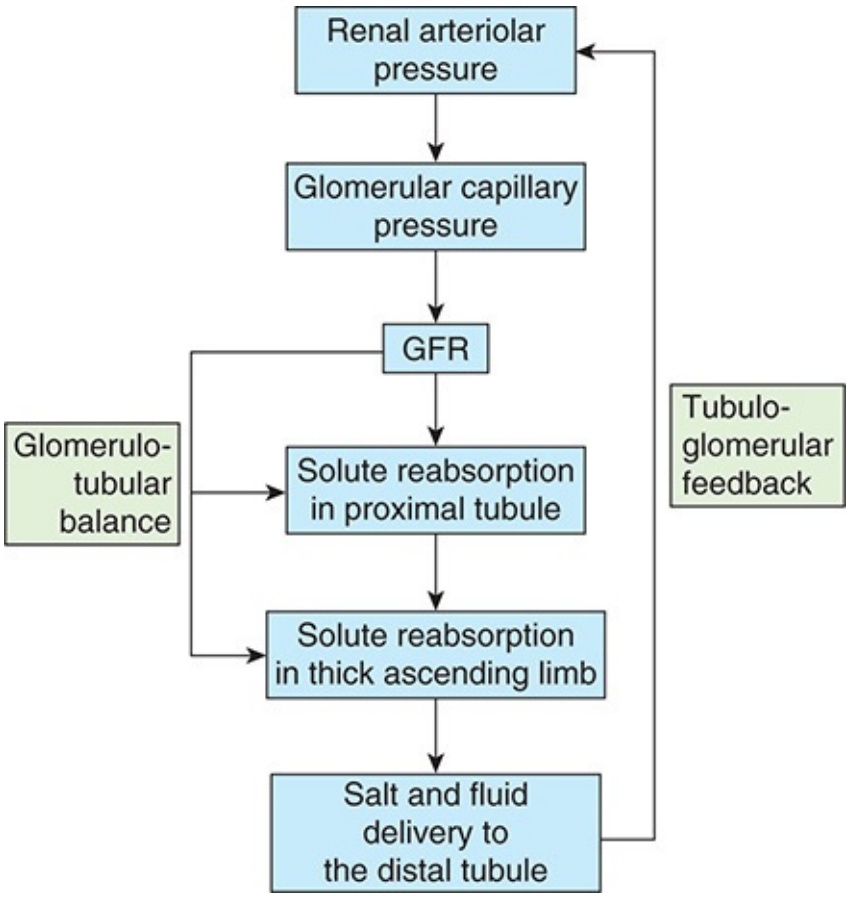


FIGURE 37–12 Mechanisms of glomerulotubular balance and tubuloglomerular feedback.

The sensor for this response is the **macula densa.** The amount of fluid entering the distal tubule at the end of the thick ascending limb of the loop of Henle depends on the amount of Na^+ and Cl^- in it. The Na^+ and Cl^- enter the macula densa cells via the Na–K–2Cl cotransporter in their apical membranes. The increased Na^+ causes increased Na, K ATPase activity and the resultant increased ATP hydrolysis causes more adenosine to be formed. Presumably, adenosine is secreted from the basal membrane of the cells. It acts via adenosine A_1 receptors on the macula densa cells to increase their release of Ca^{2+} to the vascular smooth muscle in the afferent arterioles. This causes afferent vasoconstriction and a resultant decrease in GFR. Presumably, a similar mechanism generates a signal that decreases renin secretion by the adjacent juxtaglomerular cells in the afferent arteriole (see Chapter 38), but this remains unsettled.

Conversely, an increase in GFR causes an increase in the reabsorption of

solutes, and consequently of water, primarily in the proximal tubule, so that in general the percentage of the solute reabsorbed is held constant. This process is called **glomerulotubular balance,** and it is particularly prominent for Na^+. The change in Na^+ reabsorption occurs within seconds after a change in filtration, so it seems unlikely that an extrarenal humoral factor is involved. Alternatively, one mediating factor is the oncotic pressure in the peritubular capillaries. When the GFR is high, there is a relatively large increase in the oncotic pressure of the plasma leaving the glomeruli via the efferent arterioles and hence in their capillary branches. This increases the reabsorption of Na^+ from the tubule. However, other as yet unidentified intrarenal mechanisms are likely also involved.

WATER TRANSPORT

Normally, 180 L of fluid is filtered through the glomeruli each day, while the average daily urine volume is about 1 L. The same load of solute can be excreted per 24 h in a urine volume of 500 mL with a concentration of 1400 mOsm/kg or in a volume of 23.3 L with a concentration of 30 mOsm/kg **(Table 37–6)**. These figures demonstrate two important facts: First, at least 87% of the filtered water is reabsorbed, even when the urine volume is 23 L; and second, the reabsorption of the remainder of the filtered water can be varied without affecting total solute excretion. Therefore, when the urine is concentrated, water is retained in excess of solute; and when it is dilute, water is lost from the body in excess of solute. Both facts have great importance in the regulation of the osmolality of the body fluids. A key regulator of water output is vasopressin acting on the collecting ducts.

TABLE 37–6 Alterations in water metabolism produced by vasopressin in humans. In each case, the osmotic load excreted is 700 mOsm/day.

	GFR (mL/min)	Percentage of Filtered Water Reabsorbed	Urine Volume (L/day)	Urine Concentration (mOsm/kg H_2O)	Gain or Loss of Water in Excess of Solute (L/day)
Urine isotonic to plasma	125	98.7	2.4	290	…
Vasopressin (maximal antidiuresis)	125	99.7	0.5	1400	1.9 gain
No vasopressin ("complete" diabetes insipidus)	125	87.1	23.3	30	20.9 loss

AQUAPORINS

Rapid diffusion of water across cell membranes depends on the presence of water channels, integral membrane proteins called **aquaporins.** To date, 13 aquaporins have been cloned; however, only four aquaporins (aquaporin-1, -2, -3, and -4) play a key role in the kidney. The roles played by aquaporin-1 and aquaporin-2 in renal water transport are discussed below.

PROXIMAL TUBULE

Active transport of many substances occurs from the fluid in the proximal tubule, but micropuncture studies have shown that the fluid remains essentially iso-osmotic until the end of the proximal tubule (Figure 37–8). **Aquaporin-1** is localized to both the basolateral and apical membrane of the proximal tubules and its presence allows water to move rapidly out of the tubule along the osmotic gradients set up by active transport of solutes, and isotonicity is maintained. Because the ratio of the concentration in tubular fluid to the concentration in plasma (TF/P) of the nonreabsorbable substance inulin is 2.5 to 3.3 at the end of the proximal tubule, it follows that 60–70% of the filtered solute and 60–70% of the filtered water have been removed by the time the filtrate reaches this point **(Figure 37–13)**.

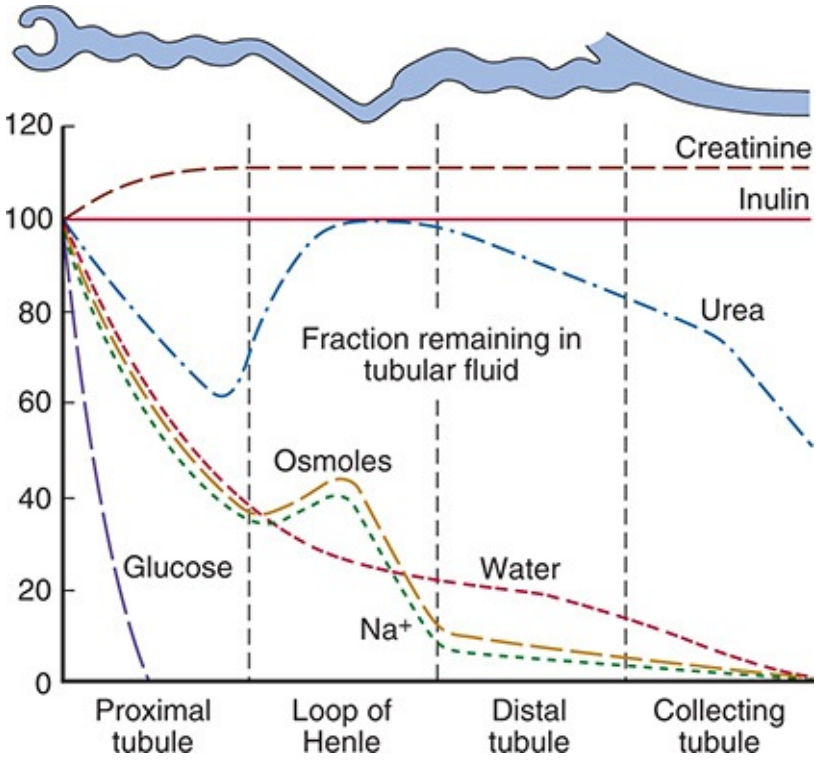


FIGURE 37–13 Changes in the percentage of the filtered amount of substances remaining in the tubular fluid along the length of the nephron in the presence of vasopressin. (Modified with permission from Sullivan LP, Grantham JJ: *Physiology of the Kidney*, 2nd ed. Lea & Febiger, 1982.)

When aquaporin-1 was knocked out in mice, proximal tubular water permeability was reduced by 80%. When the mice were subjected to dehydration, their urine osmolality did not increase (<700 mOsm/kg), even though other renal aquaporins were present. In humans with mutations that eliminate aquaporin-1 activity, the defect in water homeostasis is not as severe, though their response to dehydration is defective.

LOOP OF HENLE

As noted above, the loops of Henle of the juxtamedullary nephrons dip deeply into the medullary pyramids before draining into the distal convoluted tubules in the cortex, and all the collecting ducts descend back through the medullary pyramids to drain at the tips of the pyramids into the renal pelvis. There is a graded increase in the osmolality of the interstitium of the pyramids in humans: The osmolality at the tips of the papillae can reach about 1200 mOsm/kg of H_2O, approximately four times that of plasma. The descending limb of the loop of Henle is permeable to water, due to the presence of **aquaporin-1** in both the apical and basolateral membranes, but the ascending limb is impermeable to water. Na^+, K^+, and Cl^- are cotransported out of the thick segment of the ascending limb. Therefore, the fluid in the descending limb of the loop of Henle becomes **hypertonic** as water moves out of the tubule into the hypertonic interstitium. In the ascending limb it becomes more dilute because of the movement of Na^+ and Cl^- out of the tubular lumen, and when fluid reaches the top of the ascending limb (called the **diluting segment**) it is now **hypotonic** to plasma. In passing through the descending loop of Henle, another 15% of the filtered water is removed, so approximately 20% of the filtered water enters the distal tubule, and the TF/P of inulin at this point is about 5.

In the thick ascending limb, a carrier cotransports one Na^+, one K^+, and $2Cl^-$ from the tubular lumen into the tubular cells. This is another example of secondary active transport; the Na^+ is actively transported from the cells into the interstitium by Na, K ATPase in the basolateral membranes of the cells, keeping the intracellular Na^+ low. The Na–K–2Cl cotransporter has 12 transmembrane domains with intracellular amino and carboxyl terminals. It is a member of a

family of transporters found in many other locations, including salivary glands, the gastrointestinal tract, and the airways.

The K^+ diffuses back into the tubular lumen and back into the interstitium via ROMK and other K^+ channels. The Cl^- moves into the interstitium via ClC–Kb channels **(Figure 37–14)**.

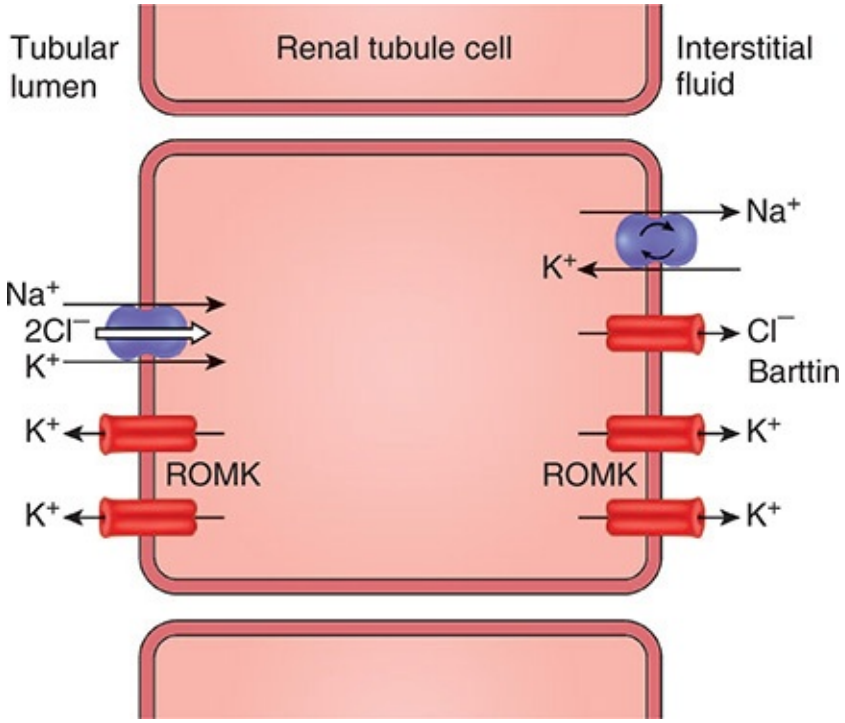


FIGURE 37–14 NaCl transport in the thick ascending limb of the loop of Henle. The Na–K–2Cl cotransporter moves these ions into the tubular cell by secondary active transport. Na^+ is transported out of the cell into the interstitium by Na, K ATPase in the basolateral membrane of the cell. Cl^- exits in basolateral ClC–Kb Cl^- channels. Barttin, a protein in the cell membrane, is essential for normal ClC–Kb function. K^+ moves from the cell to the interstitium and the tubular lumen by ROMK and other K^+ channels **(Clinical Box 37–2)**.

CLINICAL BOX 37–2

Genetic Mutations in Renal Transporters

Mutations of individual genes for many renal sodium transporters and channels cause specific syndromes such as Bartter syndrome, Liddle syndrome, and Dent disease. A large number of mutations have been

described.

Bartter syndrome is a rare but interesting condition that is due to defective transport in the thick ascending limb. It is characterized by chronic Na^+ loss in the urine, with resultant hypovolemia causing stimulation of renin and aldosterone secretion without hypertension, plus hyperkalemia and alkalosis. The condition can be caused by loss-of-function mutations in the gene for any of four key proteins: the Na–K–2Cl cotransporter, the ROMK K^+ channel, the ClC–Kb Cl^- channel, or **barttin,** a recently described integral membrane protein that is necessary for the normal function of ClC–Kb Cl^- channels.

The stria vascularis in the inner ear is responsible for maintaining the high K^+ concentration in the scala media that is essential for normal hearing. It contains both ClC–Kb and ClC–Ka Cl^- channels. Bartter syndrome associated with mutated ClC–Kb channels is not associated with deafness because the Clc–Ka channels can carry the load. However, both types of Cl^- channels are barttin-dependent, so patients with Bartter syndrome due to mutated barttin are also deaf.

Another interesting example involves the proteins polycystin-1 (PKD-1) and polycystin-2 (PKD-2). PKD-1 appears to be a Ca^{2+} receptor that activates a nonspecific ion channel associated with PKD-2. The normal function of this apparent ion channel is unknown, but both proteins are abnormal in **autosomal dominant polycystic kidney disease,** in which the renal parenchyma is progressively replaced by fluid-filled cysts until there is complete renal failure.

DISTAL TUBULE

The distal tubule, particularly its first part, is in effect an extension of the thick segment of the ascending limb. It is relatively impermeable to water, and continued removal of the solute in excess of solvent further dilutes the tubular fluid.

COLLECTING DUCTS

The collecting ducts have two portions: a cortical portion and a medullary portion. The changes in osmolality and volume in the collecting ducts depend on

the amount of vasopressin acting on the ducts. This antidiuretic hormone from the posterior pituitary gland increases the permeability of the collecting ducts to water. The key to the action of vasopressin on the collecting ducts is aquaporin-2. Unlike the other aquaporins, this aquaporin is stored in vesicles in the cytoplasm of principal cells. Vasopressin causes rapid insertion of these vesicles into the apical membrane of cells. The effect is mediated via the vasopressin V_2 receptor, cyclic adenosine 5-monophosphate (cAMP), and protein kinase A. Cytoskeletal elements are involved, including microtubule-based motor proteins (dynein and dynactin) as well as actin filament-binding proteins such as myosin-1.

In the presence of enough vasopressin to produce maximal antidiuresis, water moves out of the hypotonic fluid entering the cortical collecting ducts into the interstitium of the cortex, and the tubular fluid becomes isotonic. In this manner, as much as 10% of the filtered water is removed. The isotonic fluid then enters the medullary collecting ducts with a TF/P inulin of about 20. An additional 4.7% or more of the filtrate is reabsorbed into the hypertonic interstitium of the medulla, producing a concentrated urine with a TF/P inulin of over 300. In humans, the osmolality of urine may reach 1400 mOsm/kg of H_2O, almost five times the osmolality of plasma, with a total of 99.7% of the filtered water being reabsorbed (Table 37–6). In other species, the ability to concentrate urine is even greater. Maximal urine osmolality is about 2500 mOsm/kg in dogs, about 3200 mOsm/kg in laboratory rats, and as high as 5000 mOsm/kg in certain desert rodents.

When vasopressin is absent, the collecting duct epithelium is relatively impermeable to water. The fluid therefore remains hypotonic, and large amounts flow into the renal pelvis. In humans, the urine osmolality may be as low as 30 mOsm/kg of H_2O. The impermeability of the distal portions of the nephron is not absolute; along with the salt that is pumped out of the collecting duct fluid, about 2% of the filtered water is reabsorbed in the absence of vasopressin. However, as much as 13% of the filtered water may be excreted, and urine flow may reach 15 mL/min or more.

THE COUNTERCURRENT MECHANISM

The concentrating mechanism depends on the maintenance of a gradient of **increasing osmolality** along the medullary pyramids. This gradient is produced by the operation of the loops of Henle as **countercurrent multipliers** and

maintained by the operation of the vasa recta as **countercurrent exchangers.** A countercurrent system is a system in which the inflow runs parallel to, counter to, and in close proximity to the outflow for some distance. This occurs for both the loops of Henle and the vasa recta in the renal medulla (Figure 37–3).

The operation of each loop of Henle as a countercurrent multiplier depends on the high permeability of the thin descending limb to water (via aquaporin-1), the active transport of Na^+ and Cl^- out of the thick ascending limb, and the inflow of tubular fluid from the proximal tubule, with outflow into the distal tubule. The process can be explained using hypothetical steps leading to the normal equilibrium condition, although the steps do not occur in vivo. It is also important to remember that the equilibrium is maintained unless the osmotic gradient is washed out. These steps are summarized in **Figure 37–15** for a cortical nephron with no thin ascending limb. Assume first a condition in which osmolality is 300 mOsm/kg of H_2O throughout the descending and ascending limbs and the medullary interstitium (Figure 37–15A). Assume in addition that the pumps in the thick ascending limb can pump 100 mOsm/kg of Na^+ and Cl^- from the tubular fluid to the interstitium, increasing interstitial osmolality to 400 mOsm/kg of H_2O. Water then moves out of the thin descending limb, and its contents equilibrate with the interstitium (Figure 37–15B). However, fluid containing 300 mOsm/kg of H_2O is continuously entering this limb from the proximal tubule (Figure 37–15C), so the gradient against which the Na^+ and Cl^- are pumped is reduced and more enters the interstitium (Figure 37–15D). Meanwhile, hypotonic fluid flows into the distal tubule, and isotonic and subsequently hypertonic fluid flows into the ascending thick limb. The process keeps repeating, and the final result is a gradient of osmolality from the top to the bottom of the loop.

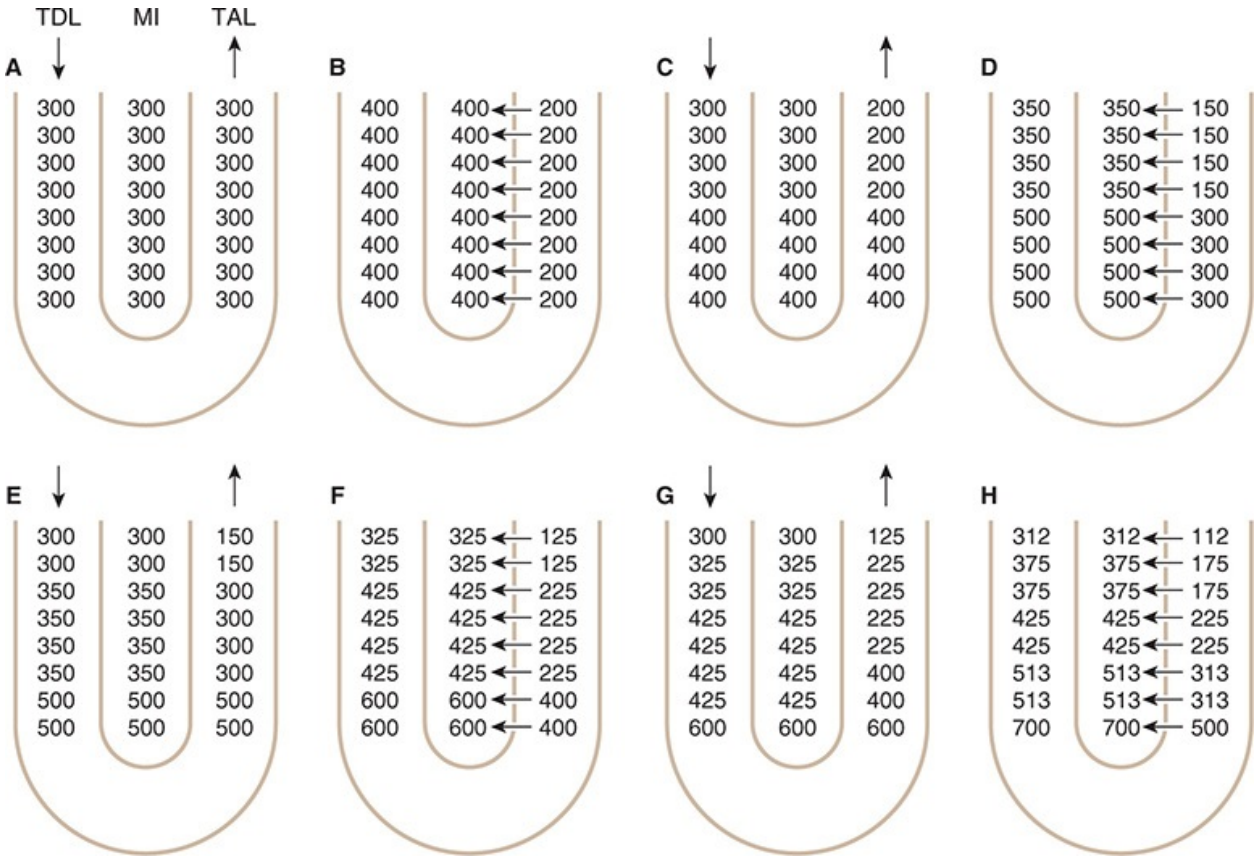


FIGURE 37–15 Operation of the loop of Henle as a countercurrent multiplier producing a gradient of hyperosmolarity in the medullary interstitium (MI). The process of generation of the gradient is illustrated as occurring in hypothetical steps, starting at A, where osmolality in both limbs and the interstitium is 300 mOsm/kg of water. The pumps in the thick ascending limb move Na^+ and Cl^- into the interstitium, increasing its osmolality to 400 mOsm/kg, and this equilibrates with the fluid in the thin descending limb. However, isotonic fluid continues to flow into the thin descending limb and hypotonic fluid out of the thick ascending limb. Continued operation of the pumps makes the fluid leaving the thick ascending limb even more hypotonic, while hypertonicity accumulates at the apex of the loop. TAL, thick ascending limb; TDL, thin descending limb.

In juxtamedullary nephrons with longer loops and thin ascending limbs, the osmotic gradient is spread over a greater distance and the osmolality at the tip of the loop is greater. This is because the thin ascending limb is relatively impermeable to water but permeable to Na^+ and Cl^-. Therefore, Na^+ and Cl^- move down their concentration gradients into the interstitium, and there is additional passive countercurrent multiplication. The greater the length of the

loop of Henle, the greater the osmolality that can be reached at the tip of the medulla.

The osmotic gradient in the medullary pyramids would not last long if the Na^+ and urea in the interstitial spaces were removed by the circulation. These solutes remain in the pyramids primarily because the vasa recta operate as countercurrent exchangers **(Figure 37–16)**. The solutes diffuse out of the vessels conducting blood toward the cortex and into the vessels descending into the pyramid. Conversely, water diffuses out of the descending vessels and into the fenestrated ascending vessels. Therefore, the solutes tend to recirculate in the medulla and water tends to bypass it, so that hypertonicity is maintained. The water removed from the collecting ducts in the pyramids is also removed by the vasa recta and enters the general circulation. Countercurrent exchange is a passive process; it depends on movement of water and could not maintain the osmotic gradient along the pyramids if the process of counter-current multiplication in the loops of Henle were to cease.

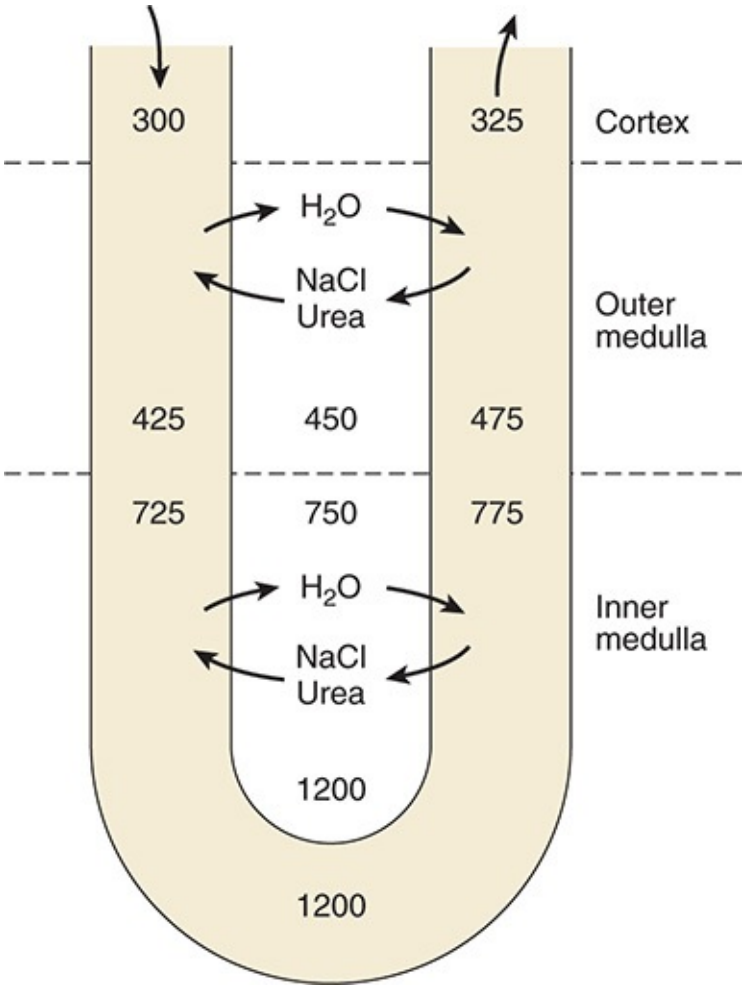


FIGURE 37–16 Operation of the vasa recta as countercurrent exchangers in the kidney. NaCl and urea diffuse out of the ascending limb of the vessel and into the descending limb, whereas water diffuses out of the descending and into the ascending limb of the vascular loop. (Modified with permission from Pitts RF: *Physiology of the Kidney and Body Fluid*, 3rd ed. Chicago: Yearbook Medical Publications, 1974.)

It is worth noting that there is a very large osmotic gradient in the loop of Henle and, in the presence of vasopressin, in the collecting ducts. It is the countercurrent system that makes this gradient possible by spreading it along a system of tubules 1 cm or more in length, rather than across a single layer of cells that is only a few micrometers thick. There are other examples of the operation of countercurrent exchangers in animals. One is the heat exchange between the arteries and venae comitantes of the limbs. To a minor degree in humans, but to a major degree in mammals living in cold water, heat is transferred from the arterial blood flowing into the limbs to the adjacent veins draining blood back into the body, making the tips of the limbs cold while conserving body heat (see Chapter 33).

ROLE OF UREA

Urea contributes to the establishment of the osmotic gradient in the medullary pyramids and to the ability to form a concentrated urine in the collecting ducts. Urea transport is mediated by urea transporters, presumably by facilitated diffusion. There are at least four isoforms of the transport protein UT-A in the kidneys (UT-A1 to UT-A4); UT-B is found in erythrocytes and in the descending limbs of the vasa recta. Urea transport in the collecting duct is mediated by UT-A1 and UT-A3, and both are regulated by vasopressin. During antidiuresis, when vasopressin is high, the amount of urea deposited in the medullary interstitium increases, thus increasing the concentrating capacity of the kidney. In addition, the amount of urea in the medullary interstitium and, consequently, in the urine varies with the amount of urea filtered, and this in turn varies with the dietary intake of protein. Therefore, a high-protein diet increases the ability of the kidneys to concentrate the urine and a low-protein diet reduces the kidneys' ability to concentrate the urine.

OSMOTIC DIURESIS

The presence of large quantities of unreabsorbed solutes in the renal tubules causes an increase in urine volume called **osmotic diuresis.** Solutes that are not reabsorbed in the proximal tubules exert an appreciable osmotic effect as the volume of tubular fluid decreases and their concentration rises. Therefore, they "hold water in the tubules." In addition, the concentration gradient against which Na^+ can be pumped out of the proximal tubules is limited. Normally, the movement of water out of the proximal tubule prevents any appreciable gradient from developing, but Na^+ concentration in the fluid falls when water reabsorption is decreased because of the presence in the tubular fluid of increased amounts of unreabsorbable solutes. The limiting concentration gradient is reached, and further proximal reabsorption of Na^+ is prevented; more Na^+ remains in the tubule, and water stays with it. The result is that the loop of Henle is presented with a greatly increased volume of isotonic fluid. This fluid has a decreased Na^+ concentration, but the total amount of Na^+ reaching the loop per unit time is increased. In the loop, reabsorption of water and Na^+ is decreased because the medullary hypertonicity is decreased. The decrease is due primarily to decreased reabsorption of Na^+, K^+, and Cl^- in the ascending limb of the loop because the limiting concentration gradient for Na^+ reabsorption is reached. More fluid passes through the distal tubule, and because of the decrease in the osmotic gradient along the medullary pyramids, less water is reabsorbed in the collecting ducts. The result is a marked increase in urine volume and excretion of Na^+ and other electrolytes.

Osmotic diuresis is produced by the administration of compounds such as mannitol and related polysaccharides that are filtered but not reabsorbed. It is also produced by naturally occurring substances when they are present in amounts exceeding the capacity of the tubules to reabsorb them. For example, in **diabetes mellitus,** if blood glucose is high, glucose in the glomerular filtrate is high, thus the filtered load will exceed the Tm_G and glucose will remain in the tubules causing polyuria. Osmotic diuresis can also be produced by the infusion of large amounts of sodium chloride or urea.

It is important to recognize the difference between osmotic diuresis and water diuresis. In water diuresis, the amount of water reabsorbed in the proximal portions of the nephron is normal, and the maximal urine flow that can be produced is about 16 mL/min. In osmotic diuresis, increased urine flow is due to decreased water reabsorption in the proximal tubules and loops and very large urine flows can be produced. As the load of excreted solute is increased, the concentration of the urine approaches that of plasma **(Figure 37–17)** in spite of maximal vasopressin secretion, because an increasingly large fraction of the

excreted urine is isotonic proximal tubular fluid. If osmotic diuresis is produced in an animal with diabetes insipidus, the urine concentration rises for the same reason.

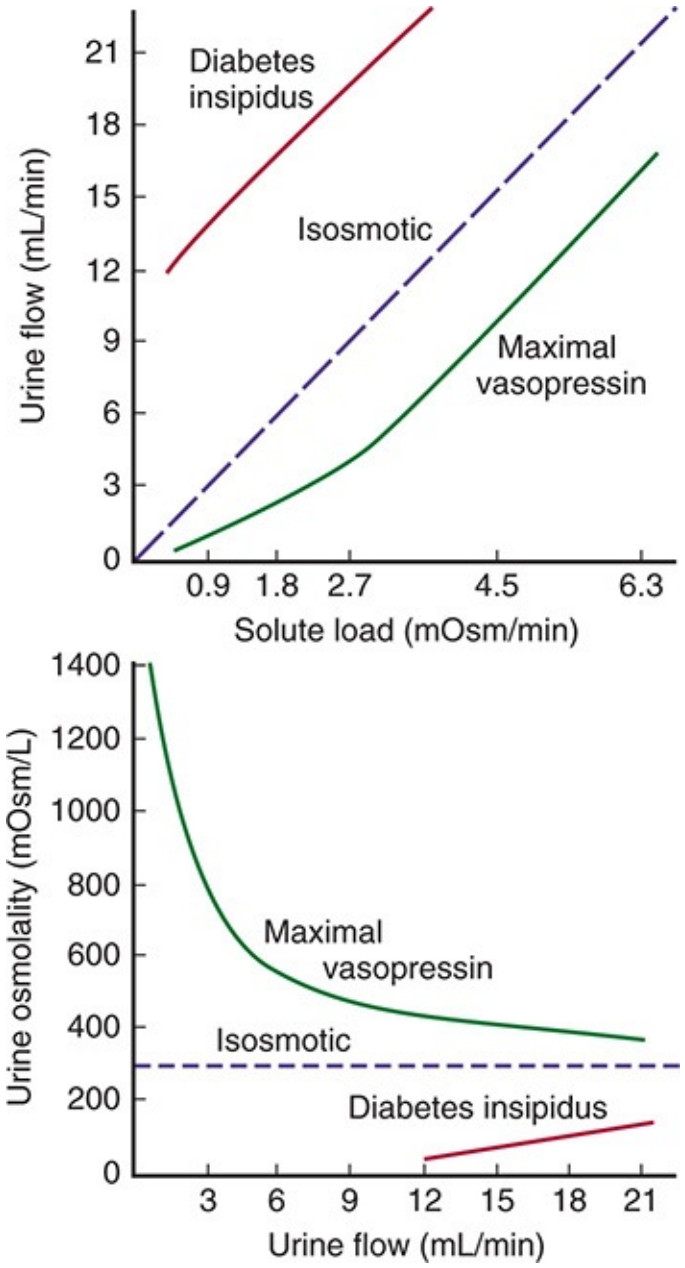


FIGURE 37–17 Approximate relationship between urine concentration and urine flow in osmotic diuresis in humans. The dashed line in the lower diagram indicates the concentration at which the urine is isosmotic with plasma. (Reproduced with permission from Brobeck JR (ed): *Best and Taylor's Physiological Basis of Medical Practice*, 9th ed. Philadelphia, PA: Williams &

Wilkins; 1979.)

RELATION OF URINE CONCENTRATION TO GFR

The magnitude of the osmotic gradient along the medullary pyramids is increased when the rate of flow of fluid through the loops of Henle is decreased. A reduction in GFR such as that caused by dehydration produces a decrease in the volume of fluid presented to the countercurrent mechanism, so that the rate of flow in the loops declines and the urine becomes more concentrated. When the GFR is low, the urine can become quite concentrated in the absence of vasopressin. If one renal artery is constricted in an animal with diabetes insipidus, the urine excreted on the side of the constriction becomes hypertonic because of the reduction in GFR, whereas that excreted on the opposite side remains hypotonic.

"FREE WATER CLEARANCE"

In order to quantitate the gain or loss of water by excretion of a concentrated or dilute urine, the "free water clearance" (C_{H2O}) is sometimes calculated. This is the difference between the urine volume and the clearance of osmoles (C_{Osm}):

$$C_{H_2O} = \dot{V} - \frac{U_{Osm}\,\dot{V}}{P_{Osm}}$$

where $\dot{V}$ is the urine flow rate and U_{Osm} and P_{Osm} the urine and plasma osmolality, respectively. C_{Osm} is the amount of water necessary to excrete the osmotic load in a urine that is isotonic with plasma. Therefore, C_{H2O} is negative when the urine is hypertonic and positive when the urine is hypotonic. For example, using the data in Table 37–6, the values for C_{H2O} are –1.3 mL/min (–1.9 L/d) during maximal antidiuresis and 14.5 mL/min (20.9 L/d) in the absence of vasopressin.

REGULATION OF NA^+ EXCRETION

Na^+ is filtered in large amounts, but it is actively transported out of all portions of the tubule except the descending thin limb of the loop of Henle. Normally about 99% of the filtered Na^+ is reabsorbed. Because Na^+ is the most abundant cation in ECF and because Na^+ salts account for over 90% of the osmotically active solute in the plasma and interstitial fluid, the amount of Na^+ in the body is a prime determinant of the ECF volume. Therefore, it is not surprising that multiple regulatory mechanisms have evolved in terrestrial animals to control the excretion of this ion. Through the operation of these regulatory mechanisms, the amount of Na^+ excreted is adjusted to equal the amount ingested over a wide range of dietary intakes, and the individual stays in Na^+ balance. When Na intake is high, or saline is infused, natriuresis occurs, whereas when ECF is reduced (eg, fluid loss following vomiting or diarrhea) a decrease in Na^+ excretion occurs. Thus, urinary Na^+ output ranges from less than 1 mEq/day on a low-salt diet to 400 mEq/day or more when the dietary Na^+ intake is high.

MECHANISMS

Variations in Na^+ excretion are brought about by changes in GFR **(Table 37–7)** and changes in tubular reabsorption, primarily in the 3% of filtered Na^+ that reaches the collecting ducts. The factors affecting GFR, including tubuloglomerular feedback, have been discussed previously. Factors affecting Na^+ reabsorption include the circulating level of aldosterone and other adrenocortical hormones, the circulating level of ANP and other natriuretic hormones, and the rate of tubular secretion of H^+ and K^+.

TABLE 37–7 Changes in Na^+ excretion that would occur as a result of changes in GFR if there were no concomitant changes in Na^+ reabsorption.

GFR (mL/min)	Plasma Na^+ (µEq/mL)	Amount Filtered (µEq/min)	Amount Reabsorbed (µEq/min)	Amount Excreted (µEq/min)
125	145	18,125	18,000	125
127	145	18,415	18,000	415
124.1	145	18,000	18,000	0

GFR, glomerular filtration rate.

EFFECTS OF ADRENOCORTICAL STEROIDS

Adrenal mineralocorticoids such as aldosterone increase tubular reabsorption of Na^+ in association with secretion of K^+ and H^+ and also Na^+ reabsorption with Cl^-. When these hormones are injected into adrenalectomized animals, a latent period of 10–30 min occurs before their effects on Na^+ reabsorption become manifest, because of the time required for the steroids to alter protein synthesis via their action on DNA. Mineralocorticoids may also have more rapid membrane-mediated effects, but these are not apparent in terms of Na^+ excretion in the whole animal. The mineralocorticoids act primarily in the collecting ducts to increase the number of active epithelial sodium channels (ENaCs) in this part of the nephron. The molecular mechanisms believed to be involved are discussed in Chapter 20 and summarized in **Figure 37–18**.

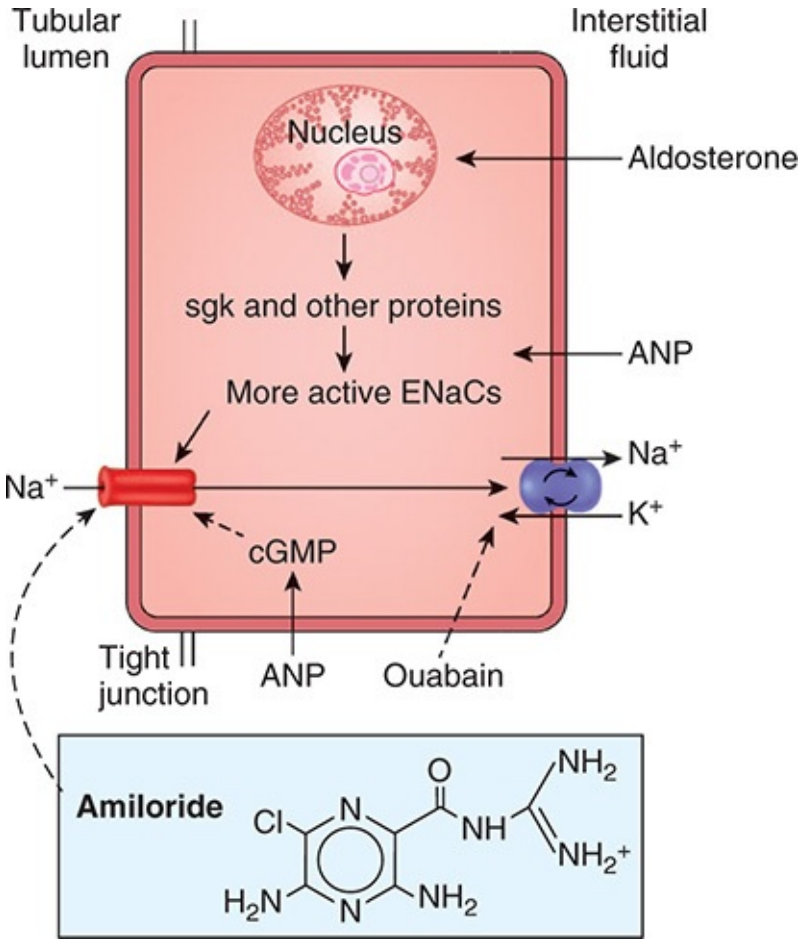


FIGURE 37–18 Renal principal cell. Na^+ enters via the ENaCs in the apical membrane and is pumped into the interstitial fluid by Na, K ATPases in the

basolateral membrane. Aldosterone activates the genome to produce serum- and glucocorticoid-regulated kinase (sgk) and other proteins, and the number of active ENaCs is increased.

In Liddle syndrome, mutations in the genes that code for the β subunit and less commonly the γ subunit of ENaC cause the channels to become constitutively active in the kidney. This leads to Na^+ retention and hypertension.

OTHER HUMORAL EFFECTS

Reduction of dietary intake of salt increases aldosterone secretion (see Figure 20–24), producing marked but slowly developing decreases in Na^+ excretion. A variety of other humoral factors affect Na^+ reabsorption. PGE_2 causes a natriuresis, possibly by inhibiting Na, K ATPase and possibly by increasing intracellular Ca^{2+}, which in turn inhibits Na^+ transport via ENaCs. Endothelin and IL-1 cause natriuresis, probably by increasing the formation of PGE_2. ANP and related molecules increase intracellular cyclic 3′,5′-guanosine monophosphate (cGMP), and this inhibits transport via ENaC. Inhibition of Na, K ATPase by another natriuretic hormone, which appears to be endogenously produced ouabain, also increases Na^+ excretion. Angiotensin II increases reabsorption of Na^+ and HCO_3^- by an action on the proximal tubules. There is an appreciable amount of angiotensin-converting enzyme in the kidneys, and the kidneys convert 20% of the circulating angiotensin I to angiotensin II. In addition, angiotensin I is generated in the kidneys.

Prolonged exposure to high levels of circulating mineralocorticoids does not cause edema in otherwise normal individuals because eventually the kidneys escape from the effects of the steroids. This **escape phenomenon,** which may be due to increased secretion of ANP, is discussed in Chapter 20. It appears to be reduced or absent in nephrosis, cirrhosis, and heart failure, and patients with these diseases continue to retain Na^+ and become edematous when exposed to high levels of mineralocorticoids.

REGULATION OF WATER EXCRETION

WATER DIURESIS

The feedback mechanism controlling vasopressin secretion and the way

vasopressin secretion is stimulated by a rise and inhibited by a drop in the effective osmotic pressure of the plasma are discussed in Chapter 17. The **water diuresis** produced by drinking large amounts of hypotonic fluid begins about 15 min after ingestion of a water load and reaches its maximum in about 40 min. The act of drinking produces a small decrease in vasopressin secretion before the water is absorbed, but most of the inhibition is produced by the decrease in plasma osmolality after the water is absorbed.

WATER INTOXICATION

During excretion of an average osmotic load, the maximal urine flow that can be produced during a water diuresis is about 16 mL/min. If water is ingested at a higher rate than this for any length of time, swelling of the cells becomes severe because of the uptake of water from the hypotonic ECF and, rarely, the symptoms of **water intoxication** may develop. Swelling of the cells in the brain causes convulsions and coma and leads eventually to death. Water intoxication can also occur when water intake is not reduced after administration of exogenous vasopressin or when secretion of endogenous vasopressin occurs in response to nonosmotic stimuli such as surgical trauma. Administration of oxytocin after parturition (to contract the uterus) can also lead to water intoxication if water intake is not monitored carefully.

REGULATION OF K^+ EXCRETION

Much of the filtered K^+ is removed from the tubular fluid by active reabsorption in the proximal tubules, and K^+ is then secreted into the fluid by the distal tubular cells. The rate of K^+ secretion is proportional to the rate of flow of the tubular fluid through the distal portions of the nephron, because with rapid flow there is less opportunity for the tubular K^+ concentration to rise to a value that stops further secretion. In the absence of complicating factors, the amount secreted is approximately equal to the K^+ intake, and K^+ balance is maintained. In the collecting ducts, Na^+ is generally reabsorbed and K^+ is secreted. There is no rigid one-for-one exchange, and much of the movement of K^+ is passive. However, there is electrical coupling in the sense that intracellular migration of Na^+ from the lumen tends to lower the potential difference across the tubular cell, and this favors movement of K^+ into the tubular lumen. K^+ excretion is decreased when the amount of Na^+ reaching the distal tubule is small. In

addition, if H^+ secretion is increased, K^+ excretion will decrease as K^+ is reabsorbed in collecting duct cells in exchange for H^+, via the action of the H, K-ATPase.

DIURETICS

Although a detailed discussion of diuretic agents is beyond the scope of this book, consideration of their mechanisms of action constitutes an informative review of the factors affecting urine volume and electrolyte excretion. These mechanisms are summarized in **Table 37–8**. Water, alcohol, osmotic diuretics, xanthines, and acidifying salts have limited clinical usefulness, and the vasopressin antagonists are currently undergoing clinical trials. However, many of the other agents on the list are used extensively in medical practice.

TABLE 37–8 Mechanism of action of various diuretics.

Agent	Mechanism of Action
Water	Inhibits vasopressin secretion
Ethanol	Inhibits vasopressin secretion
Antagonists of V_2 vasopressin receptors such as astolvaptan	Inhibit action of vasopressin on collecting duct
Large quantities of osmotically active substances such as mannitol and glucose	Produce osmotic diuresis
Xanthines such as caffeine and theophylline	Decrease tubular reabsorption of Na^+ and increase GFR
Acidifying salts such as $CaCl_2$ and NH_4Cl	Supply acid load; H^+ is buffered, but an anion is excreted with Na^+ when the ability of the kidneys to replace Na^+ with H^+ is exceeded
Carbonic anhydrase inhibitors such as acetazolamide (Diamox)	Decrease H^+ secretion, with resultant increase in Na^+ and K^+ excretion
Metolazone (Zaroxolyn), thiazides such as chlorothiazide (Diuril)	Inhibit the Na–Cl cotransporter in the early portion of the distal tubule
Loop diuretics such as furosemide (Lasix), ethacrynic acid (Edecrin), and bumetanide	Inhibit the Na–K–2Cl cotransporter in the medullary thick ascending limb of the loop of Henle
K^+-retaining natriuretics such as spironolactone (Aldactone), triamterene (Dyrenium), and amiloride (Midamor)	Inhibit Na^+–K^+ "exchange" in the collecting ducts by inhibiting the action of aldosterone (spironolactone) or by inhibiting the ENaCs (amiloride)

The carbonic anhydrase-inhibiting drugs are only moderately effective as diuretic agents, but because they inhibit acid secretion by decreasing the supply of carbonic acid, they have far-reaching effects. Not only is Na^+ excretion increased because H^+ secretion is decreased, but HCO_3^- reabsorption is also depressed; and because H^+ and K^+ compete with each other and with Na^+, the decrease in H^+ secretion facilitates the secretion and excretion of K^+.

Furosemide and the other loop diuretics inhibit the Na–K–2Cl cotransporter in the thick ascending limb of the loop of Henle. They cause a marked natriuresis and kaliuresis. Thiazides act by inhibiting Na–Cl cotransport in the distal tubule. The diuresis they cause is less marked, but both loop diuretics and thiazides cause increased delivery of Na^+ (and fluid) to the collecting ducts, facilitating K^+ excretion. Thus, over time, K^+ depletion and hypokalemia are common complications in those who use them if they do not supplement their K^+ intake. On the other hand, the so-called K^+-sparing diuretics act in the collecting duct by inhibiting the action of aldosterone or blocking ENaCs.

EFFECTS OF DISORDERED KIDNEY FUNCTION

A number of abnormalities are common to many different types of renal disease. The secretion of renin by the kidneys and the relation of the kidneys to hypertension are discussed in Chapter 38. A frequent finding in various forms of renal disease is the presence in the urine of protein, leukocytes, red cells, and **casts,** which are proteinaceous material precipitated in the tubules and washed into the bladder. Other important consequences of renal disease are loss of the ability to concentrate or dilute the urine, uremia, acidosis, and abnormal retention of Na^+ **(Clinical Box 37–3)**.

LOSS OF CONCENTRATING & DILUTING ABILITY

In kidney disease, the urine becomes less concentrated and urine volume is often increased, producing the symptoms of **polyuria** and **nocturia** (waking up at night to void). The ability to form a dilute urine is often retained, but in advanced kidney disease, the osmolality of the urine becomes fixed at about that of plasma, indicating that the diluting and concentrating functions of the kidney have both been lost. The loss is due in part to disruption of the countercurrent mechanism, but a more important cause is a loss of functioning nephrons. When one kidney is removed surgically, the number of functioning nephrons is halved. The number of osmoles to be excreted is not reduced to this extent, and so the remaining nephrons must each be filtering and excreting more osmotically active substances, producing what is in effect an osmotic diuresis. In osmotic diuresis, the osmolality of the urine approaches that of plasma. The same thing happens when the number of functioning nephrons is reduced by disease. The increased

filtration in the remaining nephrons eventually damages them, and thus more nephrons are lost. The damage resulting from increased filtration may be due to progressive fibrosis in the proximal tubule cells, but this is unsettled. However, the eventual result of this positive feedback is loss of so many nephrons that complete kidney failure with **oliguria,** or even **anuria,** results.

CLINICAL BOX 37–3

Proteinuria

In many kidney diseases and in one benign condition, the permeability of the glomerular capillaries is increased, and protein is found in the urine in more than the usual trace amounts **(proteinuria).** Most of this protein is **albumin,** and the defect is commonly called albuminuria. The relation of charges on the glomerular membrane to albuminuria has been discussed above. The amount of protein in the urine may be very large, and especially in nephrosis, the urinary protein loss may exceed the rate at which the liver can synthesize plasma proteins. The resulting hypoproteinemia reduces the oncotic pressure, and the plasma volume declines, sometimes to dangerously low levels, while edema fluid accumulates in the tissues.

A benign condition that causes proteinuria is a poorly understood change in renal hemodynamics, which in some otherwise normal individuals, causes protein to appear in urine when they are in the standing position **(orthostatic albuminuria).** Urine formed when these individuals are lying down is protein-free.

UREMIA

When the breakdown products of protein metabolism accumulate in the blood, the syndrome known as **uremia** develops. The symptoms of uremia include lethargy, anorexia, nausea and vomiting, mental deterioration and confusion, muscle twitching, convulsions, and coma. The blood urea nitrogen (BUN) and creatinine levels are high, and the blood levels of these substances are used as an index of the severity of the uremia. It probably is not the accumulation of urea and creatinine per se but rather the accumulation of other toxic substances—possibly organic acids or phenols—that produces the symptoms of uremia.

The toxic substances that cause the symptoms of uremia can be removed by

dialyzing the blood of uremic patients against a bath of suitable composition in an artificial kidney **(hemodialysis).** Patients can be kept alive and in reasonable health for many months on dialysis, even when they are completely anuric or have had both kidneys removed. However, the treatment of choice today is certainly transplantation of a kidney from a suitable donor.

Other features of chronic kidney disease include anemia, which is caused primarily by failure to produce erythro- poietin, and secondary hyperparathyroidism due to 1,25- dihydroxycholecalciferol deficiency (see Chapter 21).

ACIDOSIS

Acidosis is common in chronic kidney disease because of failure to excrete the acid products of digestion and metabolism (see Chapter 39). In the rare syndrome of **renal tubular acidosis,** there is specific impairment of the ability to make the urine acidic, and other renal functions are usually normal. However, in most cases of chronic kidney disease the urine is maximally acidified, and acidosis develops because the total amount of H^+ that can be secreted is reduced because of impaired renal tubular production of NH_4^+.

ABNORMAL NA^+ HANDLING

Many patients with kidney disease retain excessive amounts of Na^+ and become edematous. Na^+ retention in kidney disease has at least three causes. In acute glomerulonephritis, a disease that affects primarily the glomeruli, the amount of Na^+ filtered is decreased markedly. In the nephrotic syndrome, an increase in aldosterone secretion contributes to salt retention. The plasma protein level is low in this condition, and so fluid moves from the plasma into the interstitial spaces and the plasma volume falls. The decline in plasma volume triggers an increase in aldosterone secretion via the renin–angiotensin system. A third cause of Na^+ retention and edema in kidney disease is **heart failure.** Kidney disease predisposes to heart failure, partly because of the hypertension it frequently produces.

THE BLADDER

FILLING

The walls of the ureters contain smooth muscle arranged in spiral, longitudinal, and circular bundles, but distinct layers of muscle are not seen. Regular peristaltic contractions occurring one to five times per minute move the urine from the renal pelvis to the bladder, where it enters in spurts synchronous with each peristaltic wave. The ureters pass obliquely through the bladder wall and, although there are no ureteral sphincters as such, the oblique passage tends to keep the ureters closed except during peristaltic waves, preventing reflux of urine from the bladder.

EMPTYING

The smooth muscle of the bladder, like that of the ureters, is arranged in spiral, longitudinal, and circular bundles. Contraction of the circular muscle, which is called the **detrusor muscle,** is mainly responsible for emptying the bladder during urination (micturition). Muscle bundles pass on either side of the urethra, and these fibers are sometimes called the **internal urethral sphincter,** although they do not encircle the urethra. Farther along the urethra is a sphincter of skeletal muscle, the sphincter of the membranous urethra **(external urethral sphincter).** The bladder epithelium is made up of a superficial layer of flat cells and a deep layer of cuboidal cells. The innervation of the bladder is summarized in **Figure 37–19**.

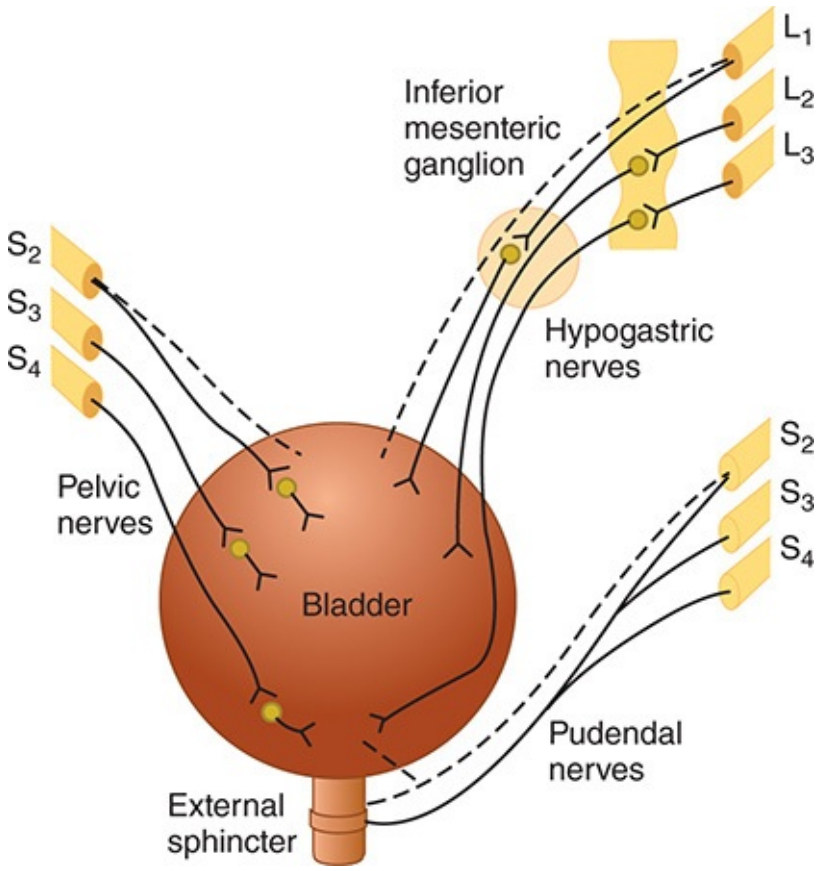


FIGURE 37–19 Innervation of the bladder. Dashed lines indicate sensory nerves. Parasympathetic innervation is shown at the left, sympathetic at the upper right, and somatic at the lower right.

The physiology of bladder emptying and the physiologic basis of its disorders are subjects about which there is much confusion. Micturition is fundamentally a spinal reflex facilitated and inhibited by higher brain centers and, like defecation, subject to voluntary facilitation and inhibition. Urine enters the bladder without producing much increase in intravesical pressure until the viscus is well filled. In addition, like other types of smooth muscle, the bladder muscle has the property of plasticity; when it is stretched, the tension initially produced is not maintained. The relation between intravesical pressure and volume can be studied by inserting a catheter and emptying the bladder, then recording the pressure while the bladder is filled with 50-mL increments of water or air **(cystometry).** A plot of intravesical pressure against the volume of fluid in the bladder is called a **cystometrogram** (Figure 37–20). The curve shows an initial slight rise in pressure when the first increments in volume are produced; a long, nearly flat segment as further increments are produced; and a sudden, sharp rise

in pressure as the micturition reflex is triggered. These three components are sometimes called segments Ia, Ib, and II. The first urge to void is felt at a bladder volume of about 150 mL, and a marked sense of fullness at about 400 mL. The flatness of segment Ib is a manifestation of the law of Laplace. This law states that the pressure in a spherical viscus is equal to twice the wall tension divided by the radius. In the case of the bladder, the tension increases as the organ fills, but so does the radius. Therefore, the pressure increase is slight until the organ is relatively full.

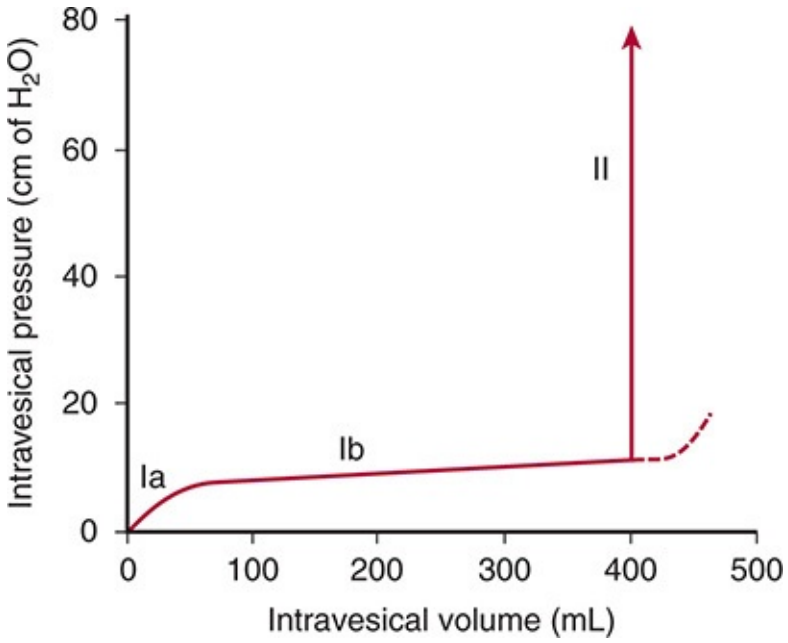


FIGURE 37–20 Cystometrogram in a normal human. The numerals identify the three components of the curve described in the text. The dashed line indicates the pressure–volume relations that would have been found had micturition not occurred and produced component II. (Modified and reproduced with permission from Tanagho EA, McAninch JW: *Smith's General Urology*, 15th ed. New York, NY: McGraw-Hill; 2000.)

During micturition, the perineal muscles and external urethral sphincter are relaxed, the detrusor muscle contracts, and urine passes out through the urethra. The bands of smooth muscle on either side of the urethra apparently play no role in micturition, and their main function in males is believed to be the prevention of reflux of semen into the bladder during ejaculation.

The mechanism by which voluntary urination is initiated remains unsettled. One of the initial events is relaxation of the muscles of the pelvic floor, and this may cause a sufficient downward tug on the detrusor muscle to initiate its

contraction. The perineal muscles and external sphincter can be contracted voluntarily, preventing urine from passing down the urethra or interrupting the flow once urination has begun. It is through the learned ability to maintain the external sphincter in a contracted state that adults are able to delay urination until the opportunity to void presents itself. After urination, the female urethra empties by gravity. Urine remaining in the urethra of the male is expelled by several contractions of the bulbocavernosus muscle.

REFLEX CONTROL

The bladder smooth muscle has some inherent contractile activity; however, when its nerve supply is intact, stretch receptors in the bladder wall initiate a reflex contraction that has a lower threshold than the inherent contractile response of the muscle. Fibers in the pelvic nerves are the afferent limb of the voiding reflex, and the parasympathetic fibers to the bladder that constitute the efferent limb also travel in these nerves. The reflex is integrated in the sacral portion of the spinal cord. In the adult, the volume of urine in the bladder that normally initiates a reflex contraction is about 300–400 mL. The sympathetic nerves to the bladder play no part in micturition, but in males they do mediate the contraction of the bladder muscle that prevents semen from entering the bladder during ejaculation.

The stretch receptors in the bladder wall have no small motor nerve system. However, the threshold for the voiding reflex, like the stretch reflexes, is adjusted by the activity of facilitatory and inhibitory centers in the brainstem. There is a facilitatory area in the pontine region and an inhibitory area in the midbrain. After transection of the brainstem just above the pons, the threshold is lowered and less bladder filling is required to trigger it, whereas after transection at the top of the midbrain, the threshold for the reflex is essentially normal. There is another facilitatory area in the posterior hypothalamus. Humans with lesions in the superior frontal gyrus have a reduced desire to urinate and difficulty in stopping micturition once it has commenced. However, stimulation experiments in animals indicate that other cortical areas also affect the process. The bladder can be made to contract by voluntary facilitation of the spinal voiding reflex when it contains only a few milliliters of urine. Voluntary contraction of the abdominal muscles aids the expulsion of urine by increasing the intra-abdominal pressure, but voiding can be initiated without straining even when the bladder is nearly empty.

EFFECTS OF DEAFFERENTATION

When the sacral dorsal roots are cut in experimental animals or interrupted by diseases of the dorsal roots, such as **tabes dorsalis** in humans, all reflex contractions of the bladder are abolished. The bladder becomes distended, thin-walled, and hypotonic, but some contractions occur because of the intrinsic response of the smooth muscle to stretch.

EFFECTS OF DENERVATION

When the afferent and efferent nerves are both destroyed, as they may be by tumors of the cauda equina or filum terminale, the bladder is flaccid and distended for a while. Gradually, however, the muscle of the "decentralized bladder" becomes active, with many contraction waves that expel dribbles of urine out of the urethra. The bladder becomes shrunken and the bladder wall hypertrophied. The reason for the difference between the small, hypertrophic bladder seen in this condition and the distended, hypotonic bladder seen when only the afferent nerves are interrupted is not known. The hyperactive state in the former condition suggests the development of denervation hypersensitization even though the neurons interrupted are preganglionic rather than postganglionic **(Clinical Box 37–4)**.

EFFECTS OF SPINAL CORD TRANSECTION

During spinal shock, the bladder is flaccid and unresponsive. It becomes overfilled, and urine dribbles through the sphincters **(overflow incontinence).** After spinal shock has passed, the voiding reflex returns, although there is, of course, no voluntary control and no inhibition or facilitation from higher centers when the spinal cord is transected. Some paraplegic patients train themselves to initiate voiding by pinching or stroking their thighs, provoking a mild mass reflex (see Chapter 12). In some instances, the voiding reflex becomes hyperactive, bladder capacity is reduced, and the wall becomes hypertrophied. This type of bladder is sometimes called the **spastic neurogenic bladder.** The reflex hyperactivity is made worse by, and may be caused by, infection in the bladder wall.

CLINICAL BOX 37–4

Abnormalities of Micturition

Three major types of bladder dysfunction are due to neural lesions: (1) due to interruption of the afferent nerves from the bladder, (2) due to interruption of both afferent and efferent nerves, and (3) due to interruption of facilitatory and inhibitory pathways descending from the brain. The bladder contracts in all three types, but the contractions are generally not sufficient to empty the viscus completely, and residual urine is left in the bladder.

CHAPTER SUMMARY

- Plasma enters the kidneys and is filtered in the glomerulus. As the filtrate passes down the nephron and through the tubules its volume is reduced and water and solutes are removed (tubular reabsorption) and waste products are secreted (tubular secretion).
- A nephron consists of an individual renal tubule and its glomerulus. Each tubule has several segments, beginning with the proximal tubule, followed by the loop of Henle (descending and ascending limbs), the distal convoluted tubule, the connecting tubule, and the collecting duct.
- The kidneys receive just under 25% of the cardiac output and renal plasma flow can be measured by infusing p-aminohippuric acid (PAH) and determining its urine and plasma concentrations.
- Renal blood flow enters the glomerulus via the afferent arteriole and leaves via the efferent arteriole (whose diameter is smaller). Renal blood flow is regulated by norepinephrine (constriction, reduction of flow), dopamine (vasodilation, increases flow), angiotensin II (constricts), prostaglandins (dilation in the renal cortex and constriction in the renal medulla), and acetylcholine (vasodilation).
- Glomerular filtration rate can be measured by a substance that is freely filtered and neither reabsorbed nor secreted in the tubules, is nontoxic, and is not metabolized by the body. Inulin meets these criteria and is extensively used to measure GFR.
- Urine is stored in the bladder before voiding (micturition). The micturition response involves reflex pathways but is under voluntary control.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. In the presence of vasopressin, the greatest fraction of filtered water is absorbed in the

   A. proximal tubule.

   B. loop of Henle.

   C. distal tubule.

   D. cortical collecting duct.

   E. medullary collecting duct.

2. In the absence of vasopressin, the greatest fraction of filtered water is absorbed in the

   A. proximal tubule.

   B. loop of Henle.

   C. distal tubule.

   D. cortical collecting duct.

   E. medullary collecting duct.

3. If the clearance of a substance which is freely filtered is less than that of inulin,

   A. there is net reabsorption of the substance in the tubules.

   B. there is net secretion of the substance in the tubules.

   C. the substance is neither secreted nor reabsorbed in the tubules.

   D. the substance becomes bound to protein in the tubules.

   E. the substance is secreted in the proximal tubule to a greater degree than in the distal tubule.

4. Glucose reabsorption occurs in the

   A. proximal tubule.

   B. loop of Henle.

   C. distal tubule.

   D. cortical collecting duct.

   E. medullary collecting duct.

5. On which of the following does aldosterone exert its greatest effect?

   A. Glomerulus

B. Proximal tubule

C. Thin portion of the loop of Henle

D. Thick portion of the loop of Henle

E. Cortical collecting duct

6. What is the clearance of a substance when its concentration in the plasma is 10 mg/dL, its concentration in the urine is 100 mg/dL, and urine flow is 2 mL/min?

A. 2 mL/min

B. 10 mL/min

C. 20 mL/min

D. 200 mL/min

E. Clearance cannot be determined from the information given

7. As urine flow increases during osmotic diuresis

A. the osmolality of urine falls below that of plasma.

B. the osmolality of urine increases because of the increased amounts of nonreabsorbable solute in the urine.

C. the osmolality of urine approaches that of plasma because plasma leaks into the tubules.

D. the osmolality of urine approaches that of plasma because an increasingly large fraction of the excreted urine is isotonic proximal tubular fluid.

E. the action of vasopressin on the renal tubules is inhibited.

CHAPTER 38

Regulation of Extracellular Fluid Composition & Volume

OBJECTIVES

After studying this chapter, you should be able to:

- Describe how the tonicity (osmolality) of the extracellular fluid is maintained by alterations in water intake and vasopressin secretion.
- Discuss the effects of vasopressin, the receptors on which it acts, and how its secretion is regulated.
- Describe how the volume of the extracellular fluid is maintained by alterations in renin and aldosterone secretion.
- Define the stimuli and the reactions that lead to the formation of angiotensin II
- List the functions of angiotensin II, and the receptors on which it acts.
- Describe the function of atrial natriuretic peptide (ANP), B-type natriuretic peptide (BNP), and C-type natriuretic peptide (CNP) and detail the receptors on which they act.
- Describe the site and mechanism of action of erythropoietin, and the feedback regulation of its secretion.

INTRODUCTION

This chapter is a review of the major homeostatic mechanisms that operate, primarily through the kidneys and the lungs, to maintain the **tonicity,** the **volume,** and the **specific ionic composition** of the extracellular fluid (ECF). The interstitial portion of this fluid is the fluid environment of the cells, and life depends on the constancy of this "internal sea" (see Chapter 1).

DEFENSE OF TONICITY

The defense of the tonicity of the ECF is primarily the function of the vasopressin-secreting and thirst mechanisms. The total body osmolality is directly proportional to the total body sodium plus the total body potassium divided by the total body water, so that changes in the osmolality of the body fluids occur when a mismatch exists between the amount of these electrolytes and the amount of water ingested or lost from the body. When the effective osmotic pressure of the plasma rises, vasopressin secretion is increased and the thirst mechanism is stimulated; water is retained in the body, diluting the hypertonic plasma; and water intake is increased **(Figure 38–1)**. Conversely, when the plasma becomes hypotonic, vasopressin secretion is decreased and "solute-free water" (water in excess of solute) is excreted. In this way, the tonicity of the body fluids is maintained within a narrow normal range. In health, plasma osmolality ranges from 280 mOsm/kg of H_2O to 295 mOsm/kg of H_2O, with vasopressin secretion maximally inhibited at 285 mOsm/kg and stimulated at higher values **(Figure 38–2)**.

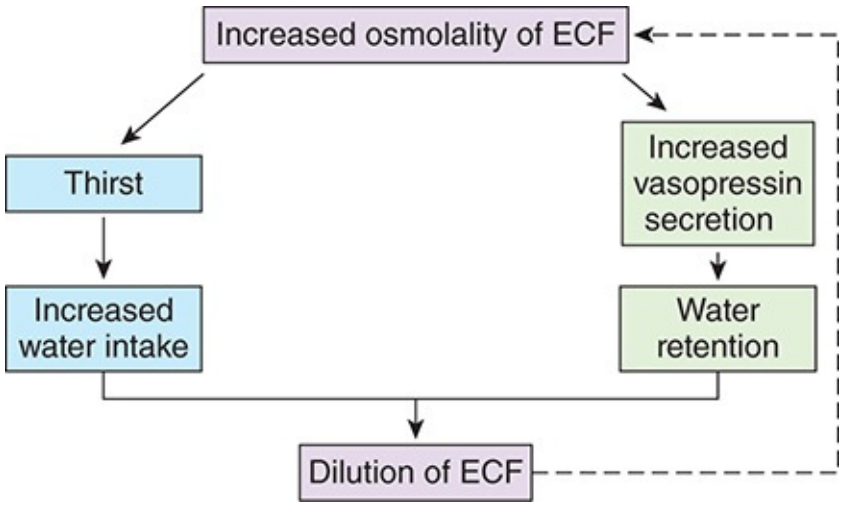


FIGURE 38–1 Mechanisms for defending extracellular fluid (ECF) tonicity. The dashed arrow indicates inhibition. (Used with permission of J Fitzsimmons.)

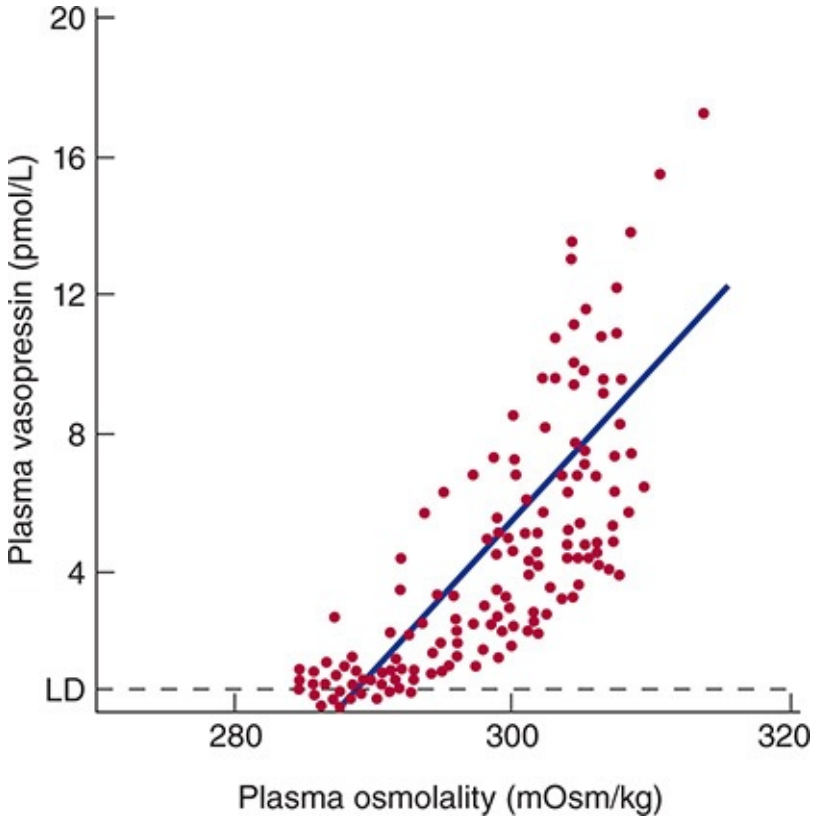


FIGURE 38–2 Relation between plasma osmolality and plasma vasopressin in healthy adult humans during infusion of hypertonic saline. LD, limit of detection. (Reproduced with permission from Thompson CJ, et al: The osmotic thresholds for thirst and vasopressin are similar in healthy humans. Clin Sci [Colch] 1986; Dec; 71:(6):651–656.)

VASOPRESSIN RECEPTORS

There are at least three kinds of vasopressin receptors: V_{1A}, V_{1B}, and V_2. All are G-protein-coupled. The V_{1A} and V_{1B} receptors act through phosphatidylinositol hydrolysis to increase the intracellular Ca^{2+} concentration. The V_2 receptors act through G_s to increase cyclic adenosine 3′, 5′-monophosphate (cAMP) levels.

EFFECTS OF VASOPRESSIN

Because one of its principal physiologic effects is the retention of water by the kidney, vasopressin is often called the **antidiuretic hormone (ADH).** It increases the permeability of the collecting ducts of the kidney, so that water enters the hypertonic interstitium of the renal pyramids. The urine becomes concentrated, and its volume decreases. The overall effect is therefore retention of water in excess of solute; consequently, the effective osmotic pressure of the body fluids is decreased. In the absence of vasopressin, the urine is hypotonic to plasma, urine volume is increased, and there is a net water loss. Consequently, the osmolality of the body fluid rises.

The mechanism by which vasopressin exerts its antidiuretic effect is activated by **V_2 receptors** and involves the insertion of aquaporin 2 into the apical (luminal) membranes of the principal cells of the collecting ducts. Movement of water across membranes by simple diffusion is now known to be augmented by movement through these water channels. These channels are stored in endosomes inside the cells, and vasopressin causes their rapid translocation to the luminal membranes.

V_{1A} receptors mediate the vasoconstrictor effect of vasopressin, and vasopressin is a potent stimulator of vascular smooth muscle in vitro. However, relatively large amounts of vasopressin are needed to raise blood pressure in vivo, because vasopressin also acts on the brain to decrease in cardiac output. The site of this action is the **area postrema,** one of the circumventricular organs (see Chapter 33). Hemorrhage is a potent stimulus for vasopressin secretion, and the blood pressure fall after hemorrhage is more marked in animals that have been treated with synthetic peptides that block the pressor action of vasopressin. Consequently, it appears that vasopressin does play a role in blood pressure homeostasis.

V_{1A} receptors are also found in the liver and the brain. Vasopressin causes glycogenolysis in the liver, and, as noted above, it is a neurotransmitter in the brain and spinal cord.

The V_{1B} receptors (also called V_3 receptors) appear to be unique to the anterior pituitary, where they mediate increased secretion of adrenocorticotropic hormone (ACTH) from the corticotropes.

METABOLISM

Circulating vasopressin is rapidly inactivated, principally in the liver and kidneys. It has a **biologic half-life** of approximately 18 min in humans.

CONTROL OF VASOPRESSIN SECRETION: OSMOTIC STIMULI

Vasopressin is stored in the posterior pituitary and released into the bloodstream in response to impulses in the nerve fibers that contain the hormone. The factors affecting its secretion are summarized in **Table 38–1**. When the effective osmotic pressure of the plasma is increased above 285 mOsm/kg, the rate of discharge of neurons containing vasopressin increases and vasopressin secretion occurs (Figure 38–2). At 285 mOsm/kg, plasma vasopressin is at or near the limits of detection by available assays, but its levels probably decrease when plasma osmolality is below this level. Vasopressin secretion is regulated by osmoreceptors located in the anterior hypothalamus. They are outside the blood-brain barrier and appear to be located in the circumventricular organs, primarily the organum vasculosum of the lamina terminalis (OVLT) (see Chapter 33). The osmotic threshold for thirst (Figure 38–1) is the same as or slightly greater than the threshold for increased vasopressin secretion (Figure 38–2), and it is still uncertain whether the same osmoreceptors mediate both effects.

TABLE 38–1 Summary of stimuli affecting vasopressin secretion.

Vasopressin Secretion Increased	Vasopressin Secretion Decreased
Increased effective osmotic pressure of plasma	Decreased effective osmotic pressure of plasma
Decreased ECF volume	Increased ECF volume
Pain, emotion, "stress," exercise	Alcohol
Nausea and vomiting	
Standing	
Clofibrate, carbamazepine	
Angiotensin II	

ECF, extracellular fluid.

Vasopressin secretion is thus controlled by a delicate feedback mechanism that operates continuously to defend the osmolality of the plasma. Significant changes in secretion occur when osmolality is changed as little as 1%. In this way, the osmolality of the plasma in normal individuals is maintained very close to 285 mOsm/L.

VOLUME EFFECTS ON VASOPRESSIN SECRETION

ECF volume also affects vasopressin secretion. Vasopressin secretion is increased when ECF volume is low and decreased when ECF volume is high (Table 38–1). There is an inverse relationship between the rate of vasopressin secretion and the rate of discharge in afferents from stretch receptors in the low- and high-pressure portions of the vascular system. The low-pressure receptors are those in the great veins, right and left atria, and pulmonary vessels; the high-pressure receptors are those in the carotid sinuses and aortic arch (see Chapter 32). The exponential increases in plasma vasopressin produced by decreases in blood pressure are documented in **Figure 38–3**. However, the low-pressure receptors monitor the fullness of the vascular system, and moderate decreases in blood volume that reduce central venous pressure without lowering arterial pressure can also increase plasma vasopressin.

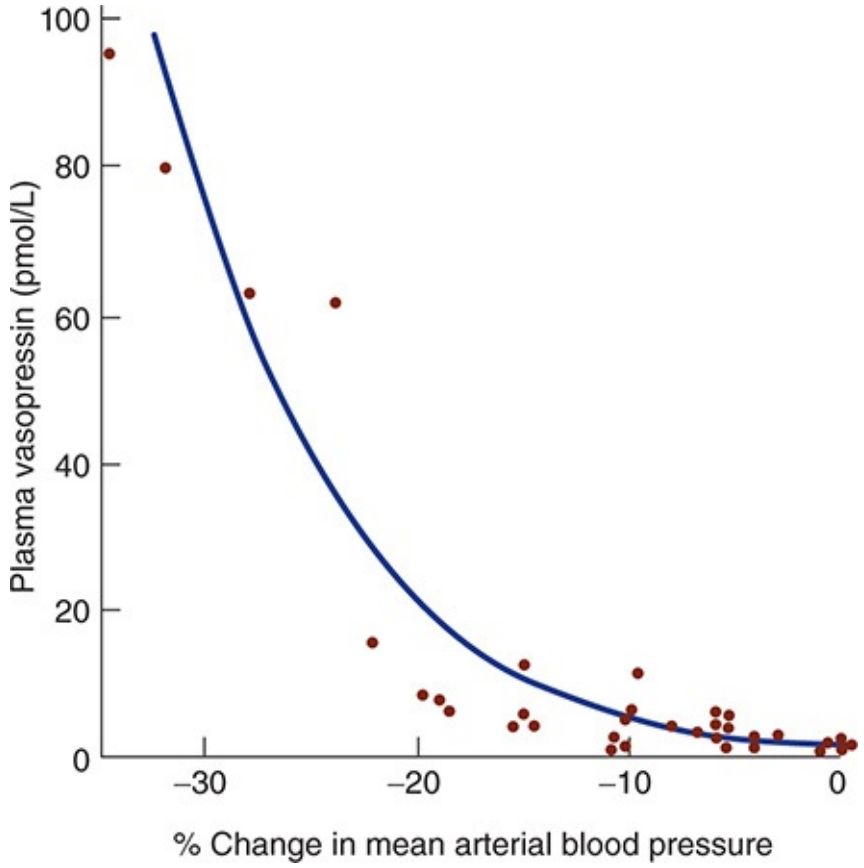


FIGURE 38–3 Relation of mean arterial blood pressure to plasma vasopressin in healthy adult humans in whom a progressive decline in blood

pressure was induced by infusion of graded doses of the ganglionic blocking drug trimethaphan. The relation is exponential rather than linear. (Data from Baylis PH: Osmoregulation and control of vasopressin secretion in healthy humans. Am J Physiol 1987;253:R671.)

Thus, the low-pressure receptors are the primary mediators of volume effects on vasopressin secretion. Impulses pass from them via the vagi to the nucleus of the tractus solitarius (NTS). An inhibitory pathway projects from the NTS to the caudal ventrolateral medulla (CVLM), and there is a direct excitatory pathway from the CVLM to the hypothalamus. Angiotensin II reinforces the response to hypovolemia and hypotension by acting on the circumventricular organs to increase vasopressin secretion (see Chapter 33).

Hypovolemia and hypotension produced by conditions such as hemorrhage release large amounts of vasopressin, and in the presence of hypovolemia, the osmotic response curve is shifted to the left **(Figure 38–4)**. Its slope is also increased. The result is water retention and reduced plasma osmolality. This includes hyponatremia, since Na^+ is the most abundant osmotically active component of the plasma.

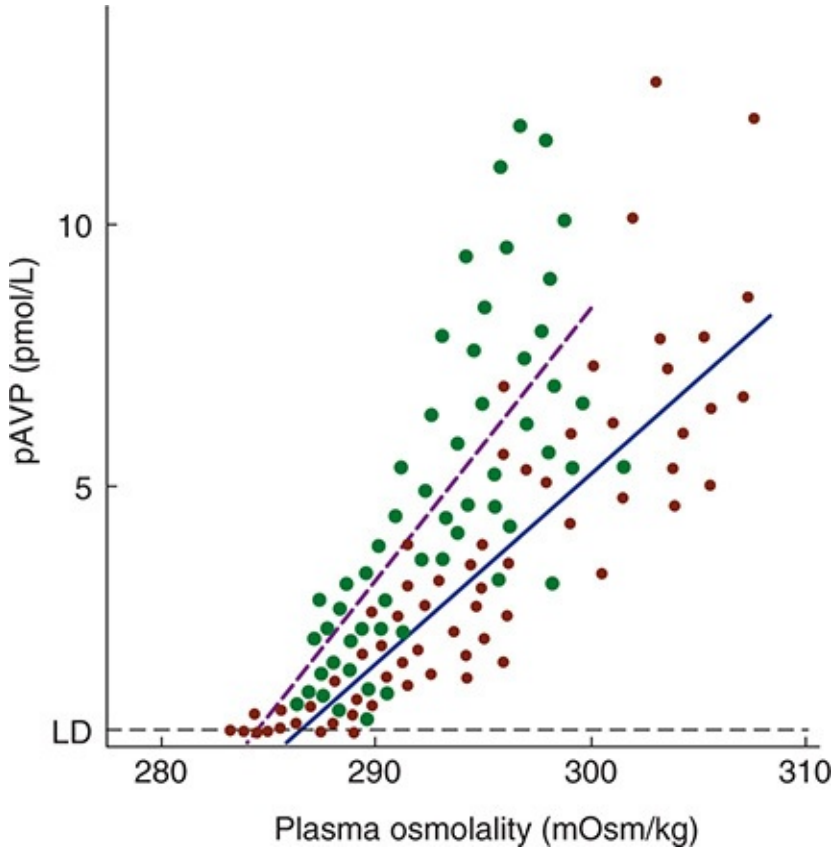


FIGURE 38–4 Effect of hypovolemia and hypervolemia on the relation between plasma vasopressin (pAVP) and plasma osmolality. Seven blood samples were drawn at various times from 10 normal men when hypovolemia was induced by water deprivation (green circles, dashed line) and again when hypervolemia was induced by infusion of hypertonic saline (red circles, solid line). Linear regression analysis defined the relationship pAVP = 0.52 (plasma osmolality—283.5) for water deprivation and pAVP = 0.38 (plasma osmolality—285.6) for hypertonic saline. LD, limit of detection. Note the steeper curve as well as the shift of the intercept to the left during hypovolemia. (Used with permission of CJ Thompson.)

OTHER STIMULI AFFECTING VASOPRESSIN SECRETION

A variety of stimuli in addition to osmotic pressure changes and ECF volume aberrations increase vasopressin secretion. These include pain, nausea, surgical stress, and some emotions (Table 38–1). Nausea is associated with particularly large increases in vasopressin secretion. Alcohol decreases vasopressin secretion.

CLINICAL IMPLICATIONS

In various clinical conditions, volume and other nonosmotic stimuli bias the osmotic control of vasopressin secretion. For example, patients who have had surgery may have elevated levels of plasma vasopressin because of pain and hypovolemia, and this may cause low plasma osmolality and dilutional hyponatremia **(see Clinical Box 38–1**).

Diabetes insipidus is the syndrome that results when there is a vasopressin deficiency **(central diabetes insipidus)** or when the kidneys fail to respond to the hormone **(nephrogenic diabetes insipidus).**

Causes of vasopressin deficiency include disease processes in the supraoptic and paraventricular nuclei, the hypothalamohypophysial tract, or the posterior pituitary gland. It has been estimated that 30% of clinical cases are due to neoplastic lesions of the hypothalamus, either primary or metastatic; 30% are posttraumatic; 30% are idiopathic; and the remainder are due to vascular lesions, infections, systemic diseases such as sarcoidosis that affect the hypothalamus, or mutations in the gene for prepropressophysin. Disease that develops after

surgical removal of the posterior lobe of the pituitary may be temporary if the distal ends of the supraoptic and paraventricular fibers are only damaged, because the fibers recover, make new vascular connections, and begin to secrete vasopressin again.

The symptoms of diabetes insipidus are passage of large amounts of dilute urine **(polyuria)** and the drinking of large amounts of fluid **(polydipsia),** provided the thirst mechanism is intact. It is the polydipsia that keeps these patients healthy. If their sense of thirst is depressed for any reason and their intake of dilute fluid decreases, dehydration that can be fatal develops.

Another cause of diabetes insipidus is inability of the kidneys to respond to vasopressin **(nephrogenic diabetes insipidus).** Two forms of this disease have been described. In one form, the gene for the V_2 receptor is mutated, making the receptor unresponsive. The V_2 receptor gene is on the X chromosome, thus this condition is X-linked and inheritance is sex-linked recessive. In the other form of the condition, mutations occur in the autosomal gene for aquaporin-2 and produce nonfunctional versions of this water channel, many of which do not reach the apical membrane of the collecting duct but are trapped in intracellular locations.

CLINICAL BOX 38–1

Syndrome of Inappropriate Antidiurectic Hormone

The **syndrome of "inappropriate" hypersecretion of antidiuretic hormone (SIADH)** occurs when vasopressin is inappropriately high relative to serum osmolality. Vasopressin is responsible not only for dilutional **hyponatremia** (serum sodium < 135 mmol/L) but also for loss of salt in the urine when water retention is sufficient to expand the ECF volume, reducing aldosterone secretion (see Chapter 20). This occurs in patients with cerebral disease ("cerebral salt wasting") and pulmonary disease ("pulmonary salt wasting"). Hypersecretion of vasopressin in patients with pulmonary diseases such as lung cancer may be due in part to the interruption of inhibitory impulses in vagal afferents from the stretch receptors in the atria and great veins.

A significant number of lung tumors and some other cancers secrete vasopressin. There is a process called **"vasopressin escape"** that counteracts the water-retaining action of vasopressin to limit the degree of hyponatremia in SIADH. Studies in rats have demonstrated that prolonged exposure to elevated levels of vasopressin can lead eventually to down-regulation of the

production of aquaporin-2. This permits urine flow suddenly to increase and plasma osmolality to fall despite exposure of the collecting ducts to elevated levels of the hormone; that is, the individual escapes from the renal effects of vasopressin.

THERAPEUTIC HIGHLIGHTS

Patients with inappropriate hypersecretion of vasopressin have been successfully treated with demeclocycline, an antibiotic that reduces the renal response to vasopressin.

SYNTHETIC AGONISTS & ANTAGONISTS

Synthetic peptides that have selective actions and are more active than naturally occurring vasopressin have been produced by altering the amino acid residues. For example, 1-deamino-8-D-arginine vasopressin (desmopressin; dDAVP) has very high antidiuretic activity with little pressor activity, making it valuable in the treatment of vasopressin deficiency.

DEFENSE OF VOLUME

The volume of the ECF is determined primarily by the total amount of osmotically active solute in the ECF. The composition of the ECF is discussed in Chapter 1. Because Na^+ and Cl^- are by far the most abundant osmotically active solutes in ECF, and because changes in Cl^- are to a great extent secondary to changes in Na^+, the amount of Na^+ in the ECF is the most important determinant of ECF volume. Therefore, the mechanisms that control Na^+ balance are the major mechanisms defending ECF volume. However, there is volume control of water excretion as well; a rise in ECF volume inhibits vasopressin secretion, and a decline in ECF volume produces an increase in the secretion of this hormone. Volume stimuli override the osmotic regulation of vasopressin secretion. Angiotensin II stimulates aldosterone and vasopressin secretion. It also causes thirst and constricts blood vessels, which help maintain blood pressure. Thus, angiotensin II plays a key role in the body's response to hypovolemia **(Figure 38–5)**. In addition, expansion of the ECF volume increases the secretion of atrial natriuretic peptide (ANP) and B-type natriuretic peptide (BNP) by the heart, and

this causes natriuresis and diuresis.

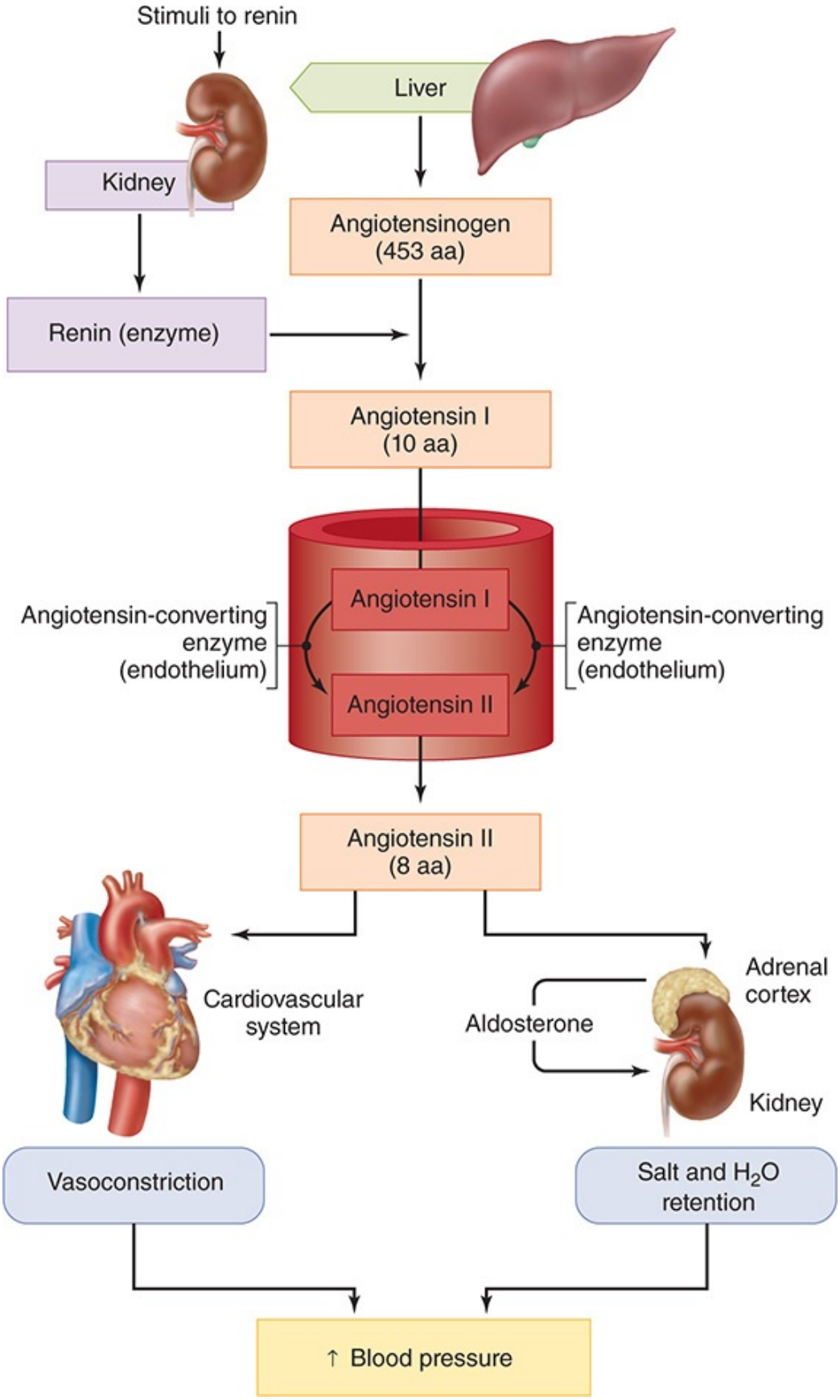

Stimuli to renin
Liver
Kidney
Angiotensinogen
(453 aa)
Renin (enzyme)
Angiotensin I
(10 aa)
Angiotensin I
Angiotensin-converting
enzyme
(endothelium)
Angiotensin-converting
enzyme
(endothelium)
Angiotensin II
Angiotensin II
(8 aa)
Cardiovascular
system
Adrenal
cortex
Aldosterone
Kidney
Vasoconstriction
Salt and H_2O
retention
↑ Blood pressure

FIGURE 38–5 Summary of the renin–angiotensin system and the stimulation of aldosterone secretion by angiotensin II. The plasma concentration of renin is the rate-limiting step in the renin–angiotensin system; therefore, it is the major determinant of plasma angiotensin II concentration.

In disease states, loss of water from the body **(dehydration)** causes a moderate decrease in ECF volume, because water is lost from both the intracellular and ECF compartments; but excessive loss of Na^+ in the stools (diarrhea), urine (severe acidosis, adrenal insufficiency), or sweat (heat prostration) decreases ECF volume markedly and eventually leads to shock. The immediate compensations in shock operate principally to maintain intravascular volume, but they also affect Na^+ balance. In adrenal insufficiency, the decline in ECF volume is not only due to loss of Na^+ in the urine but also to its movement into cells. Because of the key position of Na^+ in volume homeostasis, it is not surprising that more than one mechanism has evolved to control the excretion of this ion.

The filtration and reabsorption of Na^+ in the kidneys and the effects of these processes on Na^+ excretion are discussed in Chapter 37. When ECF volume is decreased, blood pressure falls, glomerular capillary pressure declines, and the glomerular filtration rate (GFR) therefore falls, reducing the amount of Na^+ filtered. Tubular reabsorption of Na^+ is increased, in part because the secretion of aldosterone is increased. Aldosterone secretion is controlled in part by a feedback system in which the change that initiates increased secretion is a decline in mean intravascular pressure. Other changes in Na^+ excretion occur too rapidly to be solely due to changes in aldosterone secretion. For example, rising from the supine to the standing position increases aldosterone secretion. However, Na^+ excretion is decreased within a few minutes, and this rapid change in Na^+ excretion occurs in adrenalectomized subjects. It is probably due to hemodynamic changes and possibly to decreased ANP secretion.

The kidneys produce three hormones: 1,25-dihydroxycholecalciferol (see Chapter 21), renin, and erythropoietin. Natriuretic peptides, substances secreted by the heart and other tissues, increase excretion of sodium by the kidneys, and an additional natriuretic hormone (endogenous ouabain) inhibits Na, K ATPase.

THE RENIN–ANGIOTENSIN SYSTEM

RENIN

The rise in blood pressure produced by injection of kidney extracts is due to **renin,** an acid protease secreted by the kidneys into the bloodstream. This enzyme acts in concert with angiotensin-converting enzyme (ACE) to form angiotensin II **(Figure 38–6)**. It is a glycoprotein with a molecular weight of 37,326 in humans. The molecule is made up of two lobes, or domains, between which the active site of the enzyme is located in a deep cleft. Two aspartic acid residues, one at position 104 and one at position 292 (residue numbers from human preprorenin), are juxtaposed in the cleft and are essential for activity. Thus, renin is an aspartyl protease.

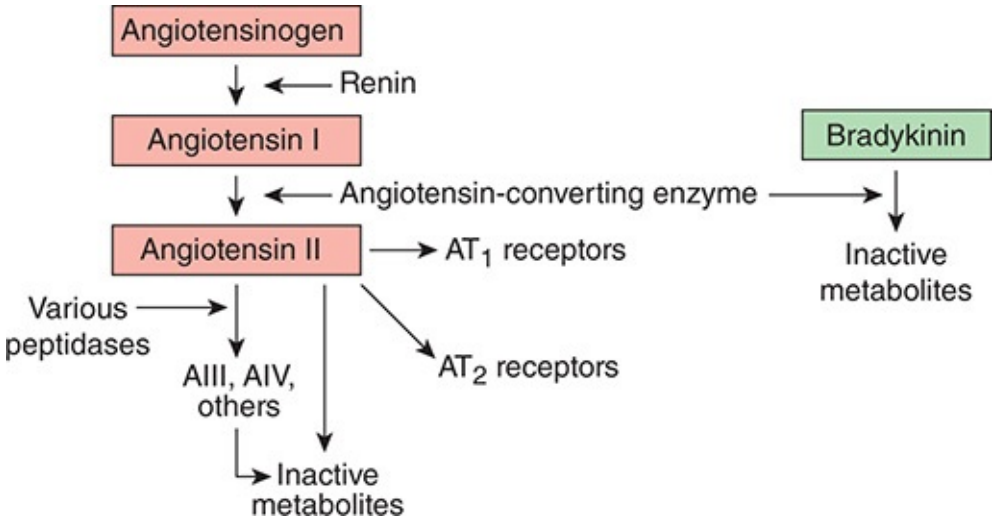


FIGURE 38–6 Formation and metabolism of circulating angiotensins.

Like other hormones, renin is synthesized as a large preprohormone. Human **preprorenin** contains 406 amino acid residues. The **prorenin** that remains after removal of a leader sequence of 23 amino acid residues from the amino terminal contains 383 amino acid residues, and after removal of the pro sequence from the amino terminal of prorenin, active **renin** contains 340 amino acid residues. Prorenin has little if any biologic activity.

Some prorenin is converted to renin in the kidneys, and some is secreted. Prorenin is also secreted by other organs, including the ovaries. After nephrectomy, the prorenin level in the circulation is usually only moderately reduced and may actually rise, but the level of active renin falls to essentially zero. Thus, very little prorenin is converted to renin in the circulation, and active renin is a product primarily, if not exclusively, of the kidneys. Prorenin is

secreted constitutively, whereas active renin is formed in the secretory granules of the granular cells in the juxtaglomerular apparatus, the same cells that produce renin (see below). Active renin has a half-life in the circulation of 80 min or less. Its only known function is to cleave the decapeptide **angiotensin I** from the amino terminal end of **angiotensinogen (renin substrate)** (Figure 38–7).

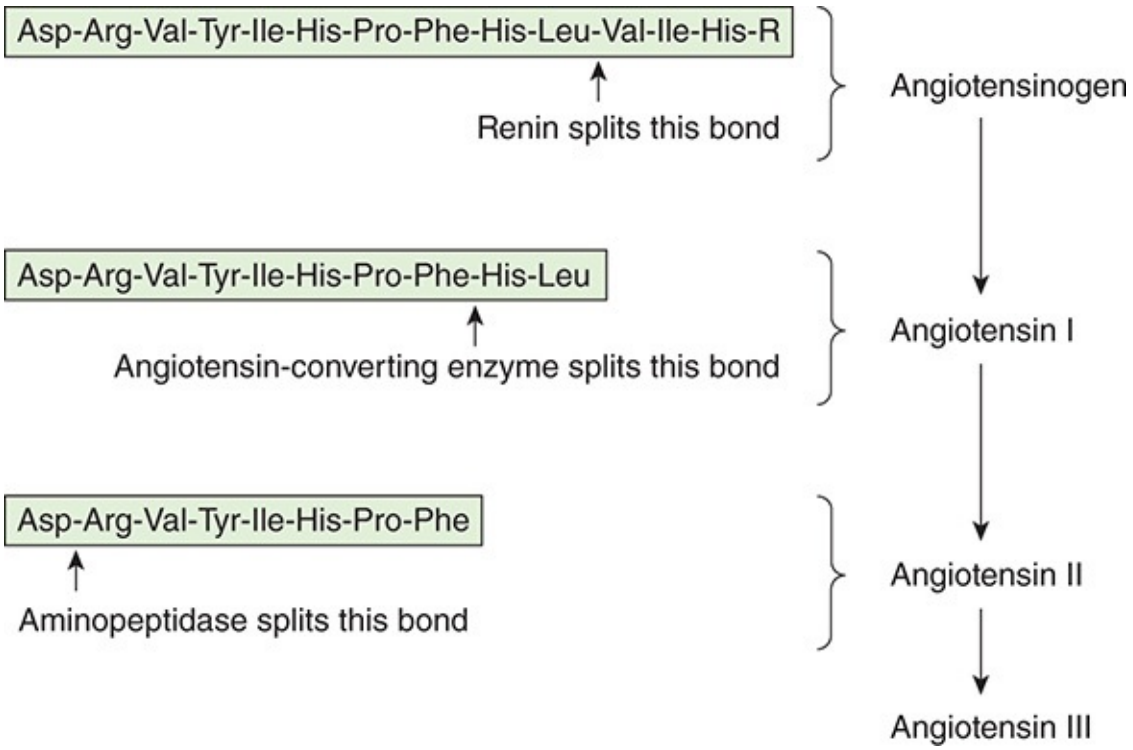


FIGURE 38–7 Structure of the amino terminal end of angiotensinogen and angiotensins I, II, and III in humans. R, remainder of protein. After removal of a 24-amino-acid leader sequence, angiotensinogen contains 453 amino acid residues. The structure of angiotensin II in dogs, rats, and many other mammals is the same as that in humans. Bovine and ovine angiotensin II have valine instead of isoleucine at position 5.

ANGIOTENSINOGEN

Circulating angiotensinogen is found in the α_2-globulin fraction of the plasma (Figure 38–6). It contains about 13% carbohydrate and is made up of 453 amino acid residues. It is synthesized in the liver with a 32-amino-acid signal sequence that is removed in the endoplasmic reticulum. Its circulating level is increased by

glucocorticoids, thyroid hormones, estrogens, several cytokines, and angiotensin II.

ANGIOTENSIN-CONVERTING ENZYME & ANGIOTENSIN II

ACE is a dipeptidyl carboxypeptidase that splits off histidyl-leucine from the physiologically inactive angiotensin I, forming the octapeptide **angiotensin II** (Figure 38–7). The same enzyme inactivates bradykinin (Figure 38–6). Increased tissue bradykinin produced when ACE is inhibited acts on B_2 receptors to produce the cough that is an annoying side effect in up to 20% of patients treated with ACE inhibitors **(see Clinical Box 38–2**). Most of the converting enzyme that forms angiotensin II in the circulation is located in endothelial cells. Much of the conversion occurs as the blood passes through the lungs, but conversion also occurs in many other parts of the body.

ACE is an ectoenzyme that exists in two forms: a **somatic** form found throughout the body and a **germinal** form found solely in postmeiotic spermatogenic cells and spermatozoa (see Chapter 23). Both forms have a single transmembrane domain and a short cytoplasmic tail. However, somatic ACE is a 170-kDa protein with two homologous extracellular domains, each containing an active site. Germinal ACE is a 90-kDa protein that has only one extracellular domain and active site. Both enzymes are formed from a single gene. However, the gene has two different promoters, producing two different mRNAs. In male mice in which the ACE gene has been knocked out, blood pressure is lower than normal, but in females it is normal. In addition, fertility is reduced in males but not in females.

CLINICAL BOX 38–2

Pharmacologic Manipulation of the Renin–Angiotensin System

It is now possible to inhibit the secretion or the effects of renin in a variety of ways. Inhibitors of prostaglandin synthesis such as **indomethacin** and β-adrenergic blocking drugs such as **propranolol** reduce renin secretion. The peptide **pepstatin** and newly developed renin inhibitors such as **enalkiren** prevent renin from generating angiotensin I. ACE inhibitors such as **captopril** and **enalapril** prevent conversion of angiotensin I to angiotensin II. **Saralasin**

and several other analogs of angiotensin II are competitive inhibitors of the action of angiotensin II on both AT_1 and AT_2 receptors. **Losartan** (DuP-753) selectively blocks AT_1 receptors, and PD-123177 and several other drugs selectively block AT_2 receptors.

METABOLISM OF ANGIOTENSIN II

Angiotensin II is metabolized rapidly; its half-life in the circulation in humans is 1–2 min. It is metabolized by various peptidases. An aminopeptidase removes the aspartic acid (Asp) residue from the amino terminal of the peptide (Figure 38–7). The resulting heptapeptide has physiologic activity and is sometimes called **angiotensin III.** Removal of a second amino terminal residue from angiotensin III produces the hexapeptide sometimes called angiotensin IV, which is also said to have some activity. Most, if not all, of the other peptide fragments that are formed are inactive. In addition, aminopeptidase can act on angiotensin I to produce (des-Asp^1) angiotensin I, and this compound can be converted directly to angiotensin III by the action of ACE. Angiotensin-metabolizing activity is found in red blood cells and many tissues. In addition, angiotensin II appears to be removed from the circulation by some sort of trapping mechanism in the vascular beds of tissues other than the lungs.

Renin is usually measured by incubating the sample to be assayed and measuring by immunoassay the amount of angiotensin I generated. This measures the **plasma renin activity (PRA)** of the sample. Deficiency of angiotensinogen as well as renin can cause low PRA values, and to avoid this problem, exogenous angiotensinogen is often added, so that **plasma renin concentration (PRC)** rather than PRA is measured. The normal PRA in supine subjects eating a normal amount of sodium is approximately 1 ng of angiotensin I generated per milliliter per hour. The plasma angiotensin II concentration in such subjects is about 25 pg/mL (approximately 25 pmol/L).

ACTIONS OF ANGIOTENSINS

Angiotensin I appears to function solely as the precursor of angiotensin II and does not have any other established action.

Angiotensin II produces arteriolar constriction and a rise in systolic and diastolic blood pressure. It is one of the most potent vasoconstrictors known,

being four to eight times as active as norepinephrine on a weight basis in normal individuals. However, its pressor activity is decreased in Na^+-depleted individuals and in patients with cirrhosis and some other diseases. In these conditions, circulating angiotensin II is increased, and this down-regulates the angiotensin receptors in vascular smooth muscle. Consequently, there is less response to injected angiotensin II.

Angiotensin II also acts directly on the adrenal cortex to increase the secretion of aldosterone, and the renin–angiotensin system is a major regulator of aldosterone secretion. Additional actions of angiotensin II include facilitation of the release of norepinephrine by a direct action on postganglionic sympathetic neurons, contraction of mesangial cells with a resultant decrease in GFR (see Chapter 37), and a direct effect on the renal tubules to increase Na^+ reabsorption.

Angiotensin II also acts on the brain to decrease the sensitivity of the baroreflex, and this potentiates the pressor effect of angiotensin II. In addition, it acts on the brain to increase water intake and increase the secretion of vasopressin and ACTH. It does not penetrate the blood-brain barrier, but it triggers these responses by acting on the circumventricular organs, four small structures in the brain that are outside the blood-brain barrier (see Chapter 33). One of these structures, the area postrema, is primarily responsible for the pressor potentiation, whereas two of the others, the subfornical organ (SFO) and the OVLT, are responsible for the increase in water intake (dipsogenic effect). It is not certain which of the circumventricular organs are responsible for the increases in vasopressin and ACTH secretion.

Angiotensin III [(des-Asp^1) angiotensin II] has about 40% of the pressor activity of angiotensin II, but 100% of the aldosterone-stimulating activity. It has been suggested that angiotensin III is the natural aldosterone-stimulating peptide, whereas angiotensin II is the blood pressure–regulating peptide. However, this appears not to be the case, and instead angiotensin III is simply a breakdown product with some biologic activity. The same is probably true of angiotensin IV, though some researchers have argued that it has unique effects in the brain.

TISSUE RENIN–ANGIOTENSIN SYSTEMS

In addition to the system that generates circulating angiotensin II, many different tissues contain independent renin–angiotensin systems that generate angiotensin II, apparently for local use. Components of the renin–angiotensin system are found in the walls of blood vessels and in the uterus, the placenta, and the fetal membranes. Amniotic fluid has a high concentration of prorenin. In addition,

tissue renin–angiotensin systems, or at least several components of the renin–angiotensin system, are present in the eyes, exocrine portion of the pancreas, heart, fat, adrenal cortex, testis, ovary, anterior and intermediate lobes of the pituitary, pineal, and brain. Tissue renin contributes very little to the circulating renin pool, because PRA falls to undetectable levels after the kidneys are removed. The functions of these tissue renin–angiotensin systems are unsettled, though evidence is accumulating that angiotensin II is a significant growth factor in the heart and blood vessels. ACE inhibitors or AT_1 receptor blockers are now the treatment of choice for heart failure, and part of their value may be due to inhibition of the growth effects of angiotensin II.

ANGIOTENSIN II RECEPTORS

There are at least two classes of angiotensin II receptors. AT_1 receptors are serpentine receptors coupled by a G-protein (G_q) to phospholipase C, and angiotensin II increases the cytosolic free Ca^{2+} level. It also activates numerous tyrosine kinases. In vascular smooth muscle, AT_1 receptors are associated with caveolae (see Chapter 2), and AII increases production of caveolin-1, one of the three isoforms of the protein that is characteristic of caveolae. In rodents, two different but closely related AT_1 subtypes, AT_{1A} and AT_{1B}, are coded by two separate genes. The AT_{1A} subtype is found in blood vessel walls, the brain, and many other organs. It mediates most of the known effects of angiotensin II. The AT_{1B} subtype is found in the anterior pituitary and the adrenal cortex. In humans, an AT_1 receptor gene is present on chromosome 3. There may be a second AT_1 type, but it is still unsettled whether distinct AT_{1A} and AT_{1B} subtypes occur.

There are also AT_2 receptors, which are coded in humans by a gene on the X chromosome. Like the AT_1 receptors, they have seven transmembrane domains, but their actions are different. They act via a G-protein to activate various phosphatases which in turn antagonize growth effects and open K^+ channels. In addition, AT_2 receptor activation increases the production of NO and therefore increases intracellular cyclic 3,5-guanosine monophosphate (cGMP). The overall physiologic consequences of these second-messenger effects are unsettled. AT_2 receptors are more plentiful in fetal and neonatal life, but they persist in the brain and other organs in adults.

The AT_1 receptors in the arterioles and the AT_1 receptors in the adrenal cortex

are regulated in opposite ways: an excess of angiotensin II down-regulates the vascular receptors, but it up-regulates the adrenocortical receptors, making the gland more sensitive to the aldosterone-stimulating effect of the peptide.

THE JUXTAGLOMERULAR APPARATUS

The renin in kidney extracts and the bloodstream is produced by the **juxtaglomerular cells (JG cells).** These epitheloid cells are located in the media of the afferent arterioles as they enter the glomeruli **(Figure 38–8)**. The membrane-lined secretory granules in them have been shown to contain renin. Renin is also found in agranular **lacis cells** that are located in the junction between the afferent and efferent arterioles, but its significance in this location is unknown.

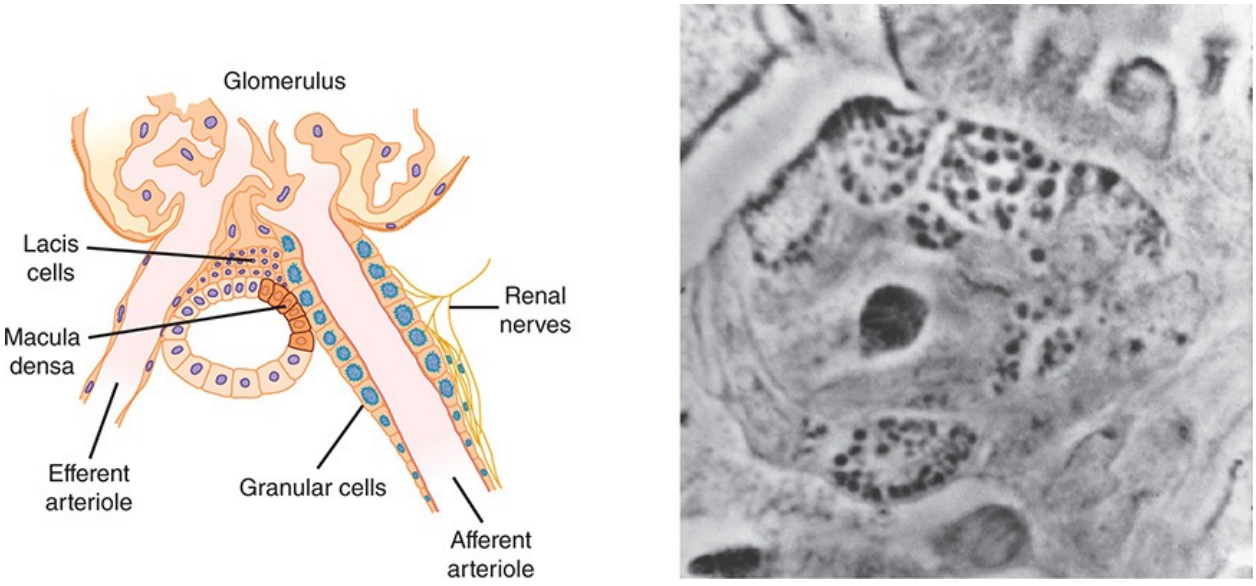


FIGURE 38–8 Left: Diagram of glomerulus, showing the juxtaglomerular apparatus. Right: Phase contrast photomicrograph of afferent arteriole in an unstained, freeze-dried preparation of the kidney of a mouse. Note the red blood cell in the lumen of the arteriole and the granular cells in the wall. (Used with permission of C Peil.)

At the point where the afferent arteriole enters the glomerulus and the efferent arteriole leaves it, the tubule of the nephron touches the arterioles of the glomerulus from which it arose. At this location, which marks the start of the distal convolution, there is a modified region of tubular epithelium called the

macula densa (Figure 38–8). The macula densa is in close proximity to the JG cells. The lacis cells, the JG cells, and the macula densa constitute the **juxtaglomerular apparatus.**

REGULATION OF RENIN SECRETION

Several different factors regulate renin secretion **(Table 38–2)**, and the rate of renin secretion at any given time is determined by the summed activity of these factors. One factor is an intra-renal baroreceptor mechanism that causes renin secretion to decrease when arteriolar pressure at the level of the JG cells increases and to increase when arteriolar pressure at this level falls. Another renin-regulating sensor is in the macula densa. Renin secretion is inversely proportional to the amount of Na^+ and Cl^- entering the distal renal tubules from the loop of Henle. Presumably, these electrolytes enter the macula densa cells via the Na–K–$2Cl^-$ transporters in their apical membranes, and the increase in some fashion triggers a signal that decreases renin secretion in the JG cells in the adjacent afferent arterioles. A possible mediator is NO, but the identity of the signal remains unsettled. Renin secretion also varies inversely with the plasma K^+ level, but the effect of K^+ appears to be mediated by the changes it produces in Na^+ and Cl^- delivery to the macula densa.

TABLE 38–2 Factors that affect renin secretion.

Stimulatory
Increased sympathetic activity via renal nerves
Increased circulating catecholamines
Prostaglandins
Inhibitory
Increased Na^+ and Cl^- reabsorption across macula densa
Increased afferent arteriolar pressure
Angiotensin II
Vasopressin

Angiotensin II feeds back to inhibit renin secretion by a direct action on the JG cells. Vasopressin also inhibits renin secretion in vitro and in vivo, although

there is some debate about whether its in vivo effect is direct or indirect.

Finally, increased activity of the sympathetic nervous system increases renin secretion. The increase is mediated both by increased circulating catecholamines and by norepinephrine secreted by postganglionic renal sympathetic nerves. The catecholamines act mainly on β_1-adrenergic receptors on the JG cells and renin release is mediated by an increase in intracellular cAMP.

The principal conditions that increase renin secretion in humans are listed in **Table 38–3**. Most of the listed conditions decrease central venous pressure, which triggers an increase in sympathetic activity, and some also decrease renal arteriolar pressure **(see Clinical Box 38–3)**. Renal artery constriction and constriction of the aorta proximal to the renal arteries produces a decrease in renal arteriolar pressure. Psychological stimuli increase the activity of the renal nerves.

TABLE 38–3 Conditions that increase renin secretion.

Na^+ depletion
Diuretics
Hypotension
Hemorrhage
Upright posture
Dehydration
Cardiac failure
Cirrhosis
Constriction of renal artery or aorta
Various psychological stimuli

HORMONES OF THE HEART & OTHER NATRIURETIC FACTORS

STRUCTURE

The existence of various **natriuretic hormones** has been postulated for some time. Two of these are secreted by the heart. The muscle cells in the atria and, to a much lesser extent in the ventricles, contain secretory granules **(Figure 38–9)**. The granules increase in number when NaCl intake is increased and ECF

expanded, and extracts of atrial tissue cause natriuresis.

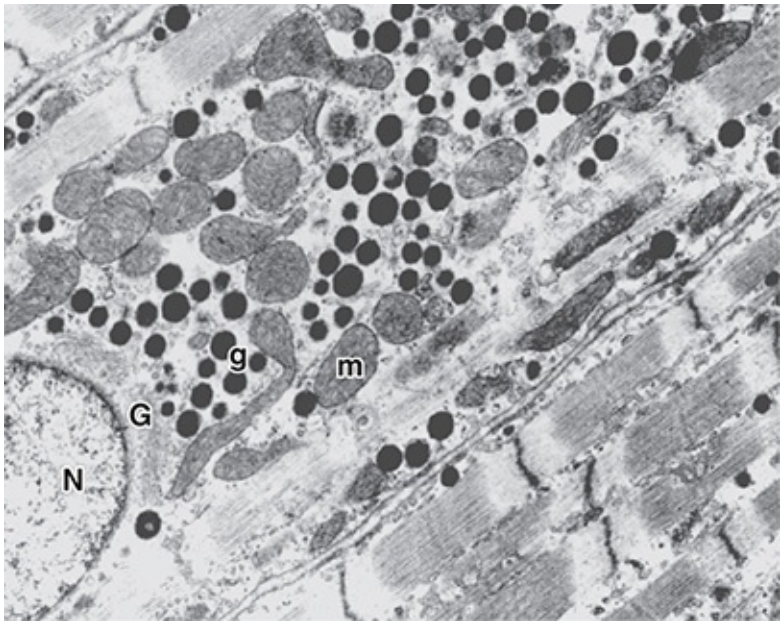


FIGURE 38–9 ANP granules (g) interspersed between mitochondria (m) in rat atrial muscle cell. G, Golgi complex; N, nucleus. The granules in human atrial cells are similar (× 17,640). (Used with permission of M Cantin.)

The first natriuretic hormone isolated from the heart was **ANP,** a polypeptide with a characteristic 17-amino-acid ring formed by a disulfide bond between two cysteines. The circulating form of this polypeptide has 28 amino acid residues. It is formed from a large precursor molecule that contains 151 amino acid residues, including a 24-amino-acid signal peptide. ANP was subsequently isolated from other tissues, including the brain, where it exists in two forms that are smaller than circulating ANP. A second natriuretic polypeptide was isolated from porcine brain and named **B-type natriuretic peptide** (also known as **brain natriuretic peptide**). It is also present in the brain in humans, but more is present in the human heart, including the ventricles. The circulating form of this hormone contains 32 amino acid residues. It has the same 17-member ring as ANP, though some of the amino acid residues in the ring are different. A third member of this family has been named **C-type natriuretic peptide (CNP)** because it was the third in the sequence to be isolated. It contains 22 amino acid residues, and there is also a larger 53-amino-acid form. CNP is present in the brain, the pituitary, the kidneys, and vascular endothelial cells. However, very little is present in the heart and the circulation, and it appears to be primarily a paracrine mediator.

CLINICAL BOX 38–3

Role of Renin in Clinical Hypertension

Constriction of one renal artery causes a prompt increase in renin secretion and the development of sustained hypertension **(renal** or **Goldblatt hypertension).** Removal of the ischemic kidney or the arterial constriction cures the hypertension if it has not persisted too long. In general, the hypertension produced by constricting one renal artery with the other kidney intact (one-clip, two-kidney Goldblatt hypertension) is associated with increased circulating renin. The clinical counterpart of this condition is **renal hypertension** due to atheromatous narrowing of one renal artery or other abnormalities of the renal circulation. However, plasma-renin activity is usually normal in one-clip one-kidney Goldblatt hypertension. The explanation of the hypertension in this situation is unsettled. However, many patients with hypertension respond to treatment with ACE inhibitors or losartan even when their renal circulation appears to be normal and they have normal or even low plasma-renin activity.

ACTIONS

ANP and BNP in the circulation act on the kidneys to increase Na^+ excretion and injected CNP has a similar effect. They appear to produce this effect by dilating afferent arterioles and relaxing mesangial cells. Both of these actions increase glomerular filtration (see Chapter 37). In addition, they act on the renal tubules to inhibit Na^+ reabsorption (see Chapter 37). Other actions include an increase in capillary permeability, leading to extravasation of fluid and a decline in blood pressure. In addition, they relax vascular smooth muscle in arterioles and venules. CNP has a greater dilator effect on veins than ANP and BNP. These peptides also inhibit renin secretion and counteract the pressor effects of catecholamines and angiotensin II.

In the brain, ANP is present in neurons, and an ANP-containing neural pathway projects from the anteromedial part of the hypothalamus to the areas in the lower brainstem that are concerned with neural regulation of the cardiovascular system. In general, the effects of ANP in the brain are opposite to those of angiotensin II, and ANP-containing neural circuits appear to be involved in counteracting the effects of vasopressin and angiotensin II,

promoting natriuresis.

NATRIURETIC PEPTIDE RECEPTORS

Three different natriuretic peptide receptors (NPRs) have been isolated and characterized. The NPR-A and NPR-B receptors both span the cell membrane and have cytoplasmic domains that are guanylyl cyclases. ANP has the greatest affinity for the NPR-A receptor, and CNP has the greatest affinity for the NPR-B receptor. The third receptor, NPR-C, binds all three natriuretic peptides but has a markedly truncated cytoplasmic domain. Some evidence suggests that it acts via G-proteins to activate phospholipase C and inhibit adenylyl cyclase. However, it has also been argued that this receptor does not trigger any intracellular change and is instead a **clearance receptor** that removes natriuretic peptides from the bloodstream and then releases them later, helping to maintain a steady blood level of the hormones.

SECRETION & METABOLISM

The concentration of ANP in plasma is about 5 fmol/mL in normal humans ingesting moderate amounts of NaCl. ANP secretion is increased when the ECF volume is increased by infusion of isotonic saline and when the atria are stretched. BNP secretion is increased when the ventricles are stretched. ANP secretion is also increased by immersion in water up to the neck **(Figure 38–10)**, a procedure that counteracts the effect of gravity on the circulation and increases central venous and consequently atrial pressure. Note that immersion also decreases the secretion of renin and aldosterone. Conversely, a small but measurable decrease in plasma ANP occurs in association with a decrease in central venous pressure on rising from the supine to the standing position. Thus, it seems clear that the atria respond directly to stretch in vivo and that the rate of ANP secretion is proportional to the degree to which the atria are stretched by increases in central venous pressure. Similarly, BNP secretion is proportional to the degree to which the ventricles are stretched. Plasma levels of both hormones are elevated in heart failure, and their measurement is being used increasingly in the diagnosis of this condition.

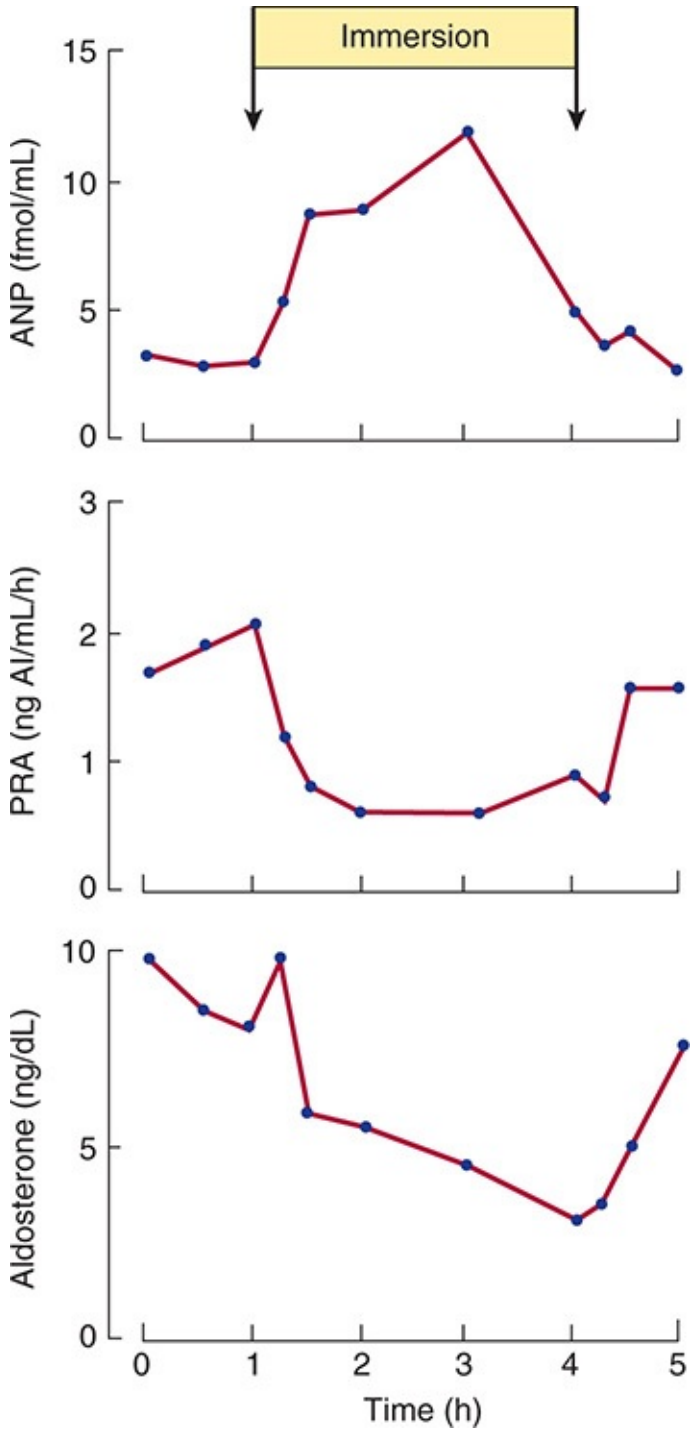


FIGURE 38–10 Effect of immersion in water up to the neck for 3 h on plasma concentrations of ANP, PRA, and aldosterone. ANP, atrial natriuretic peptide; PRA, plasma renin activity. (Reproduced with permission from Epstein M, et al: Increases in circulating atrial natriuretic factor during immersion-induced central hypervolaemia in normal humans. J Hypertension Suppl 1986 June; 4(2):S93–S99.))

Circulating ANP has a short half-life. It is metabolized by neutral endopeptidase (NEP), which is inhibited by thiorphan. Therefore, administration of thiorphan increases circulating ANP.

NA, K ATPase-INHIBITING FACTOR

Another natriuretic factor is present in blood. This factor produces natriuresis by inhibiting Na, K ATPase and raises rather than lowers blood pressure. Current evidence indicates that it may well be the digitalis-like steroid **ouabain** and that it comes from the adrenal glands. However, its physiologic significance is not yet known.

DEFENSE OF SPECIFIC IONIC COMPOSITION

Special regulatory mechanisms maintain the levels of certain specific ions in the ECF as well as the levels of glucose and other nonionized substances important in metabolism (see Chapter 1). The feedback of Ca^{2+} on the parathyroids and the calcitonin-secreting cells to adjust their secretion maintains the ionized calcium level of the ECF (see Chapter 21). The Mg^{2+} concentration is subject to close regulation, but the mechanisms controlling Mg^{+} homeostasis are incompletely understood.

The mechanisms controlling Na^{+} and K^{+} content are part of those determining the volume and tonicity of ECF and have been discussed above. The levels of these ions are also dependent on the H^{+} concentration, and pH is one of the major factors affecting the anion composition of ECF. This will be discussed in Chapter 39.

ERYTHROPOIETIN

STRUCTURE & FUNCTION

When an individual bleeds or becomes hypoxic, hemoglobin synthesis is enhanced, and production and release of red blood cells from the bone marrow **(erythropoiesis)** are increased (see Chapter 31). Conversely, when the red cell volume is increased above normal levels by transfusion, the erythropoietic activity of the bone marrow decreases. These adjustments are brought about by changes in the circulating level of **erythropoietin,** a circulating glycoprotein that

contains 165 amino acid residues and four oligosaccharide chains that are necessary for its activity in vivo. Its blood level is markedly increased in anemia **(Figure 38–11)**.

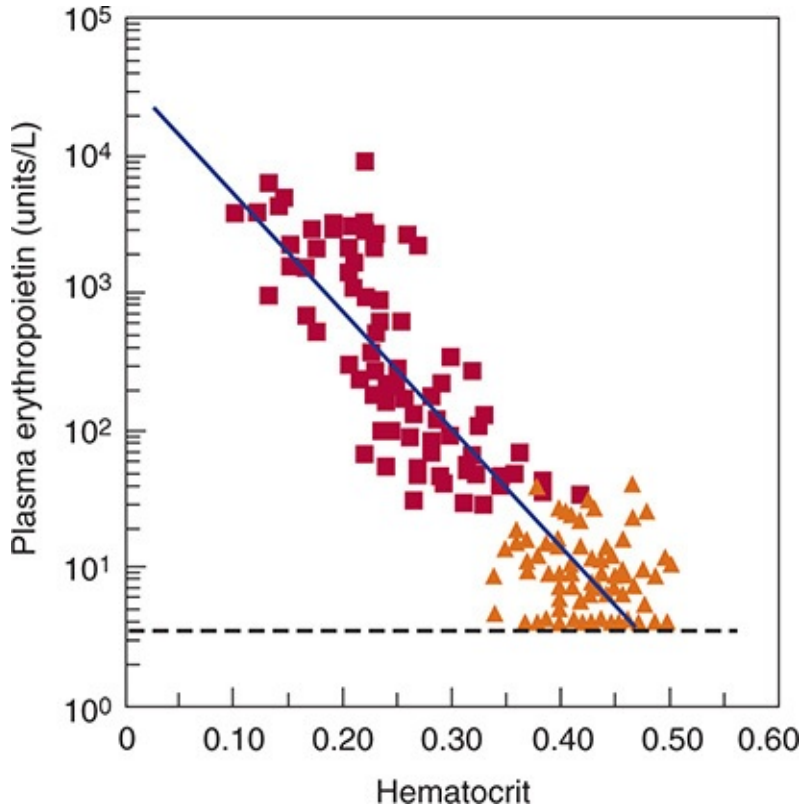


FIGURE 38–11 Plasma erythropoietin levels in normal blood donors (triangles) and patients with various forms of anemia (squares). (Reproduced with permission from Erslev AJ: Erythropoietin. N Engl J Med 1991; May 9; 324(19):1339–1344.)

Erythropoietin increases the number of erythropoietin-sensitive committed stem cells in the bone marrow that are converted to red blood cell precursors and subsequently to mature erythrocytes (see Chapter 31). The receptor for erythropoietin is a linear protein with a single transmembrane domain that is a member of the cytokine receptor superfamily (see Chapter 3). The receptor has tyrosine kinase activity, and it activates a cascade of serine and threonine kinases, resulting in inhibited apoptosis of red cells and their increased growth and development.

The principal site of inactivation of erythropoietin is the liver, and the hormone has a half-life in the circulation of about 5 h. However, the increase in circulating red cells that it triggers takes 2–3 days to appear, since red cell

maturation is a relatively slow process.

SOURCES

In adults, about 85% of the erythropoietin comes from the kidneys and 15% from the liver. Both these organs contain the mRNA for erythropoietin. Erythropoietin can also be extracted from the spleen and salivary glands, but these tissues do not contain its mRNA and consequently do not appear to manufacture the hormone. When renal mass is reduced in adults by kidney disease or nephrectomy, the liver cannot compensate and anemia develops.

Erythropoietin is produced by interstitial cells in the peritubular capillary bed of the kidneys and by perivenous hepatocytes in the liver. It is also produced in the brain, where it exerts a protective effect against excitotoxic damage triggered by hypoxia; and in the uterus and oviducts, where it is induced by estrogen and appears to mediate estrogen-dependent angiogenesis.

The gene for the hormone has been cloned, and recombinant erythropoietin produced in animal cells is available for clinical use as epoetin alfa. The recombinant erythropoietin is of value in the treatment of the anemia associated with kidney failure; 90% of the patients with end-stage renal disease who are undergoing dialysis are anemic as a result of erythropoietin deficiency. Erythropoietin is also used to stimulate red cell production in individuals who are banking a supply of their own blood in preparation for autologous transfusions during elective surgery (see Chapter 31).

REGULATION OF SECRETION

The usual stimulus for erythropoietin secretion is hypoxia, but secretion of the hormone can also be stimulated by cobalt salts and androgens. Recent evidence suggests that the O_2 sensor regulating erythropoietin secretion in the kidneys and the liver is a heme protein that in the deoxy form stimulates and in the oxy form inhibits transcription of the erythropoietin gene to form erythropoietin mRNA. Secretion of the hormone is also facilitated by the alkalosis that develops at high altitudes. Like renin secretion, erythropoietin secretion is facilitated by catecholamines via a β-adrenergic mechanism, although the renin–angiotensin system is totally separate from the erythropoietin system.

CHAPTER SUMMARY

- Total body osmolality is directly proportional to the total body sodium plus the total body potassium divided by the total body water. Changes in the osmolality of the body fluids occur when a disproportion exists between the amount of these electrolytes and the amount of water ingested or lost from the body.
- Vasopressin's main physiologic effect is the retention of water by the kidney by increasing the water permeability of the renal collecting ducts. Water is absorbed from the urine, the urine becomes concentrated, and its volume decreases.
- Vasopressin is stored in the posterior pituitary and released into the bloodstream in response to the stimulation of osmoreceptors or baroreceptors. Increases in secretion occur when osmolality is changed as little as 1%, thus keeping the osmolality of the plasma very close to 285 mOsm/L.
- The amount of Na^+ in the ECF is the most important determinant of ECF volume, and mechanisms that control Na^+ balance are the major mechanisms defending ECF volume. The main mechanism regulating sodium balance is the renin–angiotensin system, a hormone system that regulates blood pressure.
- The kidneys secrete the enzyme renin and renin acts in concert with angiotensin-converting enzyme to form angiotensin II. Angiotensin II acts directly on the adrenal cortex to increase the secretion of aldosterone. Aldosterone increases the retention of sodium from the urine via action on the renal collecting duct.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Dehydration increases the plasma concentration of all the following hormones *except*

   A. vasopressin.

   B. angiotensin II.

   C. aldosterone.

   D. norepinephrine.

E. atrial natriuretic peptide.

2. In a patient who has become dehydrated, body water should be replaced by intravenous infusion of

A. distilled water.

B. 0.9% sodium chloride solution.

C. 5% glucose solution.

D. hyperoncotic albumin.

E. 10% glucose solution.

3. Renin is secreted by

A. cells in the macula densa.

B. cells in the proximal tubules.

C. cells in the distal tubules.

D. granular cells in the juxtaglomerular apparatus.

E. cells in the peritubular capillary bed.

4. Erythropoietin is secreted by

A. cells in the macula densa.

B. cells in the proximal tubules.

C. cells in the distal tubules.

D. granular cells in the juxtaglomerular apparatus.

E. cells in the peritubular capillary bed.

5. When a woman who has been on a low-sodium diet for 8 days is given an intravenous injection of captopril, a drug that inhibits angiotensin-converting enzyme, which of the following would be expected?

A. Blood pressure to rise because cardiac output would fall

B. Blood pressure to rise because peripheral resistance would fall

C. Blood pressure to fall because cardiac output would fall

D. Blood pressure to fall because peripheral resistance would fall

E. Plasma renin activity to fall because circulating angiotensin I level would rise

6. Which of the following would *not* be expected to increase renin secretion?

A. Administration of a drug that blocks angiotensin-converting enzyme

B. Administration of a drug that blocks AT1 receptors

C. Administration of a drug that blocks β-adrenergic receptors

D. Constriction of the aorta between the celiac artery and the renal arteries

E. Administration of a drug that reduces ECF volume

7. Which of the following is *least* likely to contribute to the beneficial effects of angiotensin-converting enzyme inhibitors in the treatment of heart failure?

A. Vasodilation

B. Decreased cardiac growth

C. Decreased cardiac afterload

D. Increased plasma renin activity

E. Decreased plasma aldosterone

CHAPTER 39

Acidification of the Urine & Bicarbonate Excretion

OBJECTIVES

After studying this chapter, you should be able to:

- Explain the three processes involved in the secretion of H^+ into the tubules and discuss the significance of these processes in the regulation of acid-base balance.
- Define acidosis and alkalosis, and give (in mEq/L and pH) the normal mean and the range of H^+ concentrations in blood that are compatible with health.
- List the principal buffers in blood, interstitial fluid, and intracellular fluid, and, using the Henderson-Hasselbalch equation, describe what is unique about the bicarbonate buffer system.
- Define the changes in blood chemistry that occur during metabolic acidosis and metabolic alkalosis. Explain the integration of respiratory and renal compensations for these conditions.
- Define the changes in blood chemistry that occur during respiratory acidosis and respiratory alkalosis, and how the kidneys compensate in these conditions.

INTRODUCTION

The kidneys play a key role in the maintenance of acid-base balance and to do this they must excrete acid in the amount equivalent to the production of nonvolatile acids in the body. The production of nonvolatile acids will vary with diet, metabolism, and disease. The kidneys must also filter and reabsorb plasma bicarbonate, and thus prevent the loss of bicarbonate in the urine. Both processes are linked physiologically, due to the nephron's ability to secrete H^+ ions into the filtrate.

RENAL H^+ SECRETION

The cells of the proximal and distal tubules, like the cells of the gastric glands (see Chapter 25), secrete hydrogen ions. Hydrogen secretion also occurs in the collecting ducts. The transporter that is responsible for H^+ secretion in the proximal tubules is the Na–H exchanger (primarily NHE3) **(Figure 39–1)**. This is an example of secondary active transport; Na^+ is moved from the inside of the cell to the interstitium by Na, K ATPase on the basolateral membrane, which keeps intracellular Na^+ low, thus establishing the drive for Na^+ to enter the cell, via the Na–H exchanger, from the tubular lumen. The Na–H exchanger secretes H^+ into the lumen in exchange for Na^+.

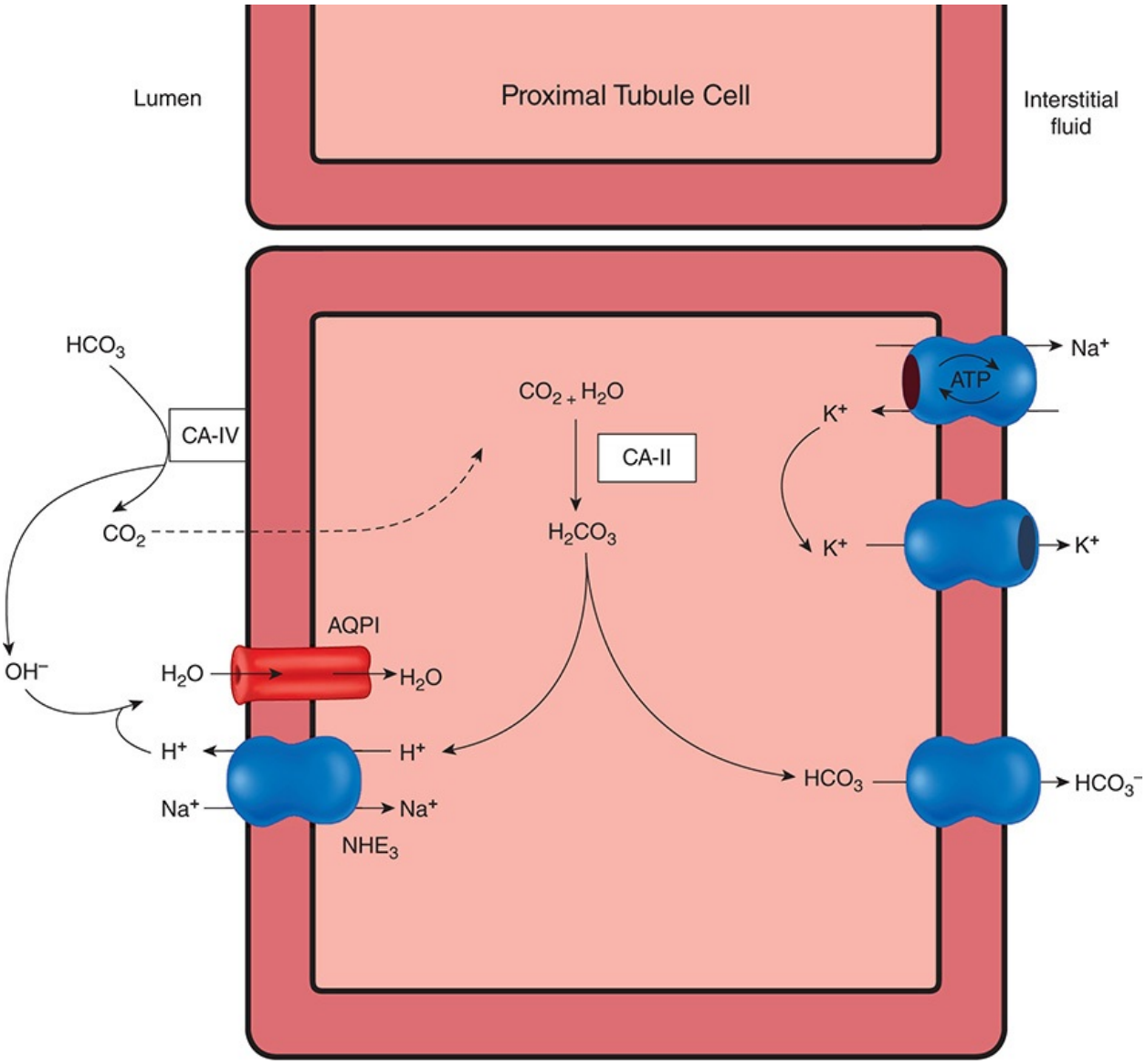


FIGURE 39–1 Secretion of acid and reabsorption of filtered bicarbonate by proximal tubular cells in the kidney. H^+ is transported into the tubular lumen by NHE_3 in exchange for Na^+. Active transport by Na, K ATPase is indicated by arrows. Dashed arrows indicate diffusion.

The secreted H^+ ion combines with filtered HCO_3^- to form H_2CO_3 and the presence of **carbonic anhydrase** on the apical membrane of the proximal tubule catalyzes the formation of H_2O and CO_2 from H_2CO_3. The apical membrane of epithelial cells lining the proximal tubule is permeable to CO_2 and H_2O, and they enter the tubule rapidly. Eighty percent of the filtered load of HCO_3^- is reabsorbed in the proximal tubule.

Inside the cell, carbonic anhydrase is also present and can catalyze the formation of H_2CO_3 from CO_2 and H_2O. H_2CO_3 dissociates into H^+ ions and HCO_3^-; the H^+ is secreted into the tubular lumen, as mentioned above, and the HCO_3^- that is formed diffuses into the interstitial fluid. Thus, for each H^+ ion secreted, one Na^+ ion and one HCO_3^- ion enter the interstitial fluid. Because carbonic anhydrase catalyzes the formation of H_2CO_3, drugs that inhibit carbonic anhydrase depress both secretion of acid by the proximal tubules and the reactions that depend on it.

Some evidence suggests that H^+ is secreted in the proximal tubules by other types of transporters, but the evidence for these additional transporters is controversial, and in any case, their contribution is small relative to that of the Na–H exchanger mechanism.

This is in contrast to what occurs in the distal tubules and collecting ducts, where H^+ secretion is relatively independent of Na^+ in the tubular lumen. In this part of the tubule, most H^+ is secreted by an ATP-driven proton pump. Aldosterone acts on this pump to increase distal H^+ secretion. The I cells in this part of the renal tubule secrete acid and, like the parietal cells in the stomach, contain abundant carbonic anhydrase and numerous tubulovesicular structures. There is evidence that the H^+-translocating ATPase that produces H^+ secretion is located in these vesicles as well as in the apical cell membrane and that, in acidosis, the number of H^+ pumps is increased by insertion of these tubulovesicles into the apical cell membrane. Some of the H^+ is additionally secreted by $H–K^+$ ATPase. The I cells also contain **anion exchanger 1 (AE1,** formerly known as **Band 3**), an anion exchange protein, in their basolateral cell membranes. This protein may function as a Cl/HCO_3 exchanger for the transport of HCO_3^- to the interstitial fluid.

FATE OF H^+ IN THE URINE

The amount of acid secreted depends on the subsequent events that modify the composition of the tubular urine. The maximal H^+ gradient against which the transport mechanisms can secrete in humans corresponds to a urine pH of about 4.5; that is, an H^+ concentration in the urine that is 1000 times the concentration in plasma. pH 4.5 is thus the **limiting pH.** This is normally reached in the collecting ducts. If there were no buffers that “tied up” H^+ in the urine, this pH

would be reached rapidly, and H^+ secretion would stop. However, three important reactions in the tubular fluid remove free H^+, permitting more acid to be secreted **(Figure 39–2)**. These are the reactions of H^+ with HCO_3^- to form CO_2 and H_2O (discussed above), with HPO_4^{2-} to form $H_2PO_4^-$ (titratable acids), and with NH_3 to form NH_4^+.

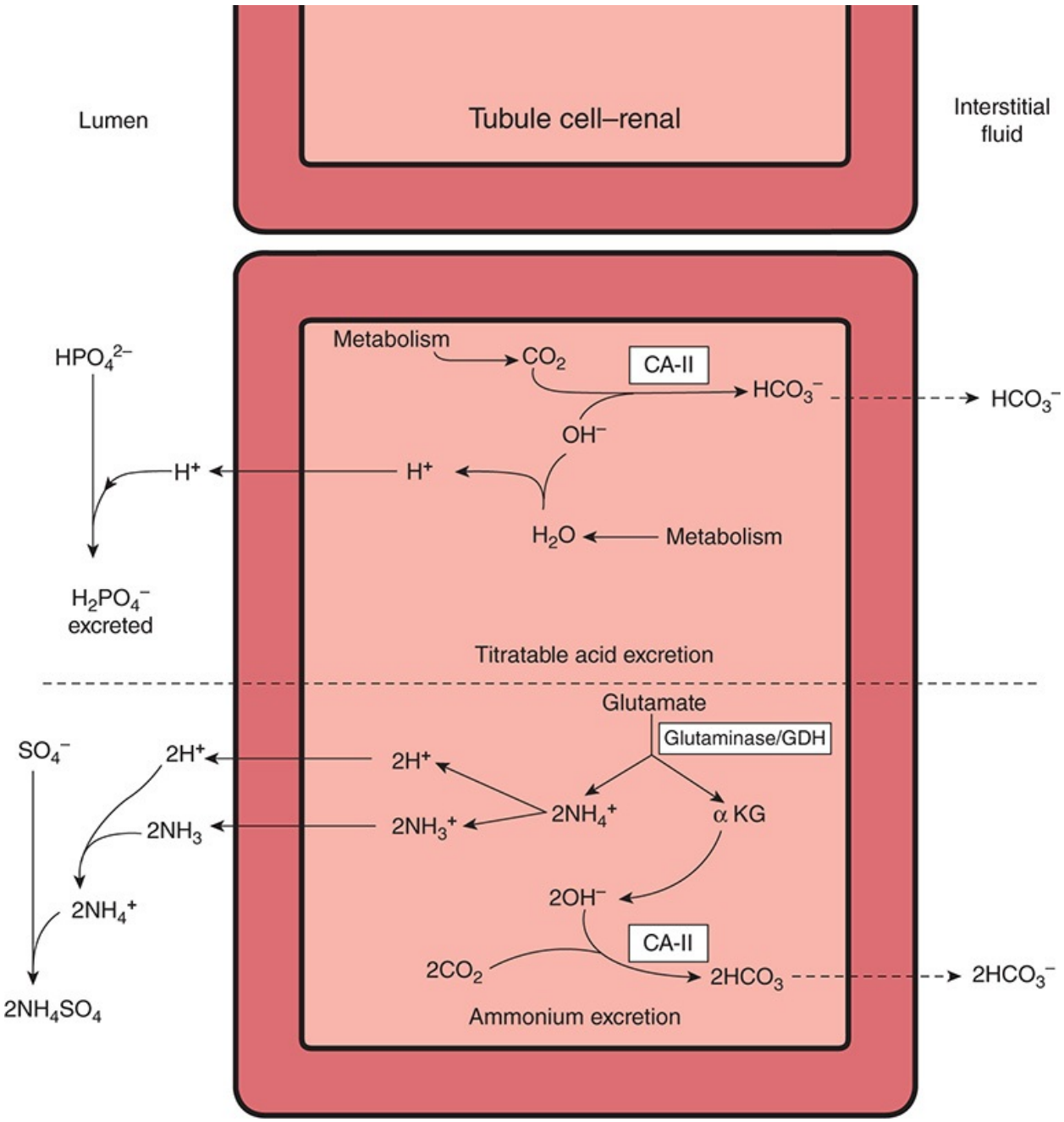


FIGURE 39–2 Titratable acid and ammonium formation. Top: Formation of

monobasic phosphate. **Bottom:** Ammonium formation. Note that in each instance one Na^+ ion and one HCO_3^- ion enter the bloodstream for each H^+ ion secreted.

REACTION WITH BUFFERS

The three buffers of importance in the renal handling of acid and its secretion into the lumen are thus bicarbonate, dibasic phosphate, and ammonia. On an average diet, approximately 40% of nonvolatile acids (about 30 mEq/day), produced by the body in the course of various metabolic reactions is excreted as **titratable acid** (ie, phosphate system) and 60% of nonvolatile acid (about 50 mEq/day) is excreted as NH_4^+. The pK′ of the bicarbonate system is 6.1, that of the dibasic phosphate system is 6.8, and that of the ammonia system is 9.0. The concentration of HCO_3^- in the plasma, and consequently in the glomerular filtrate, is normally about 24 mEq/L, whereas that of phosphate is only 1.5 mEq/L. Therefore, in the proximal tubule, most of the secreted H^+ reacts with HCO_3^- as described above, to form H_2CO_3 (Figure 39–2) and this enters the cell as CO_2 and H_2O following the action of carbonic anhydrase in the brush border of the proximal tubule cells. The CO_2 entering the tubular cells adds to the pool of CO_2 available to form H_2CO_3. Because most of the H^+ is removed from the tubule, the pH of the fluid is changed very little. This is the mechanism by which HCO_3^- is reabsorbed; for each mole of HCO_3^- removed from the tubular fluid, 1 mol of HCO_3^- diffuses from the tubular cells into the blood, although it is important to note that it is not the same mole that disappeared from the tubular fluid. About 4500 mEq of HCO_3^- are filtered and reabsorbed each day.

Secreted H^+ also reacts with dibasic phosphate (HPO_4^{2-}) to form monobasic phosphate ($H_2PO_4^-$). This happens to the greatest extent in the distal tubules and collecting ducts, because it is here that the phosphate that escapes proximal reabsorption is greatly concentrated by the reabsorption of water. H^+ ions are also known to combine to a minor degree with other buffer anions.

The ammonia buffering system allows secreted H^+ to combine with NH_3, and this occurs in the proximal tubule (where NH_3 is made, see below) and in the distal tubules. The pK′ of the ammonia system is 9.0, and the ammonia system is titrated only from the pH of the urine to pH 7.4, so it contributes very little to the

titratable acidity. Each H^+ ion that reacts with the buffers contributes to the urinary **titratable acidity,** which is measured by determining the amount of alkali that must be added to the urine to return its pH to 7.4, the pH of the glomerular filtrate. However, the titratable acidity obviously measures only a fraction of the acid secreted, since it does not account for the H_2CO_3 that has been converted to H_2O and CO_2.

The reabsorption of HCO_3^- is crucial to the maintenance of acid-base balance, as a loss of a single HCO_3^- ion in the urine would be the equivalent of adding a H^+ ion to the blood. However, the kidneys have the ability to replenish the body with new bicarbonate ions. This occurs when H^+ ions are removed from the body as NH_4^+ or titratable acid, as there is formation of new bicarbonate within the cells, and this enters the blood (ie, these bicarbonate ions are not those originally filtered, and yet they still enter the blood).

AMMONIA SECRETION

As mentioned above, reactions in the renal tubular cells produce NH_4^+ and HCO_3^-. NH_4^+ is in equilibrium with NH_3 and H^+ in the cells. Because the pK′ of this reaction is 9.0, the ratio of NH_3 to NH_4^+ at pH 7.0 is 1:100 **(Figure 39–3)**. However, NH_3 is lipid-soluble and diffuses across the cell membranes down its concentration gradient into the interstitial fluid and tubular urine. In the urine it reacts with H^+ to form NH_4^+, and the NH_4^+ remains "trapped" in the urine.

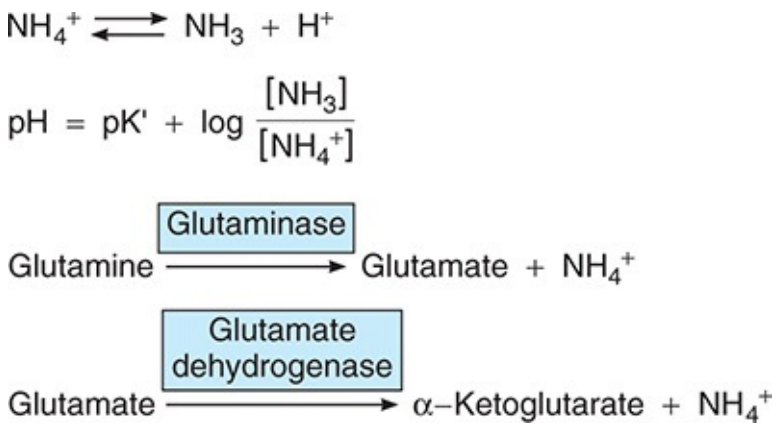


FIGURE 39–3 Major reactions involved in ammonia production in the kidneys.

The principal reaction producing NH_4^+ in cells is conversion of glutamine to glutamate. This reaction is catalyzed by the enzyme **glutaminase,** which is abundant in renal tubular cells (Figure 39–3). **Glutamate dehydrogenase** catalyzes the conversion of glutamate to α-ketoglutarate, with the production of more NH_4^+. Subsequent metabolism of α-ketoglutarate utilizes $2H^+$, freeing $2HCO_3^-$.

In chronic acidosis, the amount of NH_4^+ excreted at any given urine pH also increases, because more NH_3 enters the tubular urine. The effect of this **adaptation** of NH_3 secretion, the cause of which is unsettled, is further removal of H^+ from the tubular fluid and consequently a further enhancement of H^+ secretion by the renal tubules and excretion in the urine. Because the amount of phosphate buffer filtered at the glomerulus cannot be increased, urinary excretion of acid via the phosphate buffer system is limited. The production of NH_4^+ by the renal tubules is the only way the kidneys can remove even the normal amount, much less an increased amount, of nonvolatile acid produced in the body.

In the inner medullary cells of the collecting duct, the main process by which NH_3 is secreted into the urine and then changed to NH_4^+ is called **nonionic diffusion** (see Chapter 2), thereby maintaining the concentration gradient for diffusion of NH_3. In the proximal tubule, nonionic diffusion of NH_4^+ is less important because NH_4^+ can be secreted into the lumen, often by replacing H^+ on the Na–H exchanger.

Salicylates and a number of other drugs that are weak bases or weak acids are also secreted by nonionic diffusion. They diffuse into the tubular fluid at a rate that depends on the pH of the urine, so the amount of each drug excreted varies with the pH of the urine.

pH CHANGES ALONG THE NEPHRON

A moderate drop in pH occurs in the proximal tubular fluid, but, as noted above, most of the secreted H^+ has little effect on luminal pH because of the formation of CO_2 and H_2O from H_2CO_3. In contrast, the distal tubule has less capacity to secrete H^+, but secretion in this segment has a greater effect on urinary pH.

FACTORS AFFECTING ACID SECRETION

Renal acid secretion is altered by changes in the intracellular PCO_2, K^+ concentration, carbonic anhydrase level, and adrenocortical hormone concentration. When the PCO_2 is high **(respiratory acidosis),** more intracellular H_2CO_3 is available to buffer the hydroxyl ions and acid secretion is enhanced, whereas the reverse is true when the PCO_2 falls. K^+ depletion enhances acid secretion, apparently because the loss of K^+ causes intracellular acidosis even though the plasma pH may be elevated. Conversely, K^+ excess in the cells inhibits acid secretion. When carbonic anhydrase is inhibited, acid secretion is inhibited because the formation of H_2CO_3 is decreased. Aldosterone and the other adrenocortical steroids that enhance tubular reabsorption of Na^+ also increase the secretion of H^+ and K^+.

BICARBONATE EXCRETION

Although the process of HCO_3^- reabsorption does not involve actual transport of this ion into the tubular cells, HCO_3^- reabsorption is proportional to the amount filtered over a relatively wide range. There is no demonstrable Tm, but HCO_3^- reabsorption is decreased by an unknown mechanism when the extracellular fluid (ECF) volume is expanded **(Figure 39–4)**. When the plasma HCO_3^- concentration is low, all the filtered HCO_3^- is reabsorbed; but when the plasma HCO_3^- concentration is high; that is, above 26–28 mEq/L (the renal threshold for HCO_3^-), HCO_3^- appears in the urine and the urine becomes alkaline. Conversely, when the plasma HCO_3^- falls below about 26 mEq/L, the value at which all the secreted H^+ is being used to reabsorb HCO_3^-, more H^+ becomes available to combine with other buffer anions. Therefore, the lower the plasma HCO_3^- concentration drops, the more acidic the urine becomes and the greater its NH_4^+ content **(Clinical Box 39–1)**.

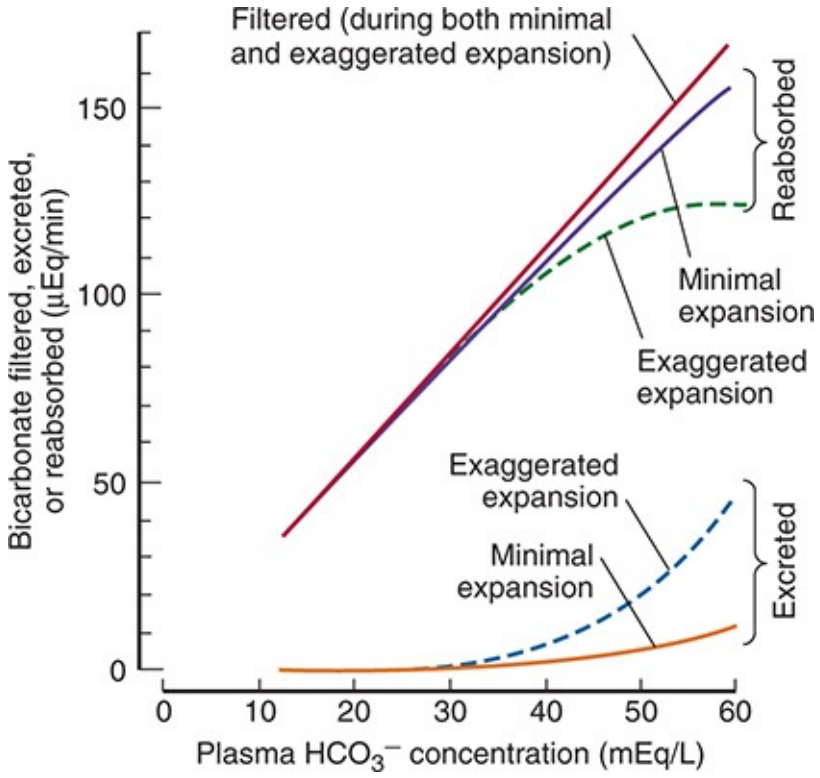


FIGURE 39–4 Effect of ECF volume on HCO_3^- filtration, reabsorption, and excretion in rats. The pattern of HCO_3^- excretion is similar in humans. The plasma HCO_3^- concentration is normally about 24 mEq/L. ECF, extracellular fluid. (Reproduced with permission from Valtin H: *Renal Function*, 2nd ed. Little, Brown; 1983.)

DEFENSE OF H^+ CONCENTRATION

The mystique that envelopes the subject of acid-base balance makes it necessary to point out that the core of the problem is not "buffer base" or "fixed cation" or the like, but simply the maintenance of the H^+ concentration of the ECF. The mechanisms regulating the composition of the ECF are particularly important as far as this specific ion is concerned, because the machinery of the cells is very sensitive to changes in H^+ concentration. Intracellular H^+ concentration, which can be measured by using microelectrodes, pH-sensitive fluorescent dyes, and phosphorus magnetic resonance, is distinct from extracellular pH and appears to be regulated by a variety of intracellular processes. However, it is sensitive to changes in ECF H^+ concentration.

The pH notation is a useful means of expressing H^+ concentrations in the

body, because the H^+ concentrations are very low relative to those of other cations. Thus, the normal Na^+ concentration of arterial plasma that has been equilibrated with red blood cells is about 140 mEq/L, whereas the H^+ concentration is 0.00004 mEq/L **(Table 39–1)**. The pH, the negative logarithm of 0.00004, is therefore 7.4. Of course, a decrease in pH of 1 unit, for example, from 7.0 to 6.0, represents a 10-fold increase in H^+ concentration. It is important to remember that the pH of blood is the pH of **true plasma**—plasma that has been in equilibrium with red cells—because the red cells contain hemoglobin, which is quantitatively one of the most important blood buffers (see Chapter 36).

TABLE 39–1 H^+ concentration and pH of body fluids.

		H^+ Concentration		
		mEq/L	mol/L	pH
Gastric HCl		150	0.15	0.8
Maximal urine acidity		0.03	3×10^{-5}	4.5
Plasma	Extreme acidosis	0.0001	1×10^{-7}	7.0
	Normal	0.00004	4×10^{-8}	7.4
	Extreme alkalosis	0.00002	2×10^{-8}	7.7
Pancreatic juice		0.00001	1×10^{-8}	8.0

CLINICAL BOX 39–1

Implications of Urinary pH Changes

Depending on the rates of the interrelated processes of acid secretion, NH_4^+ production, and HCO_3^- excretion, the pH of the urine in humans varies from 4.5 to 8.0. Excretion of urine that is at a pH different from that of the body fluids has important implications for the body's electrolyte and acid–base economy. Acids are buffered in the plasma and cells, the overall reaction being $HA + NaHCO_3 \rightarrow NaA + H_2CO_3$. The H_2CO_3 forms CO_2 and H_2O, and the CO_2 is expired, while the NaA appears in the glomerular filtrate. To the extent that the Na^+ is replaced by H^+ in the urine, Na^+ is conserved in the body. Furthermore, for each H^+ ion excreted with phosphate or as NH_4^+, there

is a net gain of one HCO_3^- ion in the blood, replenishing the supply of this important buffer anion. Conversely, when base is added to the body fluids, the OH^- ions are buffered, raising the plasma HCO_3^-. When the plasma level exceeds 28 mEq/L, the urine becomes alkaline and the extra HCO_3^- is excreted in the urine. Because the rate of maximal H^+ secretion by the tubules varies directly with the arterial PCO_2. HCO_3^- reabsorption is also affected by the PCO_2. This relationship has been discussed in more detail in the text.

THERAPEUTIC HIGHLIGHTS

Sulfonamides inhibit carbonic anhydrase and sulfonamide derivatives have been used clinically as diuretics because of their inhibitory effects on carbonic anhydrase in the kidney (see Chapter 37).

H^+ BALANCE

The pH of the arterial plasma is normally 7.40 and that of venous plasma slightly lower. Technically, **acidosis** is present whenever the arterial pH is below 7.40, and **alkalosis** is present whenever it is above 7.40, although variations of up to 0.05 pH unit occur without untoward effects. The H^+ concentrations in the ECF that are compatible with life cover an approximately fivefold range, from 0.00002 mEq/L (pH 7.70) to 0.0001 mEq/L (pH 7.00).

Amino acids are utilized in the liver for gluconeogenesis, leaving NH_4^+ and HCO_3^- as products from their amino and carboxyl groups **(Figure 39–5)**. The NH_4^+ is incorporated into urea (see Chapter 28) and the protons that are formed are buffered intracellularly by HCO_3^-, so little NH_4^+ and HCO_3^- escape into the circulation. However, metabolism of sulfur-containing amino acids produces H_2SO_4, and metabolism of phosphorylated amino acids such as phosphoserine produces H_3PO_4. These strong acids enter the circulation and present a major H^+ load to the buffers in the ECF. The H^+ load from amino acid metabolism is normally about 50 mEq/day. The CO_2 formed by metabolism in the tissues is in large part hydrated to H_2CO_3 (see Chapter 36), and the total H^+ load from this source is over 12,500 mEq/day. However, most of the CO_2 is excreted in the

lungs, and only small quantities of the H^+ remain to be excreted by the kidneys. Common sources of extra acid loads are strenuous exercise (lactic acid), diabetic ketosis (acetoacetic acid and β-hydroxybutyric acid), and ingestion of acidifying salts such as NH_4Cl and $CaCl_2$, which in effect add HCl to the body. A failure of diseased kidneys to excrete normal amounts of acid is also a cause of acidosis. Fruits are the main dietary source of alkali. They contain Na^+ and K^+ salts of weak organic acids, and the anions of these salts are metabolized to CO_2, leaving $NaHCO_3$ and $KHCO_3$ in the body. $NaHCO_3$ and other alkalinizing salts are sometimes ingested in large amounts, but a more common cause of alkalosis is loss of acid from the body as a result of vomiting of gastric juice rich in HCl. This is, of course, equivalent to adding alkali to the body.

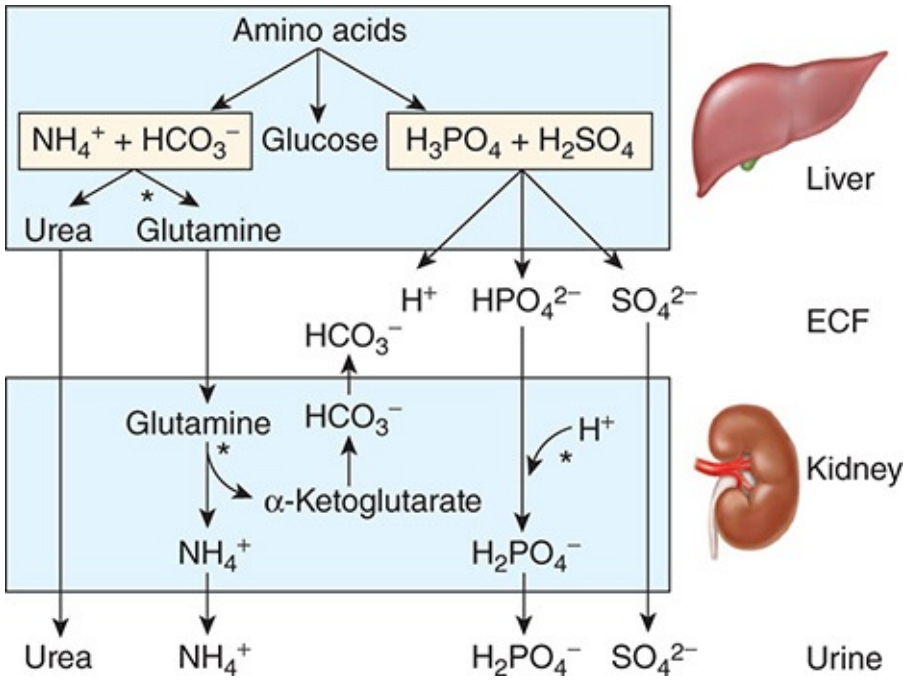


FIGURE 39–5 Role of the liver and kidneys in the handling of metabolically produced acid loads. Sites where regulation occurs are indicated by asterisks. (Modified with permission from Knepper MA, et al: Ammonium, urea, and systemic pH regulation. Am J Physiol 1987; July; 253(1 Pt 2): F199–F202.)

BUFFERING

Buffering is of key importance in maintaining H^+ homeostasis. It is defined in Chapter 1 and discussed in Chapter 36 in the context of gas transport, with an

emphasis on roles for proteins, hemoglobin, and the carbonic anhydrase system in the blood. Carbonic anhydrase is also found in high concentration in gastric acid–secreting cells (see Chapter 25) and in renal tubular cells (see Chapter 37). Carbonic anhydrase is a protein with a molecular weight of 30,000 that contains an atom of zinc in each molecule. It is inhibited by cyanide, azide, and sulfide. In vivo, buffering is, of course, not limited to the blood. The principal buffers in the blood, interstitial fluid, and intracellular fluid are listed in **Table 39–2**. The principal buffers in cerebrospinal fluid and urine are the bicarbonate and phosphate systems. In metabolic acidosis, only 15–20% of the acid load is buffered by the H_2CO_3–HCO_3^- system in the ECF, and most of the remainder is buffered in cells. In metabolic alkalosis, about 30–35% of the OH^- load is buffered in cells, whereas in respiratory acidosis and alkalosis, almost all of the buffering is intracellular.

TABLE 39–2 Principal buffers in body fluids.

Blood	$H_2CO_3 \rightleftarrows H^+ + HCO_3^-$ $HProt \rightleftarrows H^+ + Prot^-$ $HHb \rightleftarrows H^+ + Hb^-$
Interstitial fluid	$H_2CO_3 \rightleftarrows H^+ + HCO_3^-$
Intracellular fluid	$HProt \rightleftarrows H^+ + Prot^-$ $H_2PO_4^- \rightleftarrows H^+ + HPO_4^{2-}$

In animal cells, the principal regulators of intracellular pH are HCO_3^- transporters. Those characterized to date include the Cl^-–HCO_3^- exchanger **AE1,** three Na^+–HCO_3^- cotransporters, and a K^+–HCO_3^- cotransporter.

SUMMARY

When a strong acid is added to the blood, the major buffer reactions are driven to the left. The blood levels of the three "buffer anions" Hb^- (hemoglobin), $Prot^-$ (protein), and HCO_3^- consequently drop. The anions of the added acid are filtered into the renal tubules. They are accompanied ("covered") by cations, particularly Na^+, because electrochemical neutrality is maintained. By processes that have been discussed above, the tubules replace the Na^+ with H^+ and in so

doing reabsorb equimolar amounts of Na^+ and HCO_3^-, thus conserving the cations, eliminating the acid, and restoring the supply of buffer anions to normal. When CO_2 is added to the blood, similar reactions occur, except that since it is H_2CO_3 that is formed, the plasma HCO_3^- rises rather than falls.

RENAL COMPENSATION TO RESPIRATORY ACIDOSIS & ALKALOSIS

As noted in Chapter 35, a rise in arterial PCO_2 due to decreased ventilation causes **respiratory acidosis** and conversely, a decline in PCO_2 causes **respiratory alkalosis.** The initial changes shown in Figure 35–7 are those that occur independently of any compensatory mechanism; that is, they are those of **uncompensated** respiratory acidosis or alkalosis. In either situation, changes are produced in the kidneys, which then tend to **compensate** for the acidosis or alkalosis, adjusting the pH toward normal.

HCO_3^- reabsorption in the renal tubules depends not only on the filtered load of HCO_3^-, which is the product of the glomerular filtration rate and the plasma HCO_3^- level, but also on the rate of H^+ secretion by the renal tubular cells, since HCO_3^- is reabsorbed by exchange for H^+. The rate of H^+ secretion—and hence the rate of HCO_3^- reabsorption—is proportional to the arterial PCO_2, probably because the more CO_2 that is available to form H_2CO_3 in the tubular cells, the more H^+ that can be secreted. Furthermore, when the PCO_2 is high, the interior of most cells becomes more acidic. In respiratory acidosis, renal tubular H^+ secretion is therefore increased, removing H^+ from the body; and even though the plasma HCO_3^- is elevated, HCO_3^- reabsorption is increased, further raising the plasma HCO_3^-. This renal compensation for respiratory acidosis is shown graphically in the shift from acute to chronic respiratory acidosis in Figure 35–7. Cl^- excretion is increased, and plasma Cl^- falls as plasma HCO_3^- is increased. Conversely, in respiratory alkalosis, the low PCO_2 hinders renal H^+ secretion, HCO_3^- reabsorption is depressed, and HCO_3^- is excreted, further reducing the already low plasma HCO_3^- and lowering the pH toward normal.

METABOLIC ACIDOSIS

When acids stronger than Hb and the other buffer acids are added to blood, **metabolic acidosis** is produced; and when the free H^+ level falls as a result of addition of alkali or removal of acid, **metabolic alkalosis** results. Following the example from Chapter 35, if H_2SO_4 is added, the H^+ is buffered and the Hb^-, $Prot^-$, and HCO_3^- levels in plasma drop. The H_2CO_3 formed is converted to H_2O and CO_2, and the CO_2 is rapidly excreted via the lungs. This is the situation in **uncompensated** metabolic acidosis. Actually, the rise in plasma H^+ stimulates respiration, so that the PCO_2, instead of rising or remaining constant, is reduced. This **respiratory compensation** raises the pH even further. The **renal** compensatory mechanisms then bring about the excretion of the extra H^+ and return the buffer systems to normal.

RENAL COMPENSATION

The anions that replace HCO_3^- in the plasma in metabolic acidosis are filtered, each with a cation (principally Na^+), thus maintaining electrical neutrality. The renal tubular cells secrete H^+ into the tubular fluid in exchange for Na^+; and for each H^+ secreted, one Na^+ and one HCO_3^- are added to the blood. The limiting urinary pH of 4.5 would be reached rapidly and the total amount of H^+ secreted would be small if no buffers were present in the urine to "tie up" H^+. However, secreted H^+ reacts with HCO_3^- to form CO_2 and H_2O (bicarbonate reabsorption); with HPO_4^{2-} to form $H_2PO_4^-$; and with NH_3 to form NH_4^+. In this way, large amounts of H^+ can be secreted, permitting correspondingly large amounts of HCO_3^- to be returned to (in the case of bicarbonate reabsorption) or added to the depleted body stores and large numbers of the cations to be reabsorbed. It is only when the acid load is very large that cations are lost with the anions, producing diuresis and depletion of body cation stores. In chronic acidosis, glutamine synthesis in the liver is increased, using some of the NH_4^+ that usually is converted to urea (Figure 39–5), and the glutamine provides the kidneys with an additional source of NH_4^+. NH_3 secretion increases over a period of days (adaptation of NH_3 secretion), further improving the renal compensation for acidosis. In addition, the metabolism of glutamine in the kidneys produces α-

ketoglutarate, and this in turn is decarboxylated, producing HCO_3^-, which enters the bloodstream and helps buffer the acid load (Figure 39–5).

The overall reaction in blood when a strong acid such as H_2SO_4 is added is:

$$2NaHCO_3 + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2CO_3$$

For each mole of H^+ added, 1 mole of $NaHCO_3$ is lost. The kidney in effect reverses the reaction:

$$Na_2SO_4 + 2\,H_2CO_3 \rightarrow 2NaHCO_3 + 2H^+ + SO_4^{2-}$$

and the H^+ and SO_4^{2-} are excreted. Of course, H_2SO_4 is not excreted as such, the H^+ appearing in the urine as titratable acidity and NH_4^+.

In metabolic acidosis, the respiratory compensation tends to inhibit the renal response in the sense that the induced drop in PCO_2 hinders acid secretion, but it also decreases the filtered load of HCO_3^- and so its net inhibitory effect is not great.

METABOLIC ALKALOSIS

In metabolic alkalosis, the plasma HCO_3^- level and pH rise **(Figure 39–6)**. The respiratory compensation is a decrease in ventilation produced by the decline in H^+ concentration, and this elevates the PCO_2. This brings the pH back toward normal while elevating the plasma HCO_3^- level still further. The magnitude of this compensation is limited by the carotid and aortic chemoreceptor mechanisms, which drive the respiratory center if any appreciable fall occurs in the arterial PO_2. In metabolic alkalosis, more renal H^+ secretion is expended in reabsorbing the increased filtered load of HCO_3^-; and if the HCO_3^- level in plasma exceeds 26–28 mEq/L, HCO_3^- appears in the urine. The rise in PCO_2 inhibits the renal compensation by facilitating acid secretion, but its effect is relatively slight.

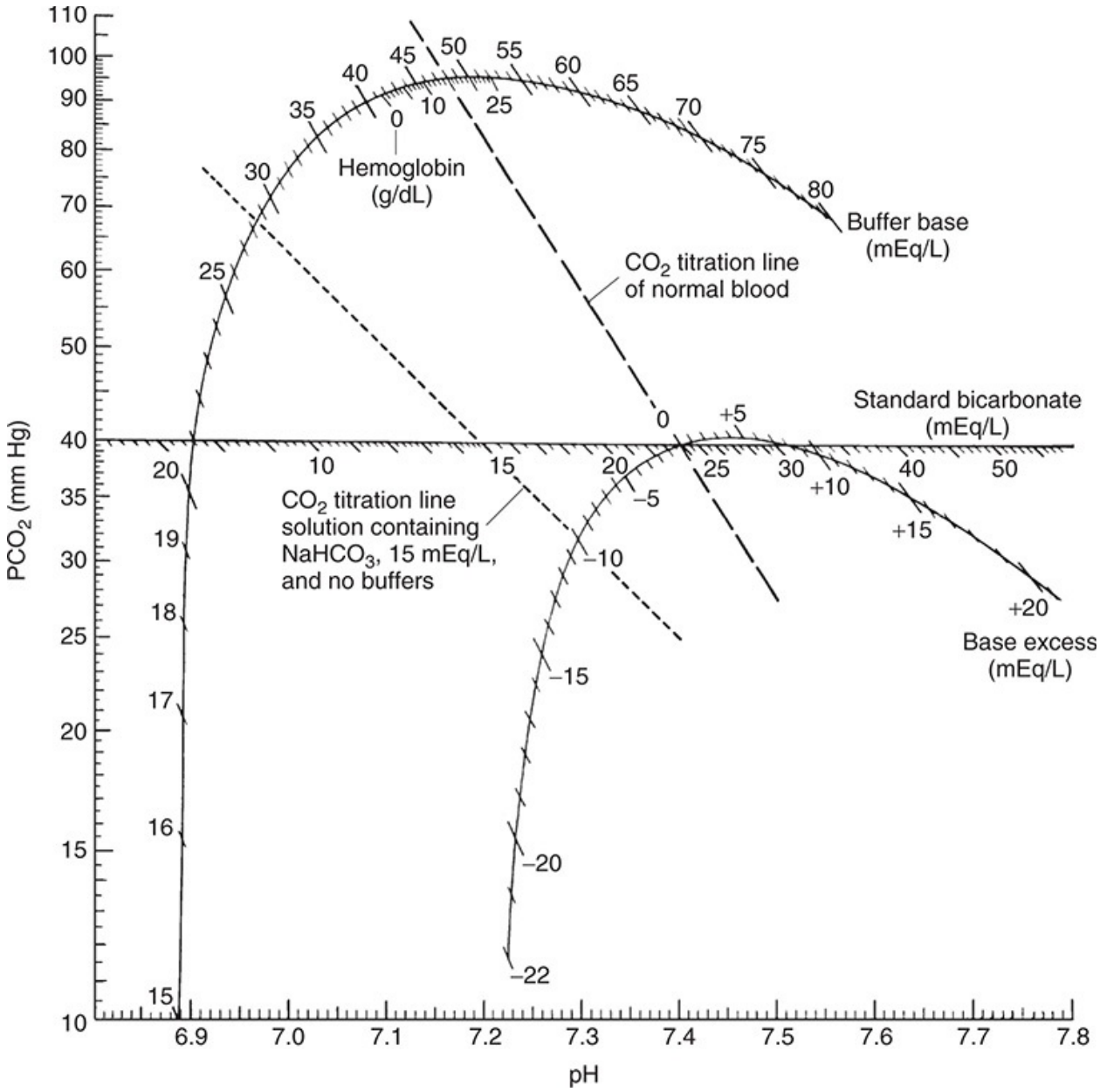


FIGURE 39–6 Siggaard–Andersen curve nomogram. (Used with permission of O Siggaard–Andersen and Radiometer, Copenhagen, Denmark.)

THE SIGGAARD–ANDERSEN CURVE NOMOGRAM

Use of the Siggaard-Andersen curve nomogram (Figure 39–6) to plot the acid–base characteristics of arterial blood is helpful in clinical situations. This nomogram has PCO_2 plotted on a log scale on the vertical axis and pH on the

horizontal axis. Thus, any point to the left of a vertical line through pH 7.40 indicates acidosis, and any point to the right indicates alkalosis. The position of the point above or below the horizontal line through a PCO_2 of 40 mm Hg defines the effective degree of hypoventilation or hyperventilation.

If a solution containing $NaHCO_3$ and no buffers were equilibrated with gas mixtures containing various amounts of CO_2, the pH and PCO_2 values at equilibrium would fall along the dashed line on the left in Figure 39–6 or a line parallel to it. If buffers were present, the slope of the line would be greater; and the greater the buffering capacity of the solution, the steeper the line. For normal blood containing 15 g of hemoglobin/dL, the CO_2 **titration line** passes through the 15-g/dL mark on the hemoglobin scale (on the underside of the upper curved scale) and the point where the PCO_2 = 40 mm Hg and pH = 7.40 lines intersect, as shown in Figure 39–6. When the hemoglobin content of the blood is low, there is significant loss of buffering capacity, and the slope of the CO_2 titration line diminishes. However, blood of course contains buffers in addition to hemoglobin, so that even the line drawn from the zero point on the hemoglobin scale through the normal PCO_2–pH intercept is steeper than the curve for a solution containing no buffers.

For clinical use, arterial blood or arterialized capillary blood is drawn anaerobically and its pH measured. The pHs of the same blood after equilibration with each of two gas mixtures containing different known amounts of CO_2 are also determined. The pH values at the known PCO_2 levels are plotted and connected to provide the CO_2 titration line for the blood sample. The pH of the blood sample before equilibration is plotted on this line, and the PCO_2 of the sample is read off the vertical scale. The **standard bicarbonate** content of the sample is indicated by the point at which the CO_2 titration line intersects the bicarbonate scale on the PCO_2 = 40 mm Hg line. The standard bicarbonate is not the actual bicarbonate concentration of the sample but, rather, what the bicarbonate concentration would be after elimination of any respiratory component. It is a measure of the alkali reserve of the blood, except that it is measured by determining the pH rather than the total CO_2 content of the sample after equilibration. Like the alkali reserve, it is an index of the degree of metabolic acidosis or alkalosis present.

Additional graduations on the upper curved scale of the nomogram (Figure 39–6) are provided for measuring **buffer base** content; the point where the CO_2 calibration line of the arterial blood sample intersects this scale shows the mEq/L

of buffer base in the sample. The buffer base is equal to the total number of buffer anions (principally $Prot^-$, HCO_3^-, and Hb^-) that can accept hydrogen ions in the blood. The normal value in an individual with 15 g of hemoglobin per deciliter of blood is 48 mEq/L.

The point at which the CO_2 calibration line intersects the lower curved scale on the nomogram indicates the **base excess.** This value, which is positive in alkalosis and negative in acidosis, is the amount of acid or base that would restore 1 L of blood to normal acid–base composition at a PCO_2 of 40 mm Hg. It should be noted that a base deficiency cannot be completely corrected simply by calculating the difference between the normal standard bicarbonate (24 mEq/L) and the actual standard bicarbonate and administering this amount of $NaHCO_3$ per liter of blood; some of the added HCO_3^- is converted to CO_2 and H_2O, and the CO_2 is lost in the lungs. The actual amount that must be added is roughly 1.2 times the standard bicarbonate deficit, but the lower curved scale on the nomogram, which has been developed empirically by analyzing many blood samples, is more accurate.

In treating acid–base disturbances, one must, of course, consider not only the blood but also all the body fluid compartments. The other fluid compartments have markedly different concentrations of buffers. It has been determined empirically that administration of an amount of acid (in alkalosis) or base (in acidosis) equal to 50% of the body weight in kilograms times the blood base excess per liter will correct the acid–base disturbance in the whole body. At least when the abnormality is severe, however, it is unwise to attempt such a large correction in a single step; instead, about half the indicated amount should be given and the arterial blood acid–base values determined again. The amount required for final correction can then be calculated and administered. It is also worth noting that, at least in lactic acidosis, $NaHCO_3$ decreases cardiac output and lowers blood pressure, so it should be used with caution.

CHAPTER SUMMARY

- The cells of the proximal and distal tubules secrete hydrogen ions. Acidification also occurs in the collecting ducts. The reaction that is primarily responsible for H^+ secretion in the proximal tubules is Na^+–H^+ exchange. Na is absorbed from the lumen of the tubule and H is excreted.
- The maximal H^+ gradient against which the transport mechanisms can

secrete in humans corresponds to a urine pH of about 4.5. However, three important reactions in the tubular fluid remove free H^+, permitting more acid to be secreted. These are the reactions with HCO_3^- to form CO_2 and H_2O, with HPO_4^{2-} to form $H_2PO_4^-$, and with NH_3 to form NH_4^+.

- Carbonic anhydrase catalyzes the formation of H_2CO_3, and drugs that inhibit carbonic anhydrase depress secretion of acid by the proximal tubules.
- Renal acid secretion is altered by changes in the intracellular PCO_2, K^+ concentration, carbonic anhydrase level, and adrenocortical hormone concentration.

MULTIPLE-CHOICE QUESTIONS

For all questions, select the single best answer unless otherwise directed.

1. Which of the following is the principal buffer in interstitial fluid?

   A. Hemoglobin
   B. Other proteins
   C. Carbonic acid
   D. H_2PO_4
   E. Compounds containing histidine

2. Increasing alveolar ventilation increases the blood pH because

   A. it activates neural mechanisms that remove acid from the blood.
   B. it makes hemoglobin a stronger acid.
   C. it increases the PO_2 of the blood.
   D. it decreases the PCO_2 in the alveoli.
   E. the increased muscle work of increased breathing generates more CO_2.

3. In uncompensated metabolic alkalosis

   A. the plasma pH, the plasma HCO_3^- concentration, and the arterial PCO_2 are all low.
   B. the plasma pH is high and the plasma HCO_3^- concentration and arterial PCO_2 are low.
   C. the plasma pH and the plasma HCO_3^- concentration are low and the arterial PCO_2 is normal.

D. the plasma pH and the plasma HCO_3^- concentration are high and the arterial PCO_2 is normal.

E. the plasma pH is low, the plasma HCO_3^- concentration is high, and the arterial PCO_2 is normal.

4. In a patient with a plasma pH of 7.10, the $[HCO_3^-]/[H_2CO_3]$ ratio in plasma is

A. 20.

B. 10.

C. 2.

D. 1.

E. 0.1.

Answers to Multiple Choice Questions

Chapter 1

1. B **2.** C **3.** B **4.** C **5.** C **6.** D **7.** E **8.** E

Chapter 2

1. A **2.** D **3.** D **4.** B **5.** C **6.** C **7.** B **8.** A

Chapter 3

1. C **2.** B **3.** D **4.** C **5.** B **6.** B **7.** E **8.** D

Chapter 4

1. C **2.** E **3.** E **4.** E **5.** C **6.** B **7.** E **8.** C **9.** D **10.** D

Chapter 5

1. B **2.** D **3.** B **4.** C **5.** C

Chapter 6

1. C **2.** C **3.** A **4.** E **5.** B **6.** D **7.** B **8.** D **9.** E

Chapter 7

1. D **2.** B **3.** D **4.** D **5.** B **6.** E **7.** E **8.** C **9.** E

Chapter 8

1. A **2.** C **3.** D **4.** E **5.** E **6.** C **7.** B **8.** D **9.** A **10.** E **11.** D

Chapter 9

1. D **2.** B **3.** D **4.** D **5.** A **6.** C **7.** E **8.** D

Chapter 10

1. D **2.** B **3.** C **4.** B **5.** E **6.** E **7.** D **8.** B **9.** D **10.** A **11.** A **12.** B **13.** B

Chapter 11

1. E **2.** D **3.** B **4.** E **5.** A **6.** C **7.** D **8.** D **9.** E **10.** D **11.** B **12.** D

Chapter 12

1. D **2.** E **3.** C **4.** C **5.** A **6.** E **7.** E **8.** B **9.** A **10.** E **11.** C **12.** D **13**. E

Chapter 13

1. A **2.** D **3.** C **4.** E **5.** D **6.** C **7.** B **8**. D

Chapter 14

1. D **2.** D **3.** C **4.** D **5.** B **6.** C **7.** A **8.** E **9.** B

Chapter 15

1. D **2.** B **3.** B **4.** A **5.** E **6.** D **7.** D **8.** B

Chapter 16

No multiple choice questions

Chapter 17

1. B **2.** E **3.** B **4.** A **5.** A **6.** B **7.** D **8.** D

Chapter 18

1. E **2.** E **3.** D **4.** C **5.** B **6.** E **7.** C **8.** A

Chapter 19

1. D **2.** B **3.** E **4.** D **5.** C **6.** D **7.** D **8.** A **9.** A

Chapter 20

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Chapter 21

1. C **2.** E **3.** D **4.** D **5.** A **6.** C **7.** E **8.** D

Chapter 22

1. C **2.** D **3.** C **4.** A

Chapter 23

1. E **2.** A **3.** C **4.** B

Chapter 24

1. E **2.** D **3.** D **4.** C **5.** E **6.** D **7.** C

Chapter 25

1. C **2.** D **3.** D **4.** C **5.** D **6.** B **7.** B **8.** C **9.** E

Chapter 26

1. C **2.** B **3.** E **4.** E **5.** A **6.** D **7.** E **8.** D **9.** D **10.** B

Chapter 27

1. D **2.** C **3.** A **4.** B **5.** E **6.** D **7.** D **8.** E **9.** C **10.** C

Chapter 28

1. C **2.** C **3.** E **4.** E **5.** D **6.** E **7.** B **8.** C **9.** E **10.** A

Chapter 29

1. C **2.** A **3.** A **4.** D **5.** D

Chapter 30

1. C **2.** E **3.** C **4.** A **5.** C **6.** C **7.** D **8.** C

Chapter 31

1. E **2.** D **3.** C **4.** A **5.** C **6.** B **7.** E **8.** D **9.** D **10.** A **11.** B **12.** B

Chapter 32

1. B **2.** D **3.** B **4.** C **5.** E **6.** C **7.** D **8.** D **9.** E

Chapter 33

1. E **2.** D **3.** B **4.** A **5.** E **6.** E **7.** D **8.** D

Chapter 34

1. D **2.** C **3.** A **4.** E **5.** D **6.** A

Chapter 35

1. E **2.** B **3.** D **4.** D

Chapter 36

1. D **2.** B **3.** B **4.** D **5.** E **6.** B **7.** C

Chapter 37

1. A **2.** A **3.** A **4.** A **5.** E **6.** C **7.** D

Chapter 38

1. E **2.** C **3.** D **4.** E **5.** D **6.** C **7.** D

Chapter 39

1. C **2.** D **3.** D **4.** B

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